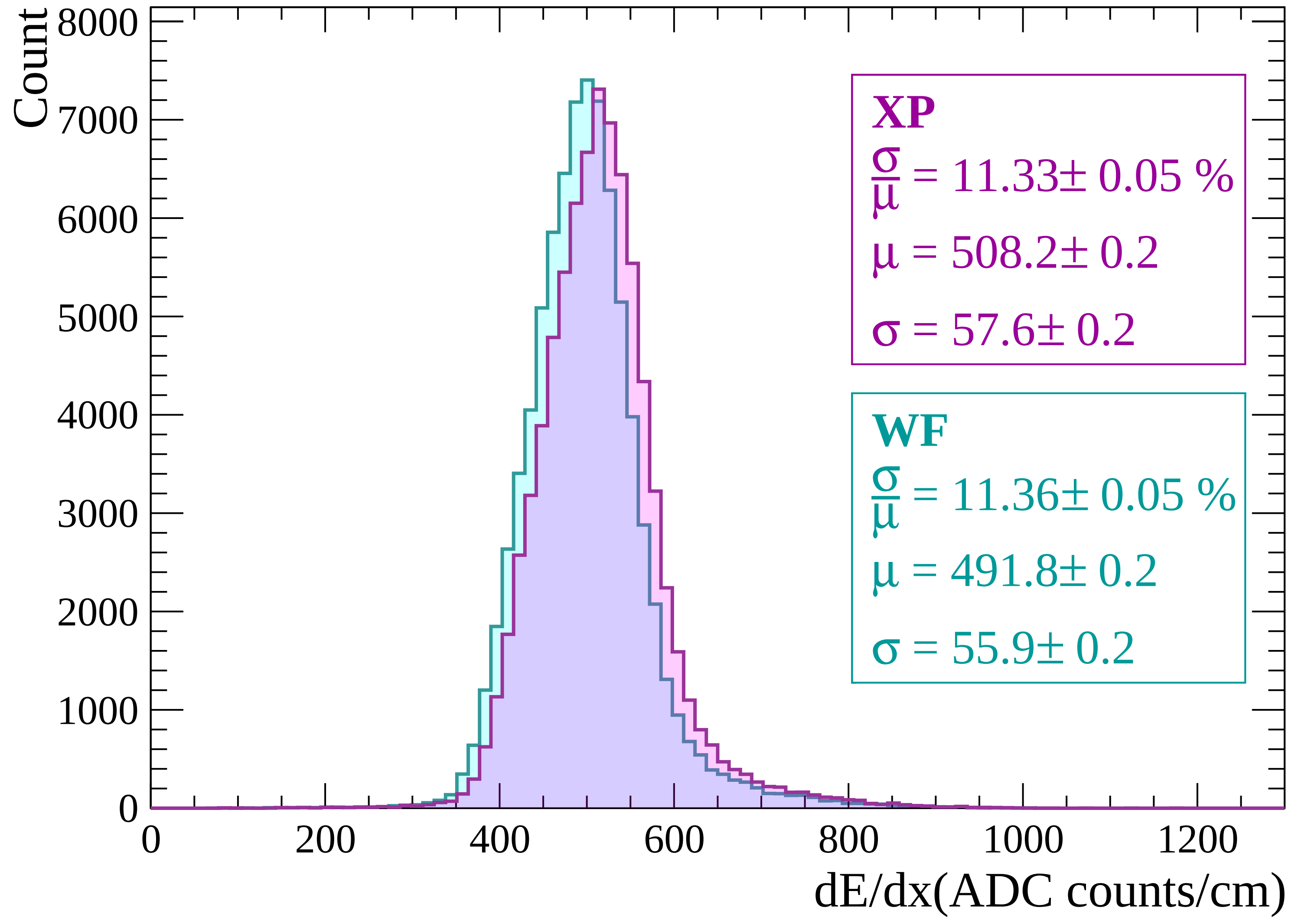
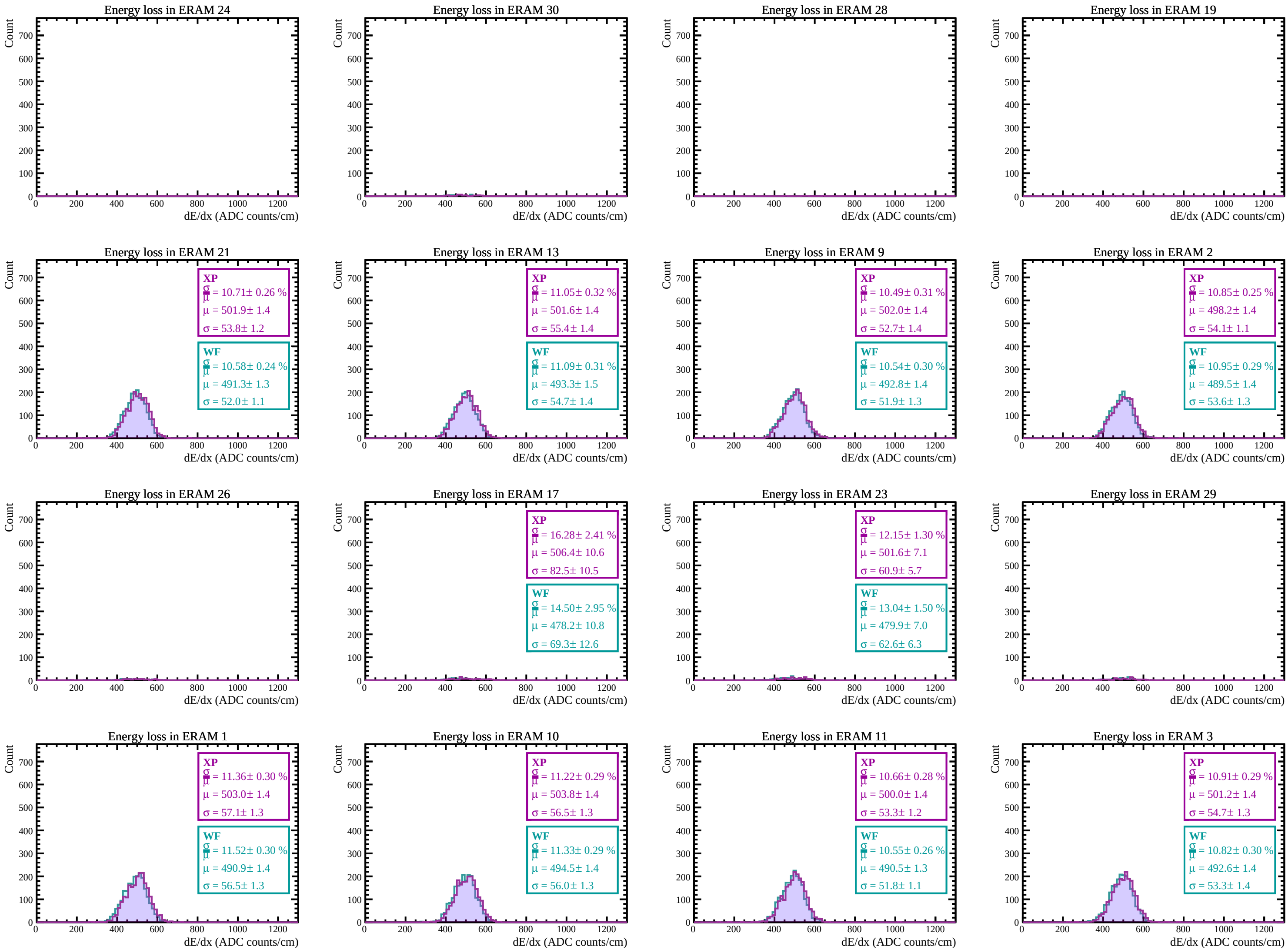
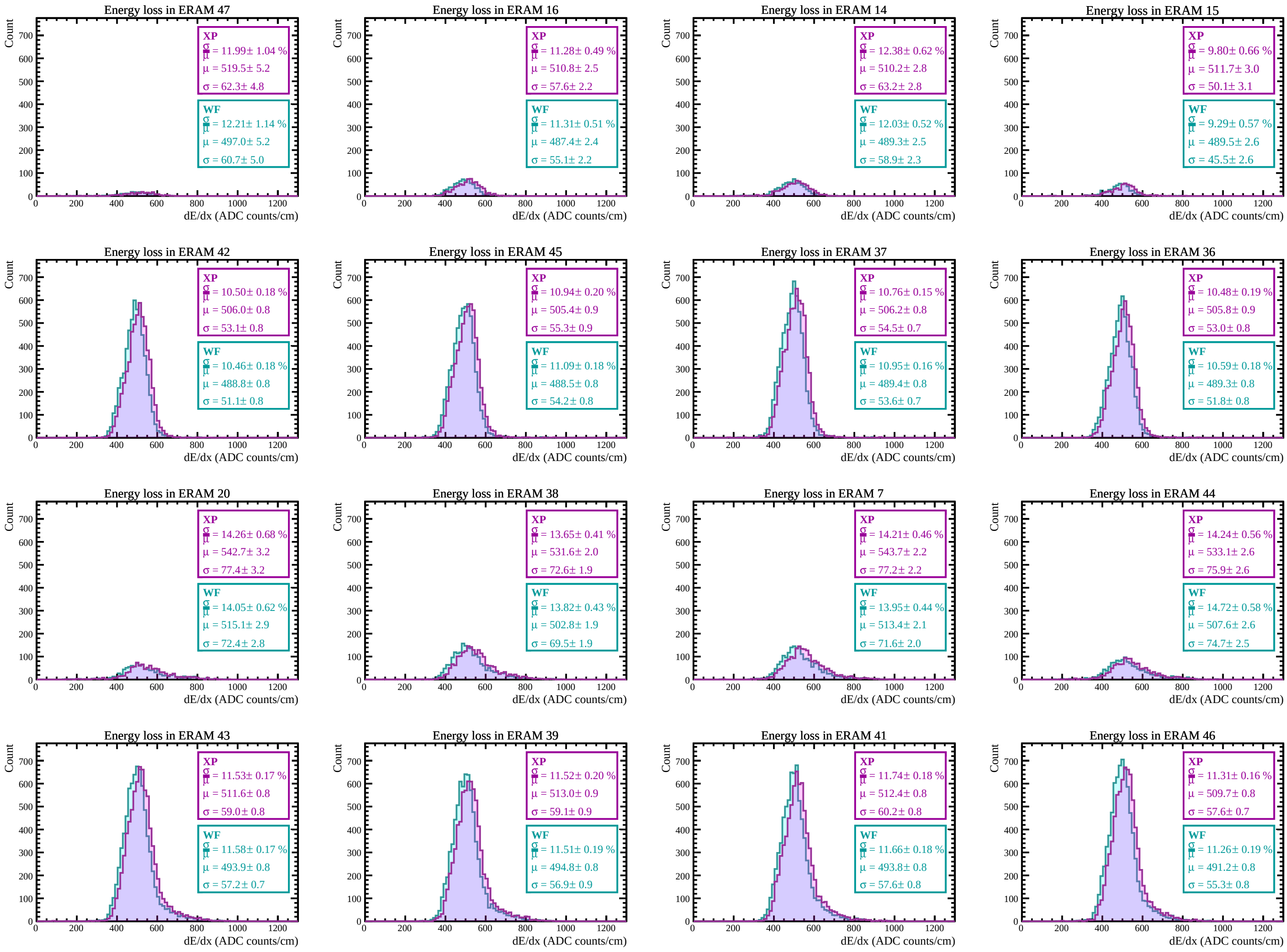
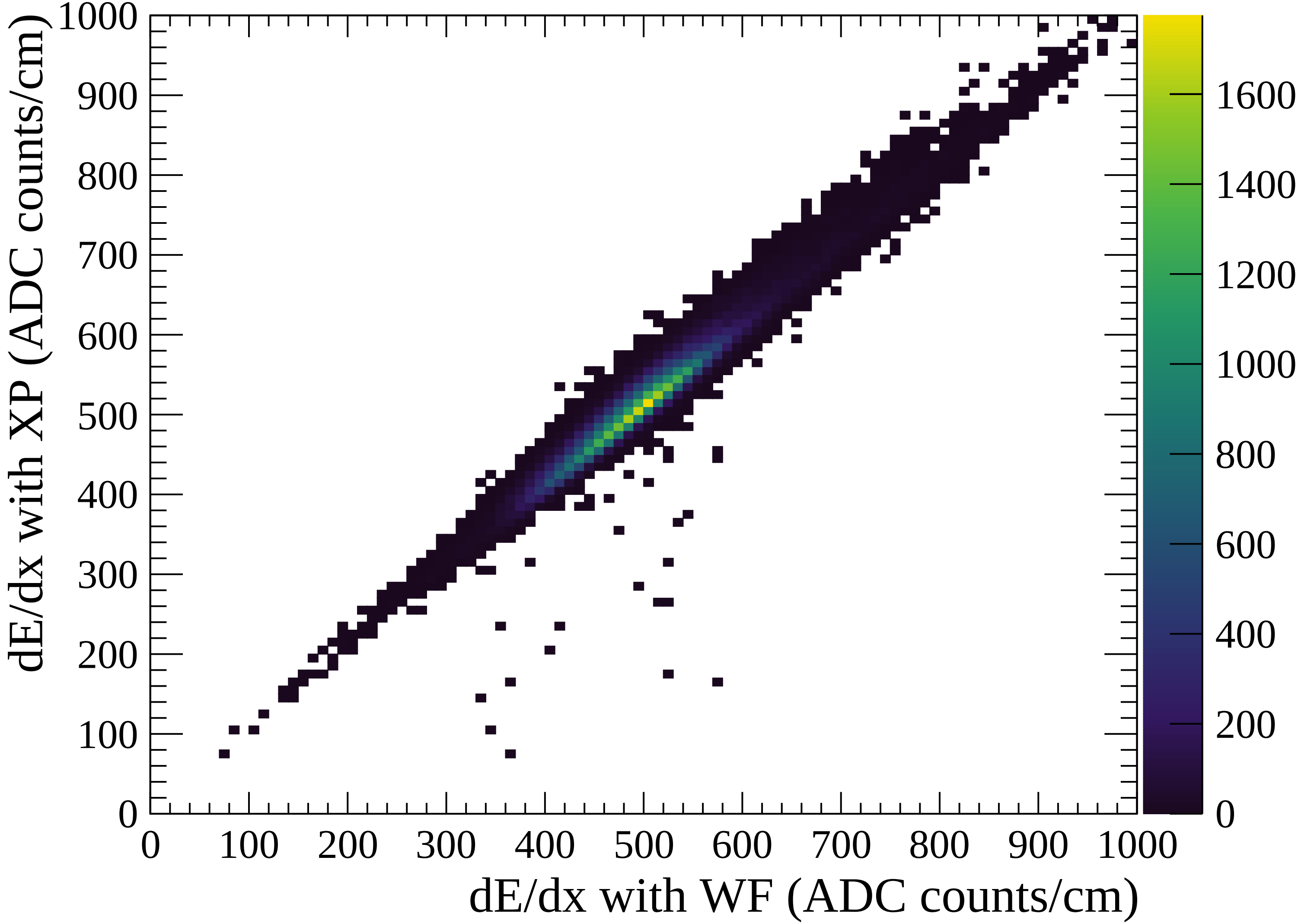

Count
8000
7000
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.33± 0.05 %
μ = 508.2± 0.2
σ = 57.6± 0.2
WF
$\frac{\sigma}{\mu}$ = 11.36± 0.05 %
μ = 491.8± 0.2
σ = 55.9± 0.2

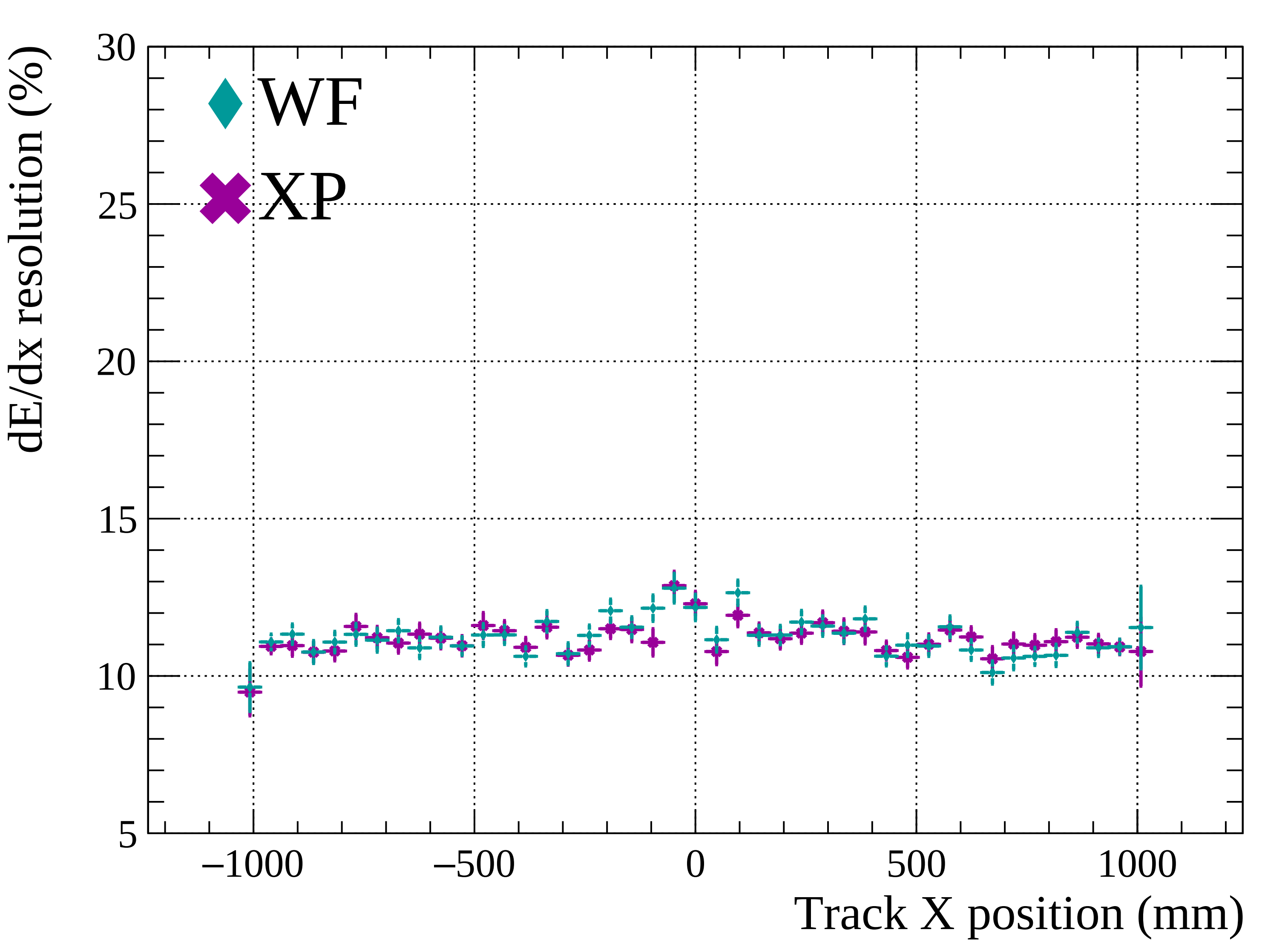


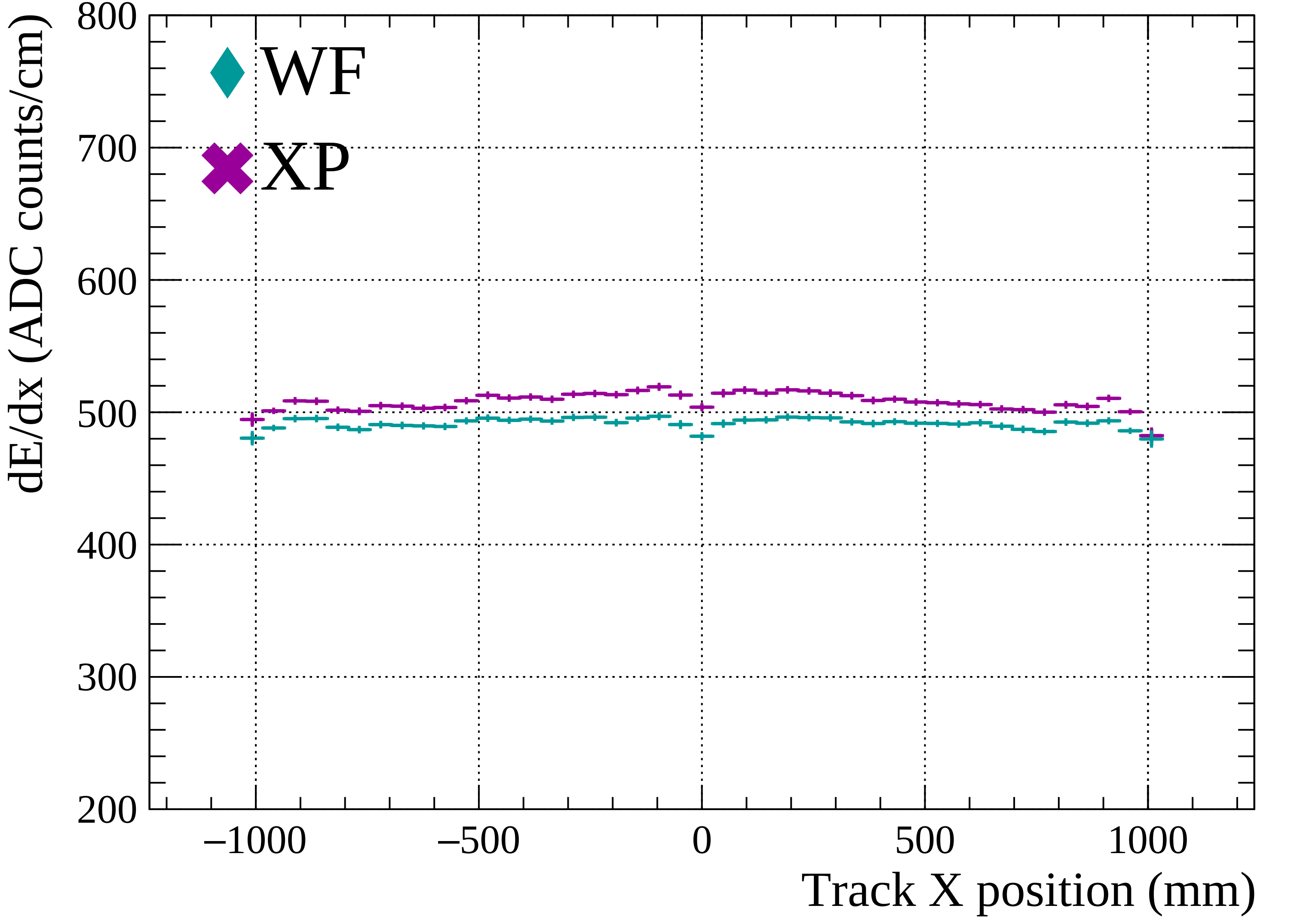
Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
Energy loss in ERAM 21
XP
σ/μ = 10.71± 0.26 %
μ = 501.9± 1.4
σ = 53.8± 1.2
WF
σ/μ = 10.58± 0.24 %
μ = 491.3± 1.3
σ = 52.0± 1.1
Energy loss in ERAM 13
XP
σ/μ = 11.05± 0.32 %
μ = 501.6± 1.4
σ = 55.4± 1.4
WF
σ/μ = 11.09± 0.31 %
μ = 493.3± 1.5
σ = 54.7± 1.4
Energy loss in ERAM 9
XP
σ/μ = 10.49± 0.31 %
μ = 502.0± 1.4
σ = 52.7± 1.4
WF
σ/μ = 10.54± 0.30 %
μ = 492.8± 1.4
σ = 51.9± 1.3
Energy loss in ERAM 2
XP
σ/μ = 10.85± 0.25 %
μ = 498.2± 1.4
σ = 54.1± 1.1
WF
σ/μ = 10.95± 0.29 %
μ = 489.5± 1.4
σ = 53.6± 1.3
Energy loss in ERAM 26
Energy loss in ERAM 17
XP
σ/μ = 16.28± 2.41 %
μ = 506.4± 10.6
σ = 82.5± 10.5
WF
σ/μ = 14.50± 2.95 %
μ = 478.2± 10.8
σ = 69.3± 12.6
Energy loss in ERAM 23
XP
σ/μ = 12.15± 1.30 %
μ = 501.6± 7.1
σ = 60.9± 5.7
WF
σ/μ = 13.04± 1.50 %
μ = 479.9± 7.0
σ = 62.6± 6.3
Energy loss in ERAM 29
Energy loss in ERAM 1
XP
σ/μ = 11.36± 0.30 %
μ = 503.0± 1.4
σ = 57.1± 1.3
WF
σ/μ = 11.52± 0.30 %
μ = 490.9± 1.4
σ = 56.5± 1.3
Energy loss in ERAM 10
XP
σ/μ = 11.22± 0.29 %
μ = 503.8± 1.4
σ = 56.5± 1.3
WF
σ/μ = 11.33± 0.29 %
μ = 494.5± 1.4
σ = 56.0± 1.3
Energy loss in ERAM 11
XP
σ/μ = 10.66± 0.28 %
μ = 500.0± 1.4
σ = 53.3± 1.2
WF
σ/μ = 10.55± 0.26 %
μ = 490.5± 1.3
σ = 51.8± 1.1
Energy loss in ERAM 3
XP
σ/μ = 10.91± 0.29 %
μ = 501.2± 1.4
σ = 54.7± 1.3
WF
σ/μ = 10.82± 0.30 %
μ = 492.6± 1.4
σ = 53.3± 1.4
Count
dE/dx (ADC counts/cm)


Energy loss in ERAM 47
XP
σ/μ = 11.99± 1.04 %
μ = 519.5± 5.2
σ = 62.3± 4.8
WF
σ/μ = 12.21± 1.14 %
μ = 497.0± 5.2
σ = 60.7± 5.0
Energy loss in ERAM 16
XP
σ/μ = 11.28± 0.49 %
μ = 510.8± 2.5
σ = 57.6± 2.2
WF
σ/μ = 11.31± 0.51 %
μ = 487.4± 2.4
σ = 55.1± 2.2
Energy loss in ERAM 14
XP
σ/μ = 12.38± 0.62 %
μ = 510.2± 2.8
σ = 63.2± 2.8
WF
σ/μ = 12.03± 0.52 %
μ = 489.3± 2.5
σ = 58.9± 2.3
Energy loss in ERAM 15
XP
σ/μ = 9.80± 0.66 %
μ = 511.7± 3.0
σ = 50.1± 3.1
WF
σ/μ = 9.29± 0.57 %
μ = 489.5± 2.6
σ = 45.5± 2.6
Energy loss in ERAM 42
XP
σ/μ = 10.50± 0.18 %
μ = 506.0± 0.8
σ = 53.1± 0.8
WF
σ/μ = 10.46± 0.18 %
μ = 488.8± 0.8
σ = 51.1± 0.8
Energy loss in ERAM 45
XP
σ/μ = 10.94± 0.20 %
μ = 505.4± 0.9
σ = 55.3± 0.9
WF
σ/μ = 11.09± 0.18 %
μ = 488.5± 0.8
σ = 54.2± 0.8
Energy loss in ERAM 37
XP
σ/μ = 10.76± 0.15 %
μ = 506.2± 0.8
σ = 54.5± 0.7
WF
σ/μ = 10.95± 0.16 %
μ = 489.4± 0.8
σ = 53.6± 0.7
Energy loss in ERAM 36
XP
σ/μ = 10.48± 0.19 %
μ = 505.8± 0.9
σ = 53.0± 0.8
WF
σ/μ = 10.59± 0.18 %
μ = 489.3± 0.8
σ = 51.8± 0.8
Energy loss in ERAM 20
XP
σ/μ = 14.26± 0.68 %
μ = 542.7± 3.2
σ = 77.4± 3.2
WF
σ/μ = 14.05± 0.62 %
μ = 515.1± 2.9
σ = 72.4± 2.8
Energy loss in ERAM 38
XP
σ/μ = 13.65± 0.41 %
μ = 531.6± 2.0
σ = 72.6± 1.9
WF
σ/μ = 13.82± 0.43 %
μ = 502.8± 1.9
σ = 69.5± 1.9
Energy loss in ERAM 7
XP
σ/μ = 14.21± 0.46 %
μ = 543.7± 2.2
σ = 77.2± 2.2
WF
σ/μ = 13.95± 0.44 %
μ = 513.4± 2.1
σ = 71.6± 2.0
Energy loss in ERAM 44
XP
σ/μ = 14.24± 0.56 %
μ = 533.1± 2.6
σ = 75.9± 2.6
WF
σ/μ = 14.72± 0.58 %
μ = 507.6± 2.6
σ = 74.7± 2.5
Energy loss in ERAM 43
XP
σ/μ = 11.53± 0.17 %
μ = 511.6± 0.8
σ = 59.0± 0.8
WF
σ/μ = 11.58± 0.17 %
μ = 493.9± 0.8
σ = 57.2± 0.7
Energy loss in ERAM 39
XP
σ/μ = 11.52± 0.20 %
μ = 513.0± 0.9
σ = 59.1± 0.9
WF
σ/μ = 11.51± 0.19 %
μ = 494.8± 0.8
σ = 56.9± 0.9
Energy loss in ERAM 41
XP
σ/μ = 11.74± 0.18 %
μ = 512.4± 0.8
σ = 60.2± 0.8
WF
σ/μ = 11.66± 0.18 %
μ = 493.8± 0.8
σ = 57.6± 0.8
Energy loss in ERAM 46
XP
σ/μ = 11.31± 0.16 %
μ = 509.7± 0.8
σ = 57.6± 0.7
WF
σ/μ = 11.26± 0.19 %
μ = 491.2± 0.8
σ = 55.3± 0.8
Count
dE/dx (ADC counts/cm)

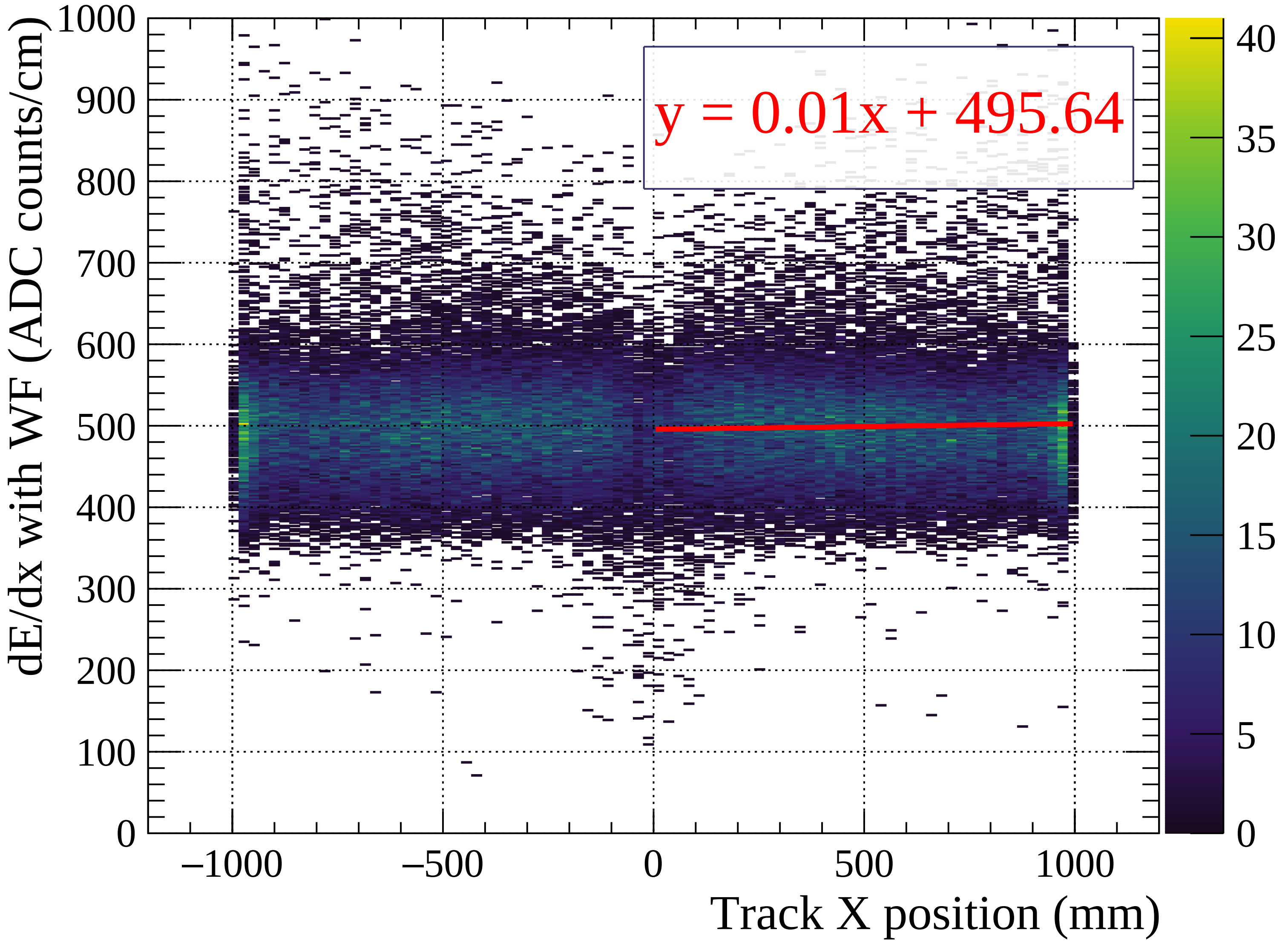


dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
1600
1400
1200
1000
800
600
400
200

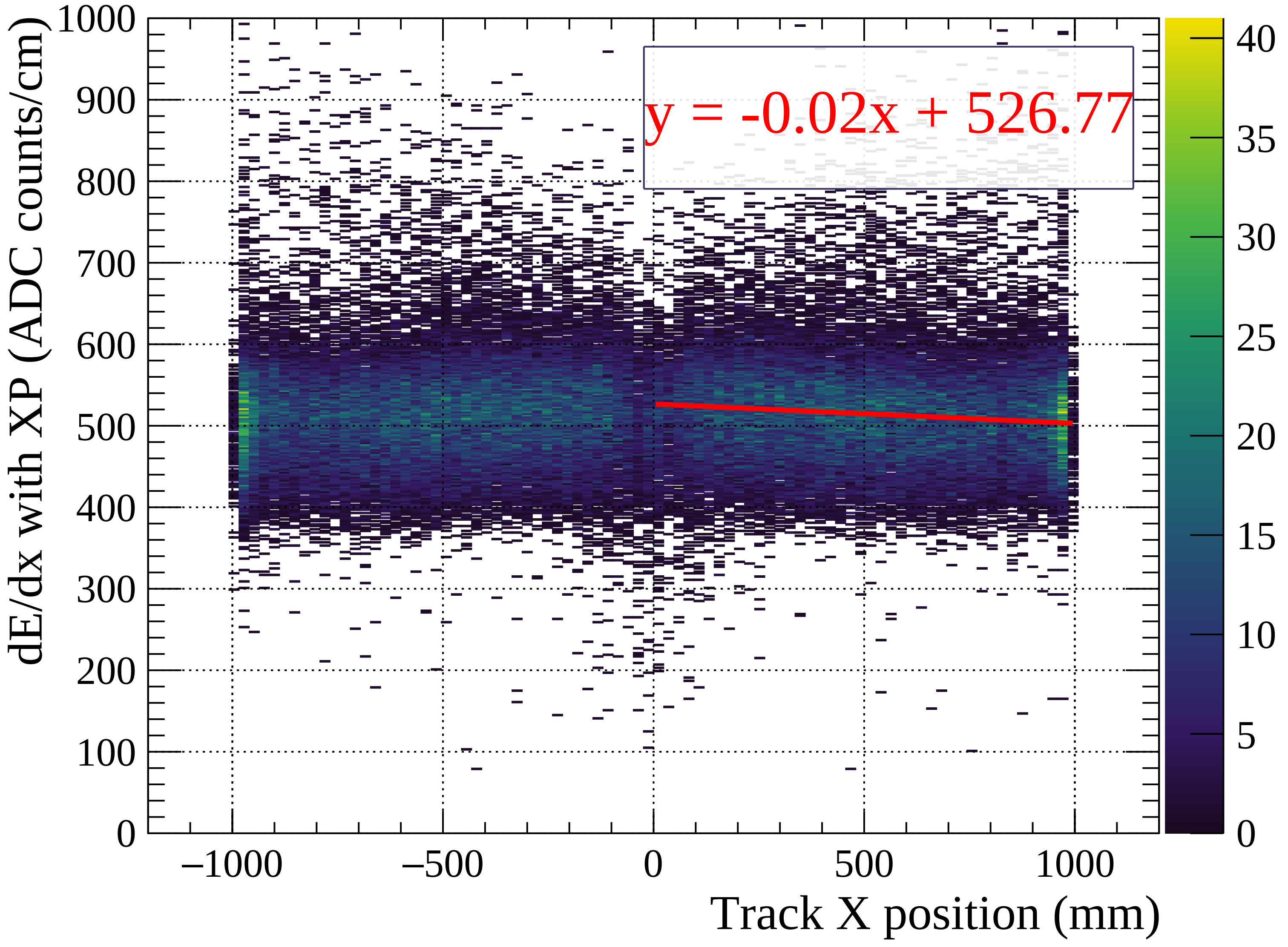

WF
XP
dE/dx resolution (%)
Track X position (mm)
30
25
20
15
10
5
−1000
−500
0
500
1000

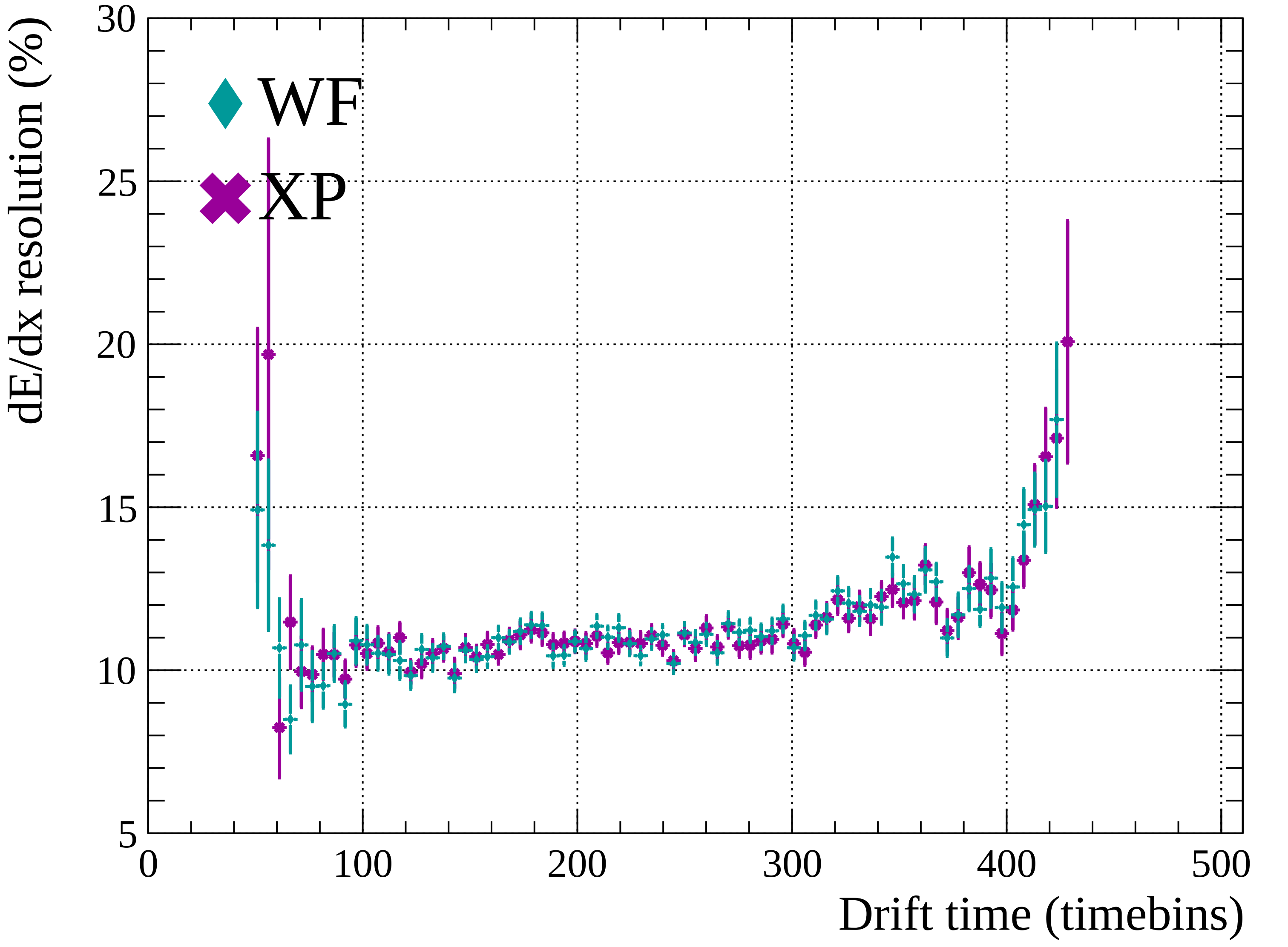


WF
XP
dE/dx (ADC counts/cm)
800
700
600
500
400
300
200
−1000
−500
0
500
1000
Track X position (mm)

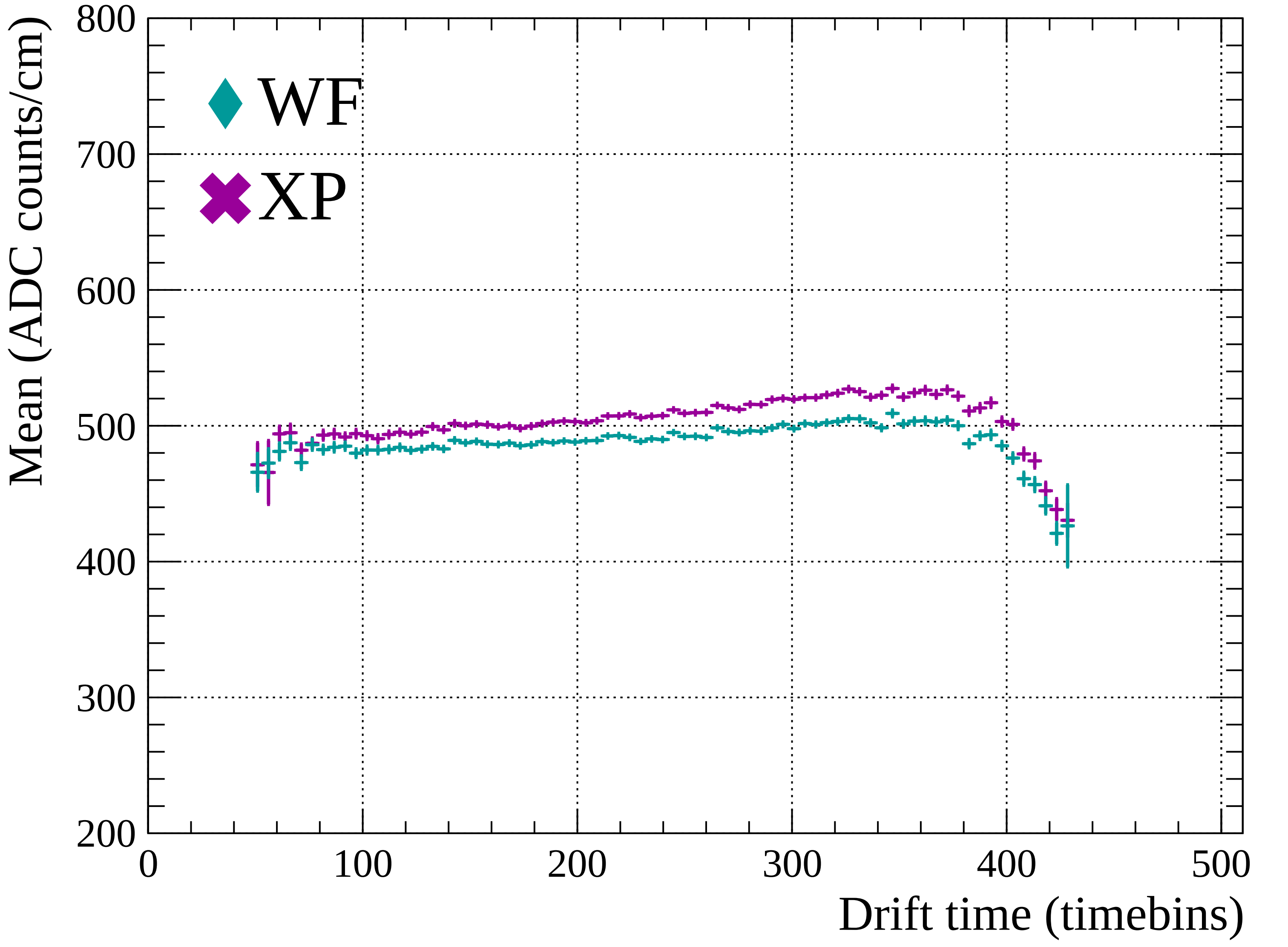


y = 0.01x + 495.64
dE/dx with WF (ADC counts/cm)
Track X position (mm)
1000
900
800
700
600
500
400
300
200
100
0
−1000
−500
0
500
1000
40
35
30
25
20
15
10
5
0

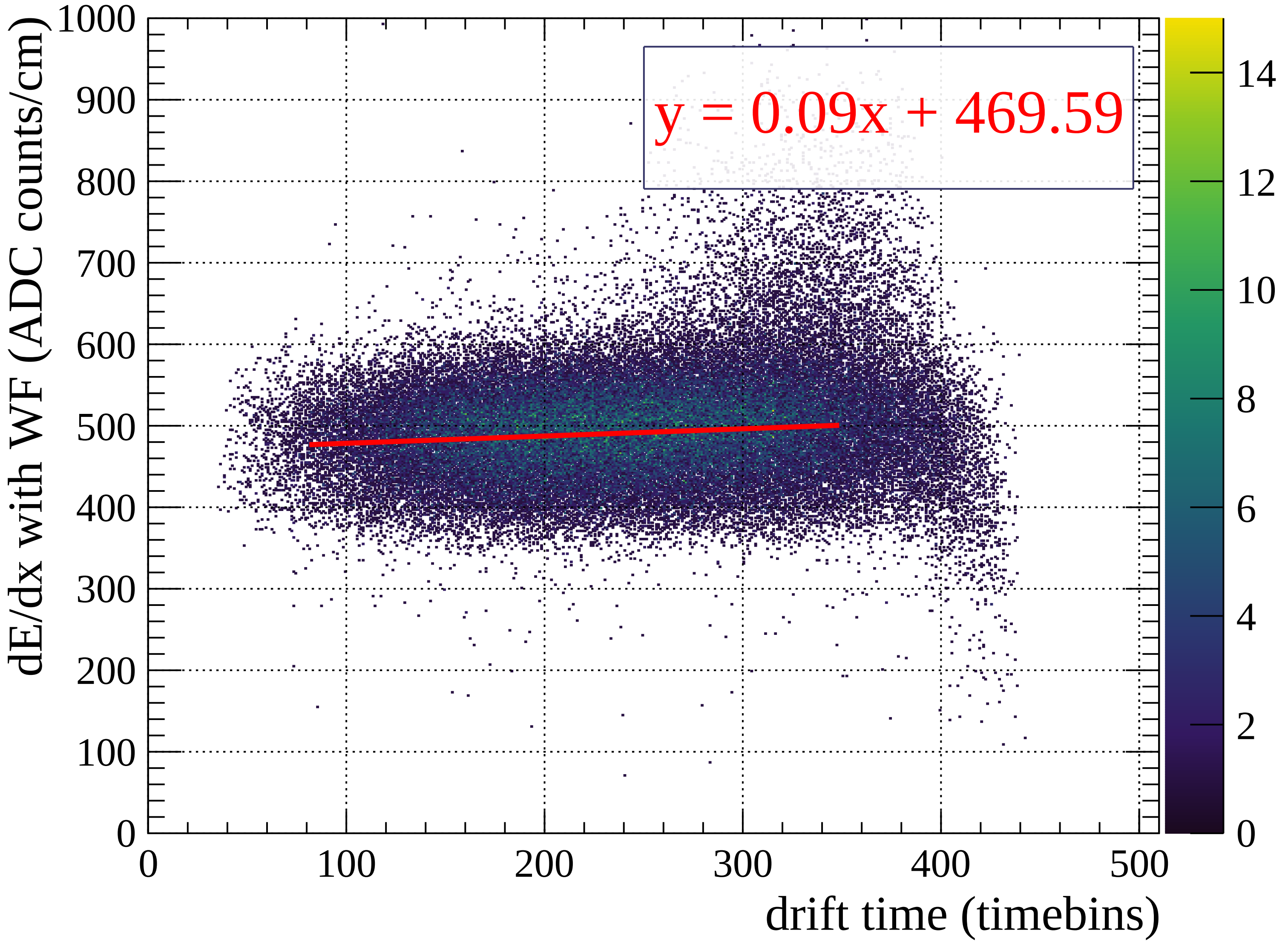

y = -0.02x + 526.77
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
5
10
15
20
25
30
35
40



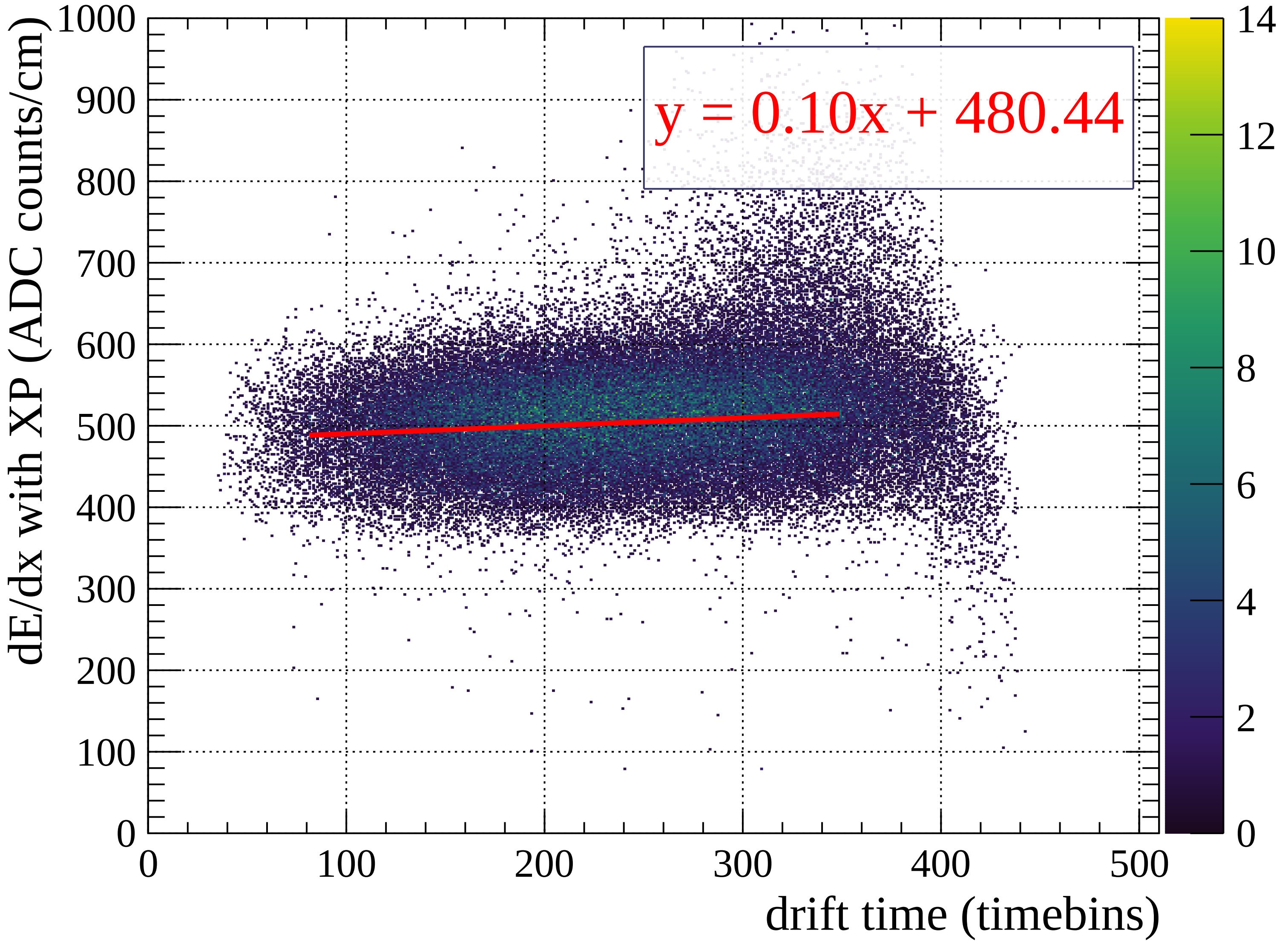
dE/dx resolution (%)
30
25
20
15
10
5
0
100
200
300
400
500
Drift time (timebins)
WF
XP

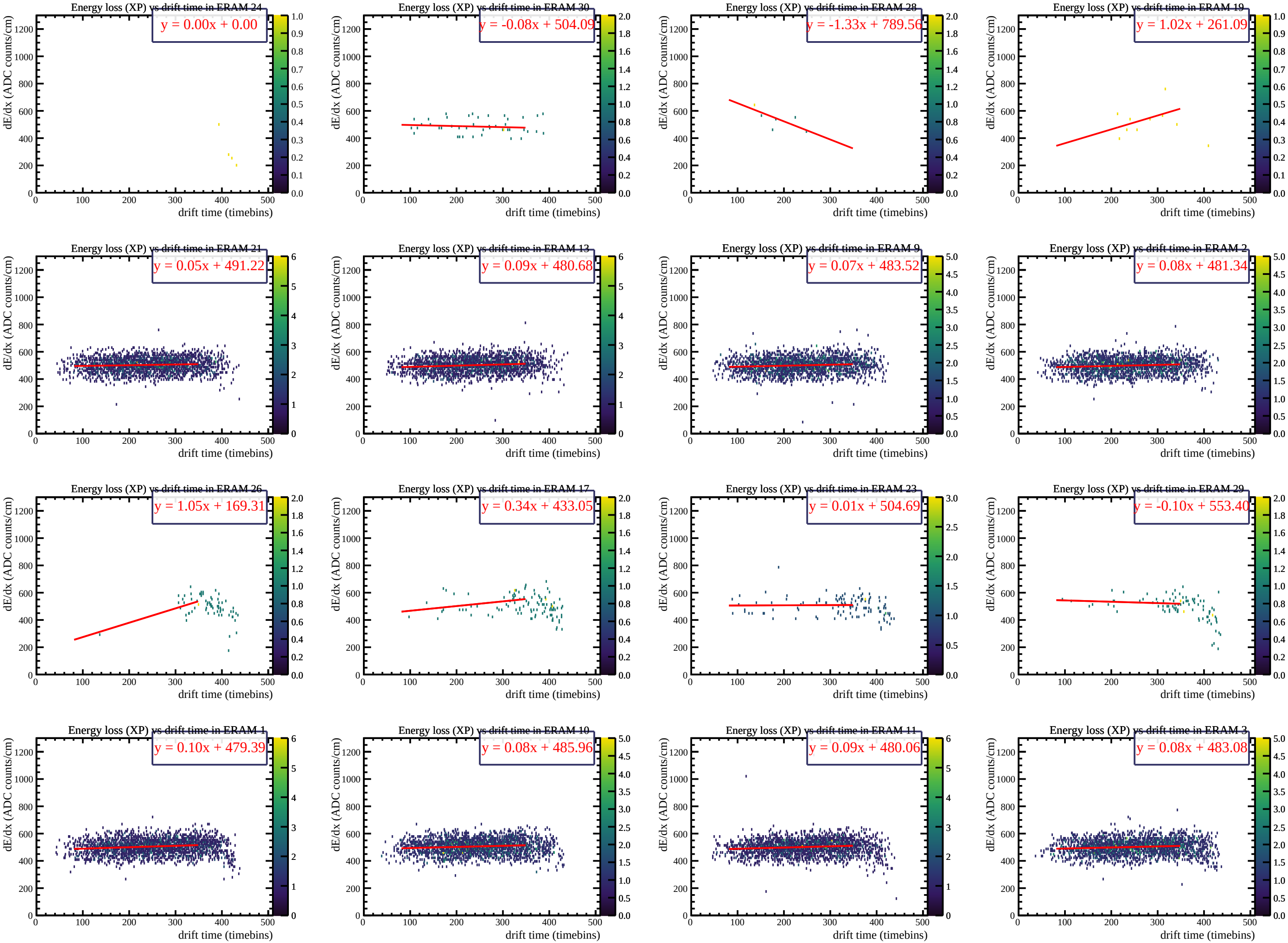


WF
XP
Mean (ADC counts/cm)
Drift time (timebins)
800
700
600
500
400
300
200
0
100
200
300
400
500

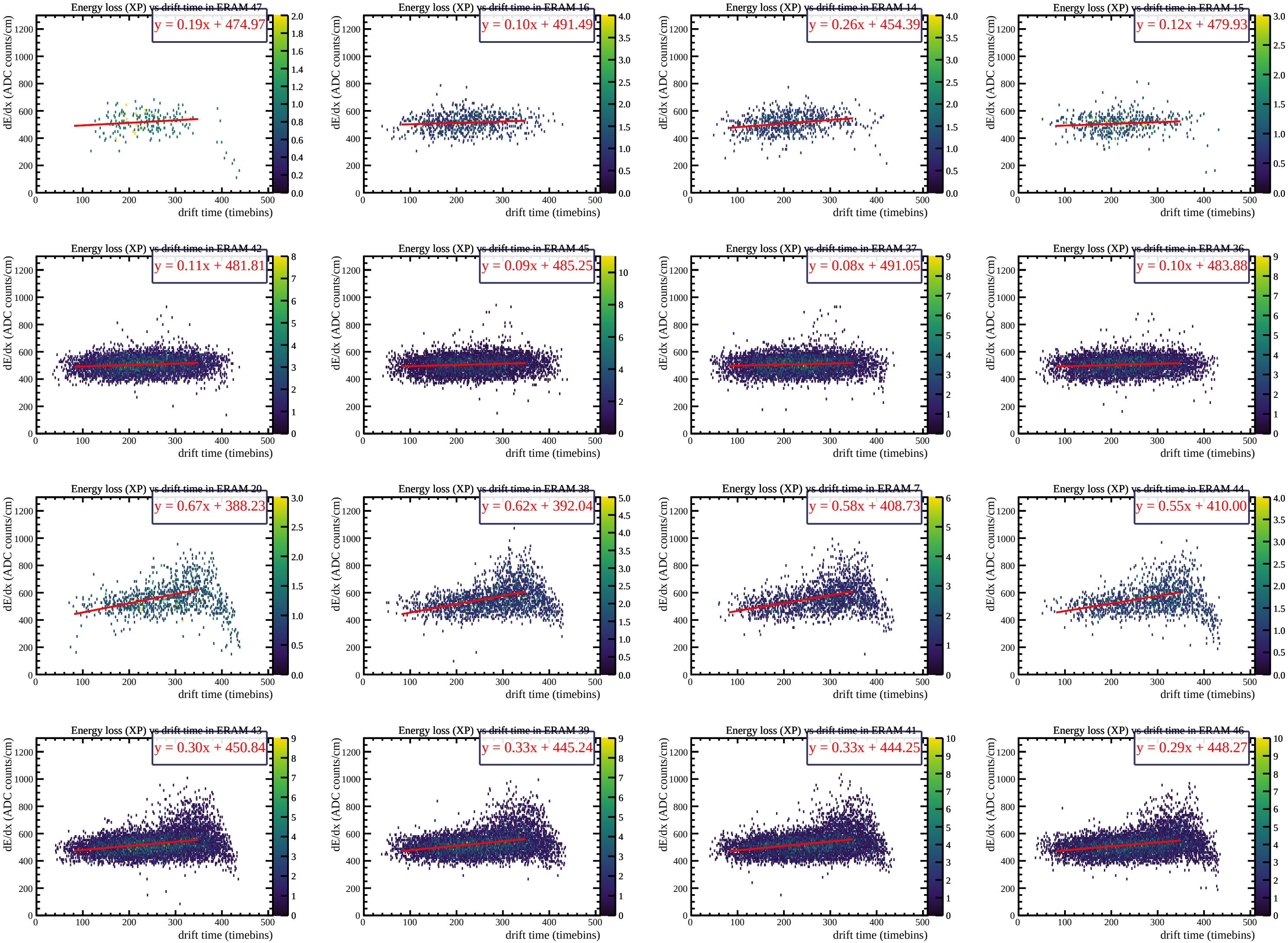


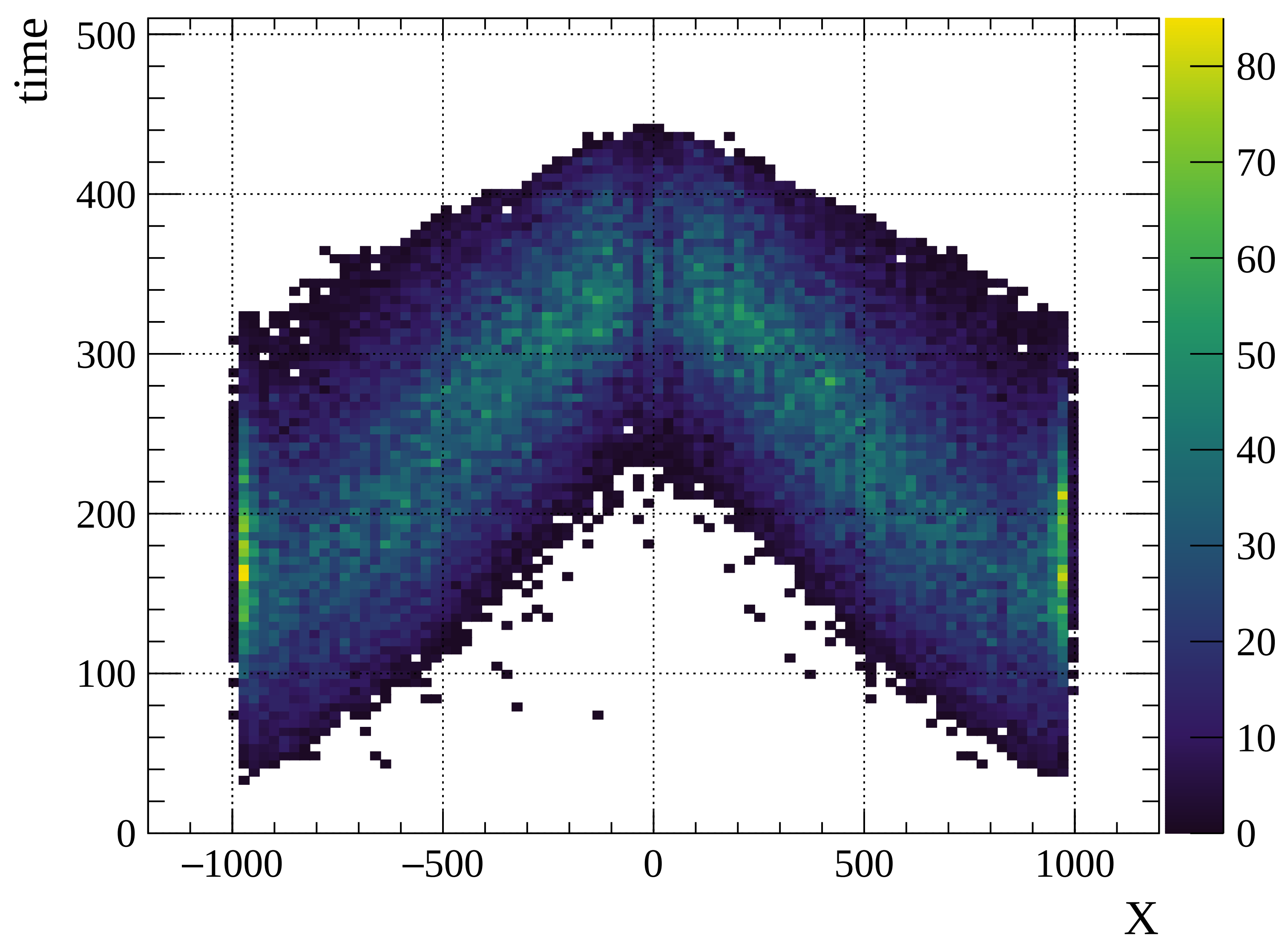
y = 0.09x + 469.59
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14


y = 0.10x + 480.44
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14

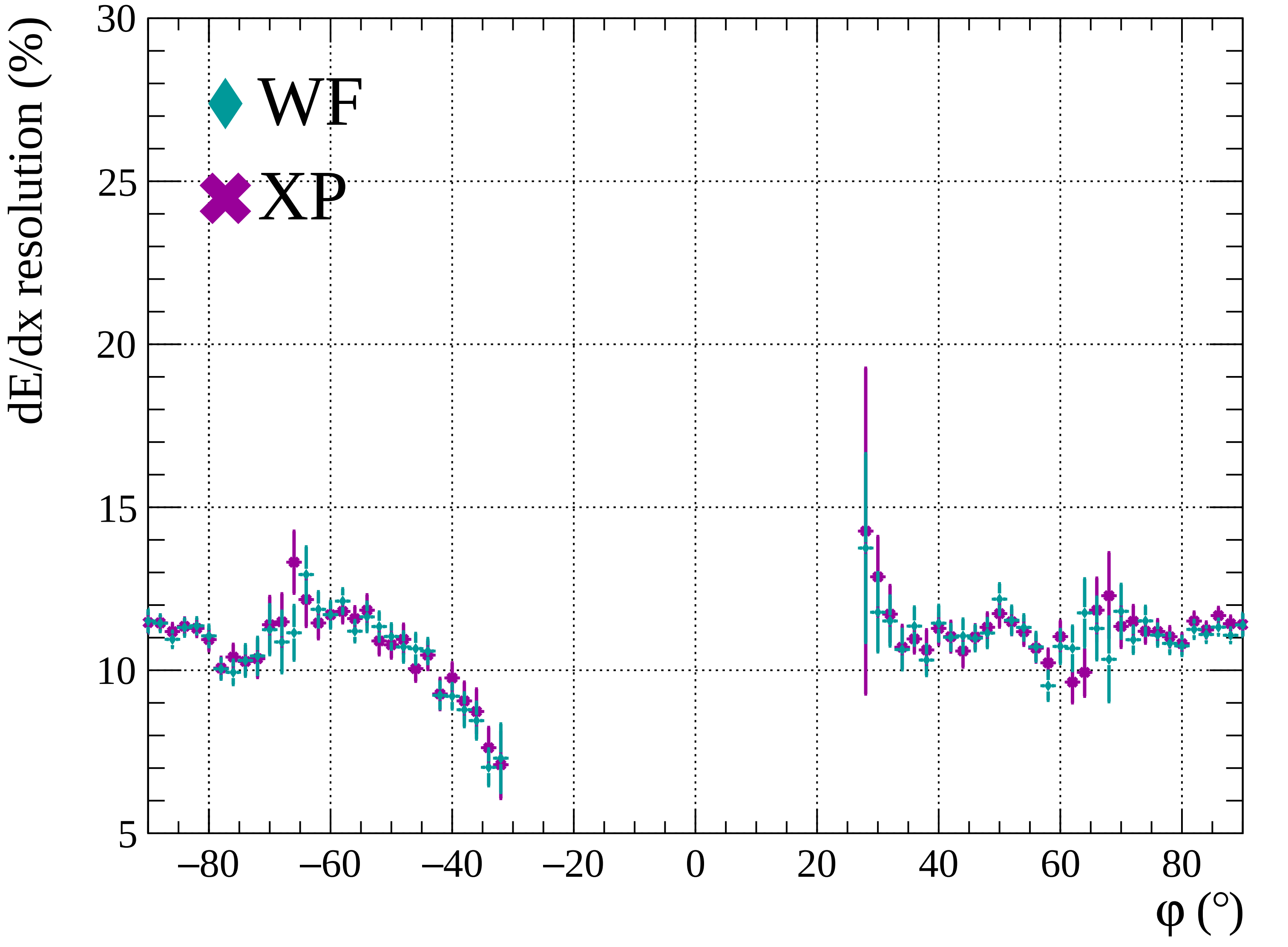


Energy loss (XP) vs drift time in ERAM 24
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 30
y = -0.08x + 504.09
Energy loss (XP) vs drift time in ERAM 28
y = -1.33x + 789.56
Energy loss (XP) vs drift time in ERAM 19
y = 1.02x + 261.09
Energy loss (XP) vs drift time in ERAM 21
y = 0.05x + 491.22
Energy loss (XP) vs drift time in ERAM 13
y = 0.09x + 480.68
Energy loss (XP) vs drift time in ERAM 9
y = 0.07x + 483.52
Energy loss (XP) vs drift time in ERAM 2
y = 0.08x + 481.34
Energy loss (XP) vs drift time in ERAM 26
y = 1.05x + 169.31
Energy loss (XP) vs drift time in ERAM 17
y = 0.34x + 433.05
Energy loss (XP) vs drift time in ERAM 23
y = 0.01x + 504.69
Energy loss (XP) vs drift time in ERAM 29
y = -0.10x + 553.40
Energy loss (XP) vs drift time in ERAM 1
y = 0.10x + 479.39
Energy loss (XP) vs drift time in ERAM 10
y = 0.08x + 485.96
Energy loss (XP) vs drift time in ERAM 11
y = 0.09x + 480.06
Energy loss (XP) vs drift time in ERAM 3
y = 0.08x + 483.08
dE/dx (ADC counts/cm)
drift time (timebins)

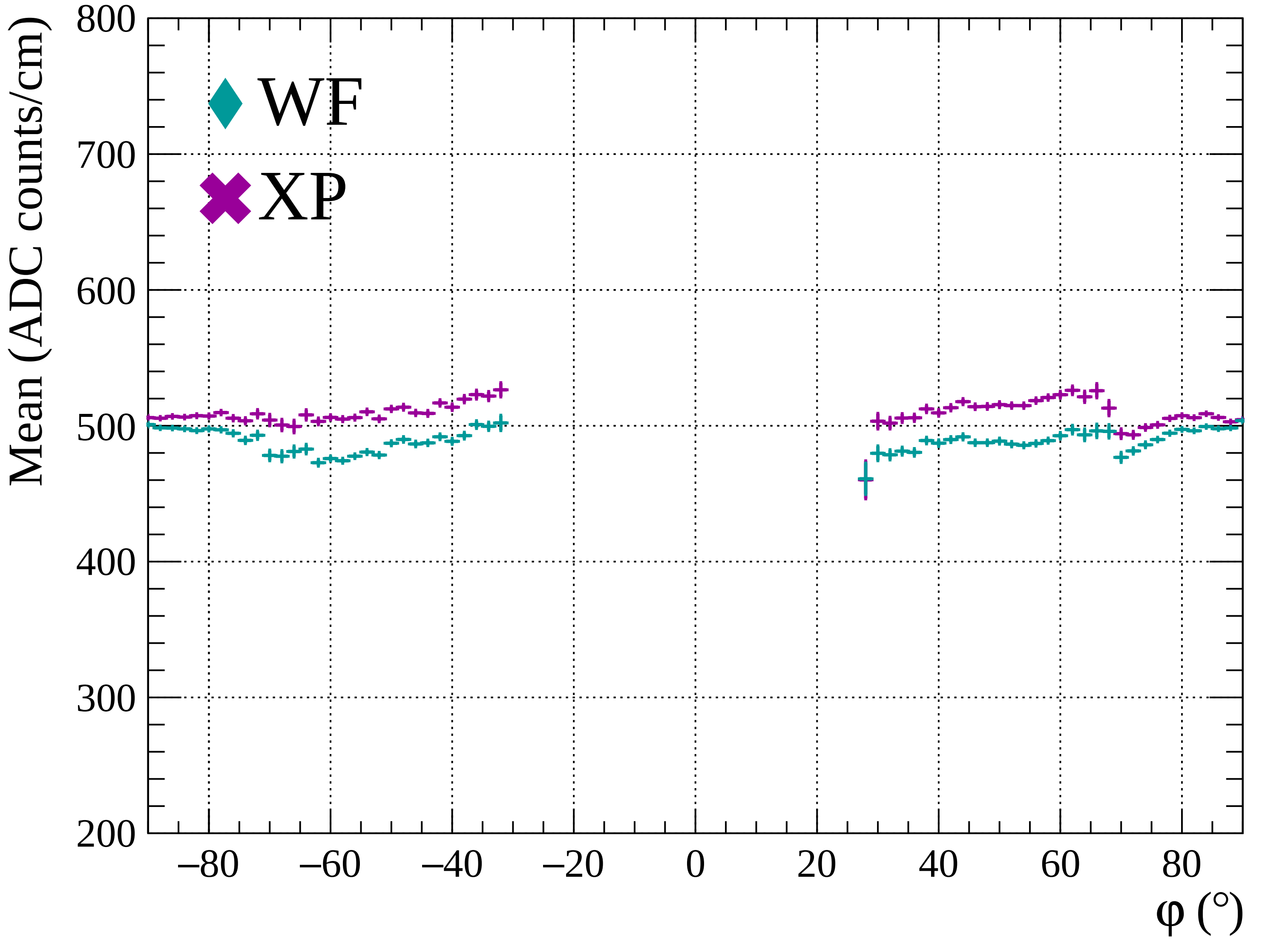

Energy loss (XP) vs drift time in ERAM 47
y = 0.19x + 474.97
Energy loss (XP) vs drift time in ERAM 16
y = 0.10x + 491.49
Energy loss (XP) vs drift time in ERAM 14
y = 0.26x + 454.39
Energy loss (XP) vs drift time in ERAM 15
y = 0.12x + 479.93
Energy loss (XP) vs drift time in ERAM 42
y = 0.11x + 481.81
Energy loss (XP) vs drift time in ERAM 45
y = 0.09x + 485.25
Energy loss (XP) vs drift time in ERAM 37
y = 0.08x + 491.05
Energy loss (XP) vs drift time in ERAM 36
y = 0.10x + 483.88
Energy loss (XP) vs drift time in ERAM 20
y = 0.67x + 388.23
Energy loss (XP) vs drift time in ERAM 38
y = 0.62x + 392.04
Energy loss (XP) vs drift time in ERAM 7
y = 0.58x + 408.73
Energy loss (XP) vs drift time in ERAM 44
y = 0.55x + 410.00
Energy loss (XP) vs drift time in ERAM 43
y = 0.30x + 450.84
Energy loss (XP) vs drift time in ERAM 39
y = 0.33x + 445.24
Energy loss (XP) vs drift time in ERAM 41
y = 0.33x + 444.25
Energy loss (XP) vs drift time in ERAM 46
y = 0.29x + 448.27
dE/dx (ADC counts/cm)
drift time (timebins)



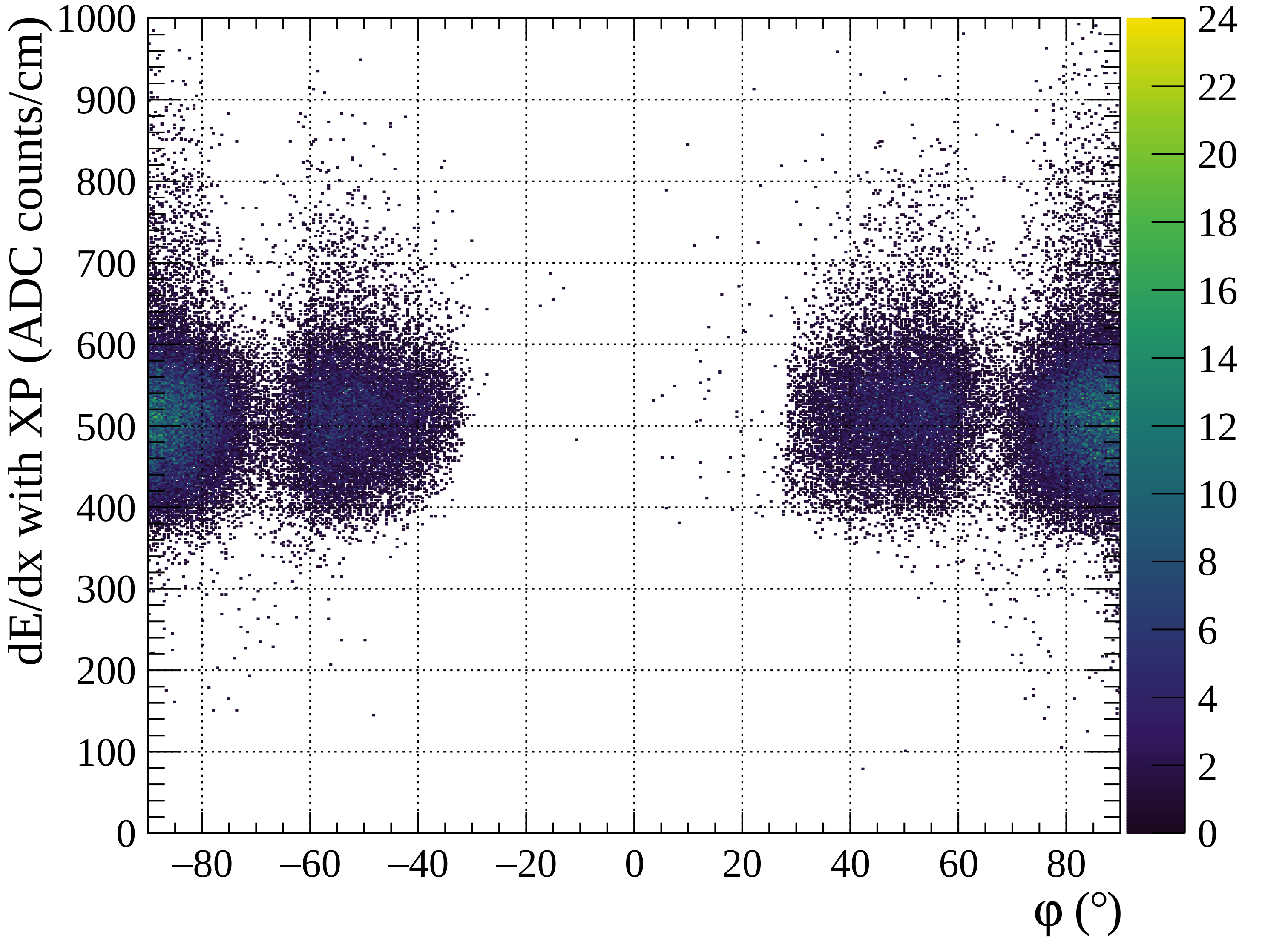
time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
80
70
60
50
40
30
20
10
0



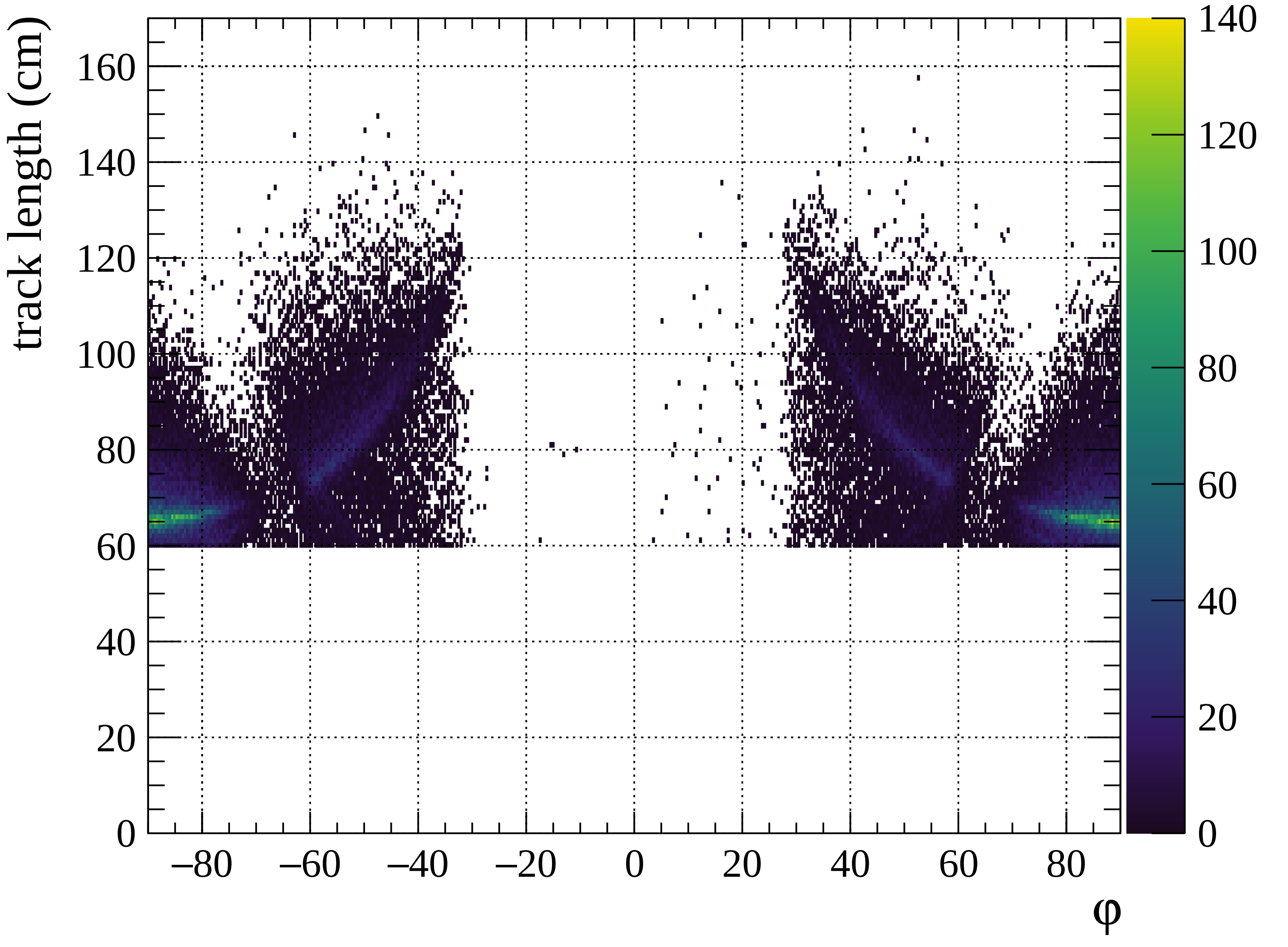
dE/dx resolution (%)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80
φ (°)
WF
XP



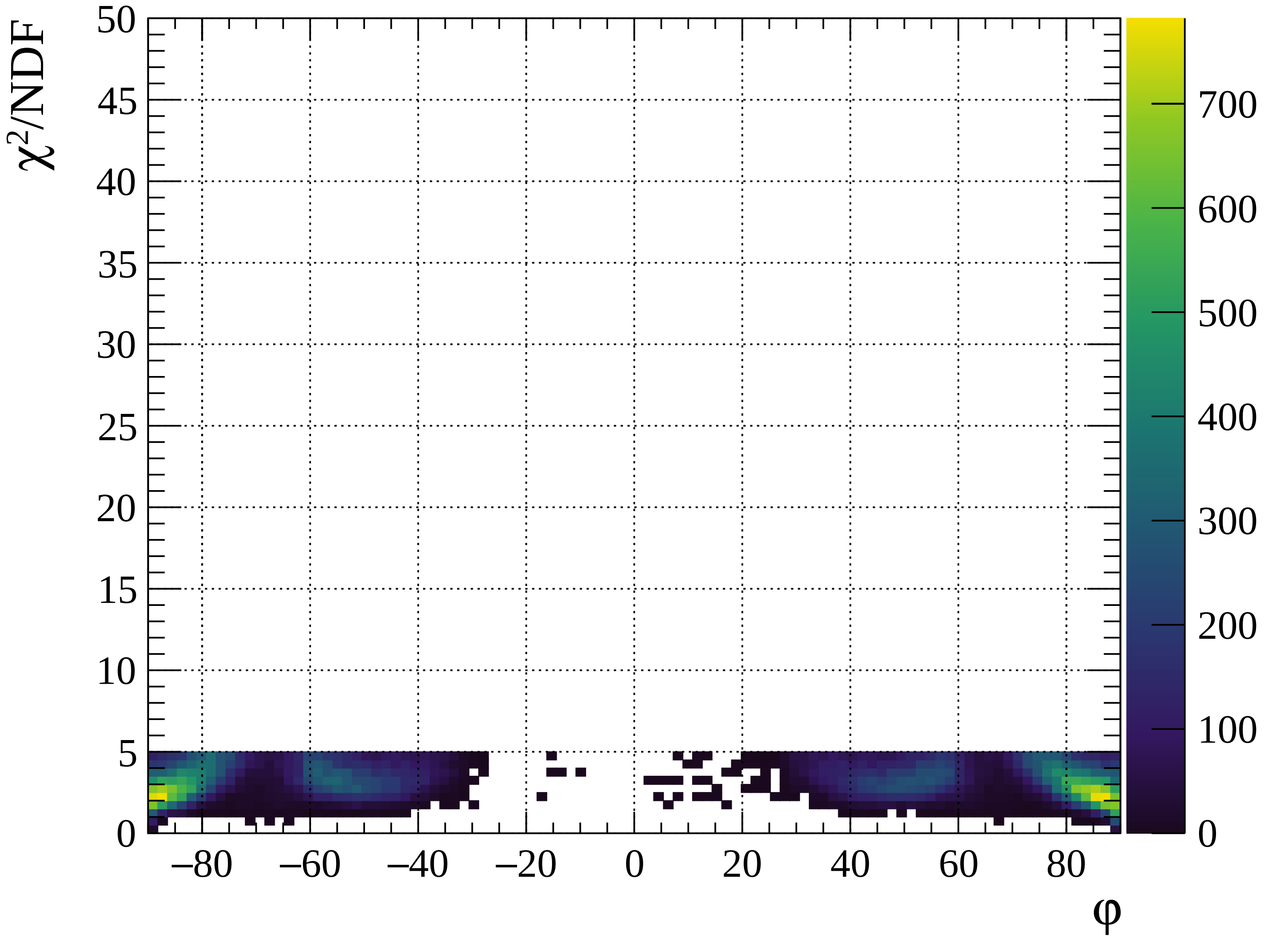
Mean (ADC counts/cm)
800
700
600
500
400
300
200
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)



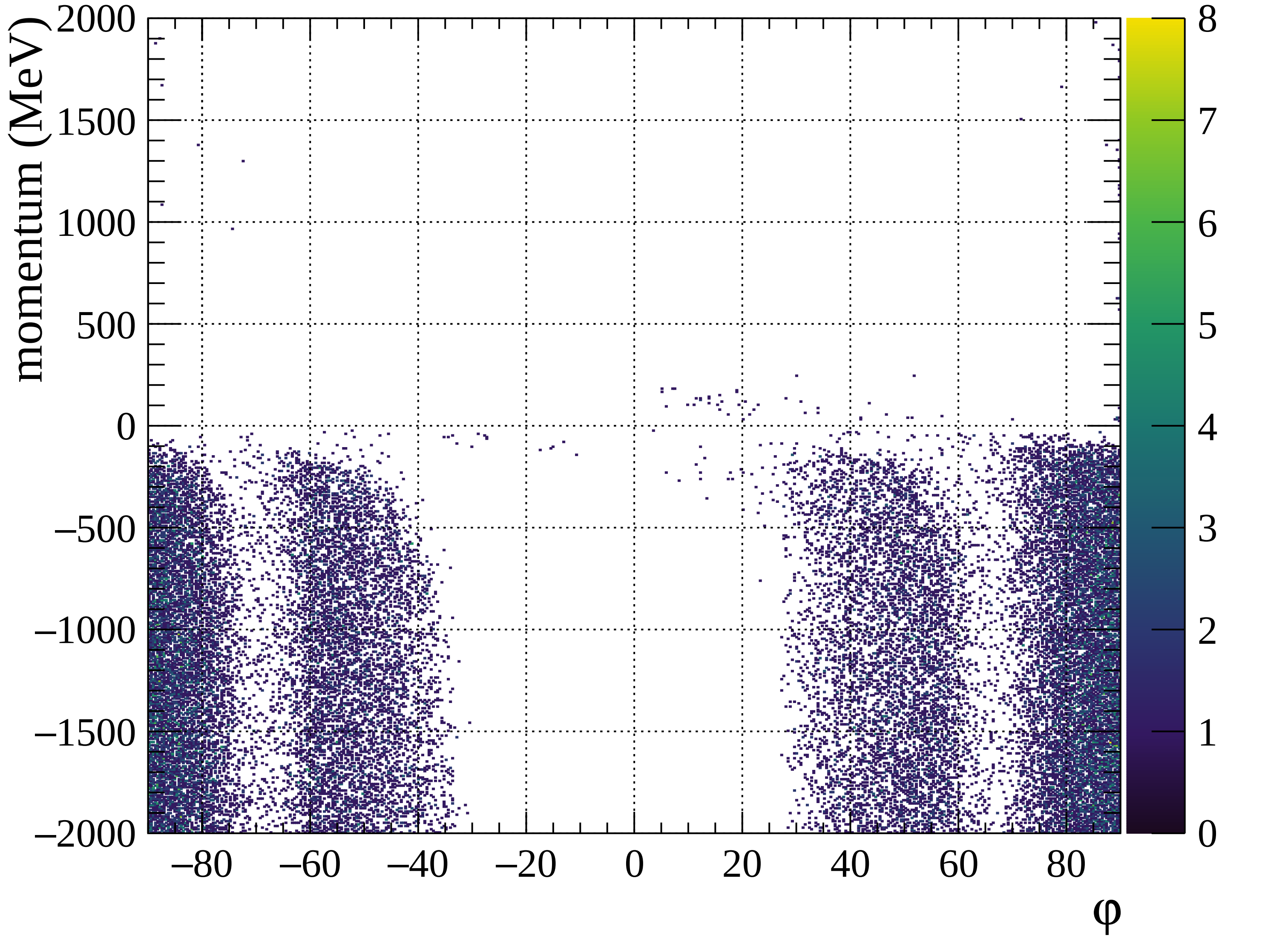
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
24
22
20
18
16
14
12
10
8
6
4
2
0



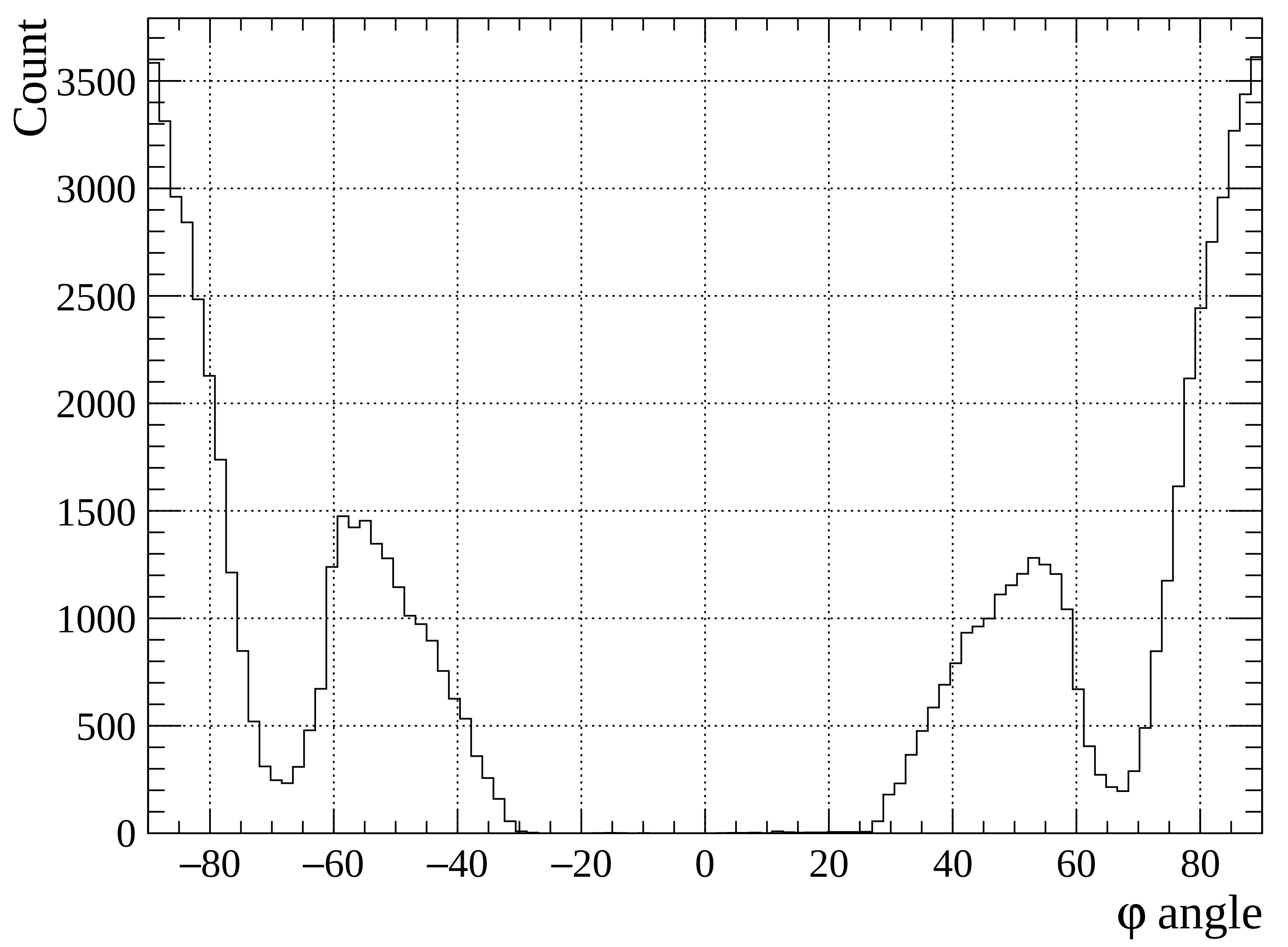
track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
φ
0
20
40
60
80
100
120
140

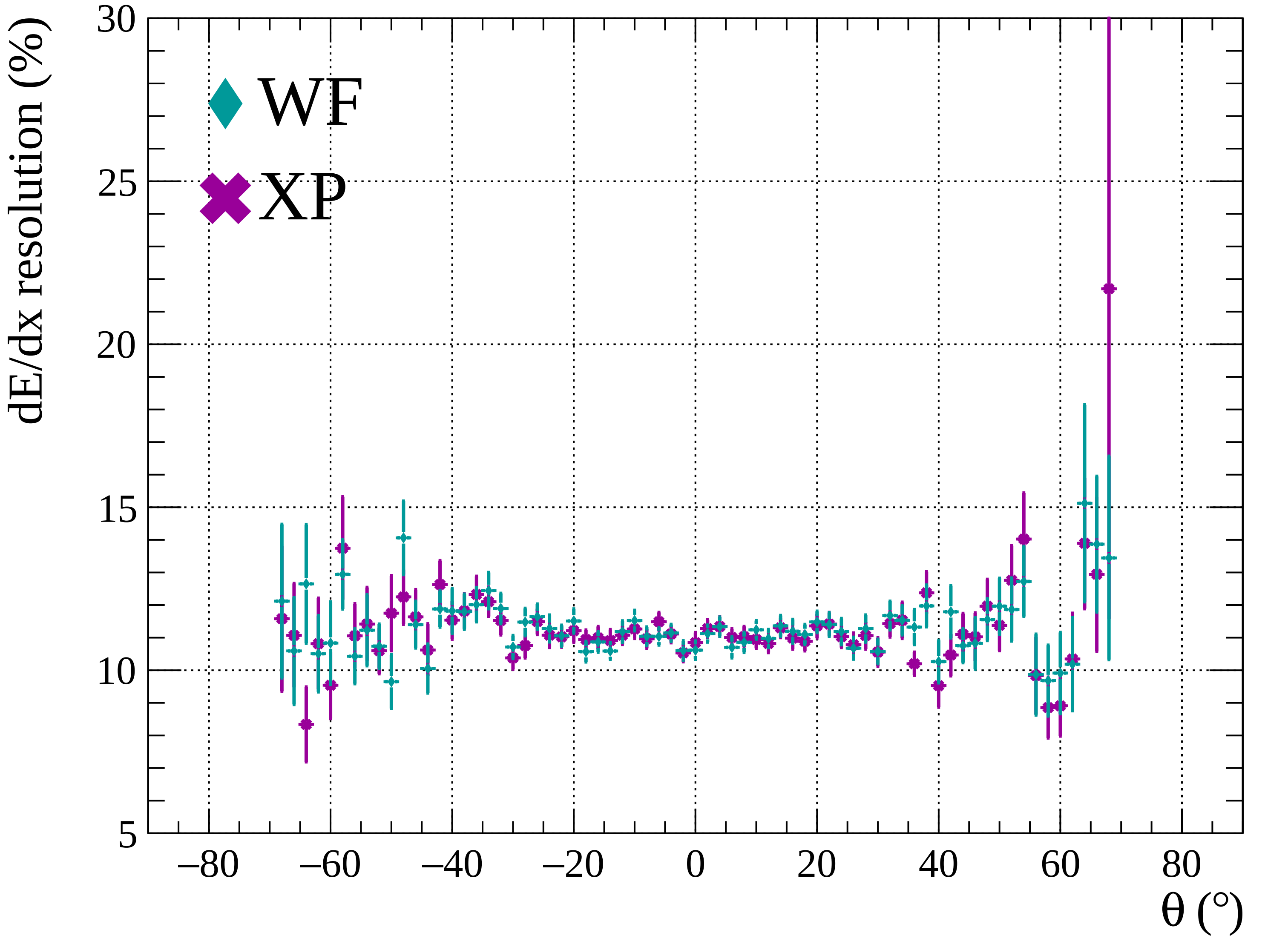


χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
700
600
500
400
300
200
100
0

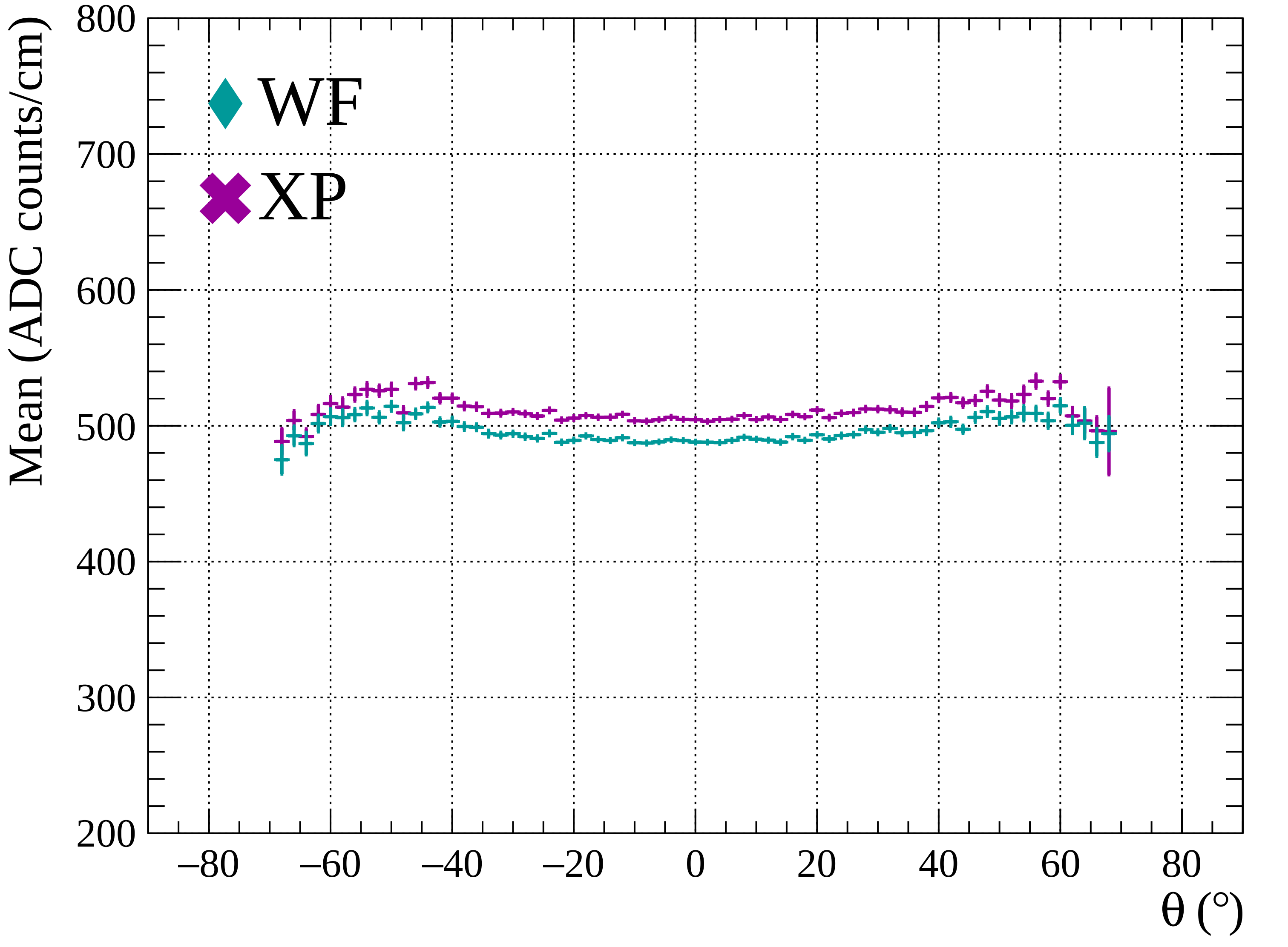


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
8
7
6
5
4
3
2
1
0

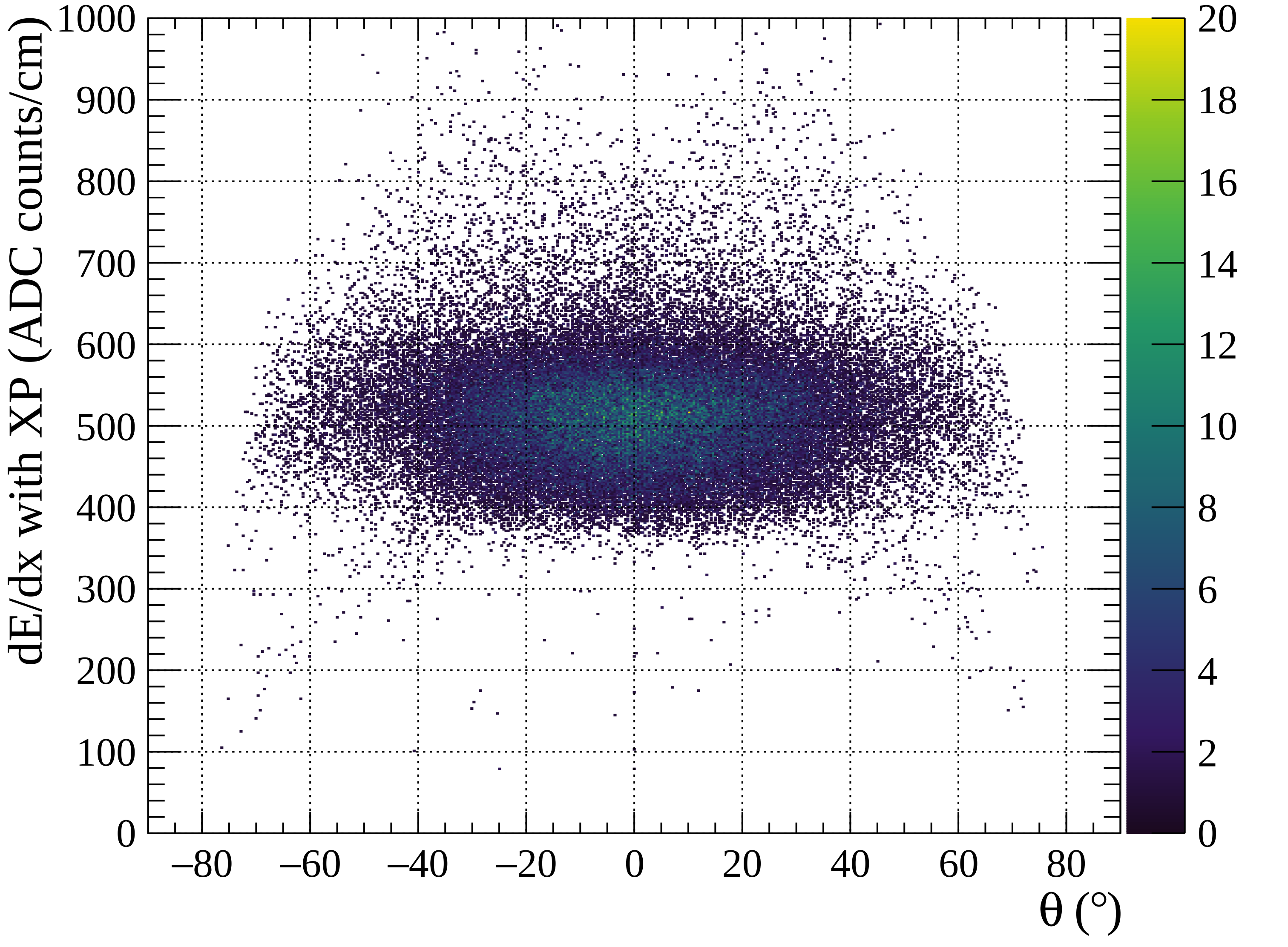

Count
3500
3000
2500
2000
1500
1000
500
0
−80
−60
−40
−20
0
20
40
60
80
φ angle



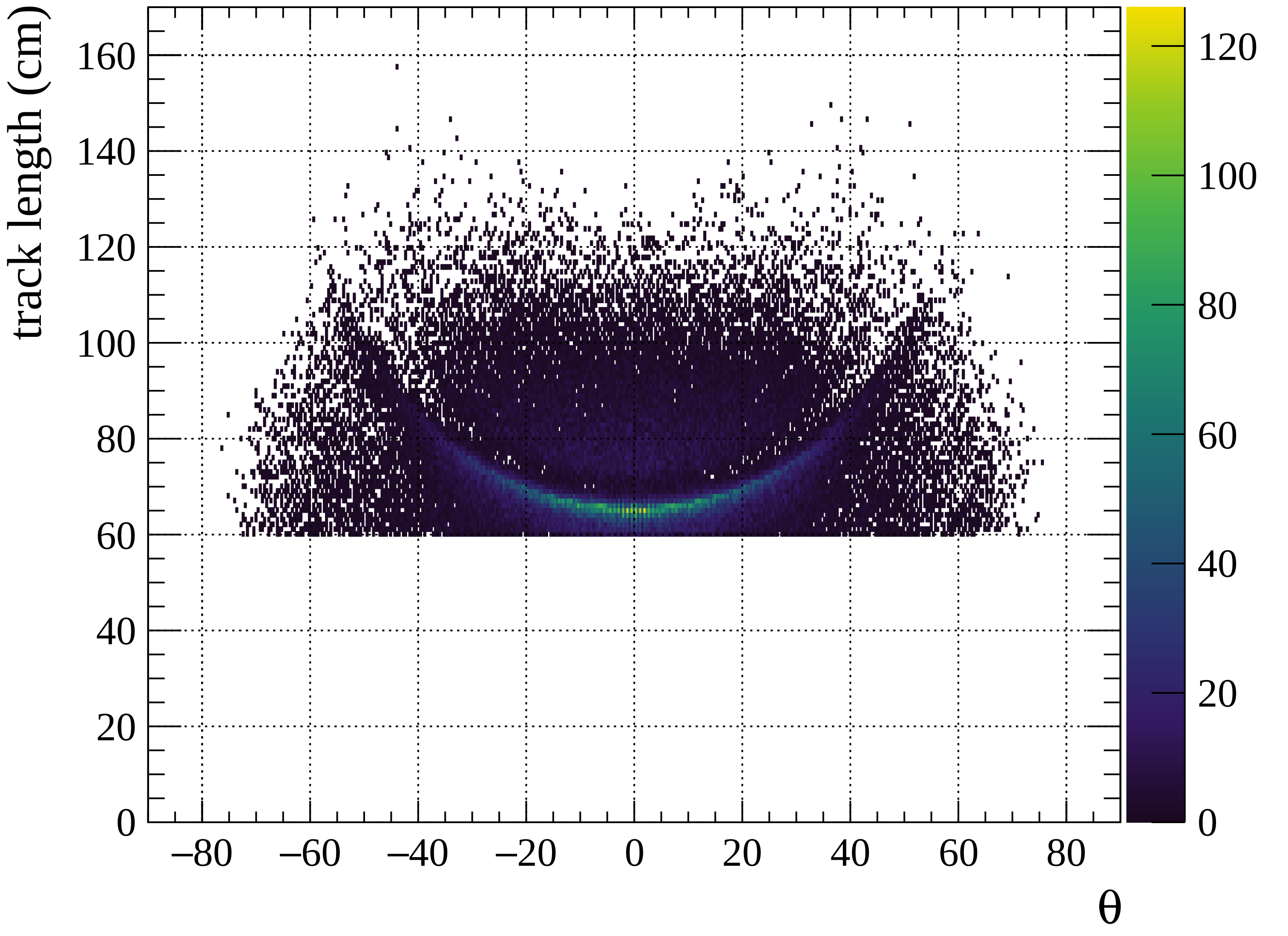
dE/dx resolution (%)
WF
XP
θ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80



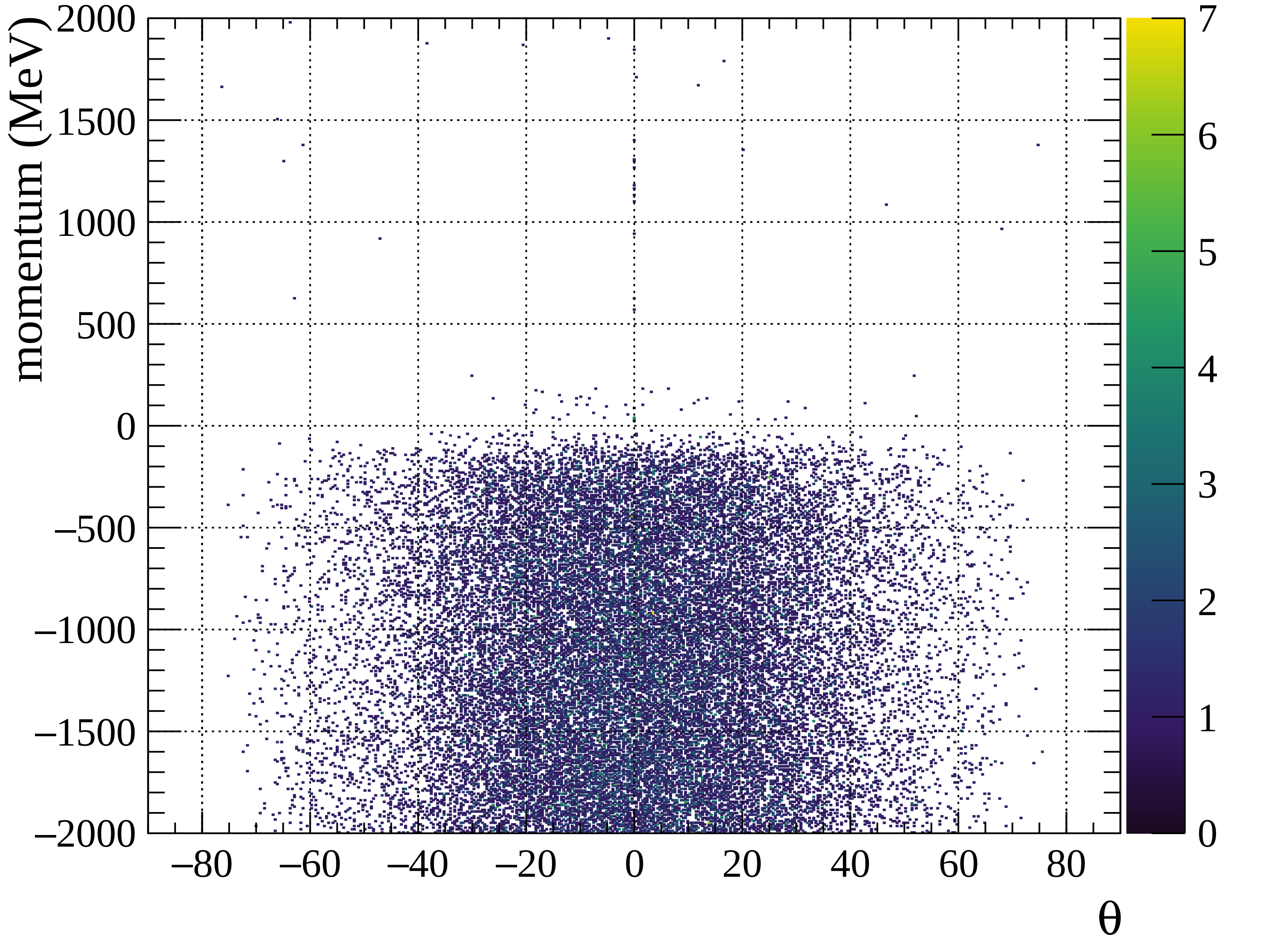
Mean (ADC counts/cm)
800
700
600
500
400
300
200
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)



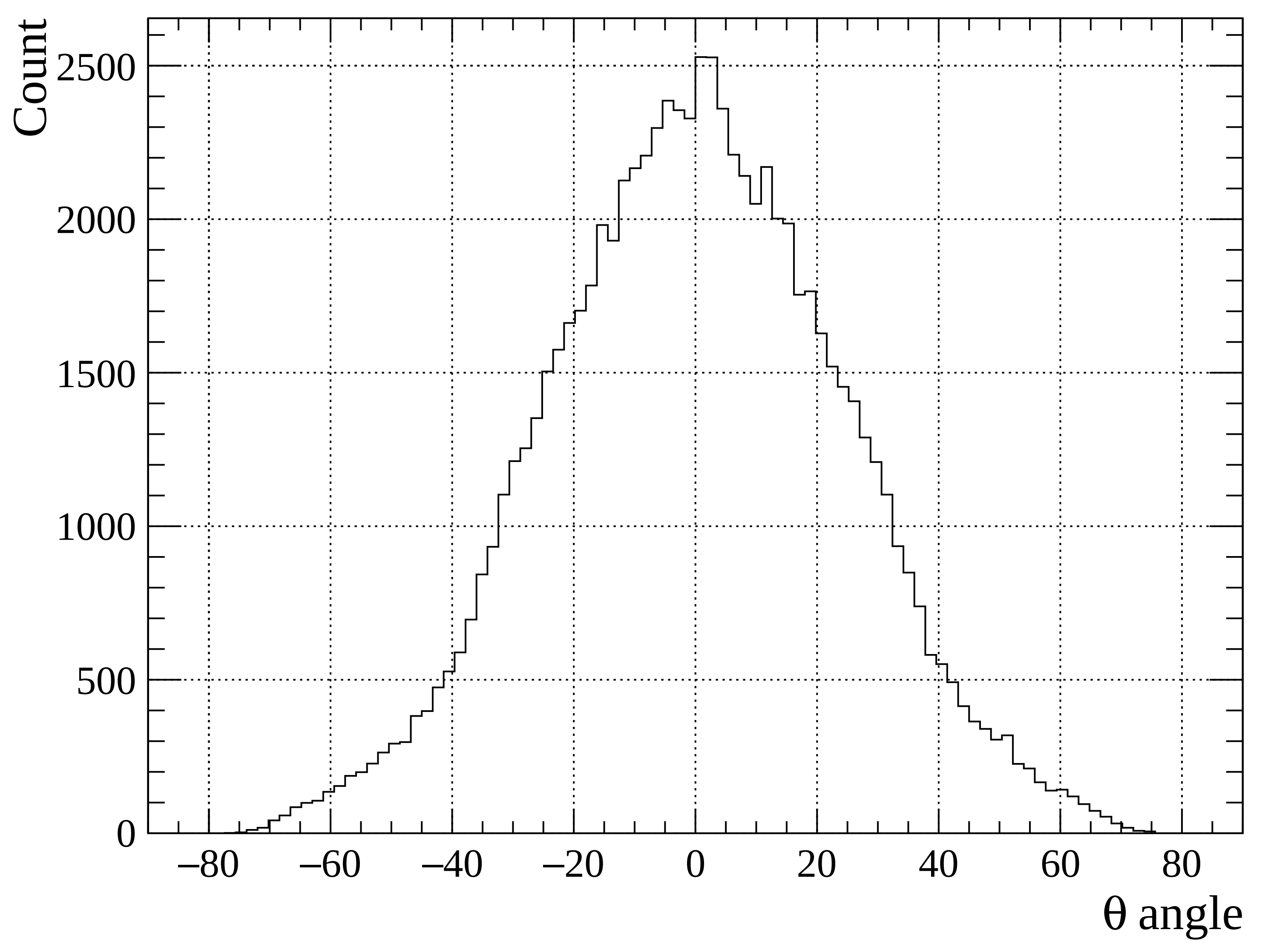
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
20
18
16
14
12
10
8
6
4
2
0



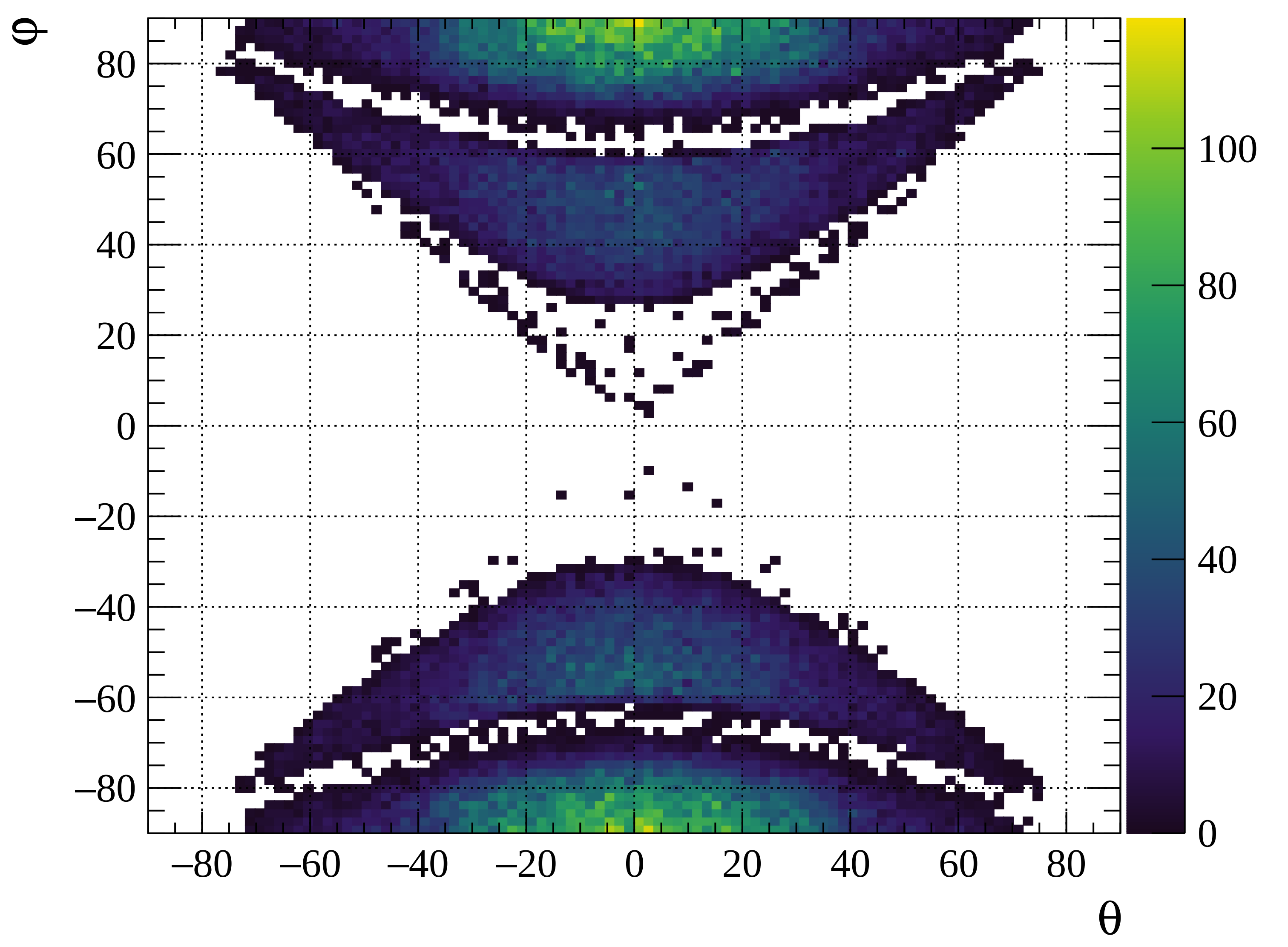
track length (cm)
θ
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
0
20
40
60
80
100
120



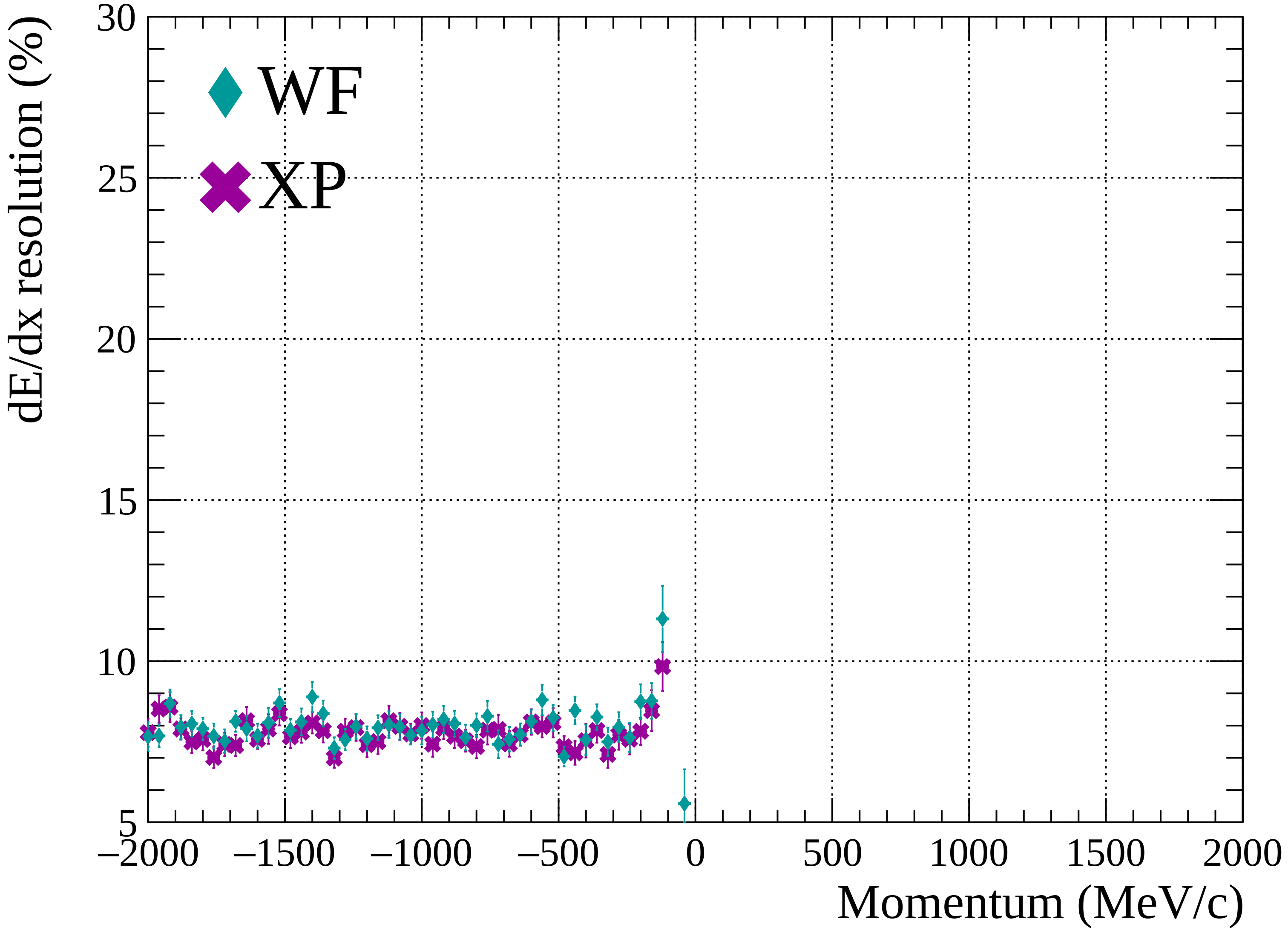
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
7
6
5
4
3
2
1
0



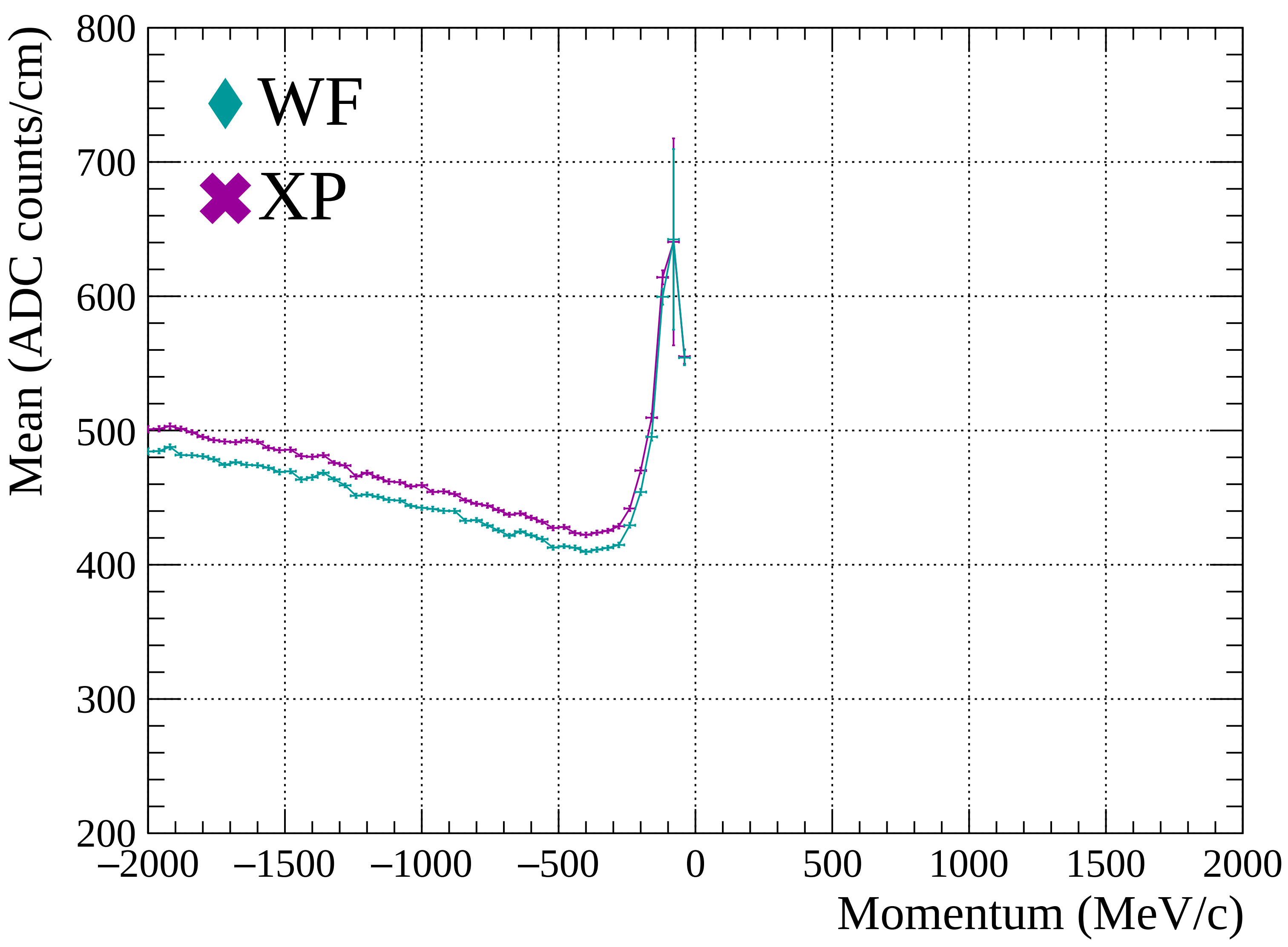
Count
2500
2000
1500
1000
500
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

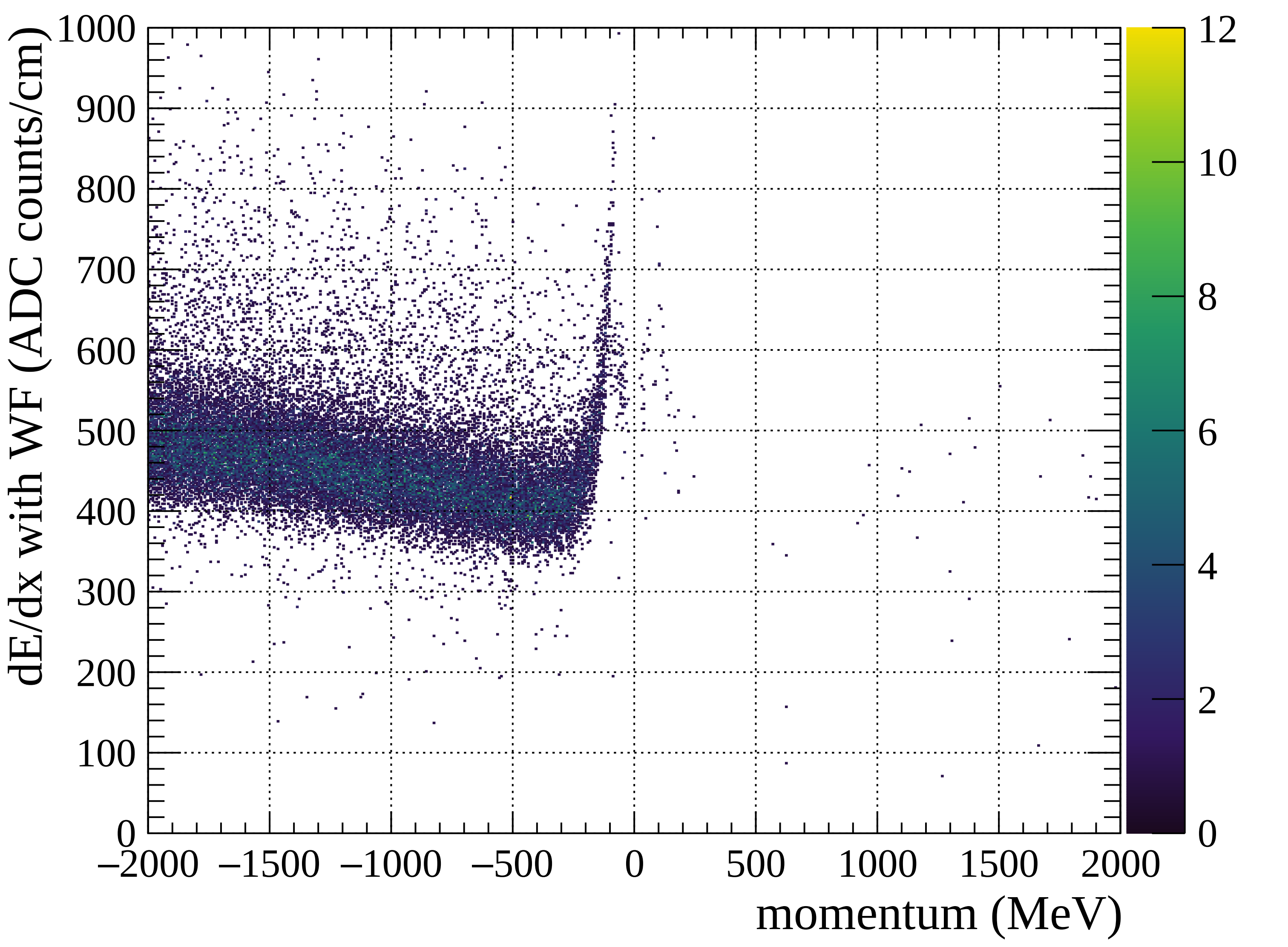


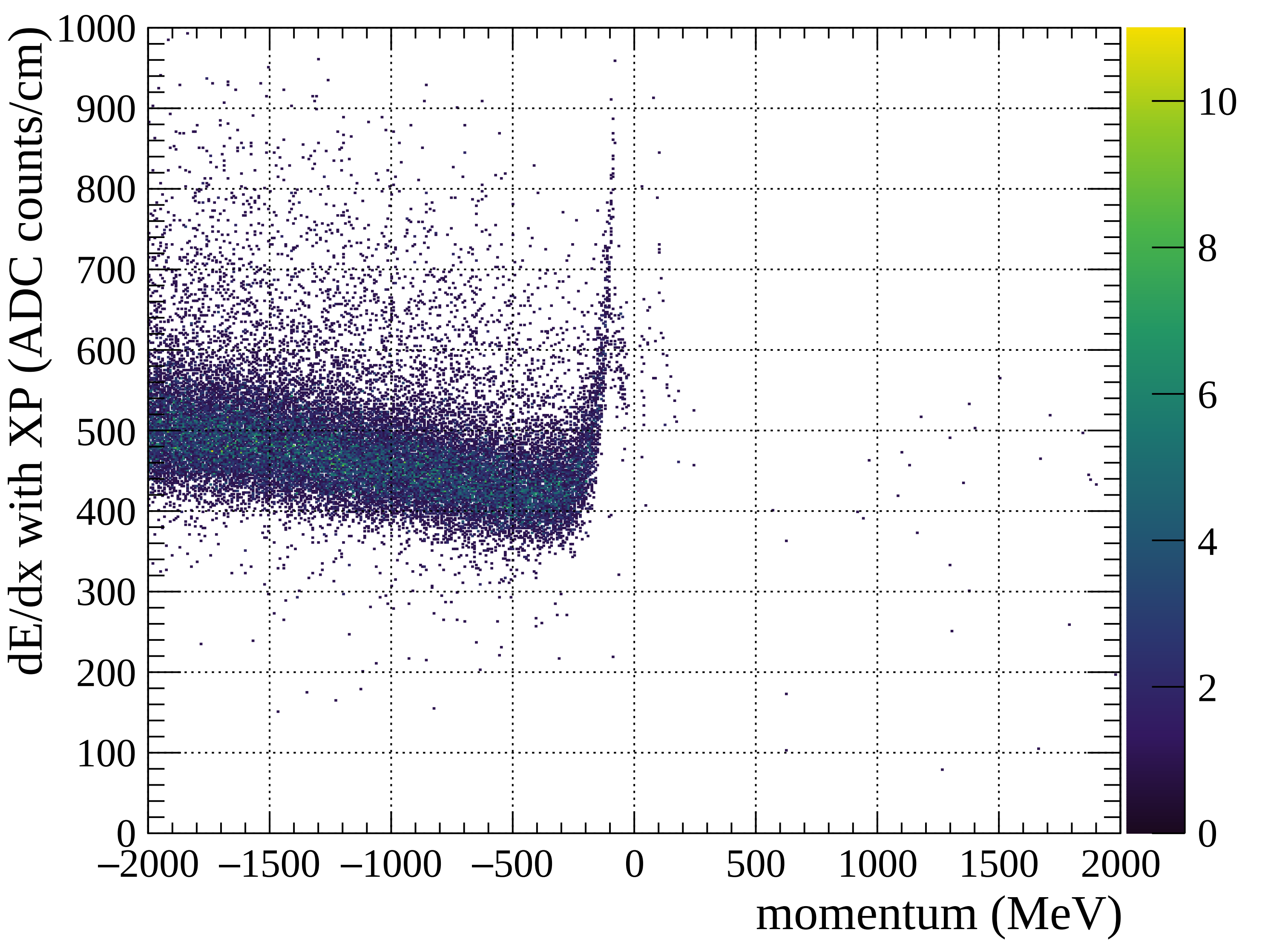
φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
100
80
60
40
20
0

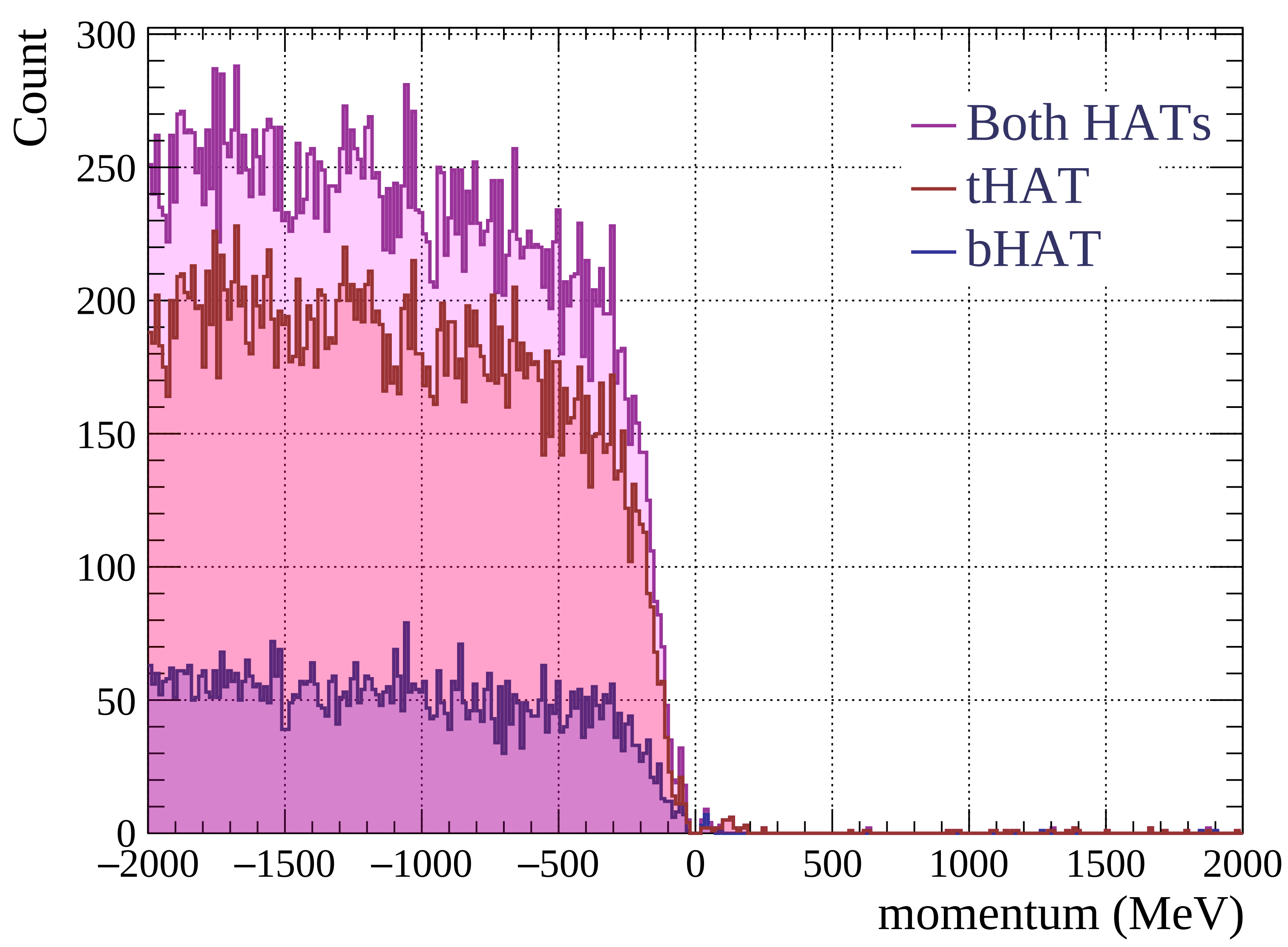


dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)

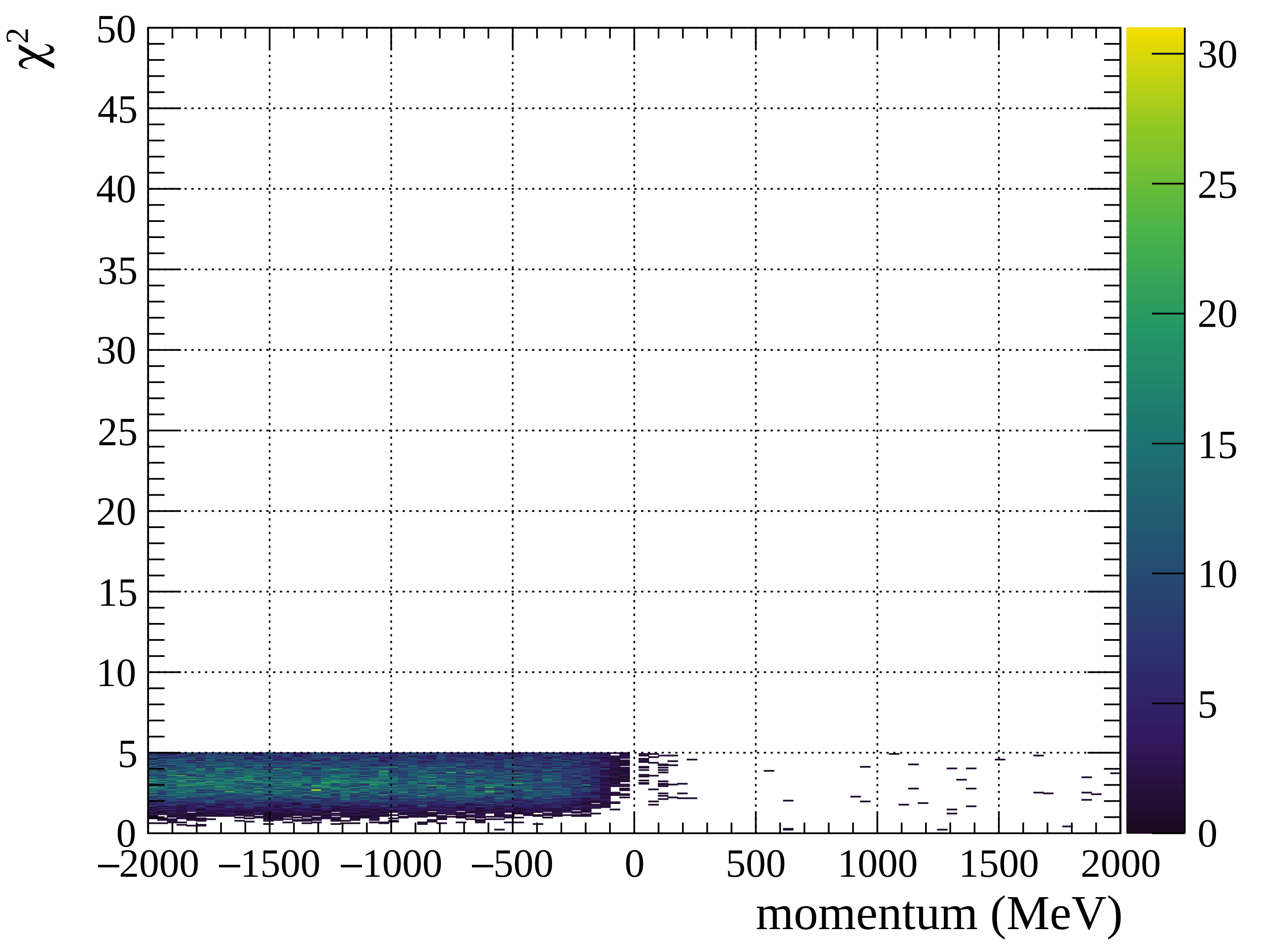

WF
XP
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)


dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
12
10
8
6
4
2
0

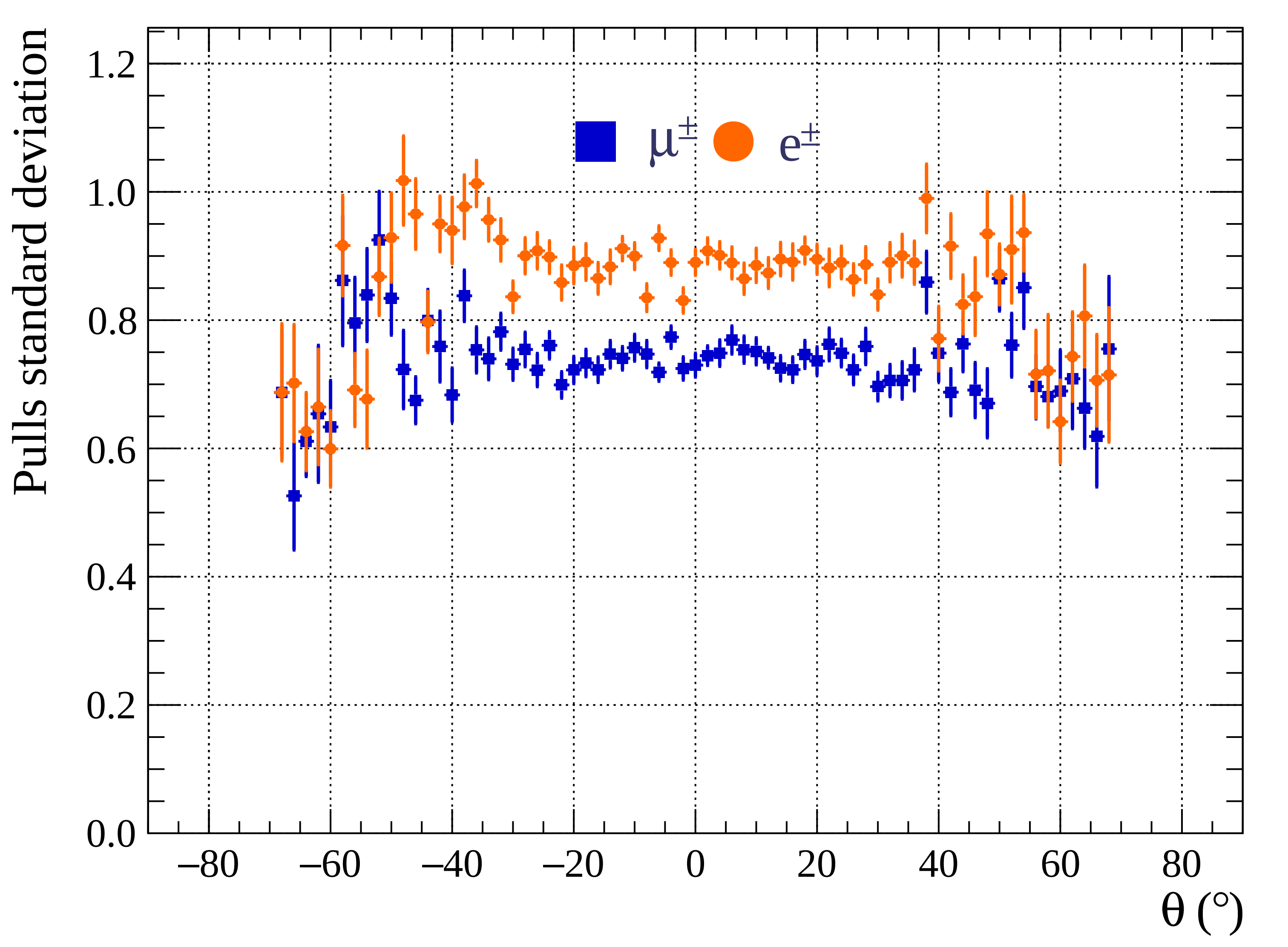

dE/dx with XP (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
10
8
6
4
2
0



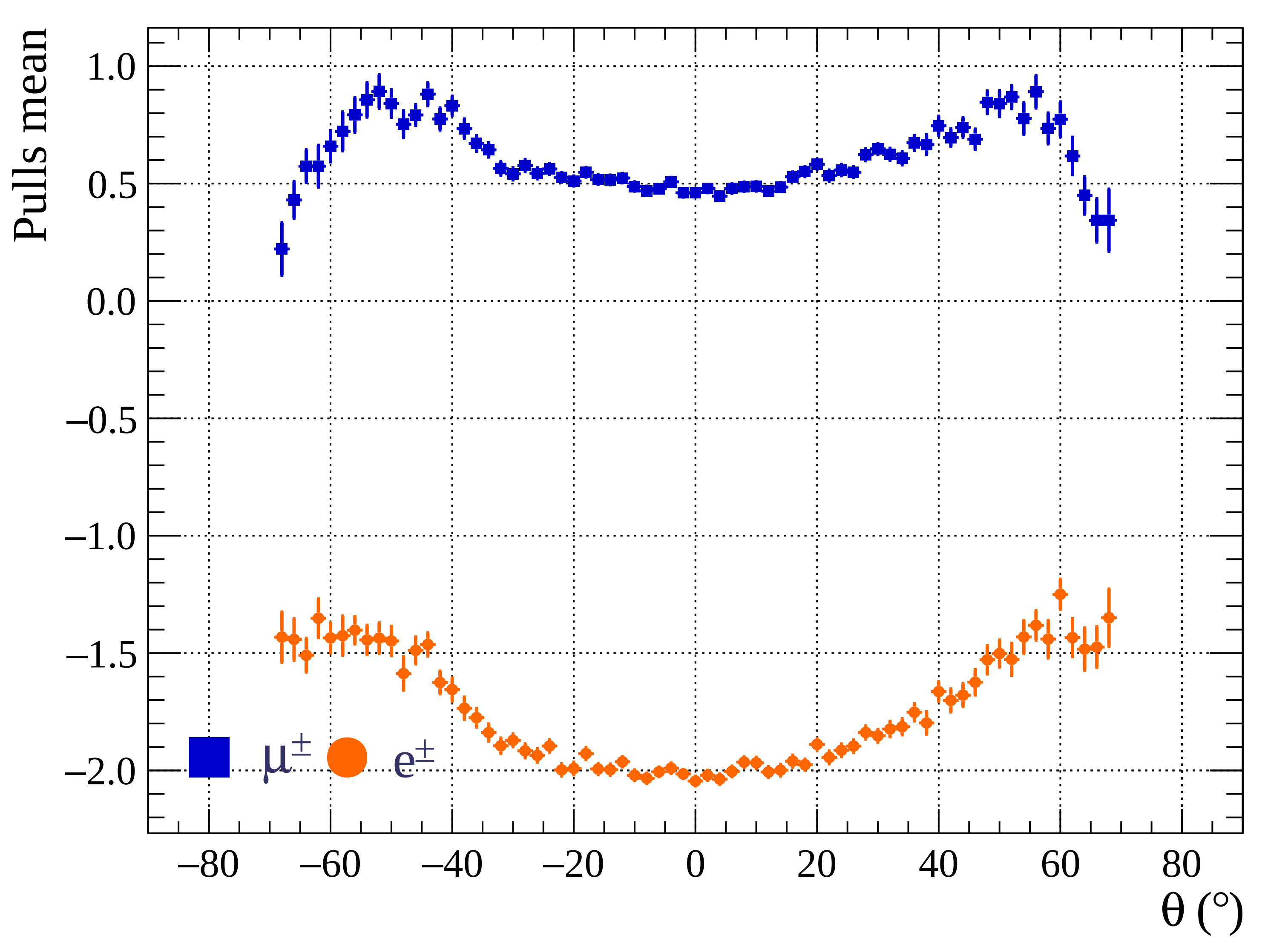
Count
300
250
200
150
100
50
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT

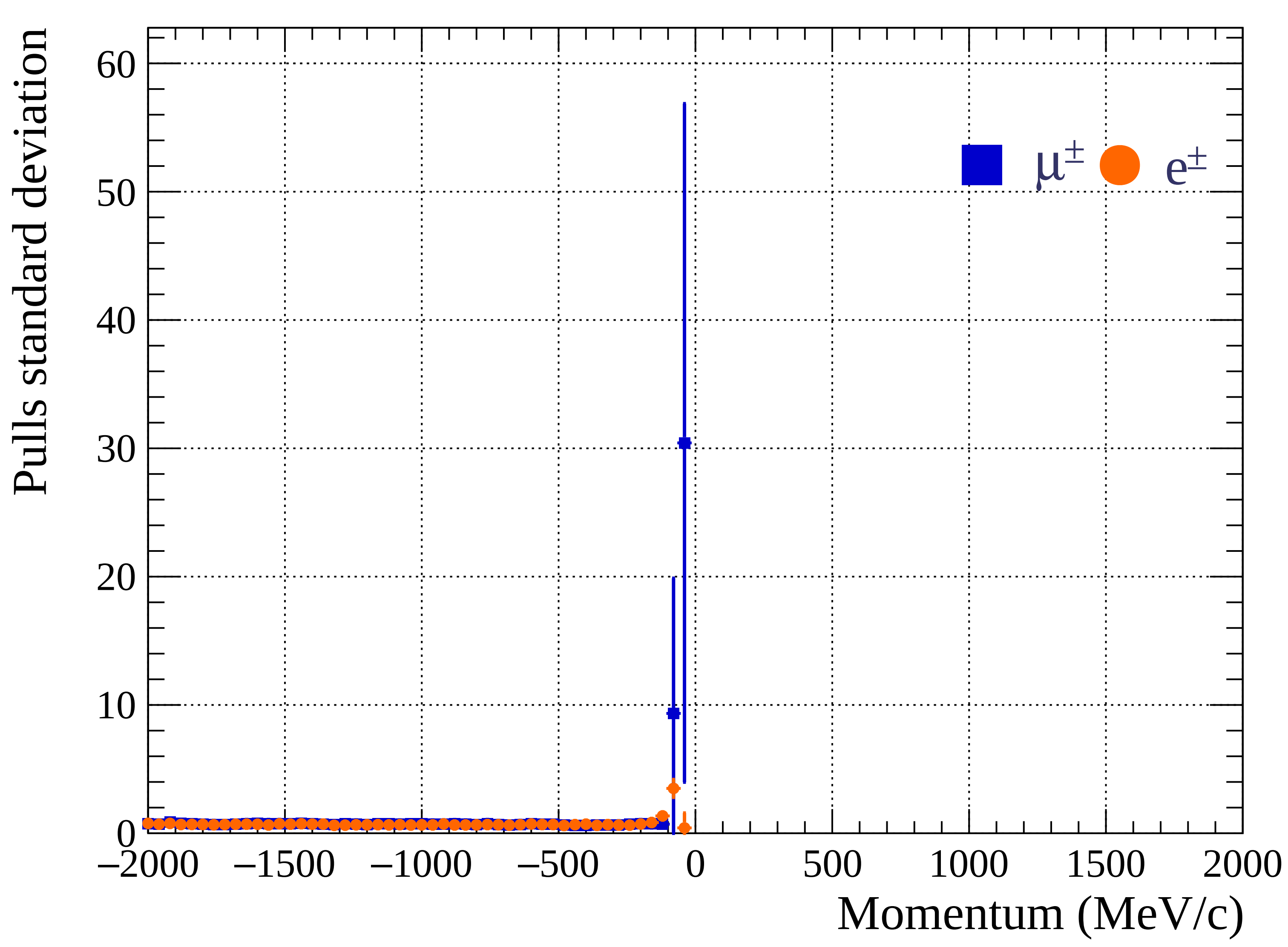


χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
30
25
20
15
10
5
0

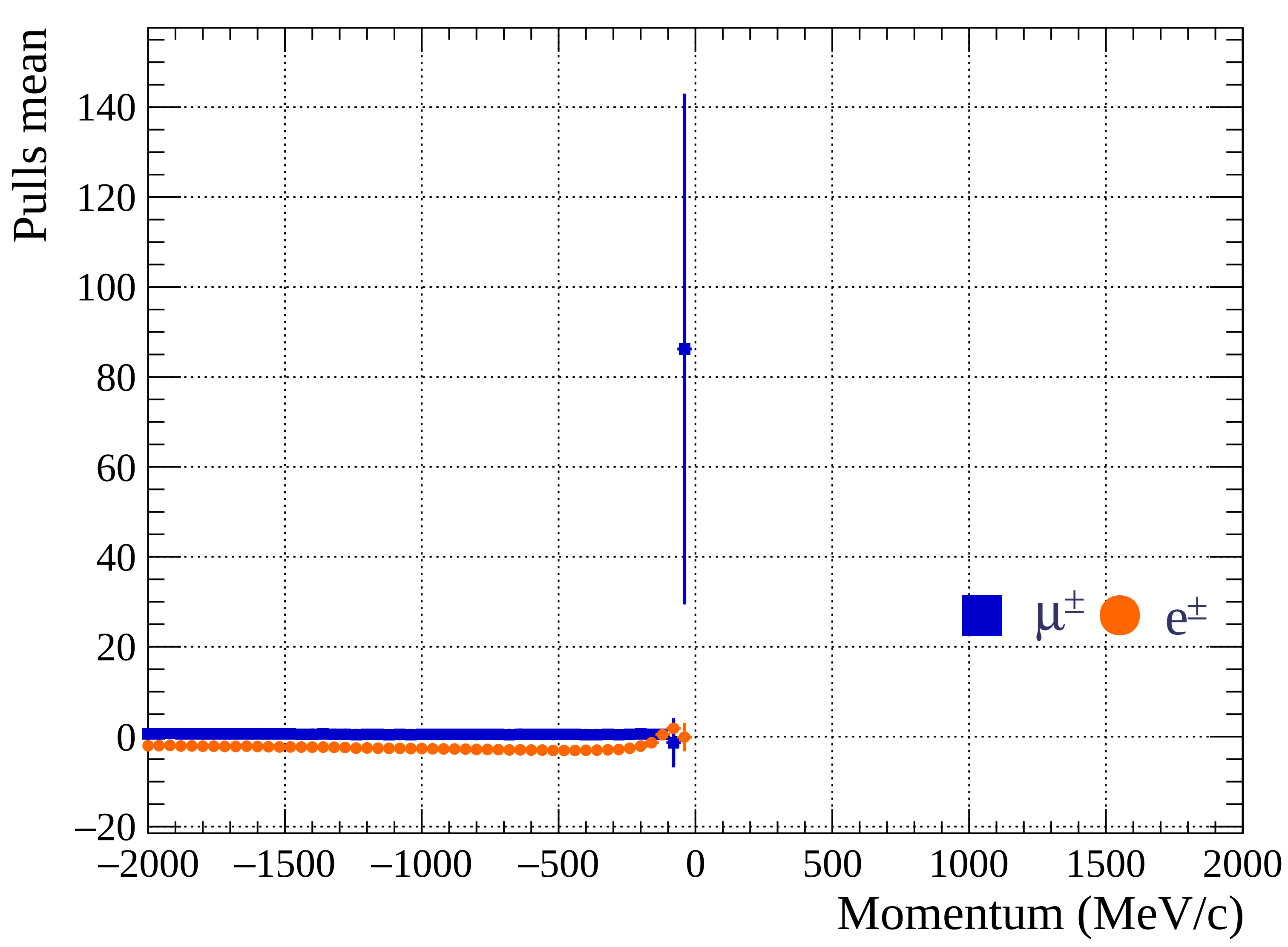


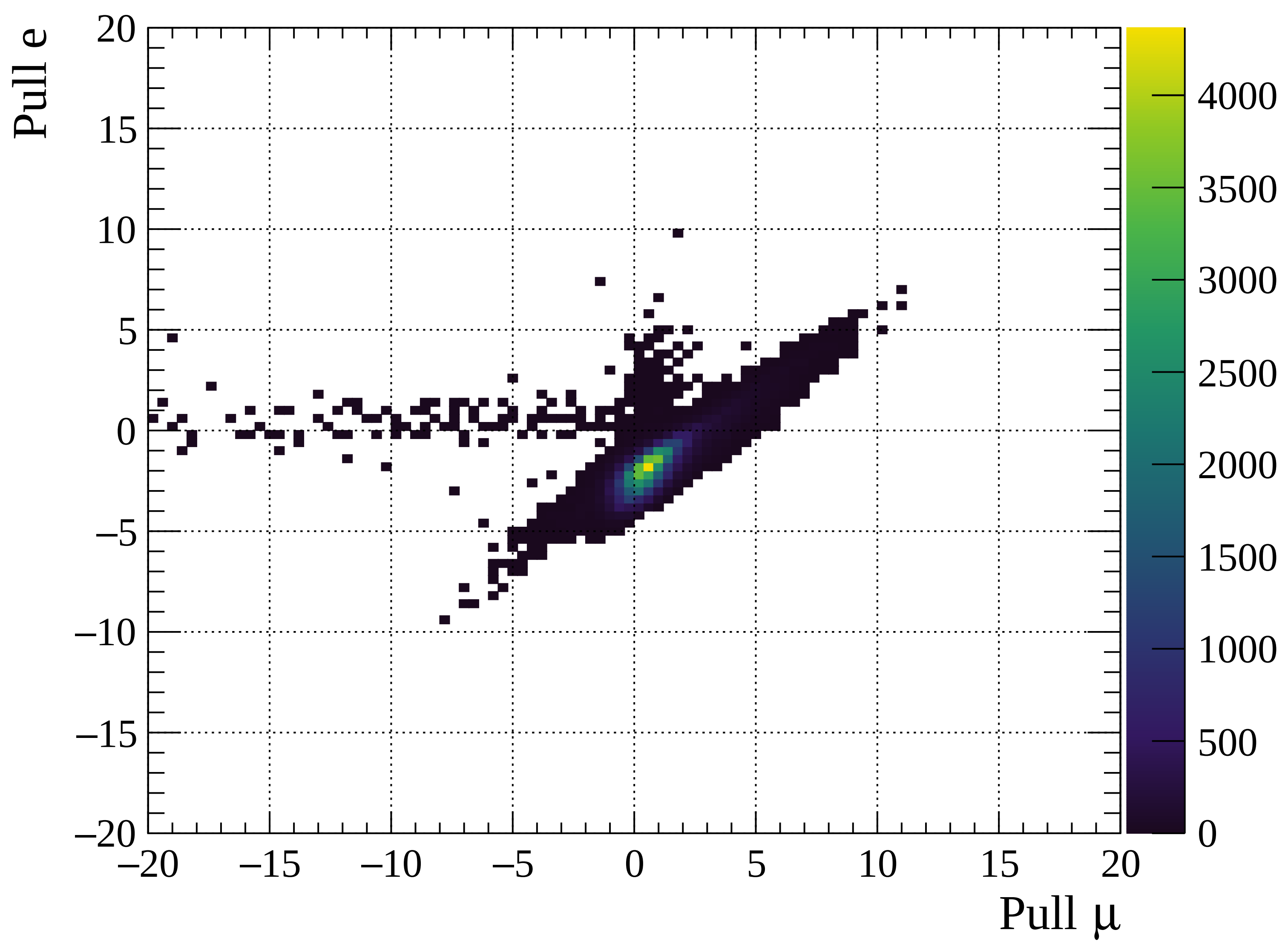
Pulls standard deviation
θ (°)
μ±
e±
0.0
0.2
0.4
0.6
0.8
1.0
1.2
−80
−60
−40
−20
0
20
40
60
80

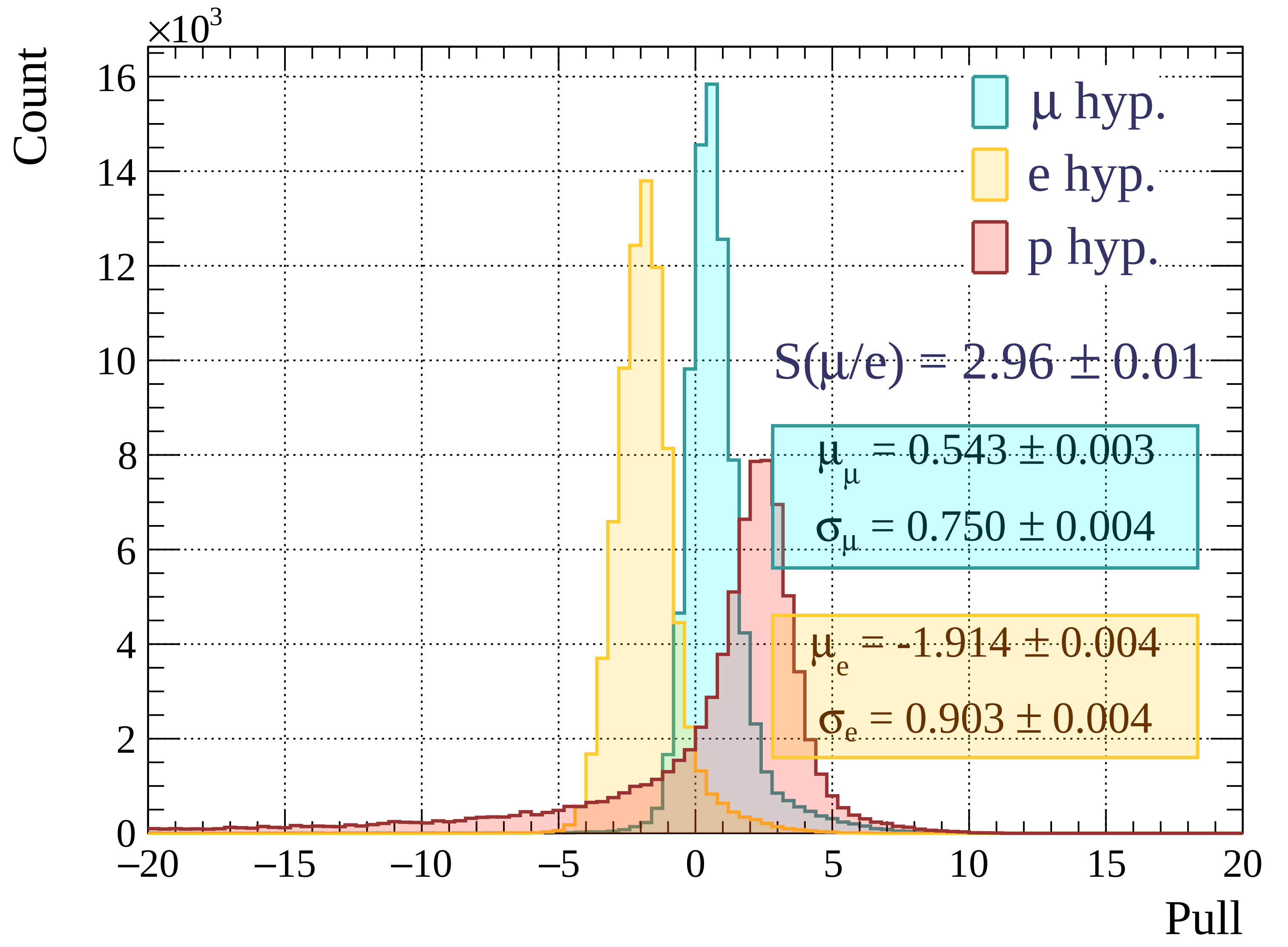

Pulls mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±



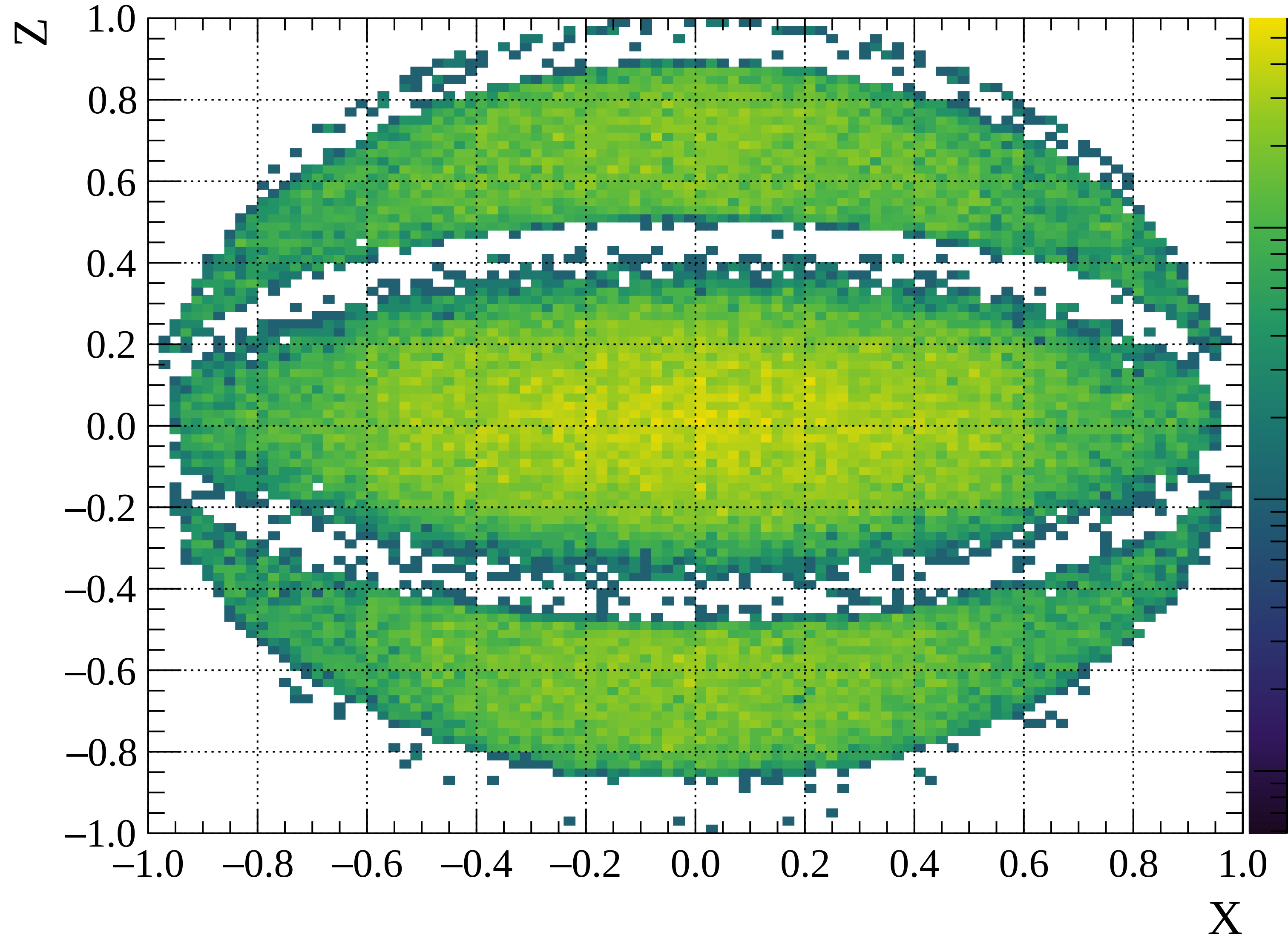
Pulls standard deviation
60
50
40
30
20
10
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$


Pulls mean
140
120
100
80
60
40
20
0
−20
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

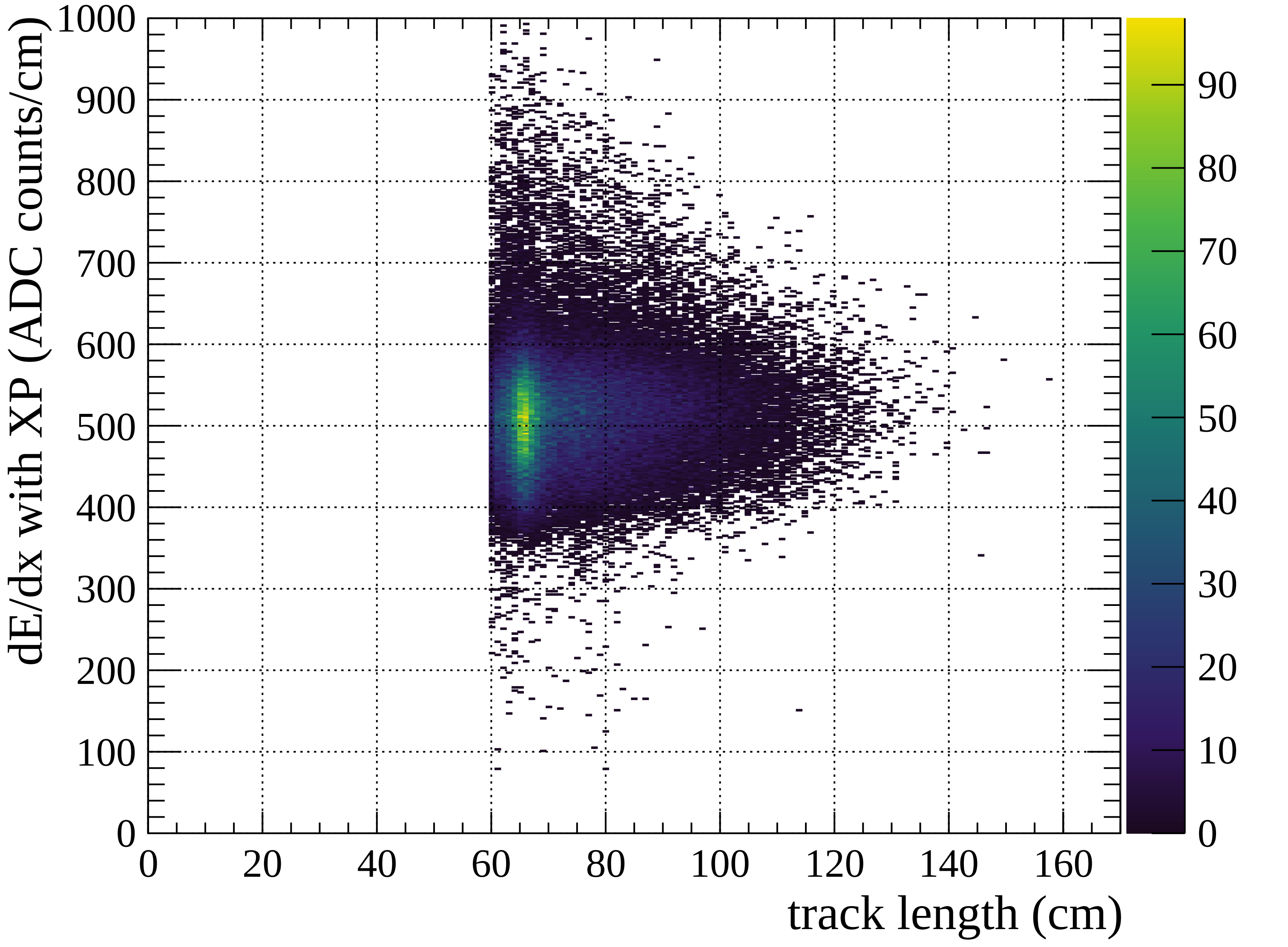

Pull e
Pull μ
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
4000
3500
3000
2500
2000
1500
1000
500
0

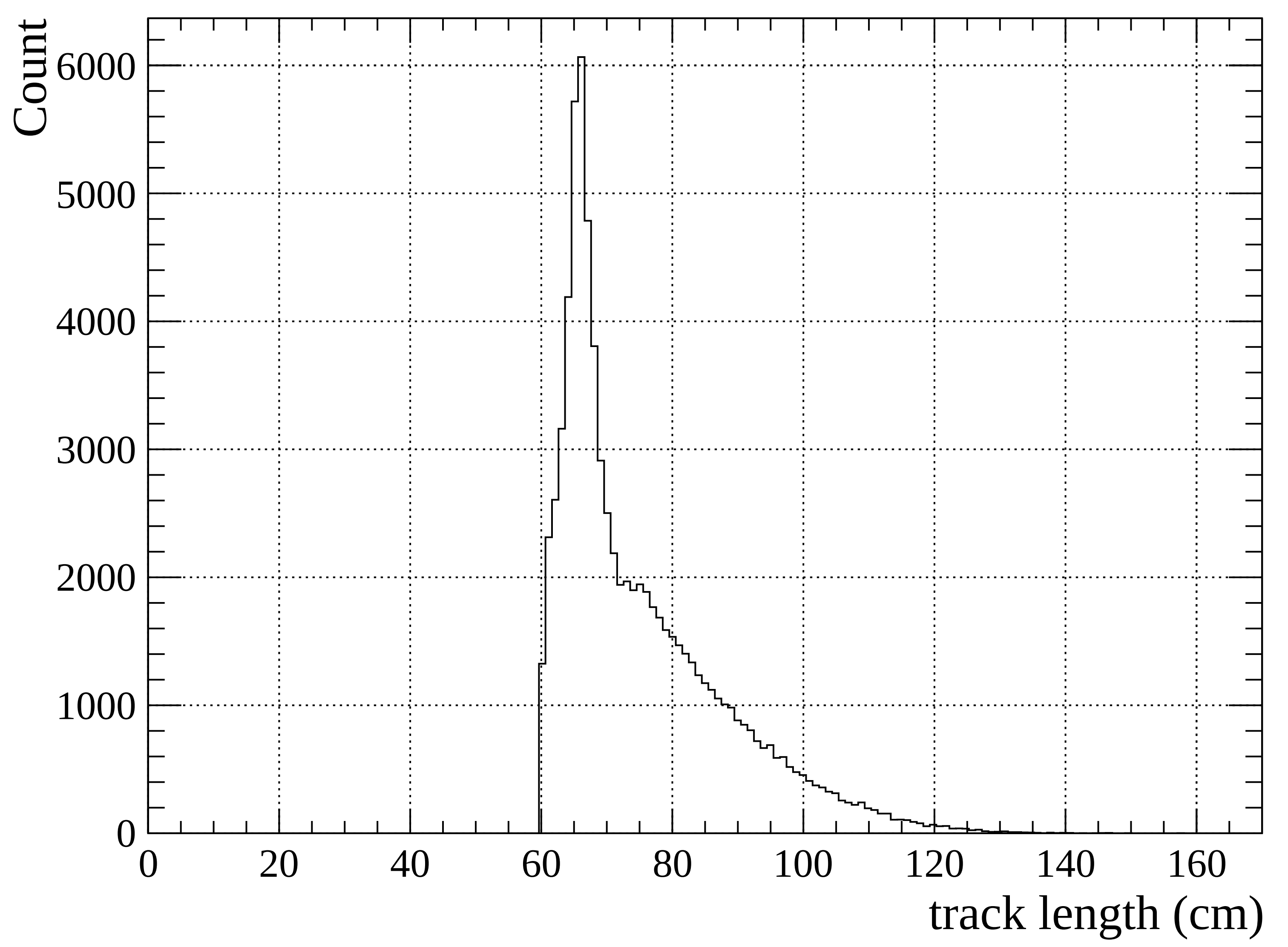


$\times 10^3$
Count
16
14
12
10
8
6
4
2
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 2.96 ± 0.01
μ_μ = 0.543 ± 0.003
σ_μ = 0.750 ± 0.004
μ_e = -1.914 ± 0.004
σ_e = 0.903 ± 0.004

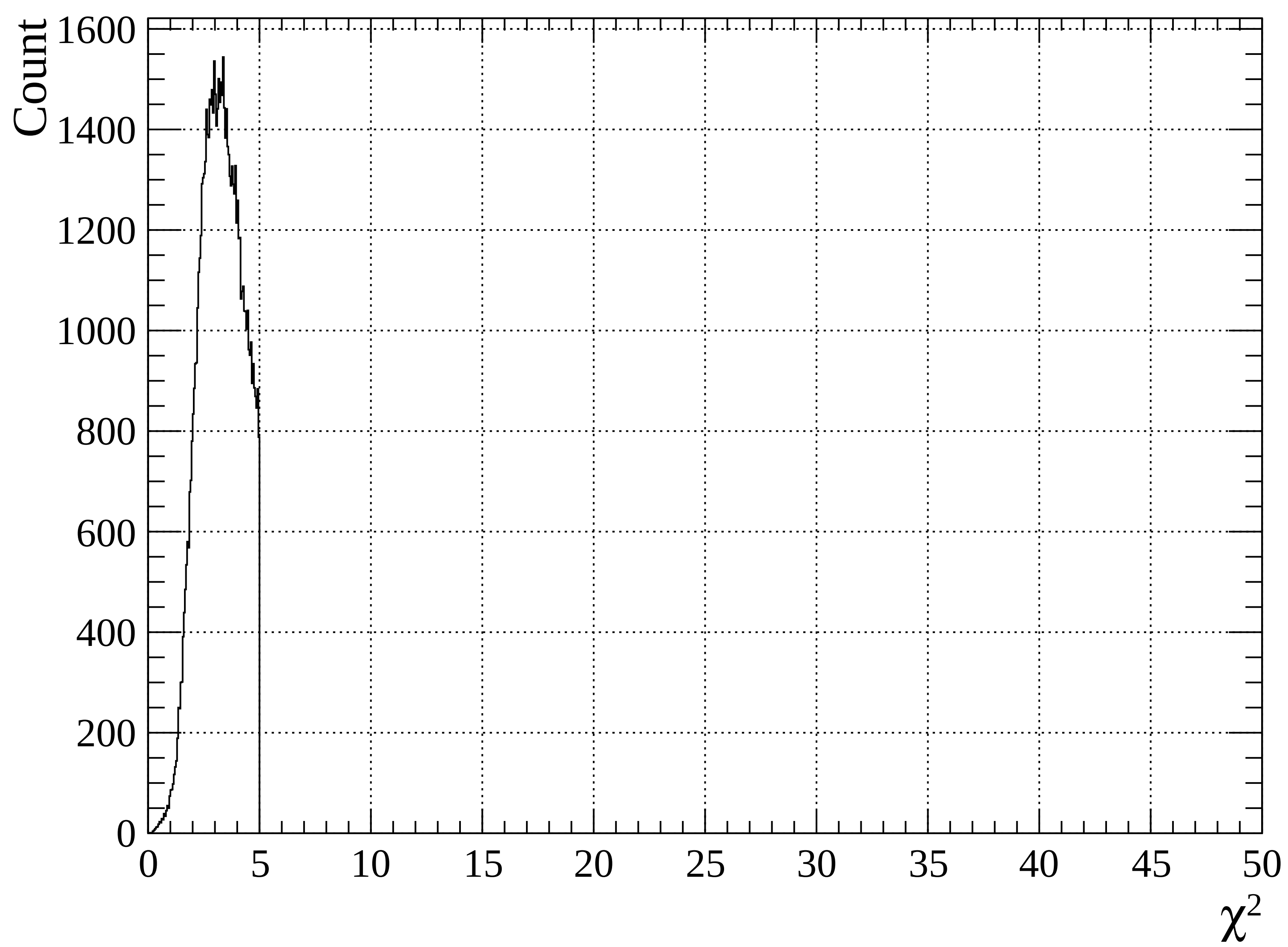


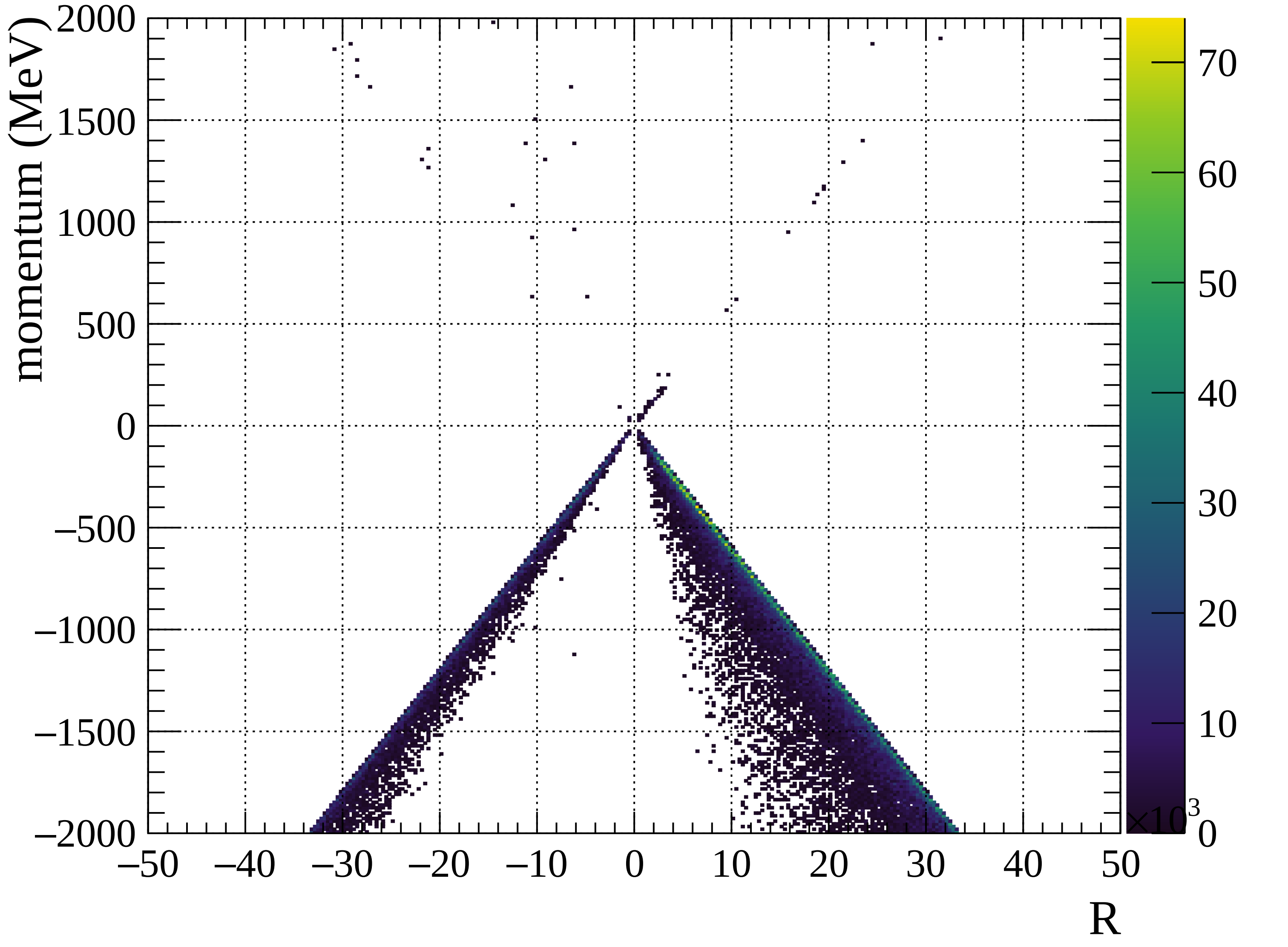
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

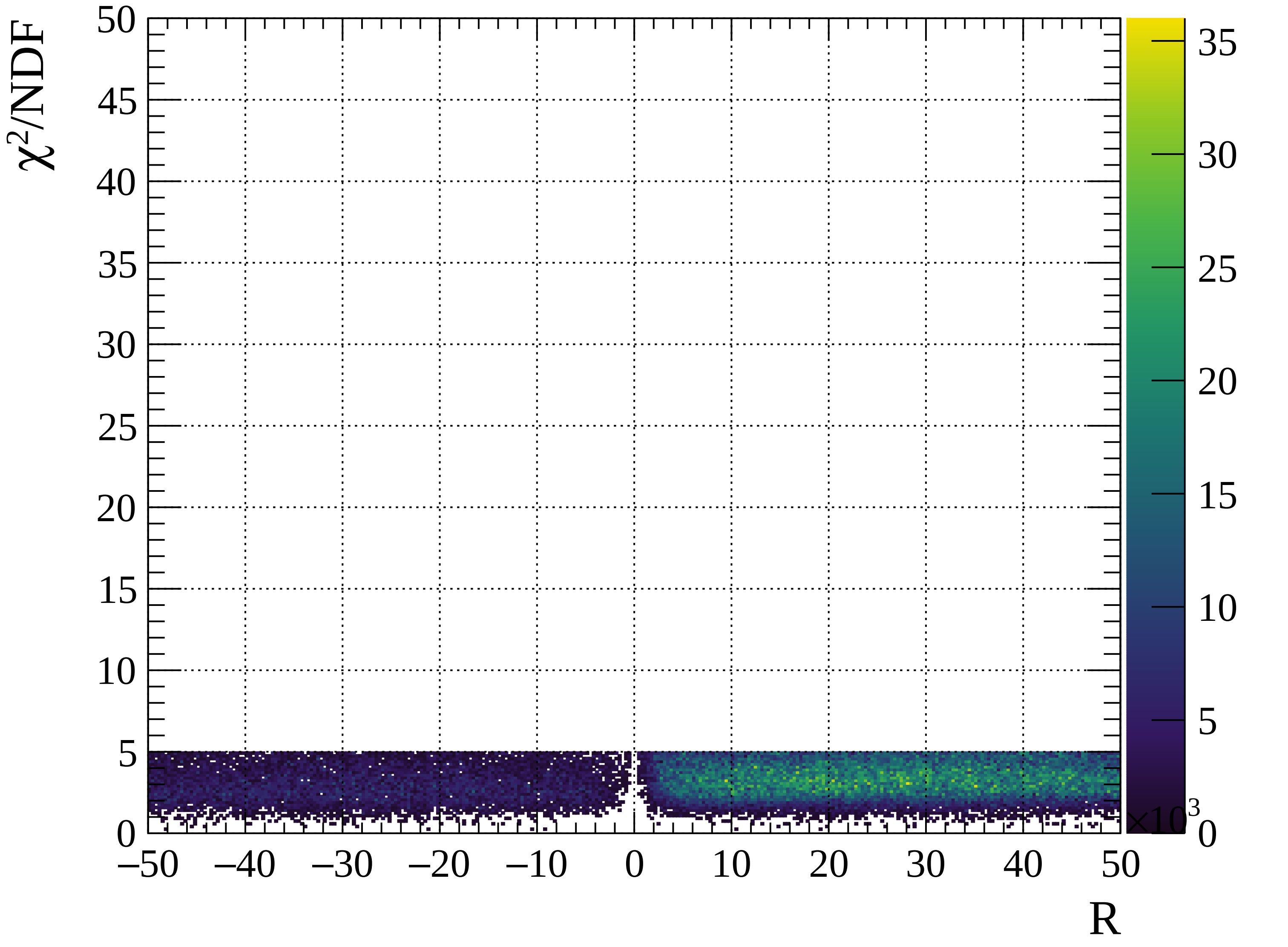

dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
10
20
30
40
50
60
70
80
90



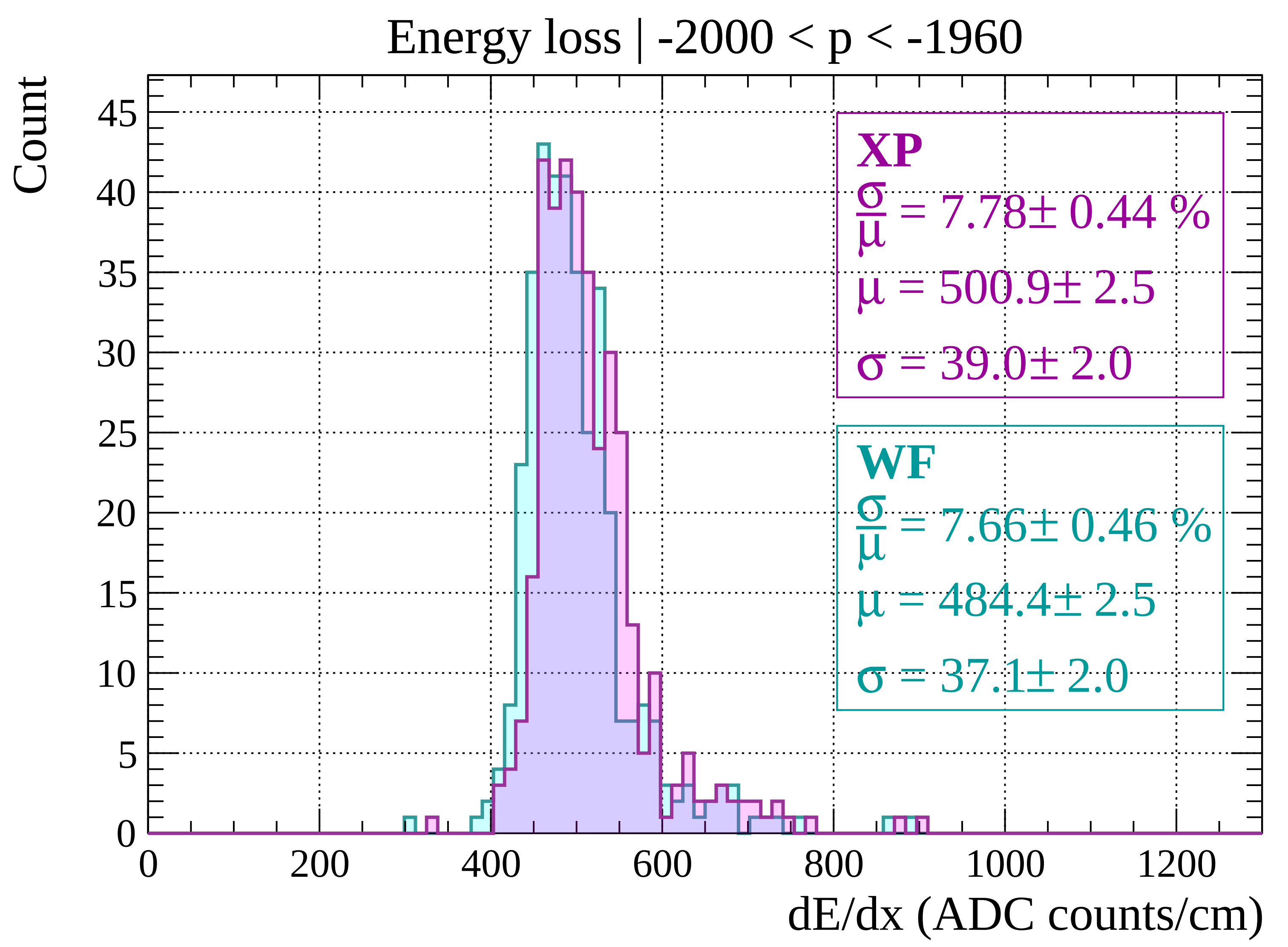
Count
6000
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
140
160
track length (cm)

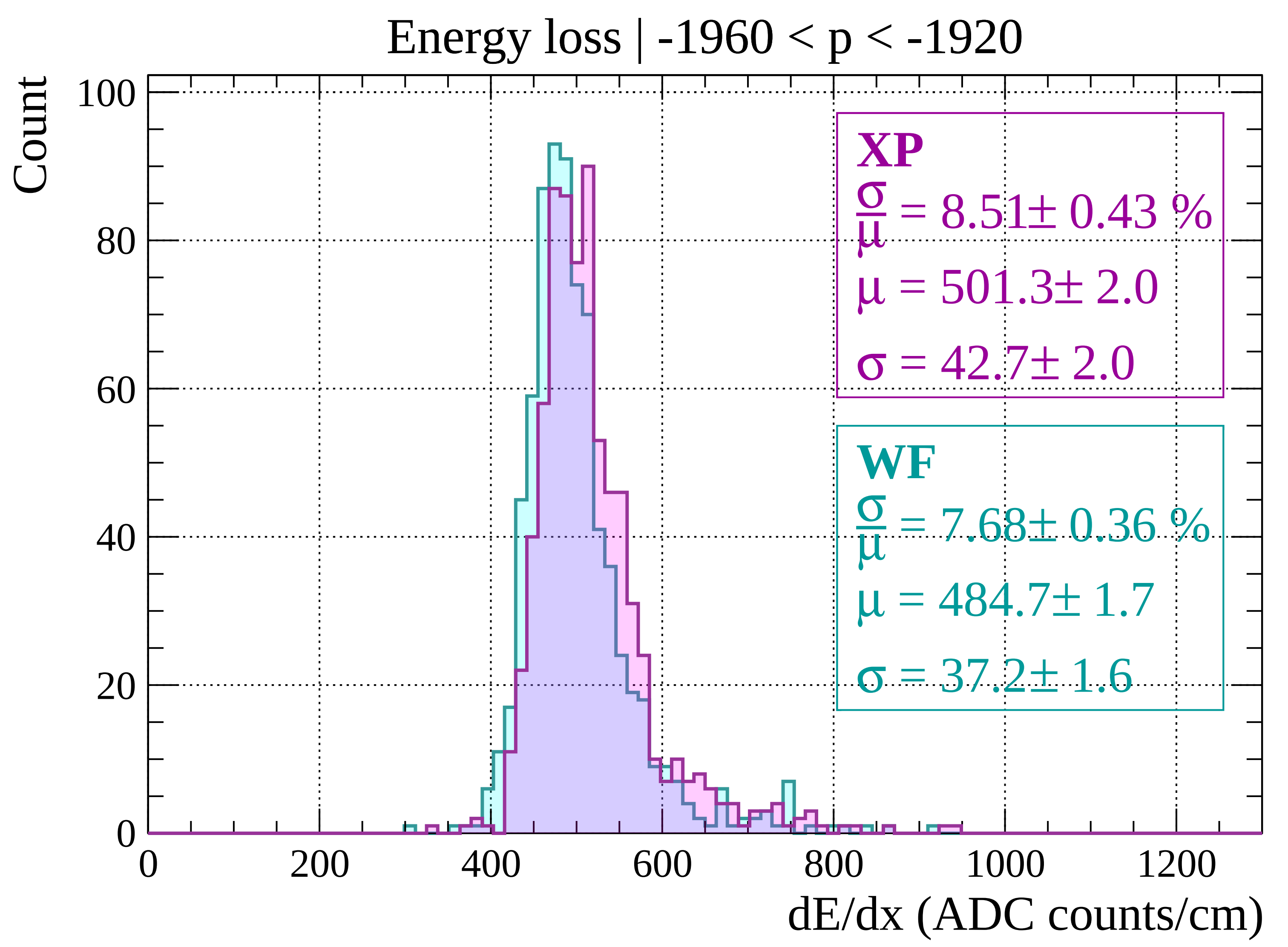

Count
1600
1400
1200
1000
800
600
400
200
0
0
5
10
15
20
25
30
35
40
45
50
χ^2

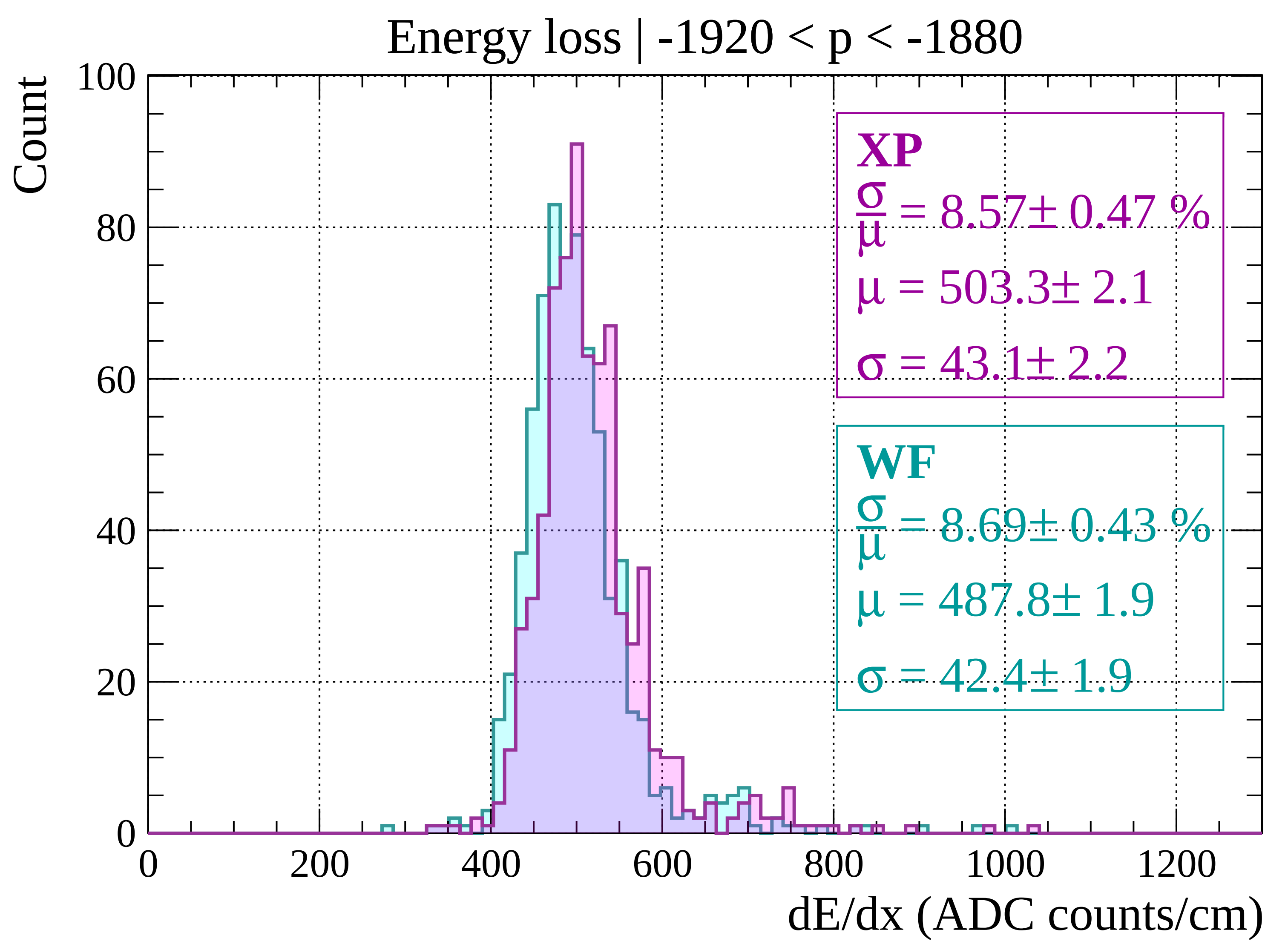

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
70
60
50
40
30
20
10
0
×10^3

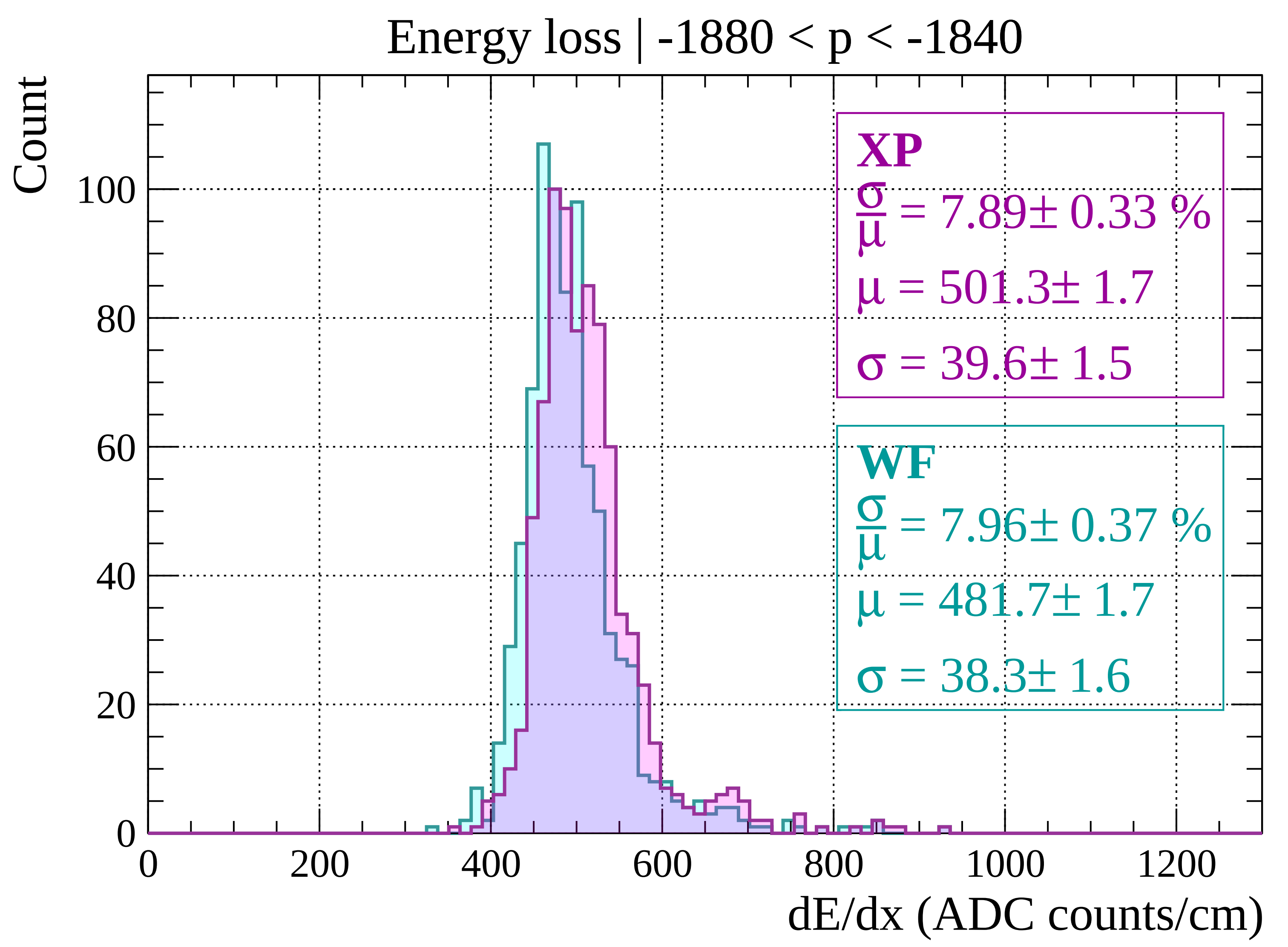


χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
35
30
25
20
15
10
5
0
$\times 10^3$

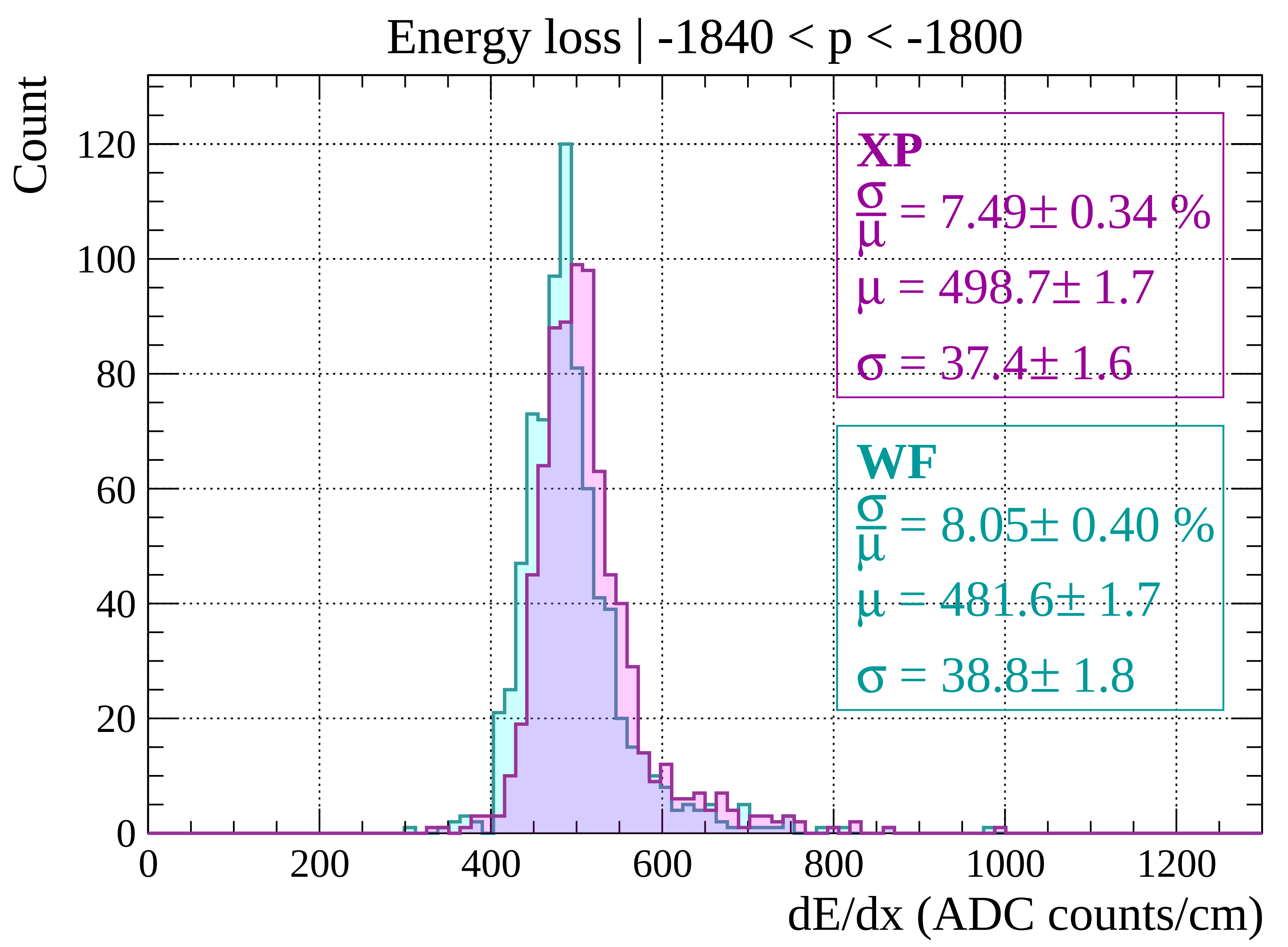

Energy loss | -2000 < p < -1960
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.78± 0.44 %
μ = 500.9± 2.5
σ = 39.0± 2.0
WF
$\frac{\sigma}{\mu}$ = 7.66± 0.46 %
μ = 484.4± 2.5
σ = 37.1± 2.0
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
45

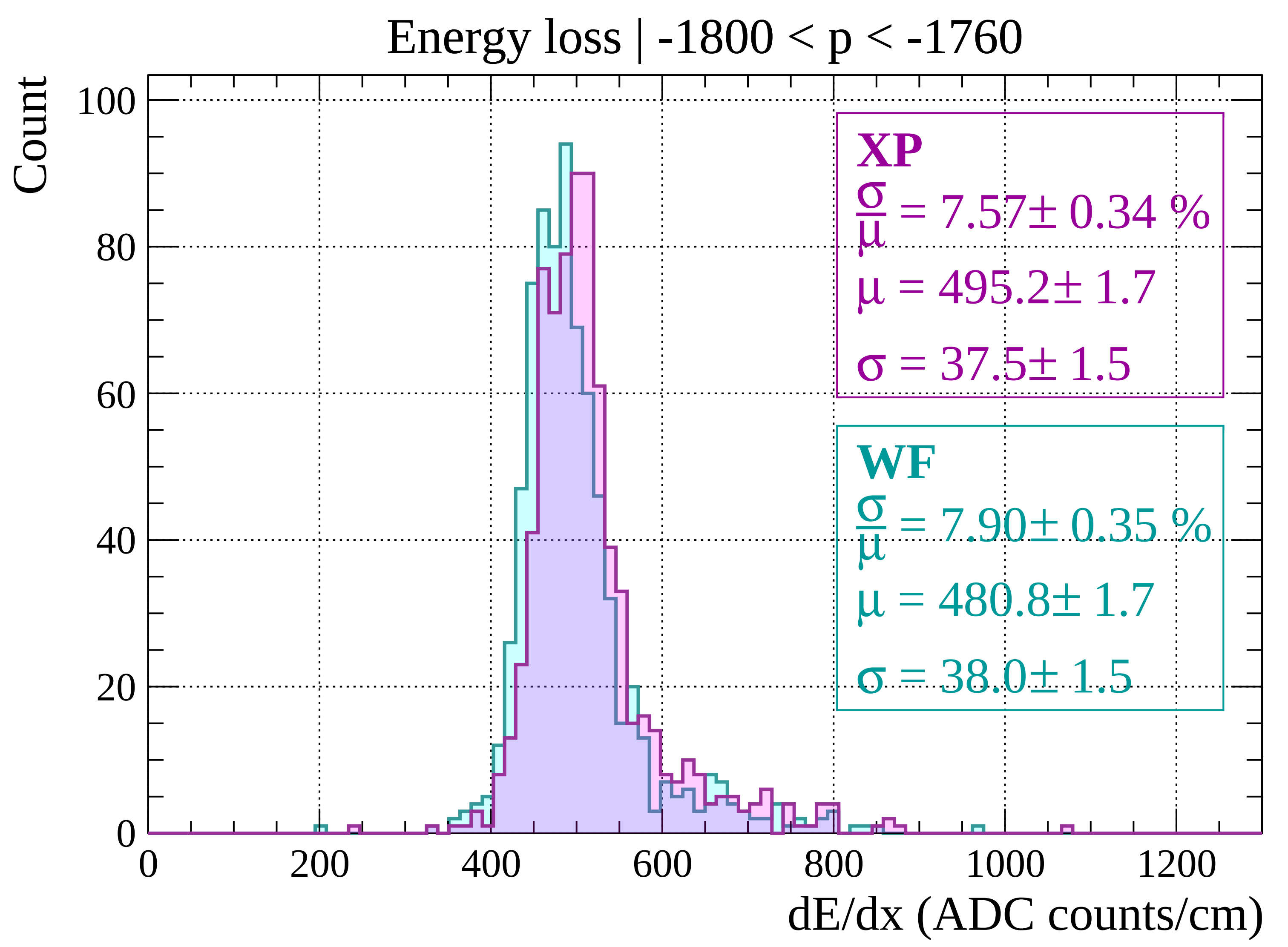

Energy loss | -1960 < p < -1920
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.51± 0.43 %
μ = 501.3± 2.0
σ = 42.7± 2.0
WF
σ/μ = 7.68± 0.36 %
μ = 484.7± 1.7
σ = 37.2± 1.6


Energy loss | -1920 < p < -1880
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.57± 0.47 %
μ = 503.3± 2.1
σ = 43.1± 2.2
WF
$\frac{\sigma}{\mu}$ = 8.69± 0.43 %
μ = 487.8± 1.9
σ = 42.4± 1.9

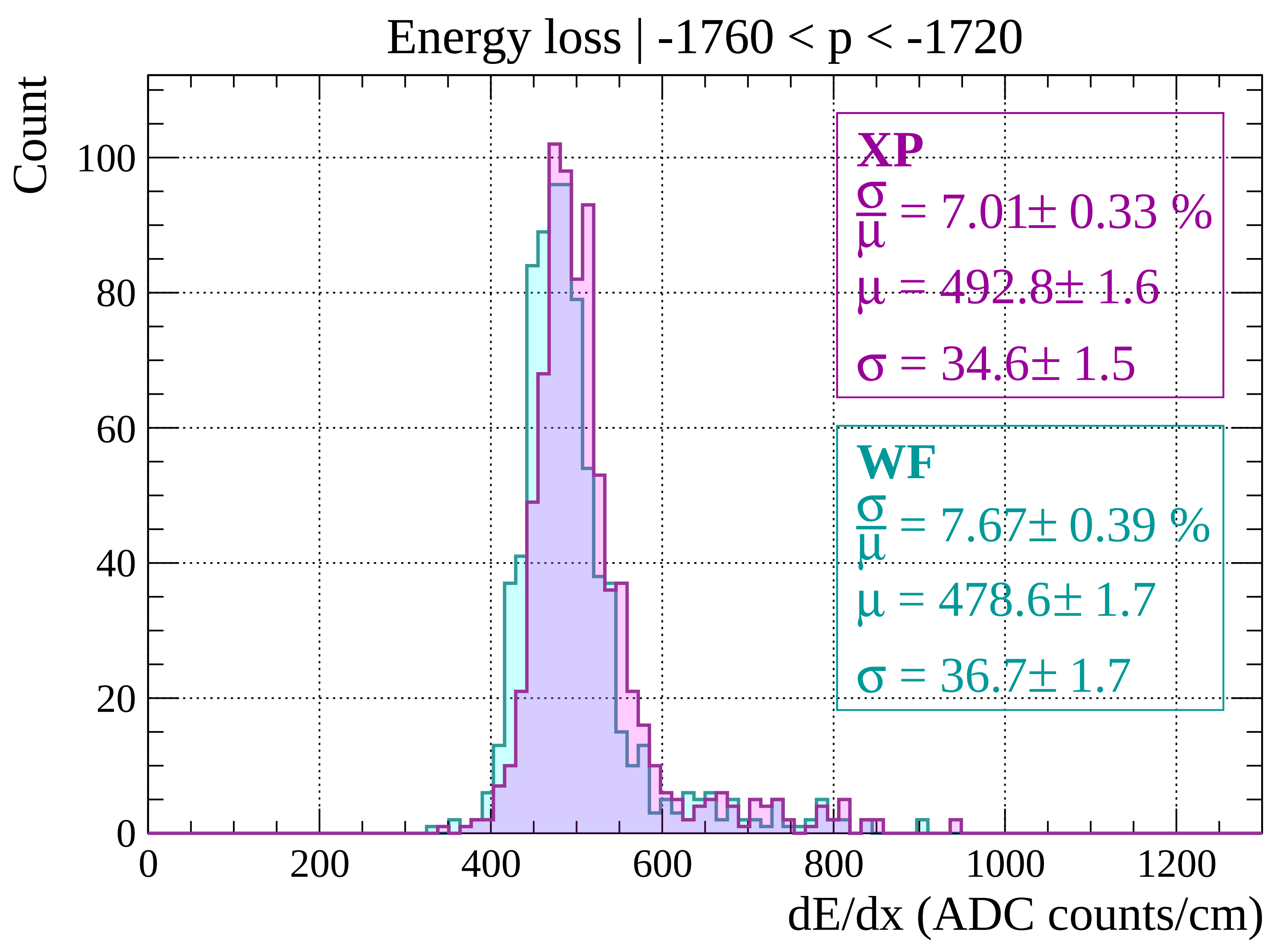


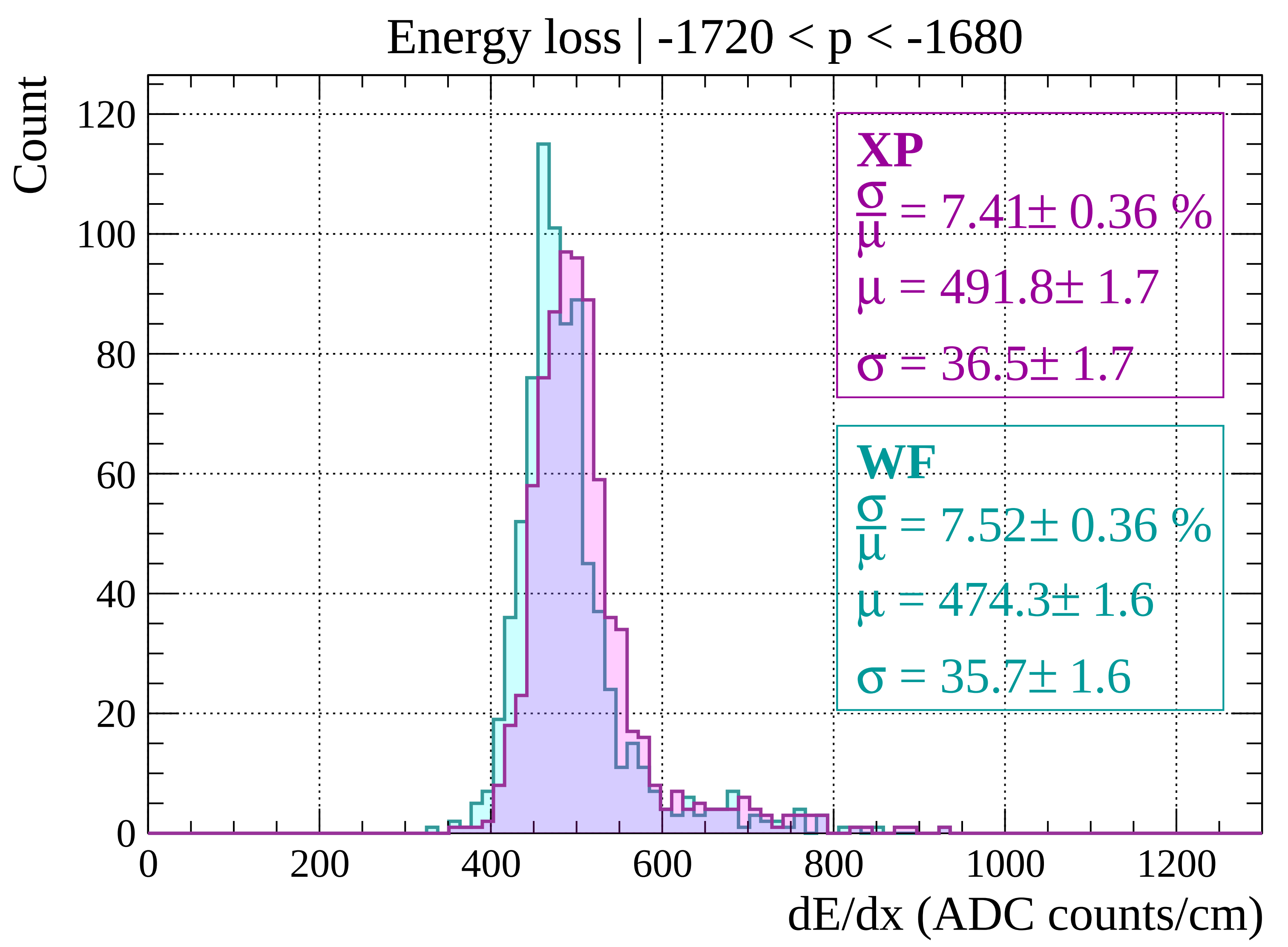
Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP
$\frac{\sigma}{\mu}$ = 7.89± 0.33 %
μ = 501.3± 1.7
σ = 39.6± 1.5
WF
$\frac{\sigma}{\mu}$ = 7.96± 0.37 %
μ = 481.7± 1.7
σ = 38.3± 1.6


Energy loss | -1840 < p < -1800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
XP
σ/μ = 7.49± 0.34 %
μ = 498.7± 1.7
σ = 37.4± 1.6
WF
σ/μ = 8.05± 0.40 %
μ = 481.6± 1.7
σ = 38.8± 1.8

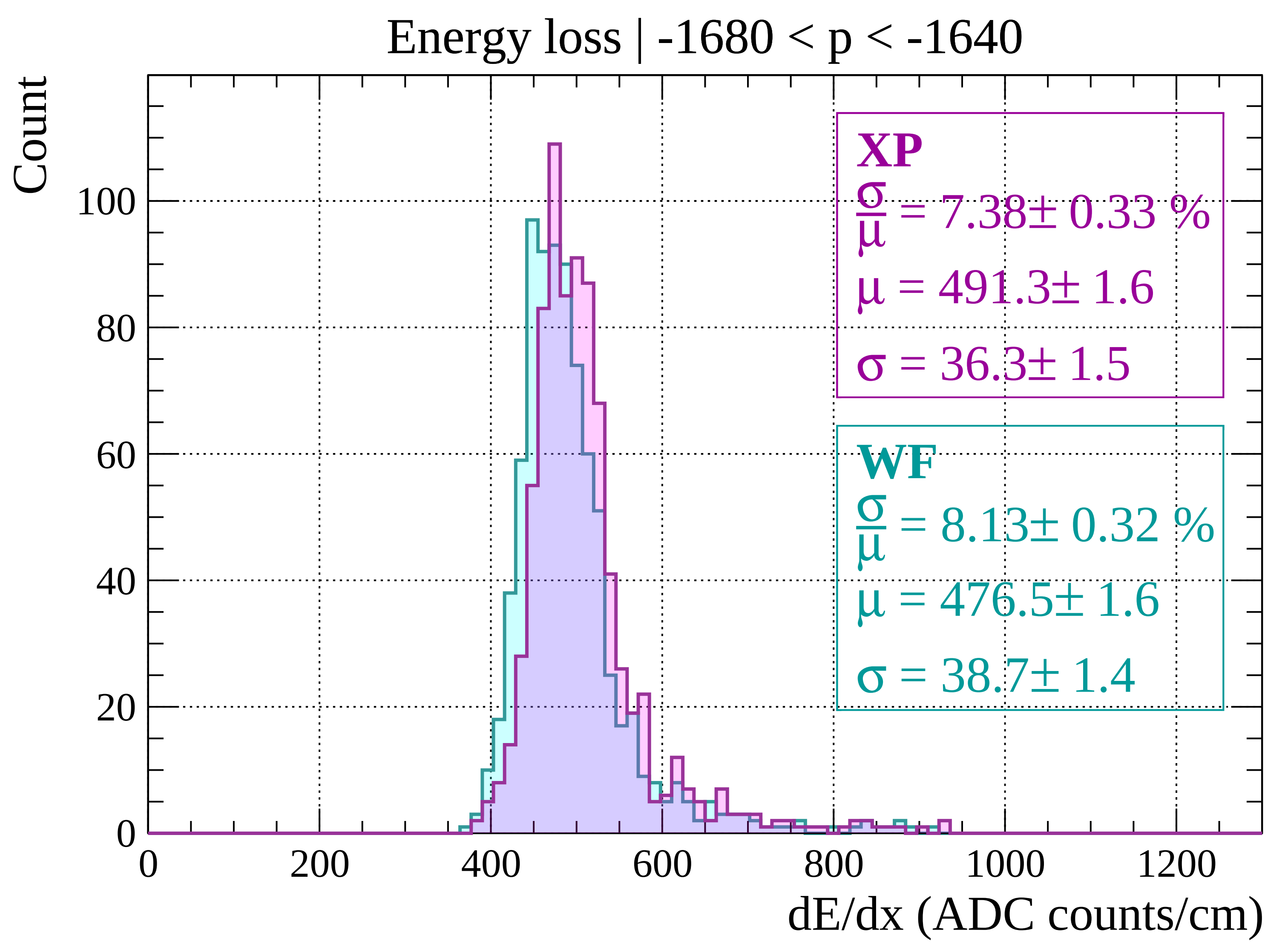


Energy loss | -1800 < p < -1760
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 7.57± 0.34 %
μ = 495.2± 1.7
σ = 37.5± 1.5
WF
σ/μ = 7.90± 0.35 %
μ = 480.8± 1.7
σ = 38.0± 1.5

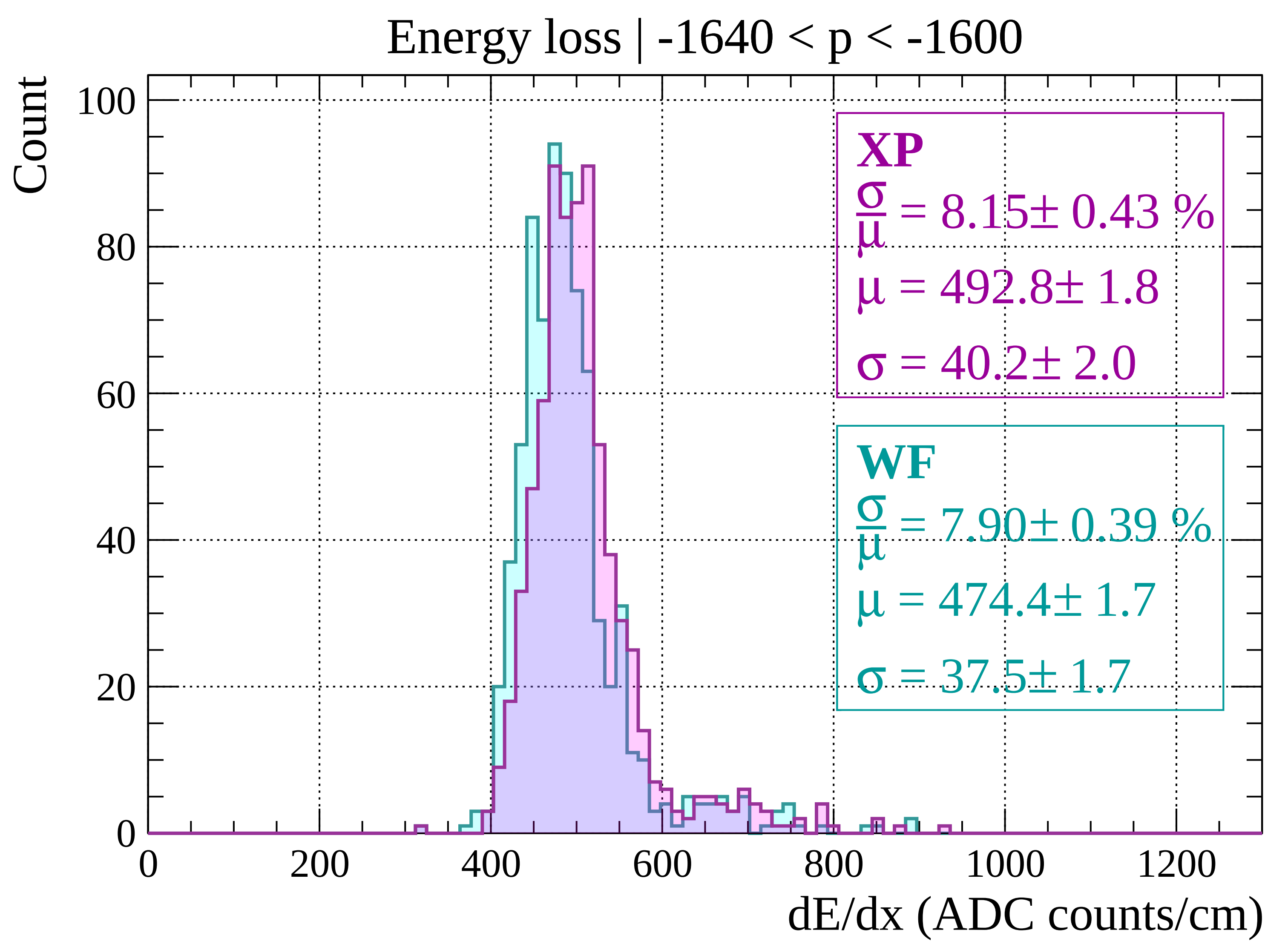

Energy loss | -1760 < p < -1720
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.01± 0.33 %
μ = 492.8± 1.6
σ = 34.6± 1.5
WF
σ/μ = 7.67± 0.39 %
μ = 478.6± 1.7
σ = 36.7± 1.7
0
20
40
60
80
100
200
400
600
800
1000
1200



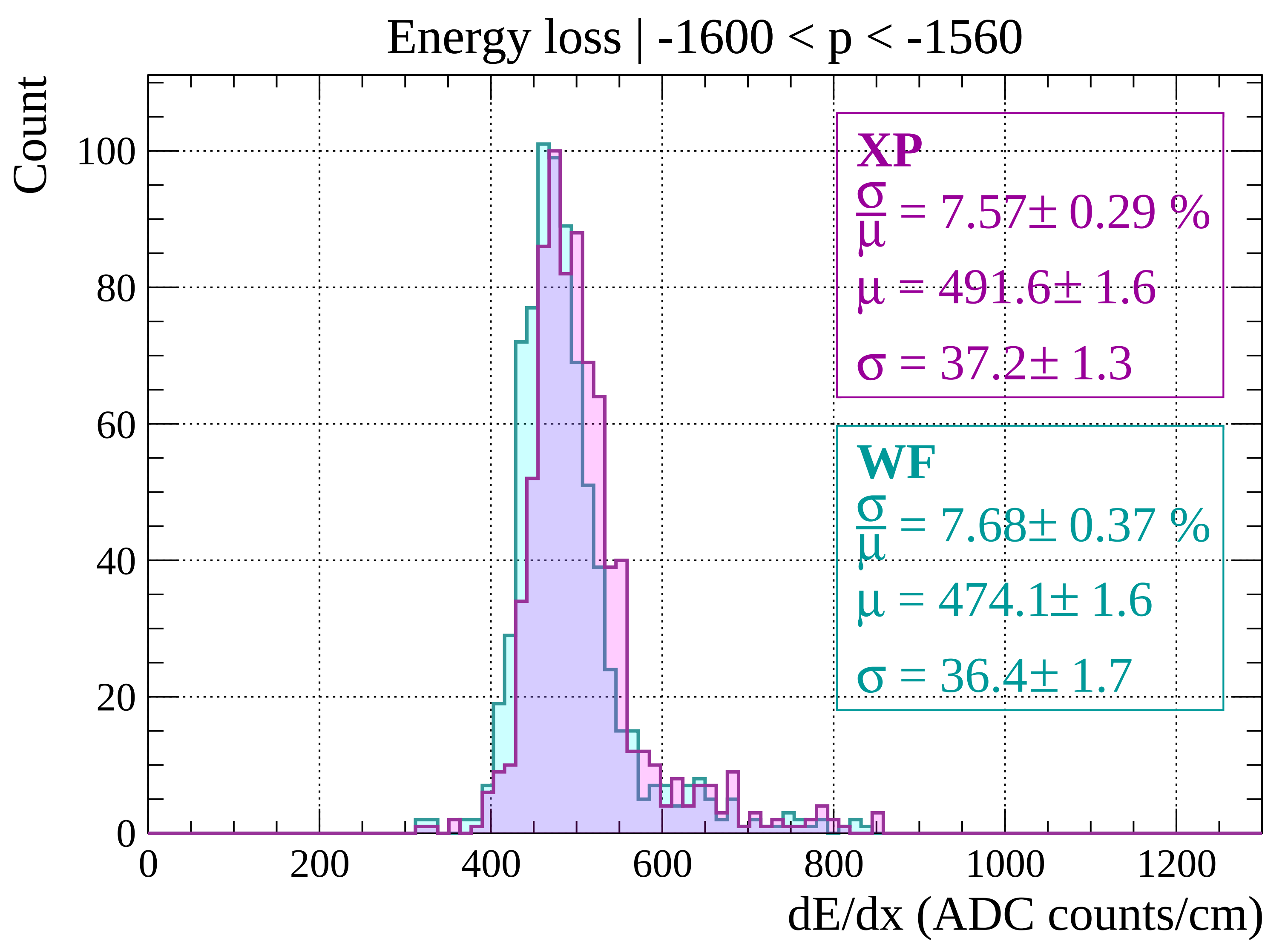
Energy loss | -1720 < p < -1680
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 7.41± 0.36 %
μ = 491.8± 1.7
σ = 36.5± 1.7
WF
σ/μ = 7.52± 0.36 %
μ = 474.3± 1.6
σ = 35.7± 1.6

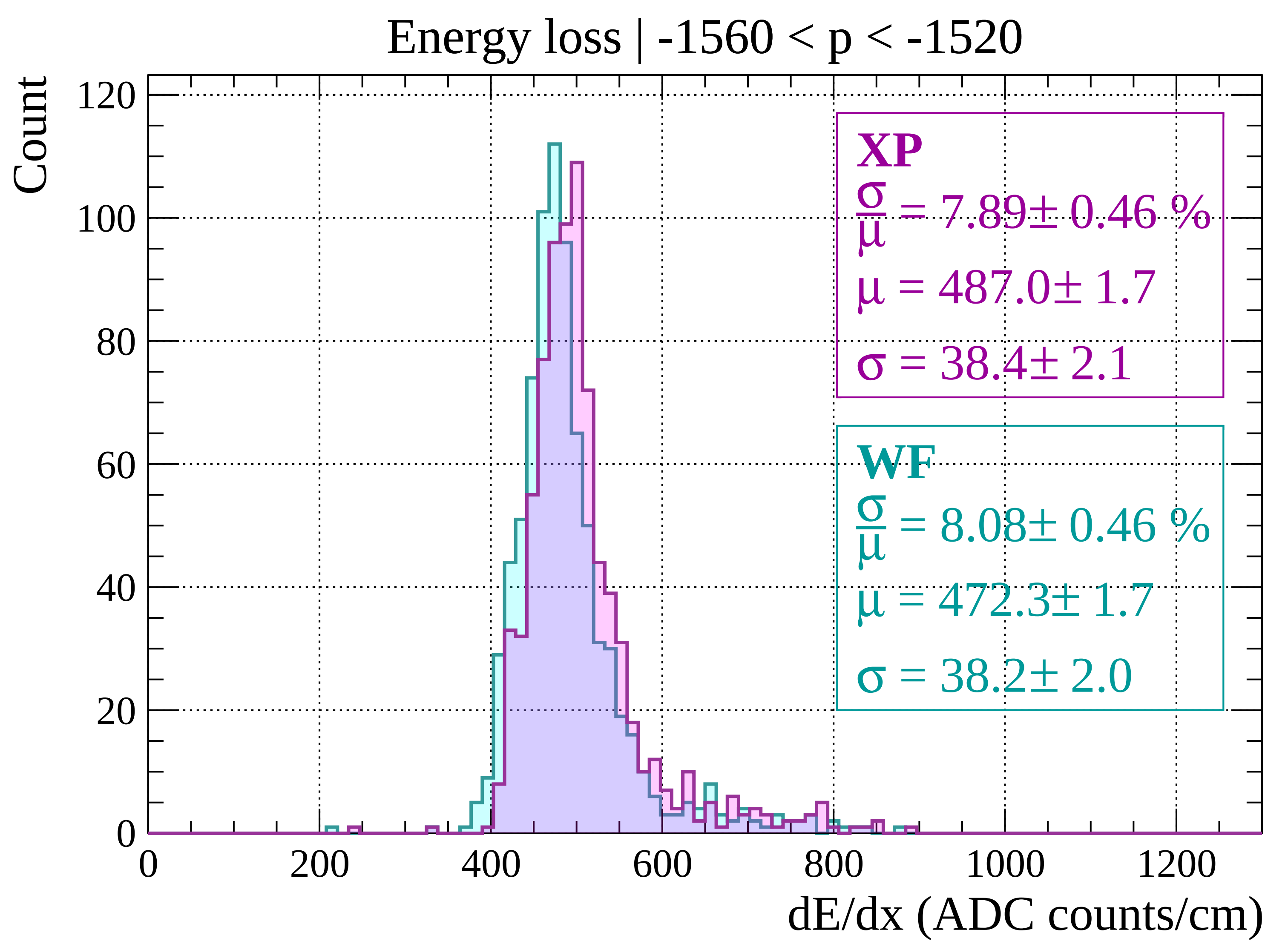


Energy loss | -1680 < p < -1640
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 7.38± 0.33 %
μ = 491.3± 1.6
σ = 36.3± 1.5
WF
σ/μ = 8.13± 0.32 %
μ = 476.5± 1.6
σ = 38.7± 1.4

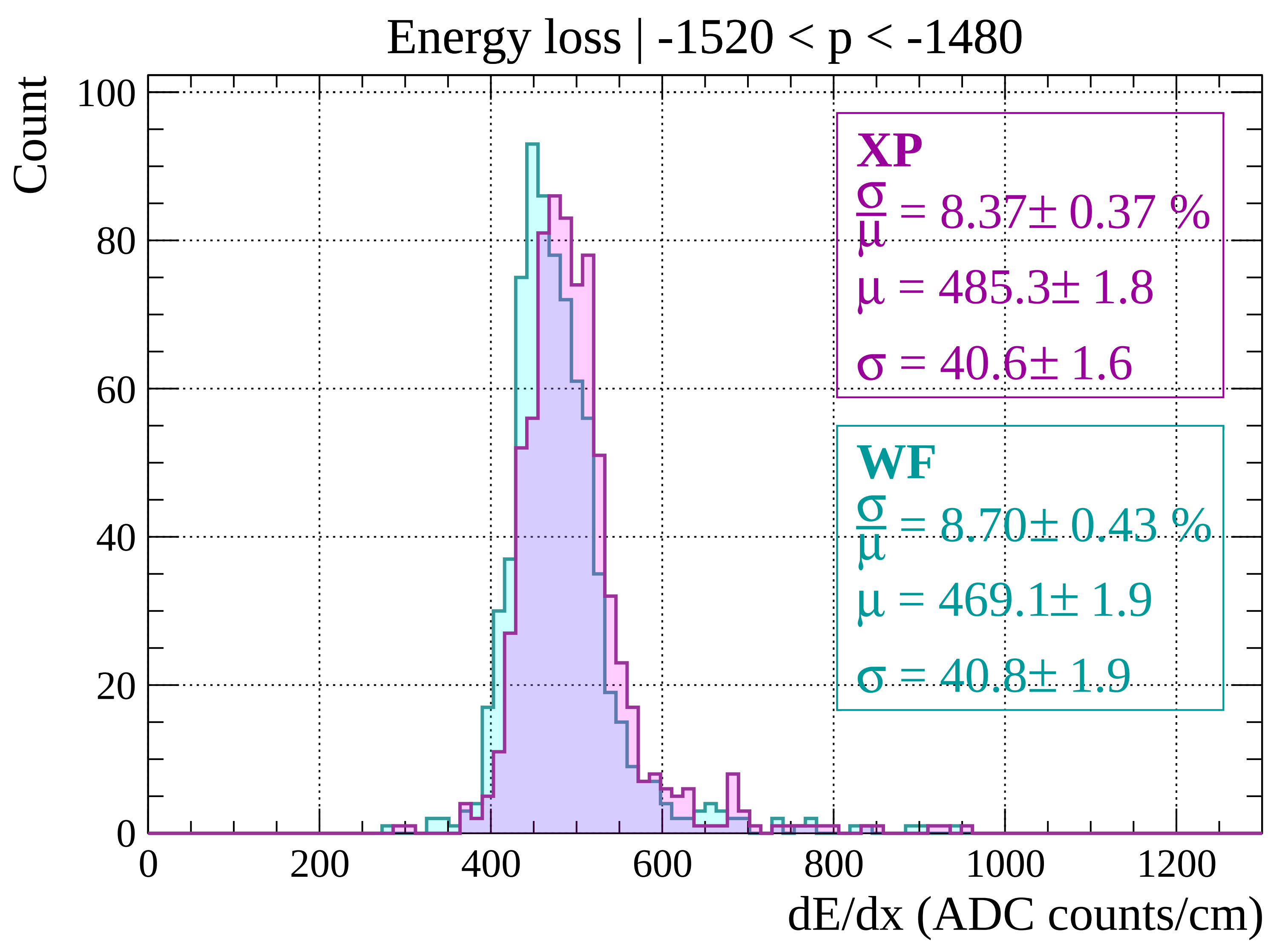


Energy loss | -1640 < p < -1600
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.15± 0.43 %
μ = 492.8± 1.8
σ = 40.2± 2.0
WF
$\frac{\sigma}{\mu}$ = 7.90± 0.39 %
μ = 474.4± 1.7
σ = 37.5± 1.7

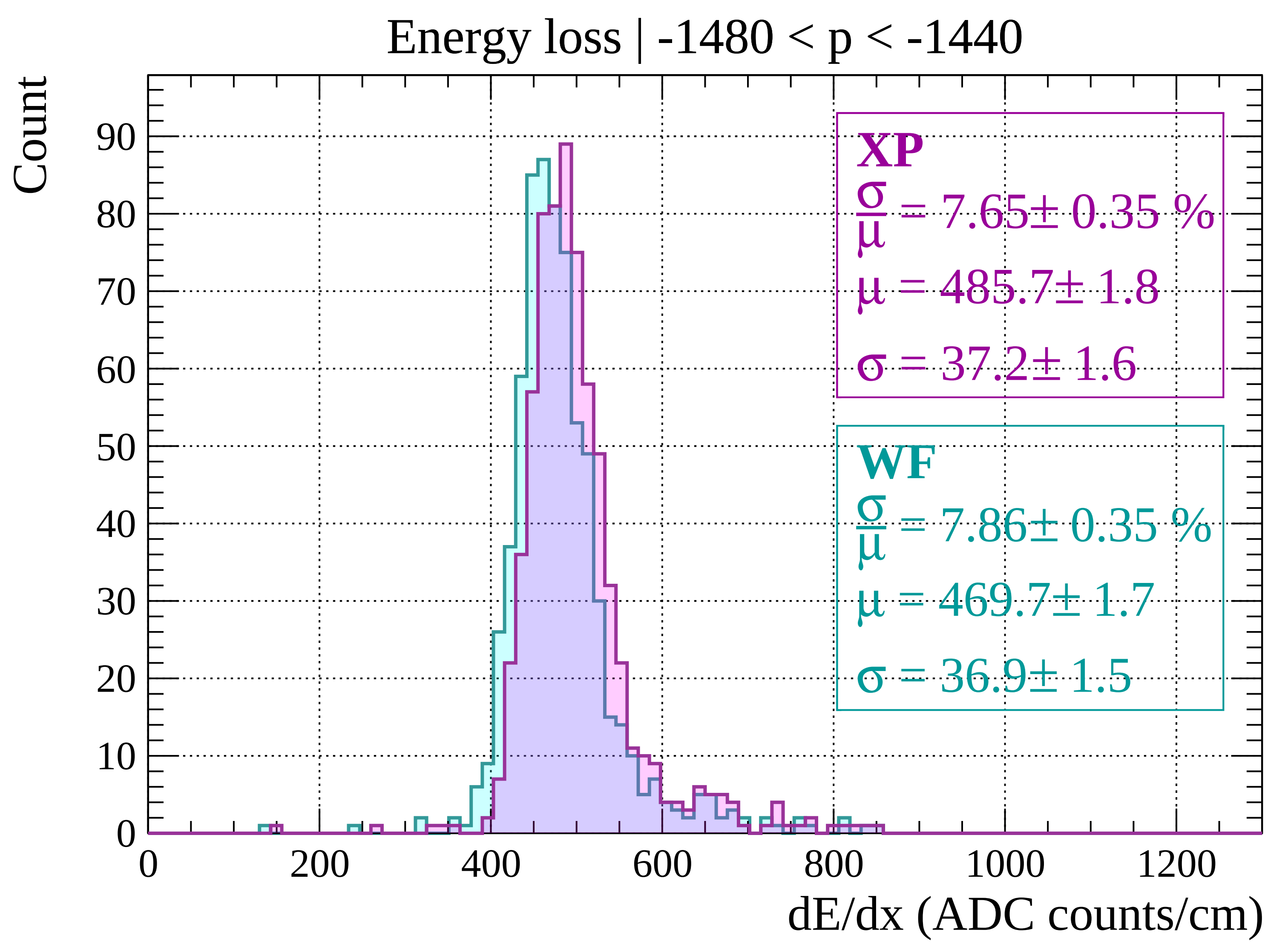

Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
σ/μ = 7.57± 0.29 %
μ = 491.6± 1.6
σ = 37.2± 1.3
WF
σ/μ = 7.68± 0.37 %
μ = 474.1± 1.6
σ = 36.4± 1.7



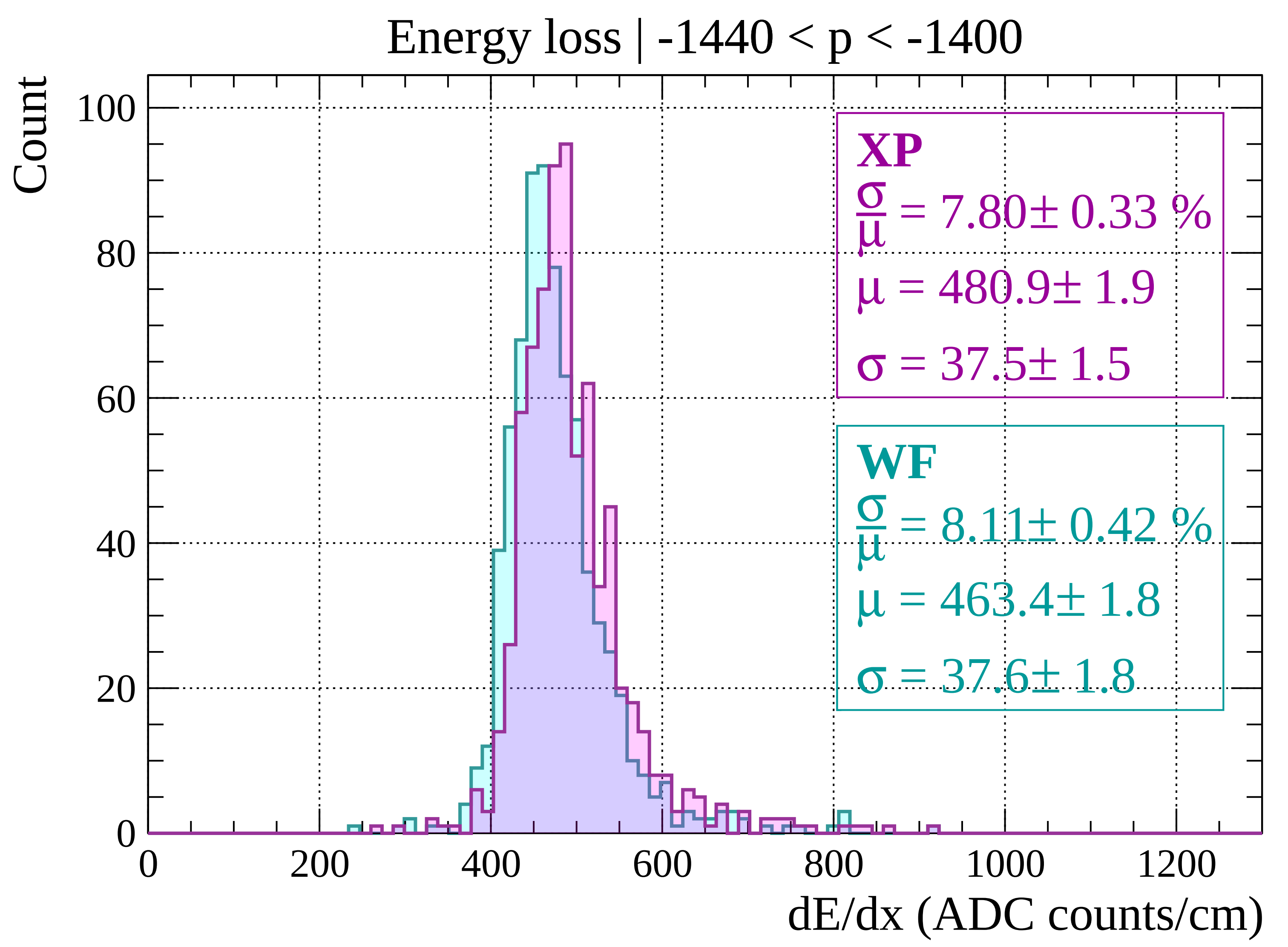
Energy loss | -1560 < p < -1520
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.89± 0.46 %
μ = 487.0± 1.7
σ = 38.4± 2.1
WF
$\frac{\sigma}{\mu}$ = 8.08± 0.46 %
μ = 472.3± 1.7
σ = 38.2± 2.0



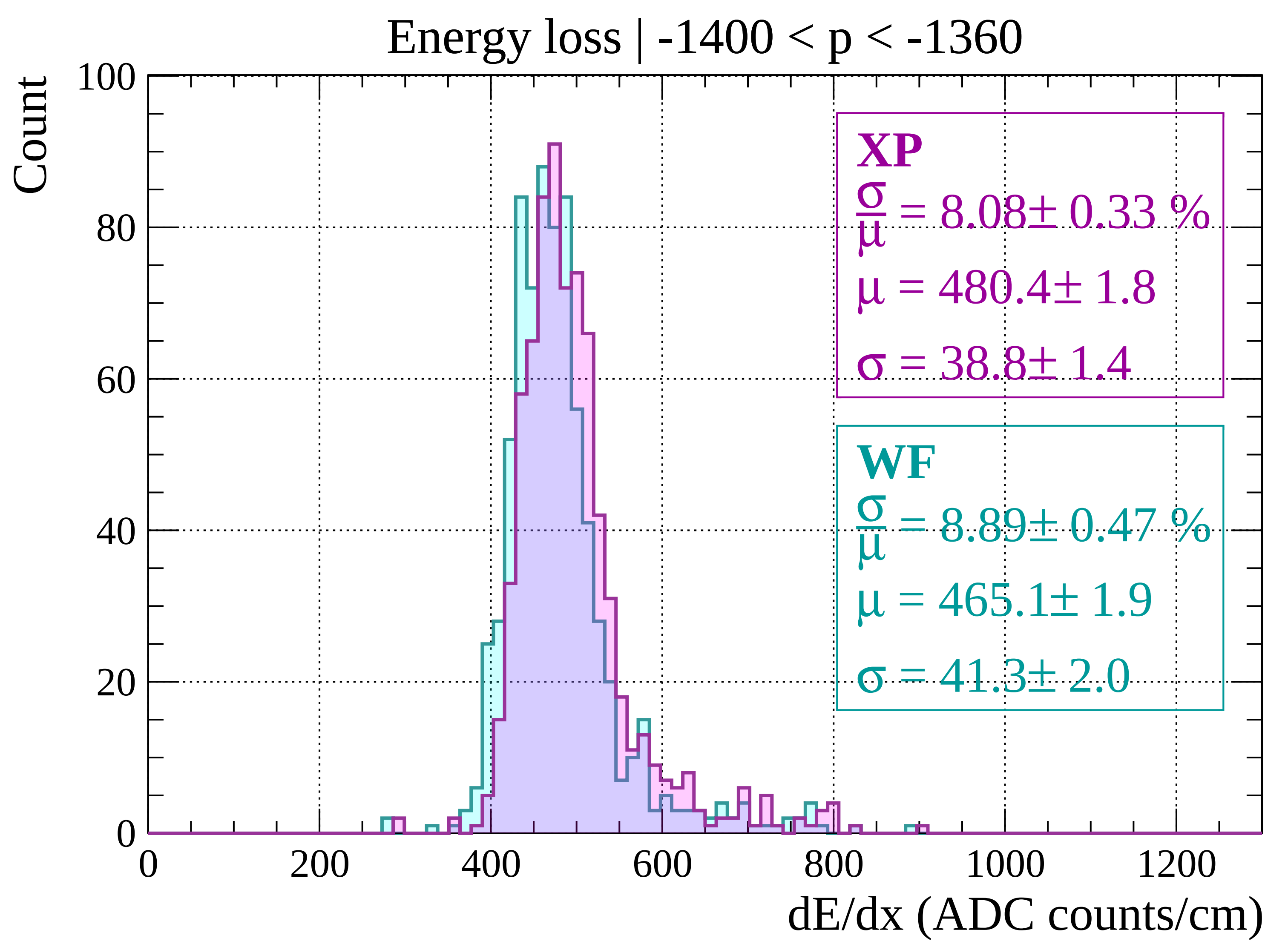
Energy loss | -1520 < p < -1480
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 8.37 \pm 0.37$ %
$\mu = 485.3 \pm 1.8$
$\sigma = 40.6 \pm 1.6$
WF
$\frac{\sigma}{\mu} = 8.70 \pm 0.43$ %
$\mu = 469.1 \pm 1.9$
$\sigma = 40.8 \pm 1.9$



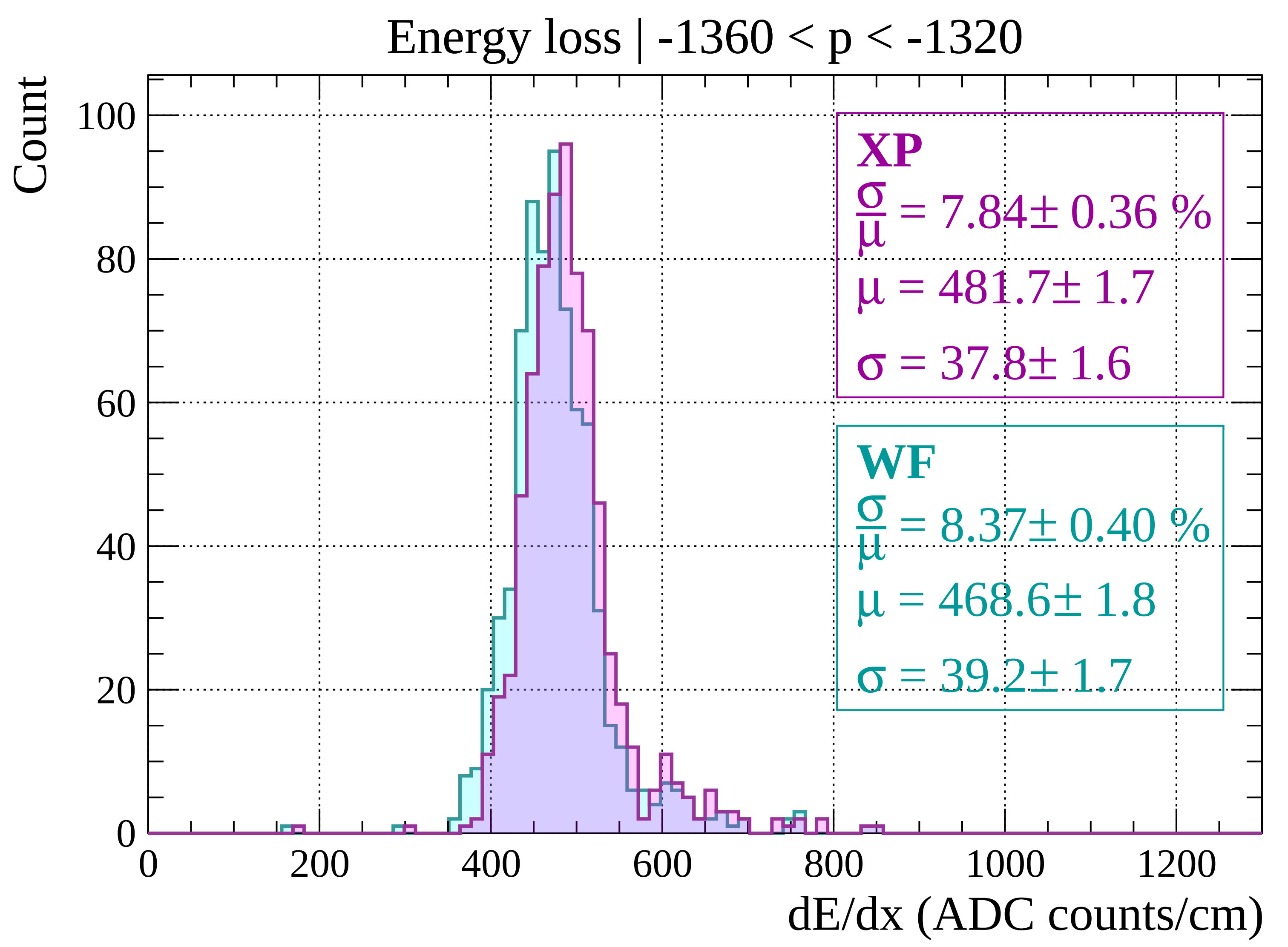
Energy loss | -1480 < p < -1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP
$\frac{\sigma}{\mu} = 7.65 \pm 0.35$ %
$\mu = 485.7 \pm 1.8$
$\sigma = 37.2 \pm 1.6$
WF
$\frac{\sigma}{\mu} = 7.86 \pm 0.35$ %
$\mu = 469.7 \pm 1.7$
$\sigma = 36.9 \pm 1.5$



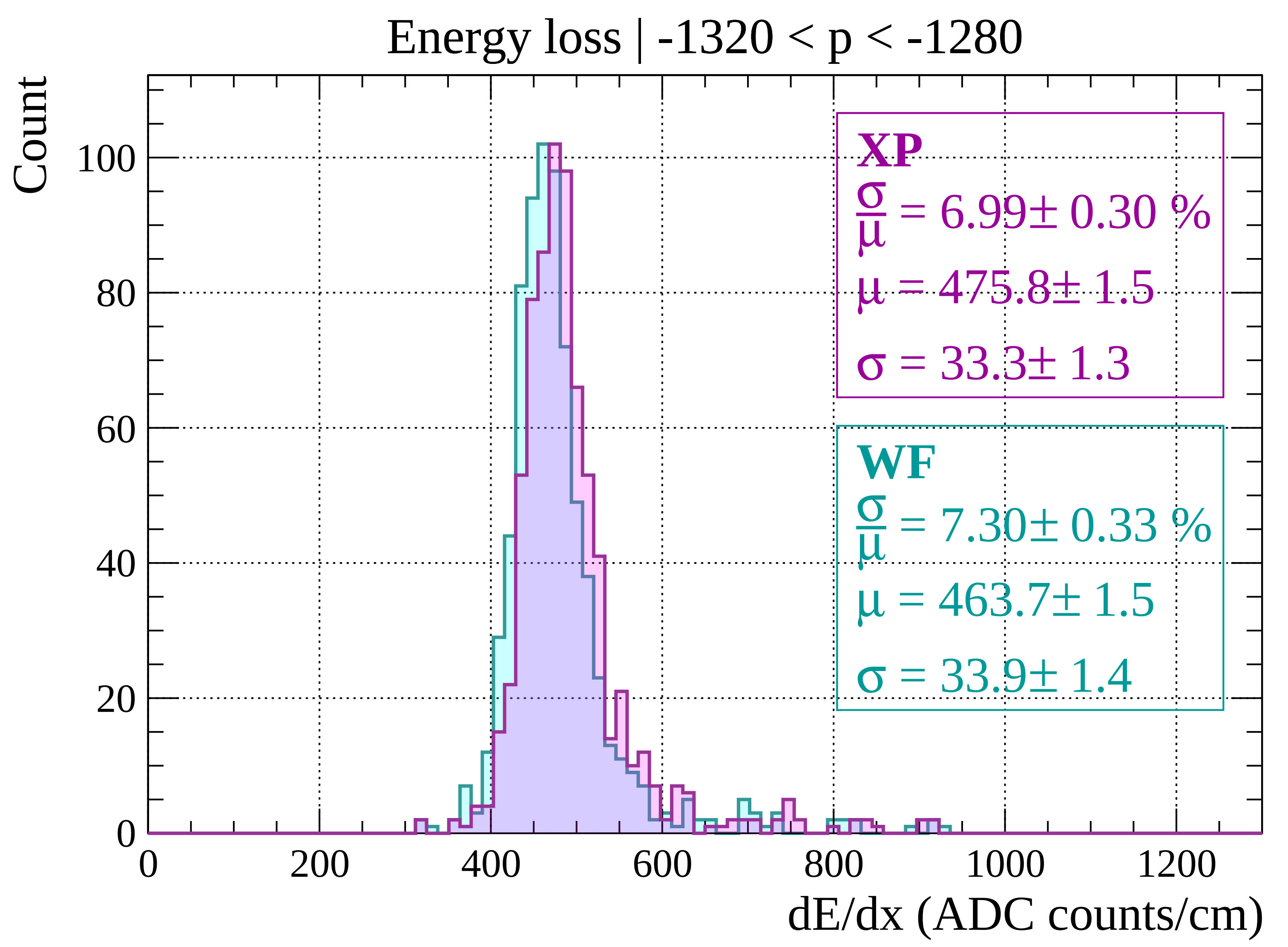
Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
σ/μ = 7.80± 0.33 %
μ = 480.9± 1.9
σ = 37.5± 1.5
WF
σ/μ = 8.11± 0.42 %
μ = 463.4± 1.8
σ = 37.6± 1.8

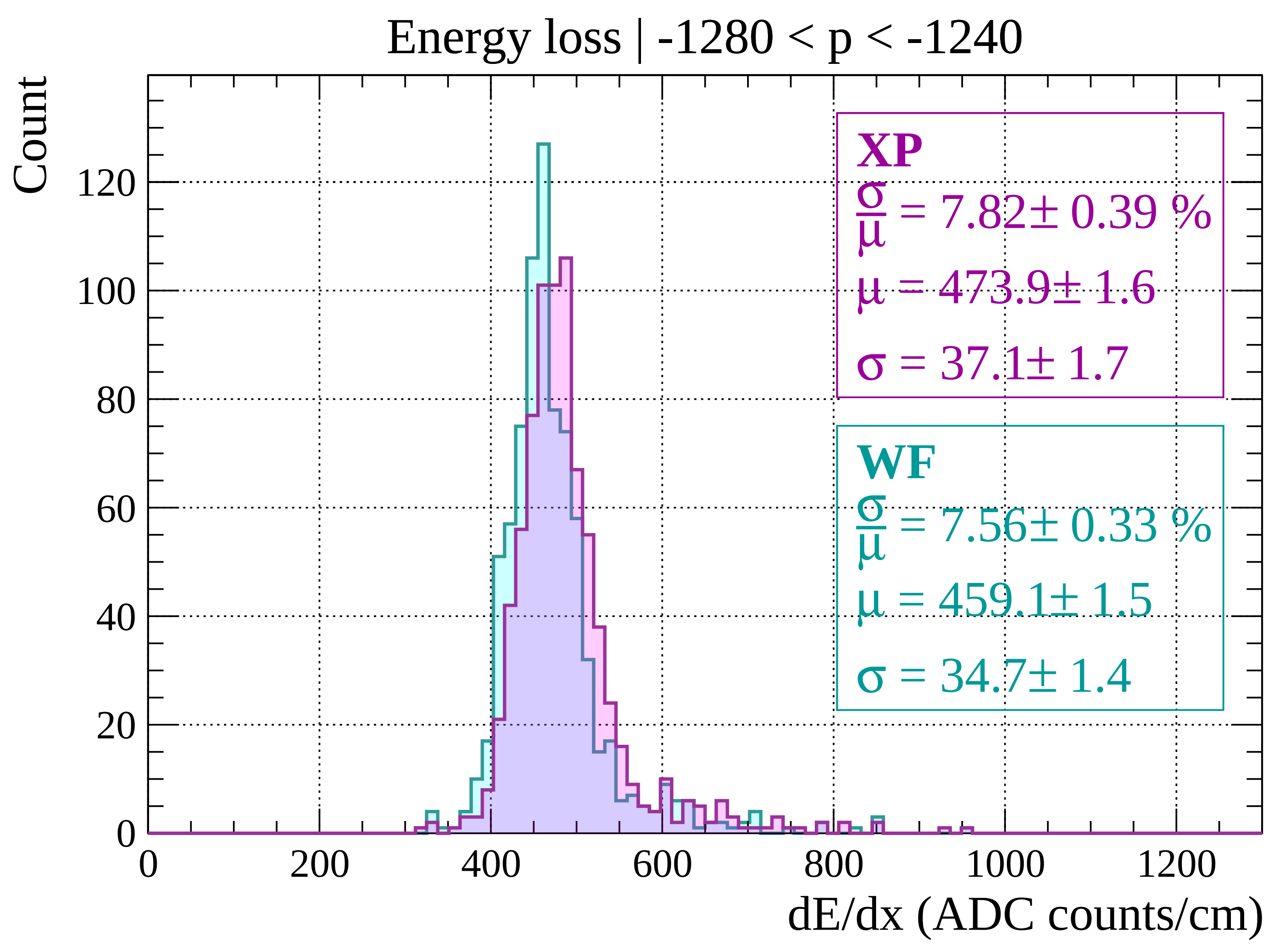


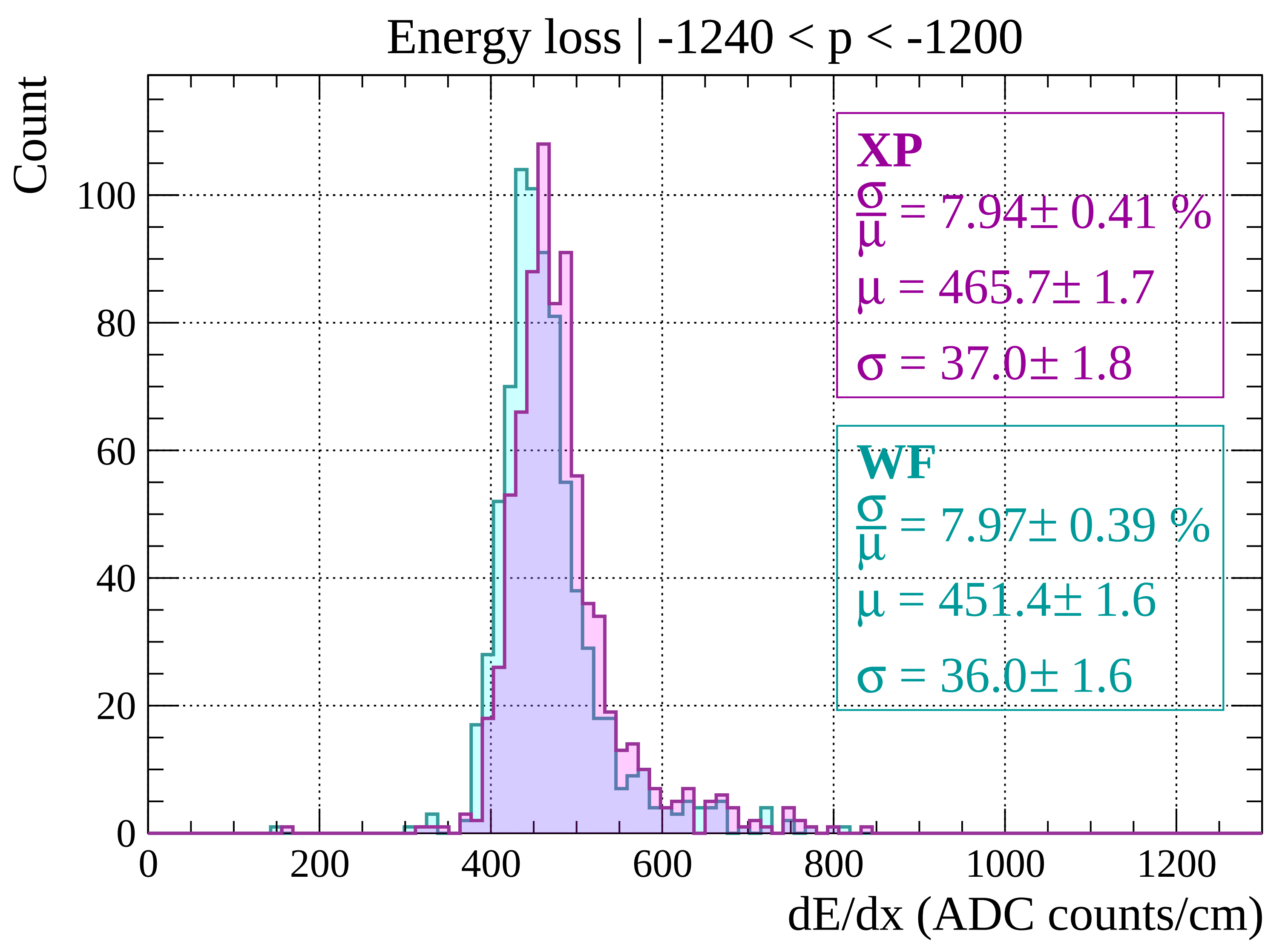
Energy loss | -1400 < p < -1360
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.08± 0.33 %
μ = 480.4± 1.8
σ = 38.8± 1.4
WF
σ/μ = 8.89± 0.47 %
μ = 465.1± 1.9
σ = 41.3± 2.0

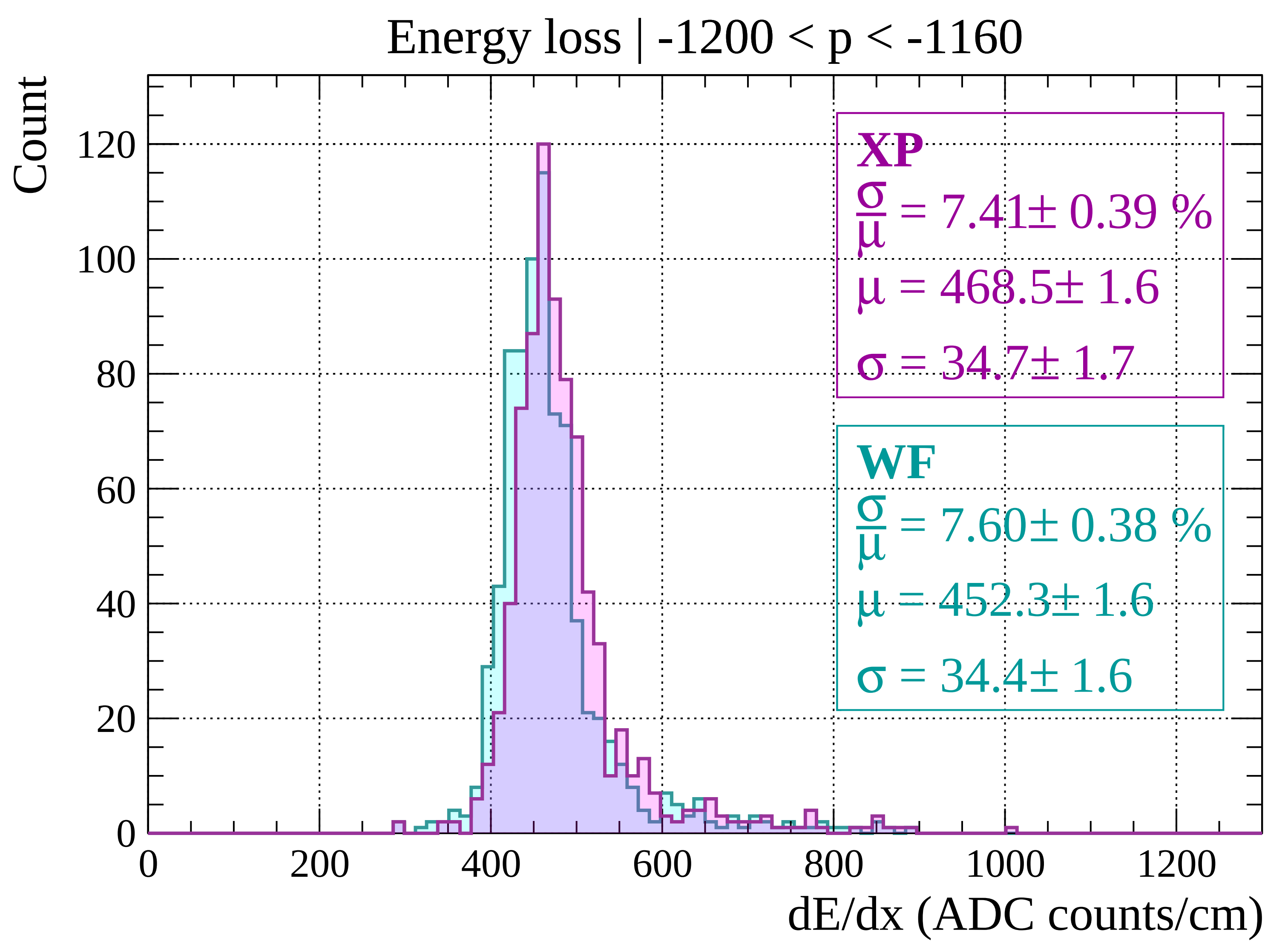


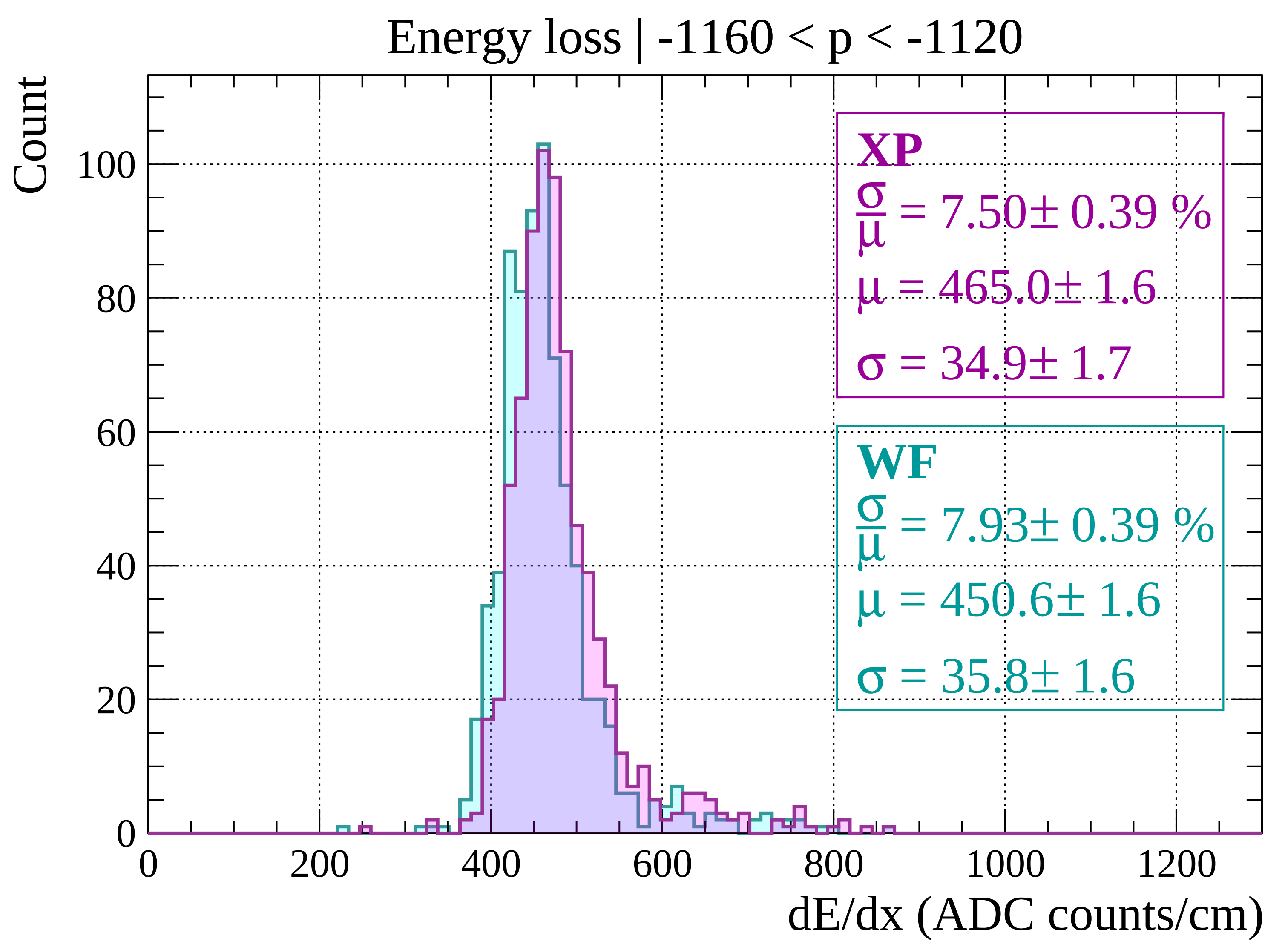
Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.84± 0.36 %
μ = 481.7± 1.7
σ = 37.8± 1.6
WF
σ/μ = 8.37± 0.40 %
μ = 468.6± 1.8
σ = 39.2± 1.7


Energy loss | -1320 < p < -1280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
σ/μ = 6.99± 0.30 %
μ = 475.8± 1.5
σ = 33.3± 1.3
WF
σ/μ = 7.30± 0.33 %
μ = 463.7± 1.5
σ = 33.9± 1.4

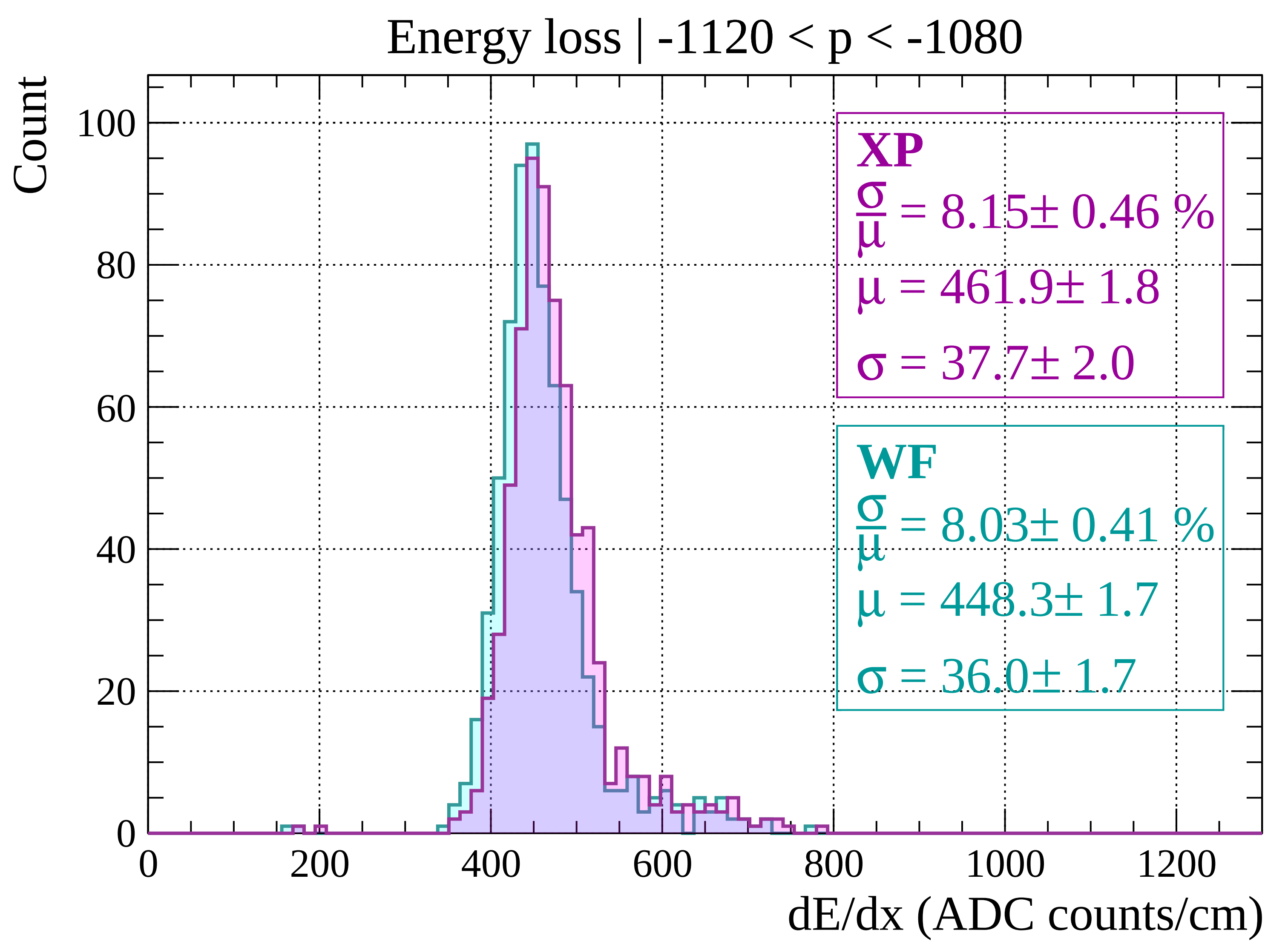

Energy loss | -1280 < p < -1240
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.82± 0.39 %
μ = 473.9± 1.6
σ = 37.1± 1.7
WF
σ/μ = 7.56± 0.33 %
μ = 459.1± 1.5
σ = 34.7± 1.4

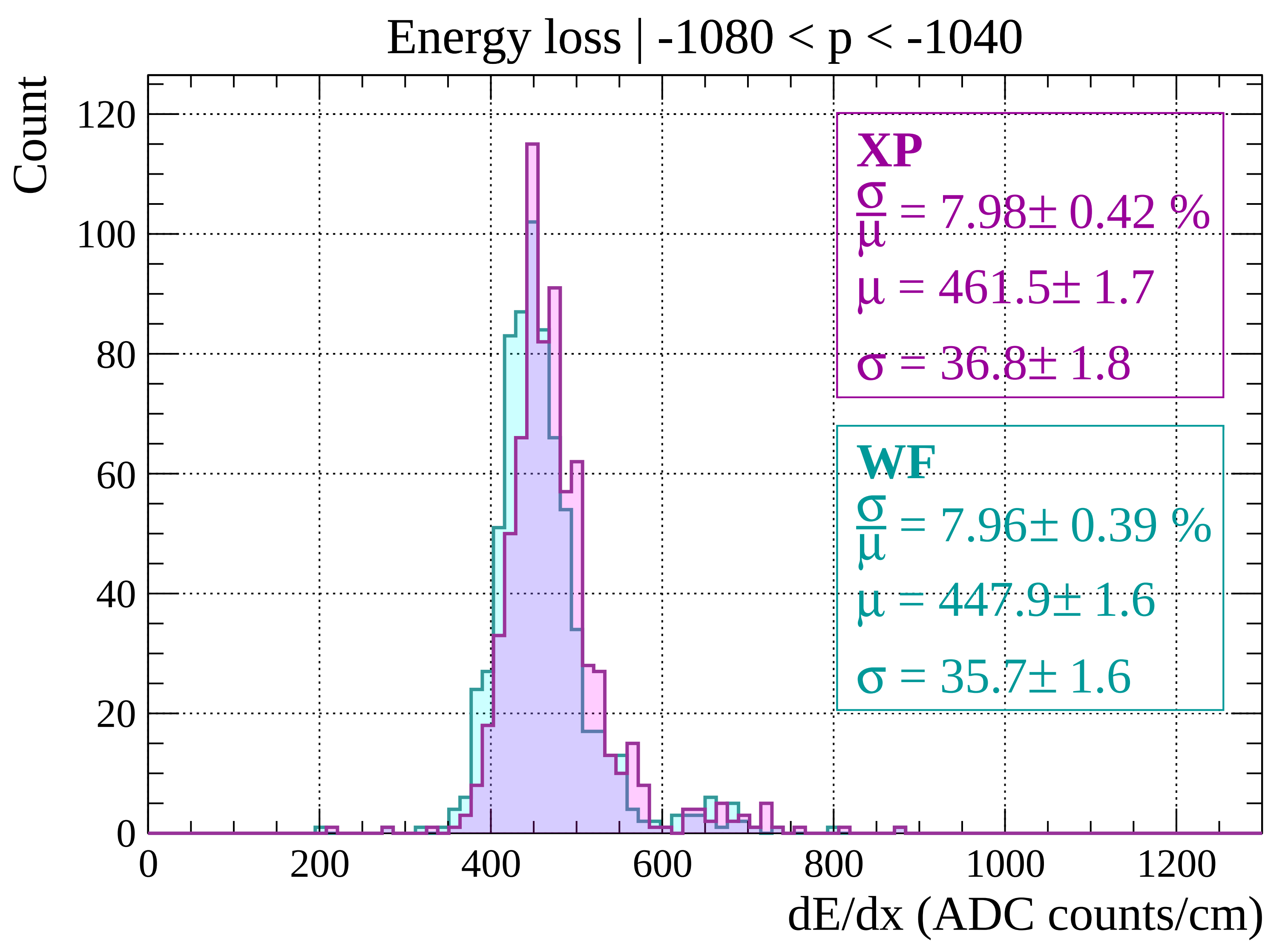

Energy loss | -1240 < p < -1200
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.94 ± 0.41 %
μ = 465.7 ± 1.7
σ = 37.0 ± 1.8
WF
$\frac{\sigma}{\mu}$ = 7.97 ± 0.39 %
μ = 451.4 ± 1.6
σ = 36.0 ± 1.6
0
20
40
60
80
100
0
200
400
600
800
1000
1200


Energy loss | -1200 < p < -1160
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.41± 0.39 %
μ = 468.5± 1.6
σ = 34.7± 1.7
WF
σ/μ = 7.60± 0.38 %
μ = 452.3± 1.6
σ = 34.4± 1.6

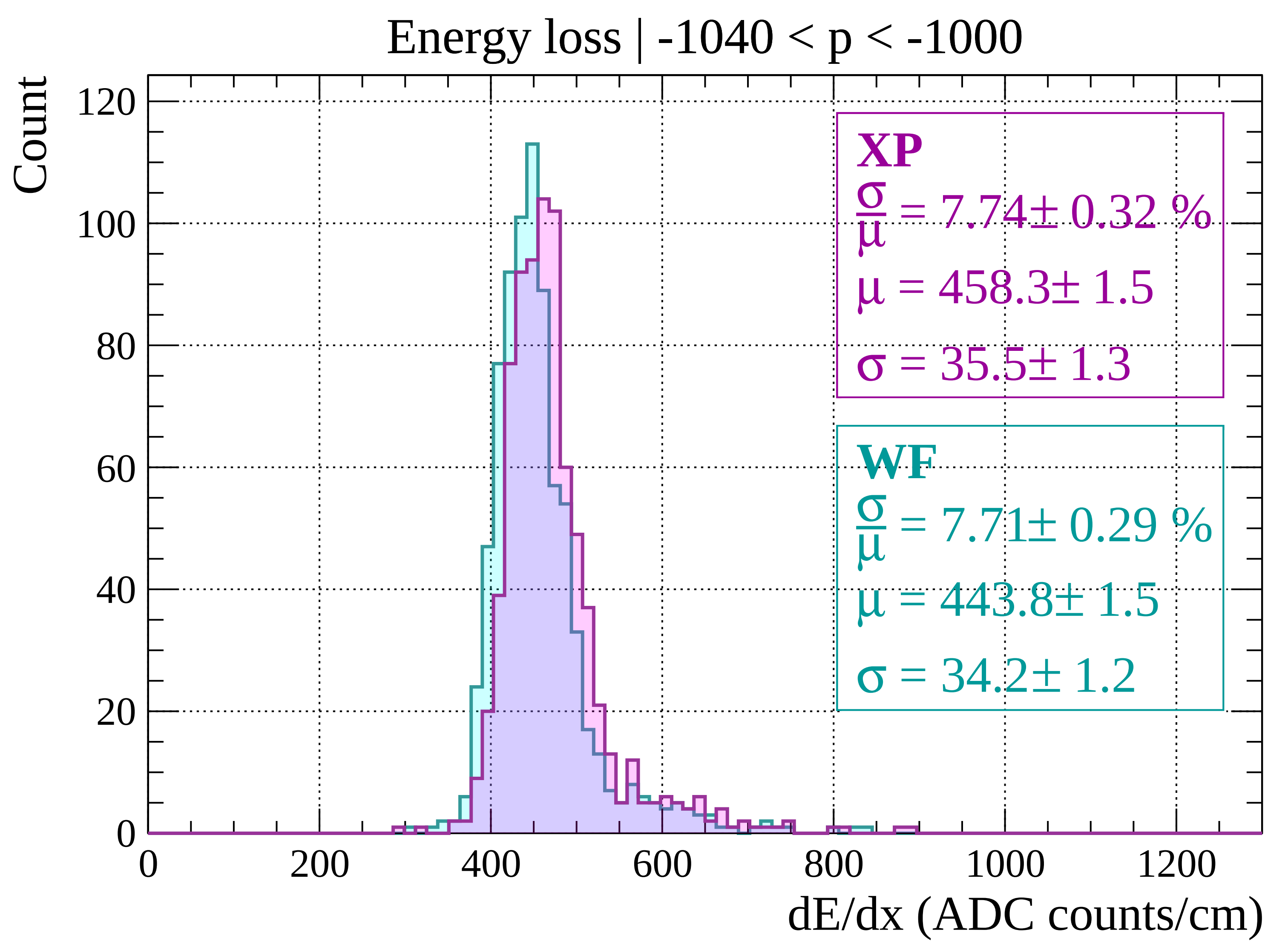


Energy loss | -1160 < p < -1120
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.50± 0.39 %
μ = 465.0± 1.6
σ = 34.9± 1.7
WF
σ/μ = 7.93± 0.39 %
μ = 450.6± 1.6
σ = 35.8± 1.6

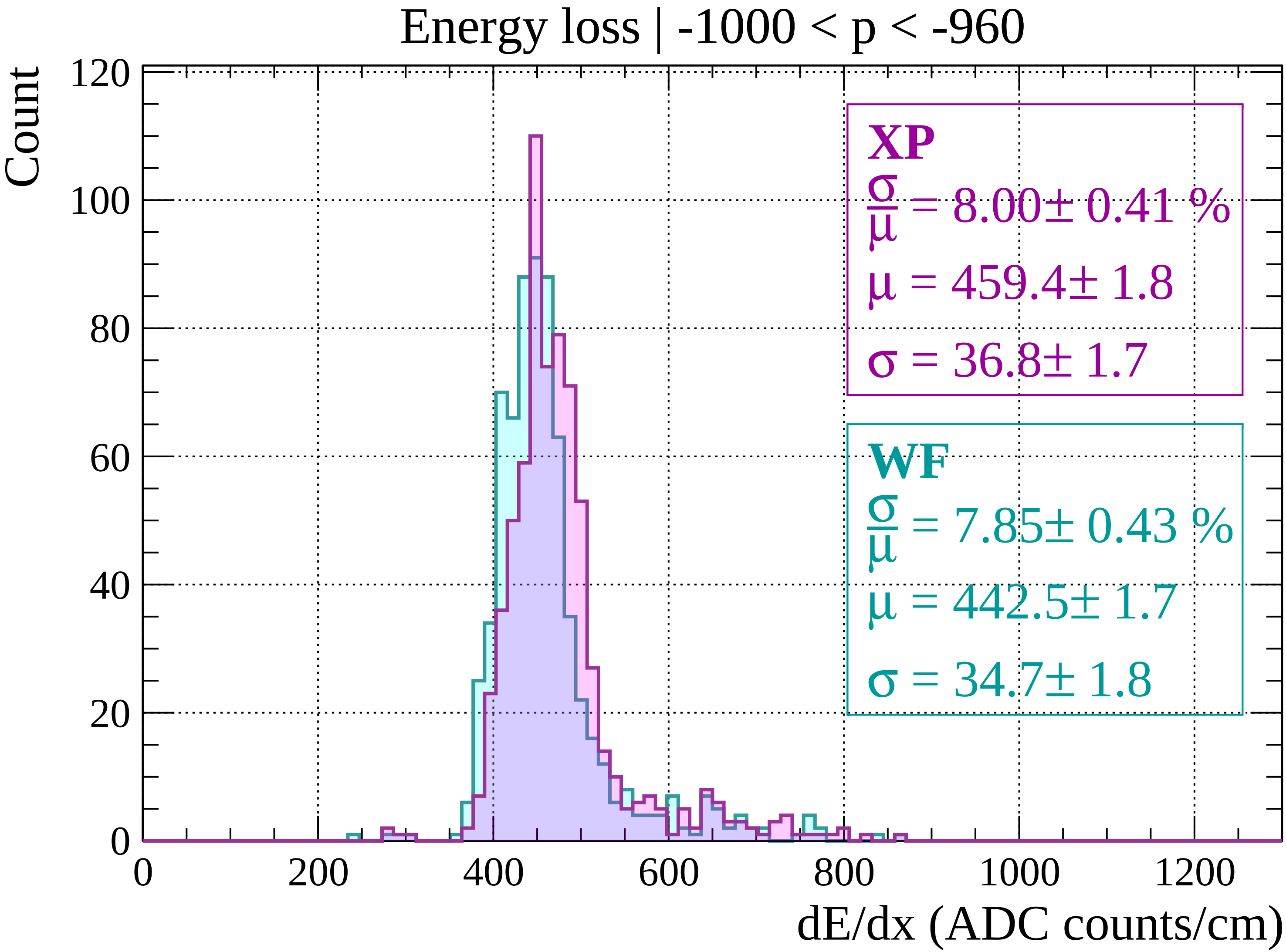

Energy loss | -1120 < p < -1080
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 8.15± 0.46 %
μ = 461.9± 1.8
σ = 37.7± 2.0
WF
σ/μ = 8.03± 0.41 %
μ = 448.3± 1.7
σ = 36.0± 1.7

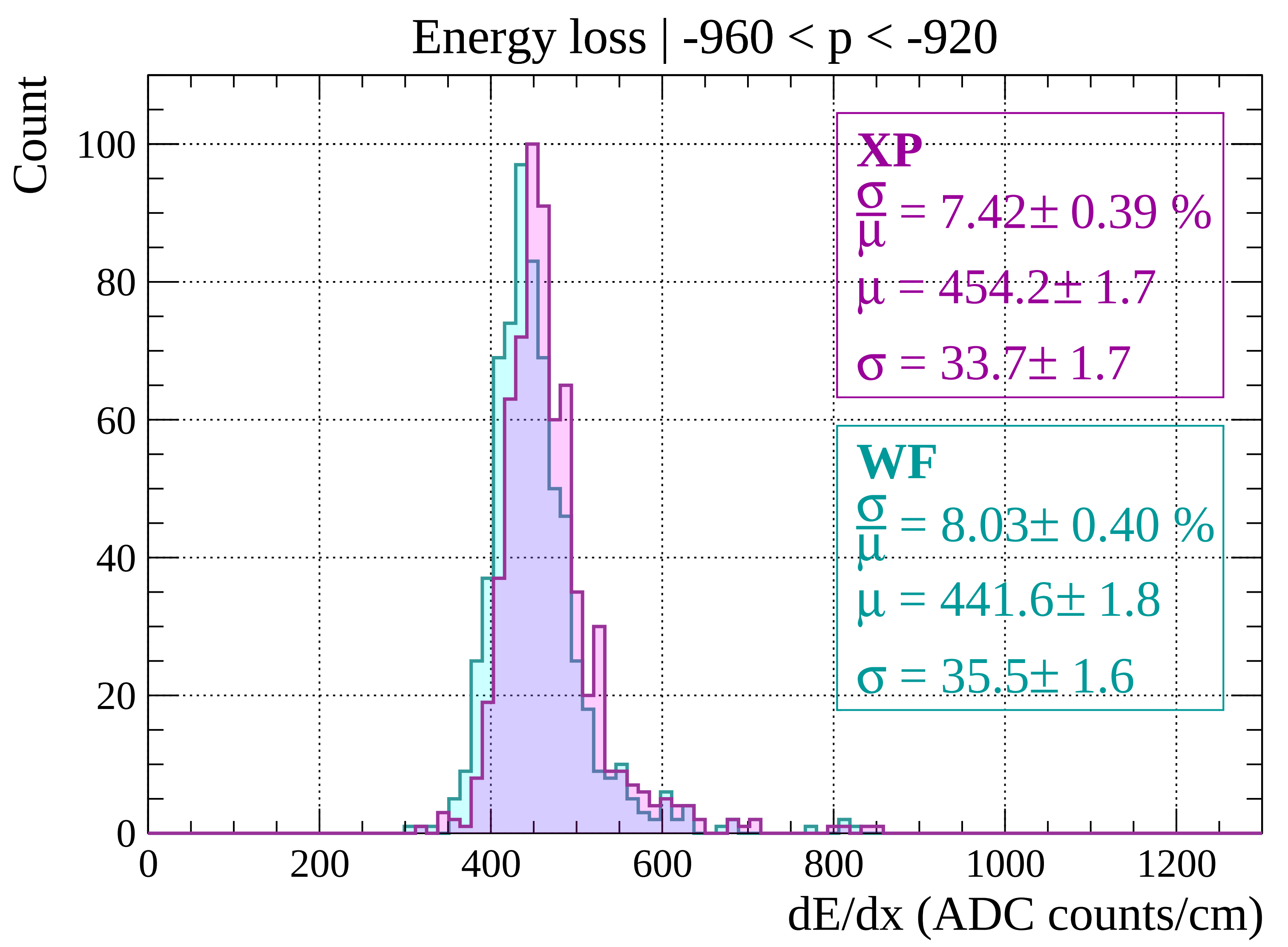


Energy loss | -1080 < p < -1040
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 7.98 \pm 0.42$ %
$\mu = 461.5 \pm 1.7$
$\sigma = 36.8 \pm 1.8$
WF
$\frac{\sigma}{\mu} = 7.96 \pm 0.39$ %
$\mu = 447.9 \pm 1.6$
$\sigma = 35.7 \pm 1.6$

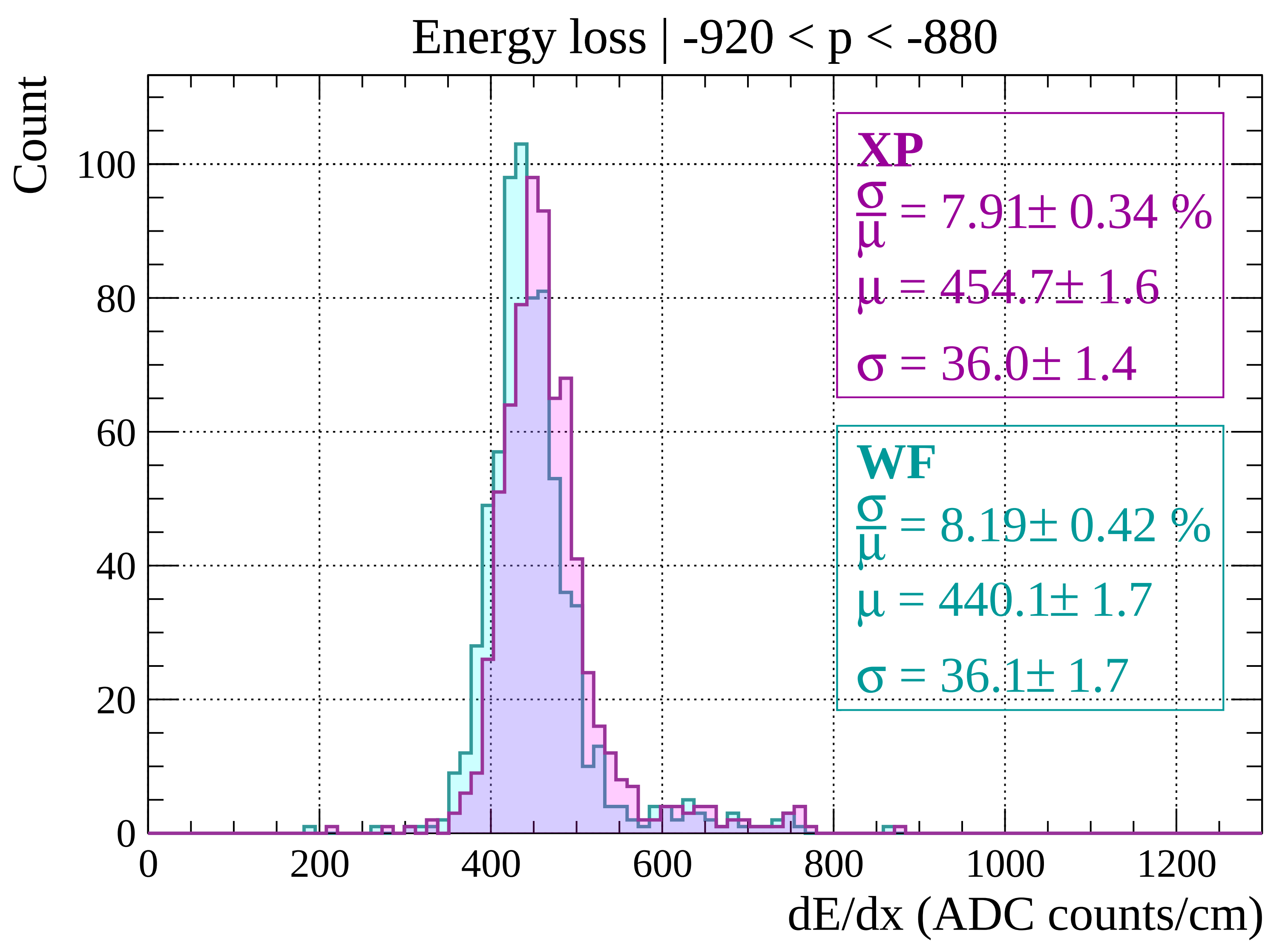


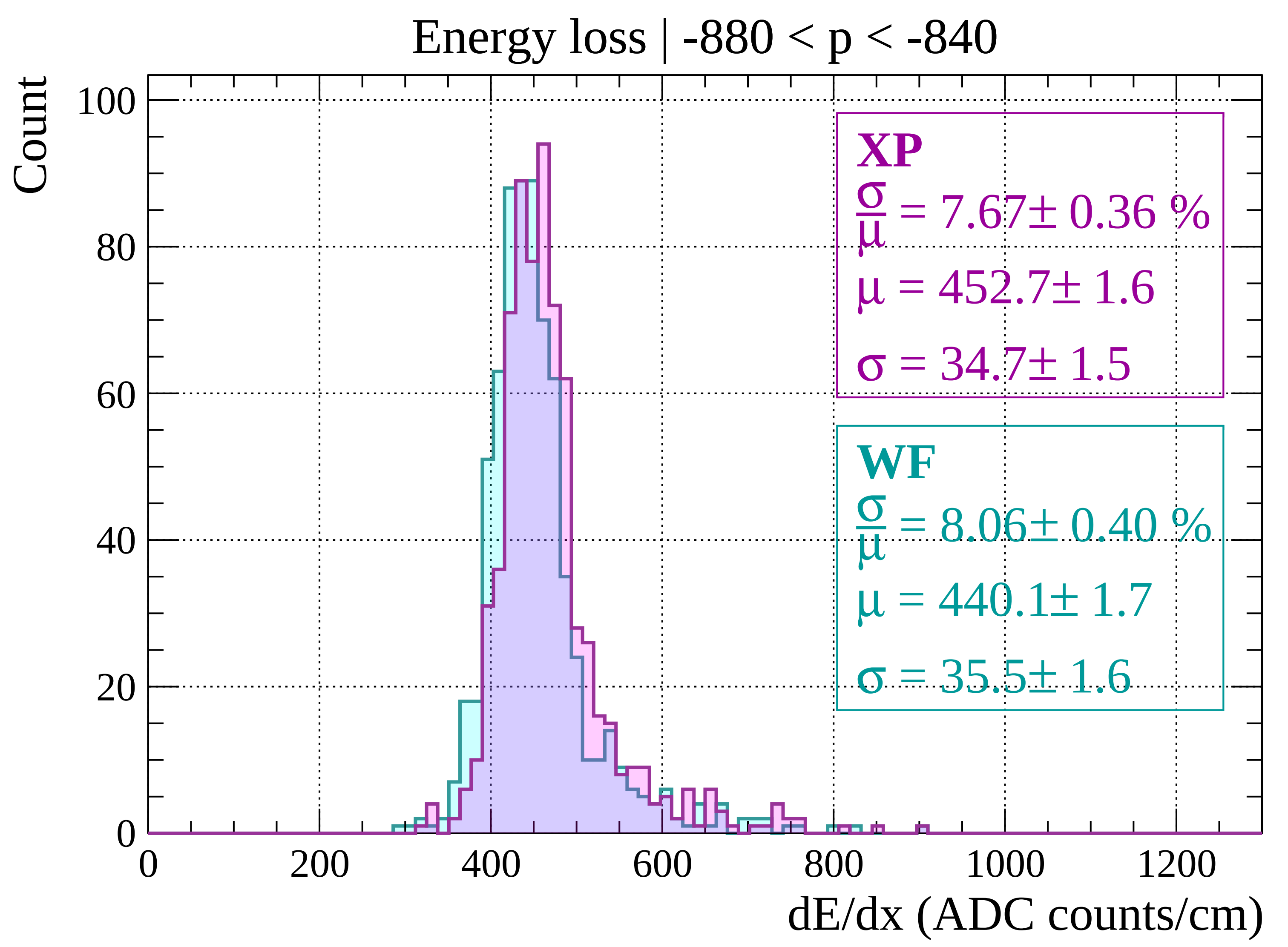
Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
$\frac{\sigma}{\mu}$ = 7.74± 0.32 %
μ = 458.3± 1.5
σ = 35.5± 1.3
WF
$\frac{\sigma}{\mu}$ = 7.71± 0.29 %
μ = 443.8± 1.5
σ = 34.2± 1.2


Energy loss | -1000 < p < -960
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 8.00± 0.41 %
μ = 459.4± 1.8
σ = 36.8± 1.7
WF
σ/μ = 7.85± 0.43 %
μ = 442.5± 1.7
σ = 34.7± 1.8

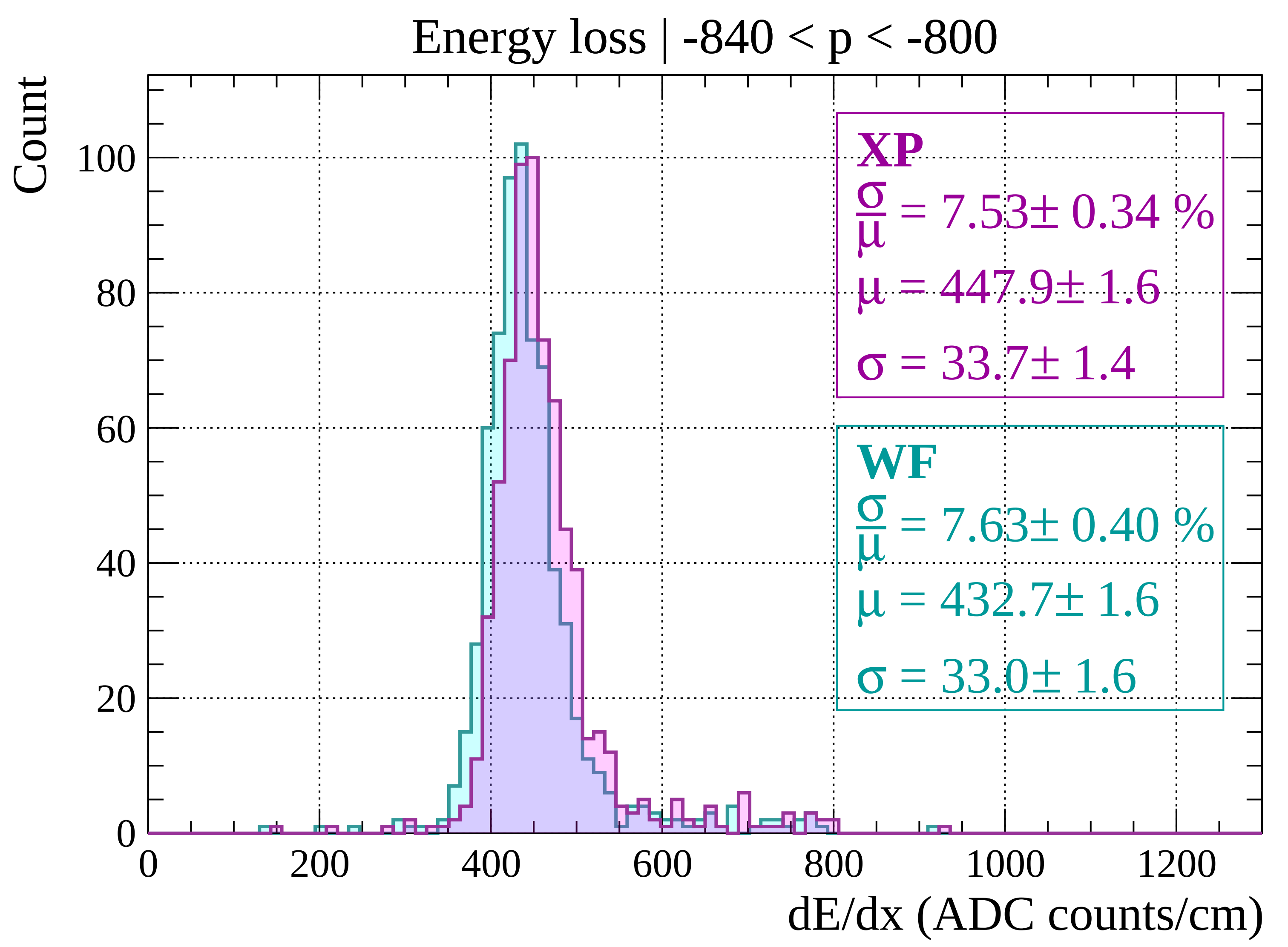


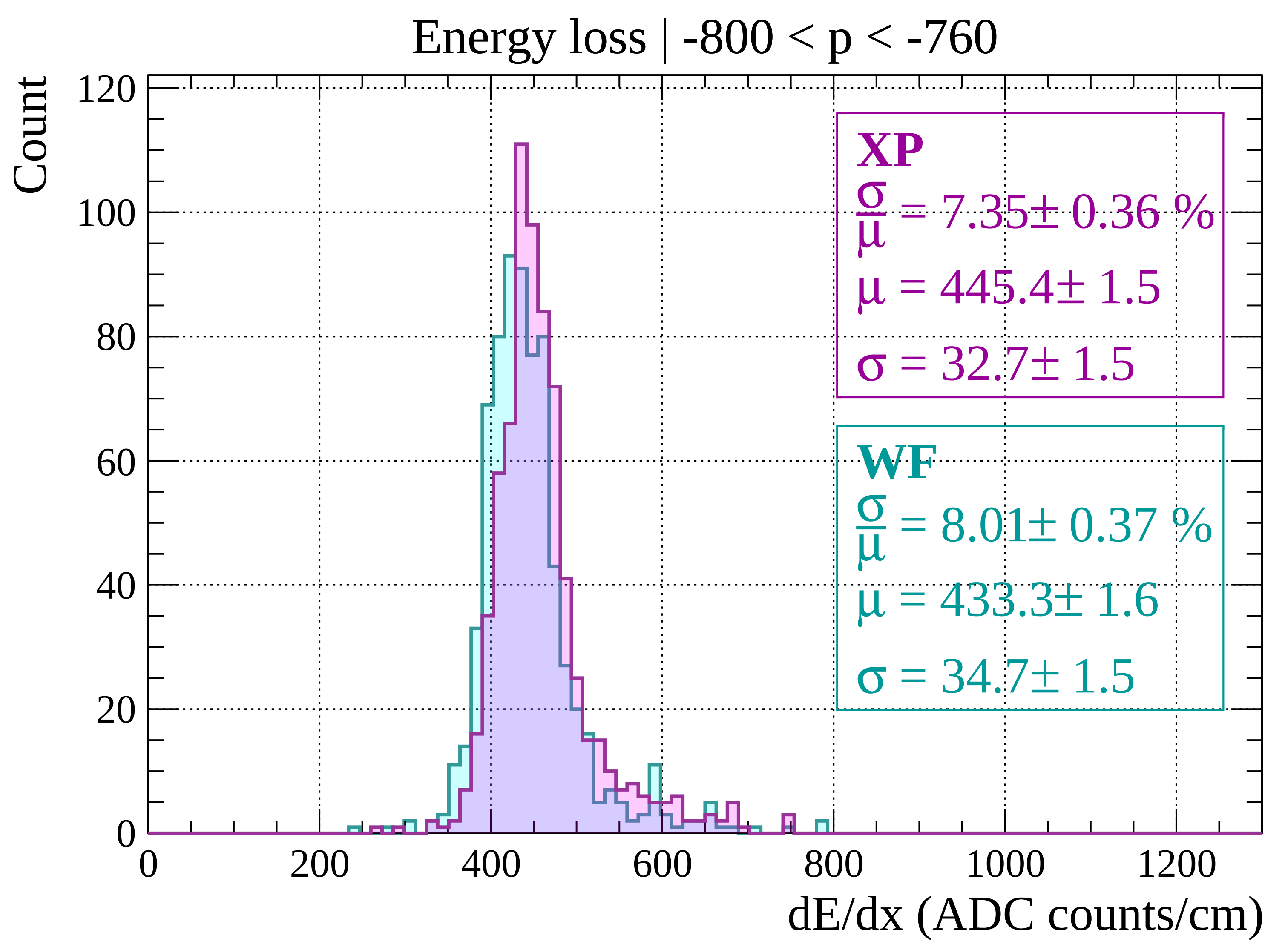
Energy loss | -960 < p < -920
Count
XP
σ/μ = 7.42± 0.39 %
μ = 454.2± 1.7
σ = 33.7± 1.7
WF
σ/μ = 8.03± 0.40 %
μ = 441.6± 1.8
σ = 35.5± 1.6
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

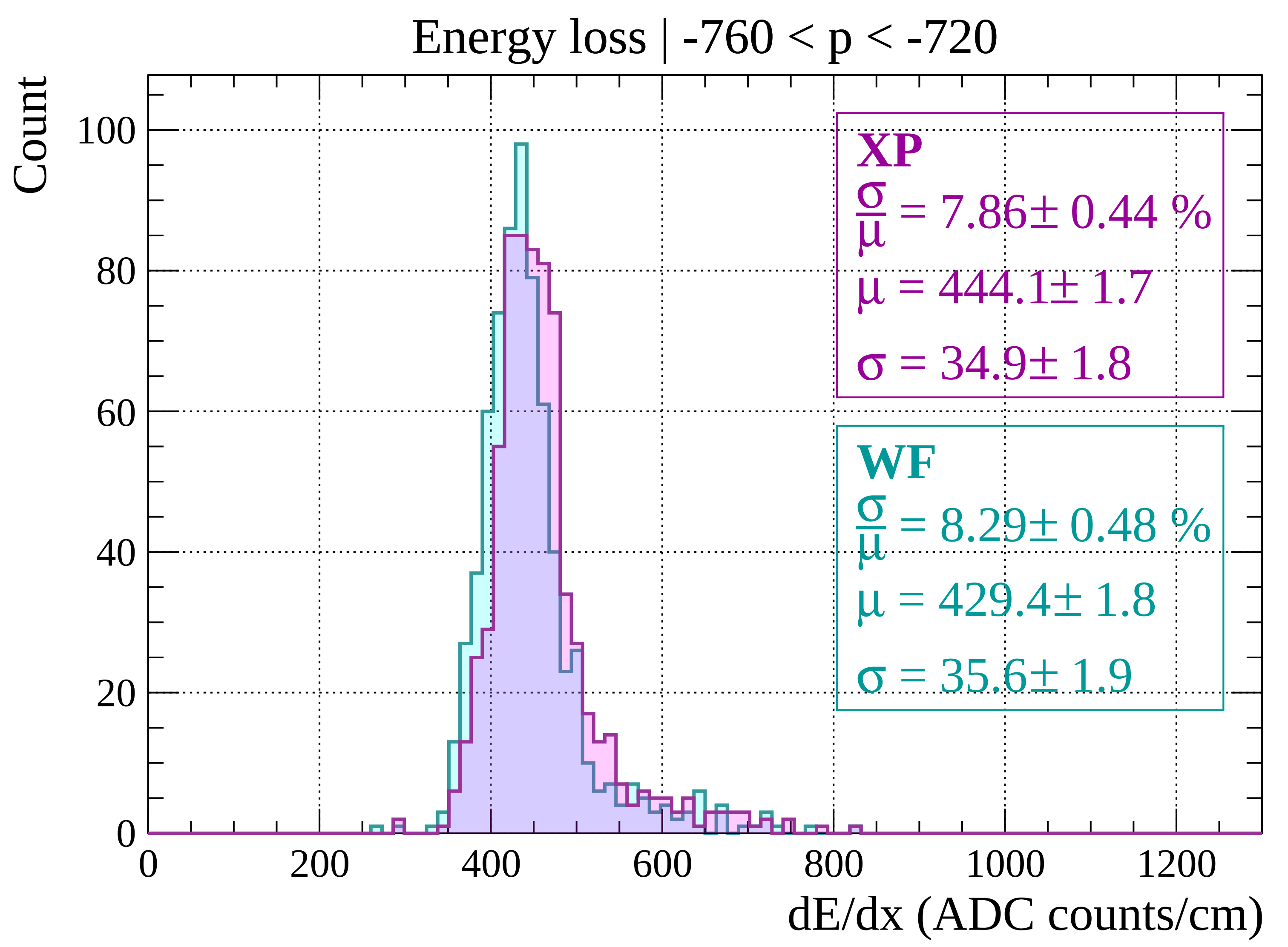

Energy loss | -920 < p < -880
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.91± 0.34 %
μ = 454.7± 1.6
σ = 36.0± 1.4
WF
$\frac{\sigma}{\mu}$ = 8.19± 0.42 %
μ = 440.1± 1.7
σ = 36.1± 1.7



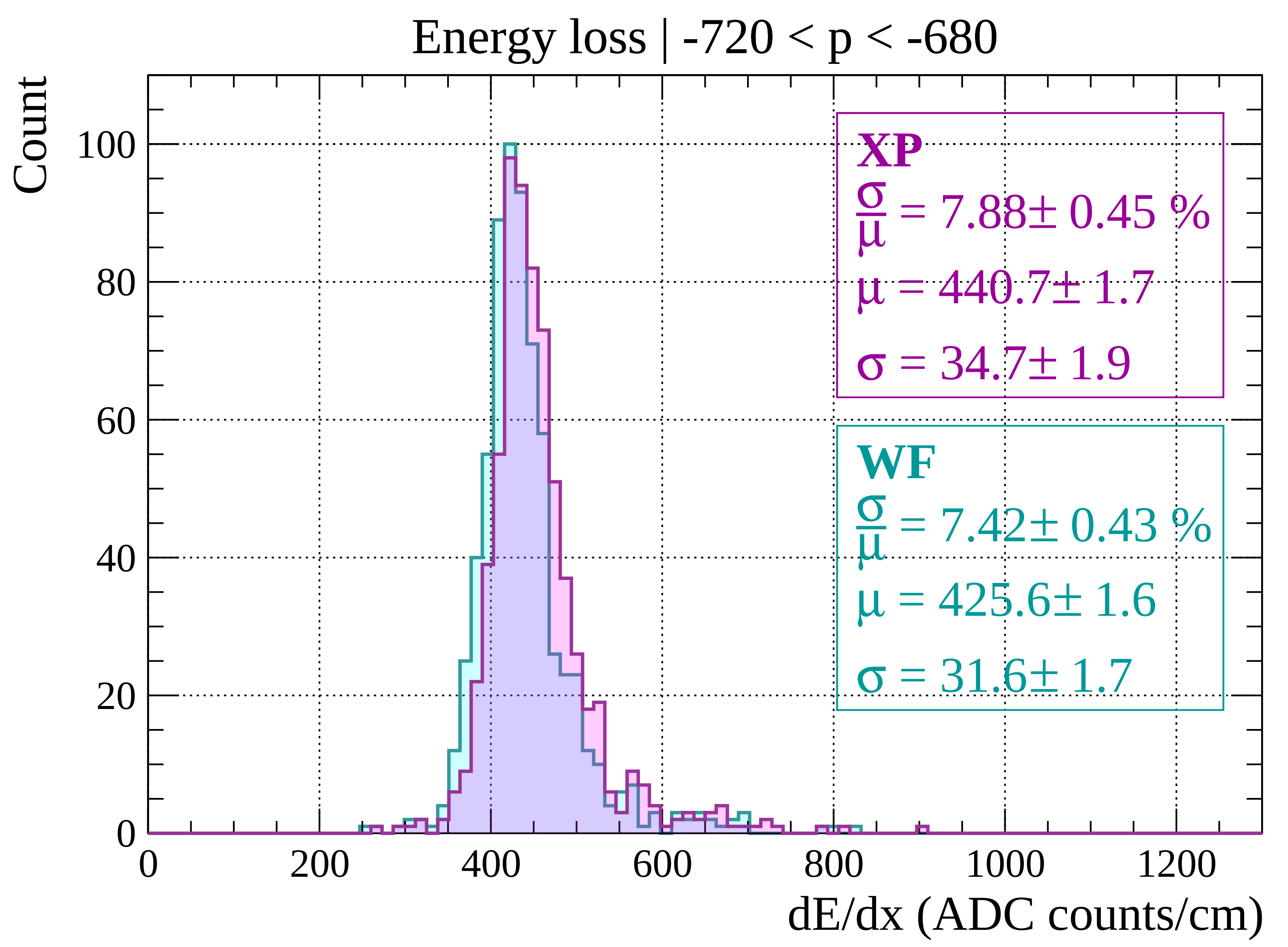
Energy loss | -880 < p < -840
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.67± 0.36 %
μ = 452.7± 1.6
σ = 34.7± 1.5
WF
σ/μ = 8.06± 0.40 %
μ = 440.1± 1.7
σ = 35.5± 1.6


Energy loss | -840 < p < -800
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 7.53± 0.34 %
μ = 447.9± 1.6
σ = 33.7± 1.4
WF
σ/μ = 7.63± 0.40 %
μ = 432.7± 1.6
σ = 33.0± 1.6

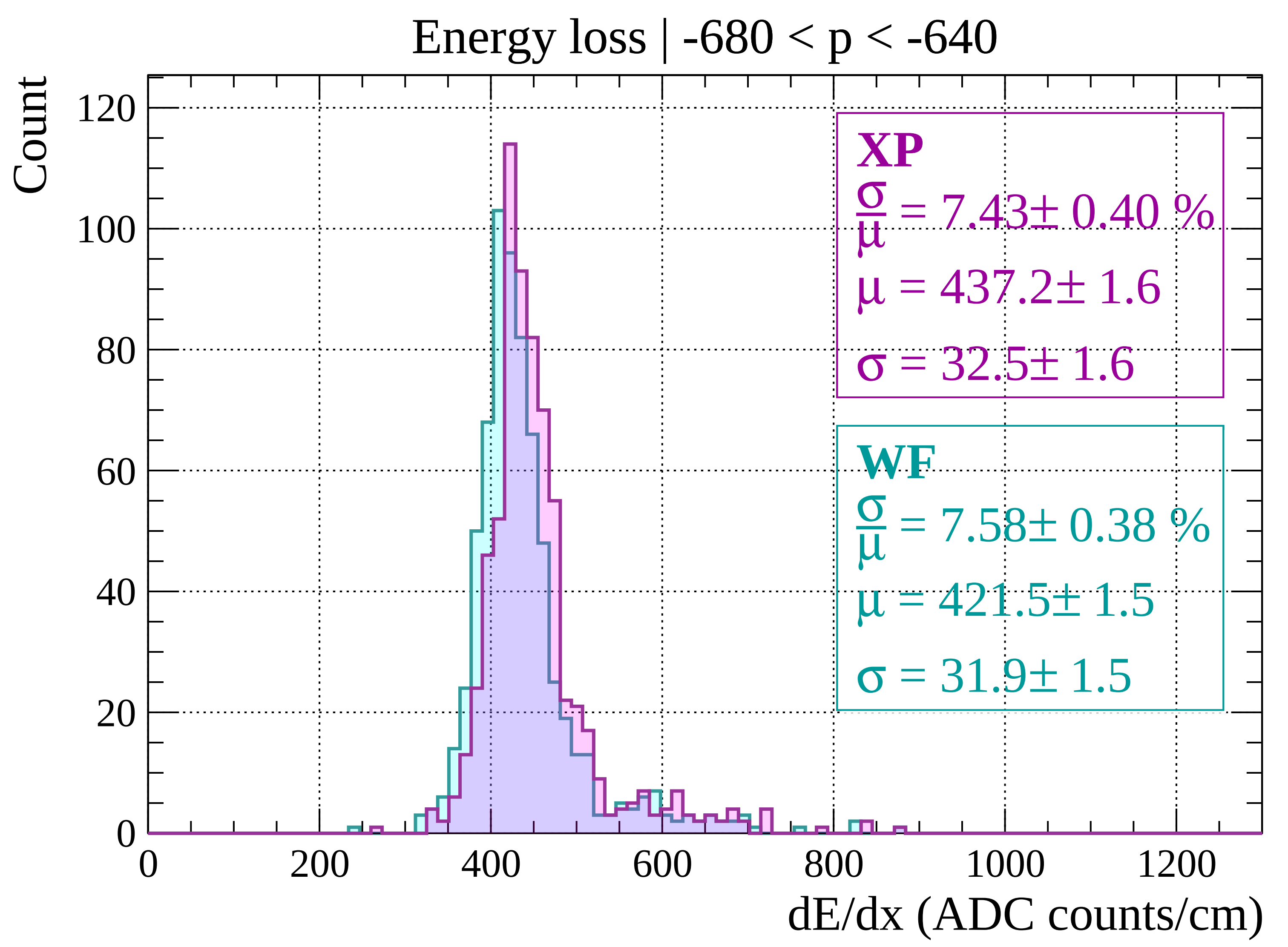

Energy loss | -800 < p < -760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
XP
σ/μ = 7.35± 0.36 %
μ = 445.4± 1.5
σ = 32.7± 1.5
WF
σ/μ = 8.01± 0.37 %
μ = 433.3± 1.6
σ = 34.7± 1.5



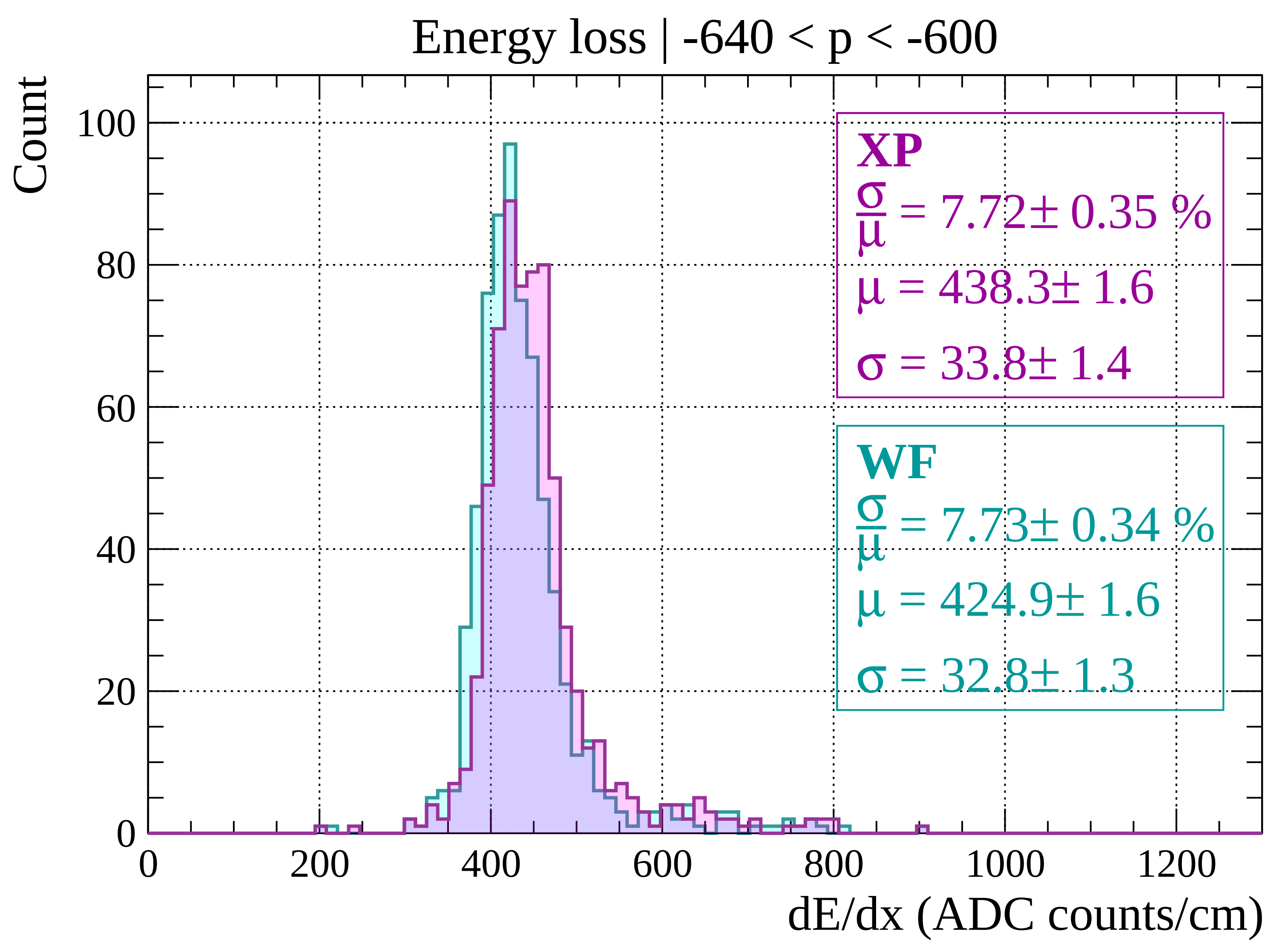
Energy loss | -760 < p < -720
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 7.86± 0.44 %
μ = 444.1± 1.7
σ = 34.9± 1.8
WF
σ/μ = 8.29± 0.48 %
μ = 429.4± 1.8
σ = 35.6± 1.9



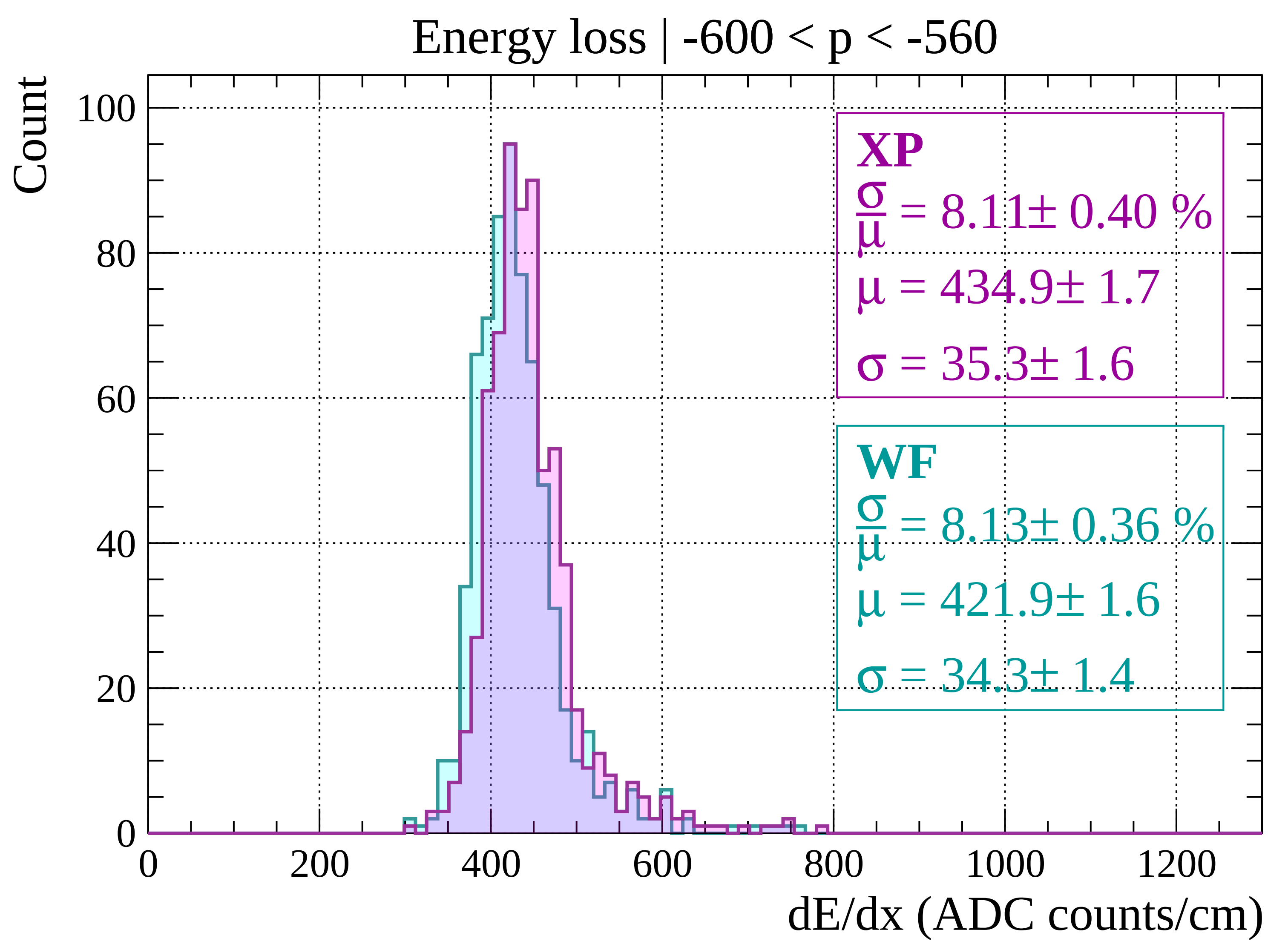
Energy loss | -720 < p < -680
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.88± 0.45 %
μ = 440.7± 1.7
σ = 34.7± 1.9
WF
σ/μ = 7.42± 0.43 %
μ = 425.6± 1.6
σ = 31.6± 1.7



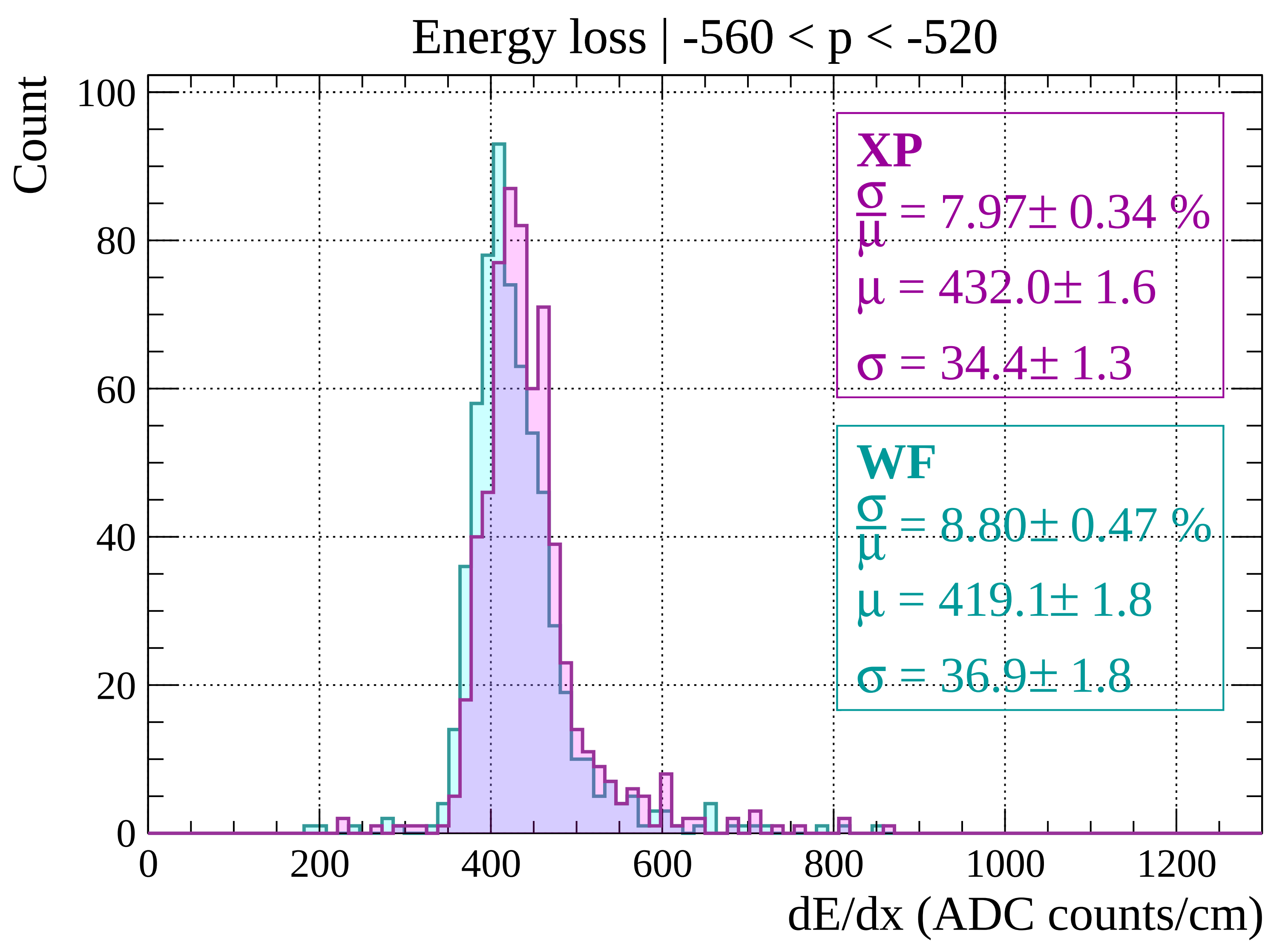
Energy loss | -680 < p < -640
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.43± 0.40 %
μ = 437.2± 1.6
σ = 32.5± 1.6
WF
$\frac{\sigma}{\mu}$ = 7.58± 0.38 %
μ = 421.5± 1.5
σ = 31.9± 1.5



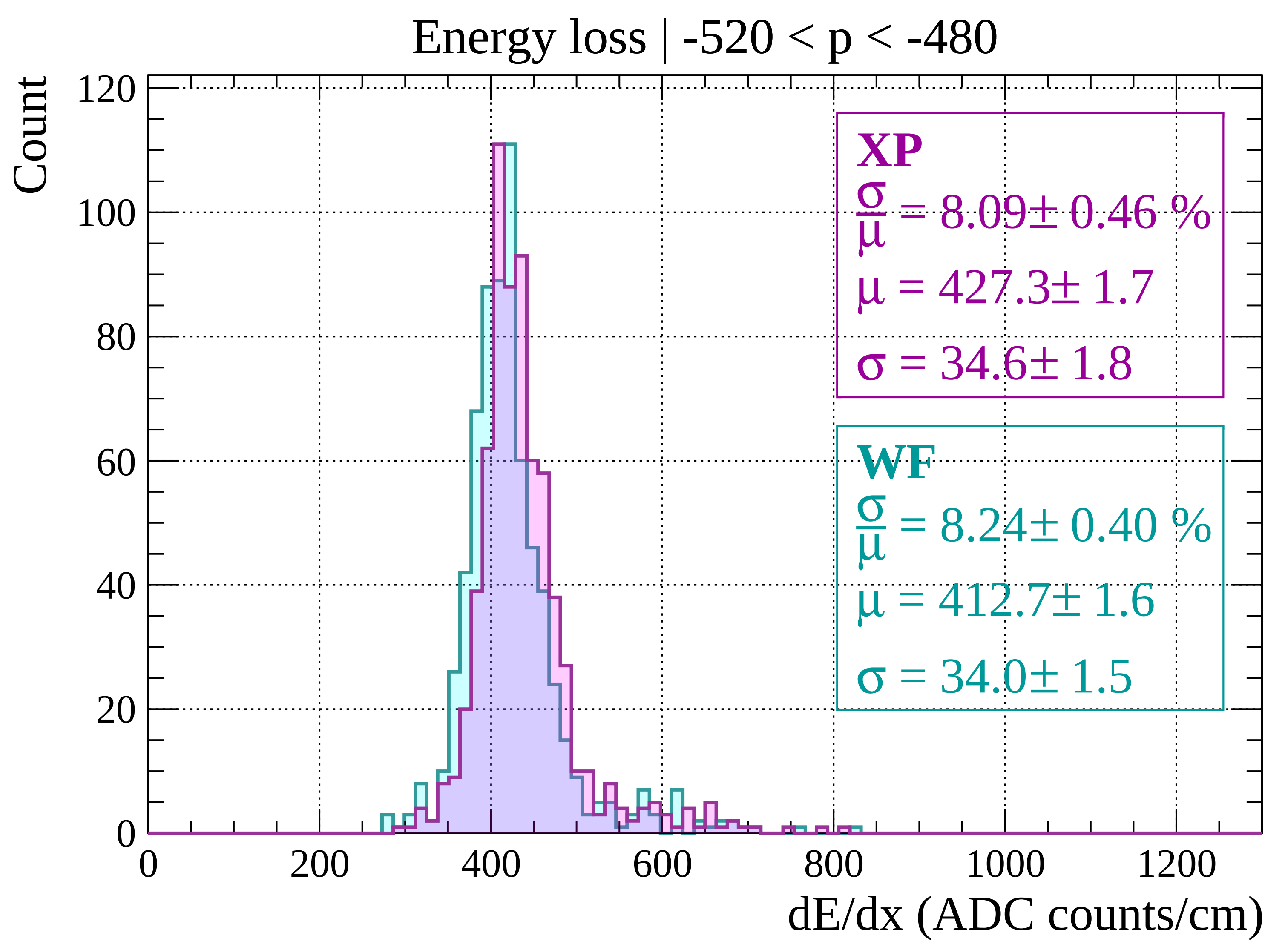
Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 7.72 \pm 0.35$ %
$\mu = 438.3 \pm 1.6$
$\sigma = 33.8 \pm 1.4$
WF
$\frac{\sigma}{\mu} = 7.73 \pm 0.34$ %
$\mu = 424.9 \pm 1.6$
$\sigma = 32.8 \pm 1.3$



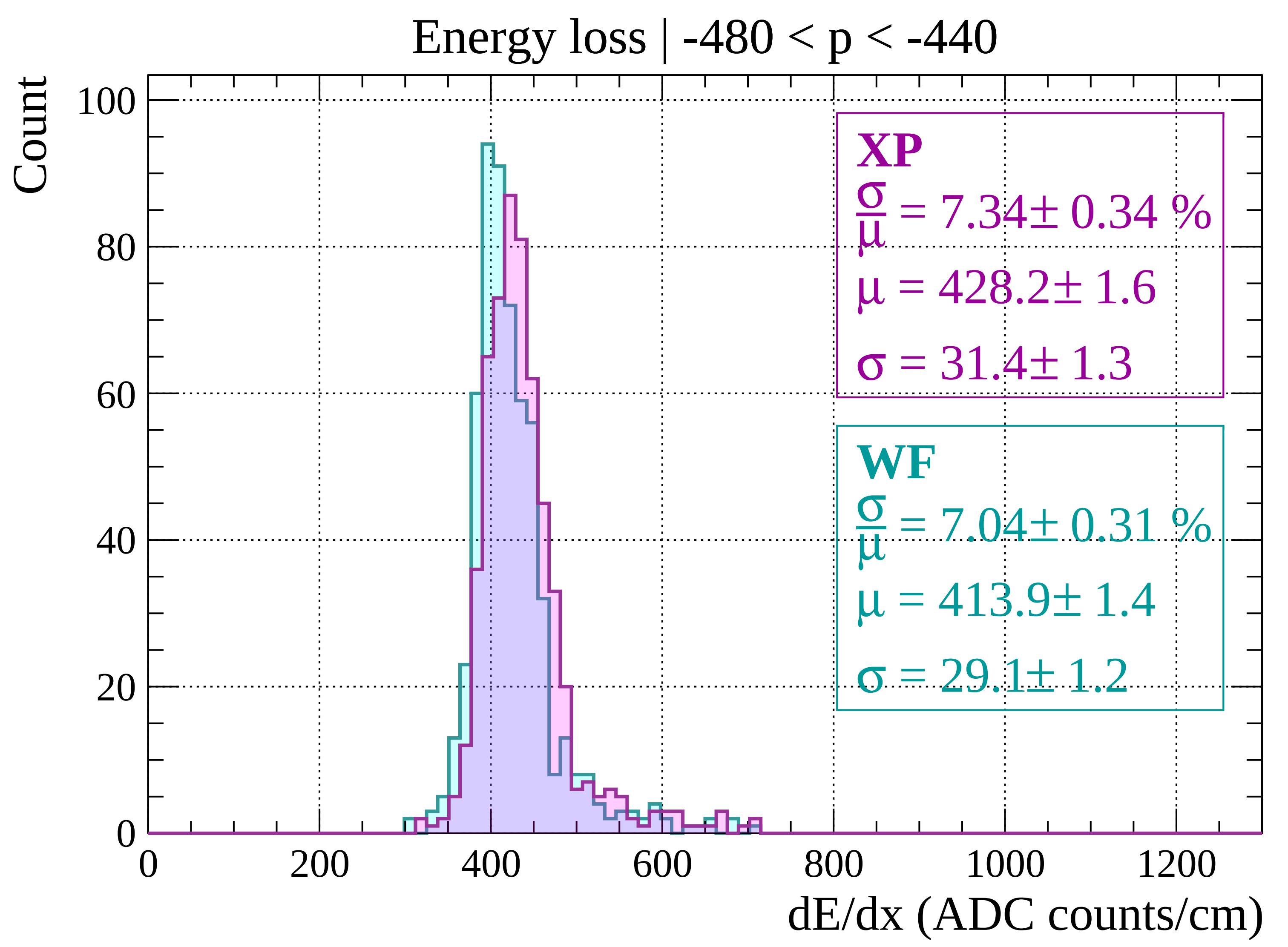
Energy loss | -600 < p < -560
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.11± 0.40 %
μ = 434.9± 1.7
σ = 35.3± 1.6
WF
σ/μ = 8.13± 0.36 %
μ = 421.9± 1.6
σ = 34.3± 1.4



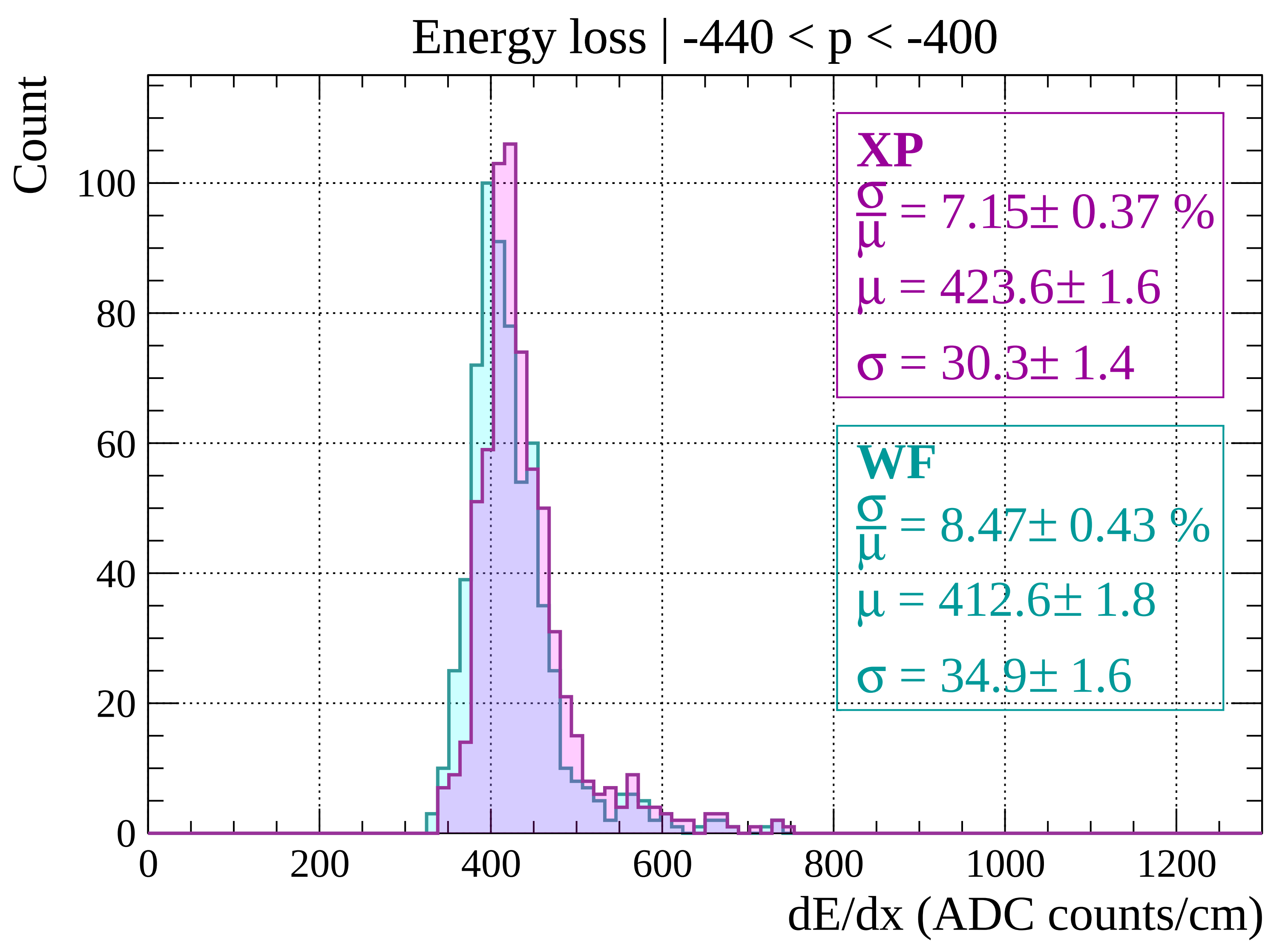
Energy loss | -560 < p < -520
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 7.97± 0.34 %
μ = 432.0± 1.6
σ = 34.4± 1.3
WF
σ/μ = 8.80± 0.47 %
μ = 419.1± 1.8
σ = 36.9± 1.8

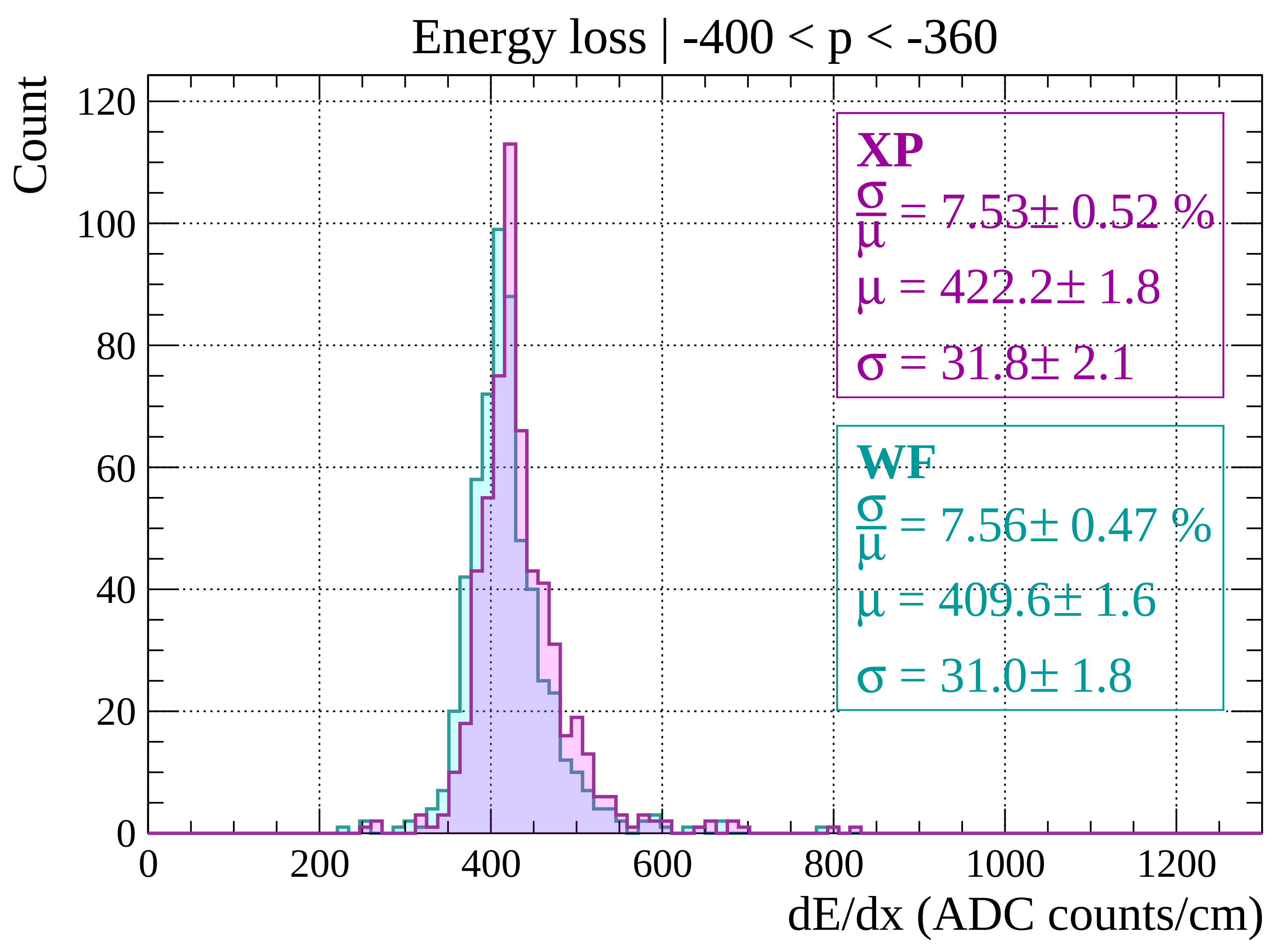


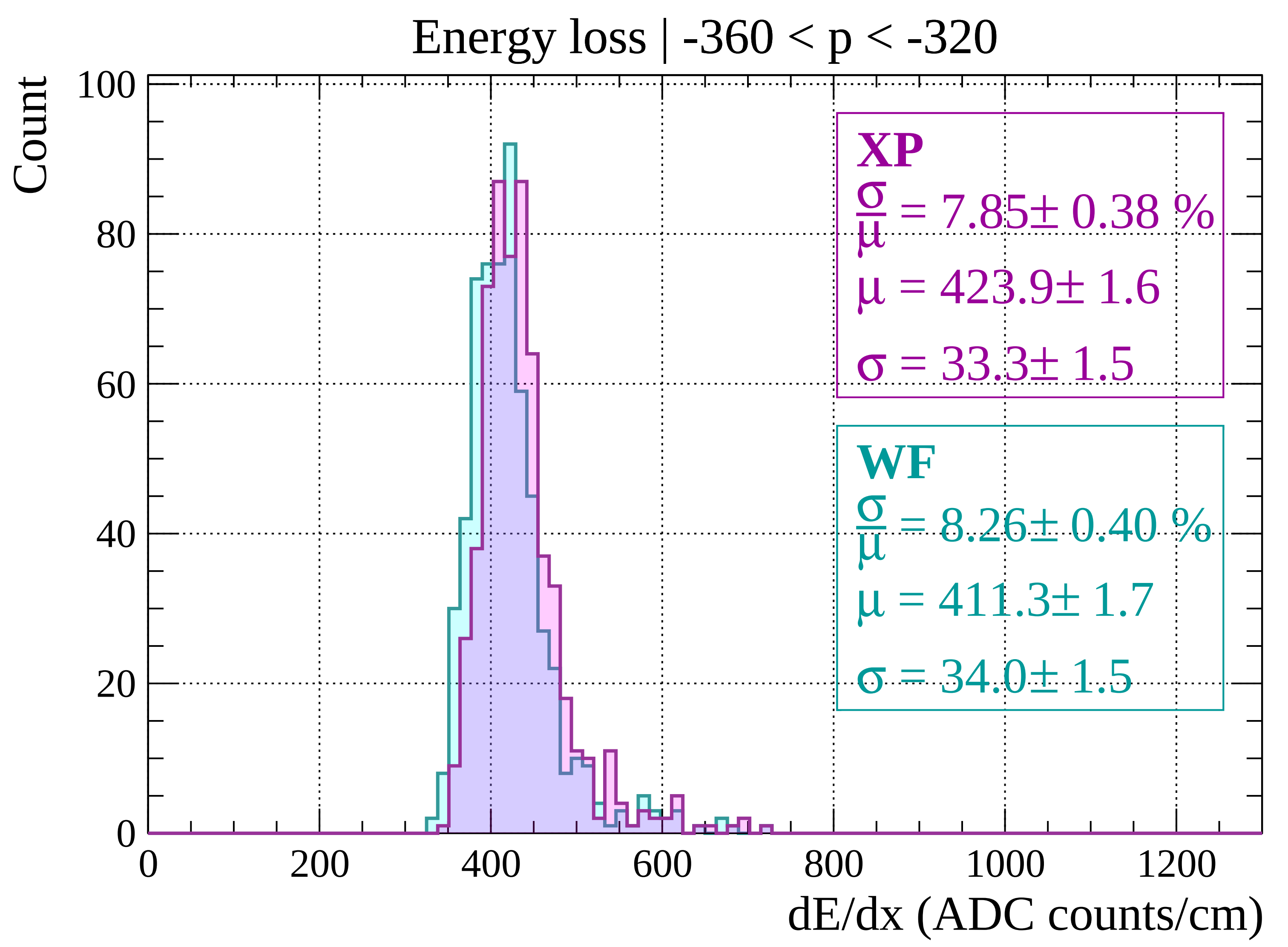
Energy loss | -520 < p < -480
Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.09± 0.46 %
μ = 427.3± 1.7
σ = 34.6± 1.8
WF
$\frac{\sigma}{\mu}$ = 8.24± 0.40 %
μ = 412.7± 1.6
σ = 34.0± 1.5



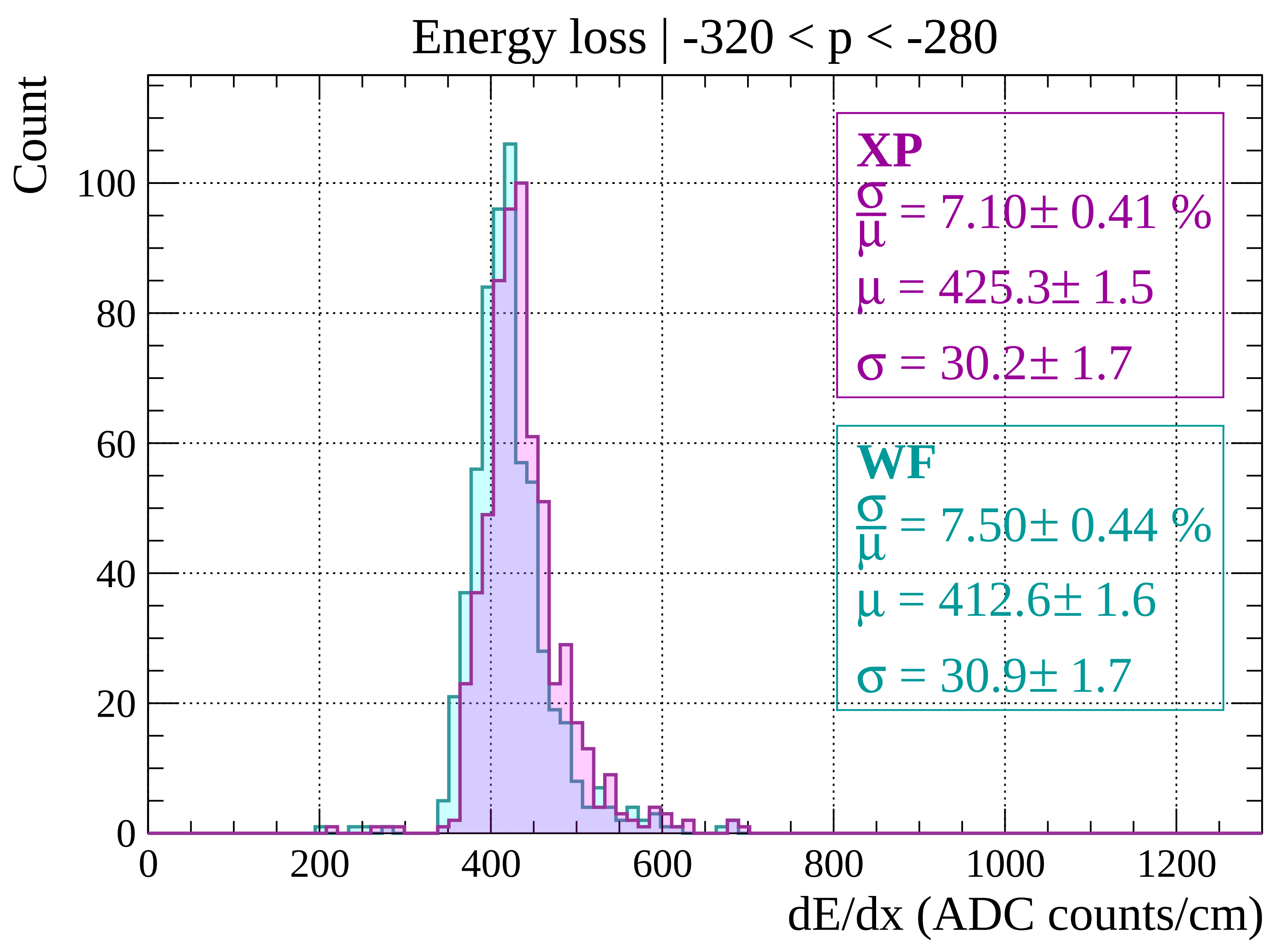
Energy loss | -480 < p < -440
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.34± 0.34 %
μ = 428.2± 1.6
σ = 31.4± 1.3
WF
σ/μ = 7.04± 0.31 %
μ = 413.9± 1.4
σ = 29.1± 1.2


Energy loss | -440 < p < -400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
$\frac{\sigma}{\mu}$ = 7.15± 0.37 %
μ = 423.6± 1.6
σ = 30.3± 1.4
WF
$\frac{\sigma}{\mu}$ = 8.47± 0.43 %
μ = 412.6± 1.8
σ = 34.9± 1.6

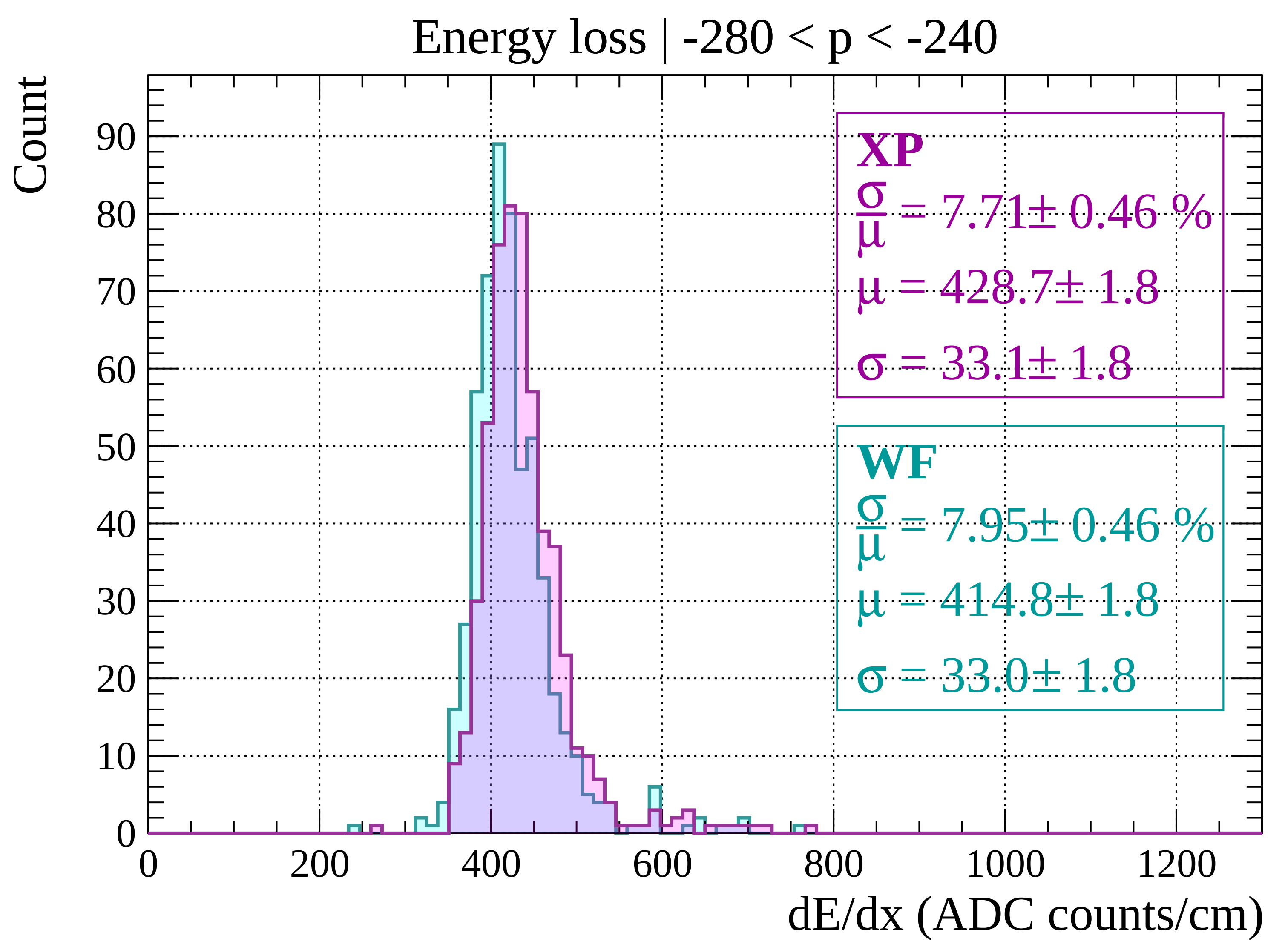

Energy loss | -400 < p < -360
Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.53± 0.52 %
μ = 422.2± 1.8
σ = 31.8± 2.1
WF
σ/μ = 7.56± 0.47 %
μ = 409.6± 1.6
σ = 31.0± 1.8



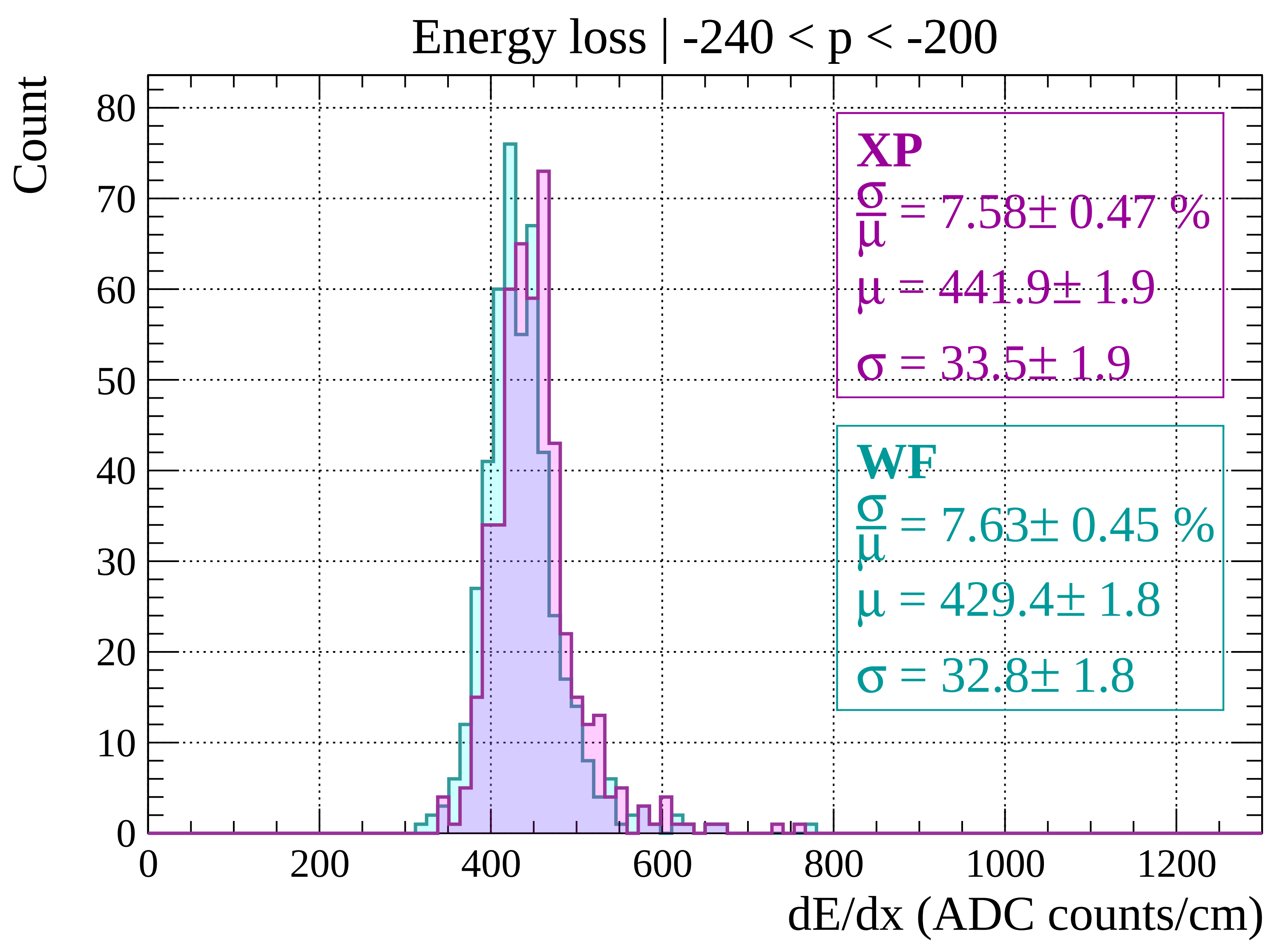
Energy loss | -360 < p < -320
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 7.85± 0.38 %
μ = 423.9± 1.6
σ = 33.3± 1.5
WF
σ/μ = 8.26± 0.40 %
μ = 411.3± 1.7
σ = 34.0± 1.5

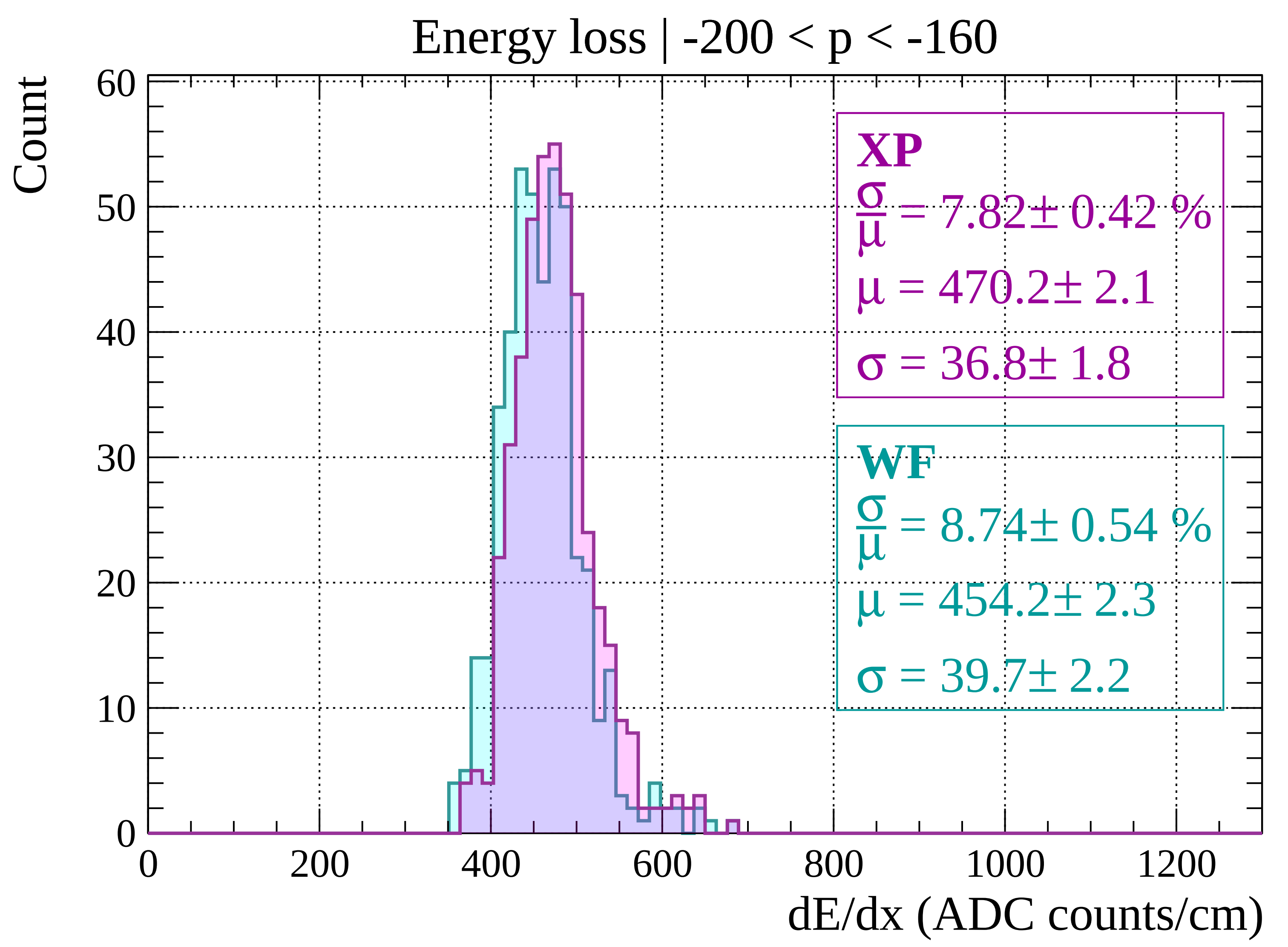


Energy loss | -320 < p < -280
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.10± 0.41 %
μ = 425.3± 1.5
σ = 30.2± 1.7
WF
$\frac{\sigma}{\mu}$ = 7.50± 0.44 %
μ = 412.6± 1.6
σ = 30.9± 1.7
0
20
40
60
80
100
200
400
600
800
1000
1200

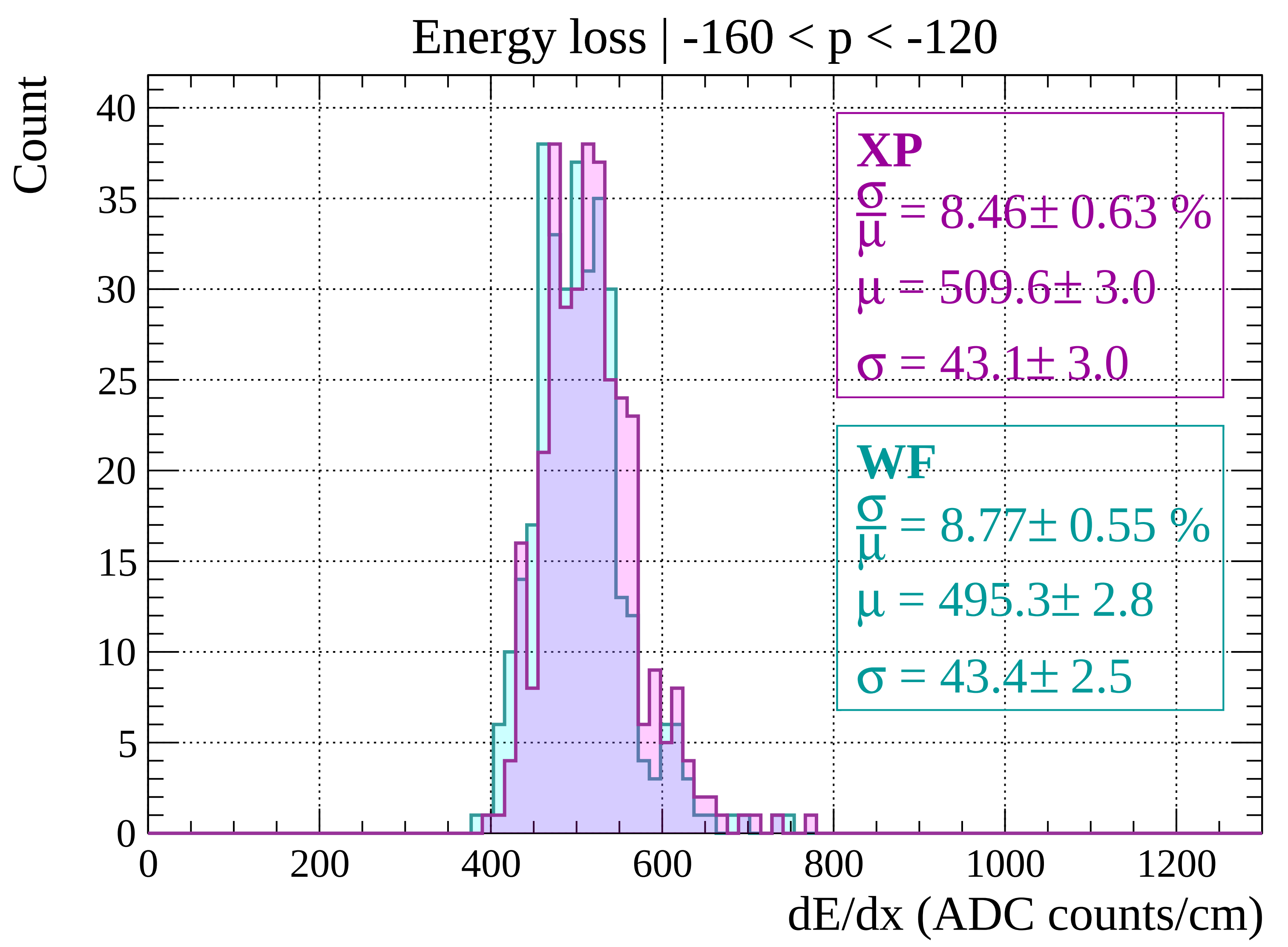


Energy loss | -280 < p < -240
Count
XP
σ/μ = 7.71± 0.46 %
μ = 428.7± 1.8
σ = 33.1± 1.8
WF
σ/μ = 7.95± 0.46 %
μ = 414.8± 1.8
σ = 33.0± 1.8
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

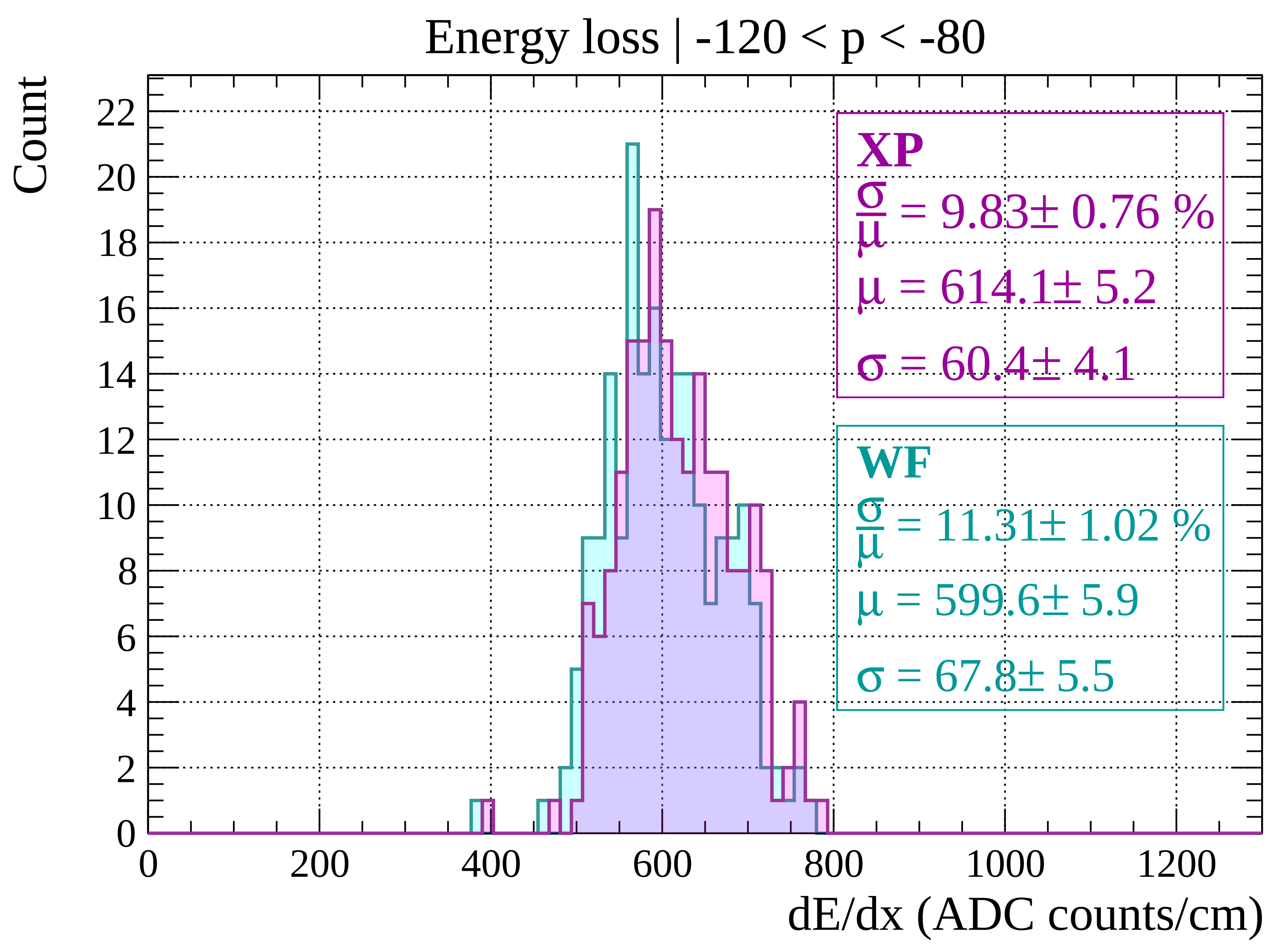

Energy loss | -240 < p < -200
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.58± 0.47 %
μ = 441.9± 1.9
σ = 33.5± 1.9
WF
σ/μ = 7.63± 0.45 %
μ = 429.4± 1.8
σ = 32.8± 1.8
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80

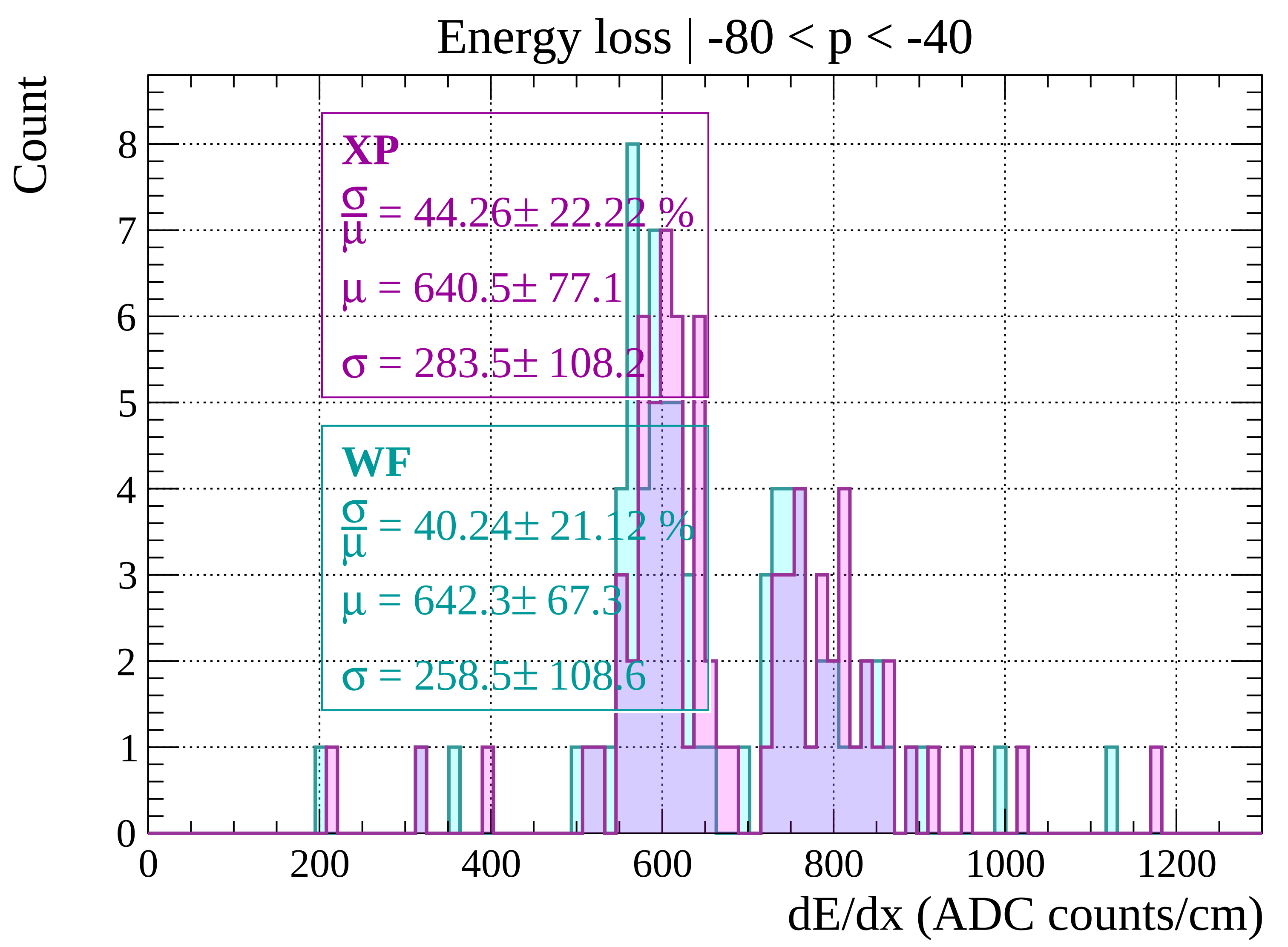


Energy loss | -200 < p < -160
Count
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.82± 0.42 %
μ = 470.2± 2.1
σ = 36.8± 1.8
WF
σ/μ = 8.74± 0.54 %
μ = 454.2± 2.3
σ = 39.7± 2.2

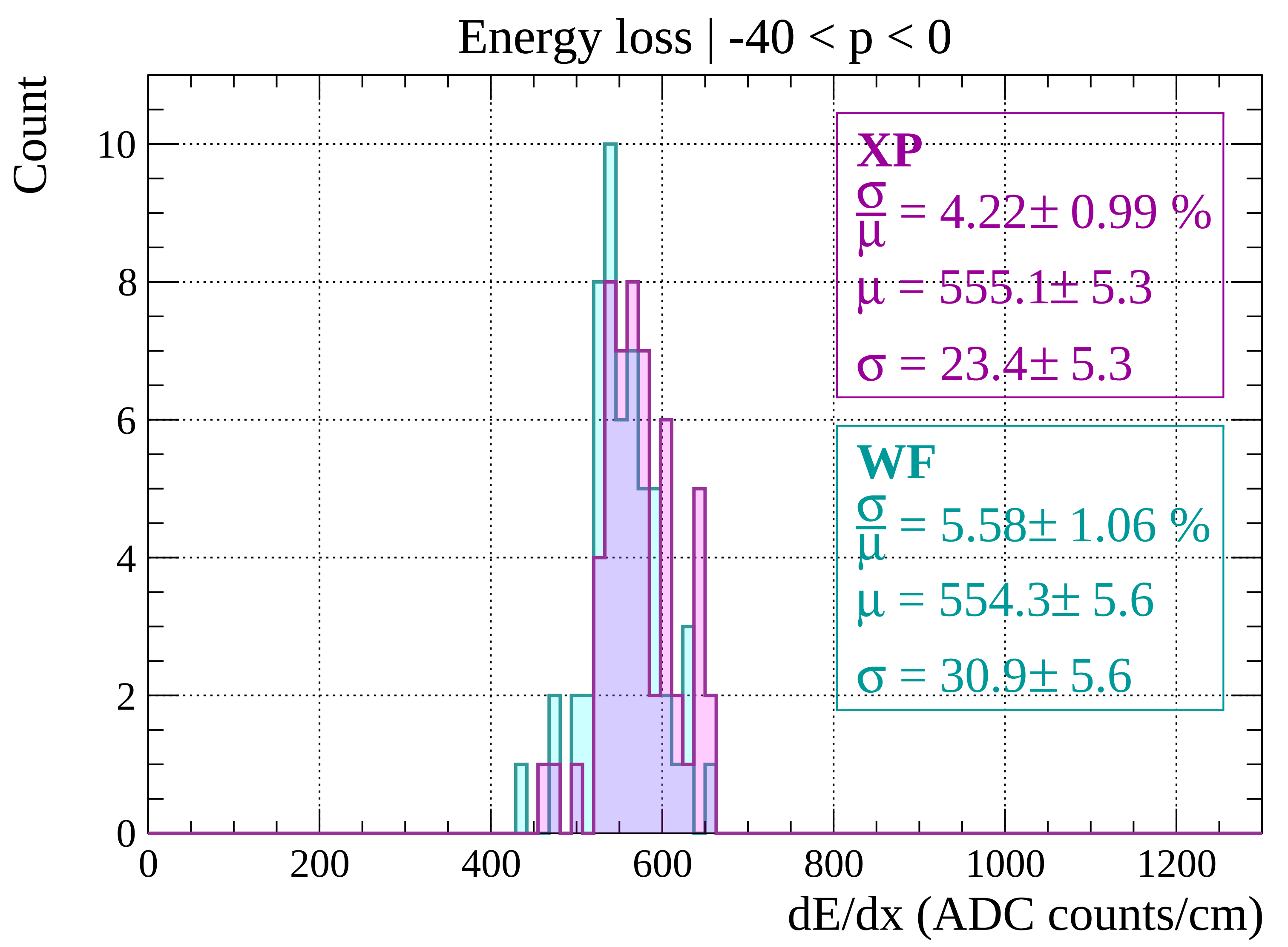


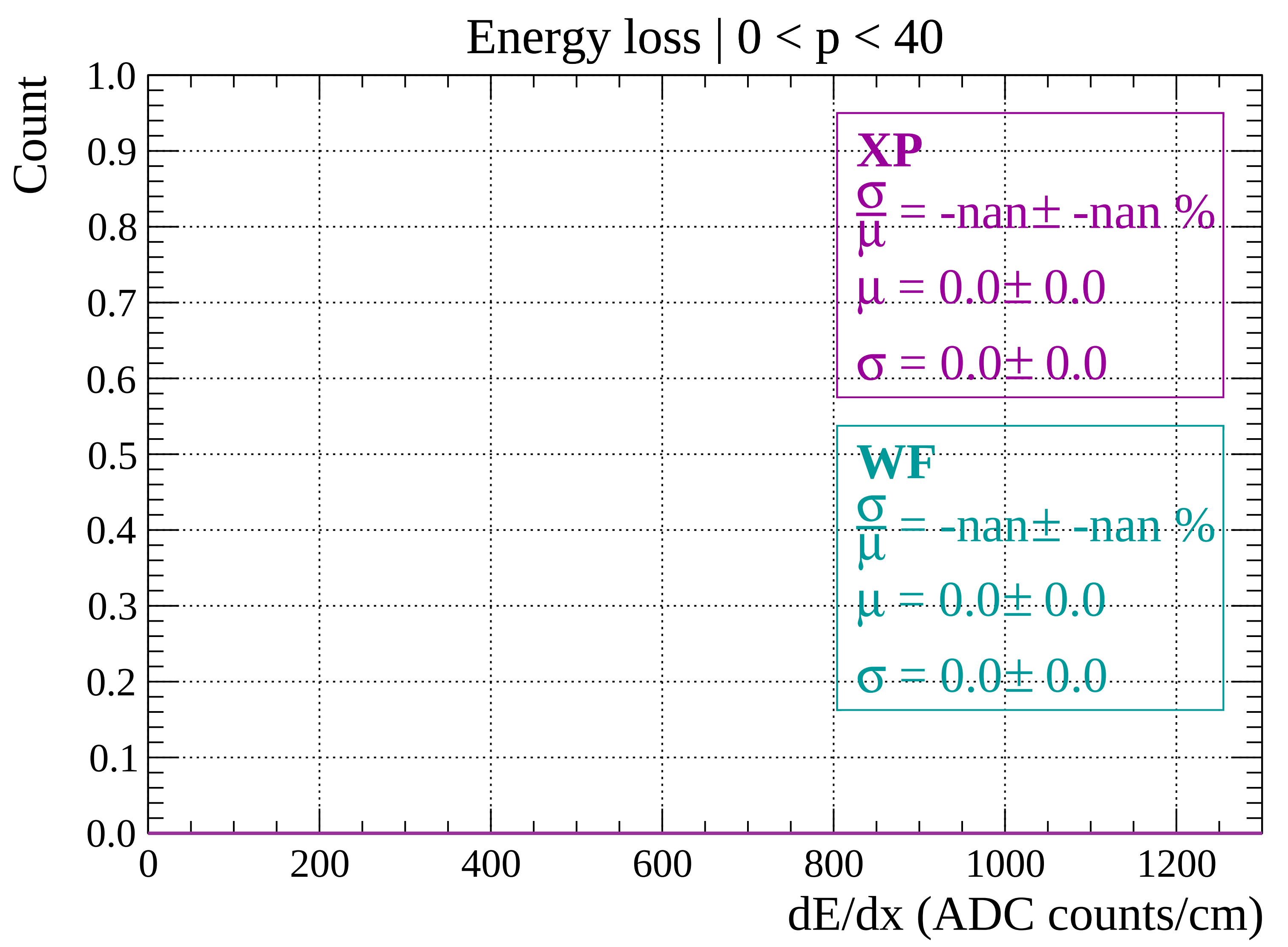
Energy loss | -160 < p < -120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
XP
$\frac{\sigma}{\mu}$ = 8.46± 0.63 %
μ = 509.6± 3.0
σ = 43.1± 3.0
WF
$\frac{\sigma}{\mu}$ = 8.77± 0.55 %
μ = 495.3± 2.8
σ = 43.4± 2.5


Energy loss | -120 < p < -80
Count
22
20
18
16
14
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.83± 0.76 %
μ = 614.1± 5.2
σ = 60.4± 4.1
WF
σ/μ = 11.31± 1.02 %
μ = 599.6± 5.9
σ = 67.8± 5.5

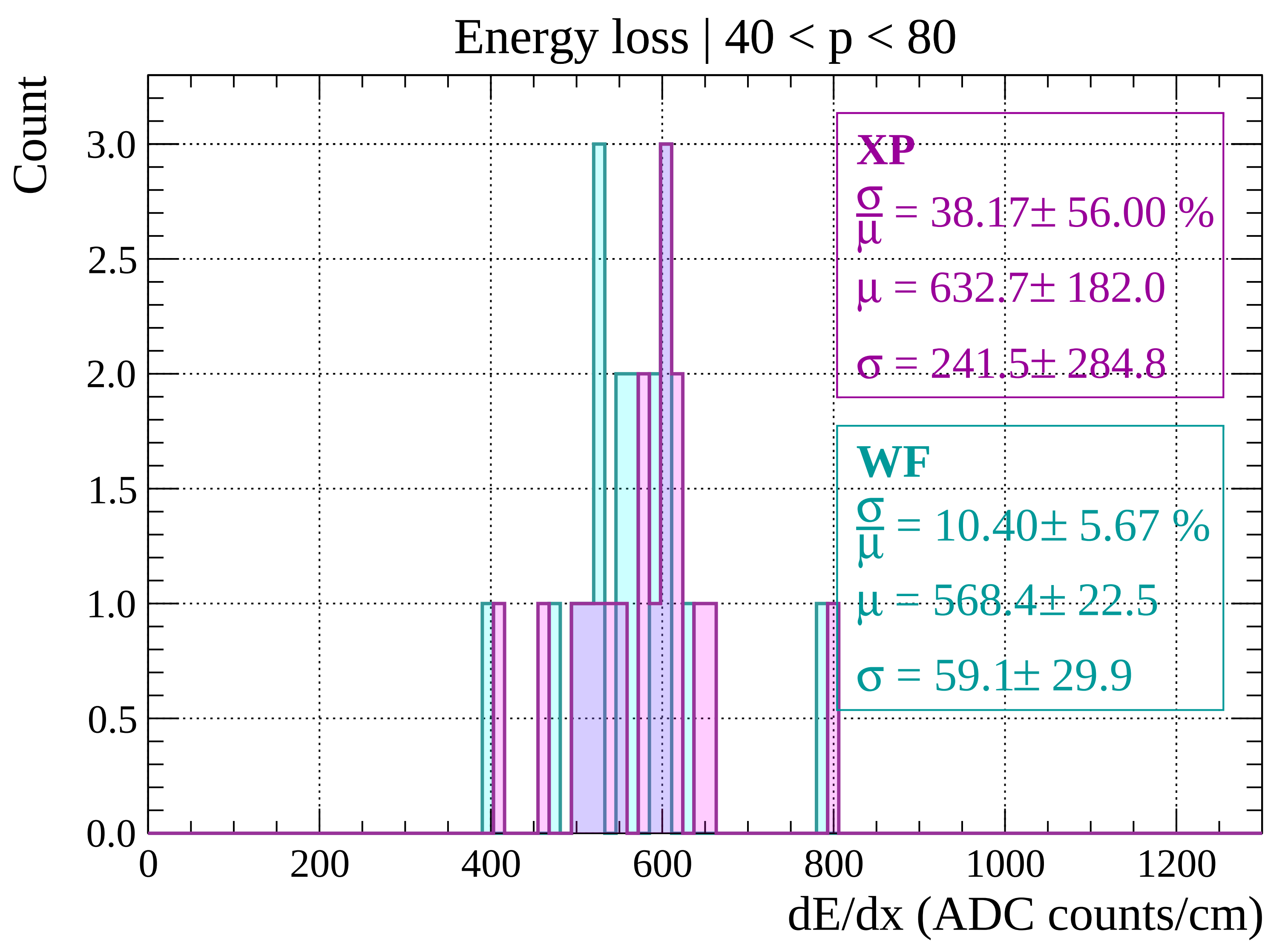


Energy loss | -80 < p < -40
Count
XP
$\frac{\sigma}{\mu}$ = 44.26± 22.22 %
μ = 640.5± 77.1
σ = 283.5± 108.2
WF
$\frac{\sigma}{\mu}$ = 40.24± 21.12 %
μ = 642.3± 67.3
σ = 258.5± 108.6
0
1
2
3
4
5
6
7
8
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

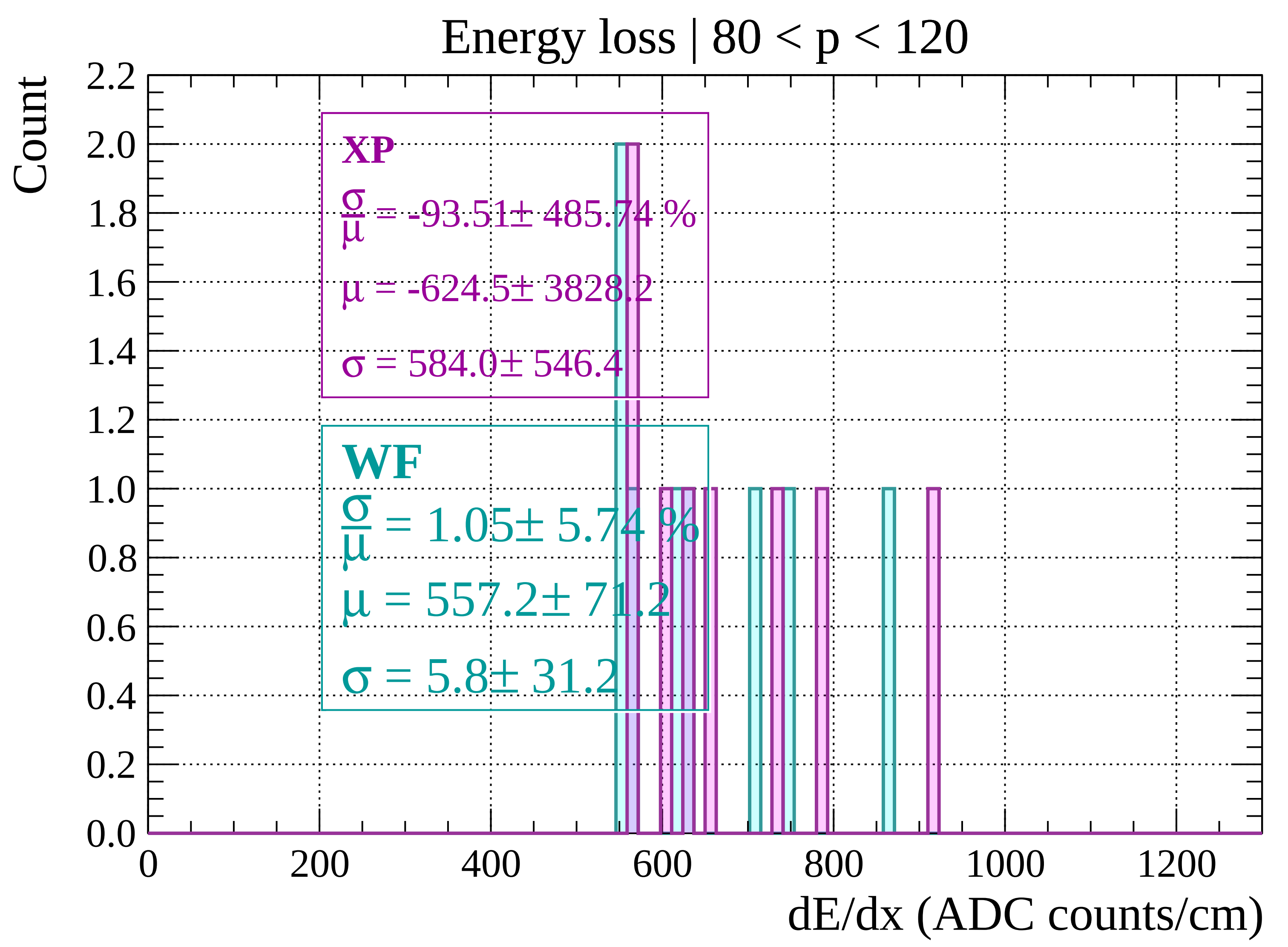

Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.22± 0.99 %
μ = 555.1± 5.3
σ = 23.4± 5.3
WF
$\frac{\sigma}{\mu}$ = 5.58± 1.06 %
μ = 554.3± 5.6
σ = 30.9± 5.6



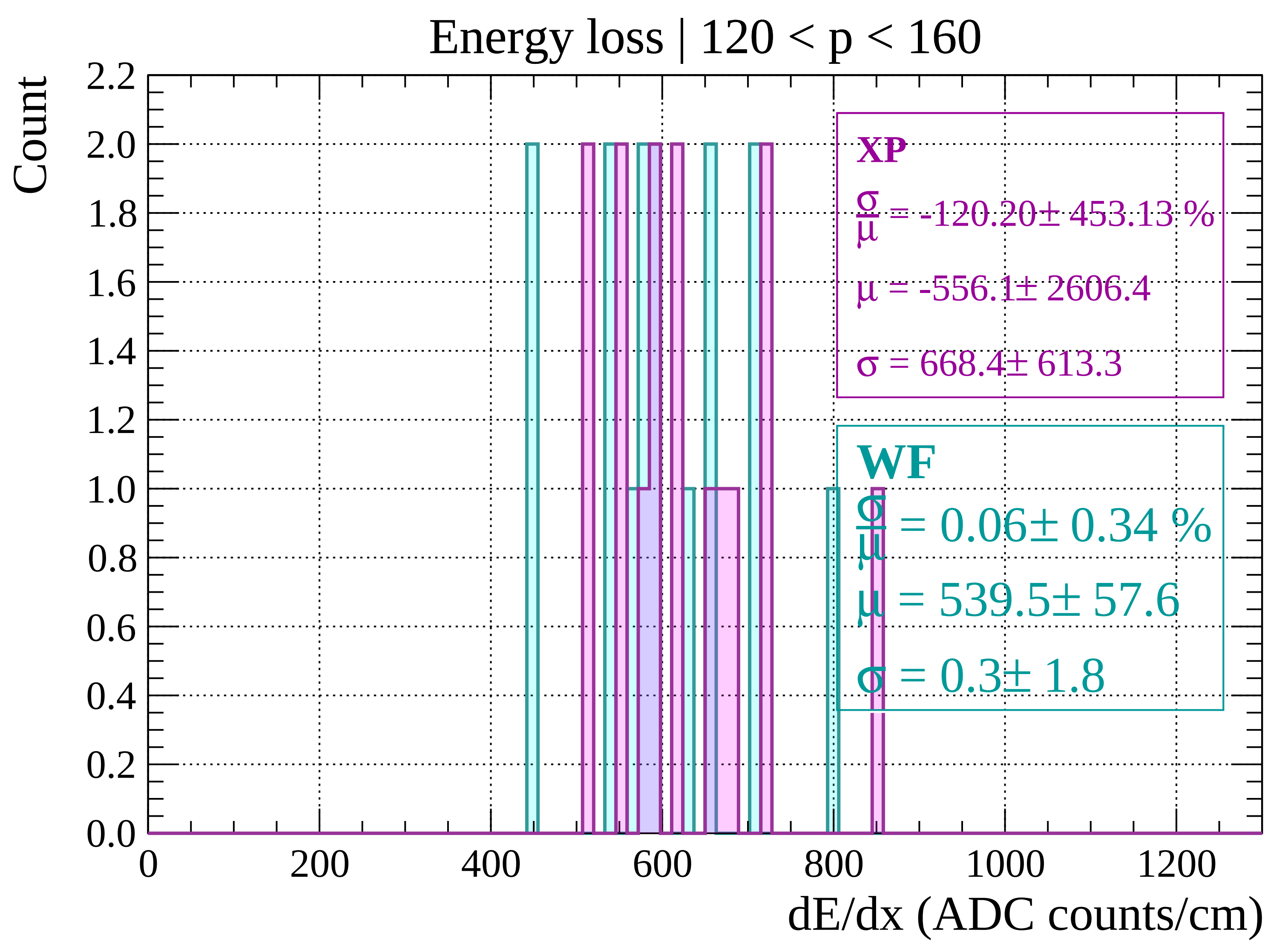
Energy loss | 0 < p < 40
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



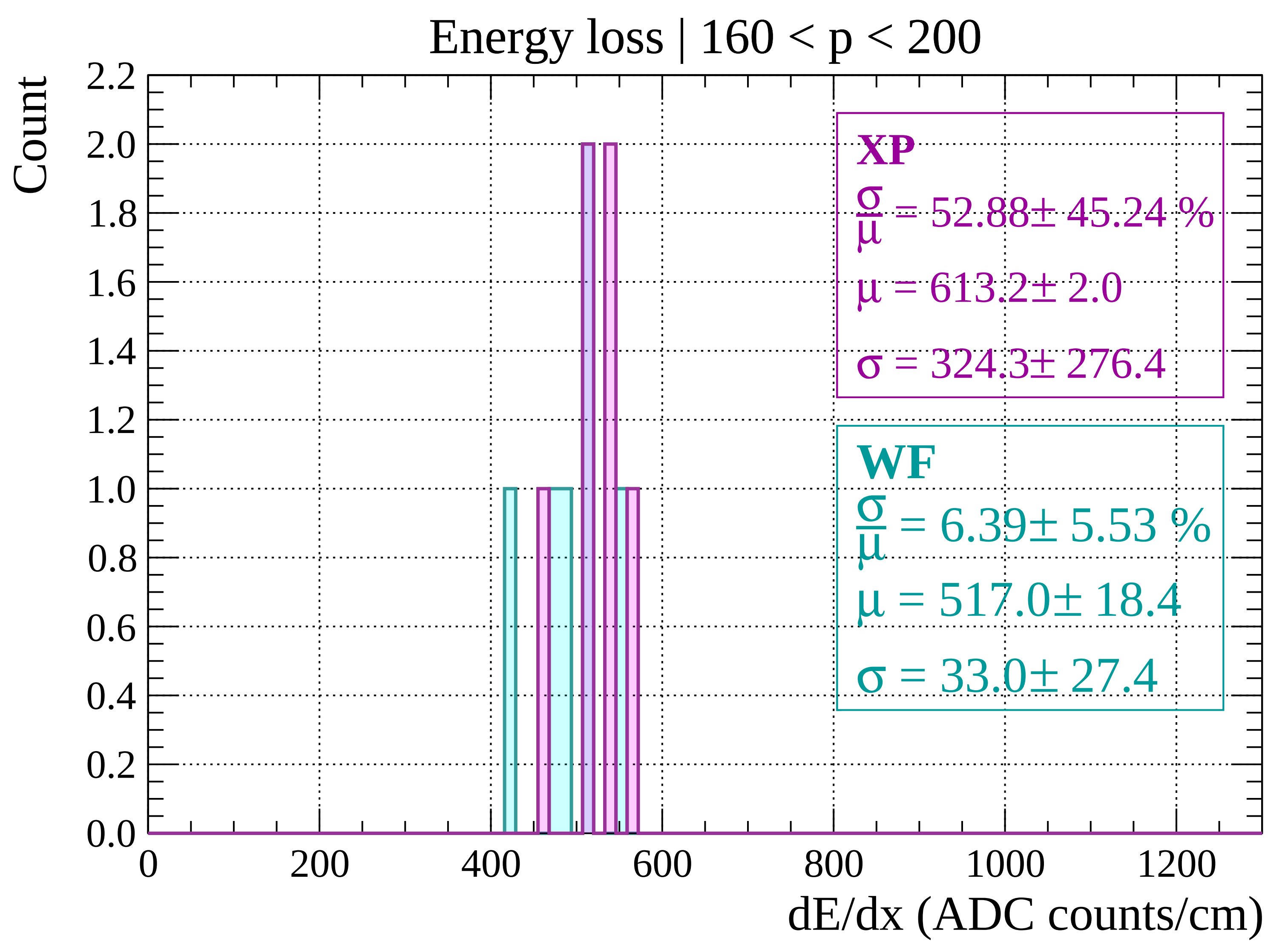
Energy loss | 40 < p < 80
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 38.17± 56.00 %
μ = 632.7± 182.0
σ = 241.5± 284.8
WF
$\frac{\sigma}{\mu}$ = 10.40± 5.67 %
μ = 568.4± 22.5
σ = 59.1± 29.9



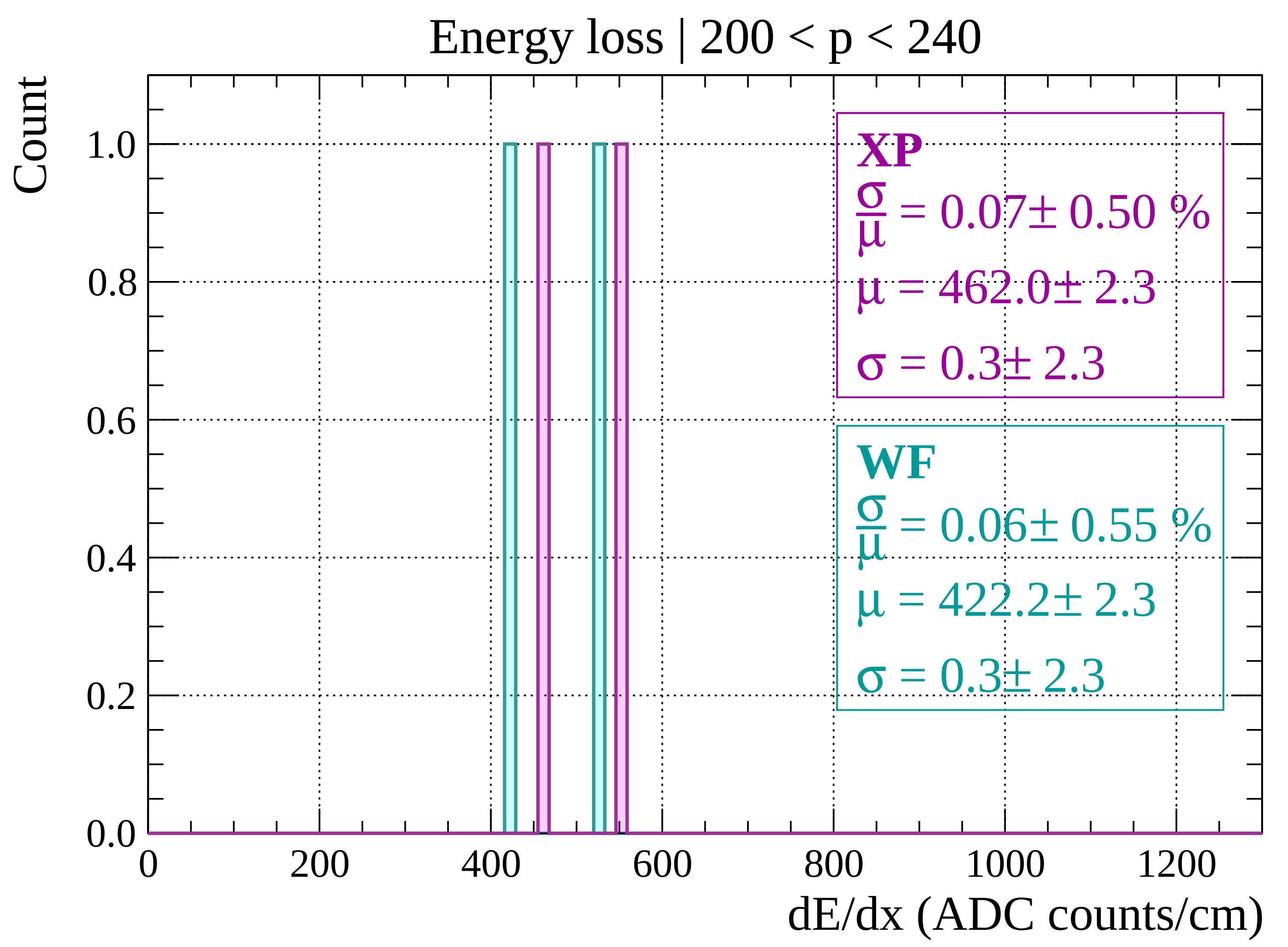
Energy loss | 80 < p < 120
XP
σ/μ = -93.51± 485.74 %
μ = -624.5± 3828.2
σ = 584.0± 546.4
WF
σ/μ = 1.05± 5.74 %
μ = 557.2± 71.2
σ = 5.8± 31.2
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200



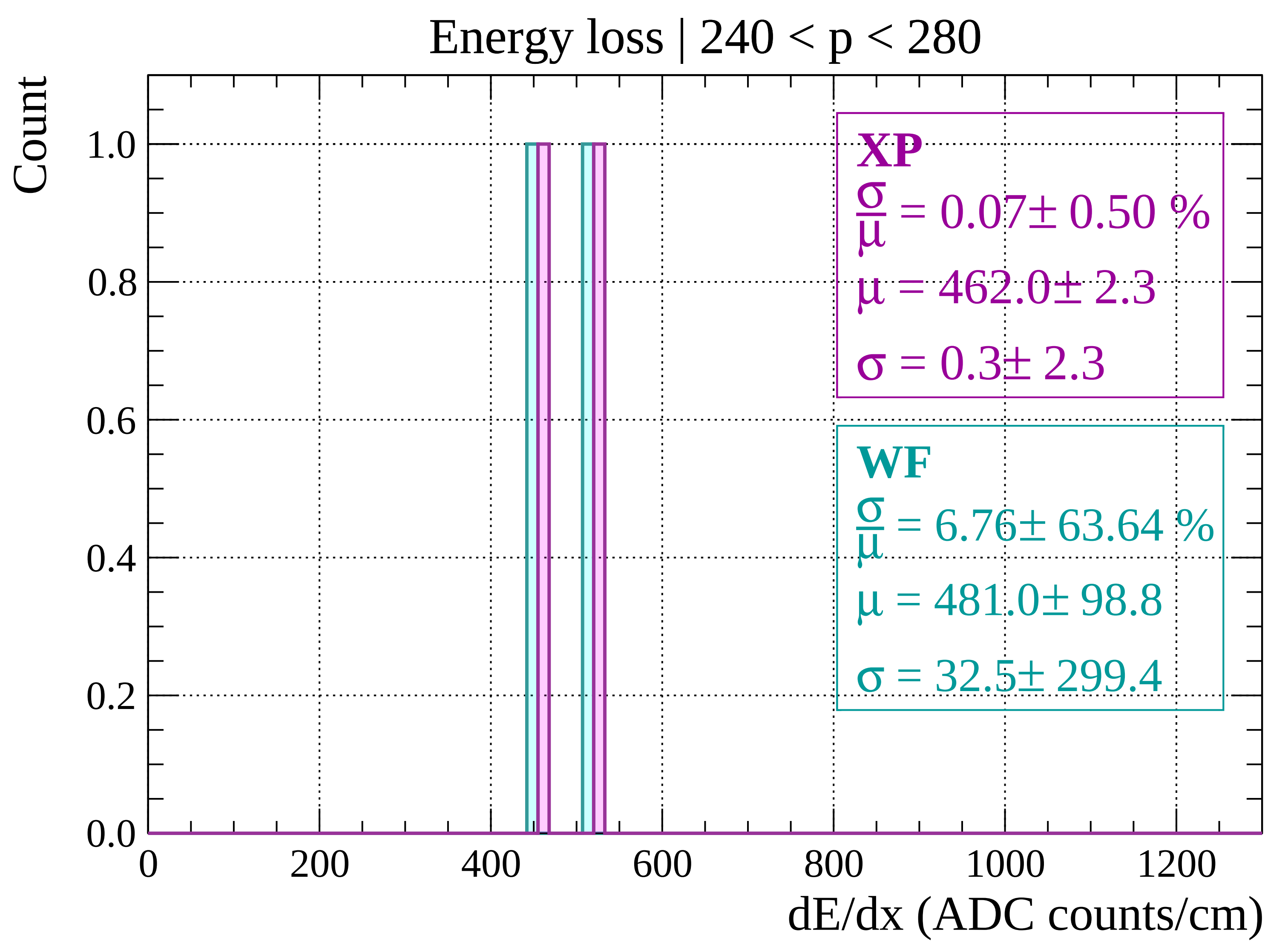
Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -120.20± 453.13 %
μ = -556.1± 2606.4
σ = 668.4± 613.3
WF
$\frac{\sigma}{\mu}$ = 0.06± 0.34 %
μ = 539.5± 57.6
σ = 0.3± 1.8



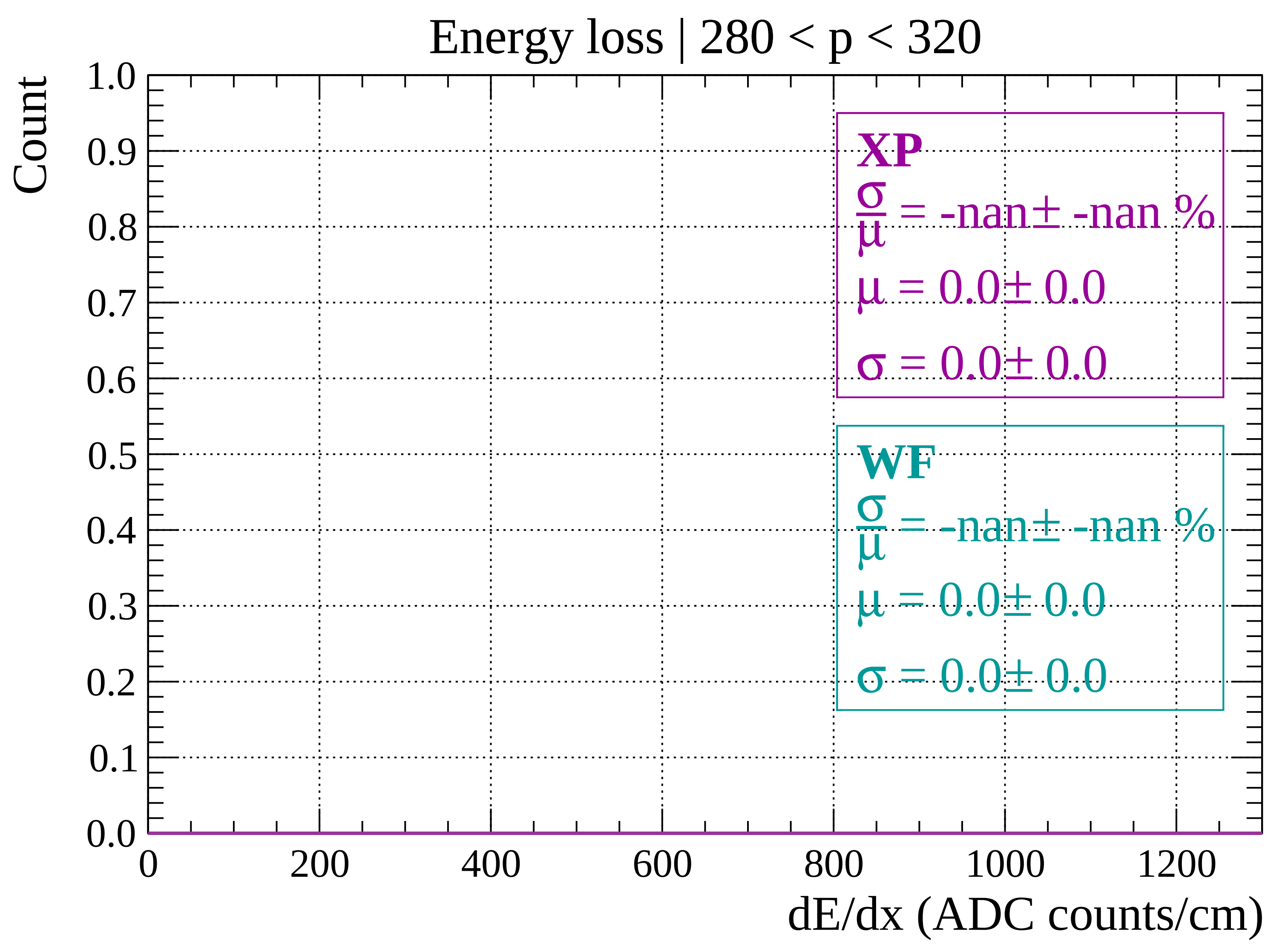
Energy loss | 160 < p < 200
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 52.88± 45.24 %
μ = 613.2± 2.0
σ = 324.3± 276.4
WF
$\frac{\sigma}{\mu}$ = 6.39± 5.53 %
μ = 517.0± 18.4
σ = 33.0± 27.4

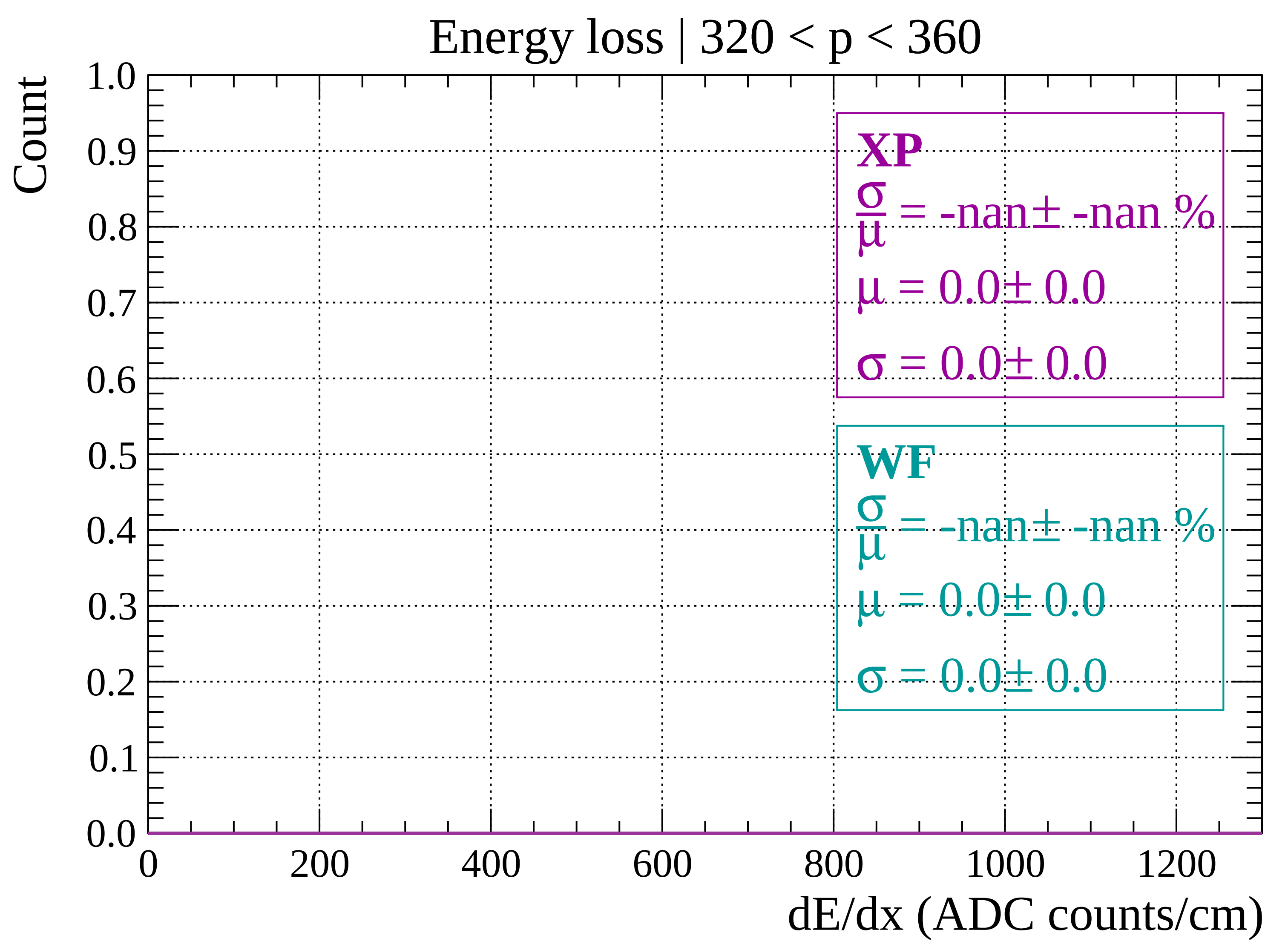


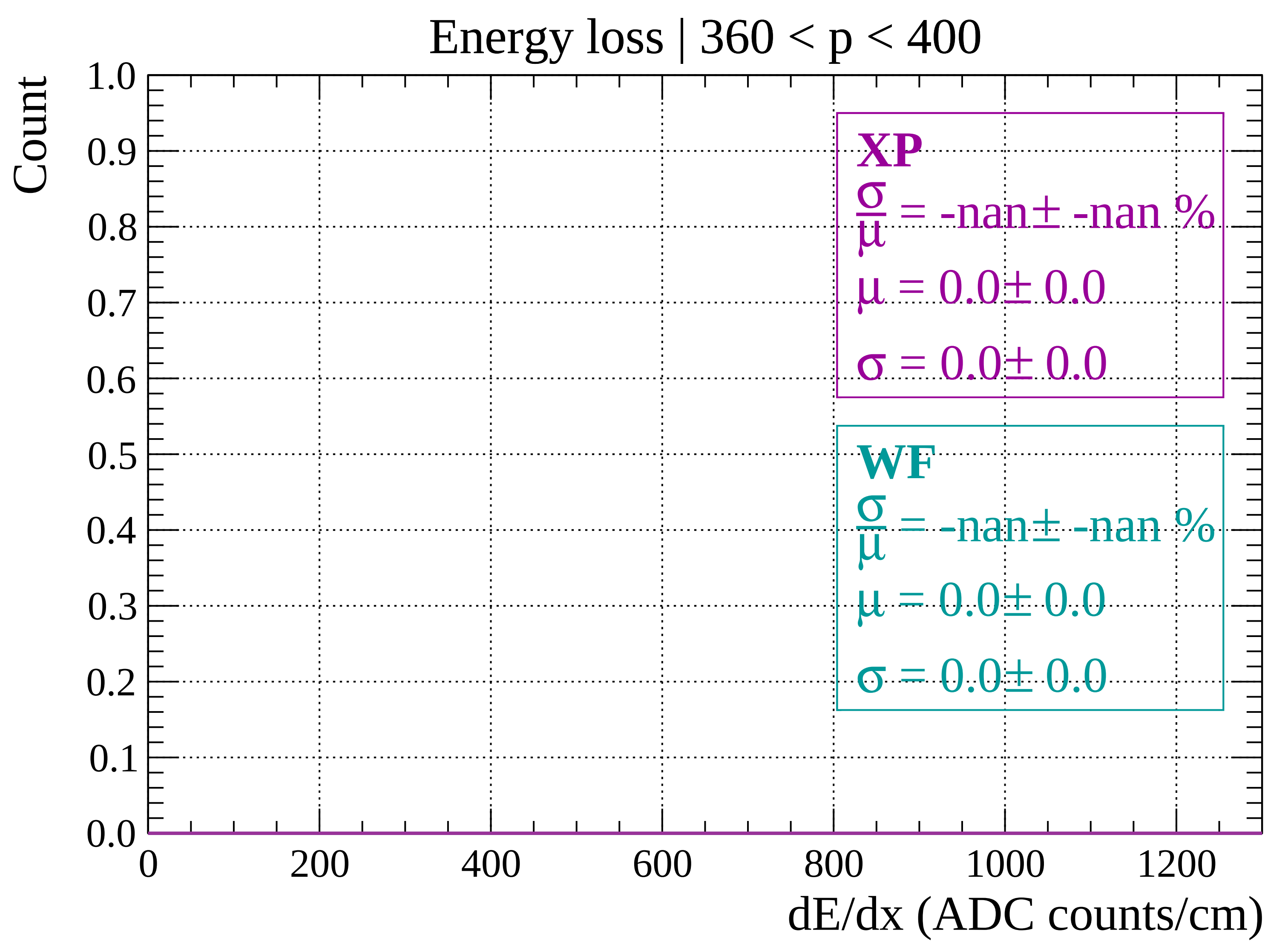
Energy loss | 200 < p < 240
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.07± 0.50 %
μ = 462.0± 2.3
σ = 0.3± 2.3
WF
σ/μ = 0.06± 0.55 %
μ = 422.2± 2.3
σ = 0.3± 2.3



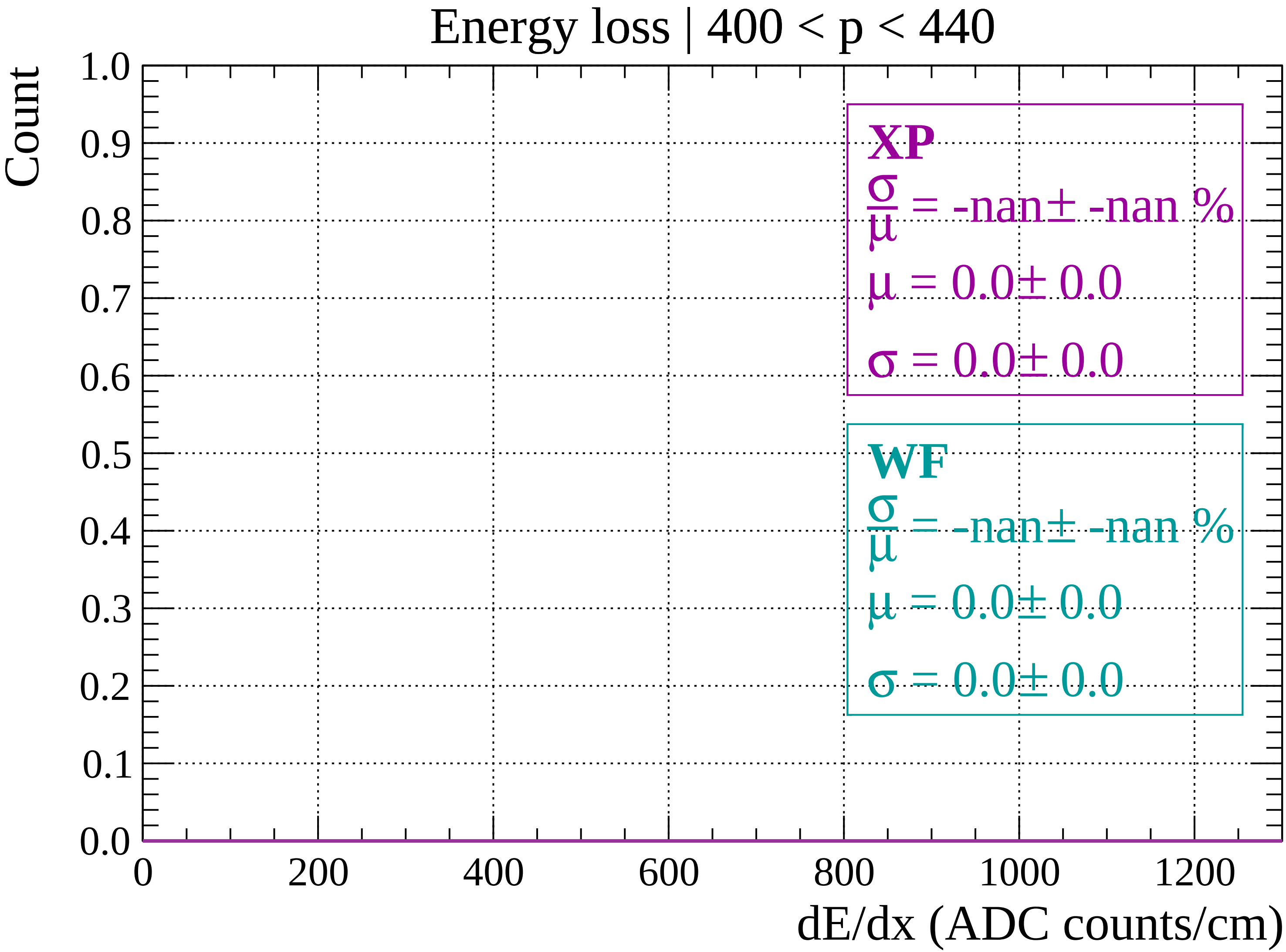
Energy loss | 240 < p < 280
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 0.07± 0.50 %
μ = 462.0± 2.3
σ = 0.3± 2.3
WF
$\frac{\sigma}{\mu}$ = 6.76± 63.64 %
μ = 481.0± 98.8
σ = 32.5± 299.4

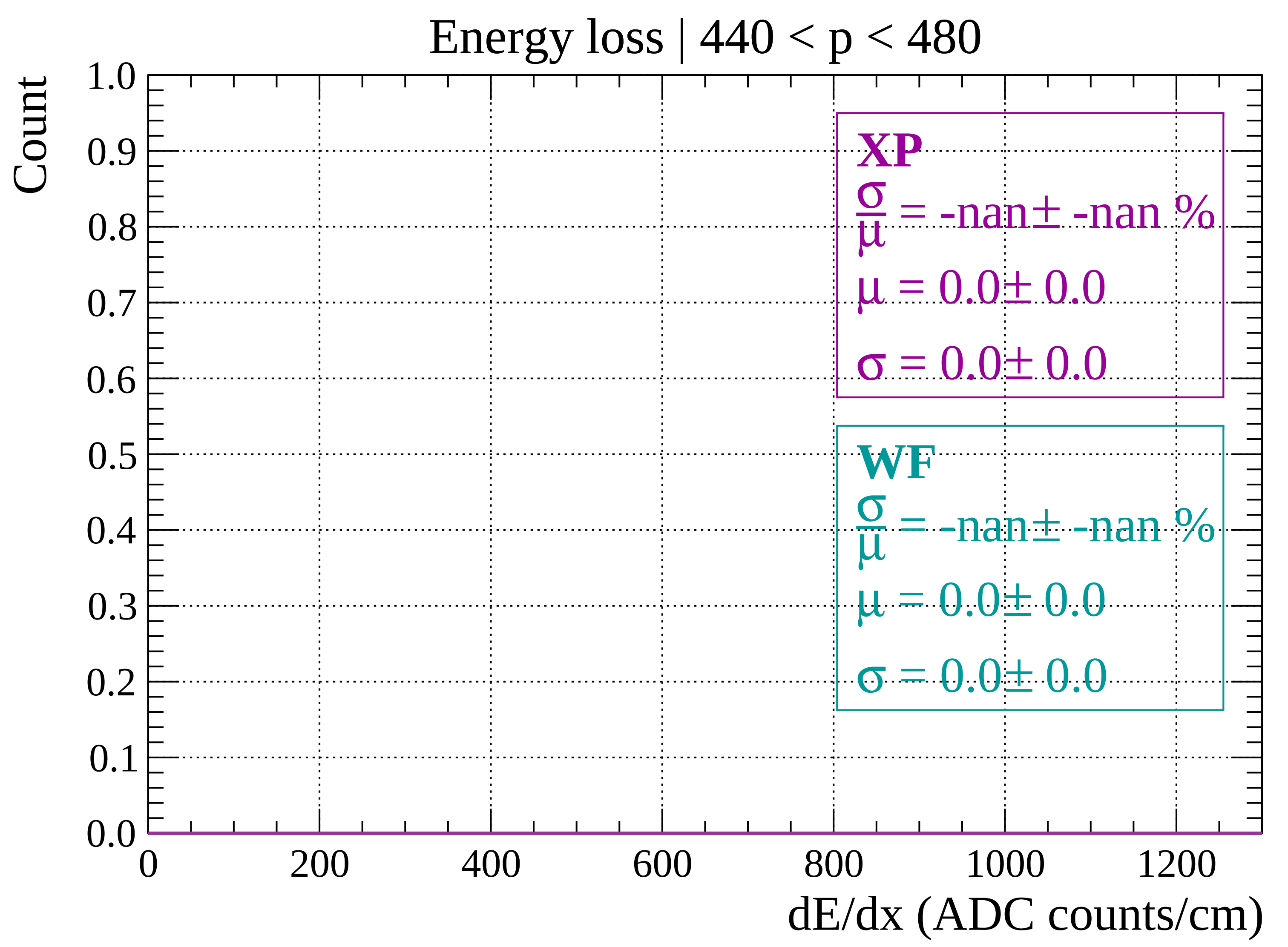

Energy loss | 280 < p < 320
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

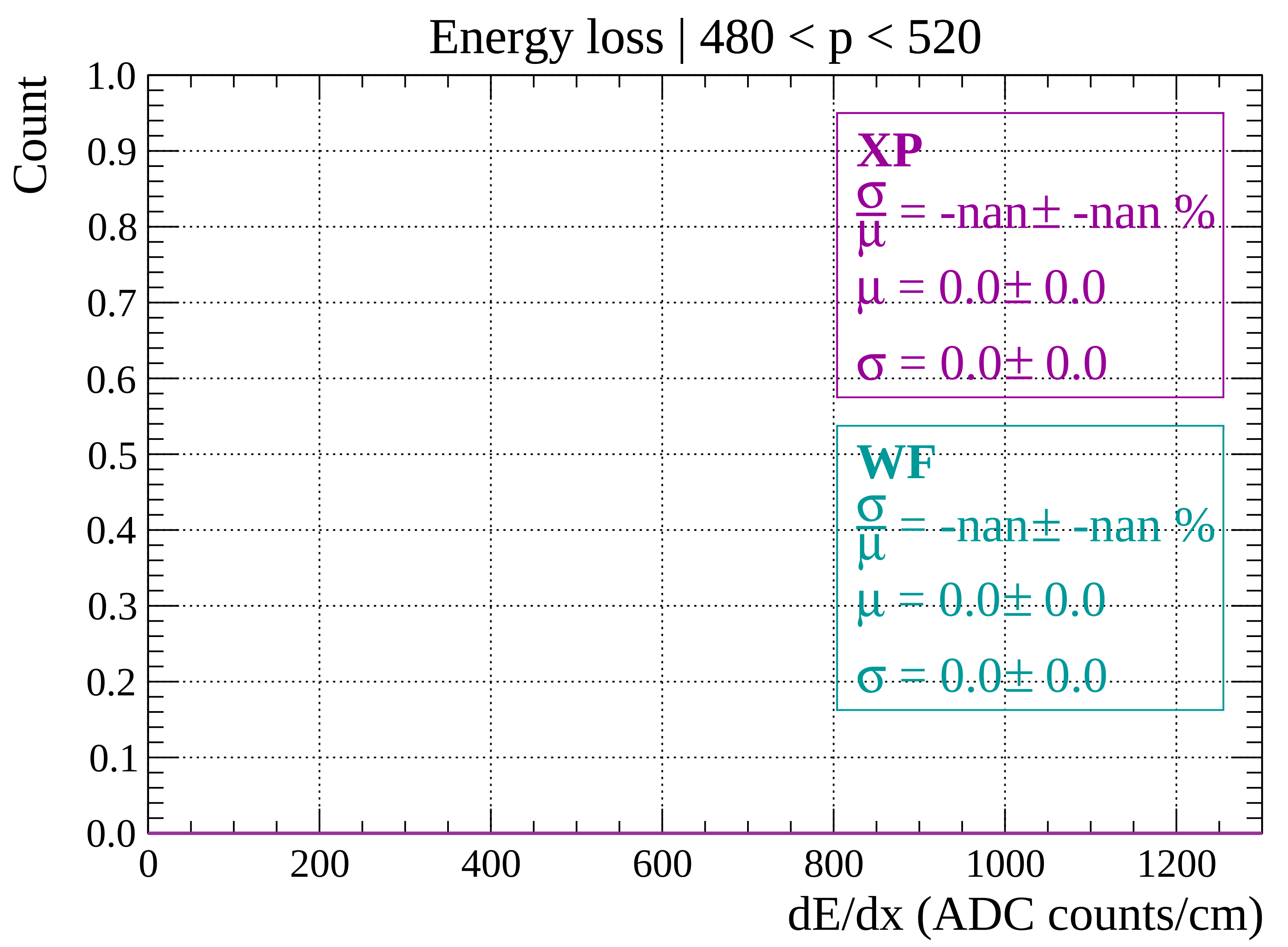

Energy loss | 320 < p < 360
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



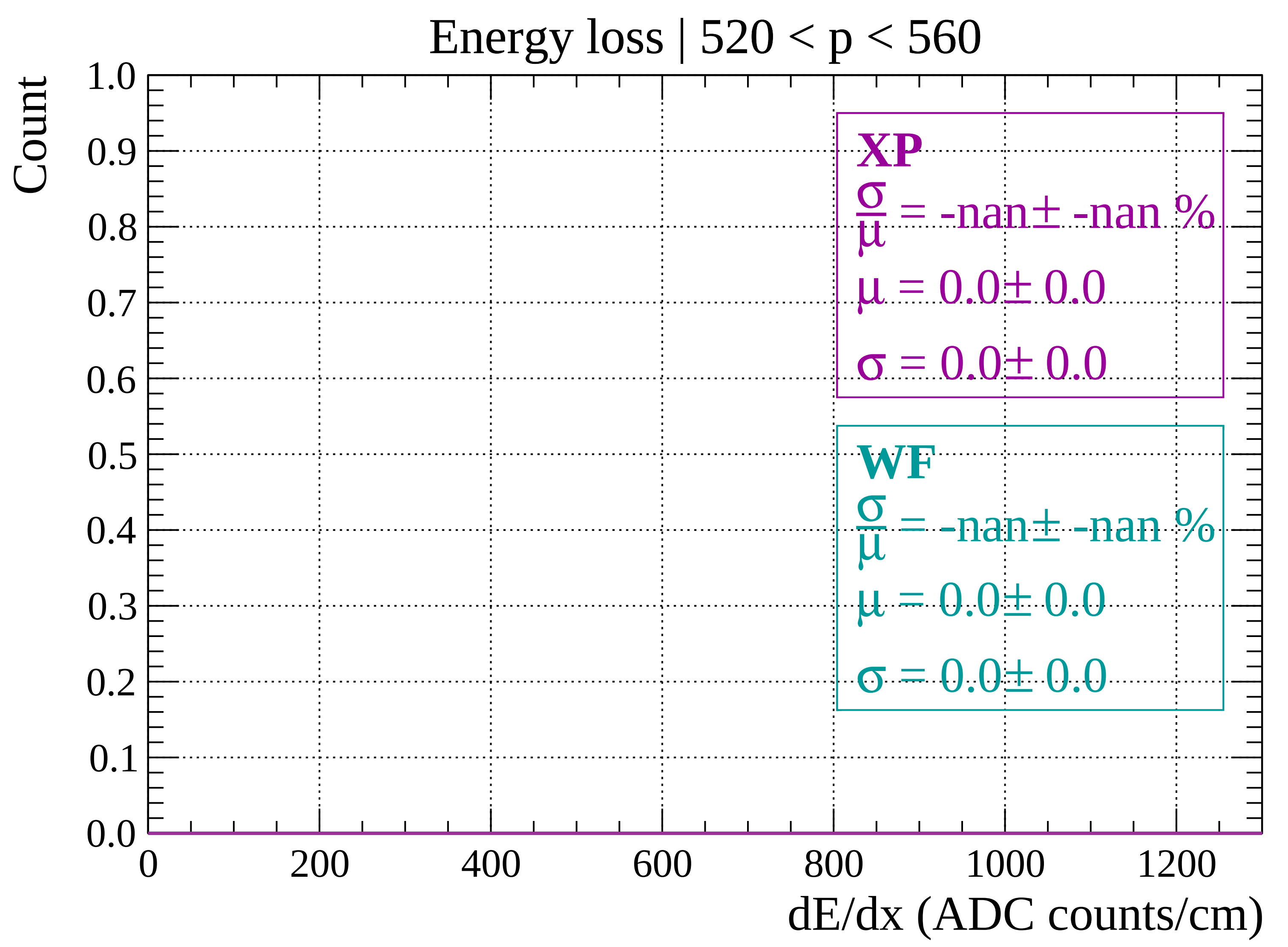
Energy loss | 360 < p < 400
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

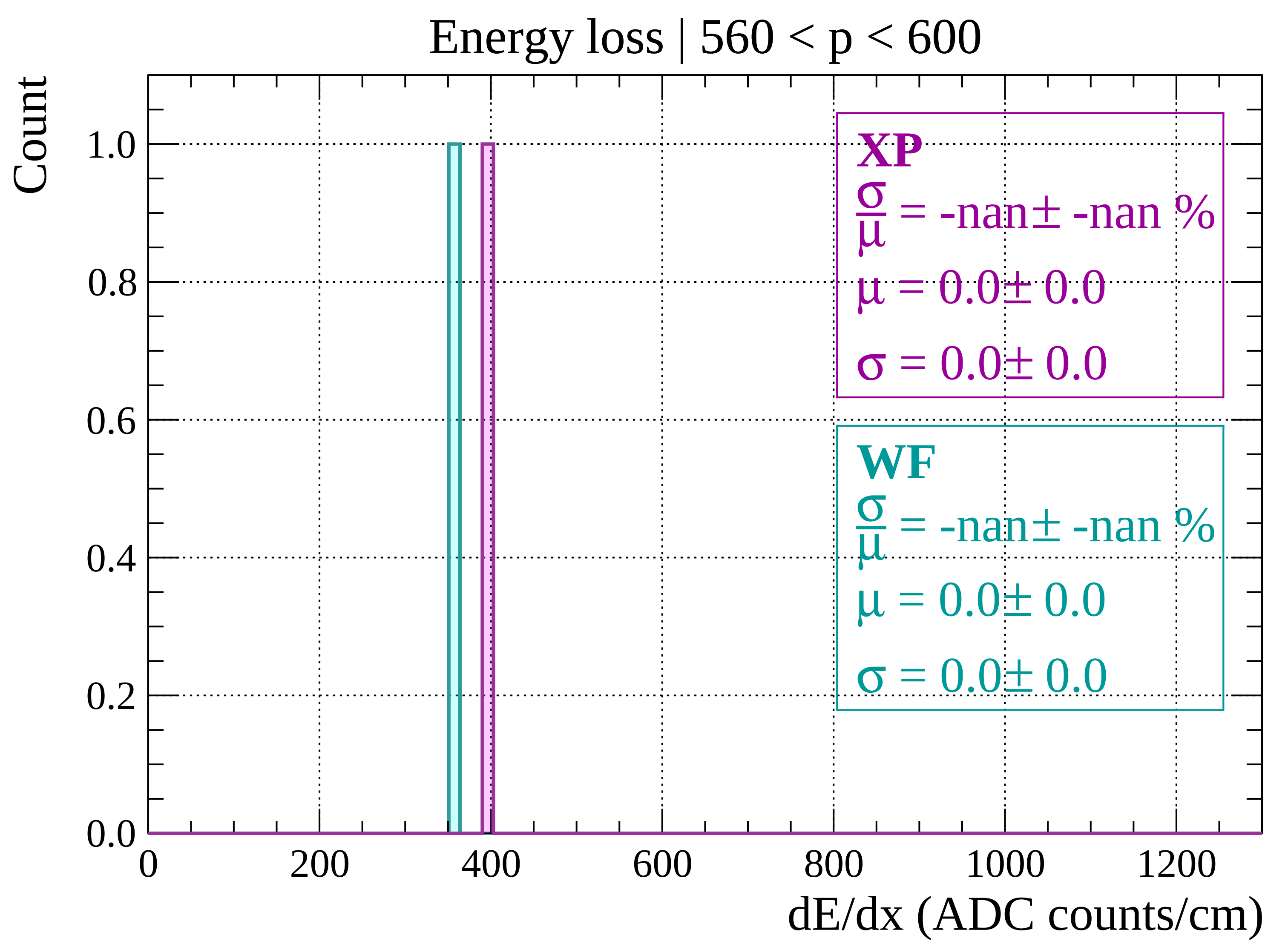

Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

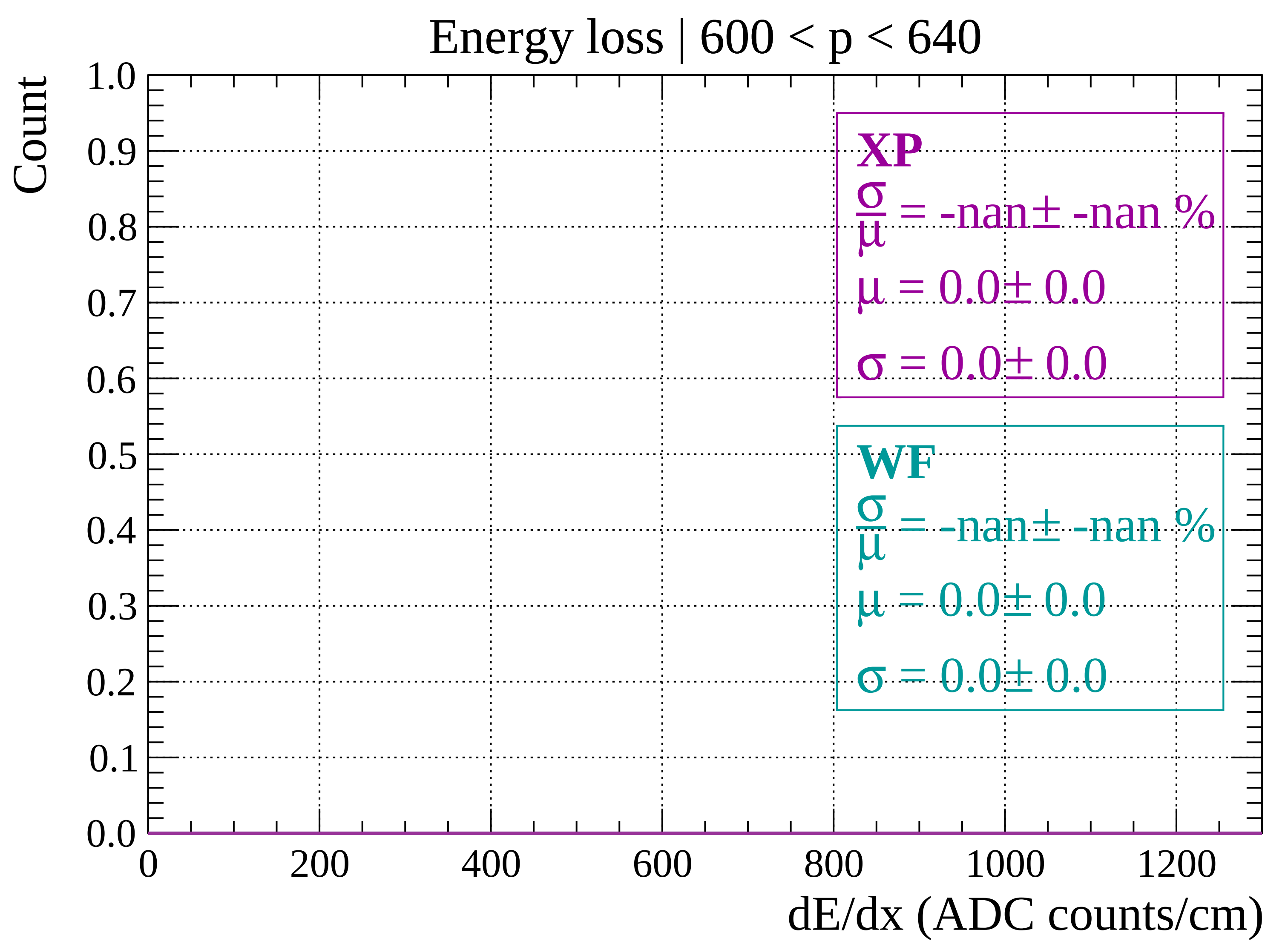

Energy loss | 440 < p < 480
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

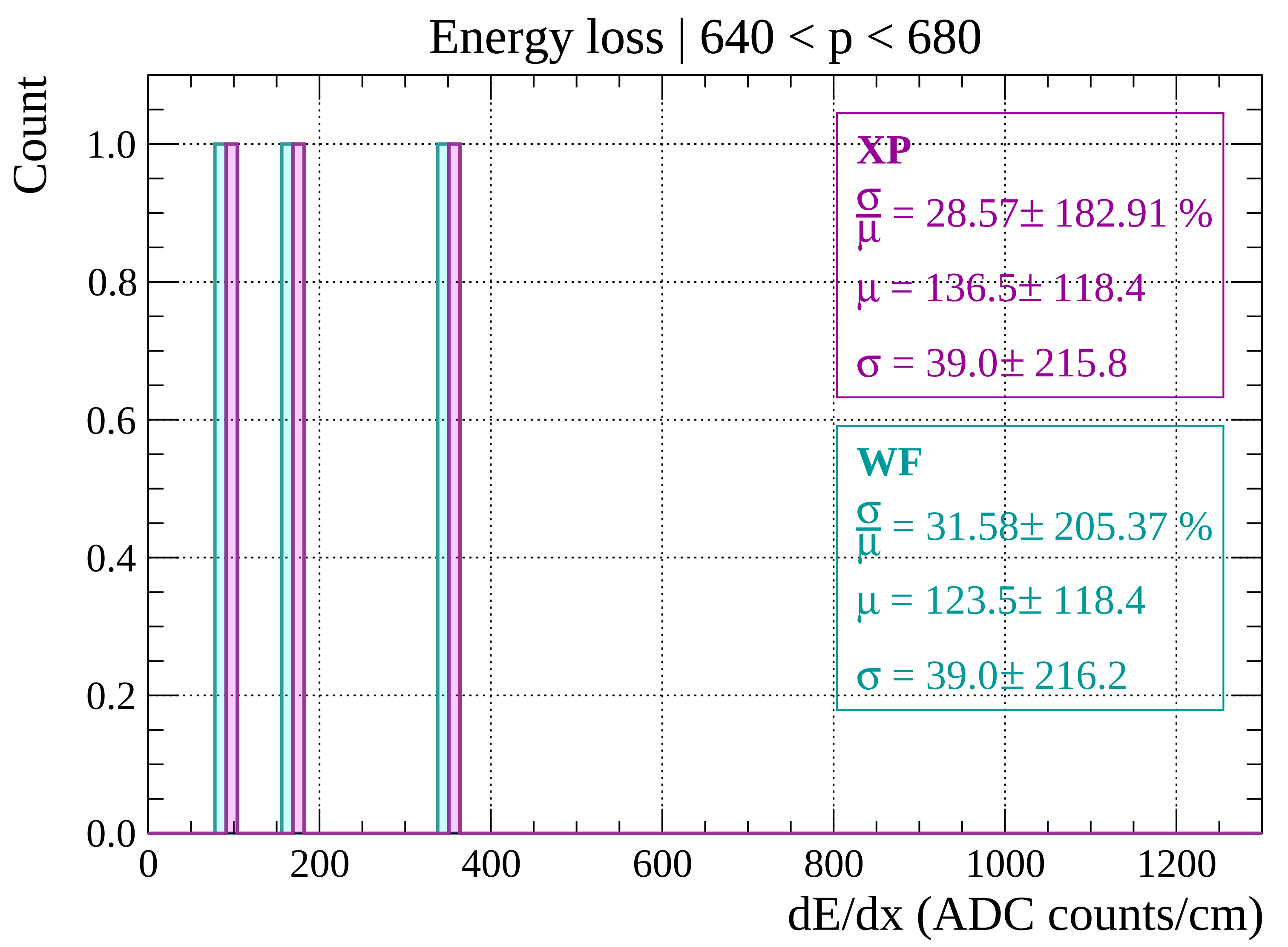


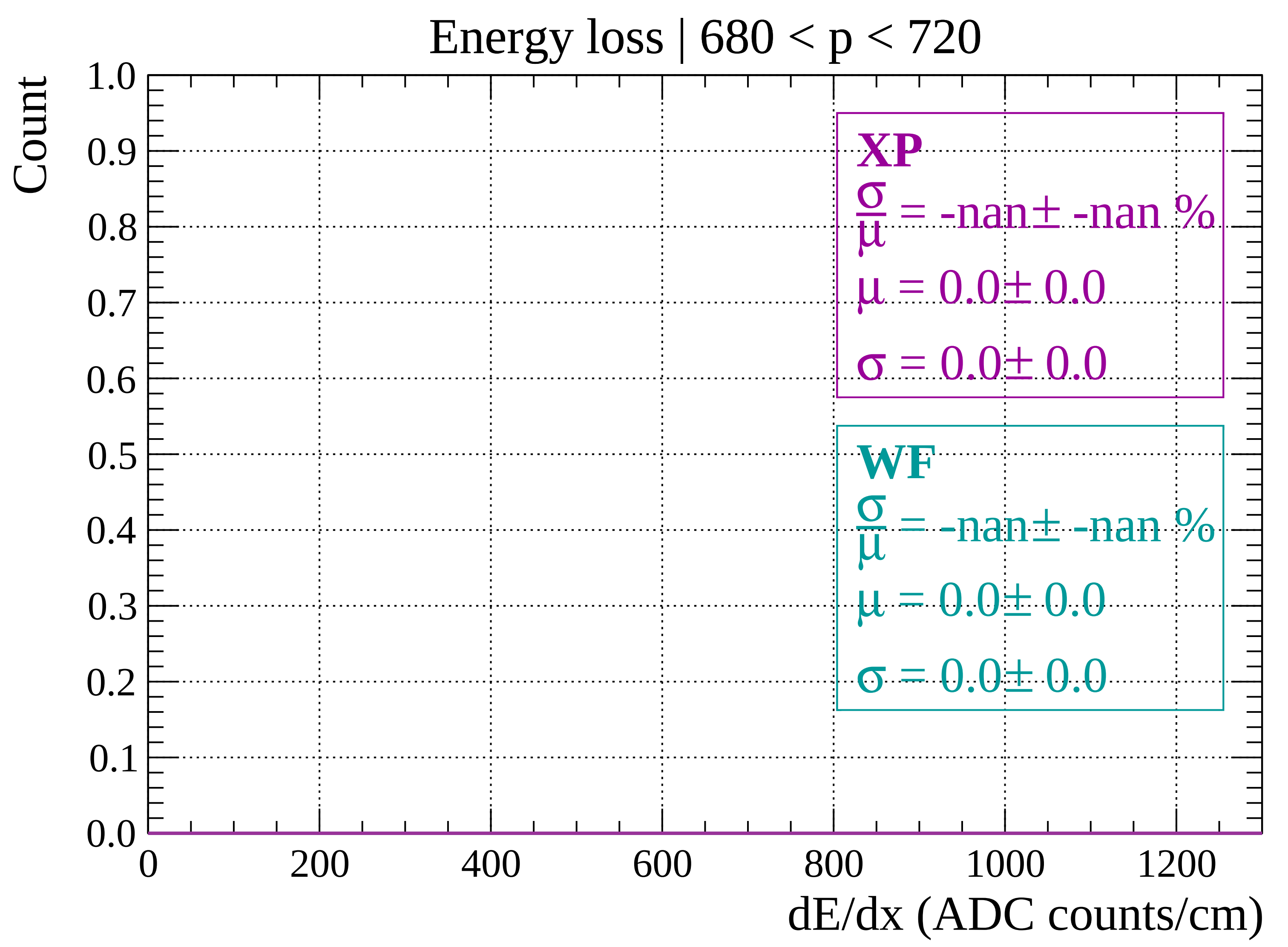
Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

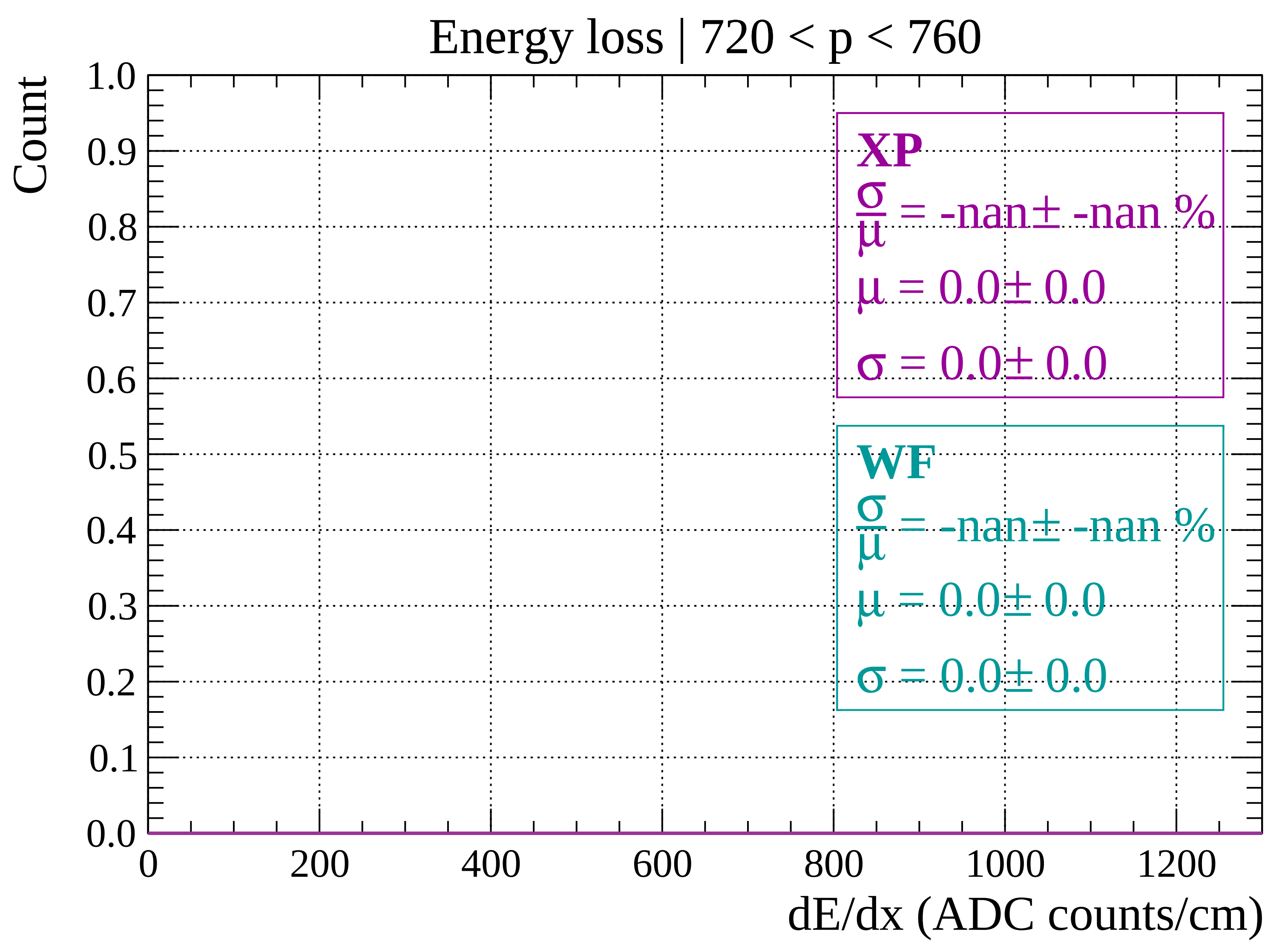

Energy loss | 520 < p < 560
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 560 < p < 600
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

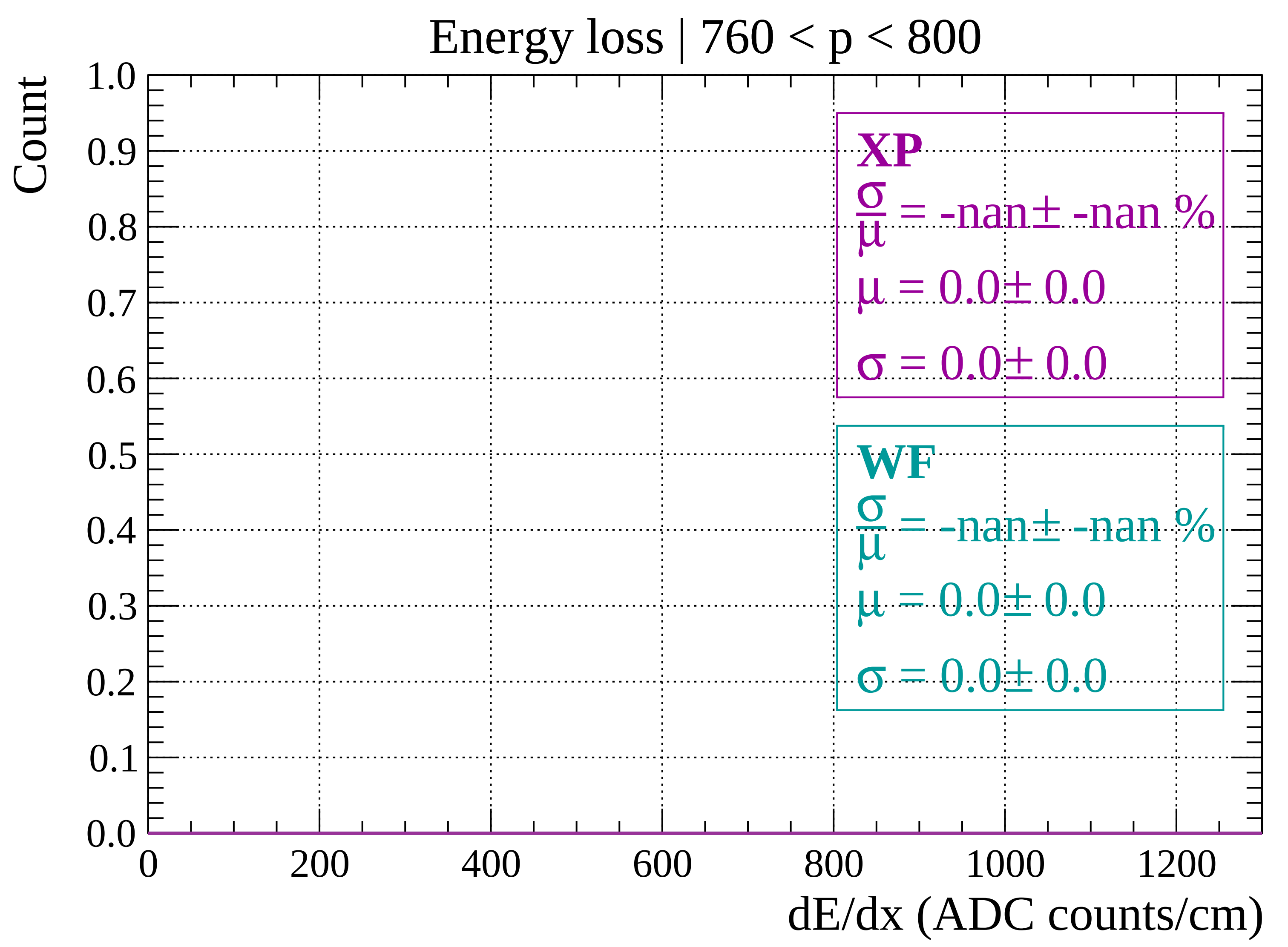

Energy loss | 600 < p < 640
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0
200
400
600
800
1000
1200
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0

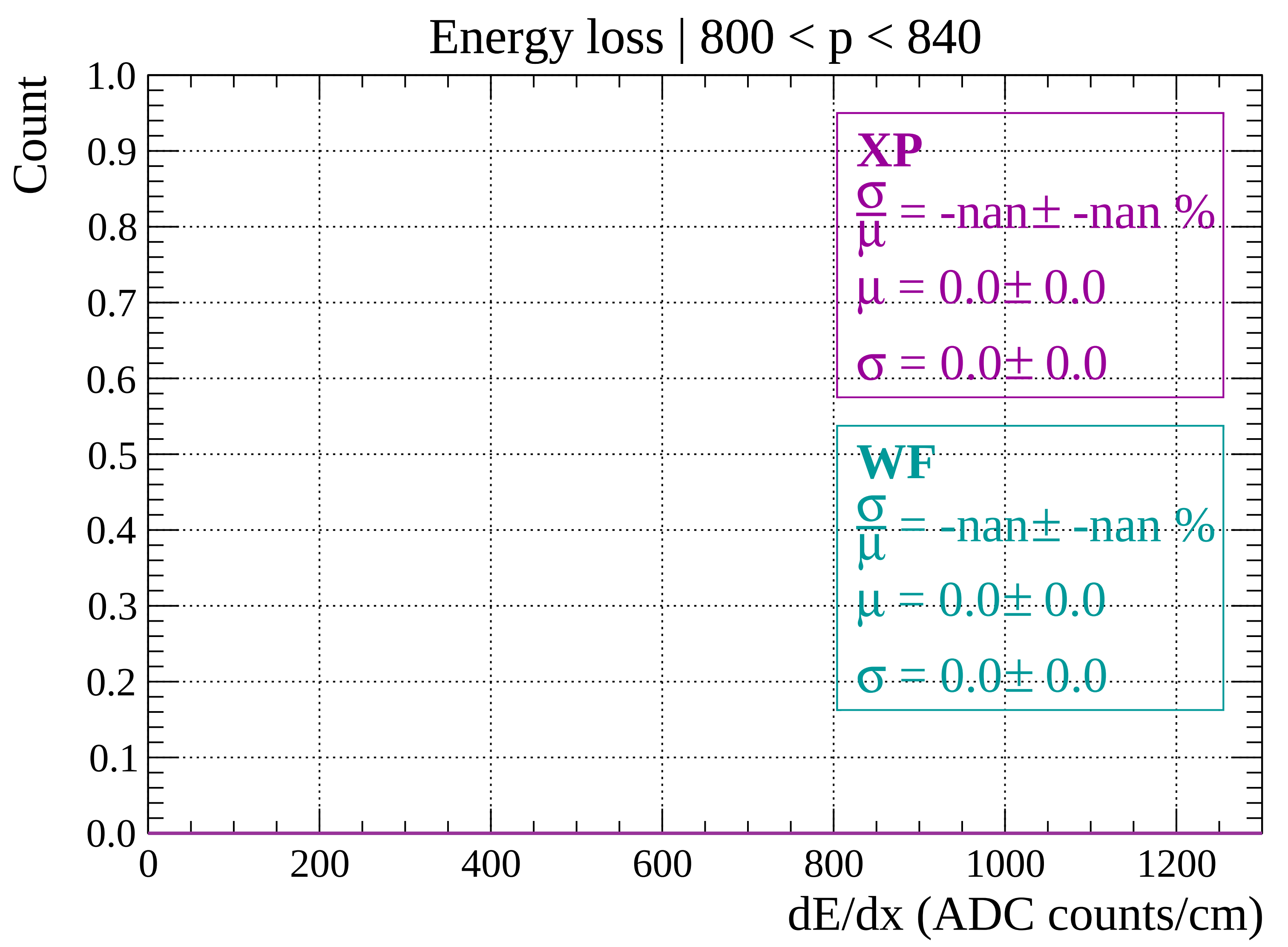

Energy loss | 640 < p < 680
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 28.57± 182.91 %
μ = 136.5± 118.4
σ = 39.0± 215.8
WF
$\frac{\sigma}{\mu}$ = 31.58± 205.37 %
μ = 123.5± 118.4
σ = 39.0± 216.2


Energy loss | 680 < p < 720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

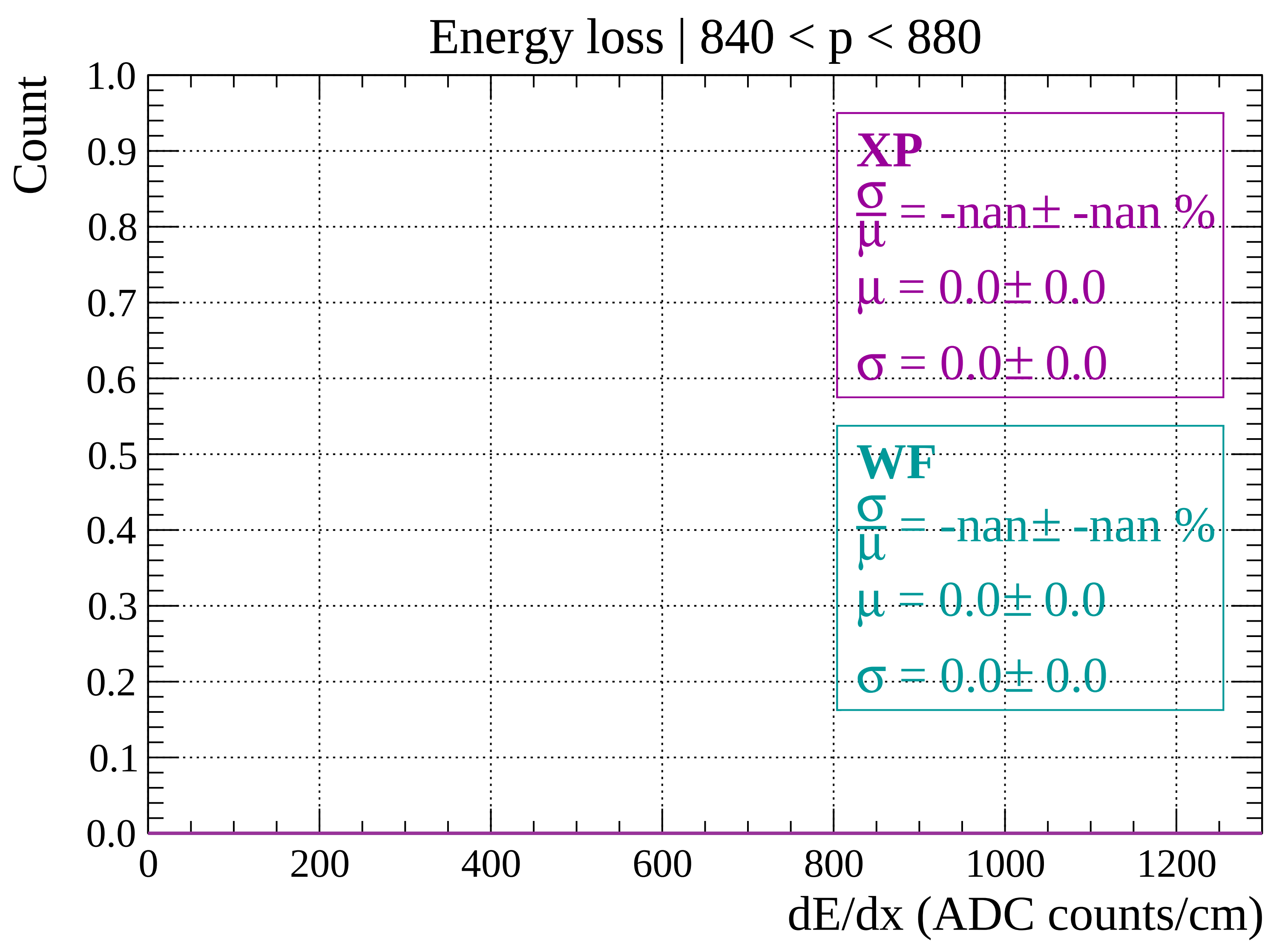


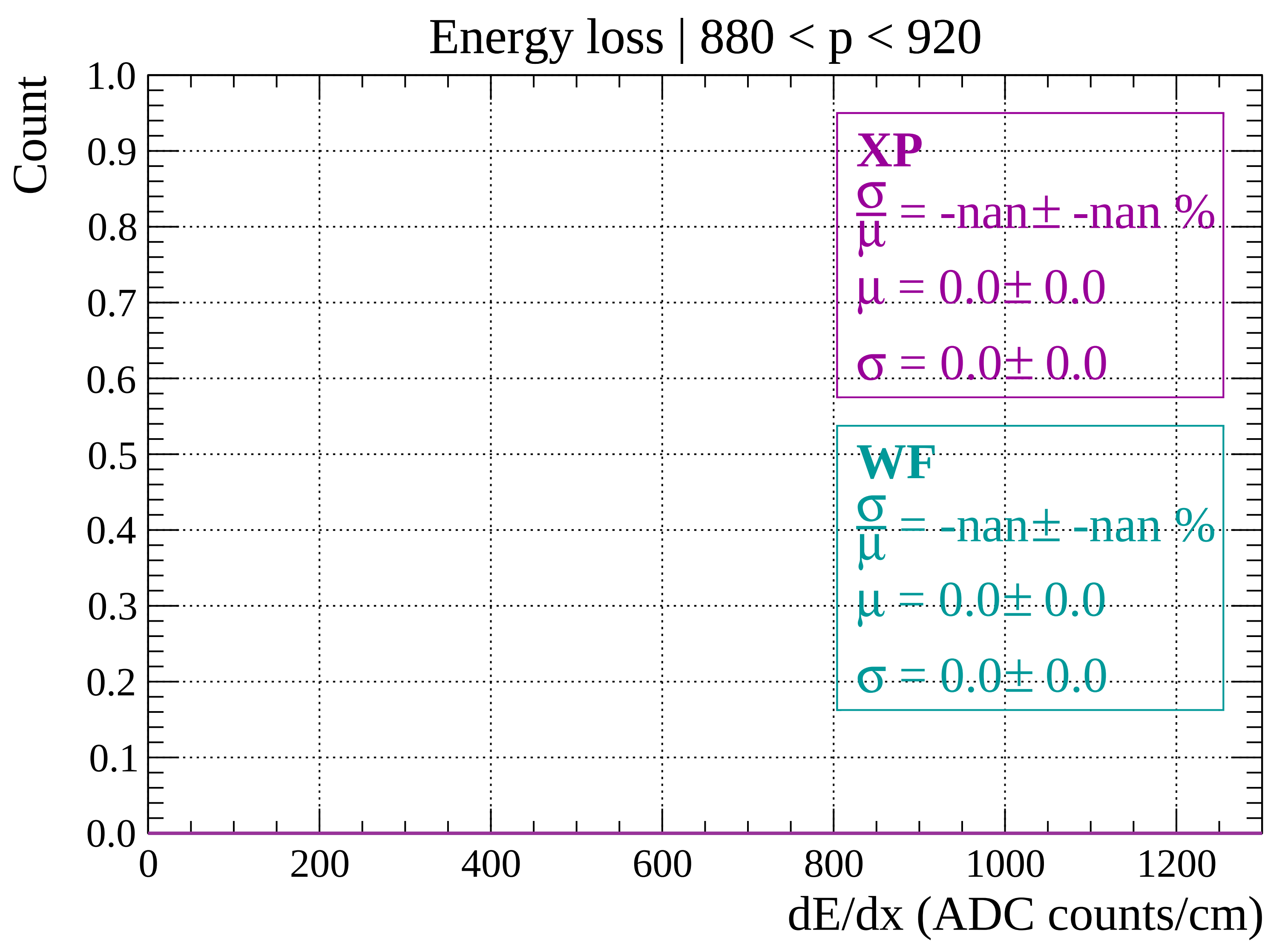
Energy loss | 720 < p < 760
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

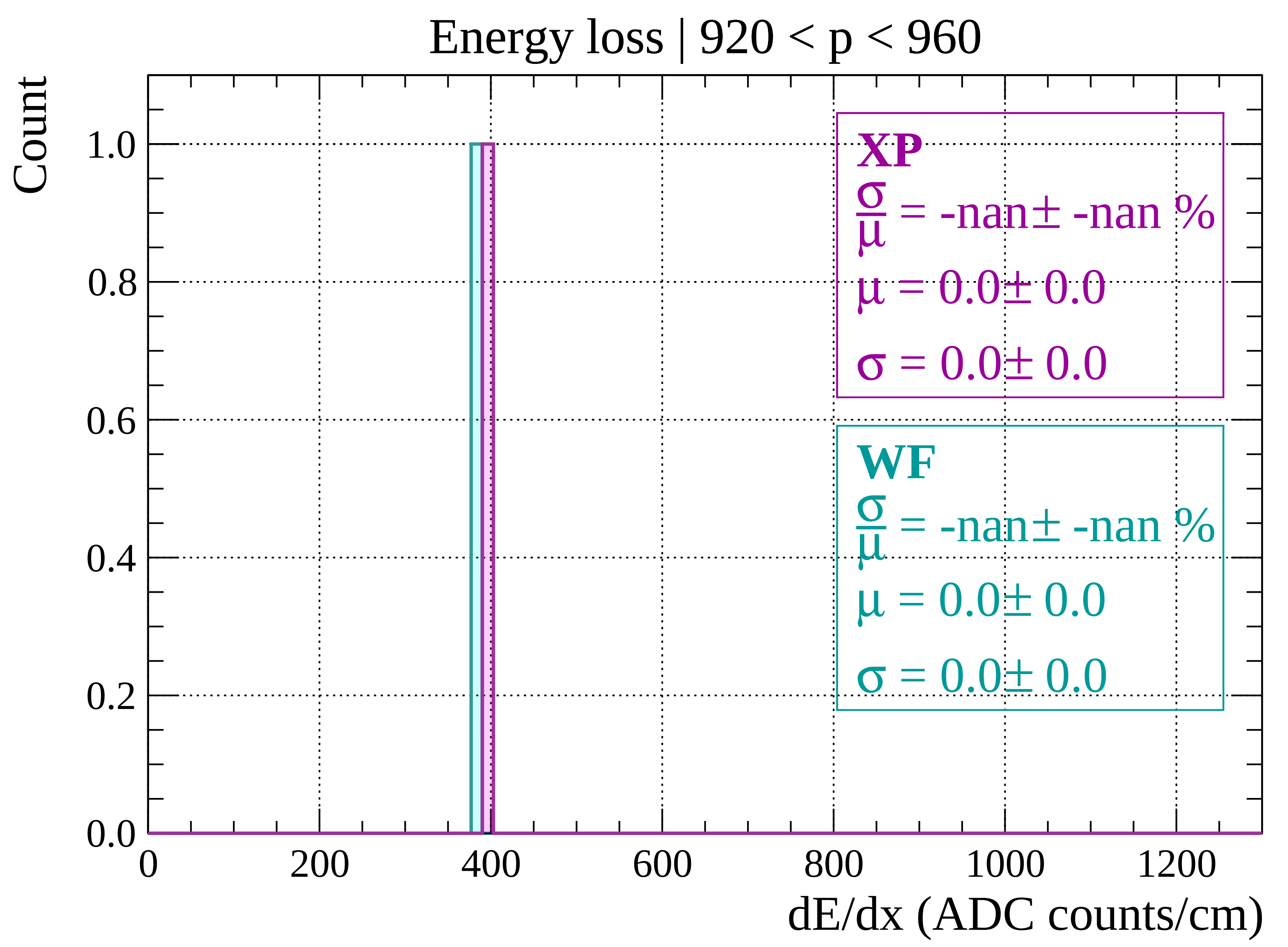

Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



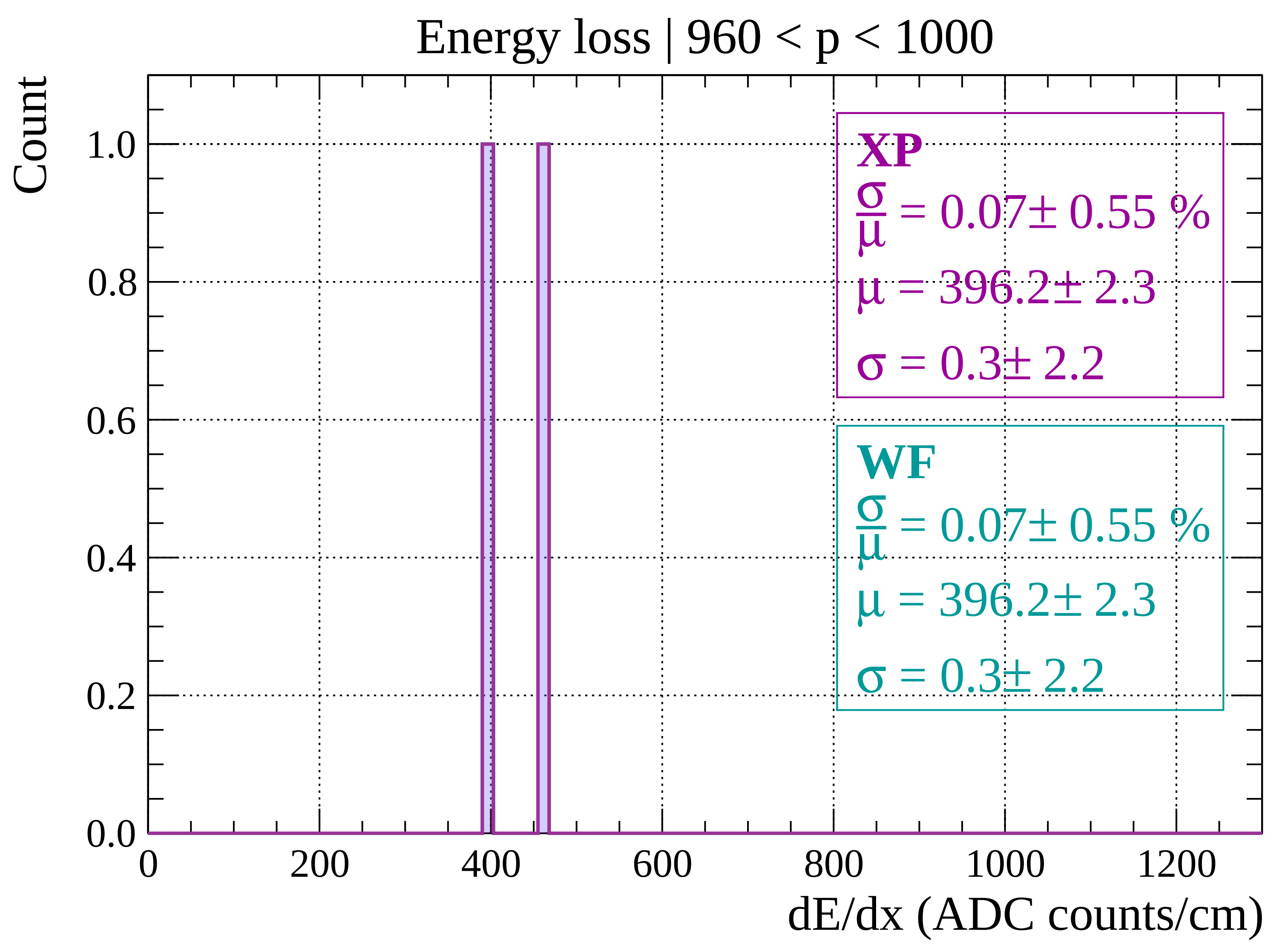
Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 840 < p < 880
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

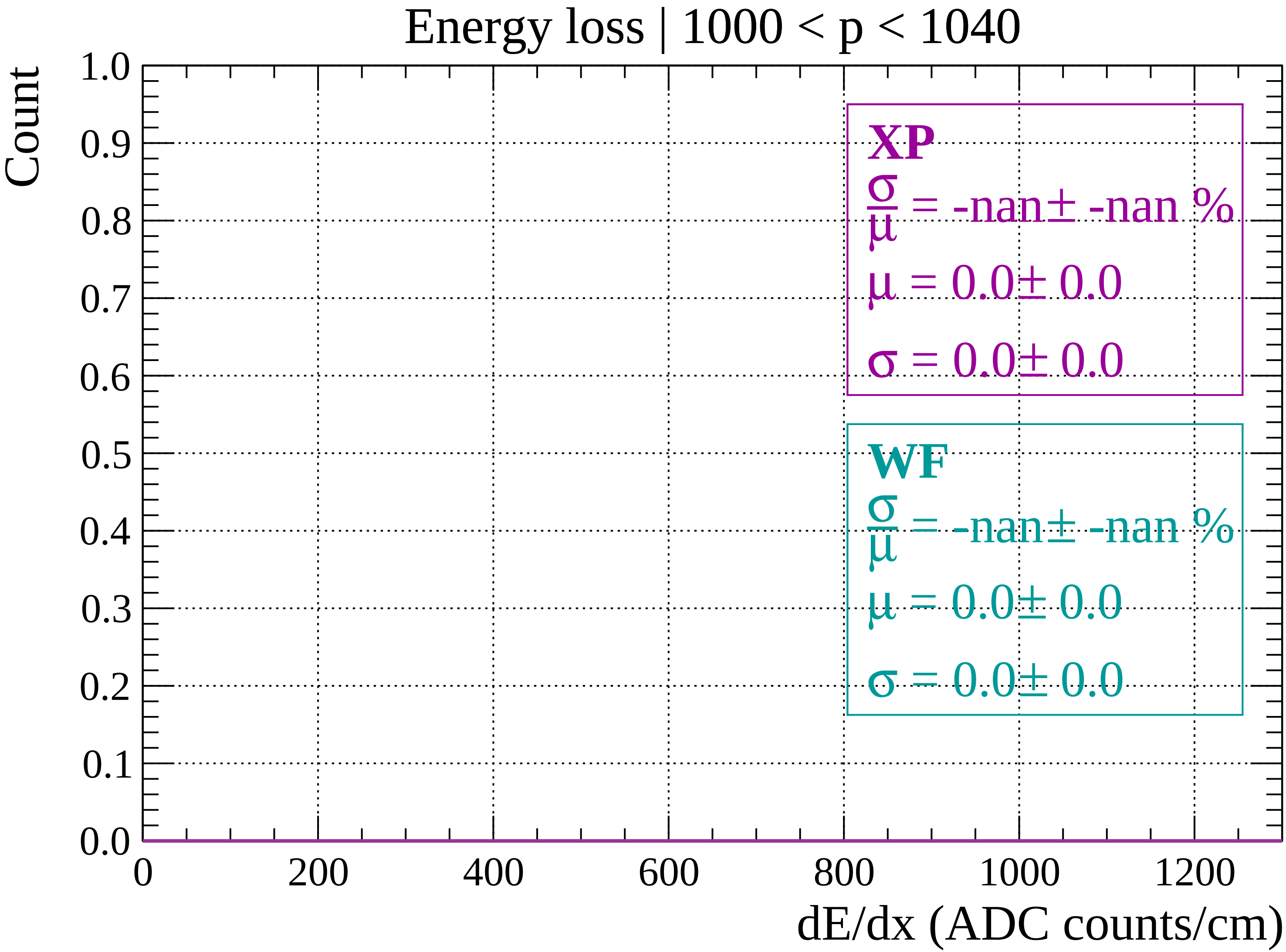

Energy loss | 880 < p < 920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



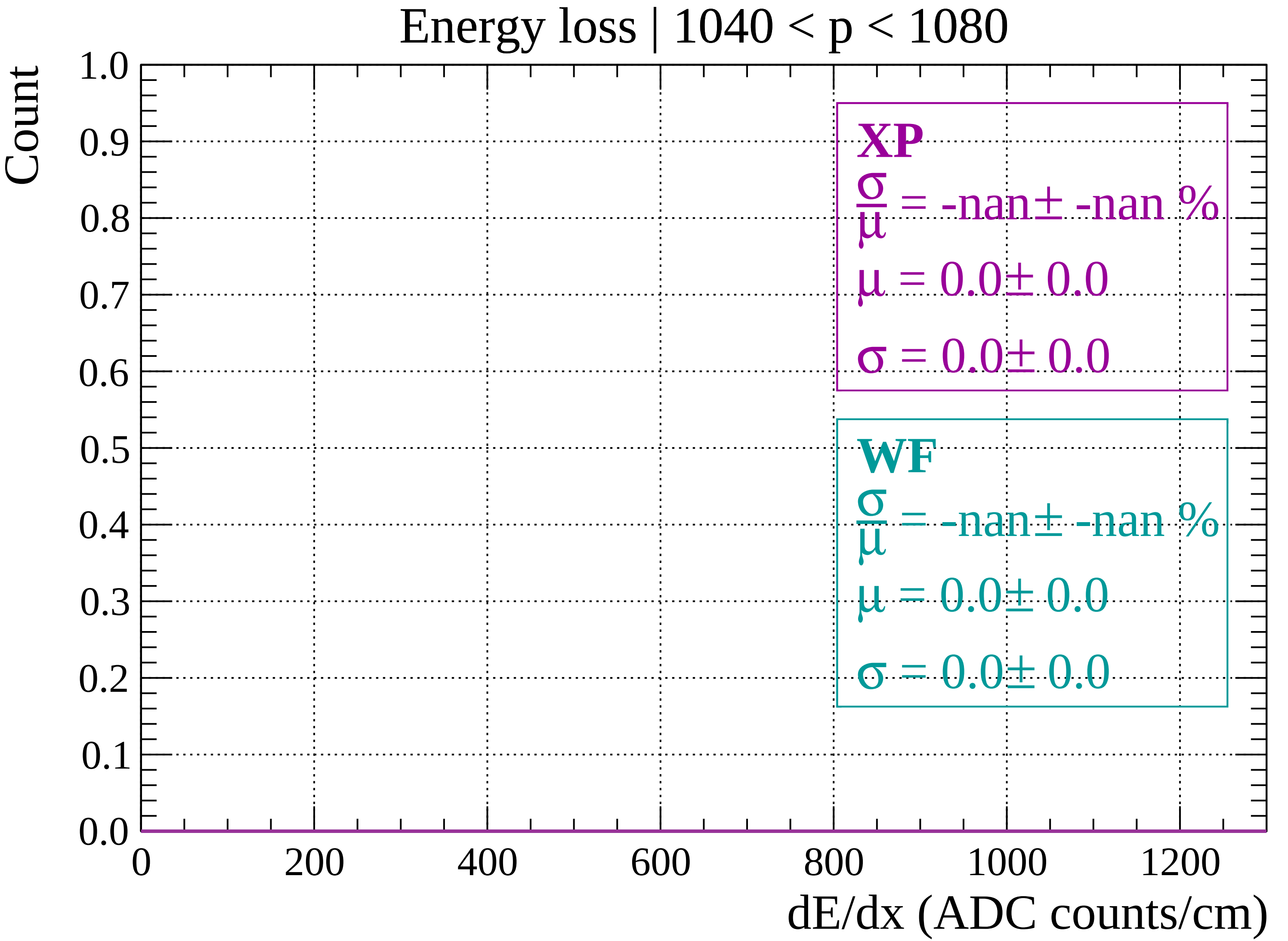
Energy loss | 920 < p < 960
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



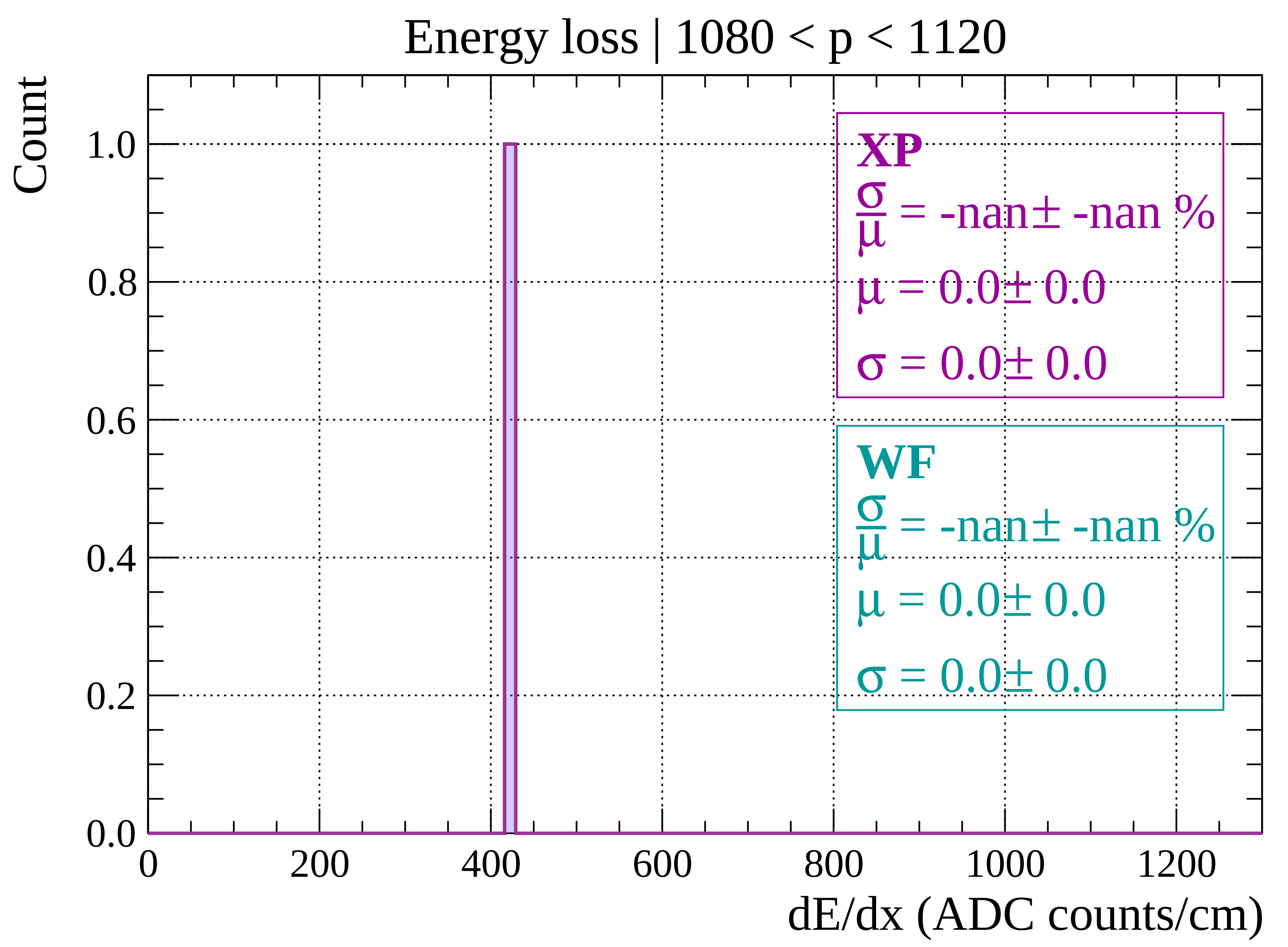
Energy loss | 960 < p < 1000
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 0.07 \pm 0.55$ %
$\mu = 396.2 \pm 2.3$
$\sigma = 0.3 \pm 2.2$
WF
$\frac{\sigma}{\mu} = 0.07 \pm 0.55$ %
$\mu = 396.2 \pm 2.3$
$\sigma = 0.3 \pm 2.2$



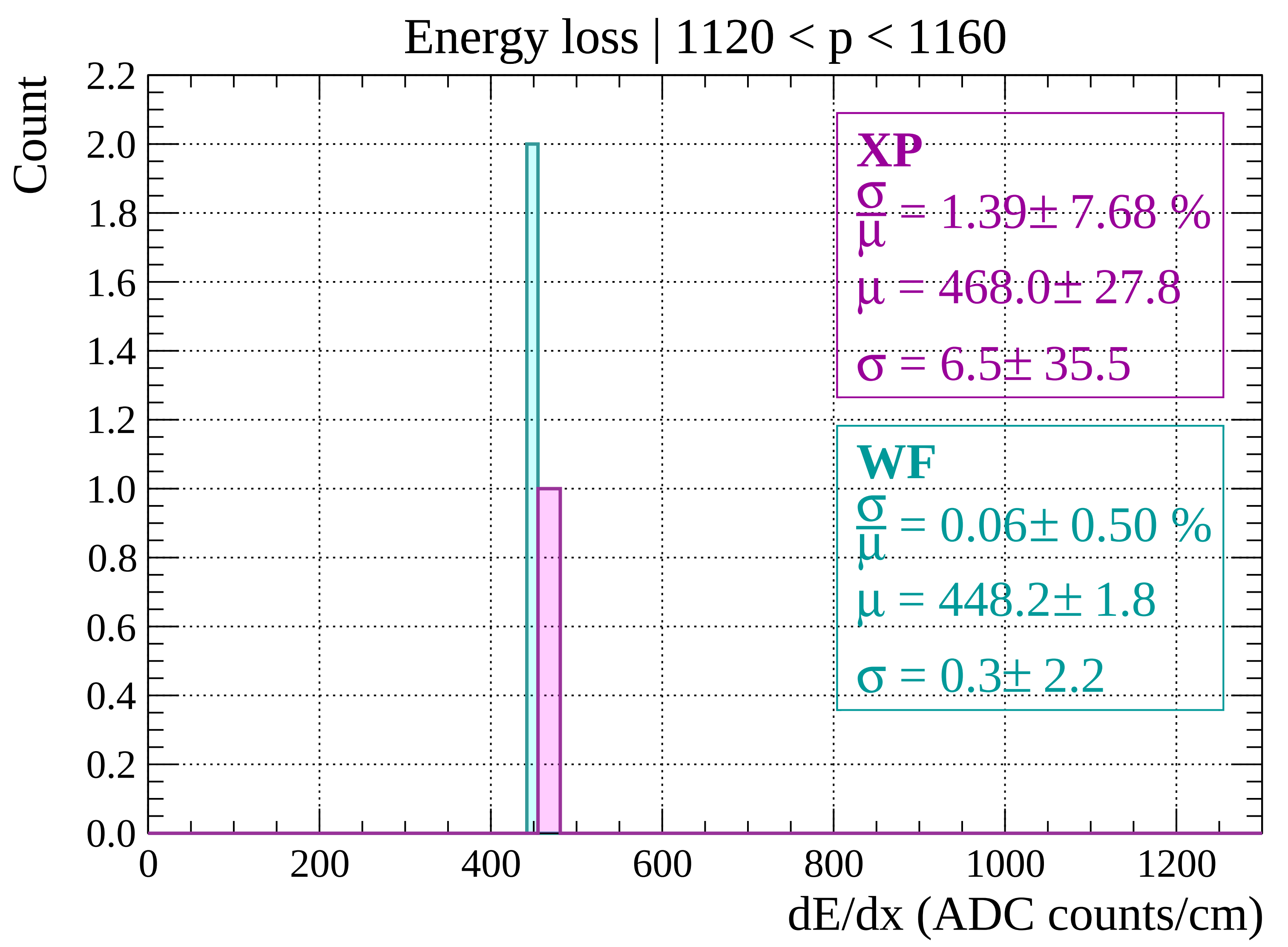
Energy loss | 1000 < p < 1040
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

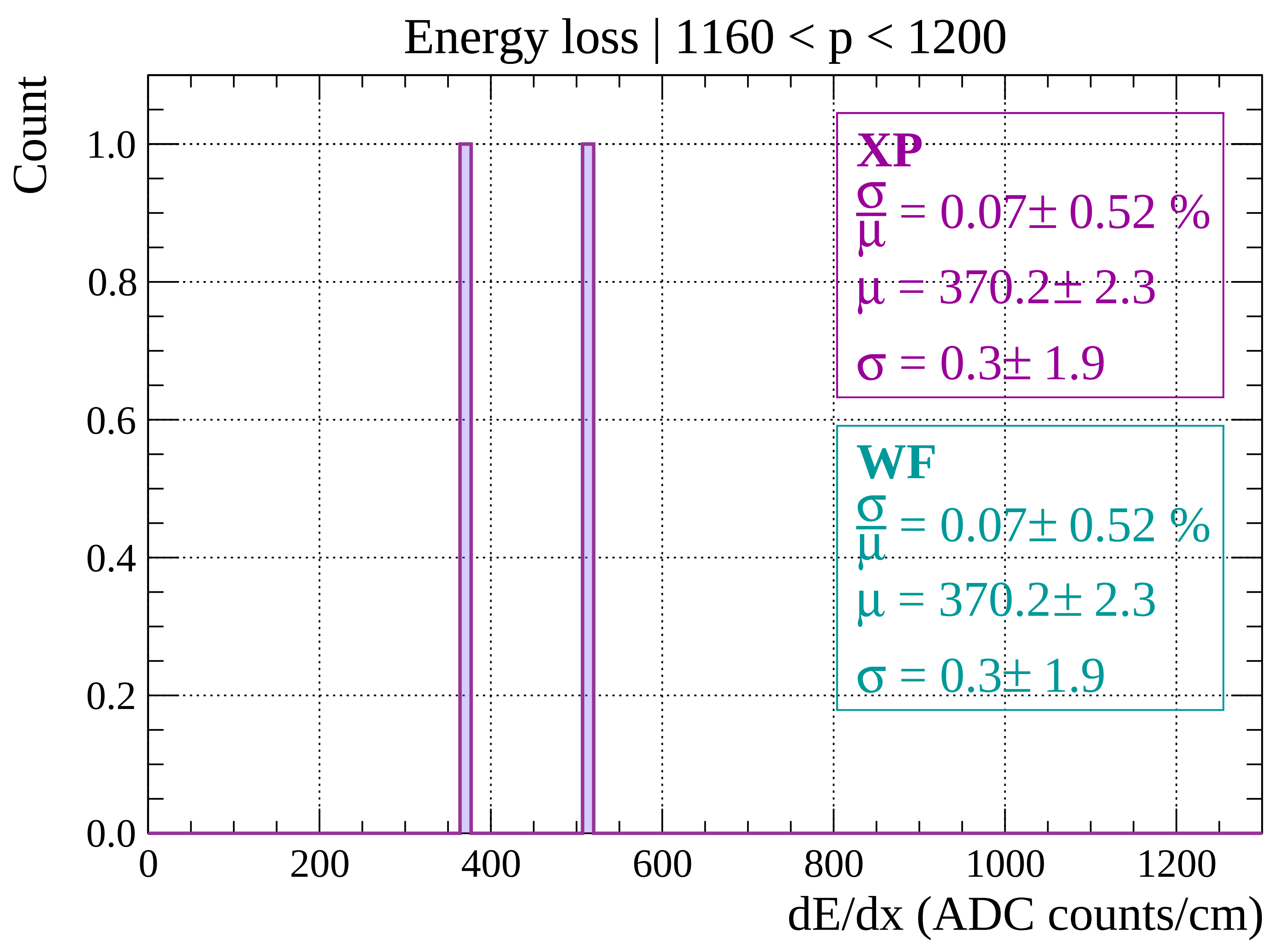


Energy loss | 1040 < p < 1080
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

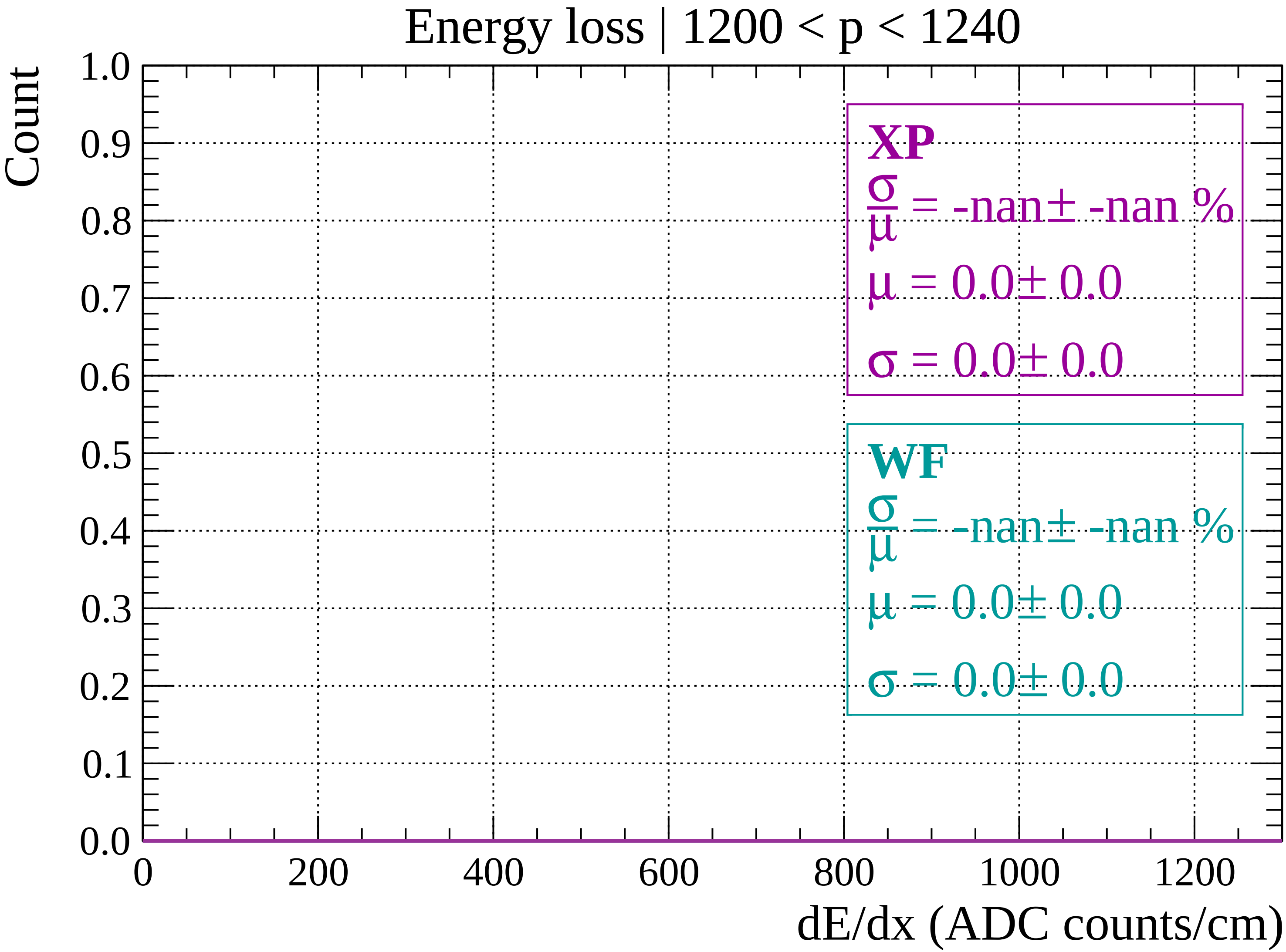


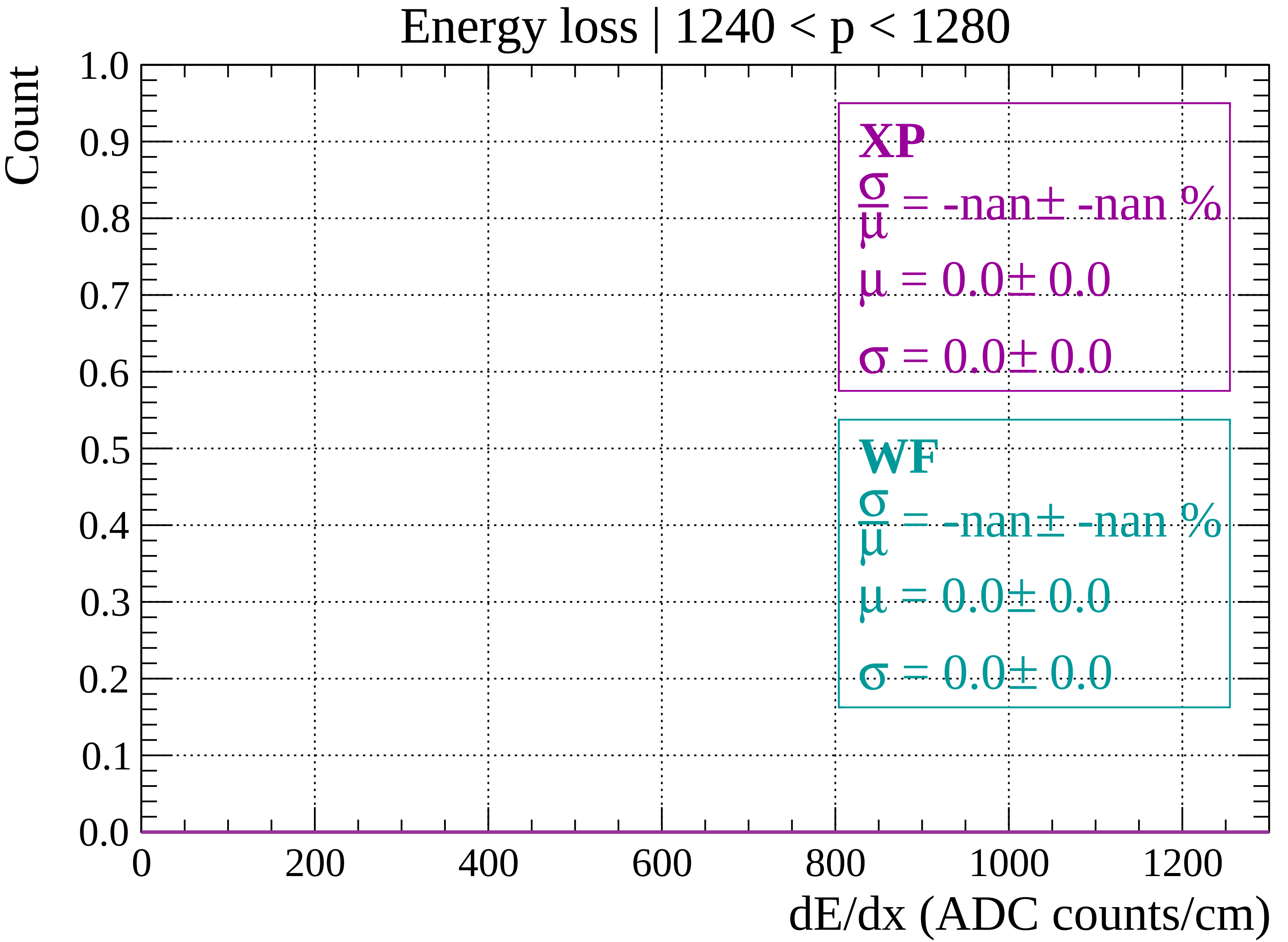
Energy loss | 1080 < p < 1120
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

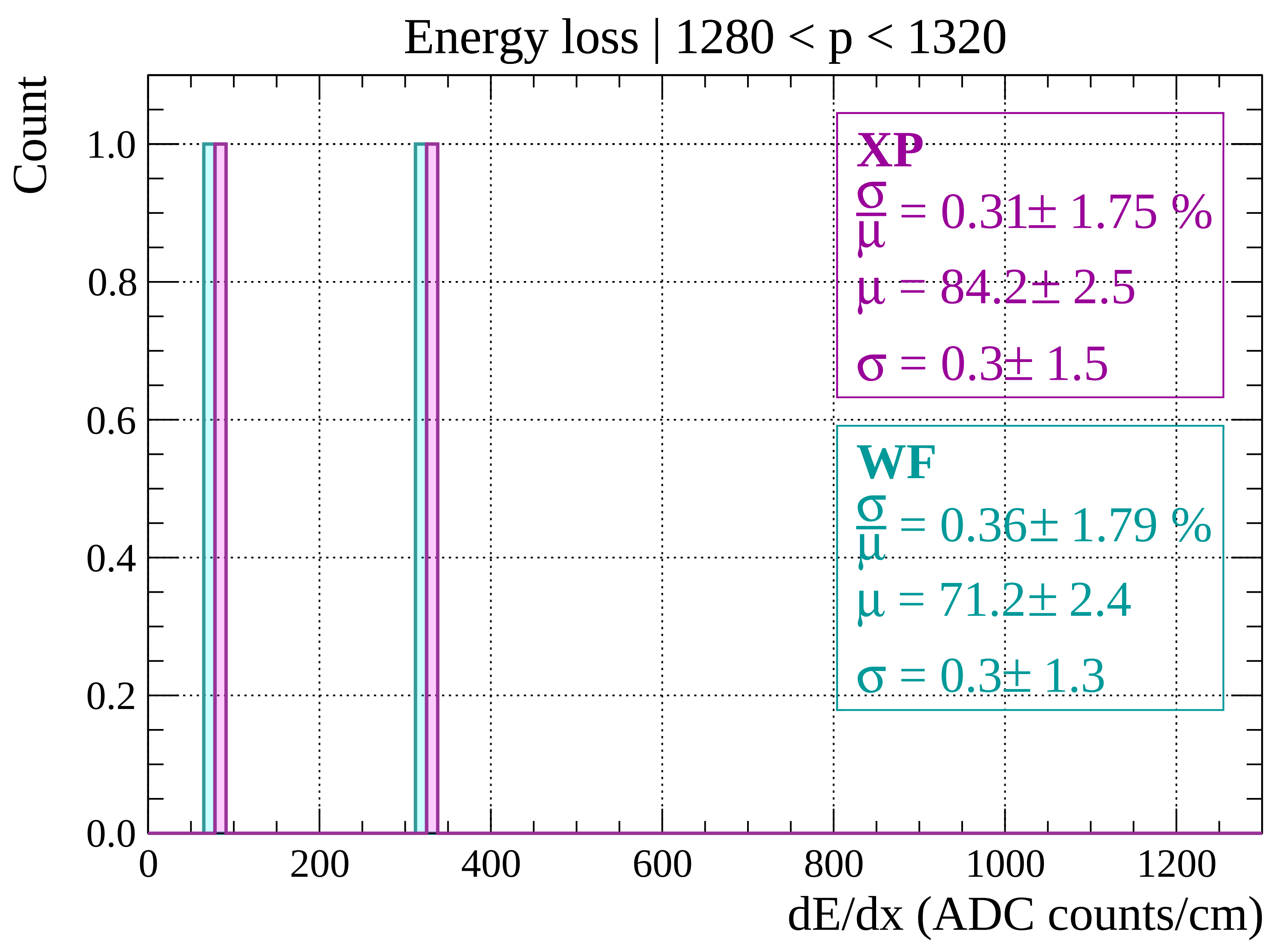

Energy loss | 1120 < p < 1160
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 1.39± 7.68 %
μ = 468.0± 27.8
σ = 6.5± 35.5
WF
σ/μ = 0.06± 0.50 %
μ = 448.2± 1.8
σ = 0.3± 2.2

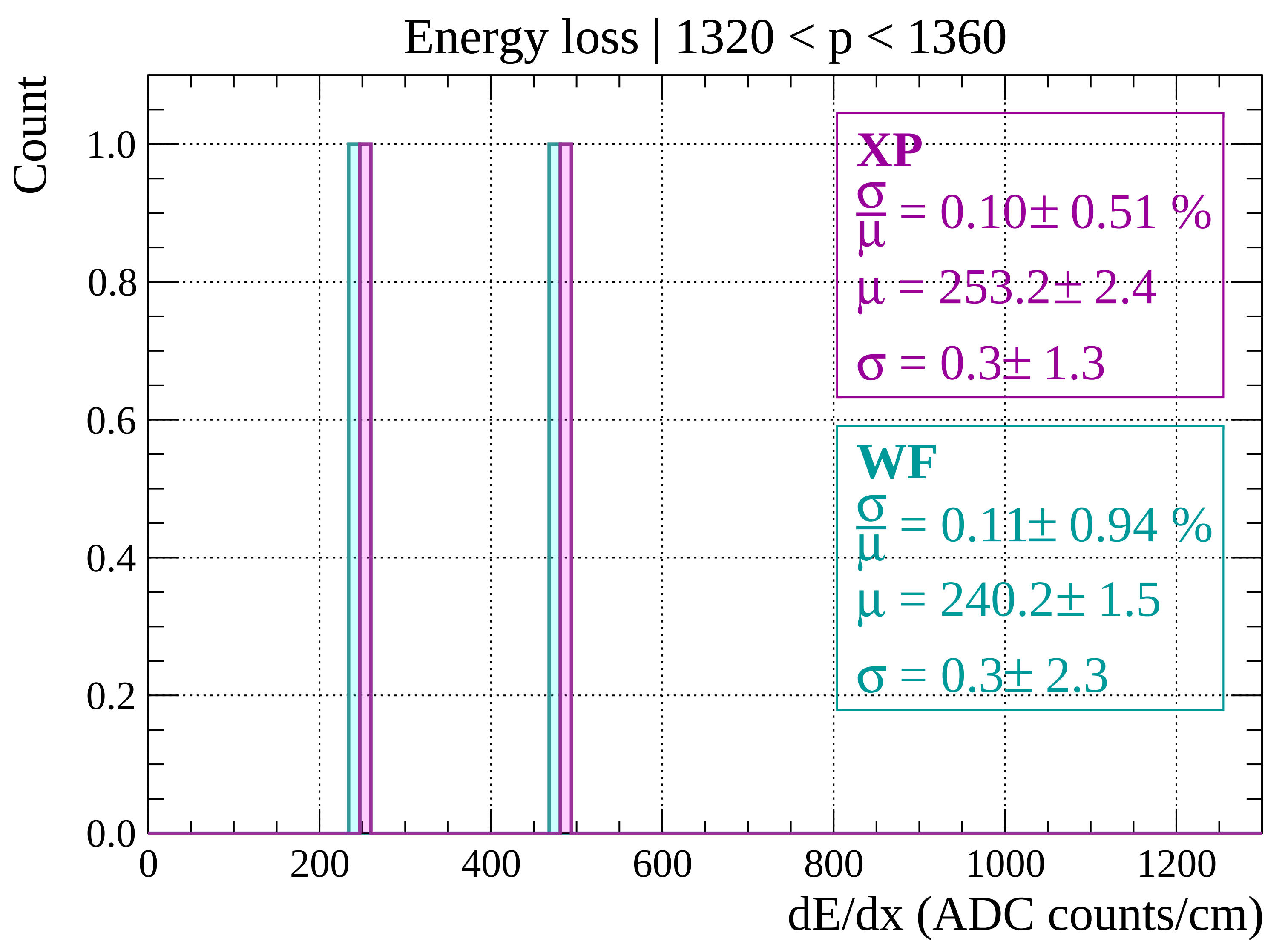


Energy loss | 1160 < p < 1200
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 0.07 \pm 0.52$ %
$\mu = 370.2 \pm 2.3$
$\sigma = 0.3 \pm 1.9$
WF
$\frac{\sigma}{\mu} = 0.07 \pm 0.52$ %
$\mu = 370.2 \pm 2.3$
$\sigma = 0.3 \pm 1.9$

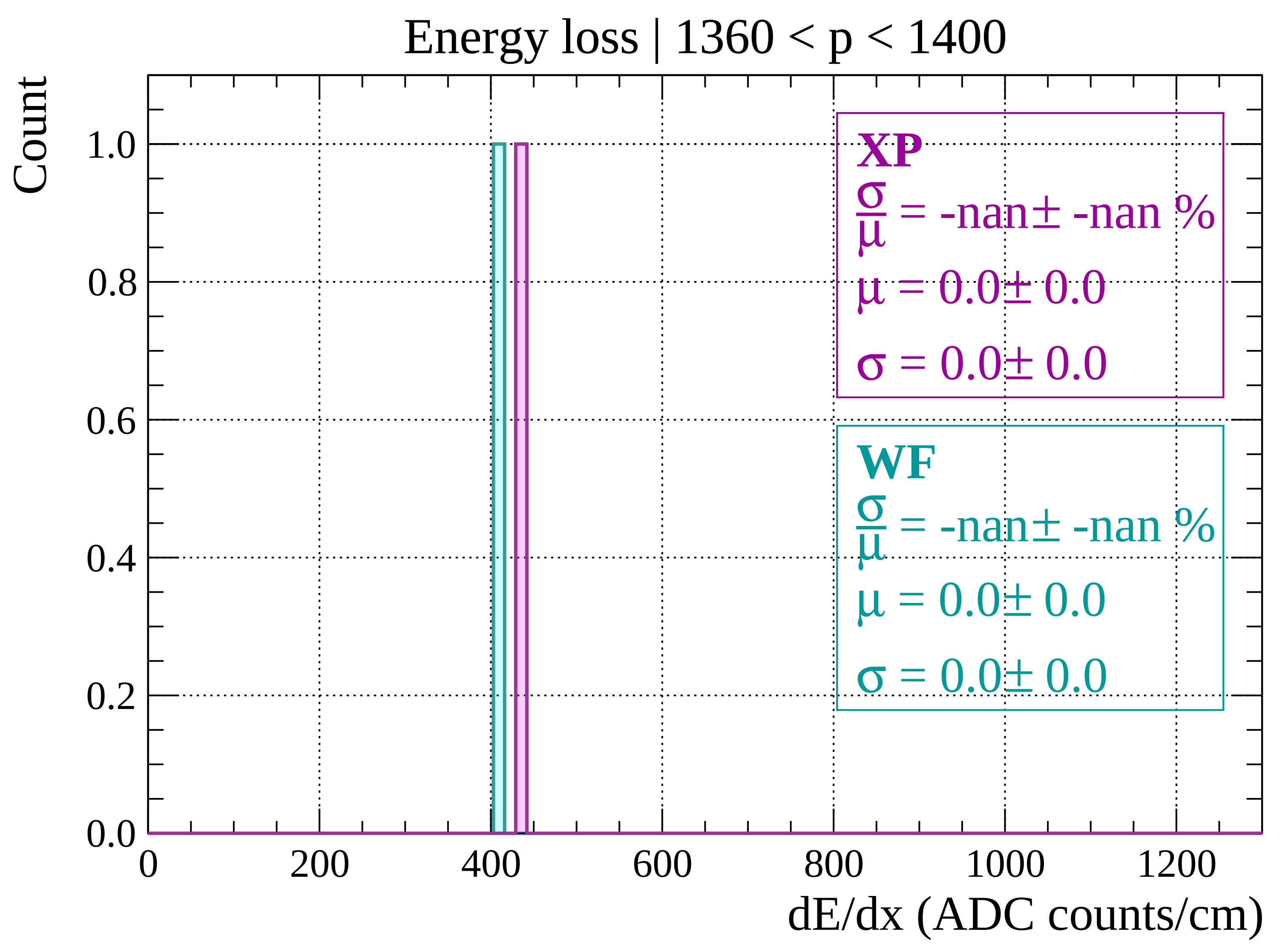

Energy loss | 1200 < p < 1240
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

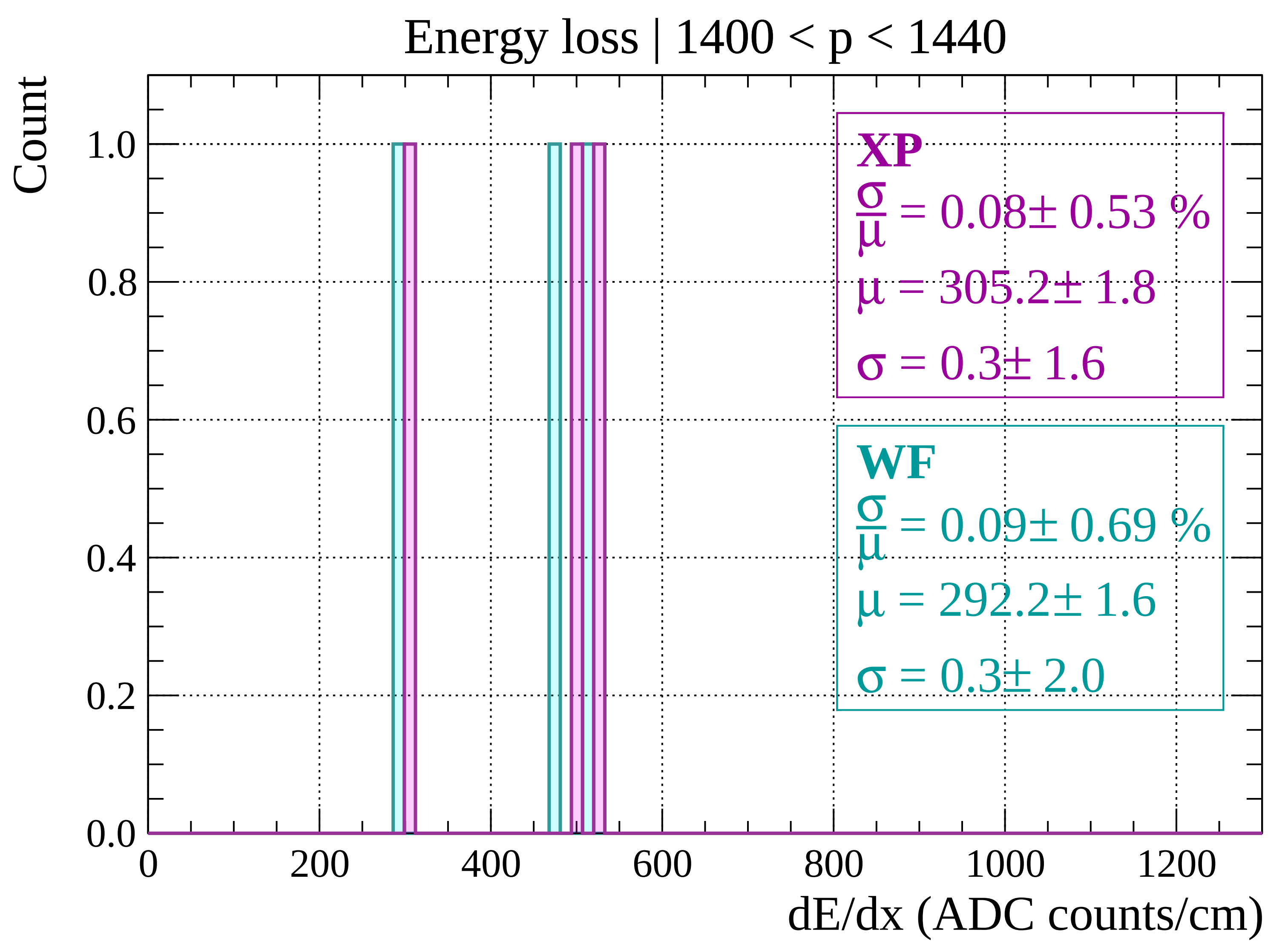

Energy loss | 1240 < p < 1280
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 1280 < p < 1320
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 0.31± 1.75 %
μ = 84.2± 2.5
σ = 0.3± 1.5
WF
$\frac{\sigma}{\mu}$ = 0.36± 1.79 %
μ = 71.2± 2.4
σ = 0.3± 1.3
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200

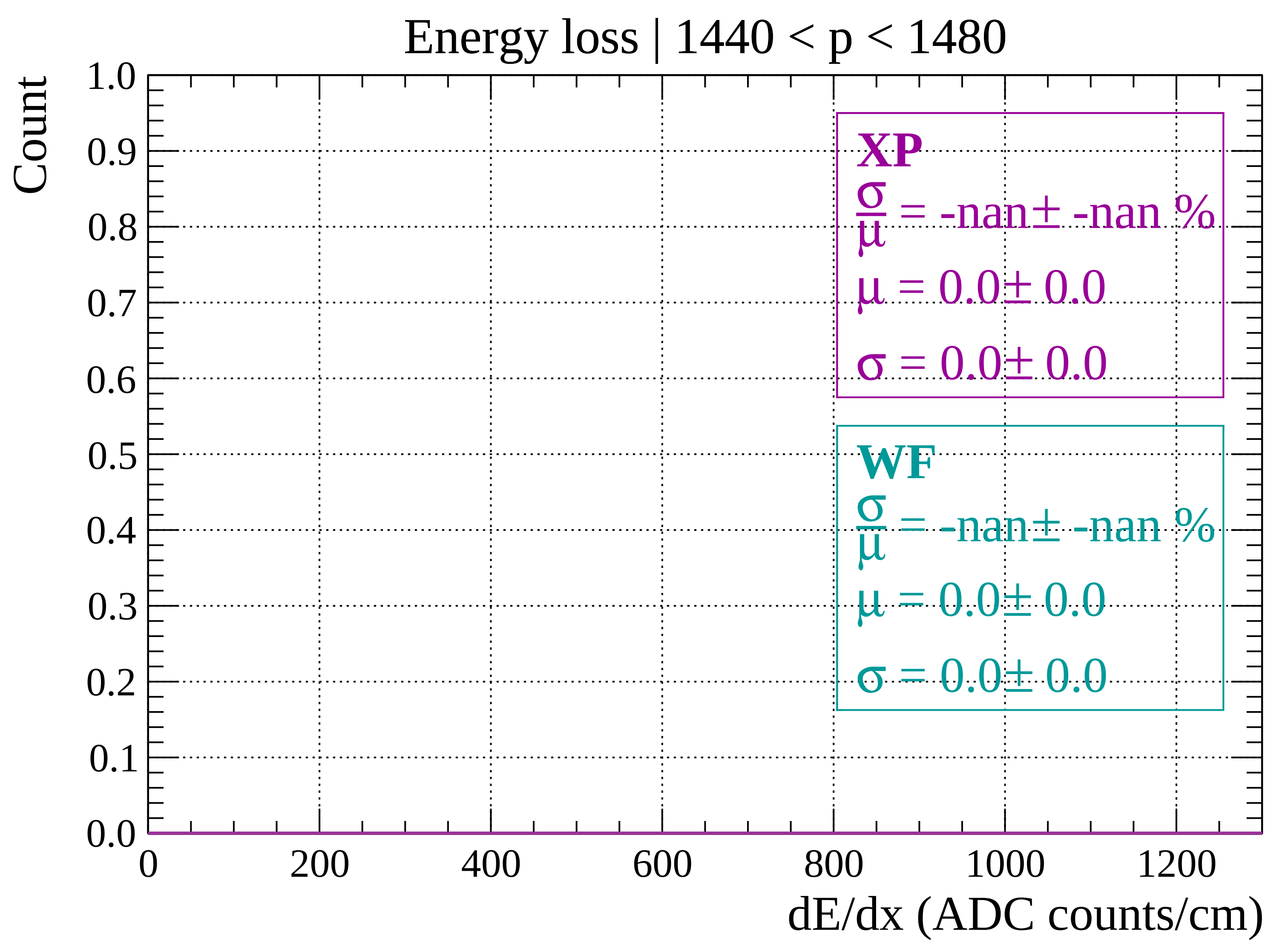


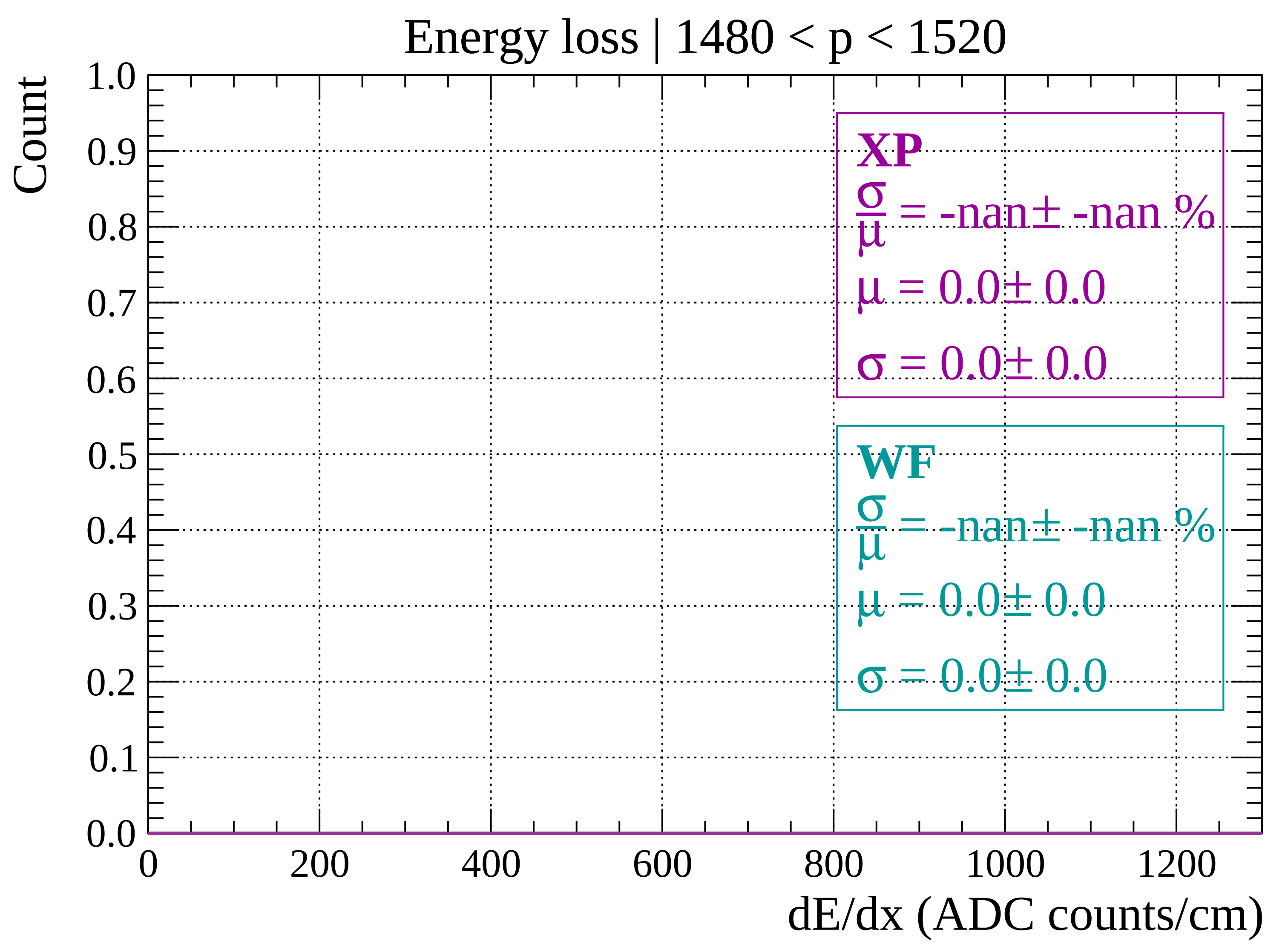
Energy loss | 1320 < p < 1360
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 0.10 \pm 0.51$ %
$\mu = 253.2 \pm 2.4$
$\sigma = 0.3 \pm 1.3$
WF
$\frac{\sigma}{\mu} = 0.11 \pm 0.94$ %
$\mu = 240.2 \pm 1.5$
$\sigma = 0.3 \pm 2.3$


Energy loss | 1360 < p < 1400
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

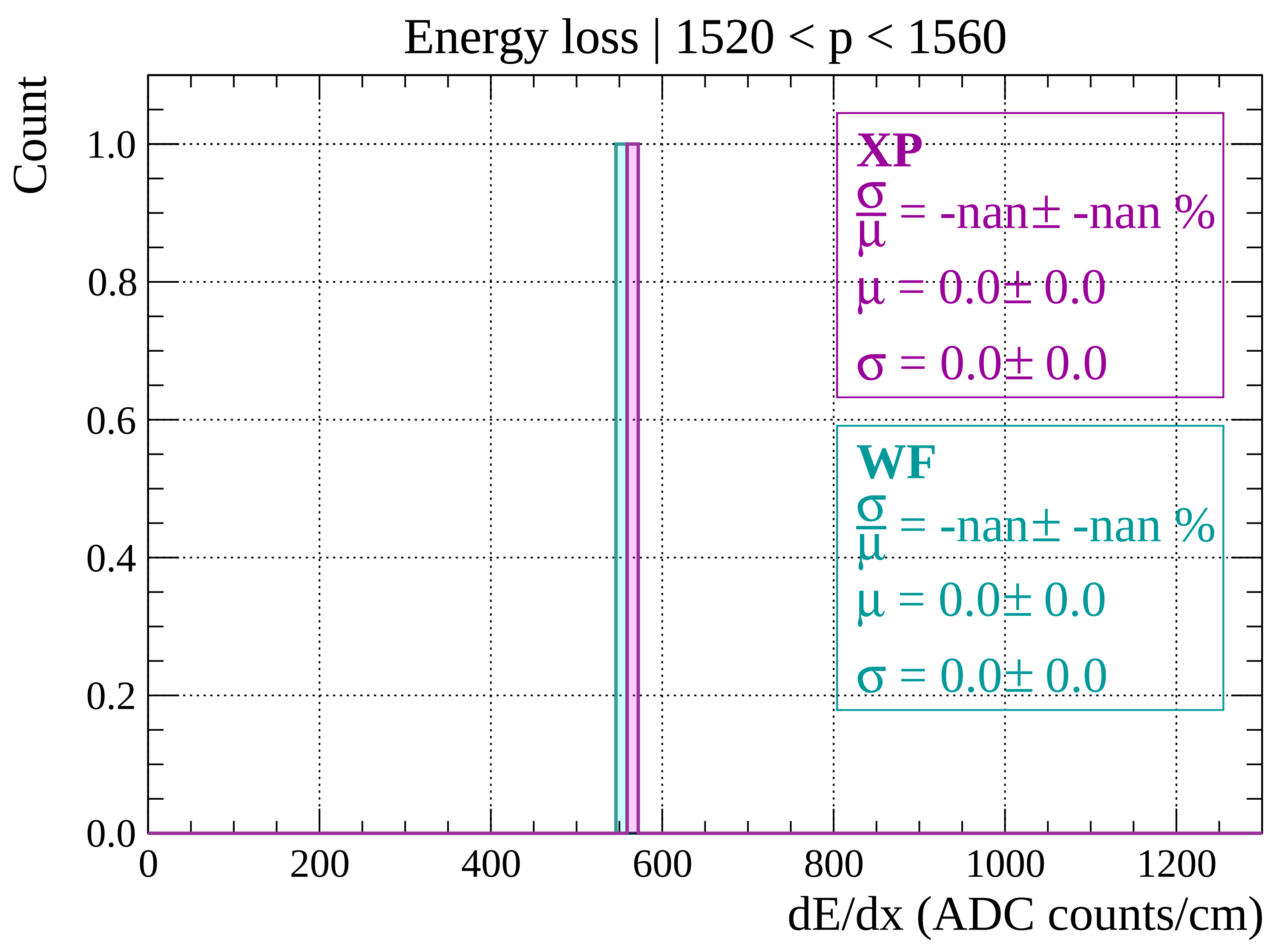


Energy loss | 1400 < p < 1440
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 0.08± 0.53 %
μ = 305.2± 1.8
σ = 0.3± 1.6
WF
$\frac{\sigma}{\mu}$ = 0.09± 0.69 %
μ = 292.2± 1.6
σ = 0.3± 2.0

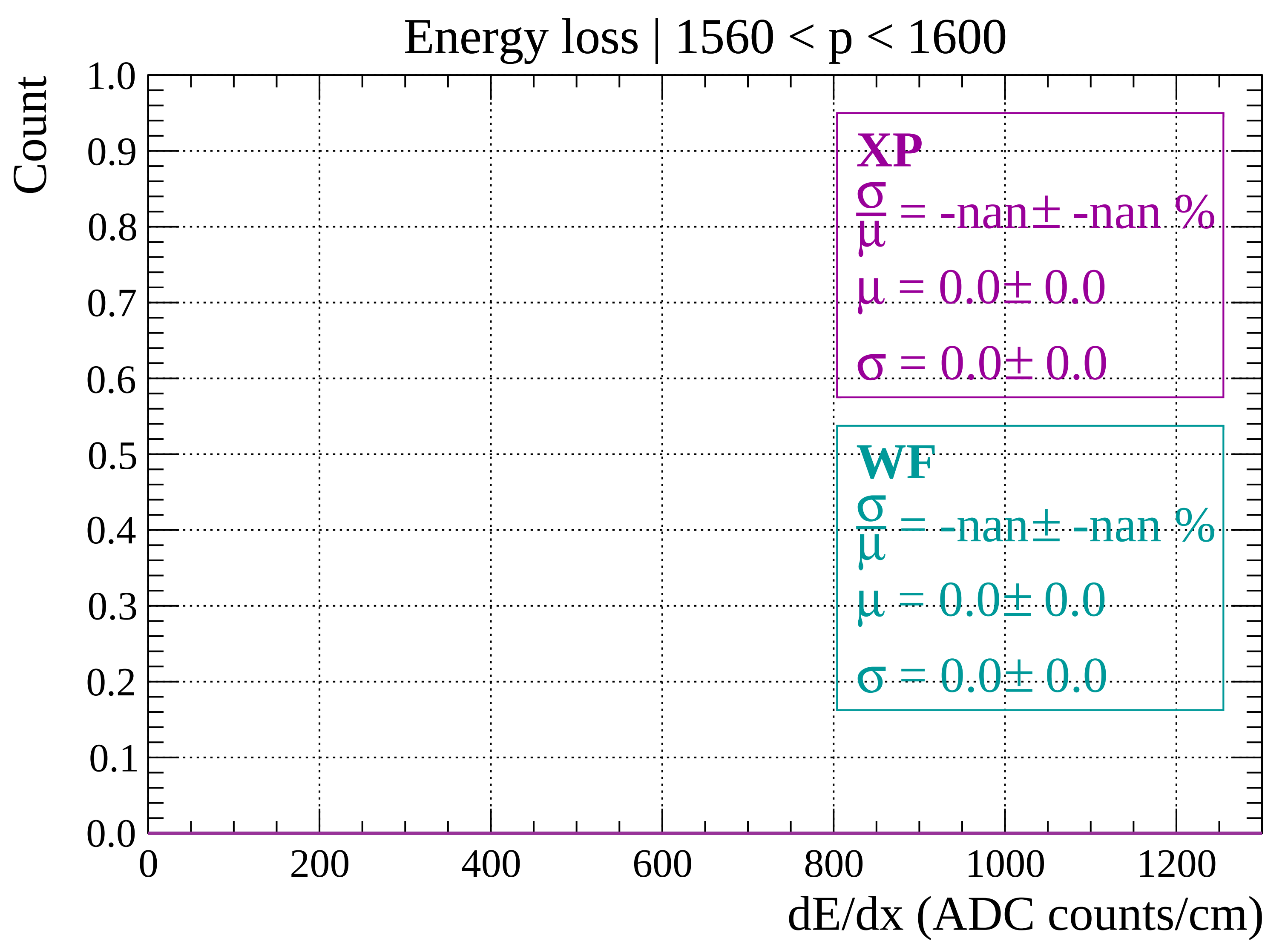

Energy loss | 1440 < p < 1480
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

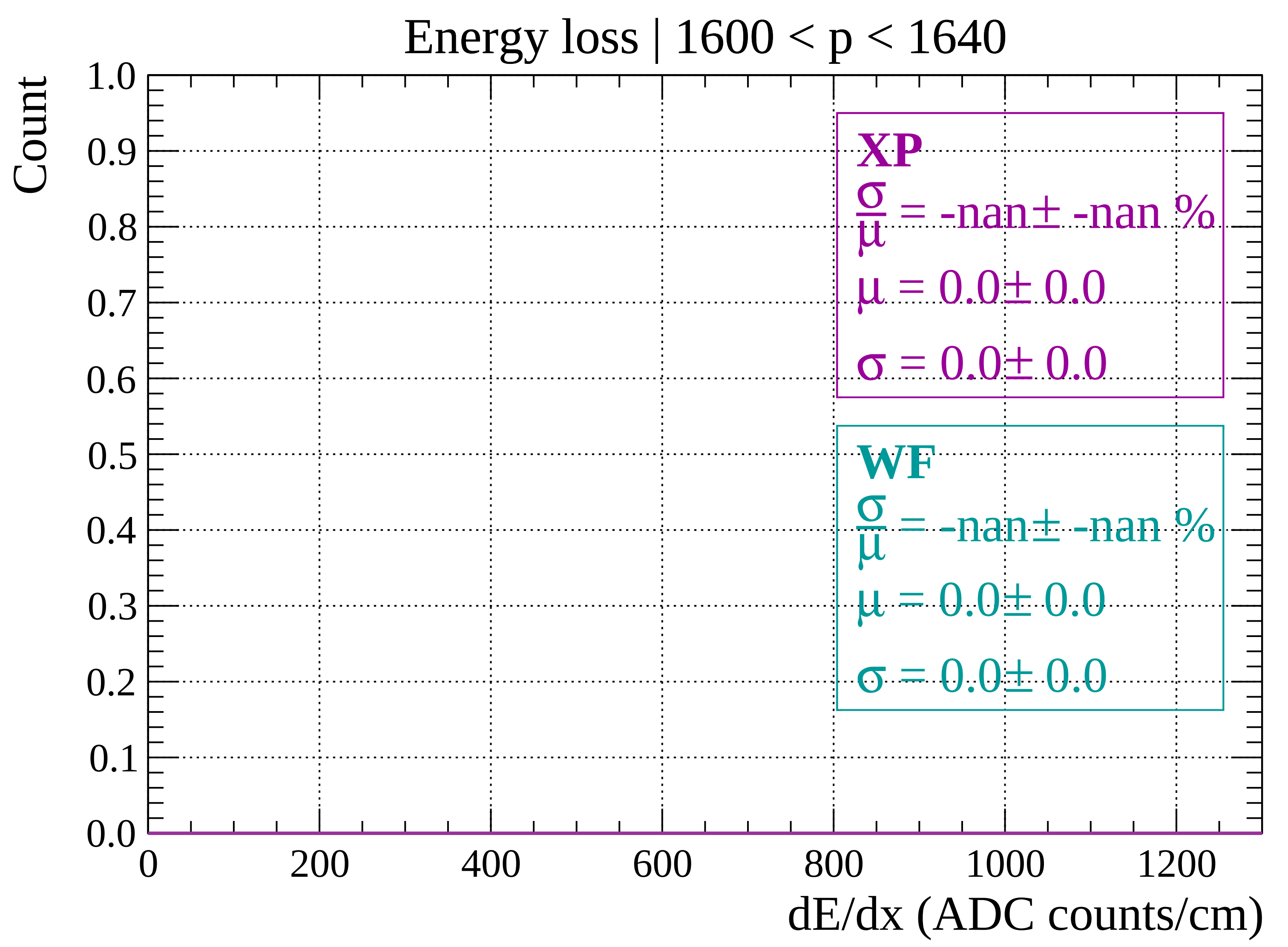


Energy loss | 1480 < p < 1520
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

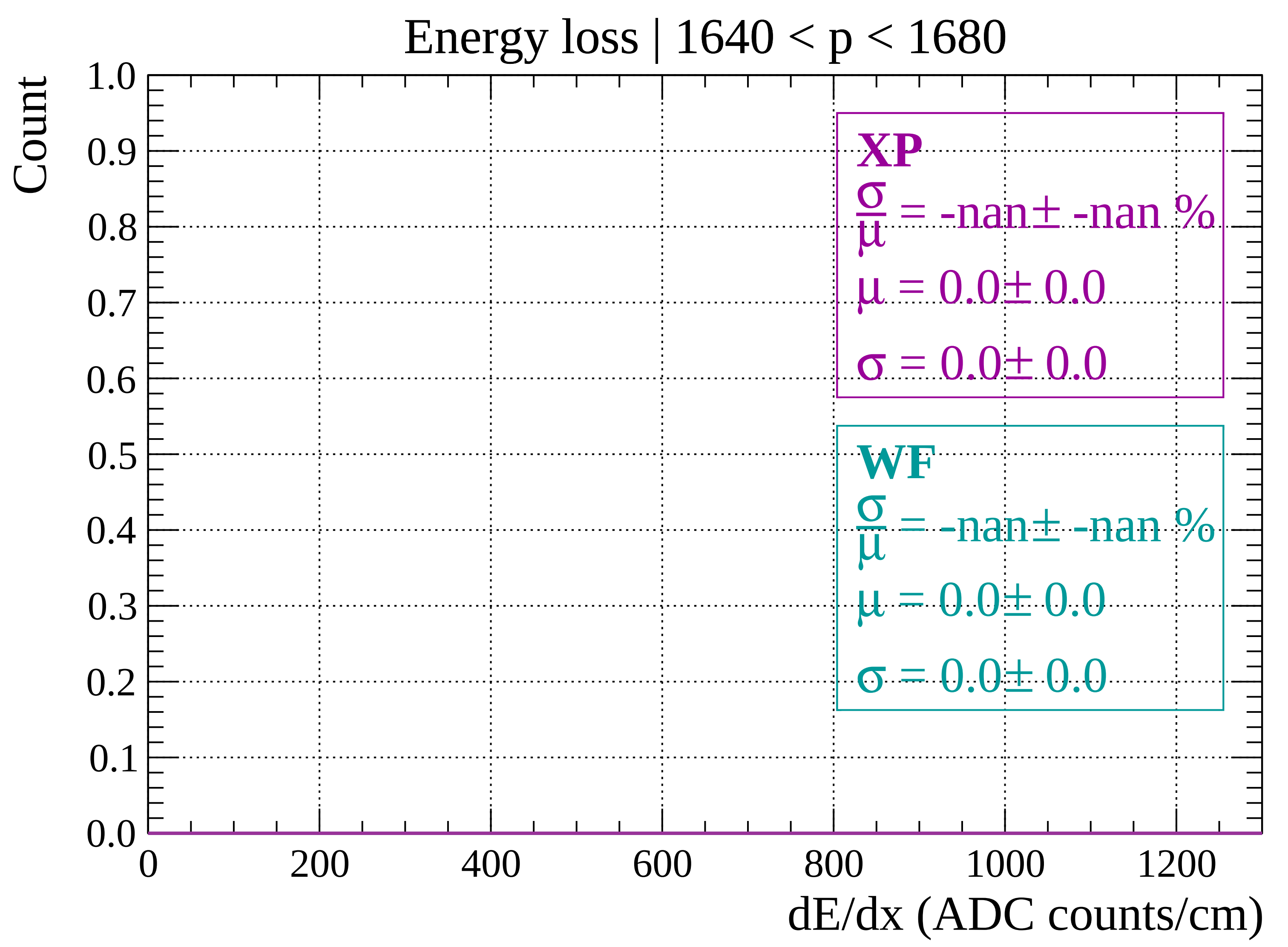


Energy loss | 1520 < p < 1560
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

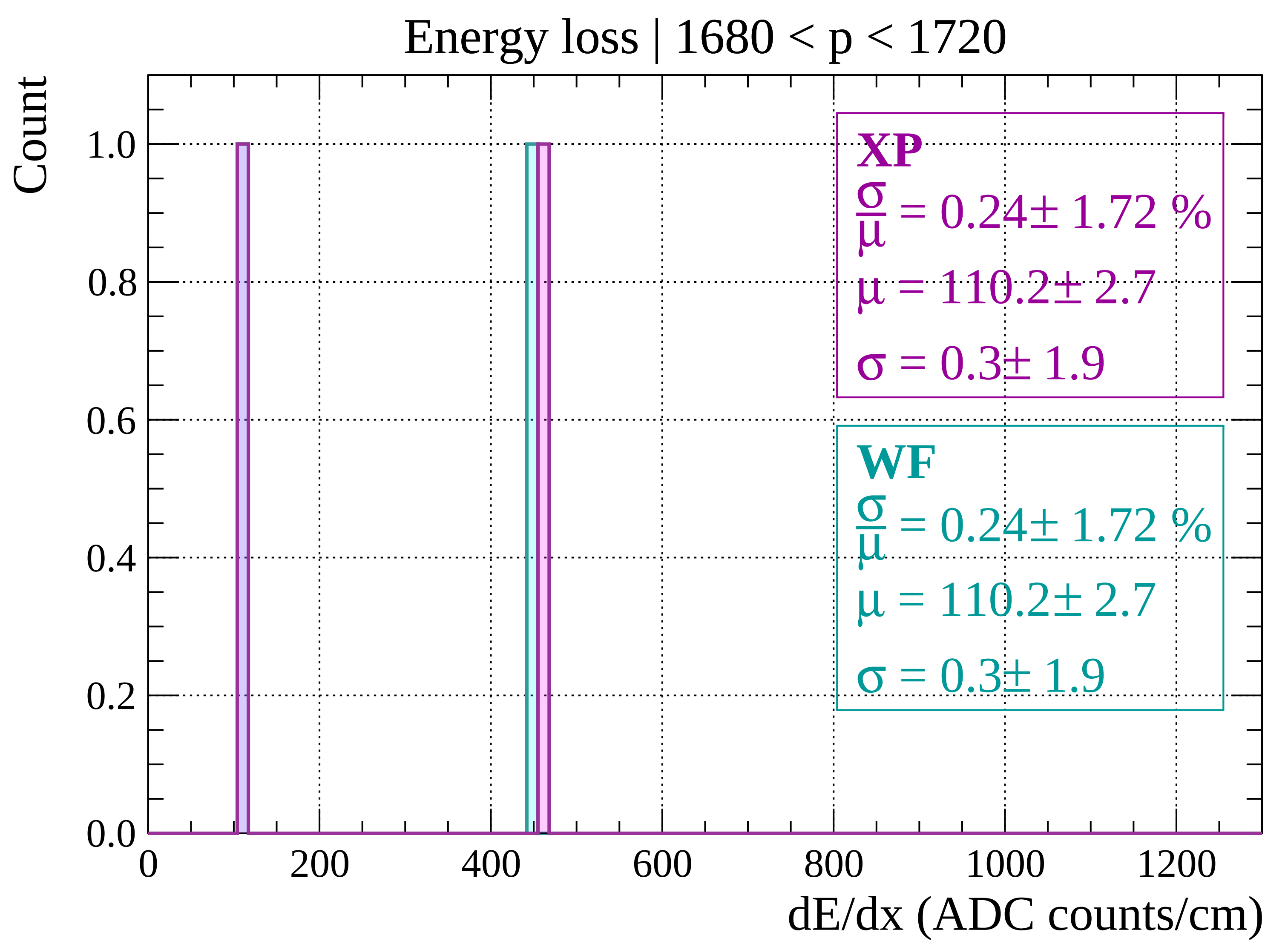

Energy loss | 1560 < p < 1600
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

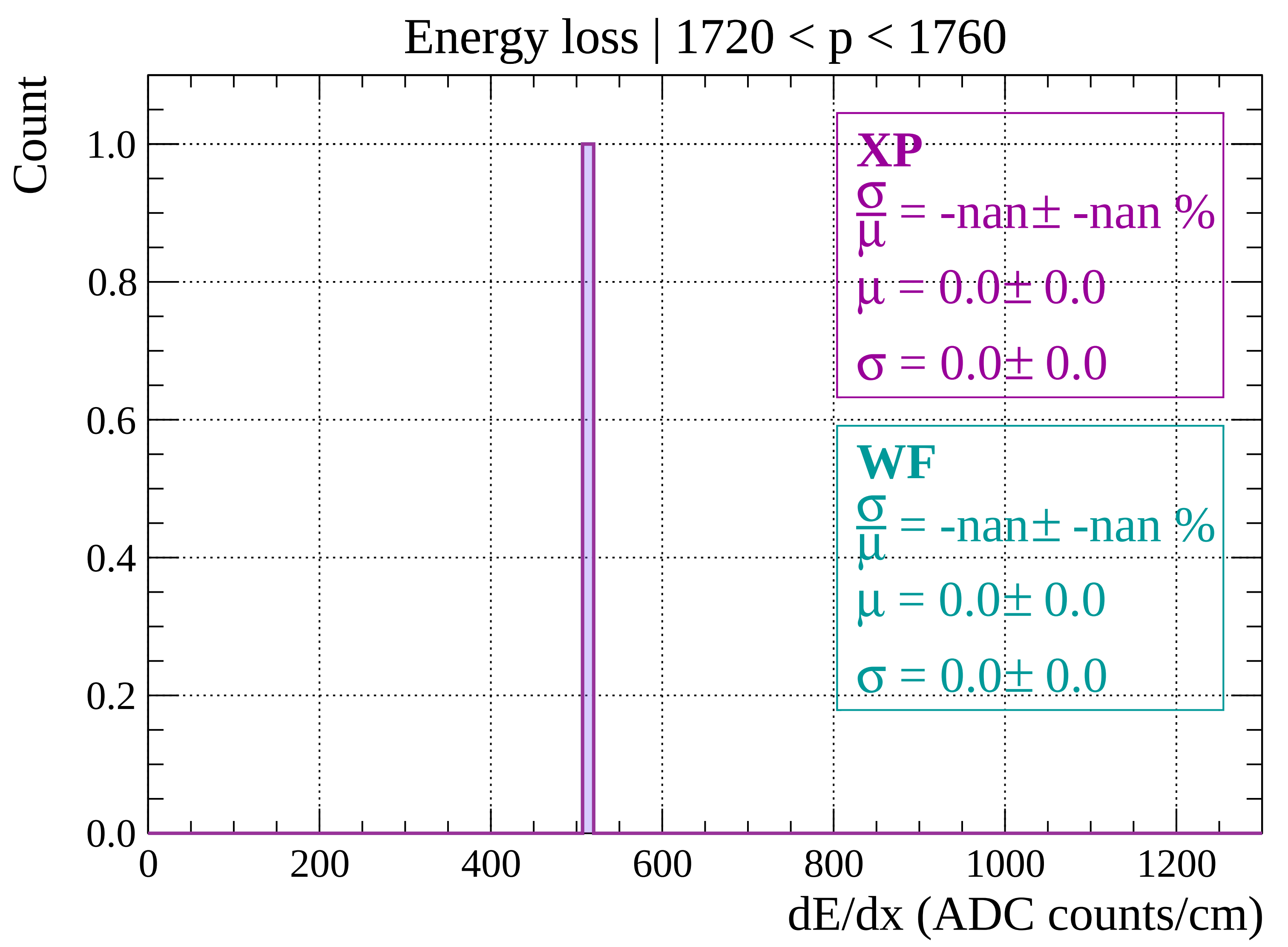


Energy loss | 1600 < p < 1640
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

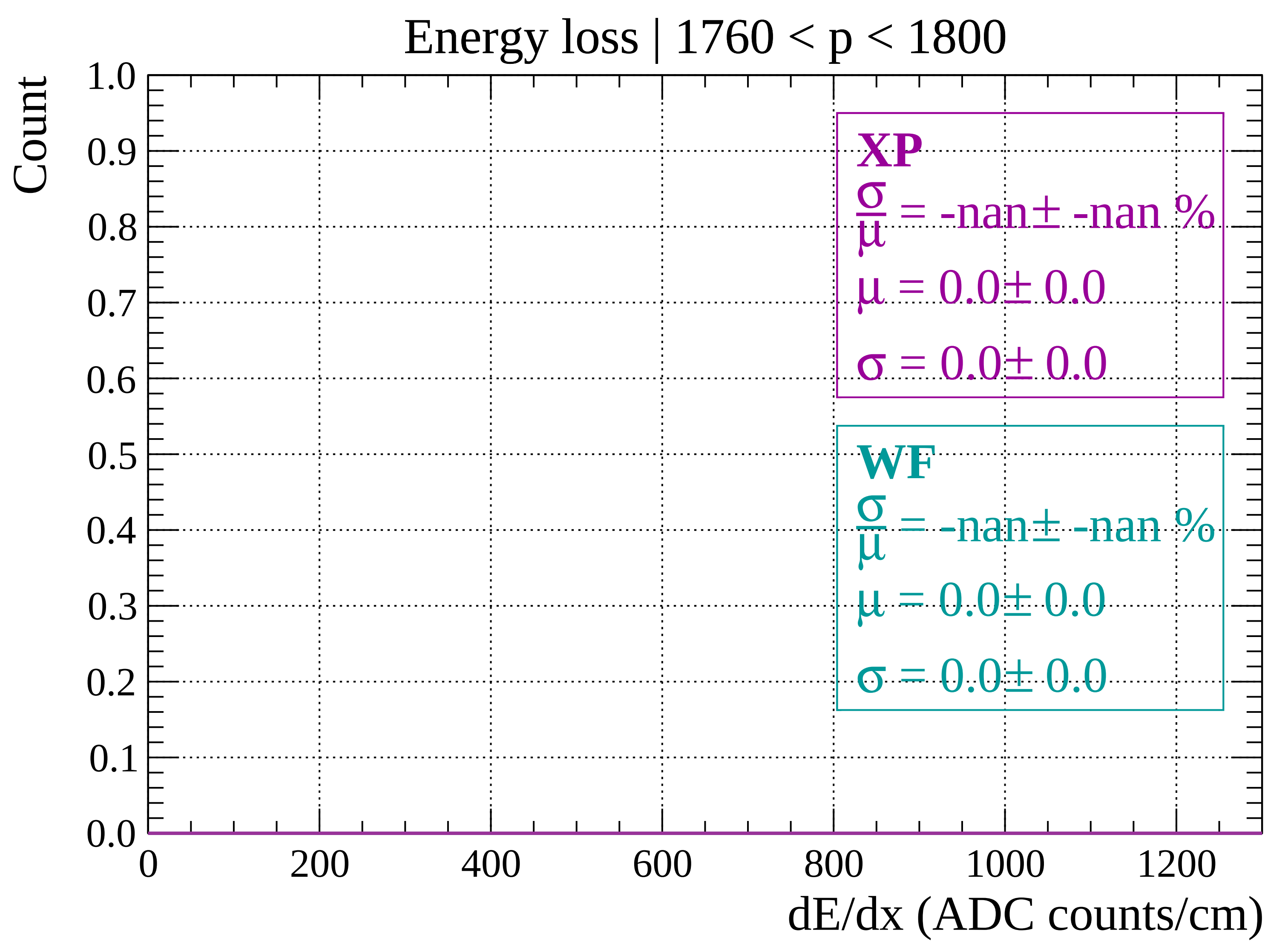


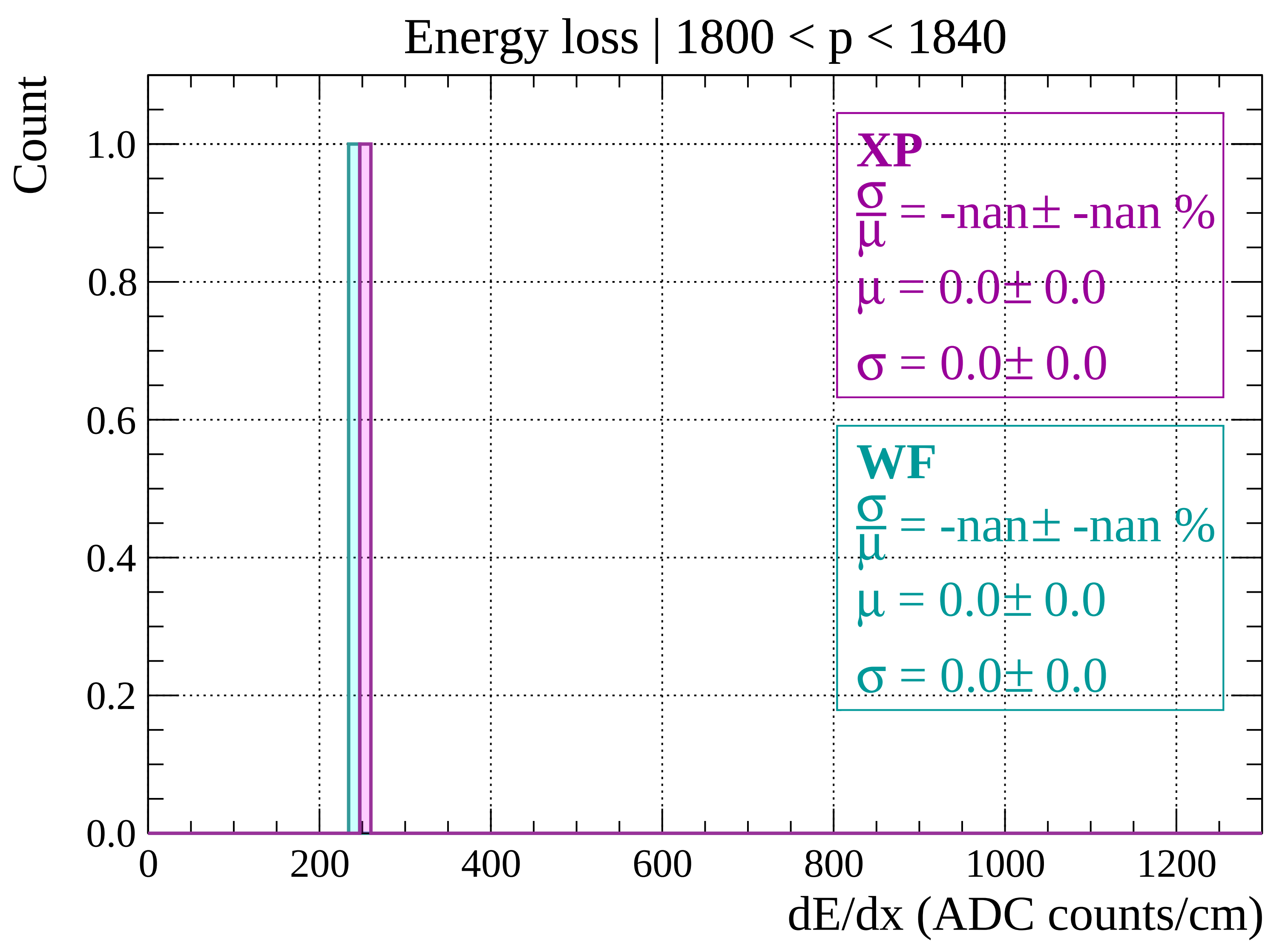
Energy loss | 1640 < p < 1680
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

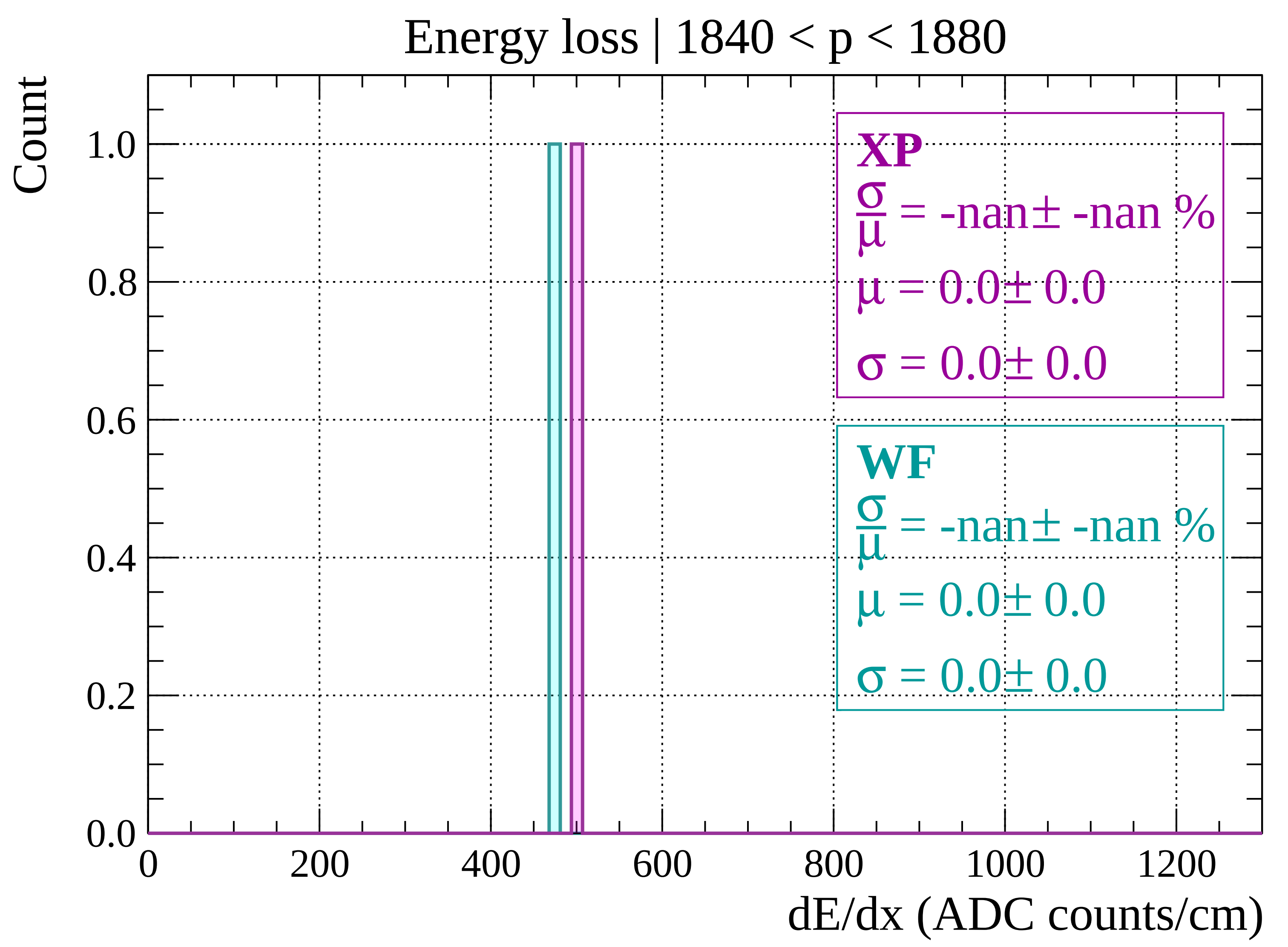

Energy loss | 1680 < p < 1720
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.24± 1.72 %
μ = 110.2± 2.7
σ = 0.3± 1.9
WF
σ/μ = 0.24± 1.72 %
μ = 110.2± 2.7
σ = 0.3± 1.9

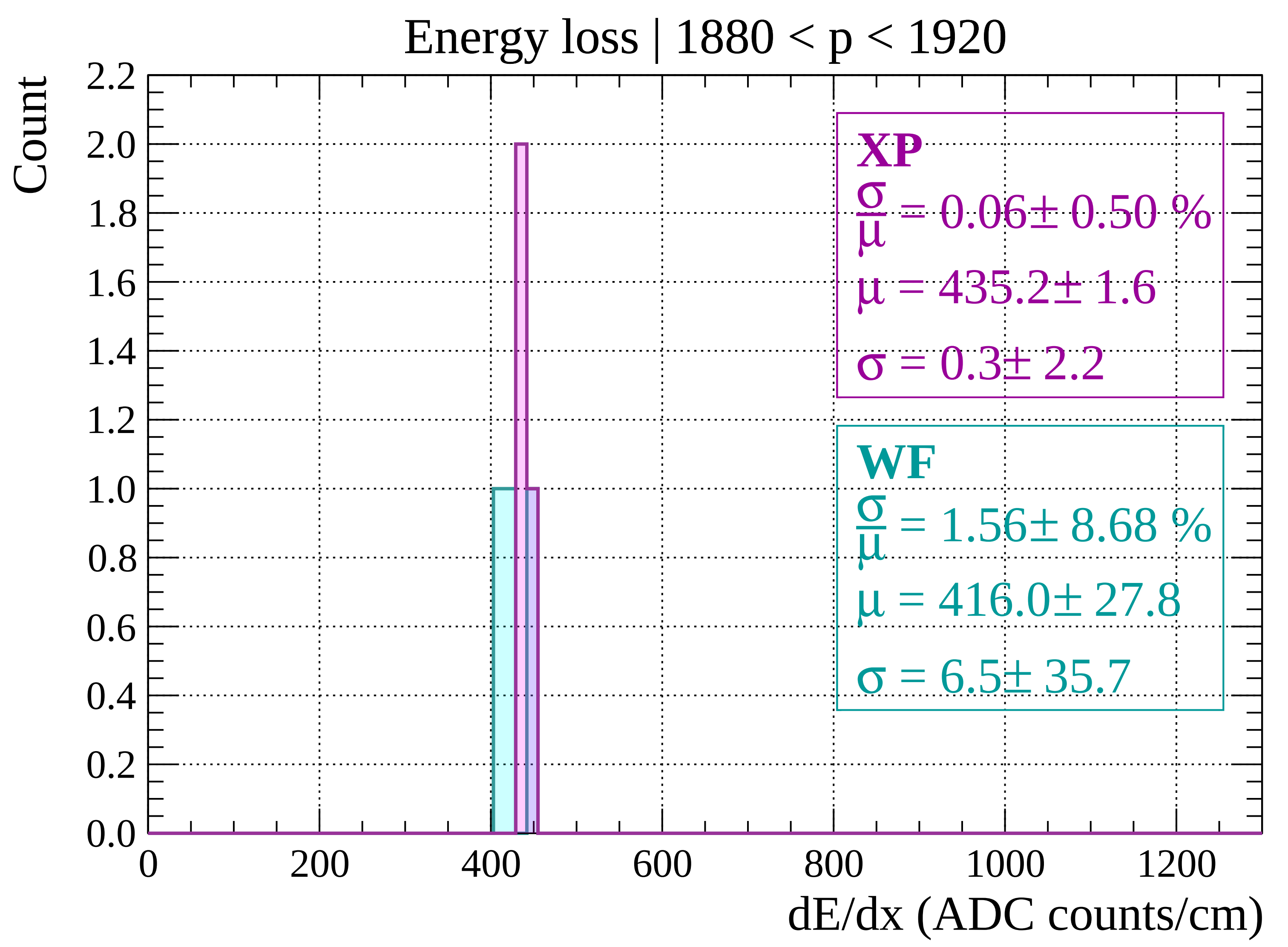


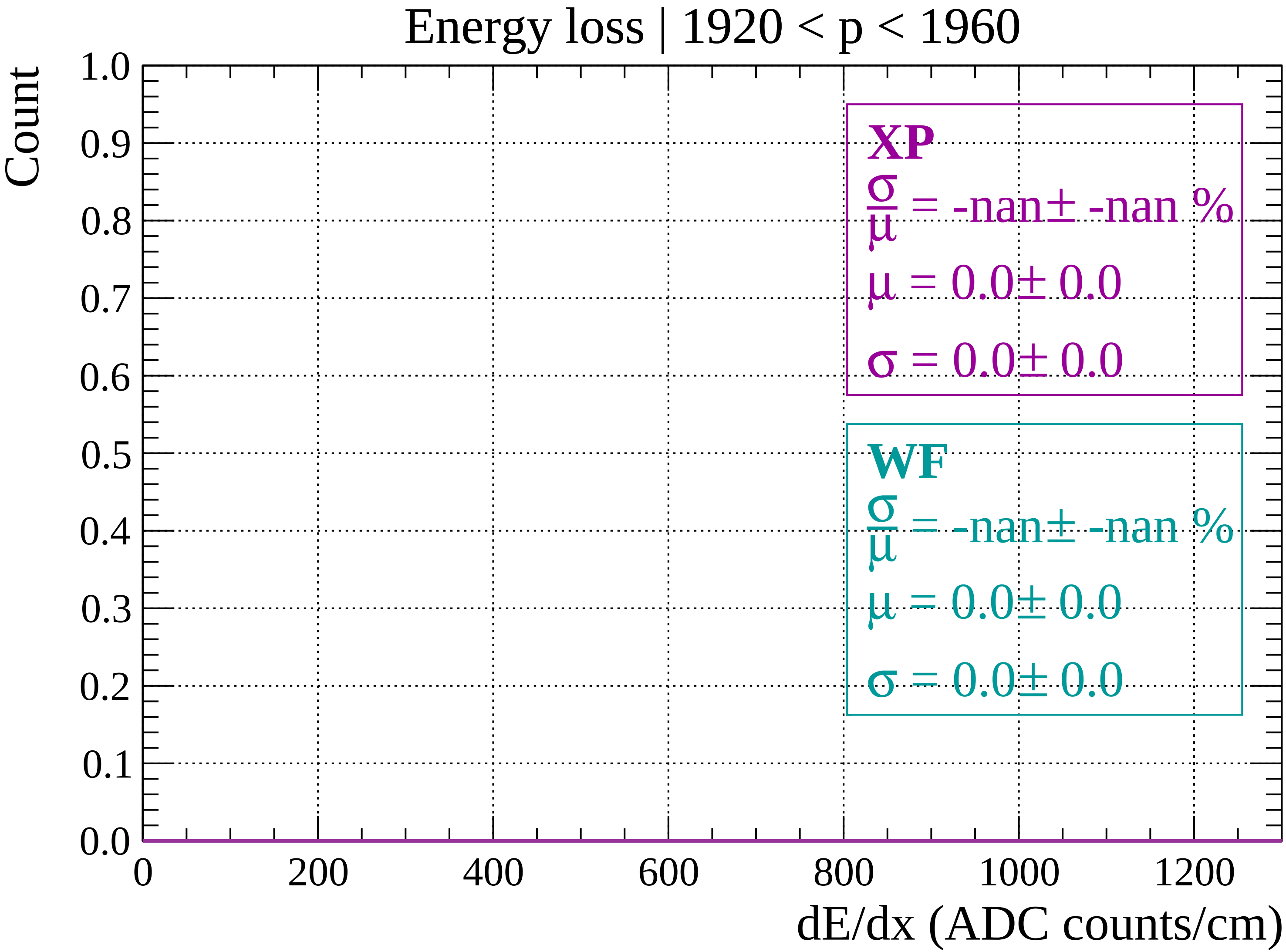
Energy loss | 1720 < p < 1760
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 1760 < p < 1800
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

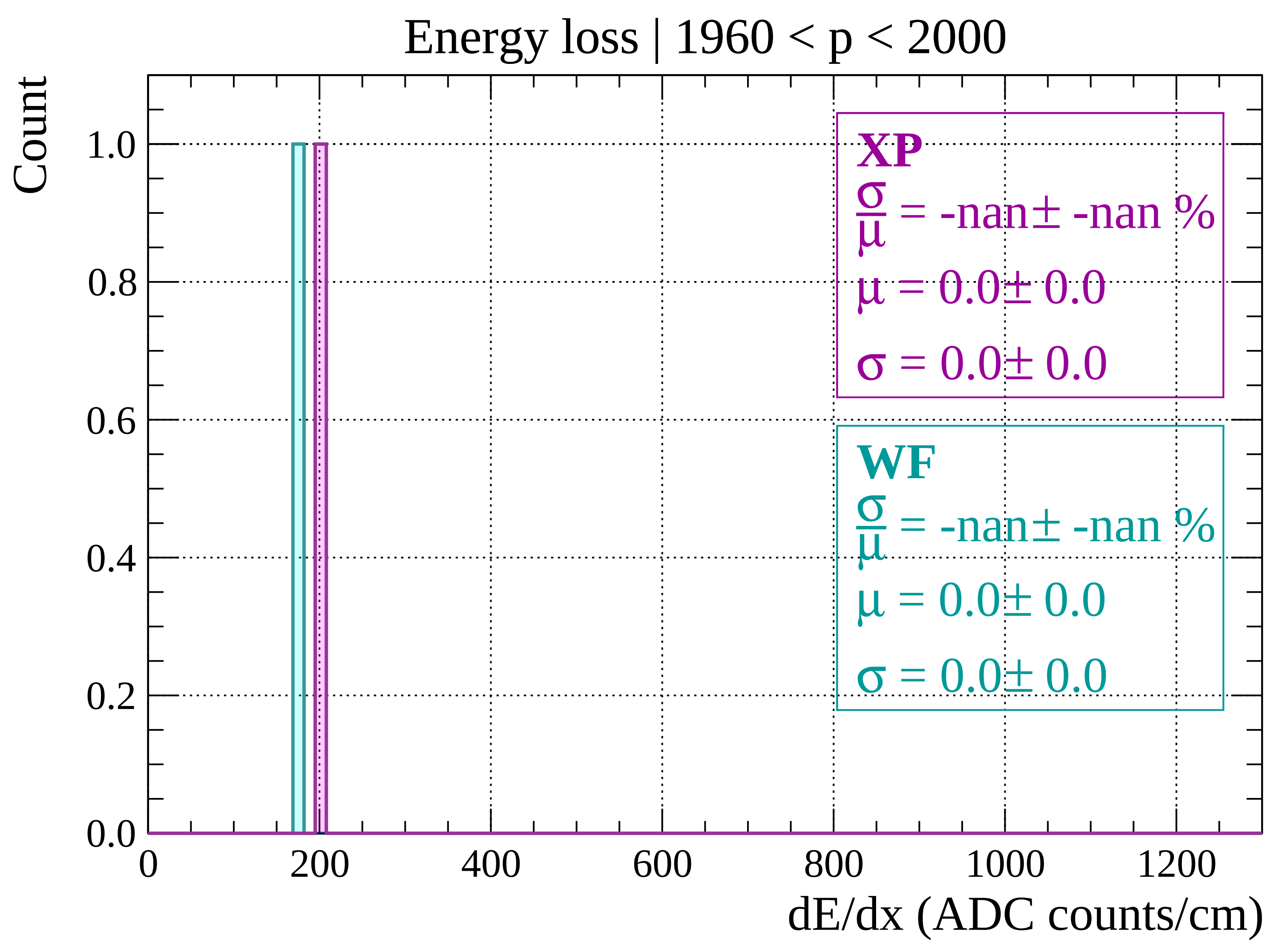

Energy loss | 1800 < p < 1840
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 1840 < p < 1880
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

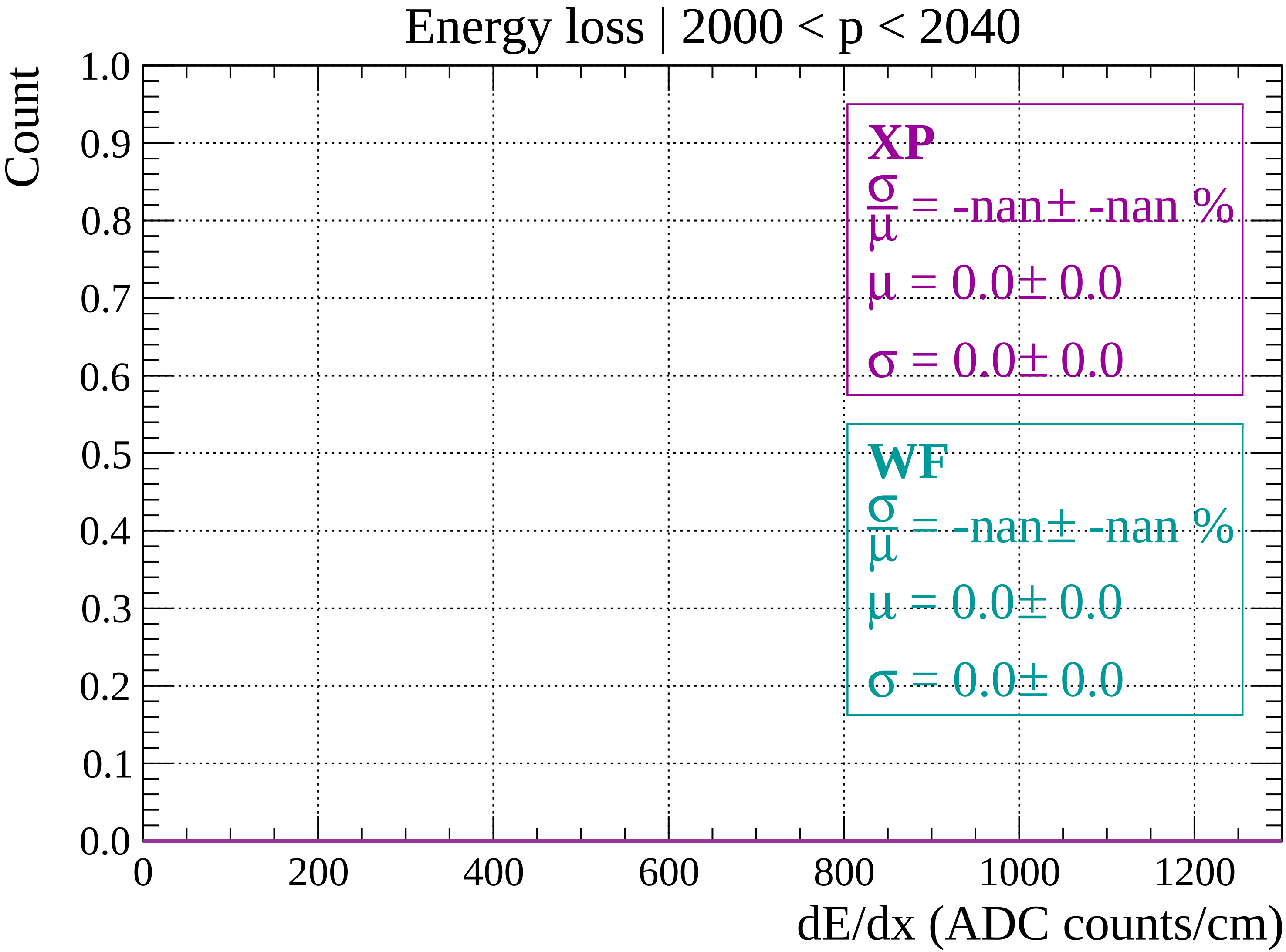

Energy loss | 1880 < p < 1920
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 0.06± 0.50 %
μ = 435.2± 1.6
σ = 0.3± 2.2
WF
σ/μ = 1.56± 8.68 %
μ = 416.0± 27.8
σ = 6.5± 35.7

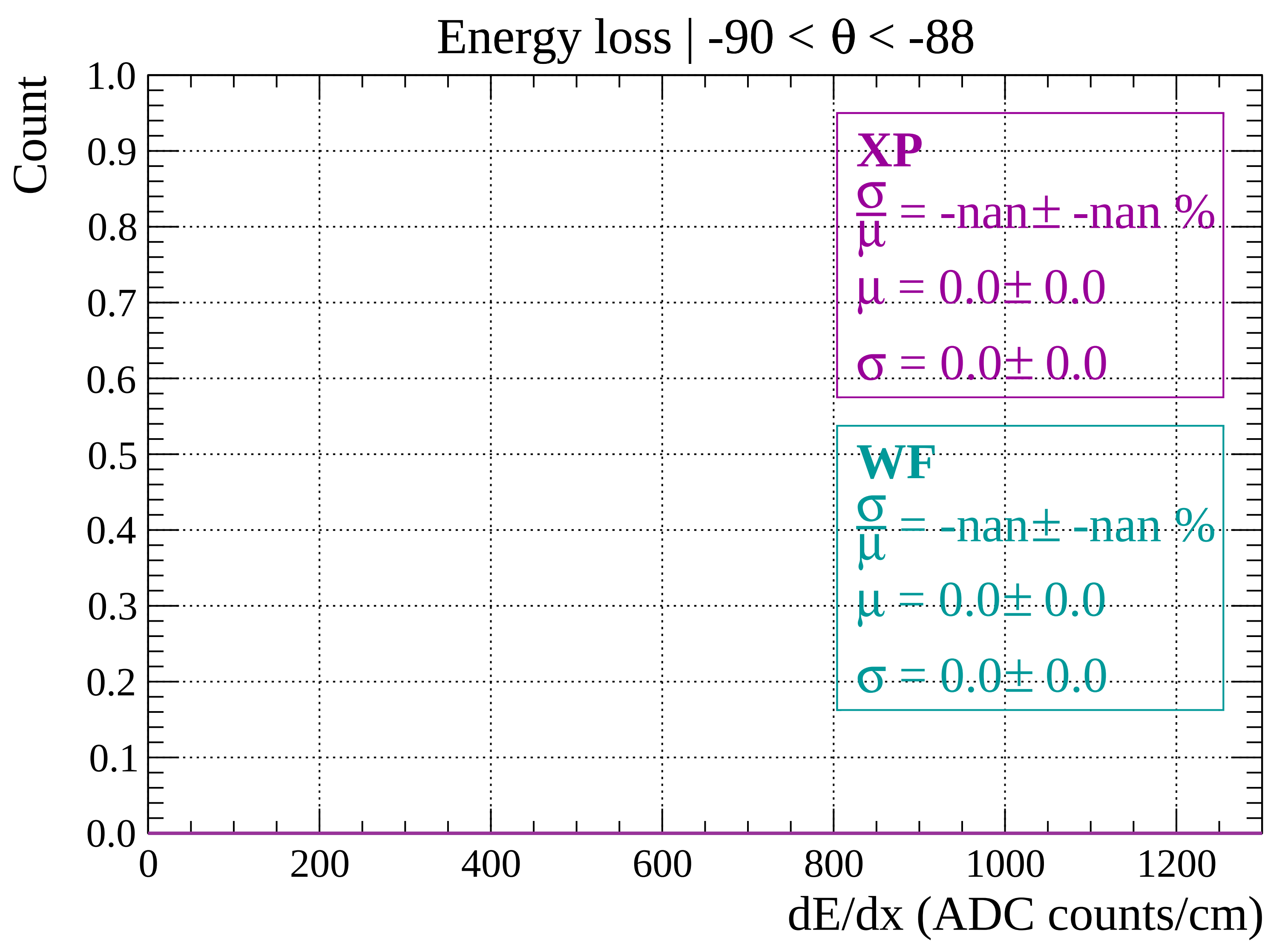


Energy loss | 1920 < p < 1960
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

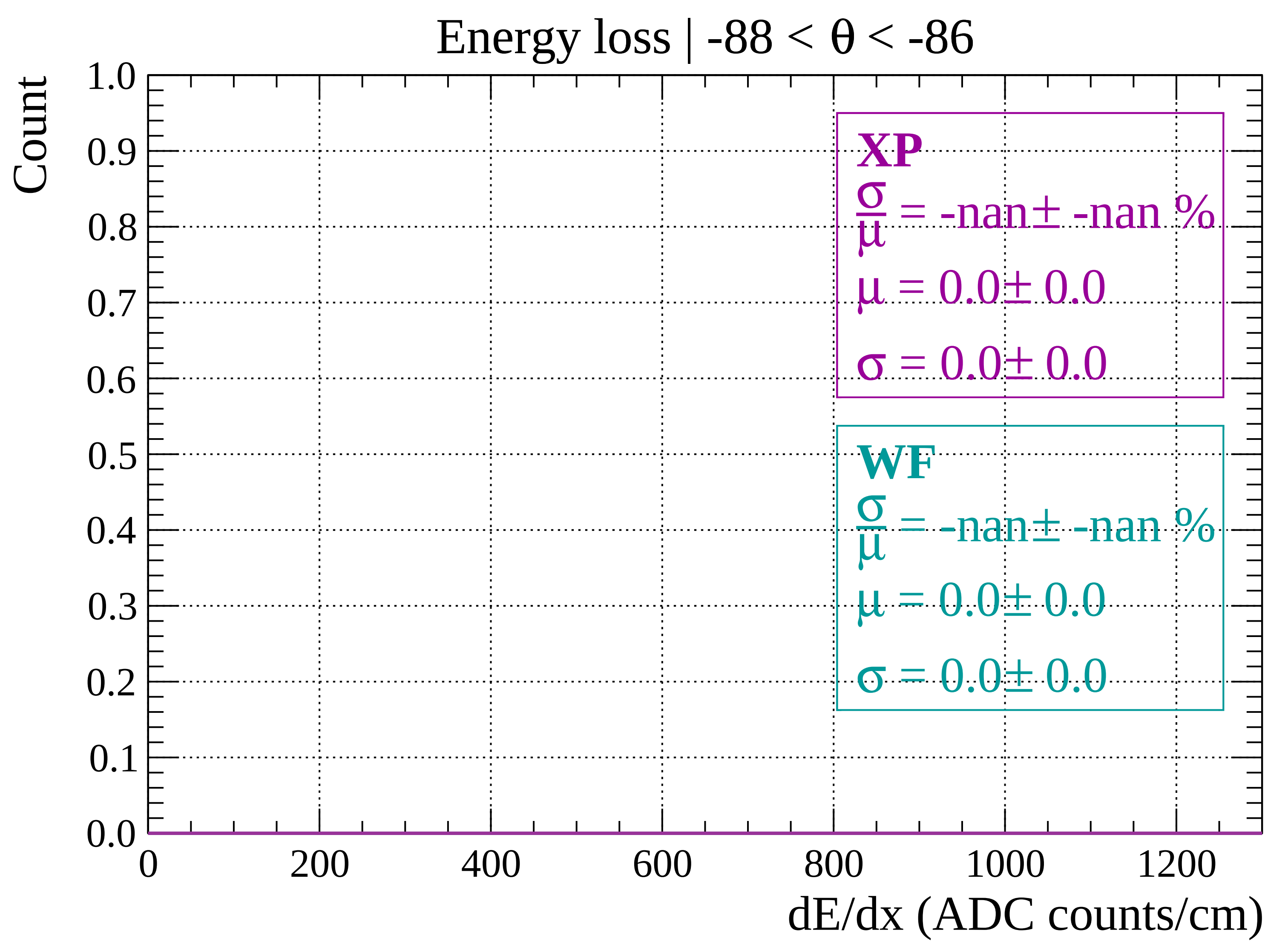


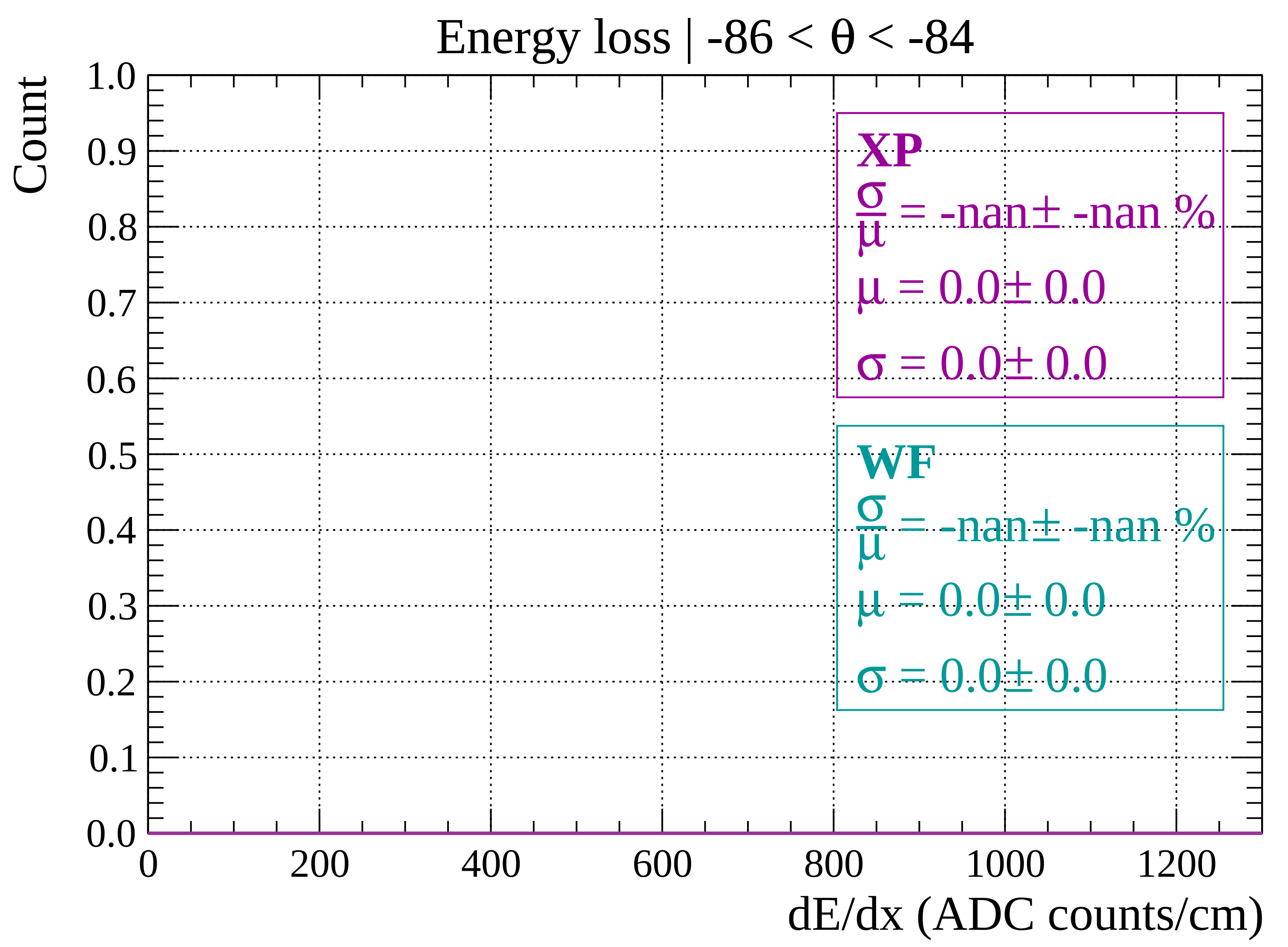
Energy loss | 1960 < p < 2000
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 2000 < p < 2040
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

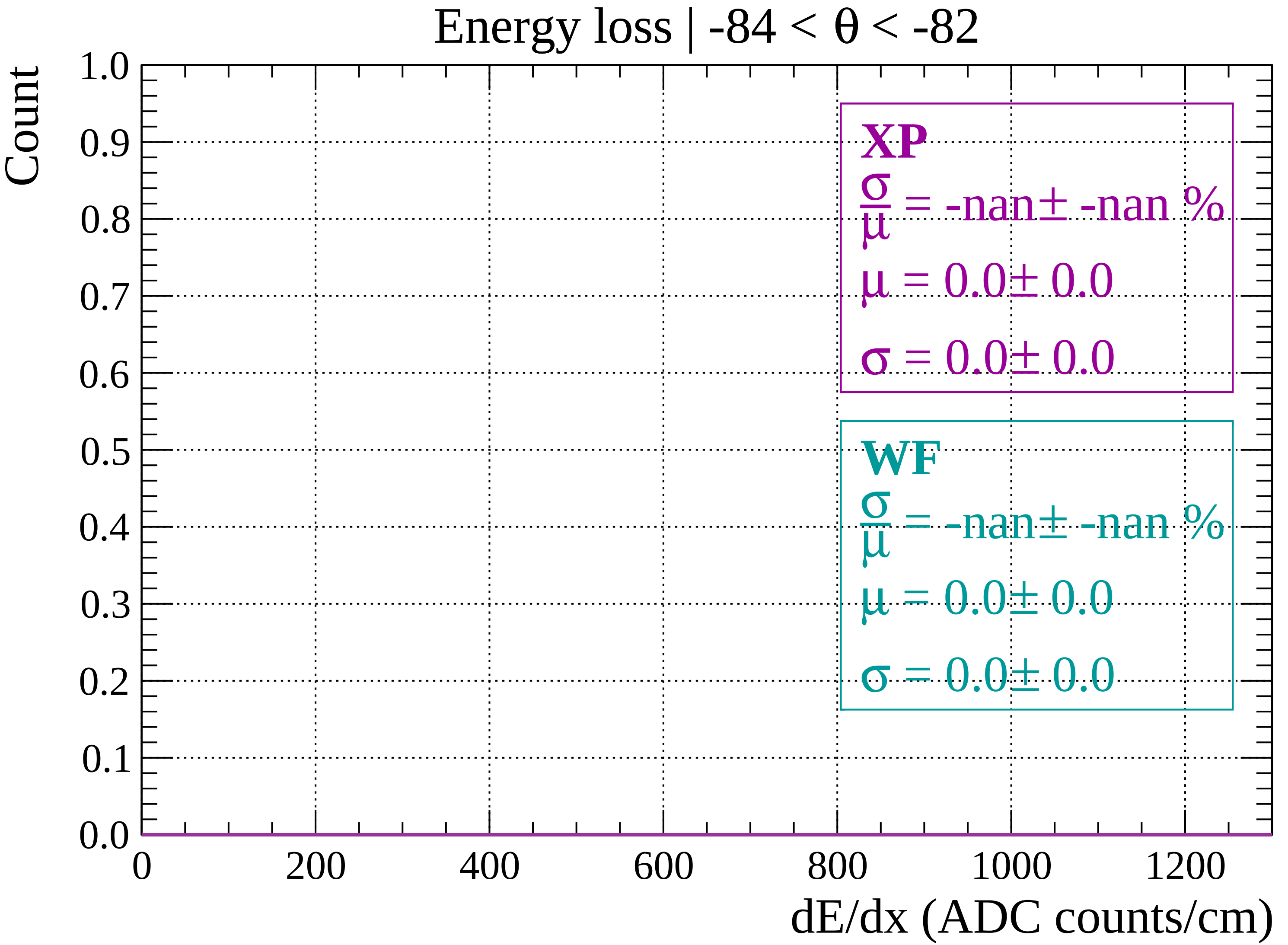


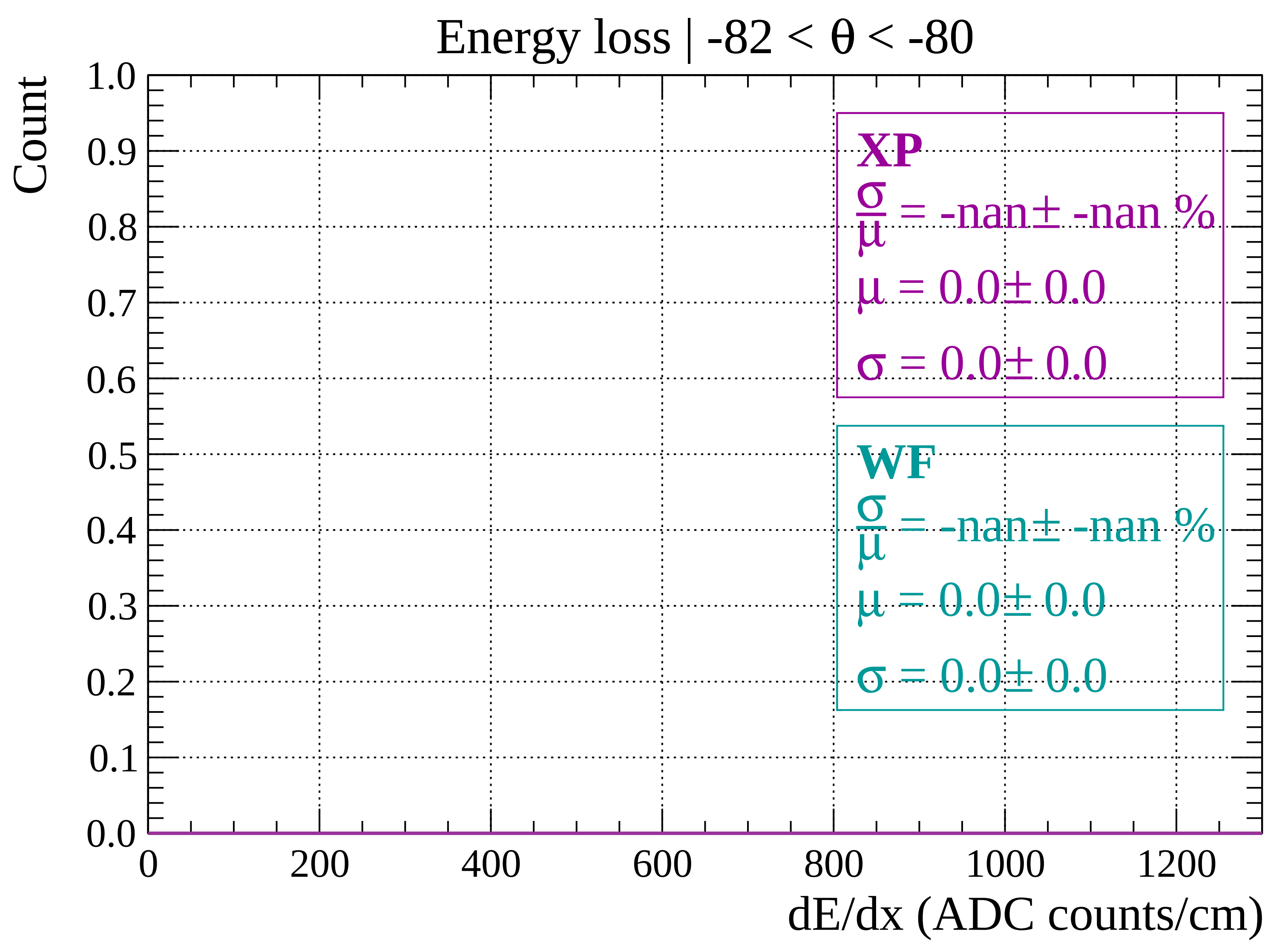
Energy loss | -90 < θ < -88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

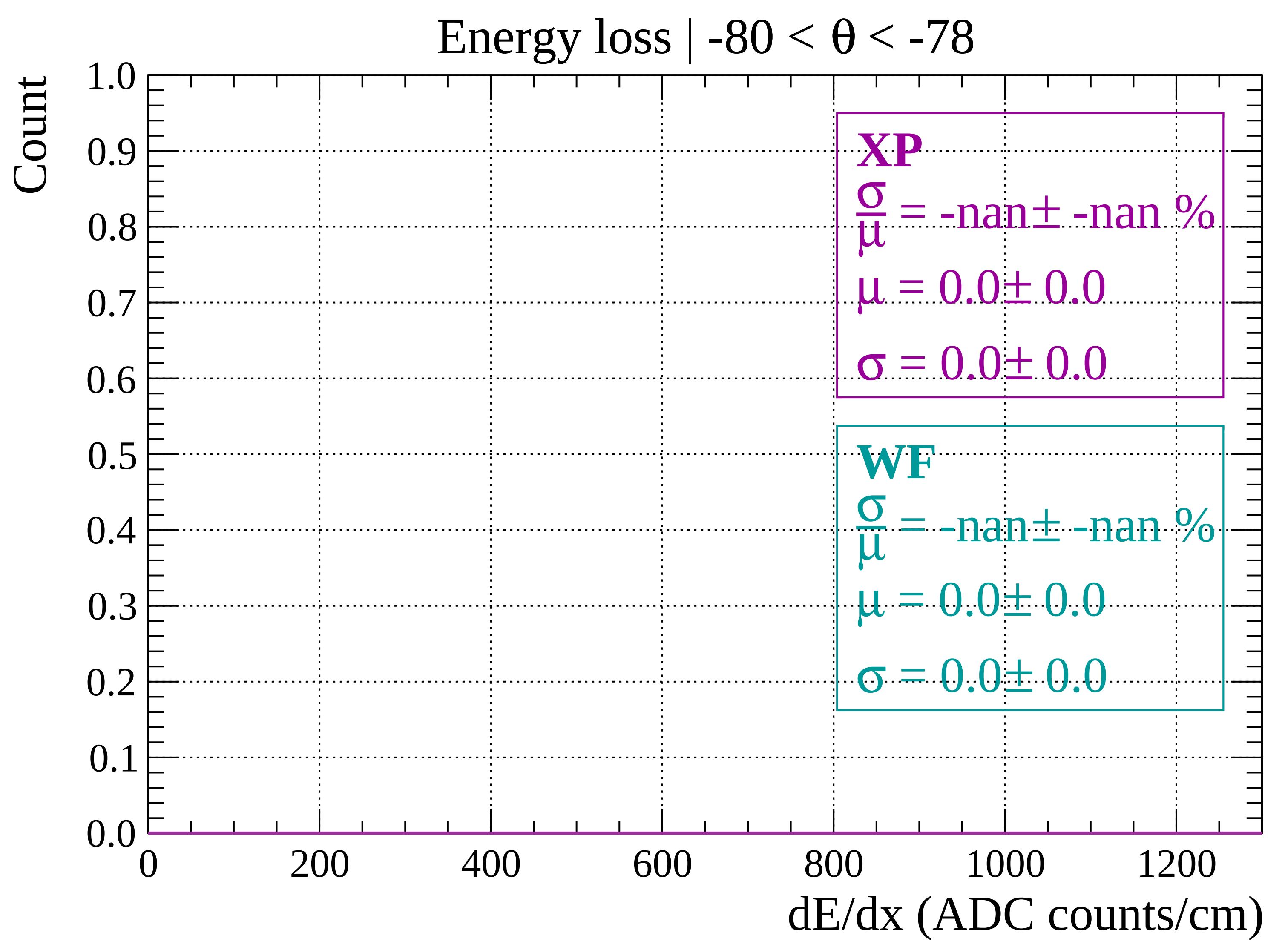

Energy loss | -88 < θ < -86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



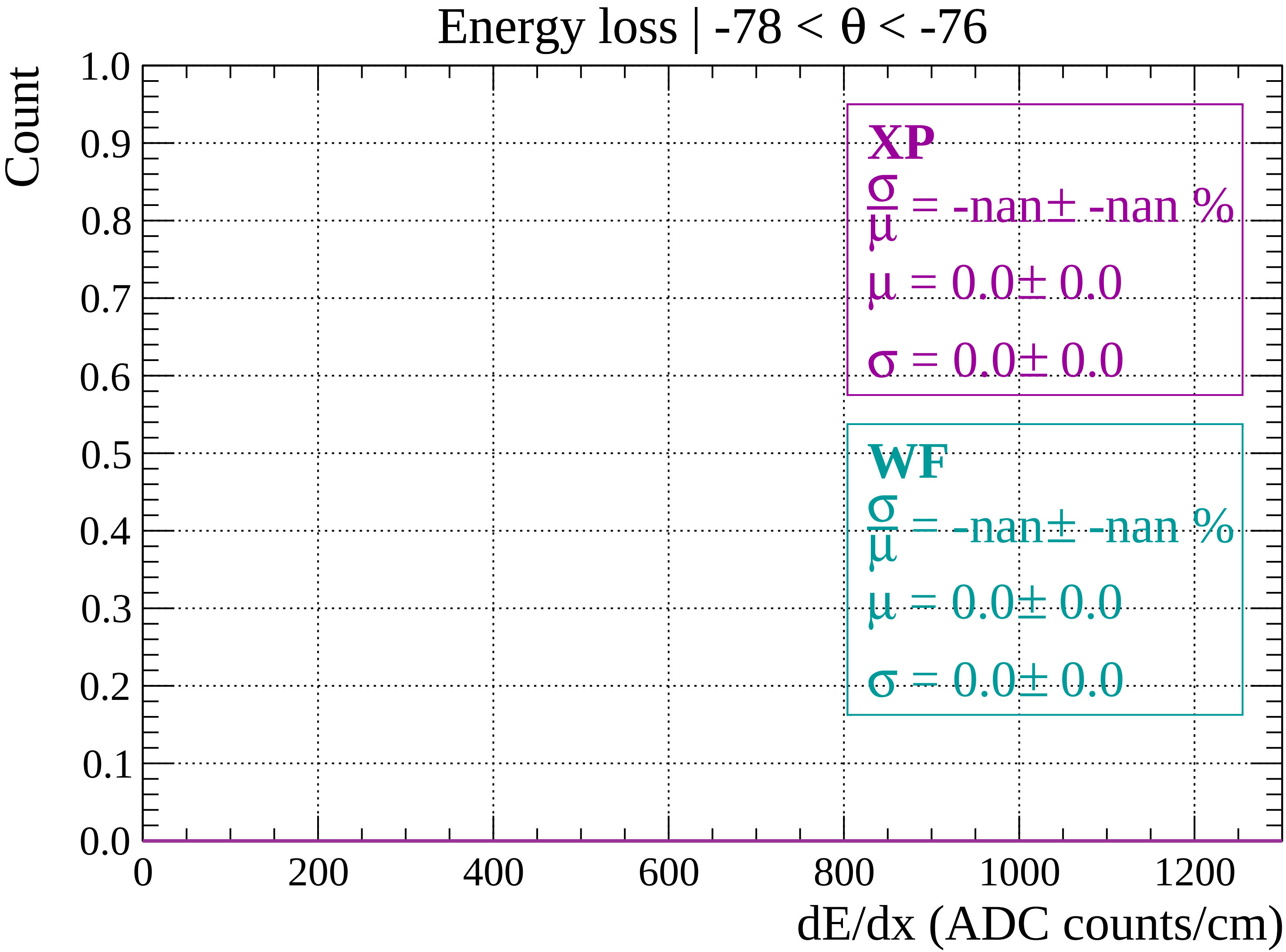
Energy loss | -86 < θ < -84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -84 < θ < -82
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

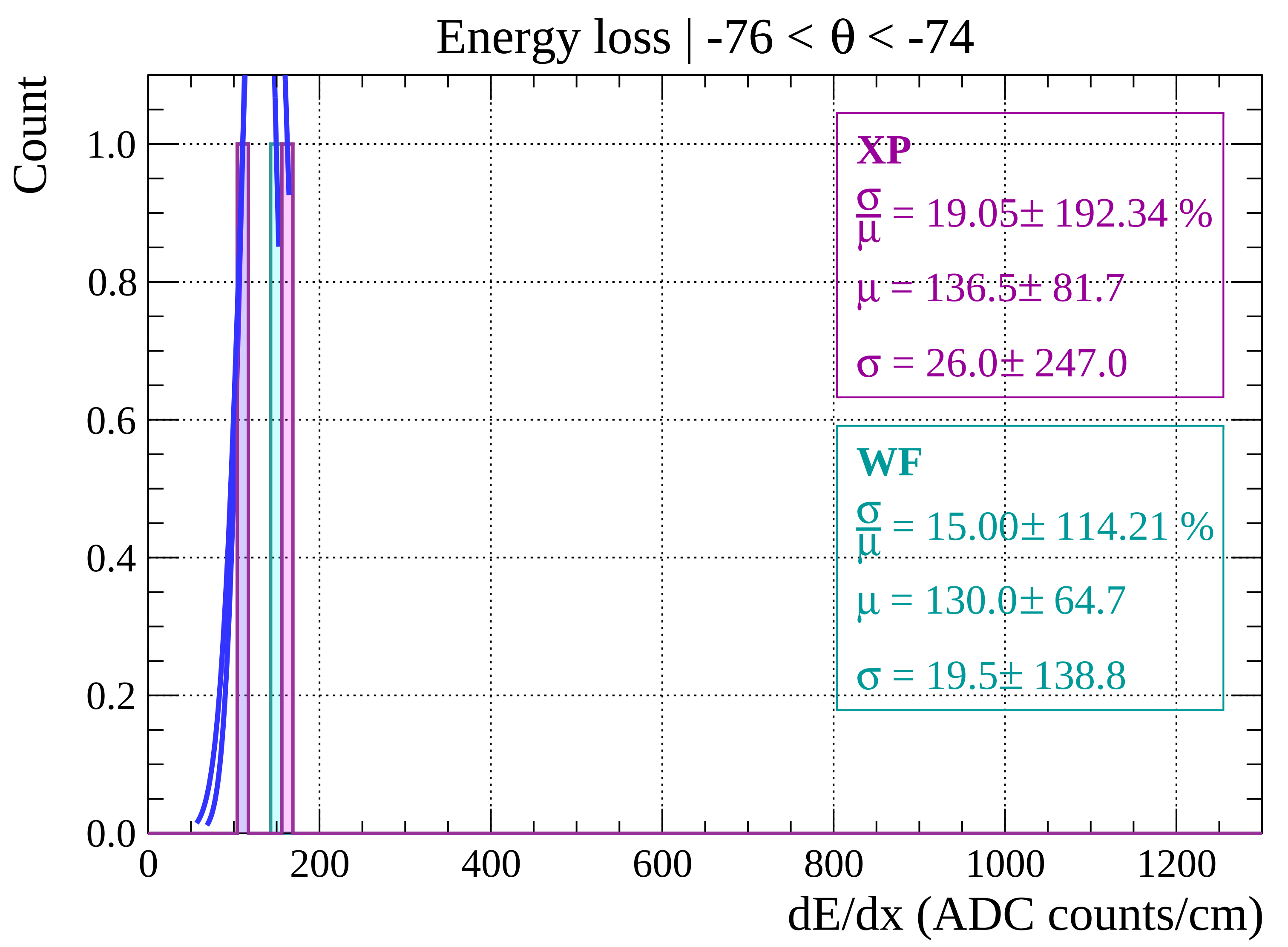

Energy loss | -82 < θ < -80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



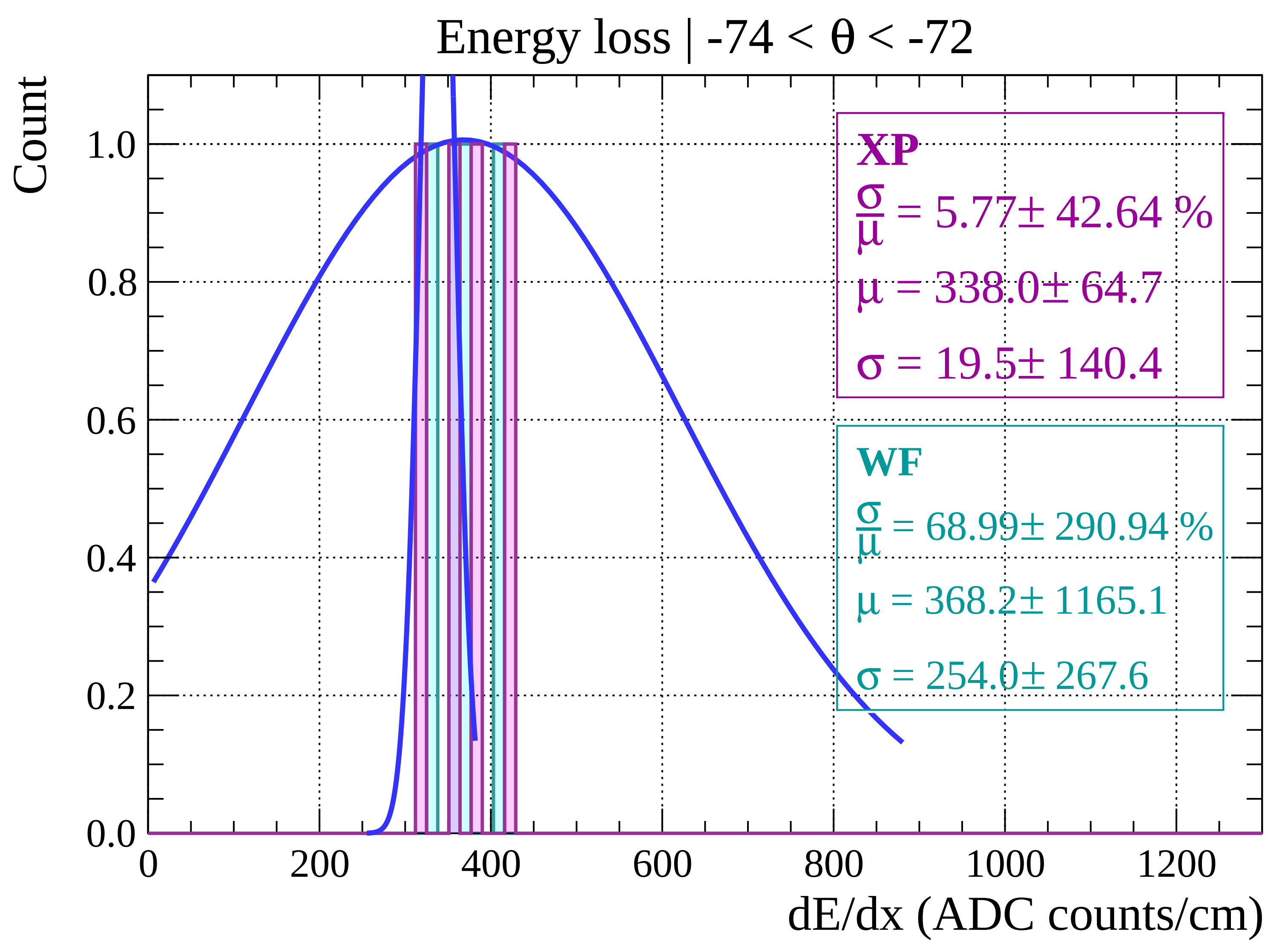
Energy loss | -80 < θ < -78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



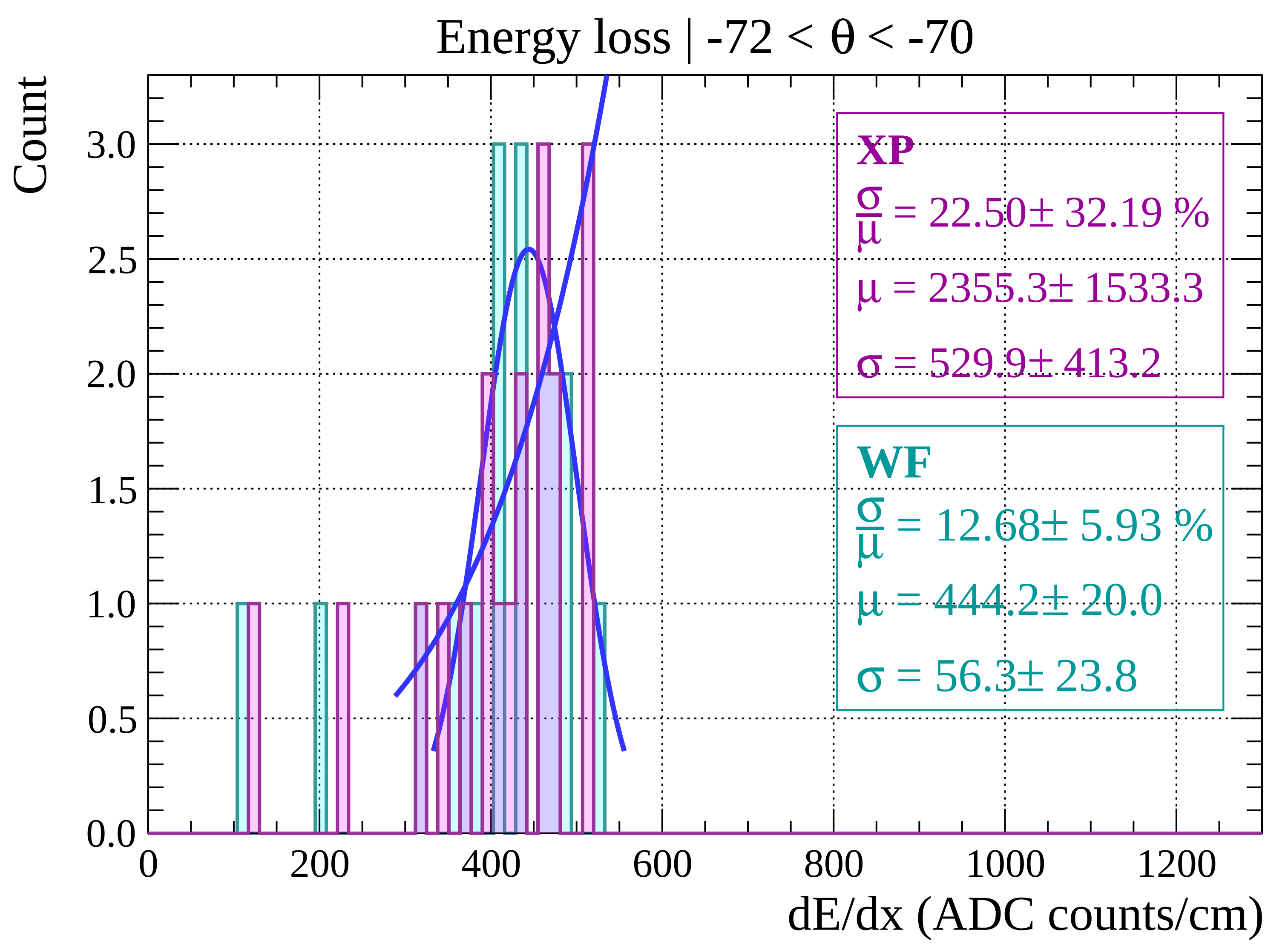
Energy loss | -78 < θ < -76
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

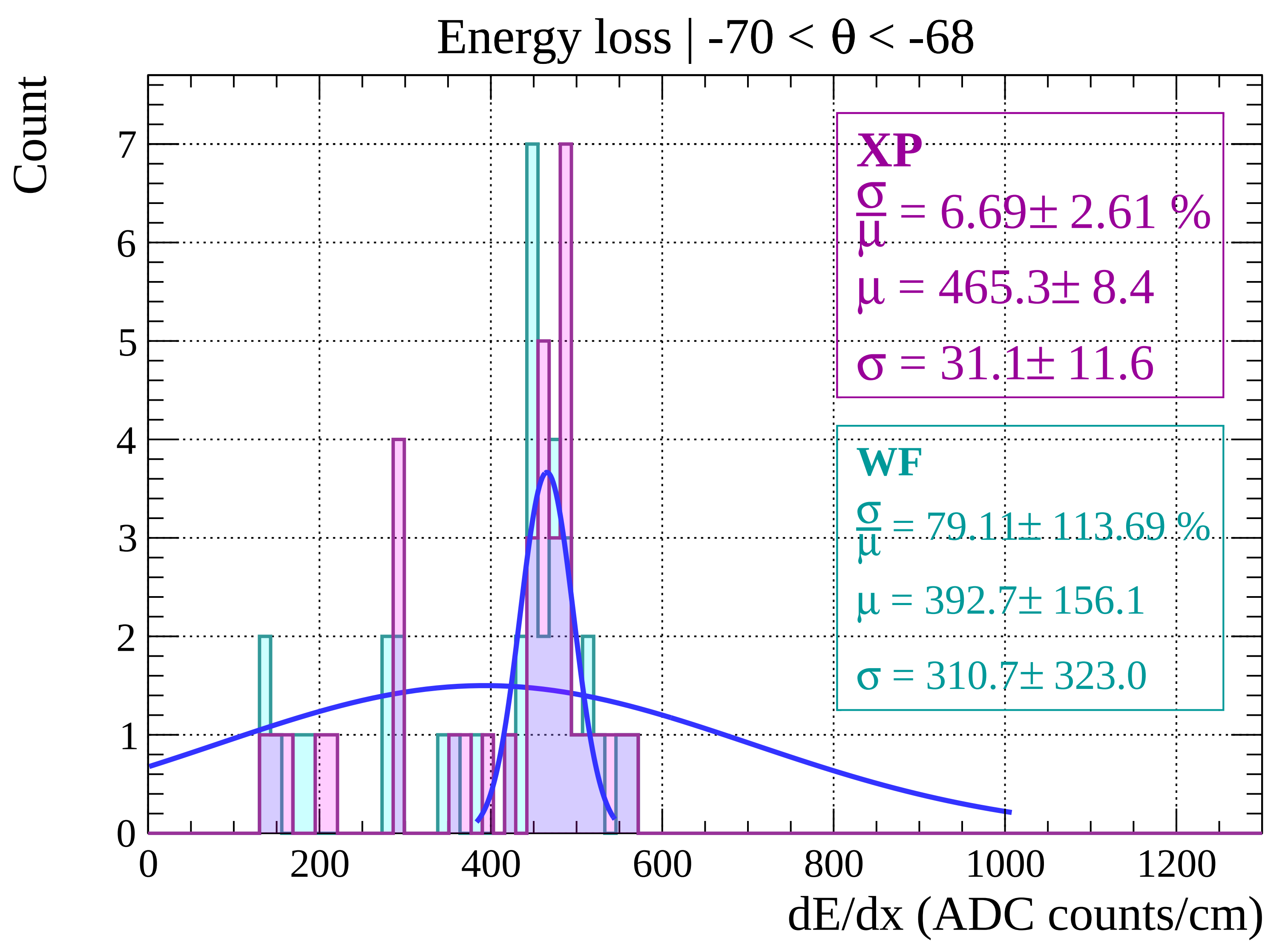


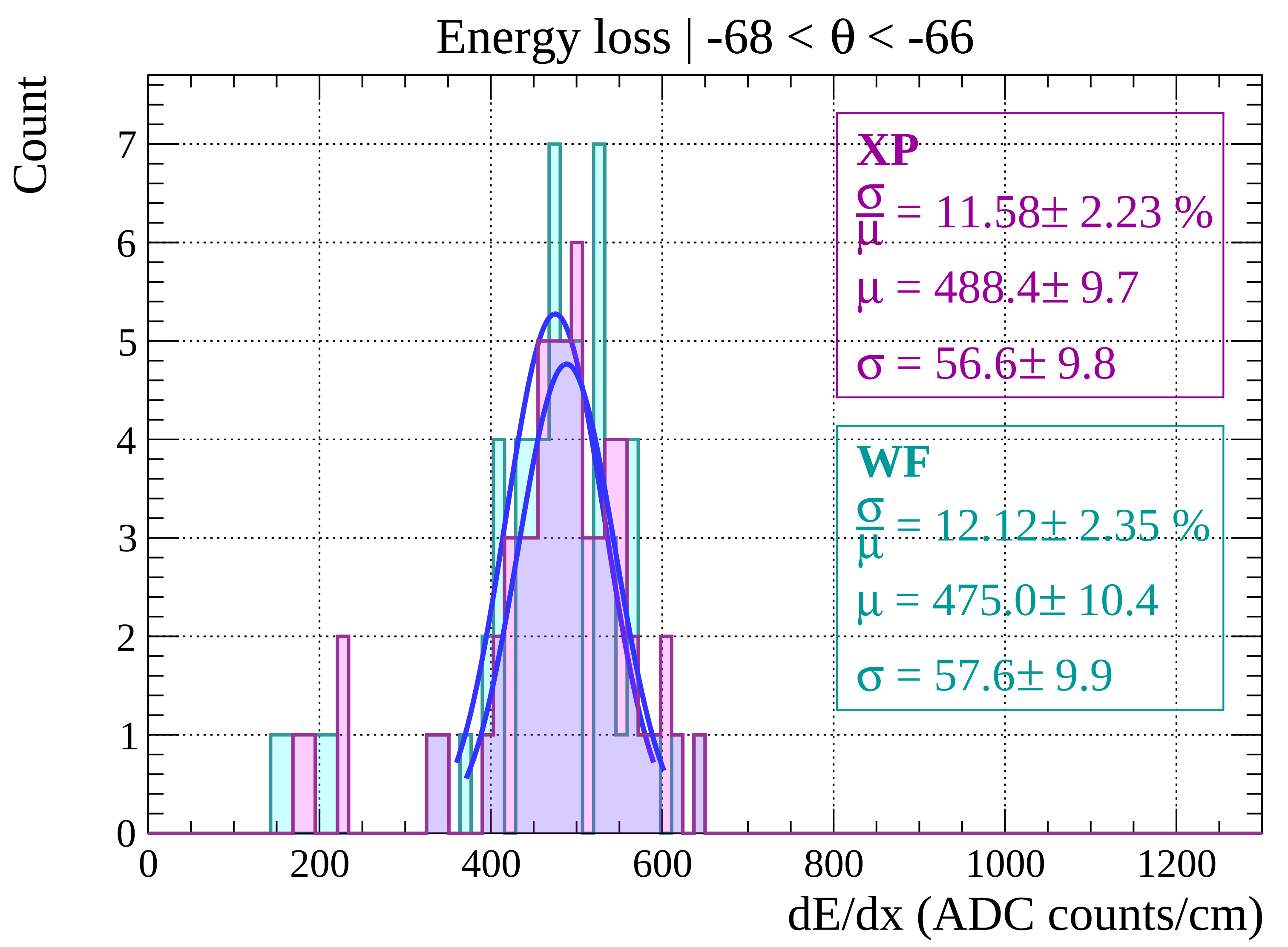
Energy loss | -76 < θ < -74
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 19.05± 192.34 %
μ = 136.5± 81.7
σ = 26.0± 247.0
WF
σ/μ = 15.00± 114.21 %
μ = 130.0± 64.7
σ = 19.5± 138.8

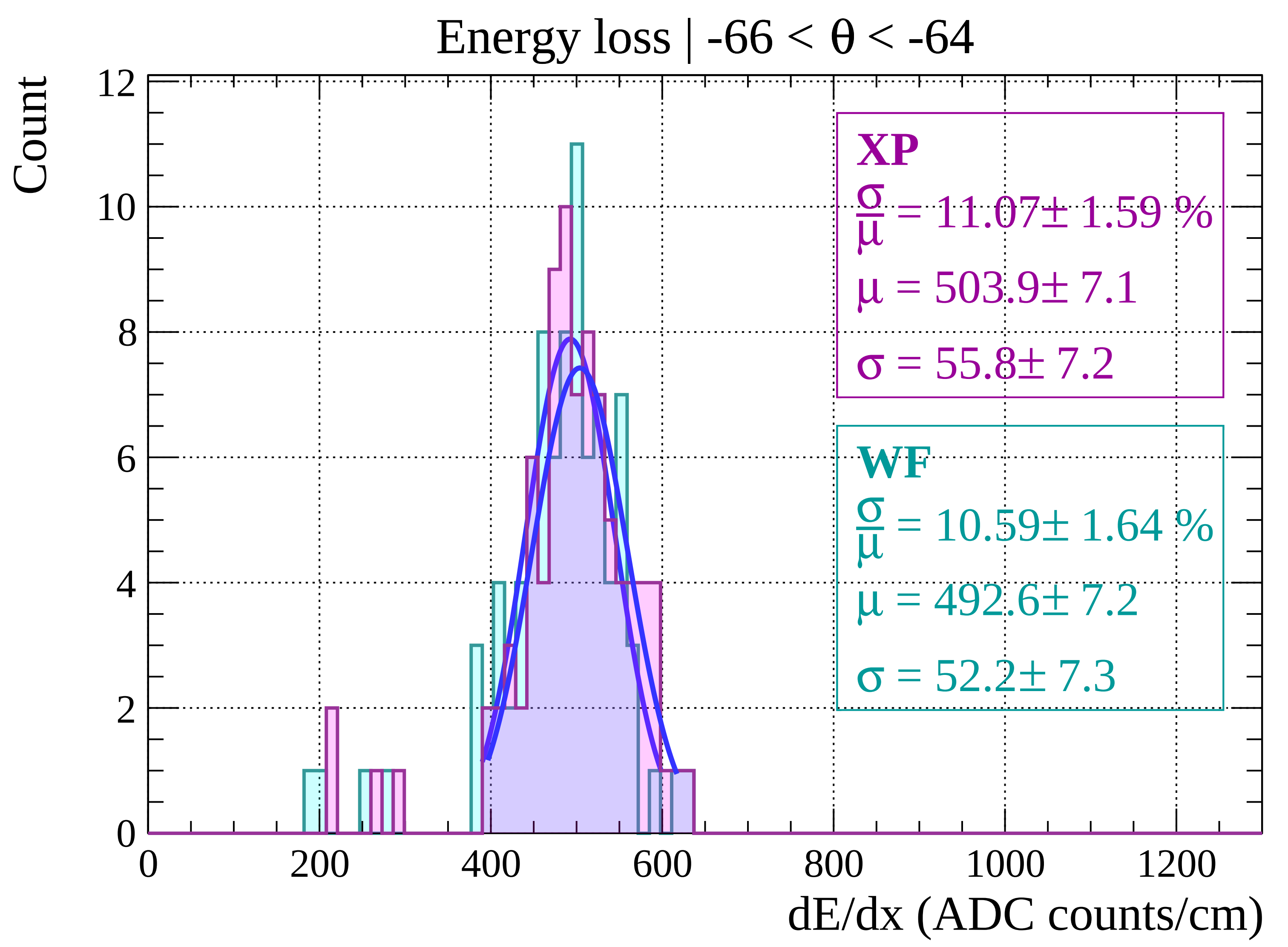


Energy loss | -74 < θ < -72
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.77± 42.64 %
μ = 338.0± 64.7
σ = 19.5± 140.4
WF
σ/μ = 68.99± 290.94 %
μ = 368.2± 1165.1
σ = 254.0± 267.6

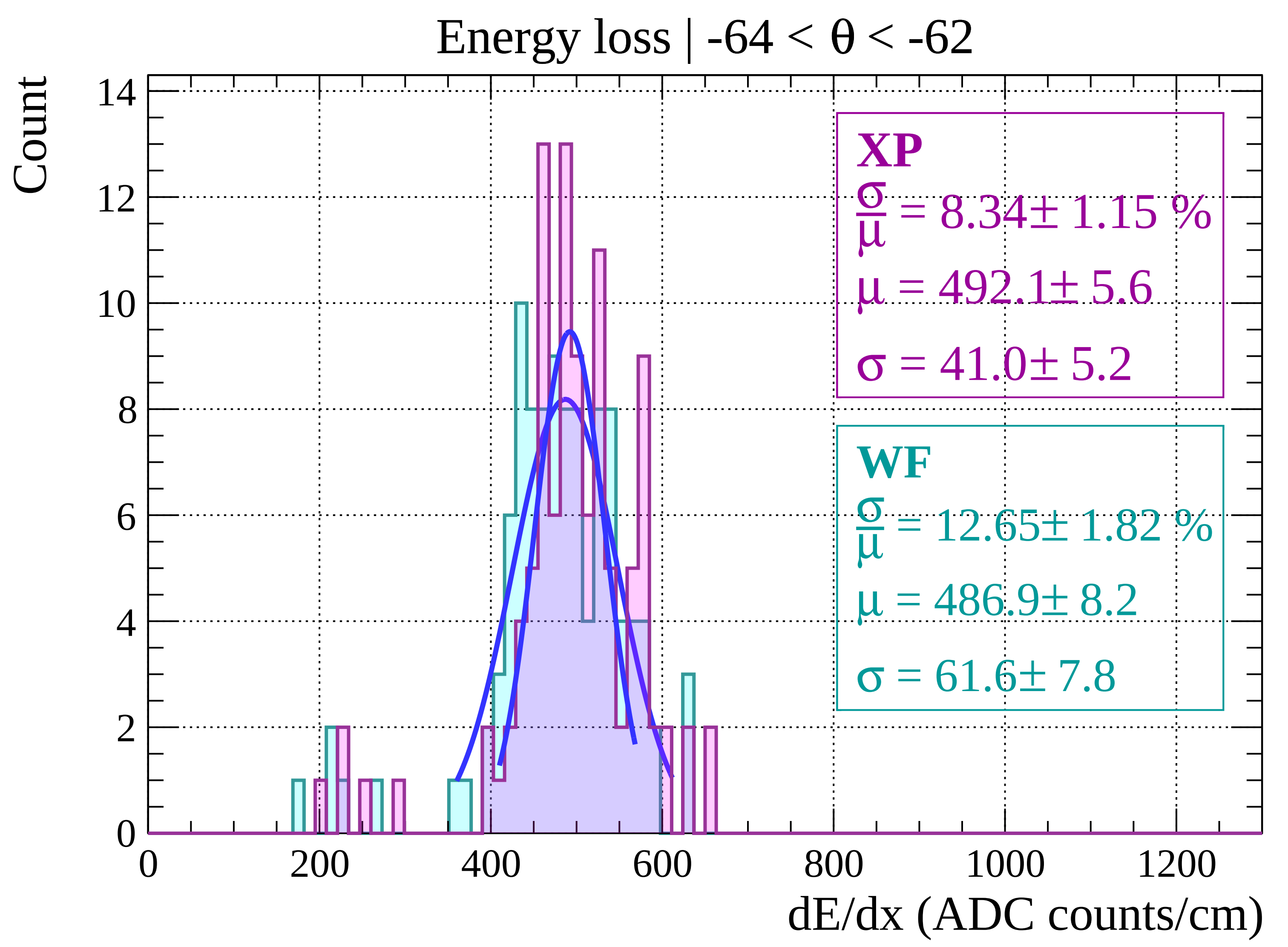

Energy loss | -72 < θ < -70
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 22.50± 32.19 %
μ = 2355.3± 1533.3
σ = 529.9± 413.2
WF
σ/μ = 12.68± 5.93 %
μ = 444.2± 20.0
σ = 56.3± 23.8


Energy loss | -70 < θ < -68
Count
7
6
5
4
3
2
1
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.69± 2.61 %
μ = 465.3± 8.4
σ = 31.1± 11.6
WF
σ/μ = 79.11± 113.69 %
μ = 392.7± 156.1
σ = 310.7± 323.0

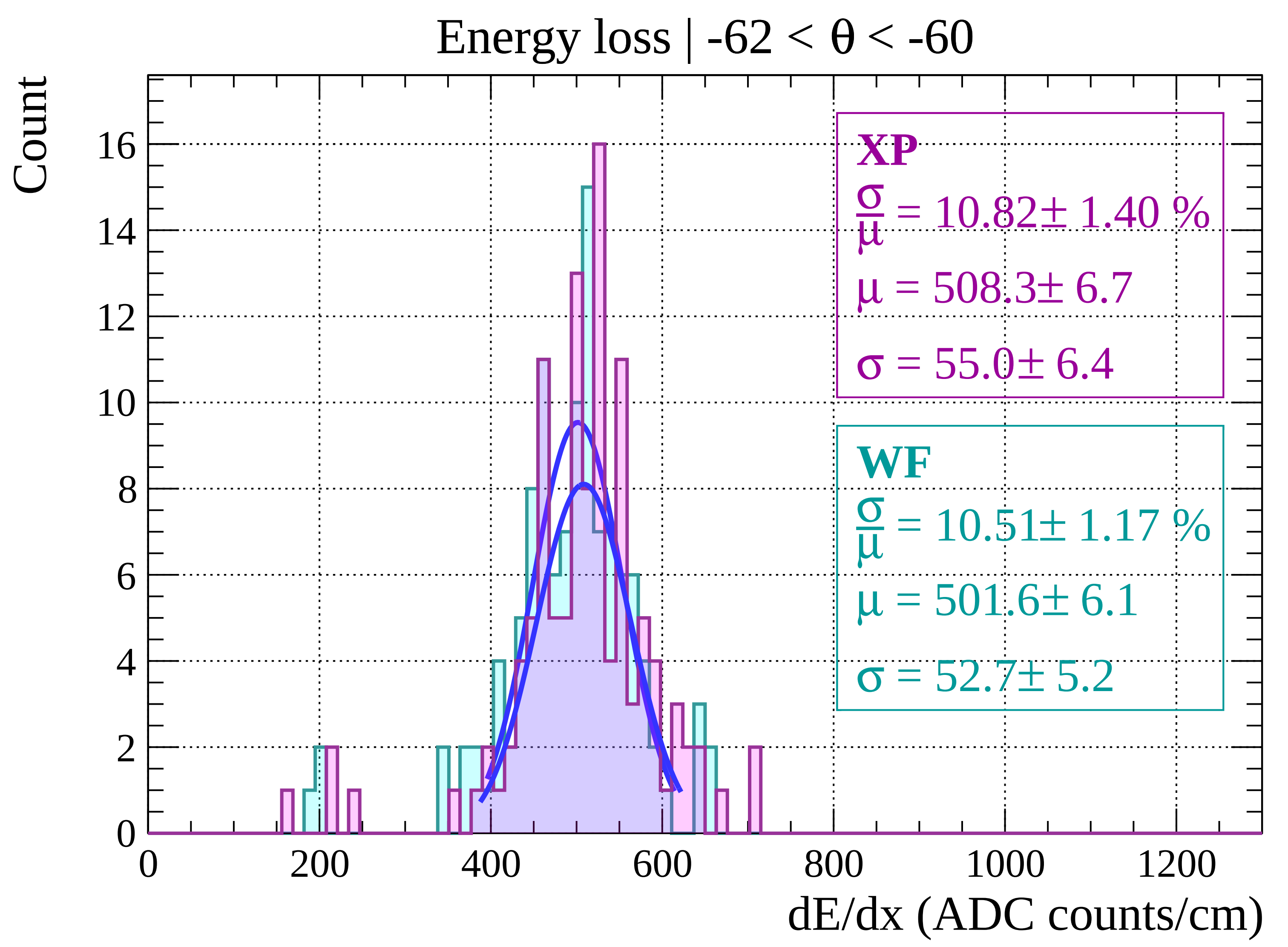

Energy loss | -68 < θ < -66
Count
XP
σ/μ = 11.58± 2.23 %
μ = 488.4± 9.7
σ = 56.6± 9.8
WF
σ/μ = 12.12± 2.35 %
μ = 475.0± 10.4
σ = 57.6± 9.9
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

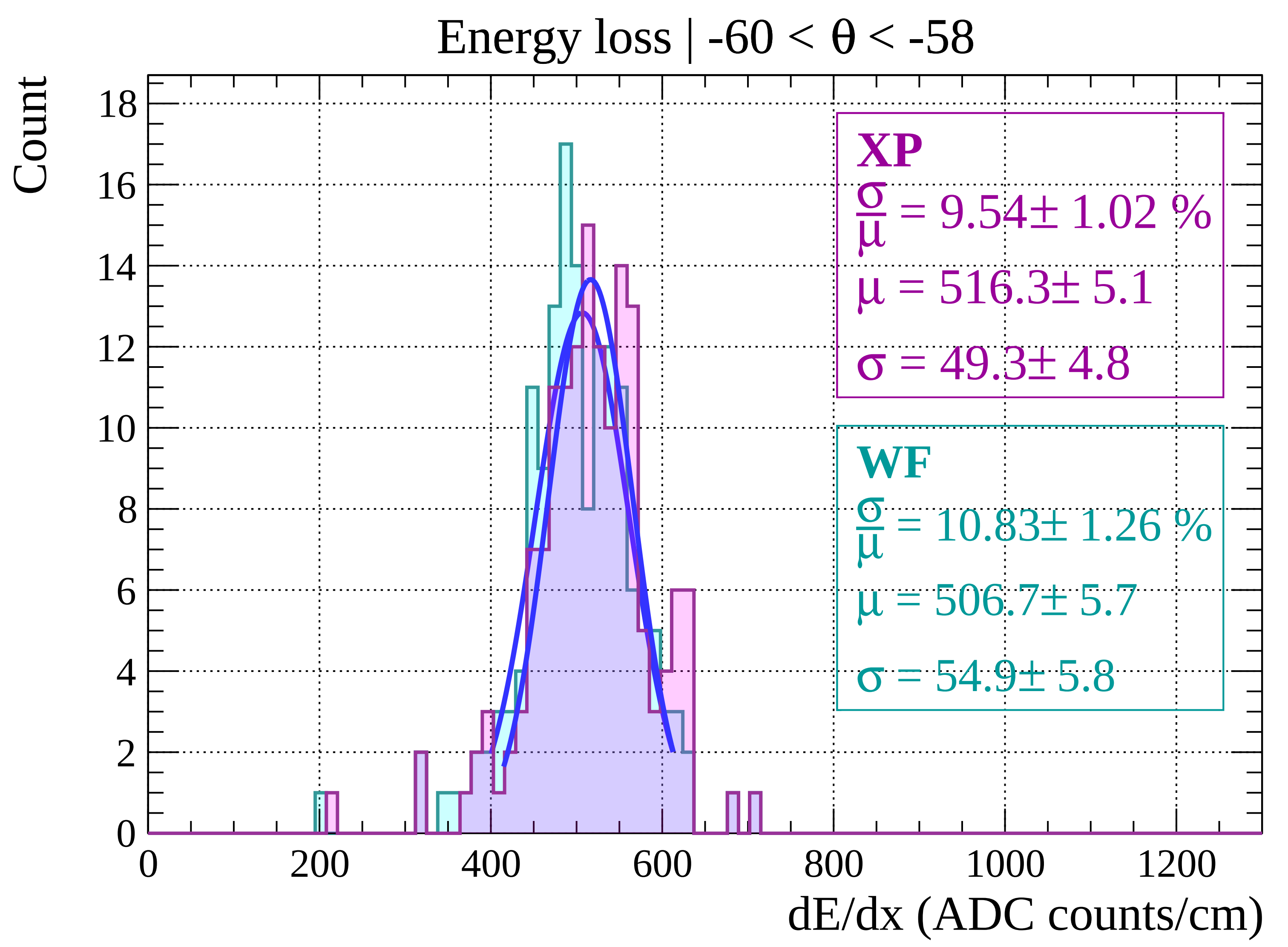


Energy loss | -66 < θ < -64
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.07± 1.59 %
μ = 503.9± 7.1
σ = 55.8± 7.2
WF
σ/μ = 10.59± 1.64 %
μ = 492.6± 7.2
σ = 52.2± 7.3

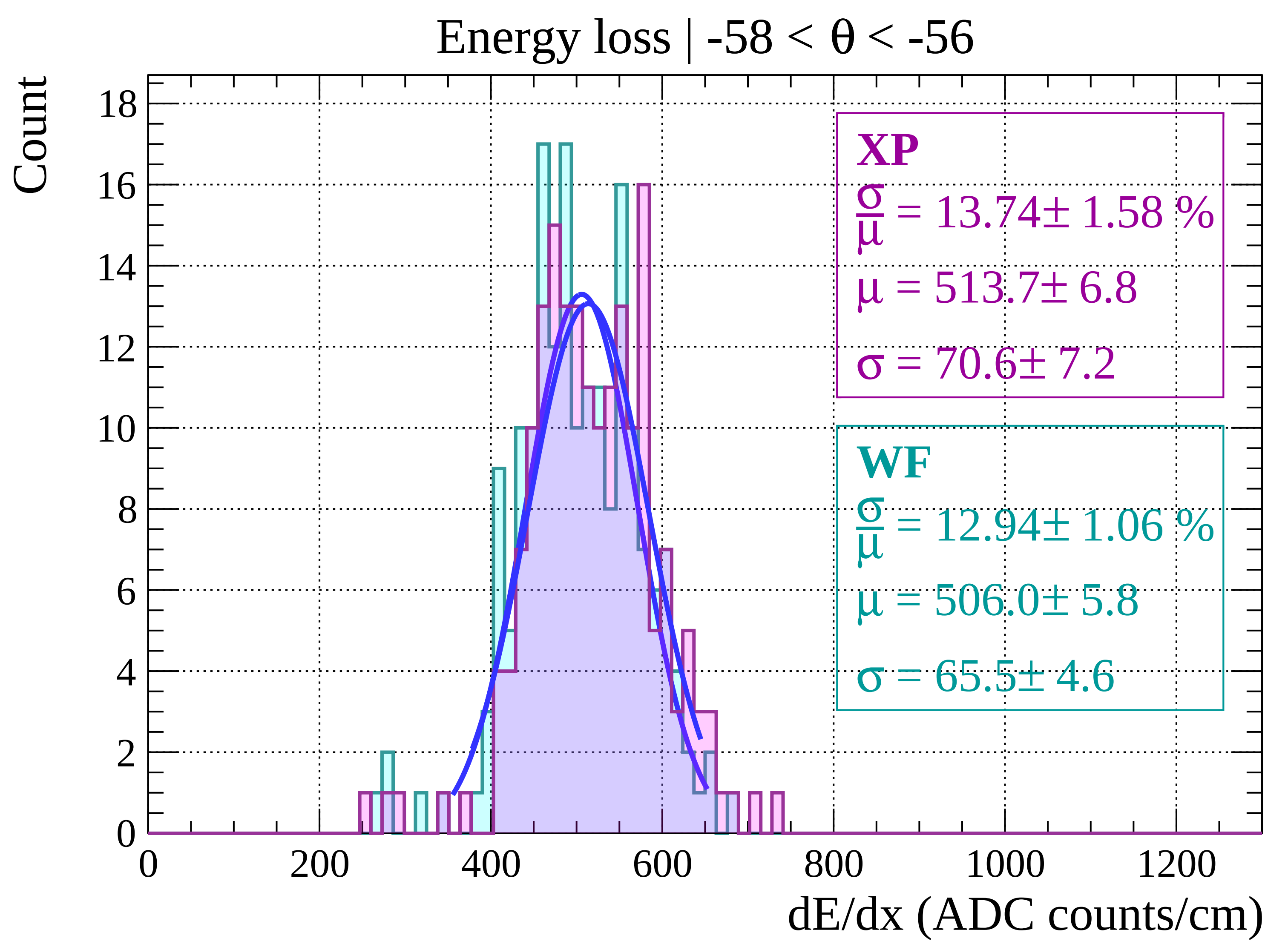


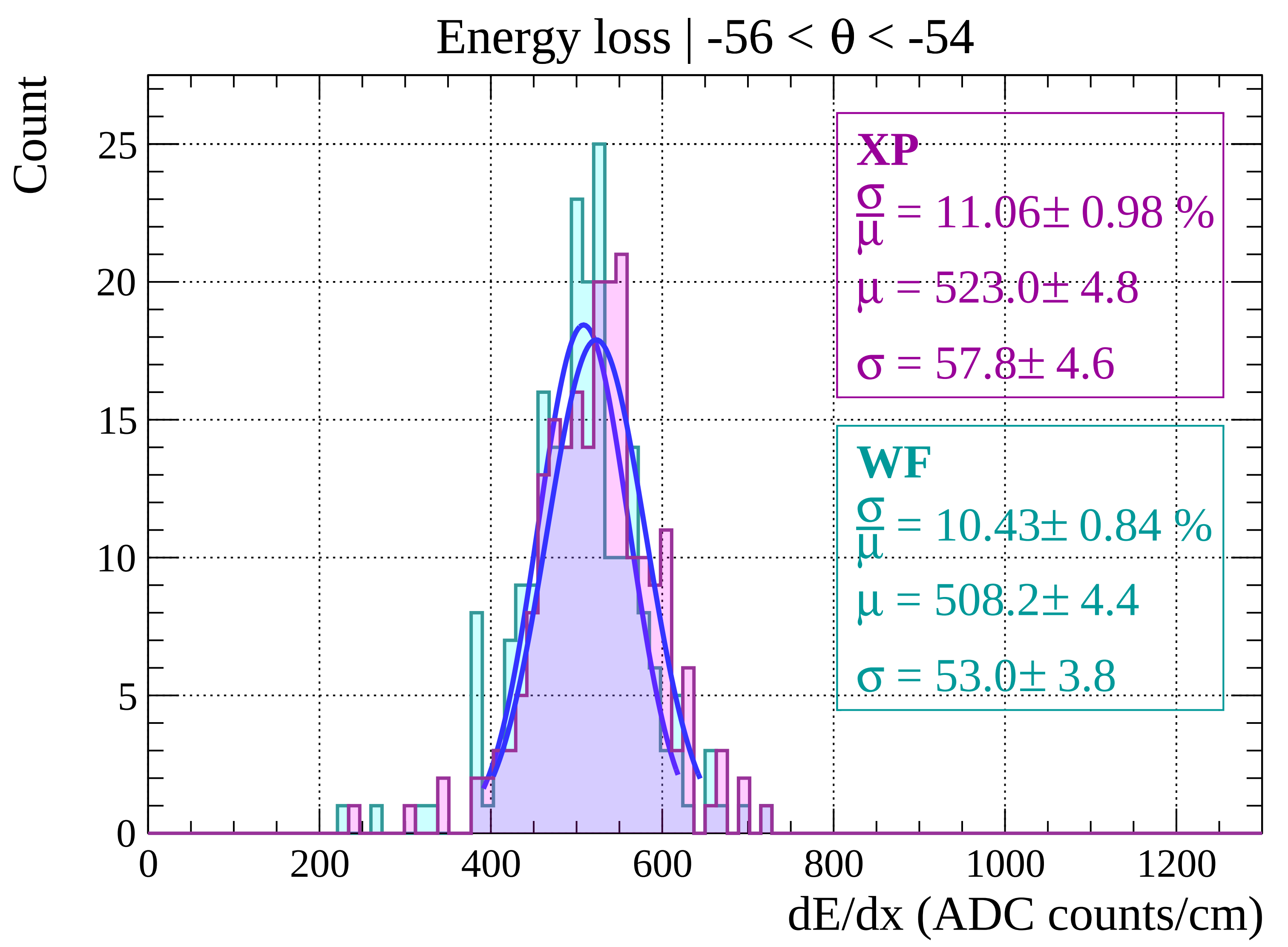
Energy loss | -64 < θ < -62
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
XP
σ/μ = 8.34± 1.15 %
μ = 492.1± 5.6
σ = 41.0± 5.2
WF
σ/μ = 12.65± 1.82 %
μ = 486.9± 8.2
σ = 61.6± 7.8


Energy loss | -62 < θ < -60
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.82± 1.40 %
μ = 508.3± 6.7
σ = 55.0± 6.4
WF
σ/μ = 10.51± 1.17 %
μ = 501.6± 6.1
σ = 52.7± 5.2
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200

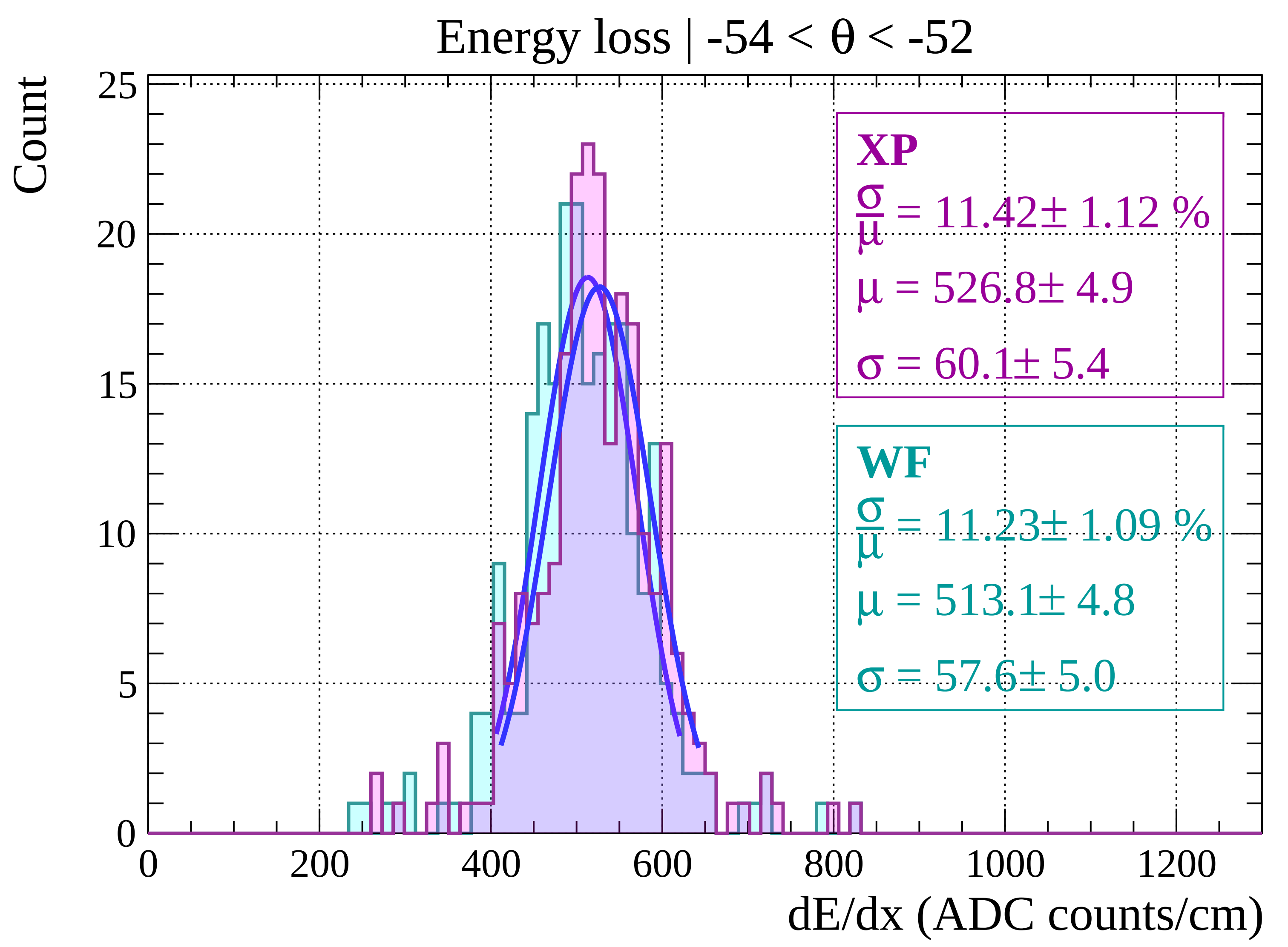


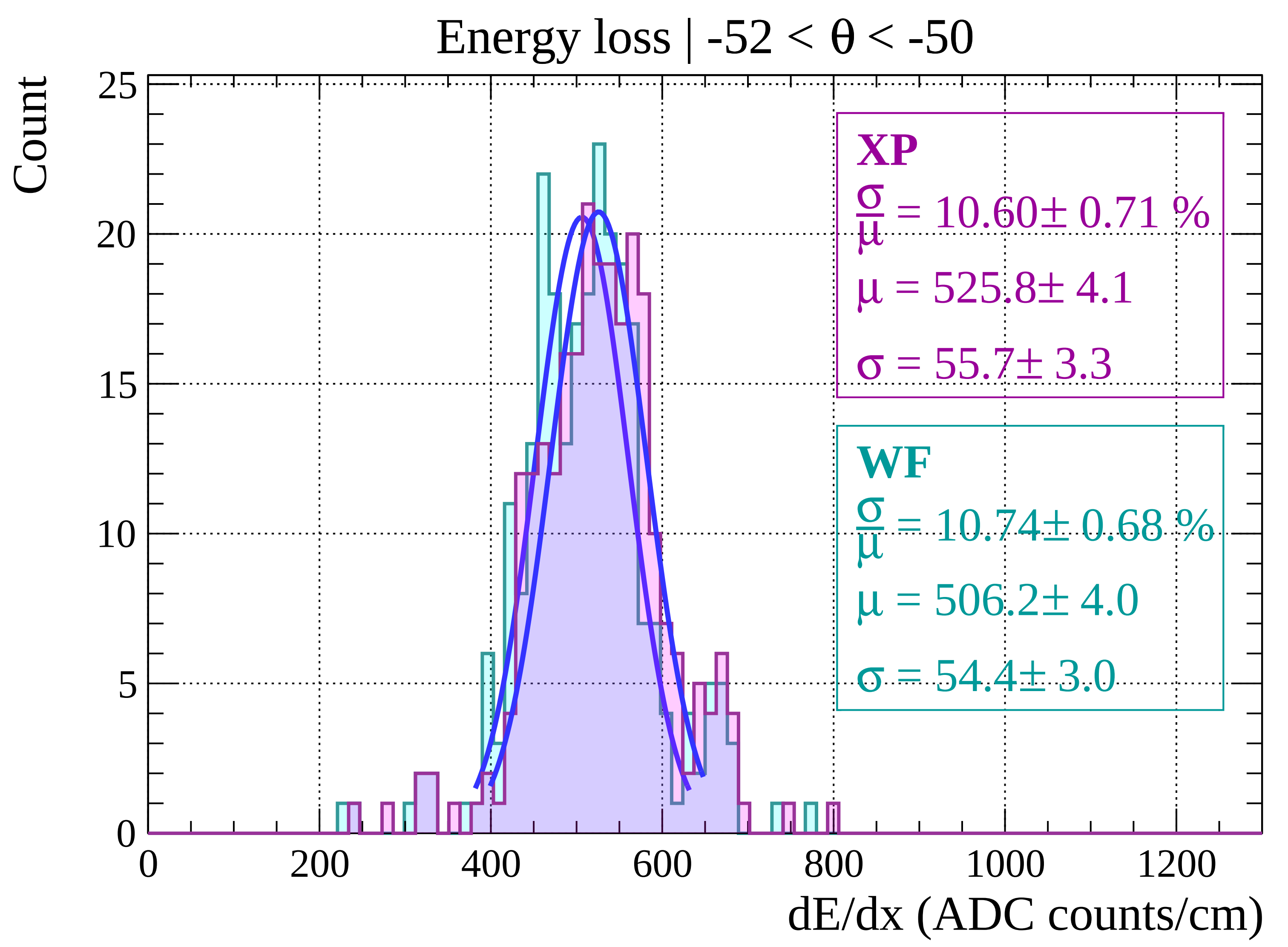
Energy loss | -60 < θ < -58
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.54± 1.02 %
μ = 516.3± 5.1
σ = 49.3± 4.8
WF
σ/μ = 10.83± 1.26 %
μ = 506.7± 5.7
σ = 54.9± 5.8
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200

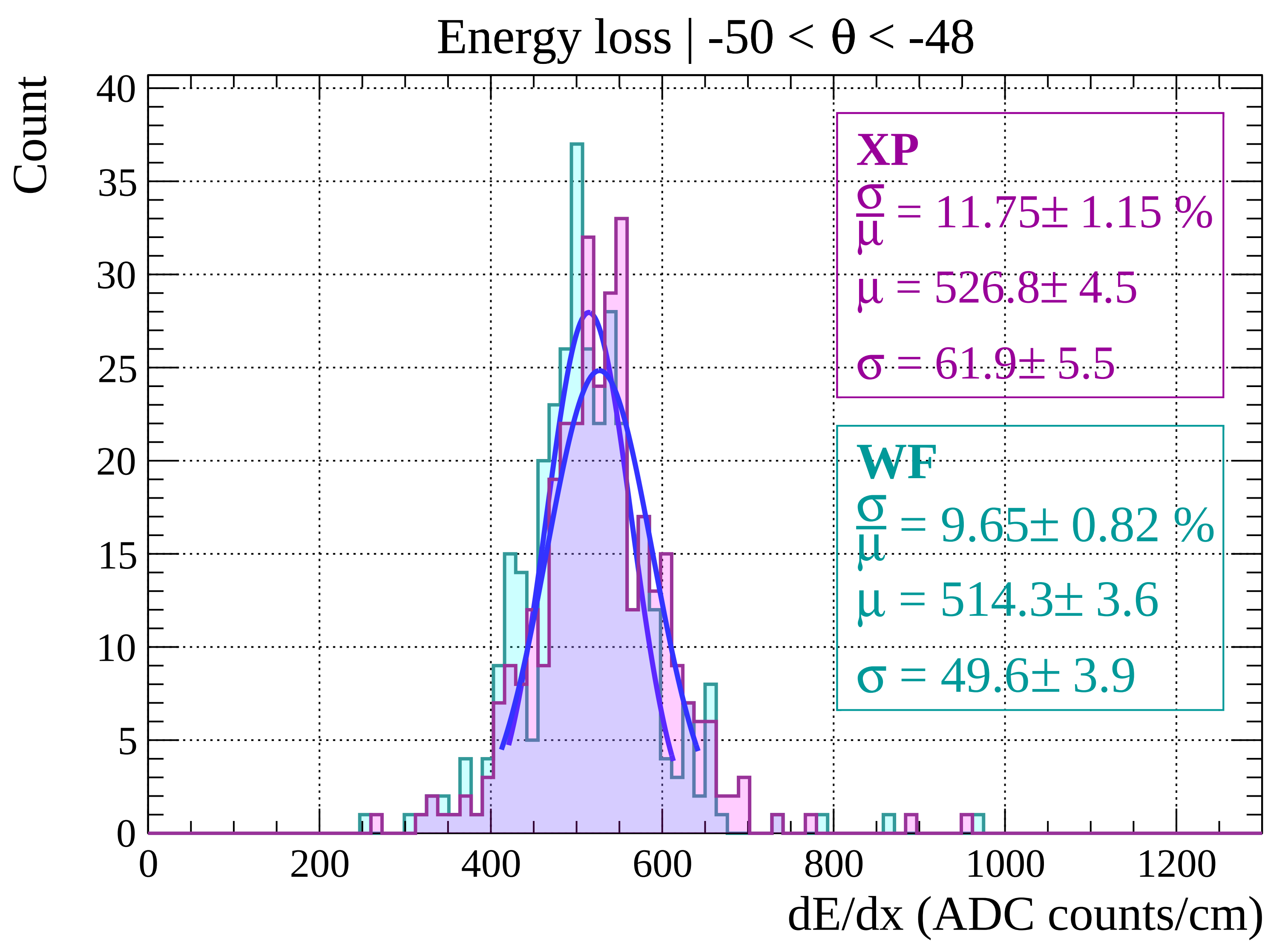

Energy loss | -58 < θ < -56
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 13.74± 1.58 %
μ = 513.7± 6.8
σ = 70.6± 7.2
WF
$\frac{\sigma}{\mu}$ = 12.94± 1.06 %
μ = 506.0± 5.8
σ = 65.5± 4.6
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18

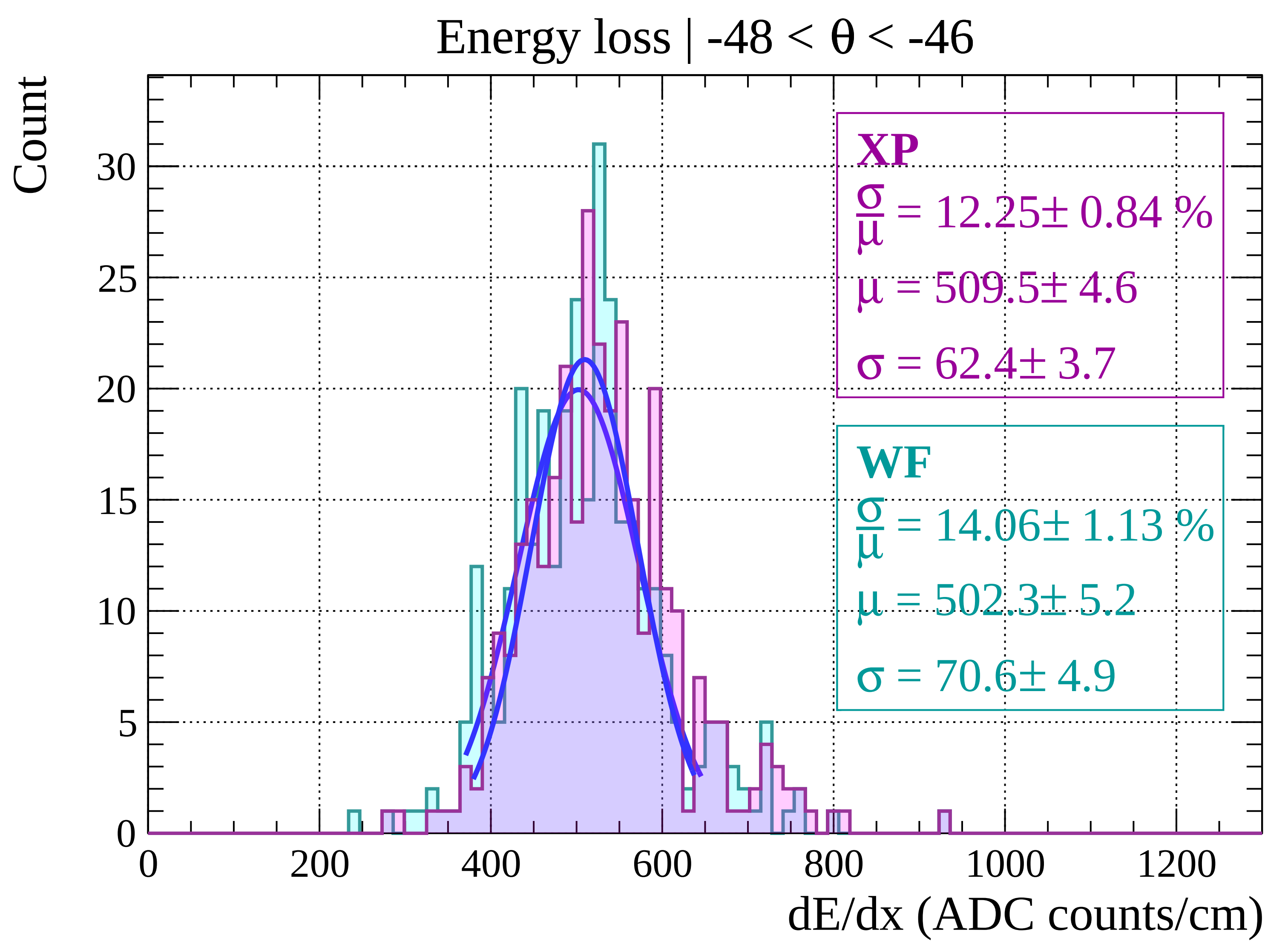


Energy loss | -56 < θ < -54
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 11.06± 0.98 %
μ = 523.0± 4.8
σ = 57.8± 4.6
WF
σ/μ = 10.43± 0.84 %
μ = 508.2± 4.4
σ = 53.0± 3.8

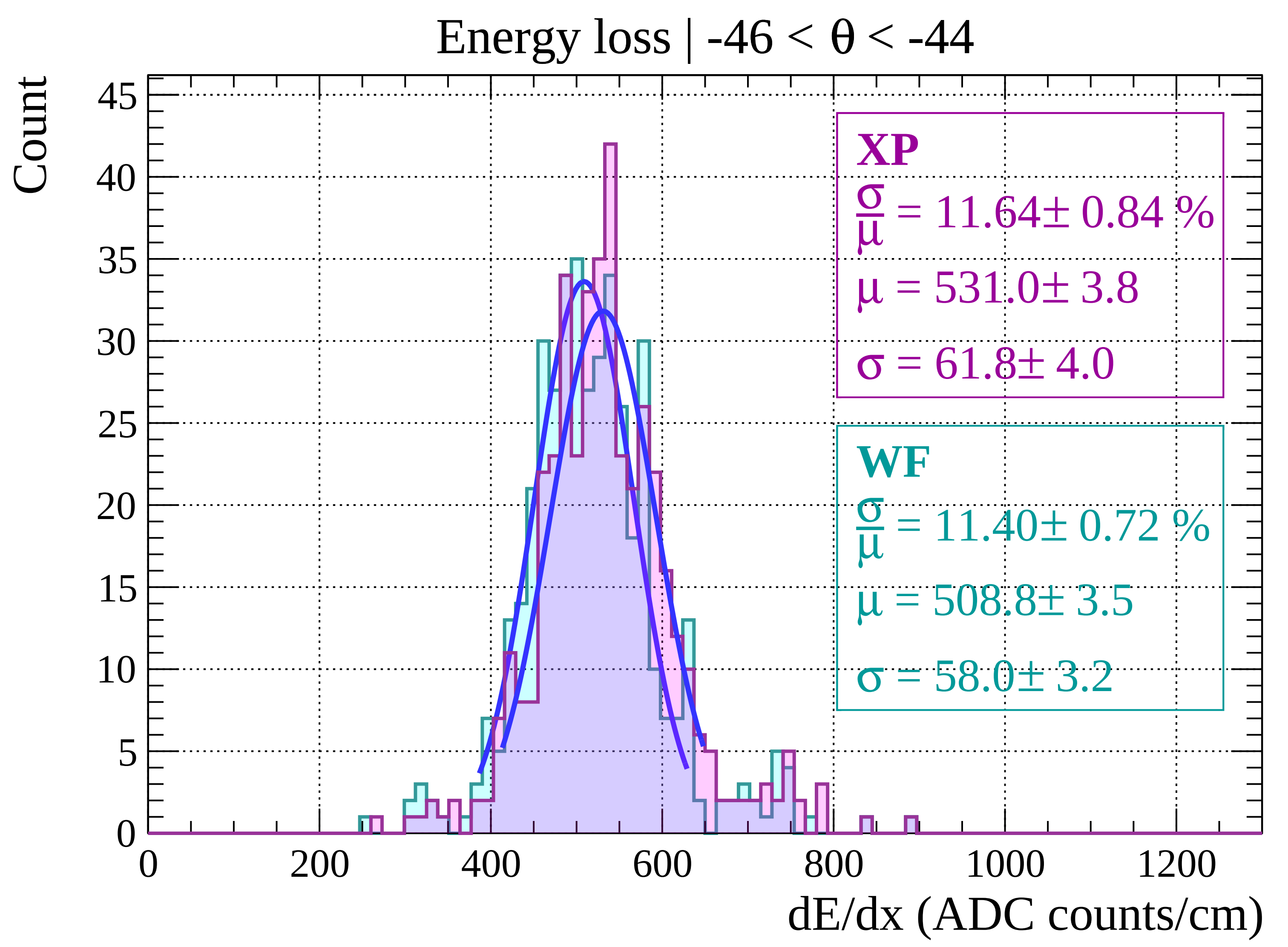

Energy loss | -54 < θ < -52
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
σ/μ = 11.42± 1.12 %
μ = 526.8± 4.9
σ = 60.1± 5.4
WF
σ/μ = 11.23± 1.09 %
μ = 513.1± 4.8
σ = 57.6± 5.0


Energy loss | -52 < θ < -50
Count
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.60± 0.71 %
μ = 525.8± 4.1
σ = 55.7± 3.3
WF
σ/μ = 10.74± 0.68 %
μ = 506.2± 4.0
σ = 54.4± 3.0

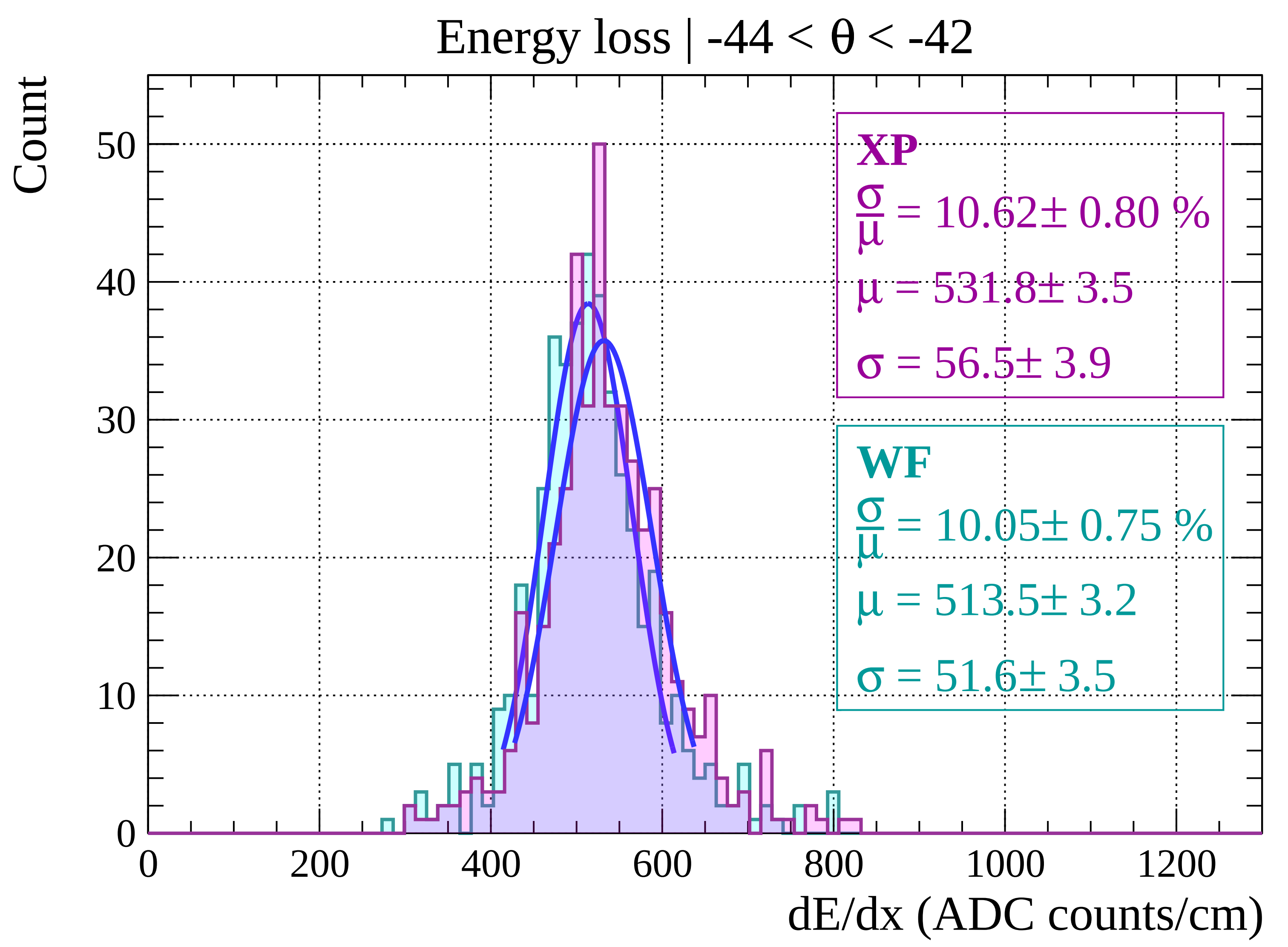

Energy loss | -50 < θ < -48
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP
σ/μ = 11.75± 1.15 %
μ = 526.8± 4.5
σ = 61.9± 5.5
WF
σ/μ = 9.65± 0.82 %
μ = 514.3± 3.6
σ = 49.6± 3.9



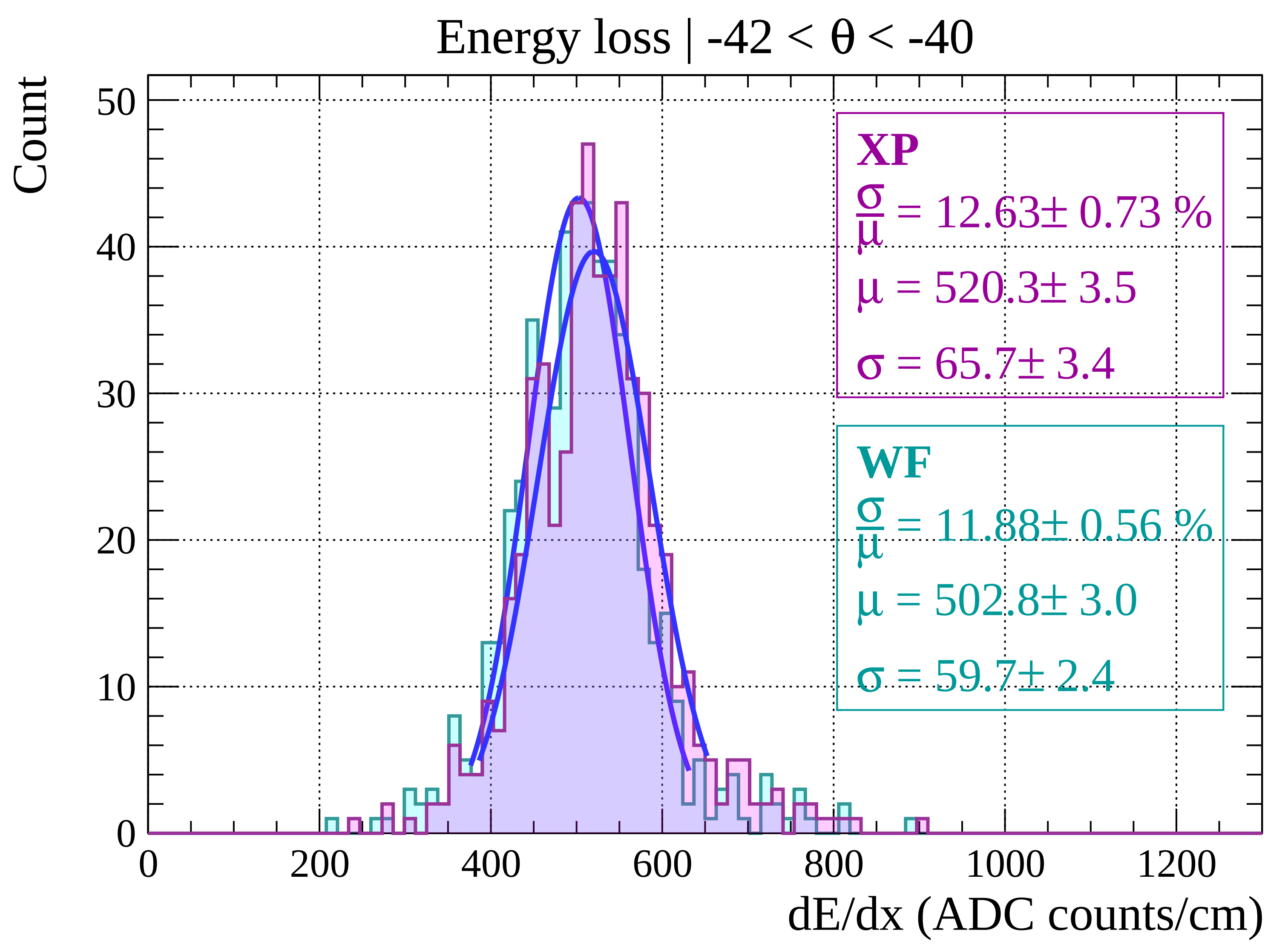
Energy loss | -48 < θ < -46
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP
σ/μ = 12.25± 0.84 %
μ = 509.5± 4.6
σ = 62.4± 3.7
WF
σ/μ = 14.06± 1.13 %
μ = 502.3± 5.2
σ = 70.6± 4.9



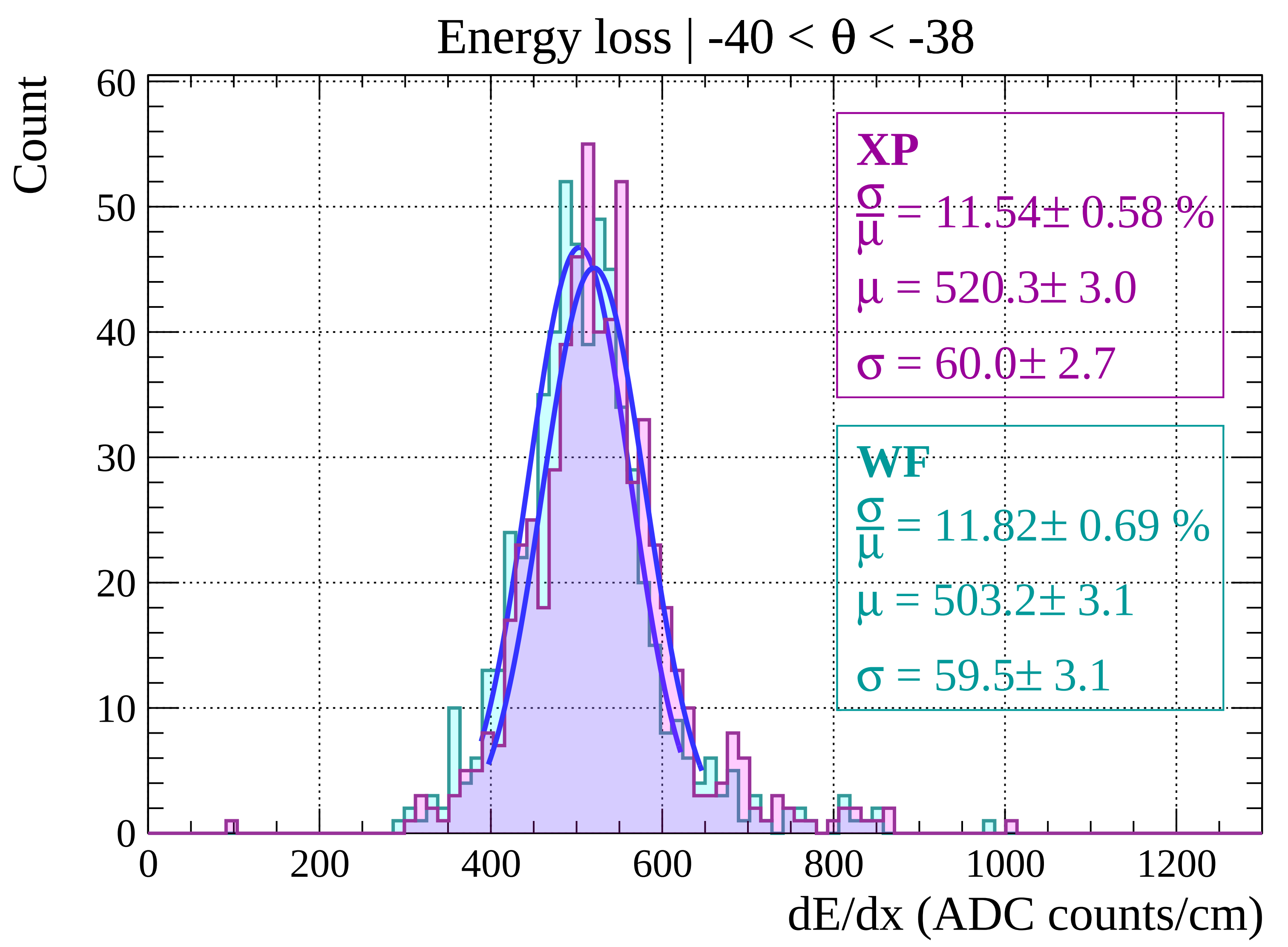
Energy loss | -46 < θ < -44
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP
σ/μ = 11.64± 0.84 %
μ = 531.0± 3.8
σ = 61.8± 4.0
WF
σ/μ = 11.40± 0.72 %
μ = 508.8± 3.5
σ = 58.0± 3.2



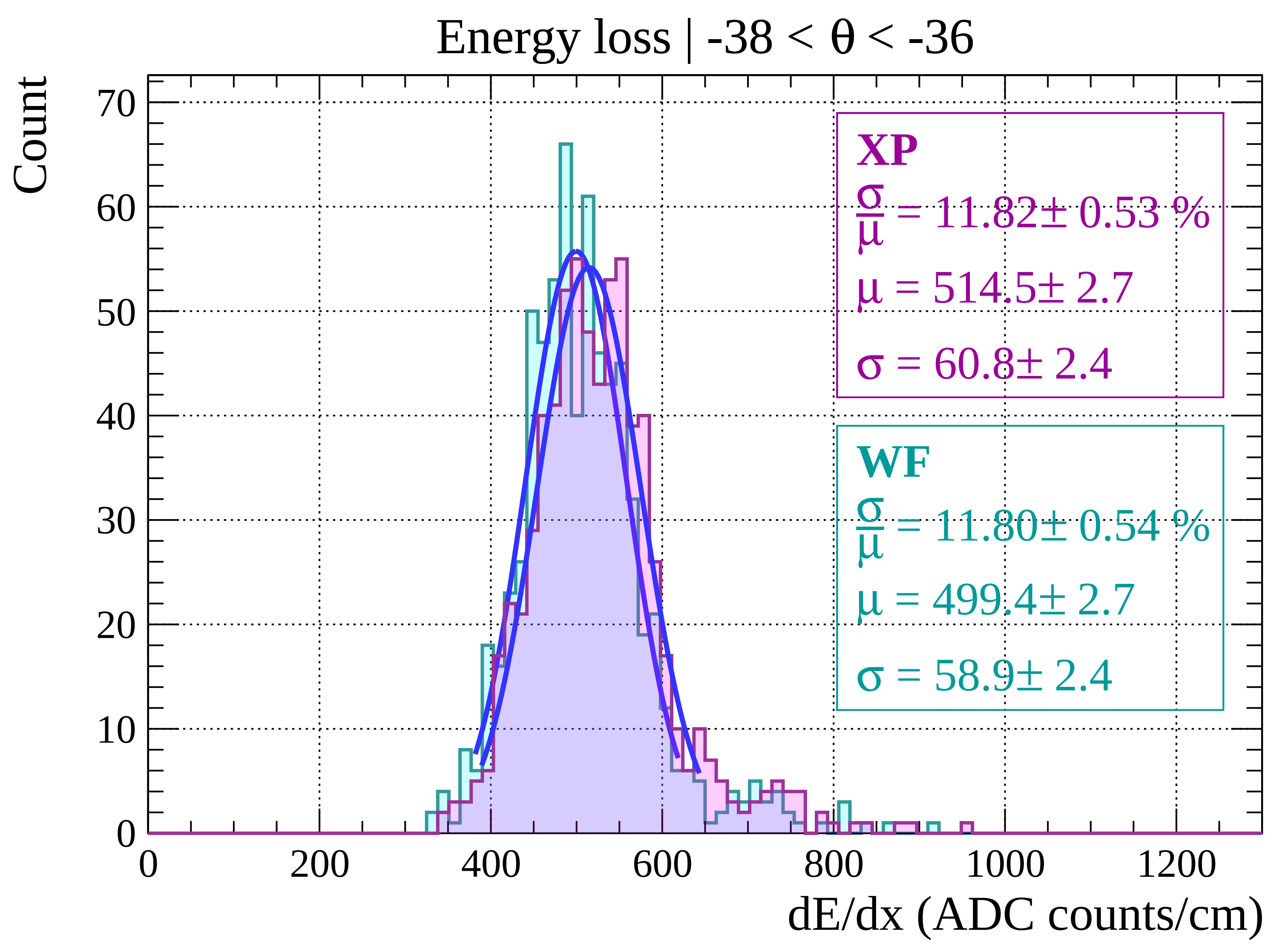
Energy loss | -44 < θ < -42
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.62± 0.80 %
μ = 531.8± 3.5
σ = 56.5± 3.9
WF
σ/μ = 10.05± 0.75 %
μ = 513.5± 3.2
σ = 51.6± 3.5
0
10
20
30
40
50
0
200
400
600
800
1000
1200



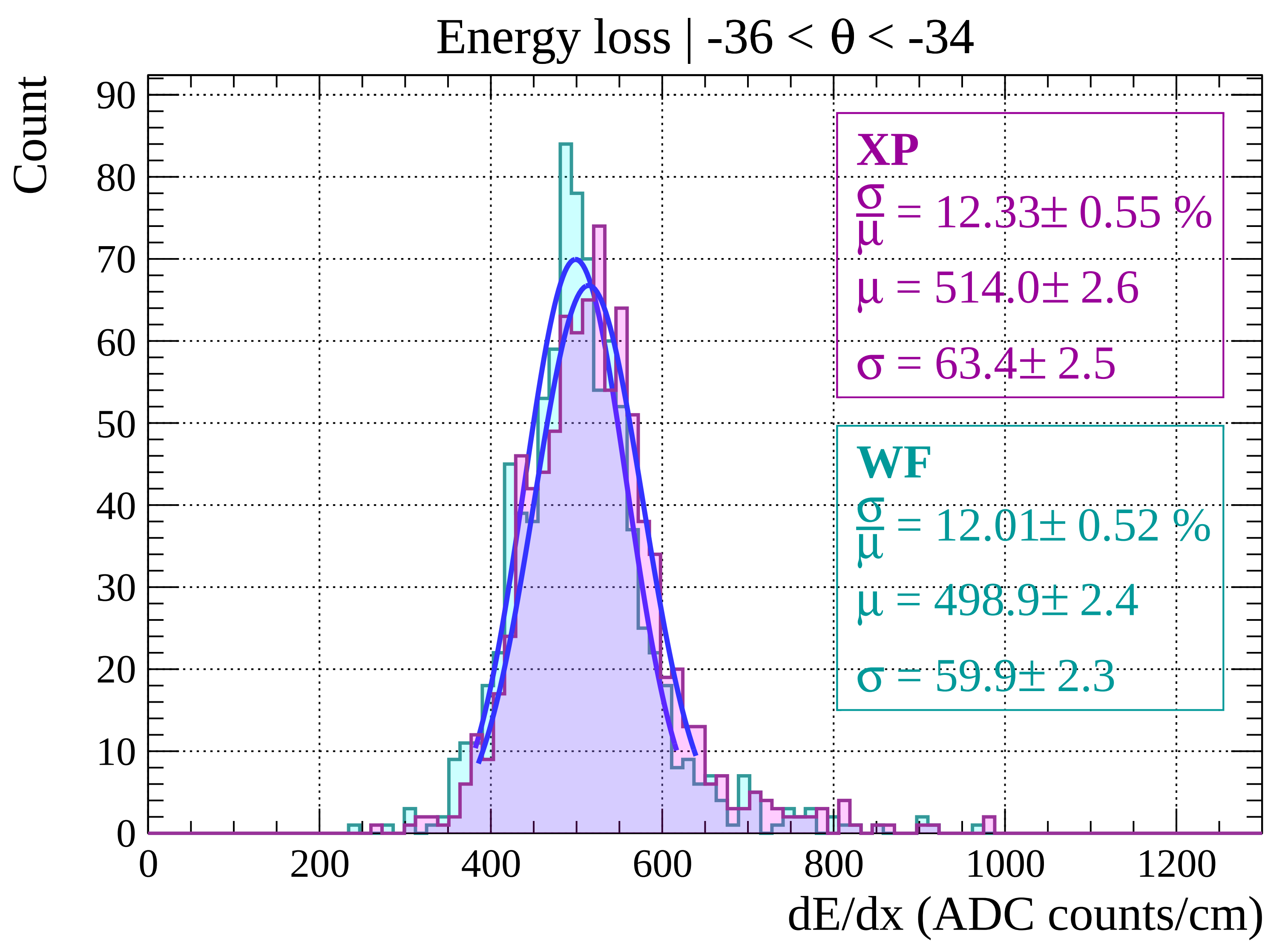
Energy loss | -42 < θ < -40
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 12.63± 0.73 %
μ = 520.3± 3.5
σ = 65.7± 3.4
WF
σ/μ = 11.88± 0.56 %
μ = 502.8± 3.0
σ = 59.7± 2.4



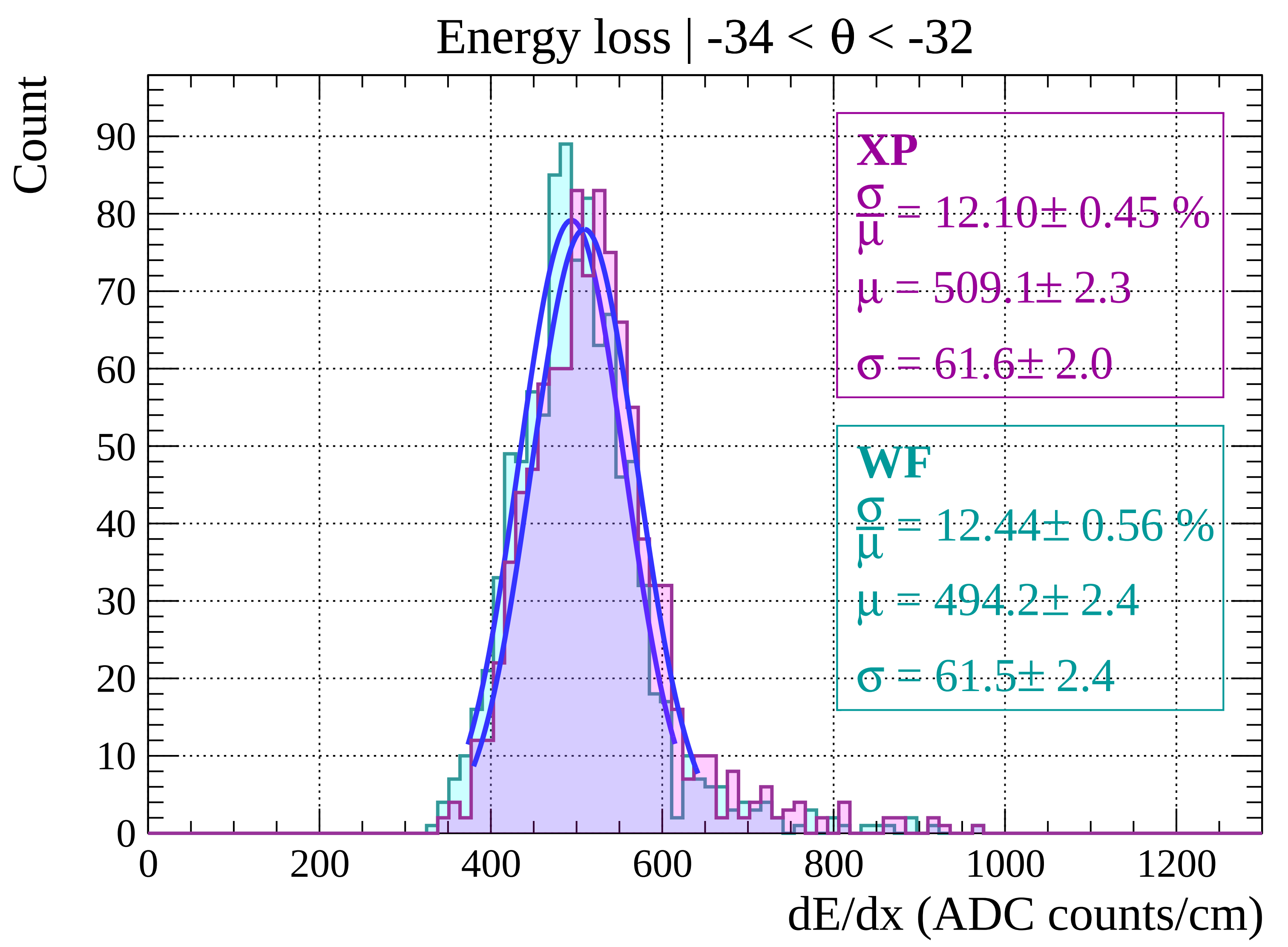
Energy loss | -40 < θ < -38
Count
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.54± 0.58 %
μ = 520.3± 3.0
σ = 60.0± 2.7
WF
$\frac{\sigma}{\mu}$ = 11.82± 0.69 %
μ = 503.2± 3.1
σ = 59.5± 3.1



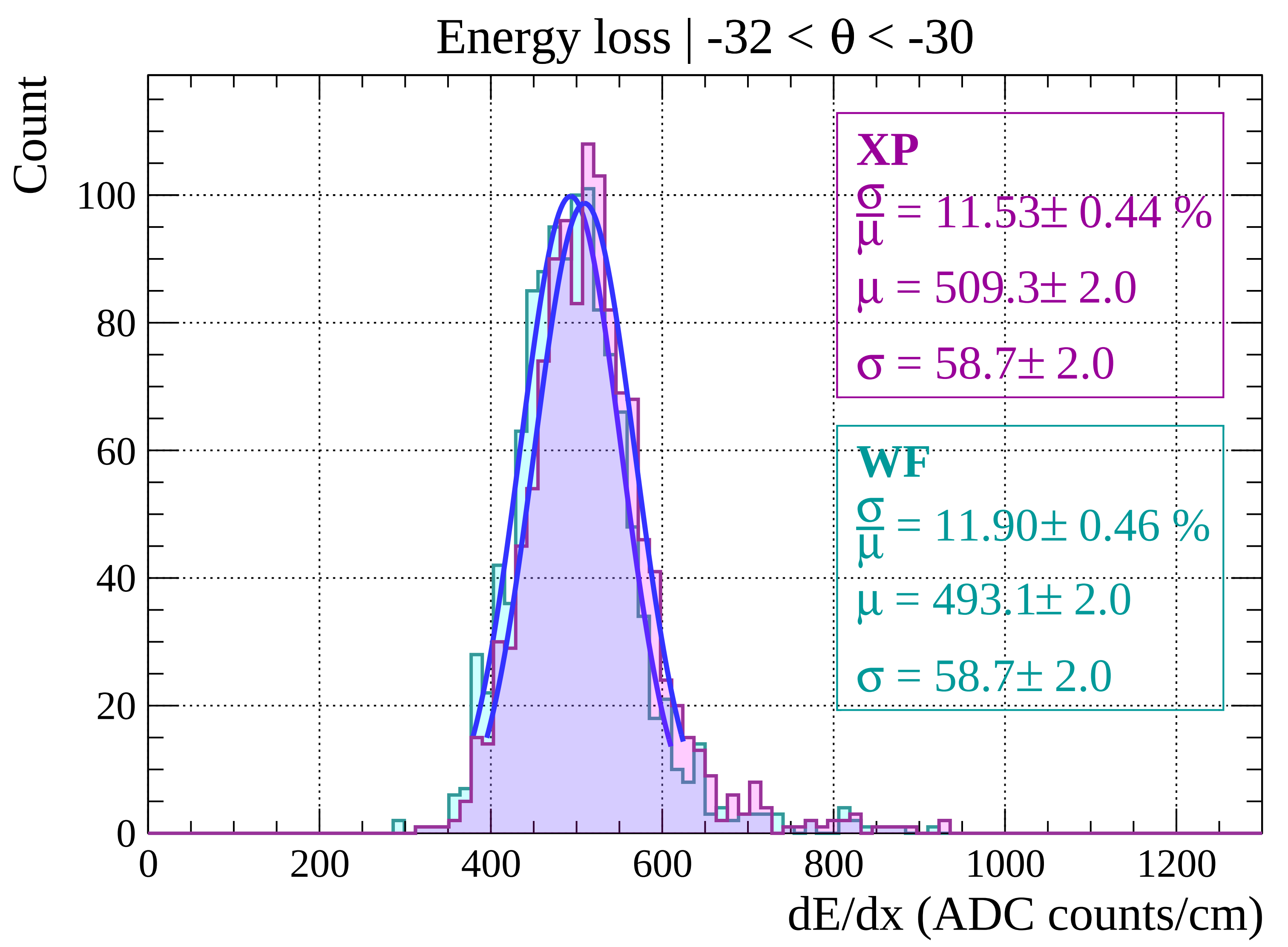
Energy loss | -38 < θ < -36
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP
σ/μ = 11.82 ± 0.53 %
μ = 514.5 ± 2.7
σ = 60.8 ± 2.4
WF
σ/μ = 11.80 ± 0.54 %
μ = 499.4 ± 2.7
σ = 58.9 ± 2.4



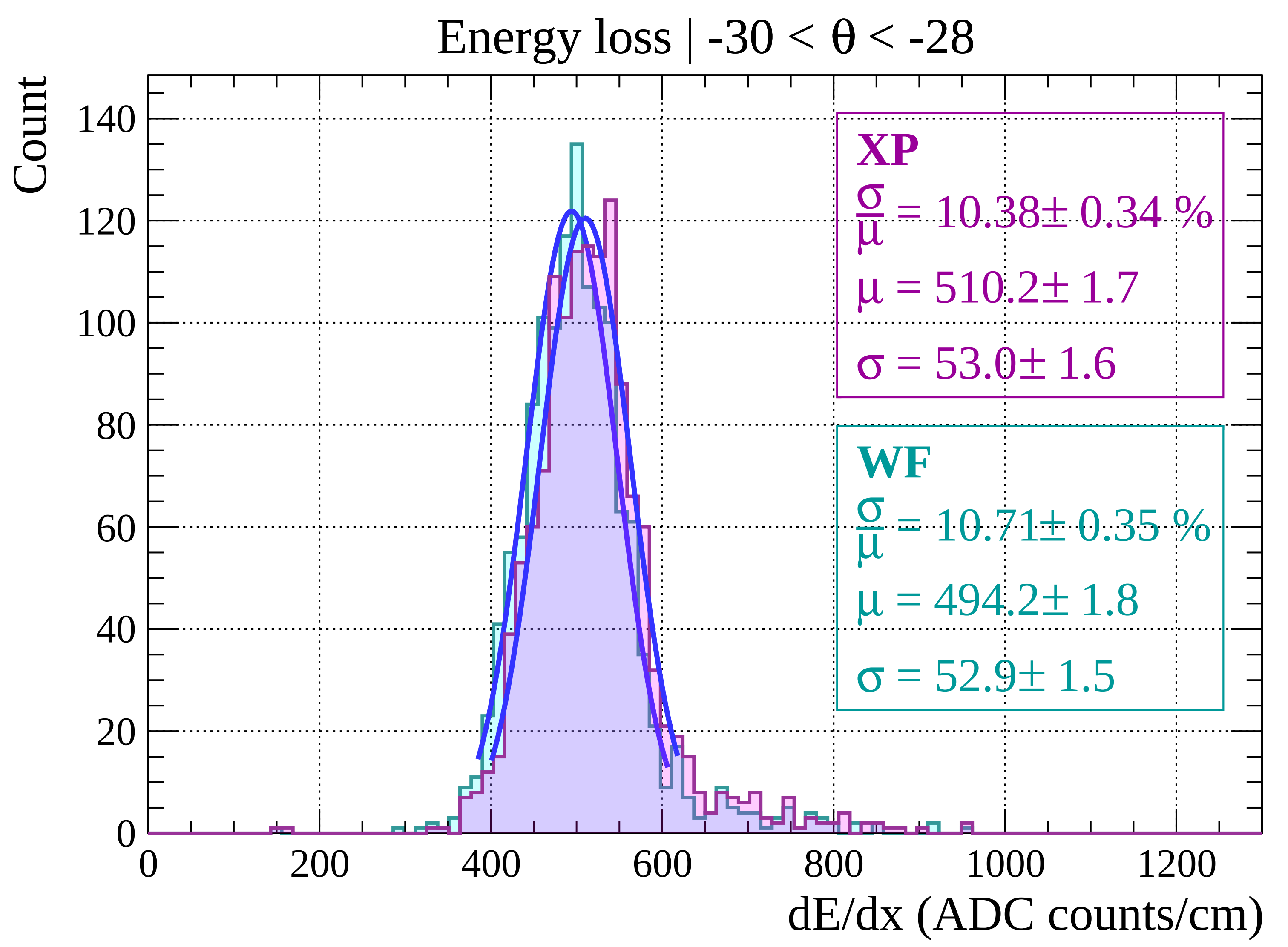
Energy loss | -36 < θ < -34
Count
90
80
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.33± 0.55 %
μ = 514.0± 2.6
σ = 63.4± 2.5
WF
σ/μ = 12.01± 0.52 %
μ = 498.9± 2.4
σ = 59.9± 2.3



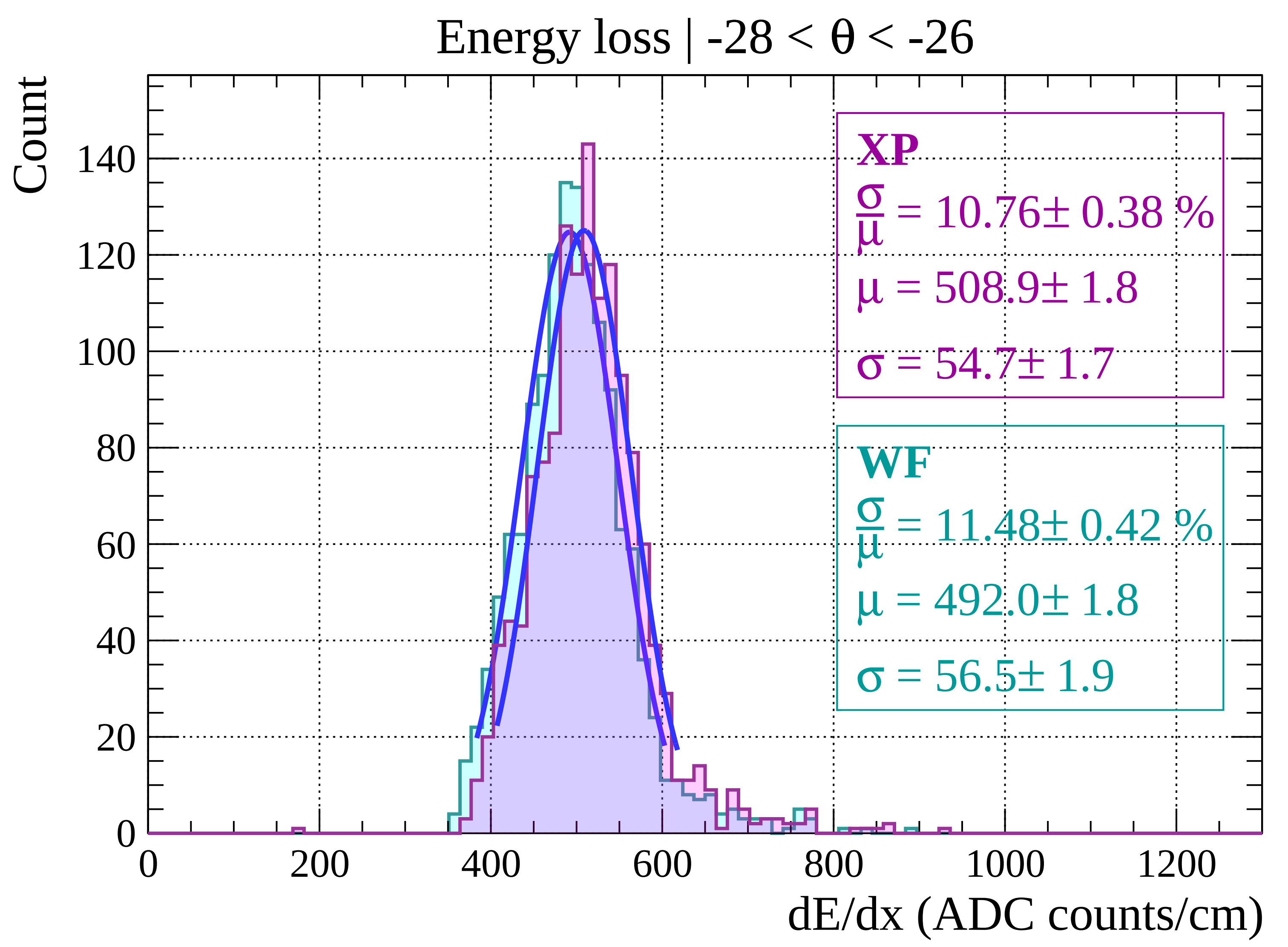
Energy loss | -34 < θ < -32
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 12.10± 0.45 %
μ = 509.1± 2.3
σ = 61.6± 2.0
WF
σ/μ = 12.44± 0.56 %
μ = 494.2± 2.4
σ = 61.5± 2.4



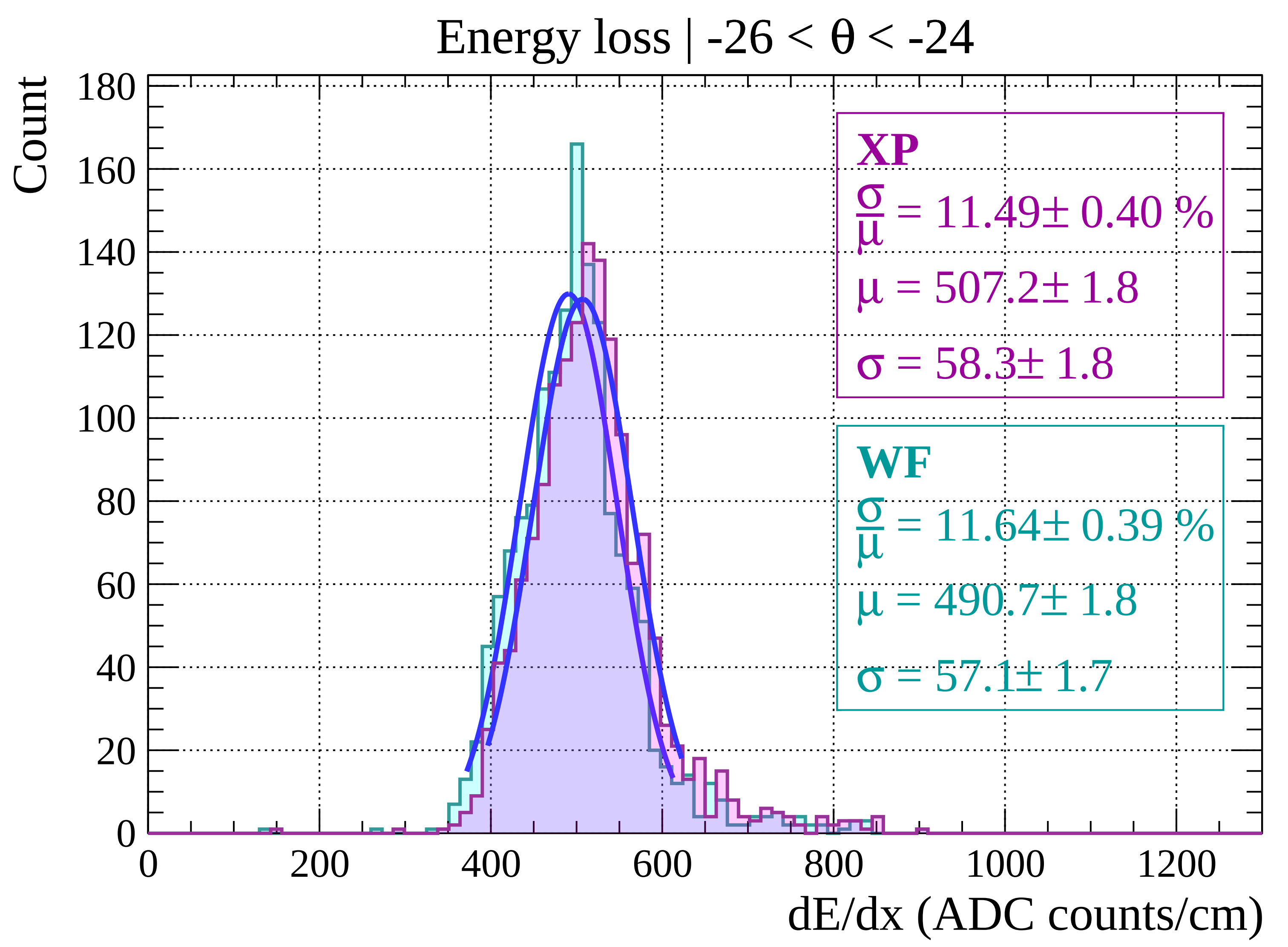
Energy loss | -32 < θ < -30
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 11.53± 0.44 %
μ = 509.3± 2.0
σ = 58.7± 2.0
WF
σ/μ = 11.90± 0.46 %
μ = 493.1± 2.0
σ = 58.7± 2.0

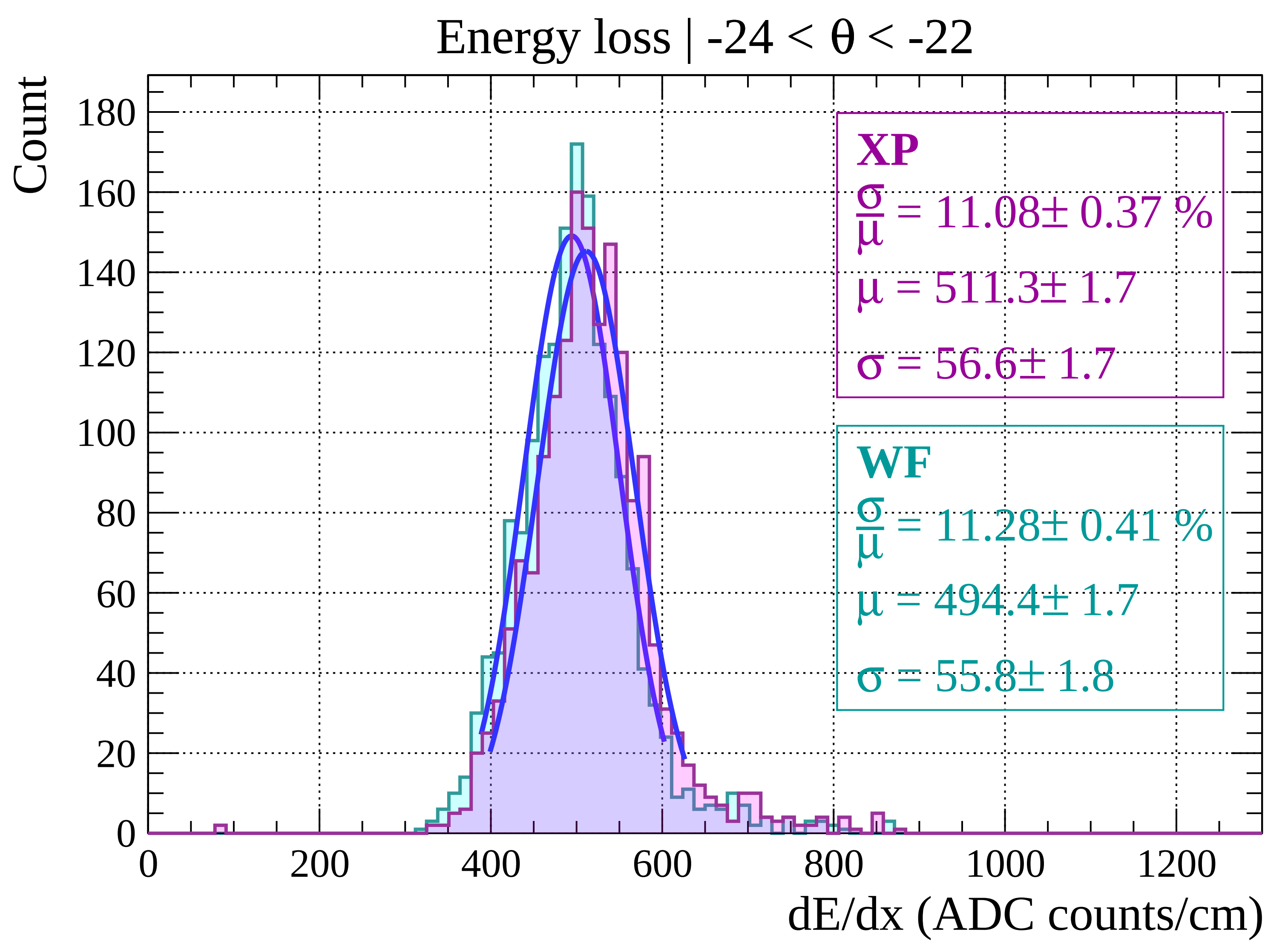


Energy loss | -30 < θ < -28
Count
XP
σ/μ = 10.38± 0.34 %
μ = 510.2± 1.7
σ = 53.0± 1.6
WF
σ/μ = 10.71± 0.35 %
μ = 494.2± 1.8
σ = 52.9± 1.5
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

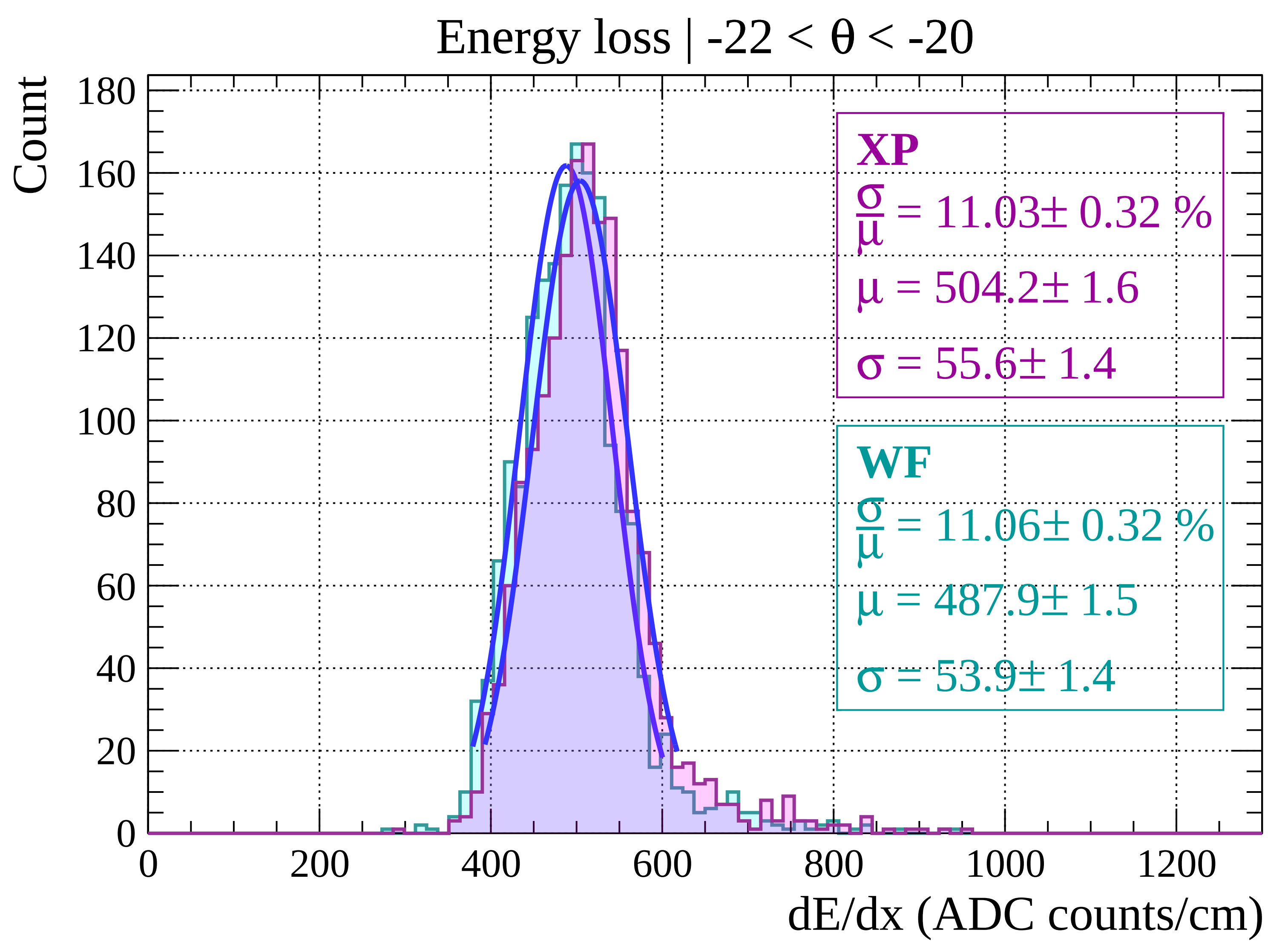


Energy loss | -28 < θ < -26
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 10.76± 0.38 %
μ = 508.9± 1.8
σ = 54.7± 1.7
WF
σ/μ = 11.48± 0.42 %
μ = 492.0± 1.8
σ = 56.5± 1.9

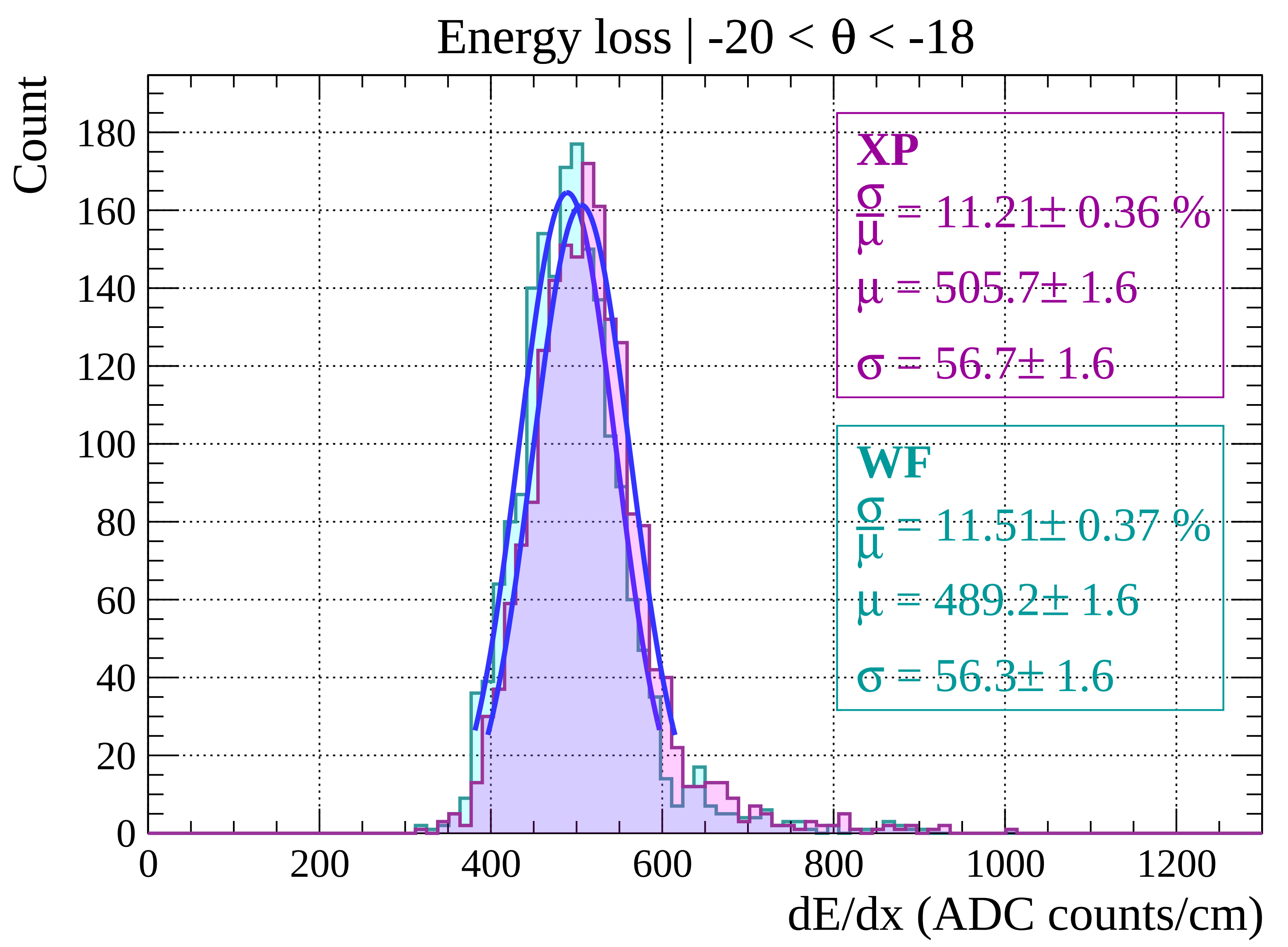

Energy loss | -26 < θ < -24
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
σ/μ = 11.49± 0.40 %
μ = 507.2± 1.8
σ = 58.3± 1.8
WF
σ/μ = 11.64± 0.39 %
μ = 490.7± 1.8
σ = 57.1± 1.7

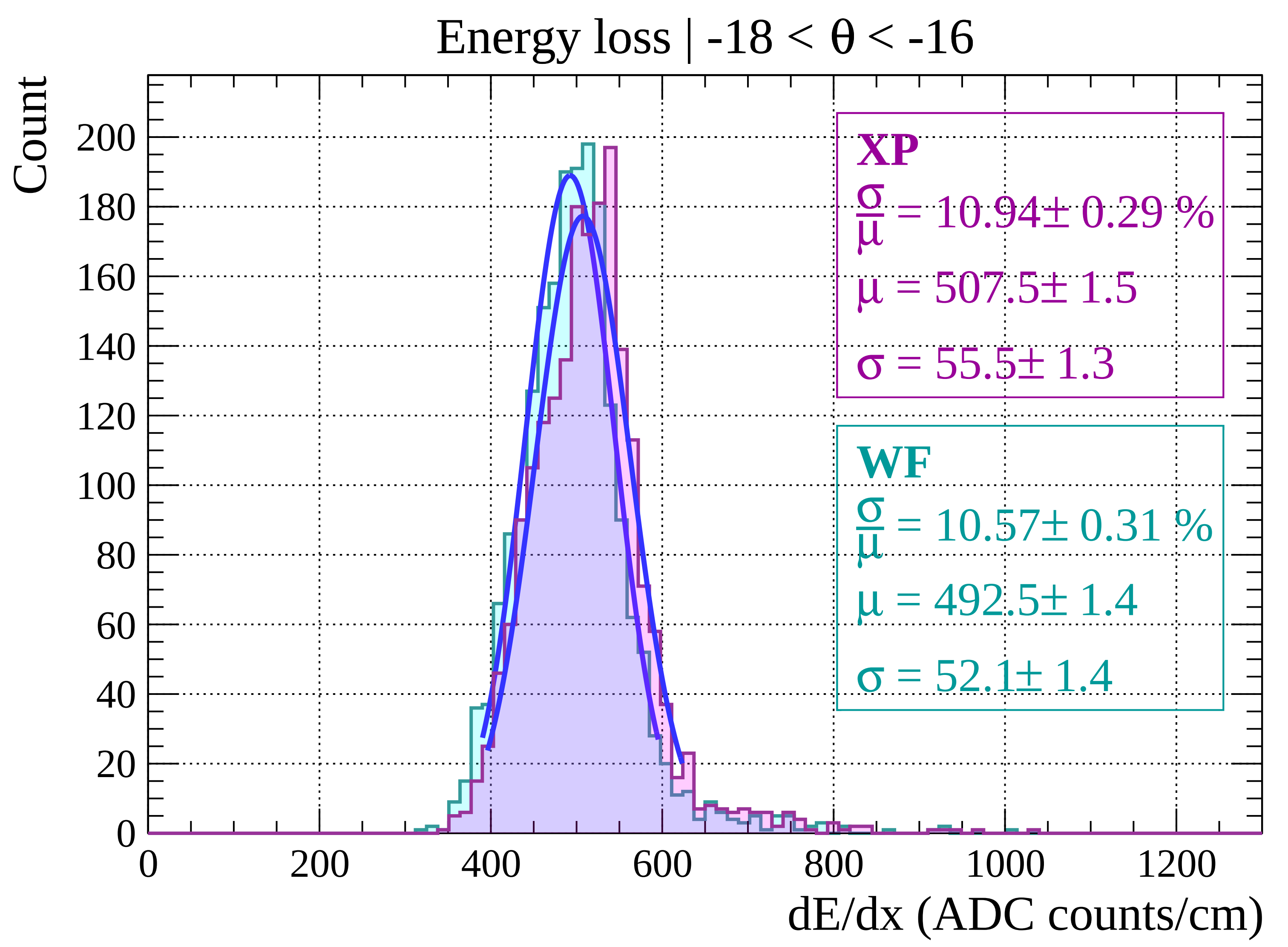


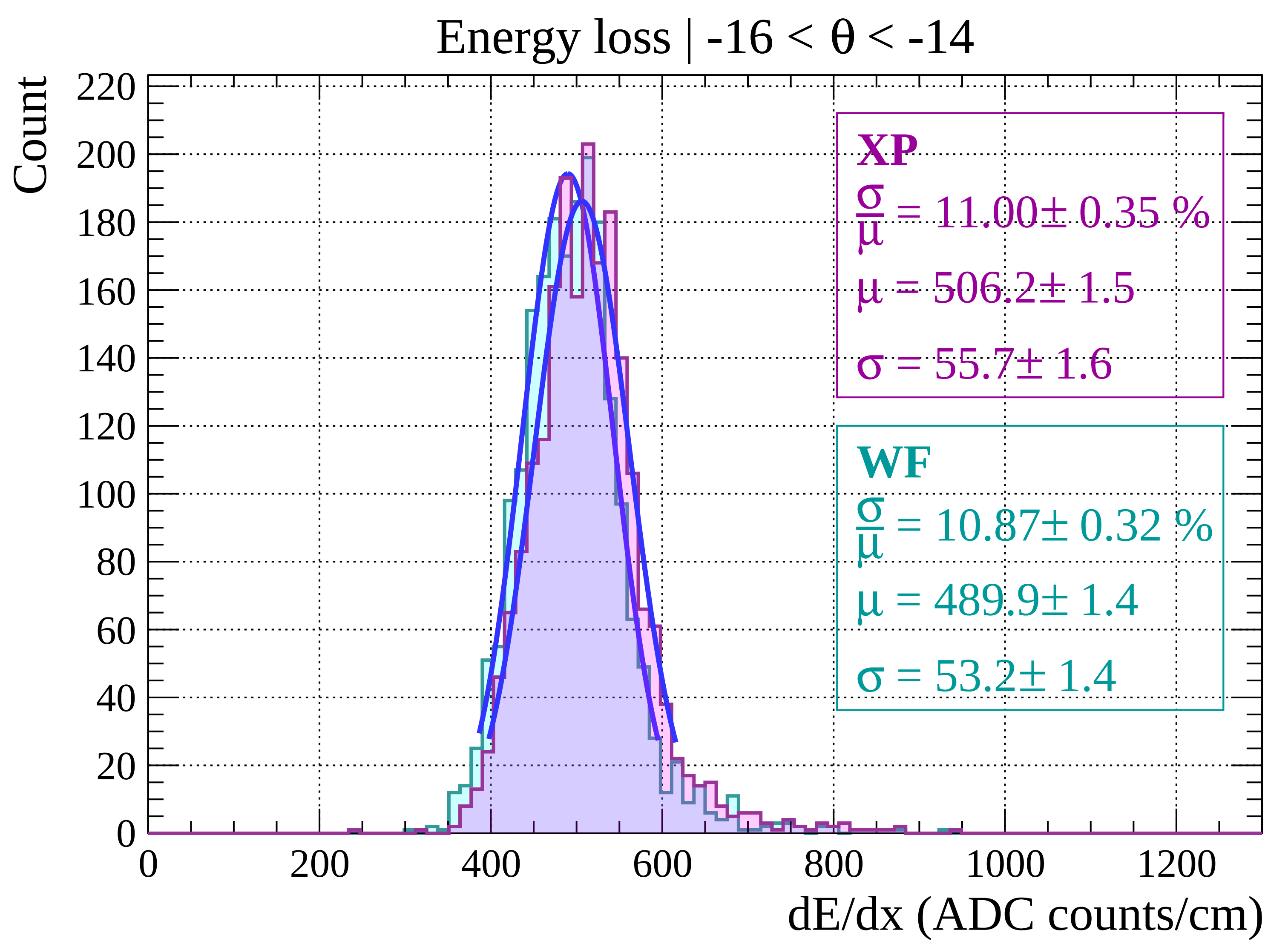
Energy loss | -24 < θ < -22
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.08± 0.37 %
μ = 511.3± 1.7
σ = 56.6± 1.7
WF
σ/μ = 11.28± 0.41 %
μ = 494.4± 1.7
σ = 55.8± 1.8
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200



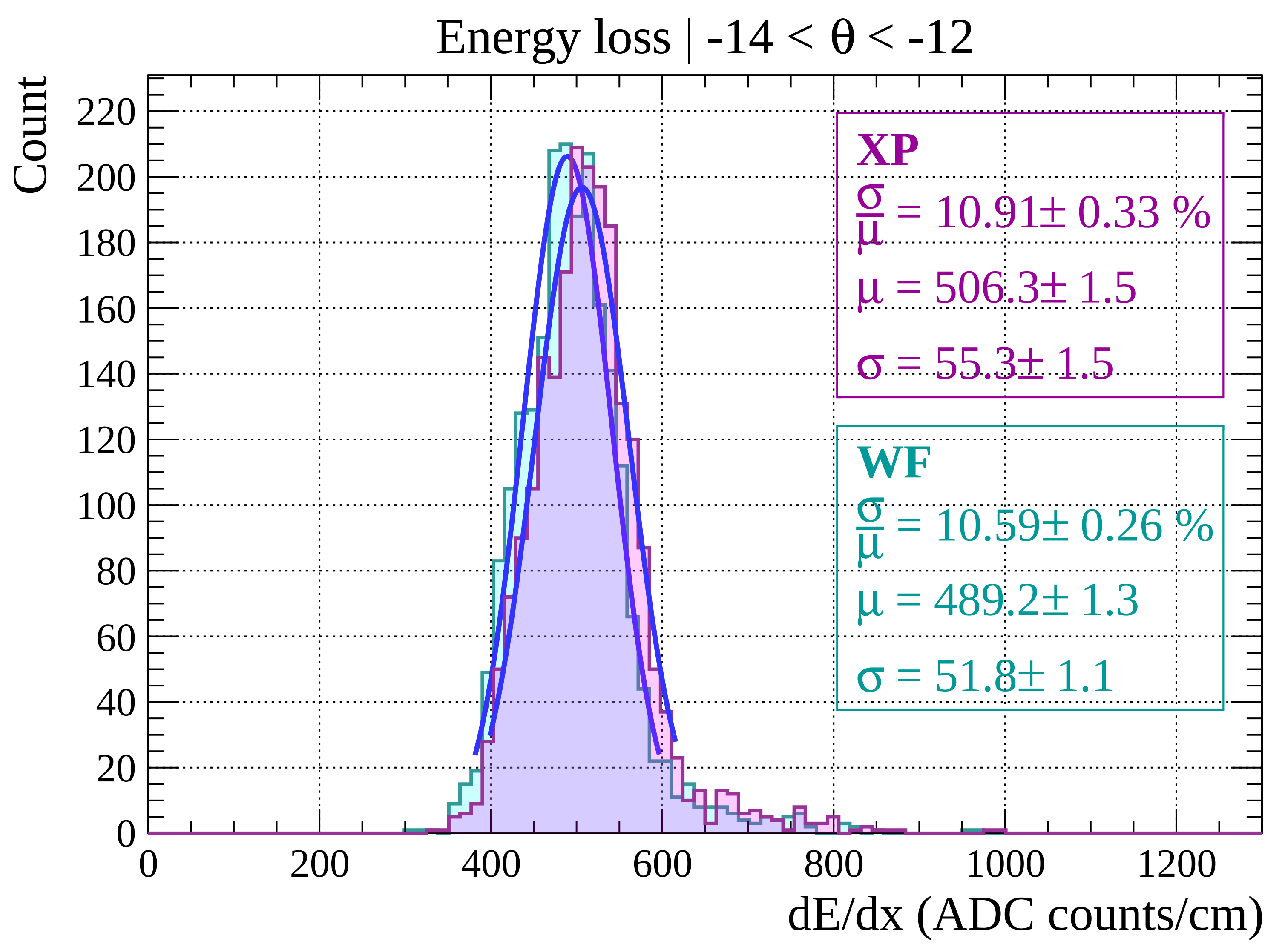
Energy loss | -22 < θ < -20
Count
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.03± 0.32 %
μ = 504.2± 1.6
σ = 55.6± 1.4
WF
σ/μ = 11.06± 0.32 %
μ = 487.9± 1.5
σ = 53.9± 1.4

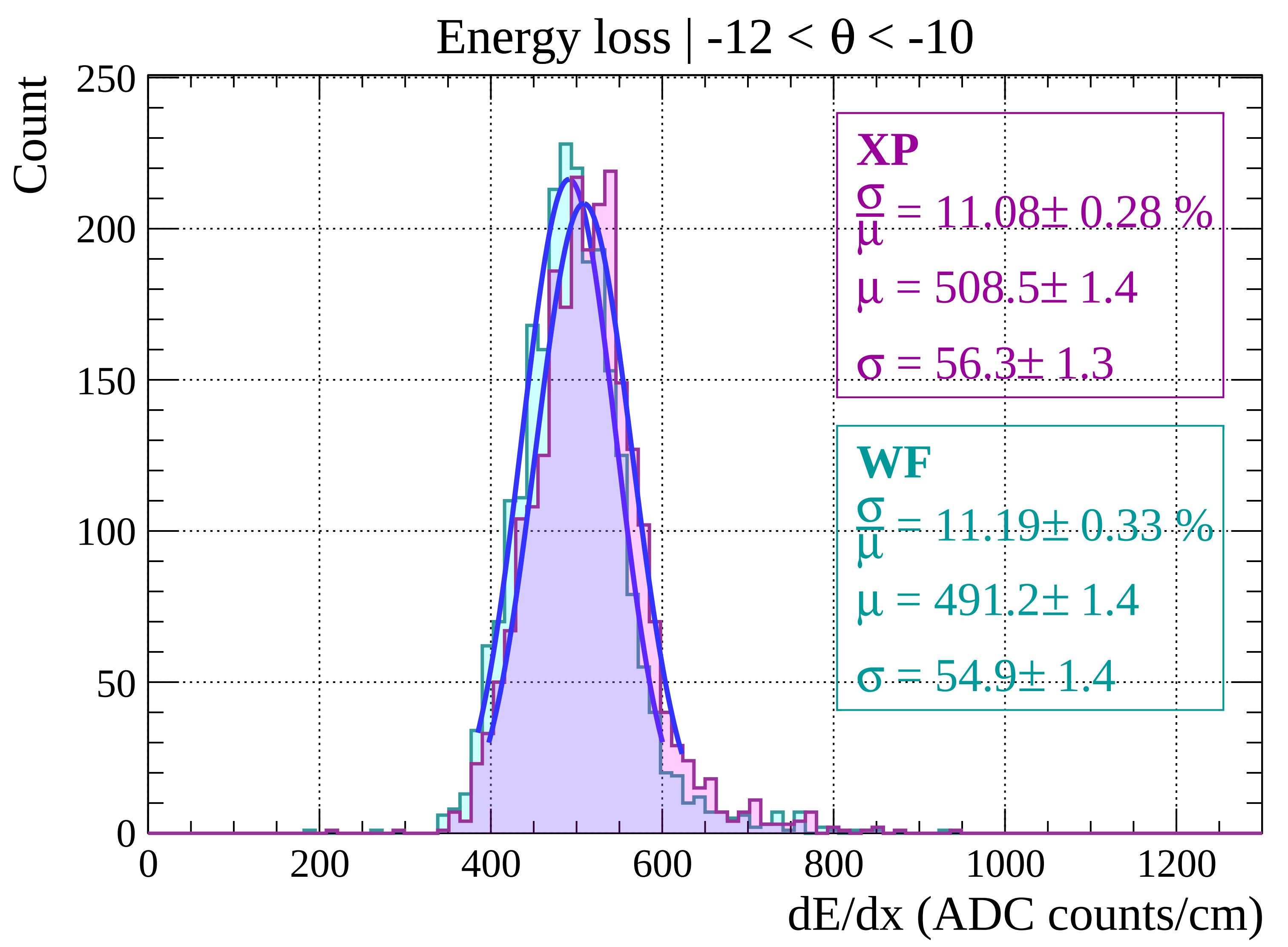

Energy loss | -20 < θ < -18
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
σ/μ = 11.21± 0.36 %
μ = 505.7± 1.6
σ = 56.7± 1.6
WF
σ/μ = 11.51± 0.37 %
μ = 489.2± 1.6
σ = 56.3± 1.6


Energy loss | -18 < θ < -16
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.94± 0.29 %
μ = 507.5± 1.5
σ = 55.5± 1.3
WF
$\frac{\sigma}{\mu}$ = 10.57± 0.31 %
μ = 492.5± 1.4
σ = 52.1± 1.4
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200

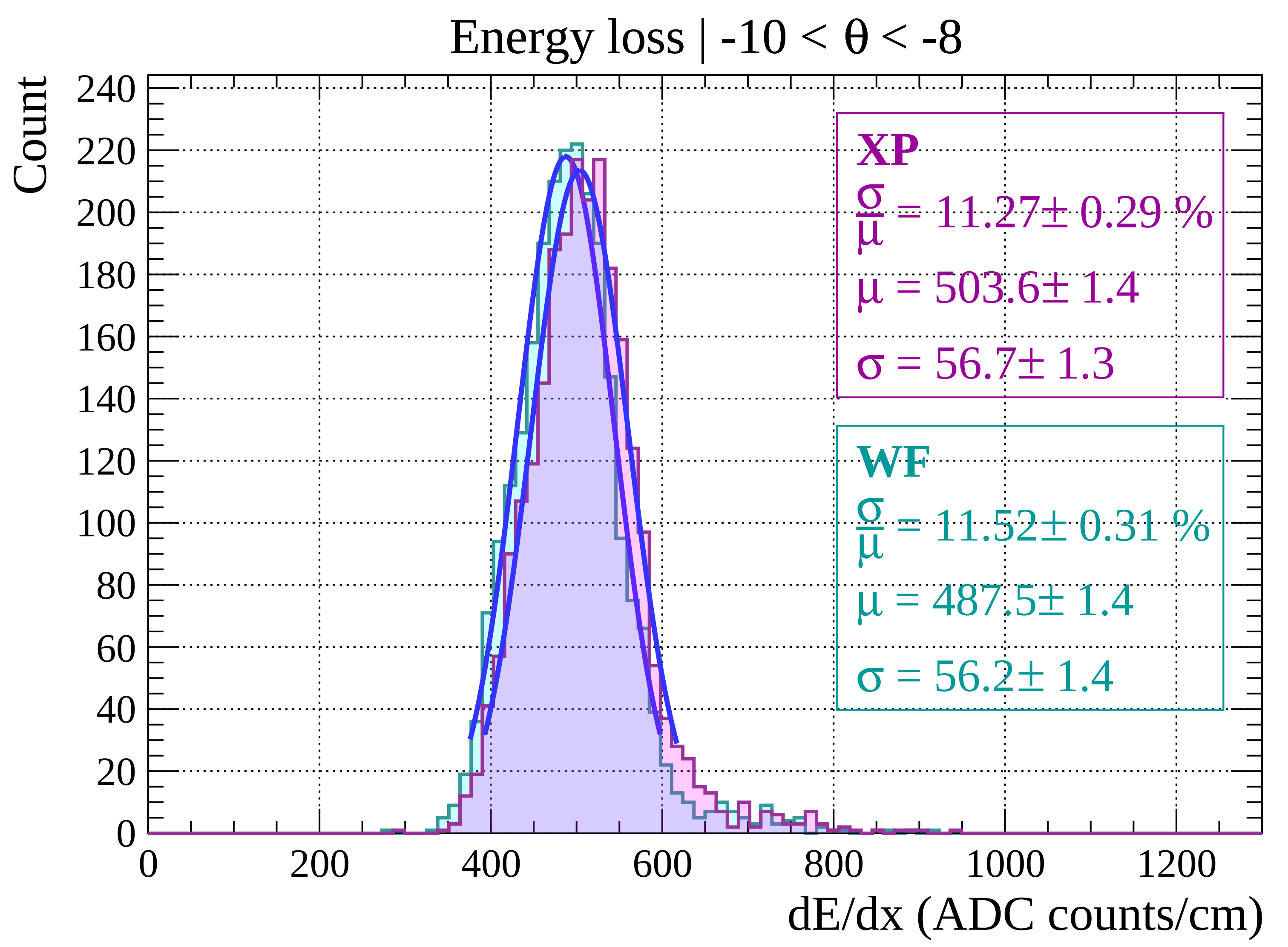


Energy loss | -16 < θ < -14
Count
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.00± 0.35 %
μ = 506.2± 1.5
σ = 55.7± 1.6
WF
σ/μ = 10.87± 0.32 %
μ = 489.9± 1.4
σ = 53.2± 1.4

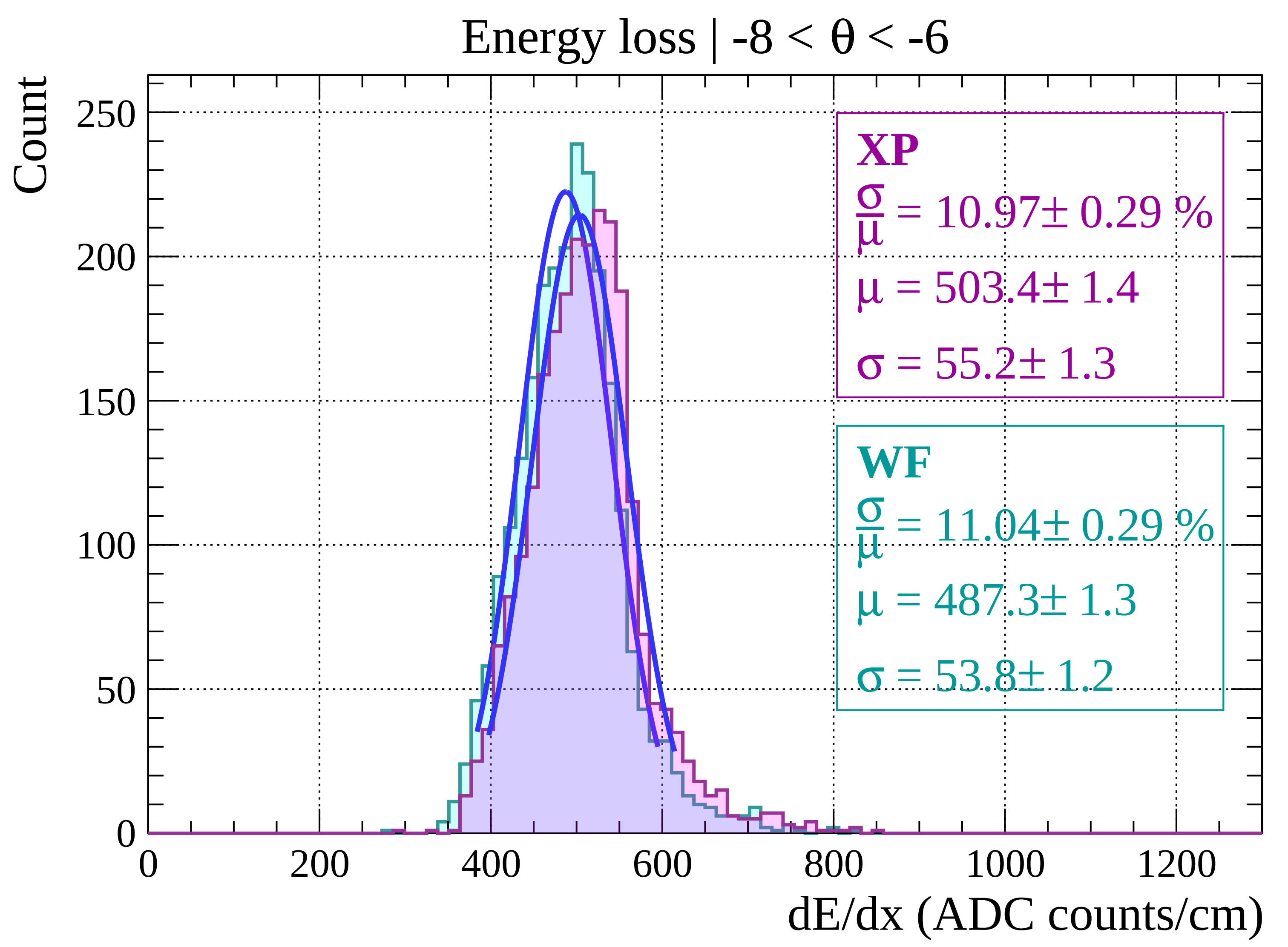

Energy loss | -14 < θ < -12
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.91± 0.33 %
μ = 506.3± 1.5
σ = 55.3± 1.5
WF
$\frac{\sigma}{\mu}$ = 10.59± 0.26 %
μ = 489.2± 1.3
σ = 51.8± 1.1
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220



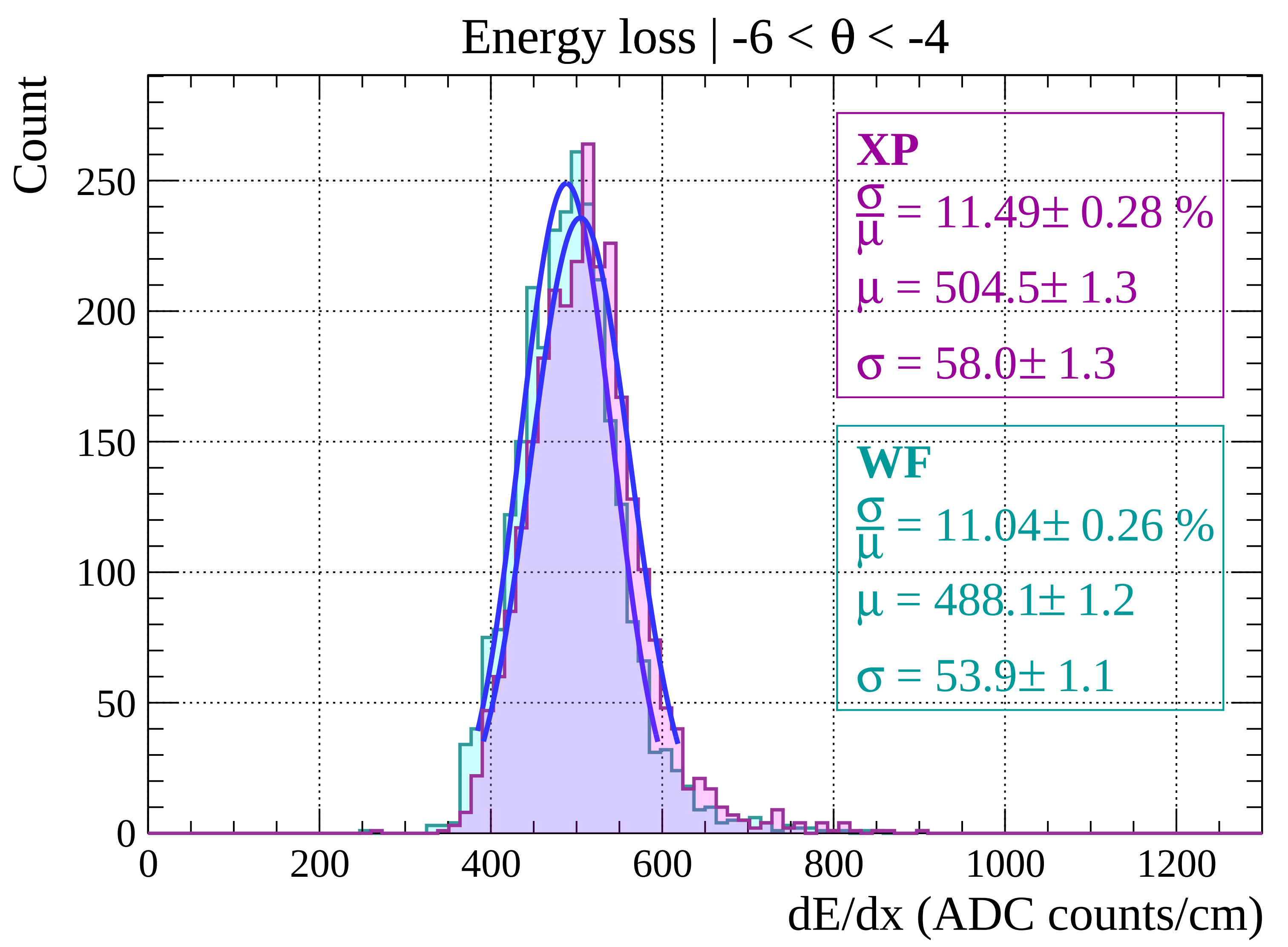
Energy loss | -12 < θ < -10
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
XP
$\frac{\sigma}{\mu}$ = 11.08± 0.28 %
μ = 508.5± 1.4
σ = 56.3± 1.3
WF
$\frac{\sigma}{\mu}$ = 11.19± 0.33 %
μ = 491.2± 1.4
σ = 54.9± 1.4



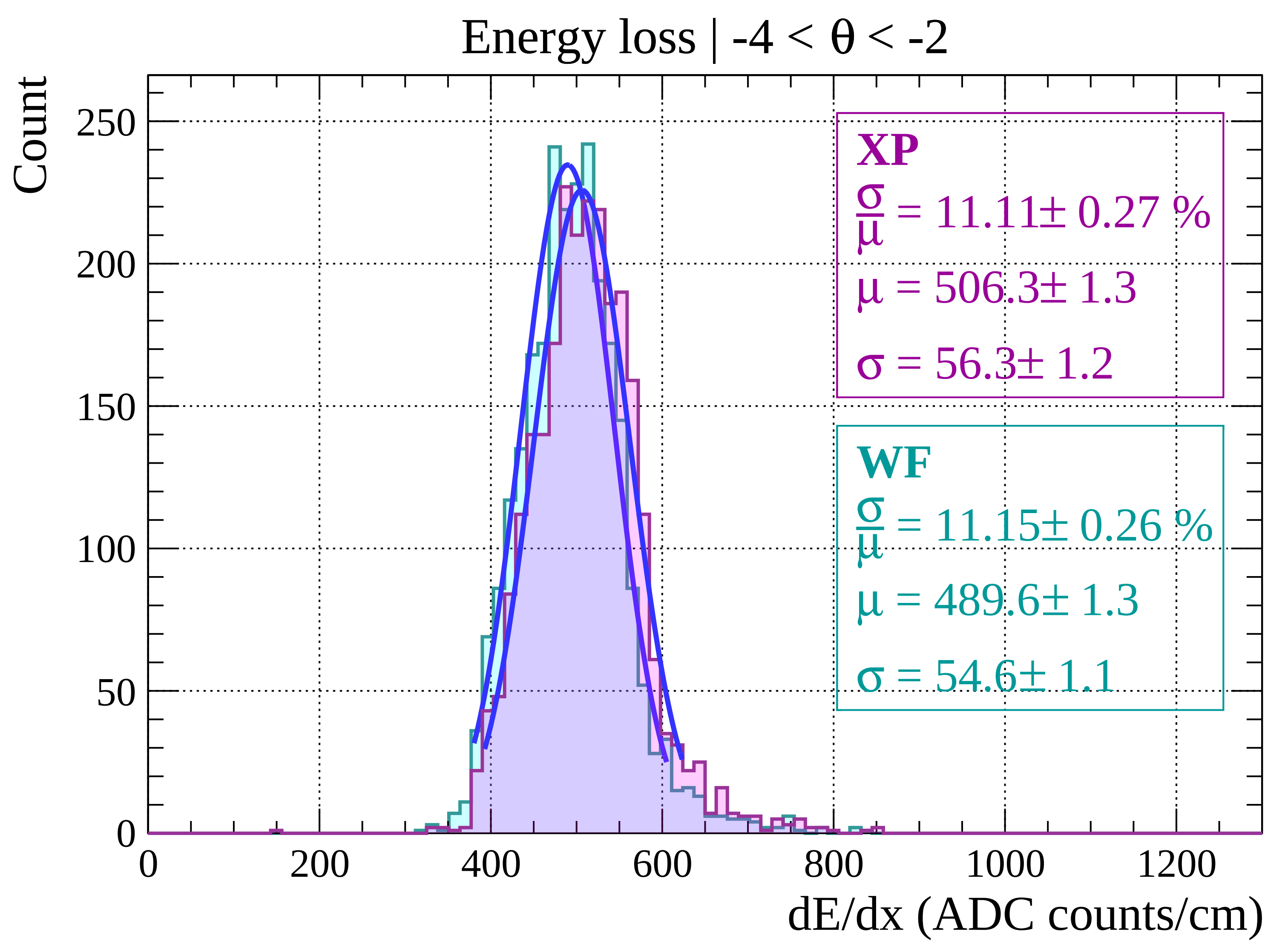
Energy loss | -10 < θ < -8
Count
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.27± 0.29 %
μ = 503.6± 1.4
σ = 56.7± 1.3
WF
$\frac{\sigma}{\mu}$ = 11.52± 0.31 %
μ = 487.5± 1.4
σ = 56.2± 1.4



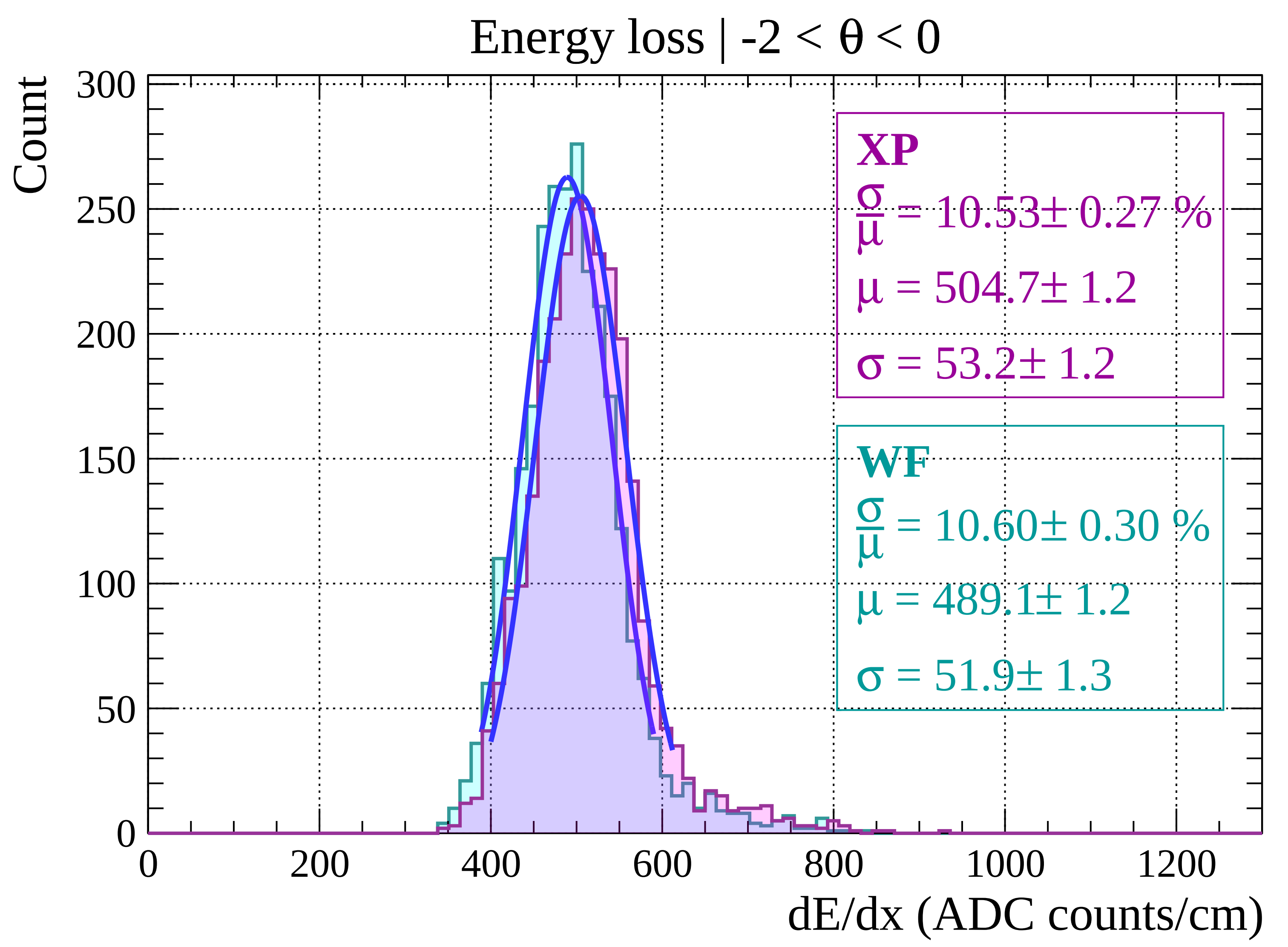
Energy loss | -8 < θ < -6
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.97± 0.29 %
μ = 503.4± 1.4
σ = 55.2± 1.3
WF
$\frac{\sigma}{\mu}$ = 11.04± 0.29 %
μ = 487.3± 1.3
σ = 53.8± 1.2

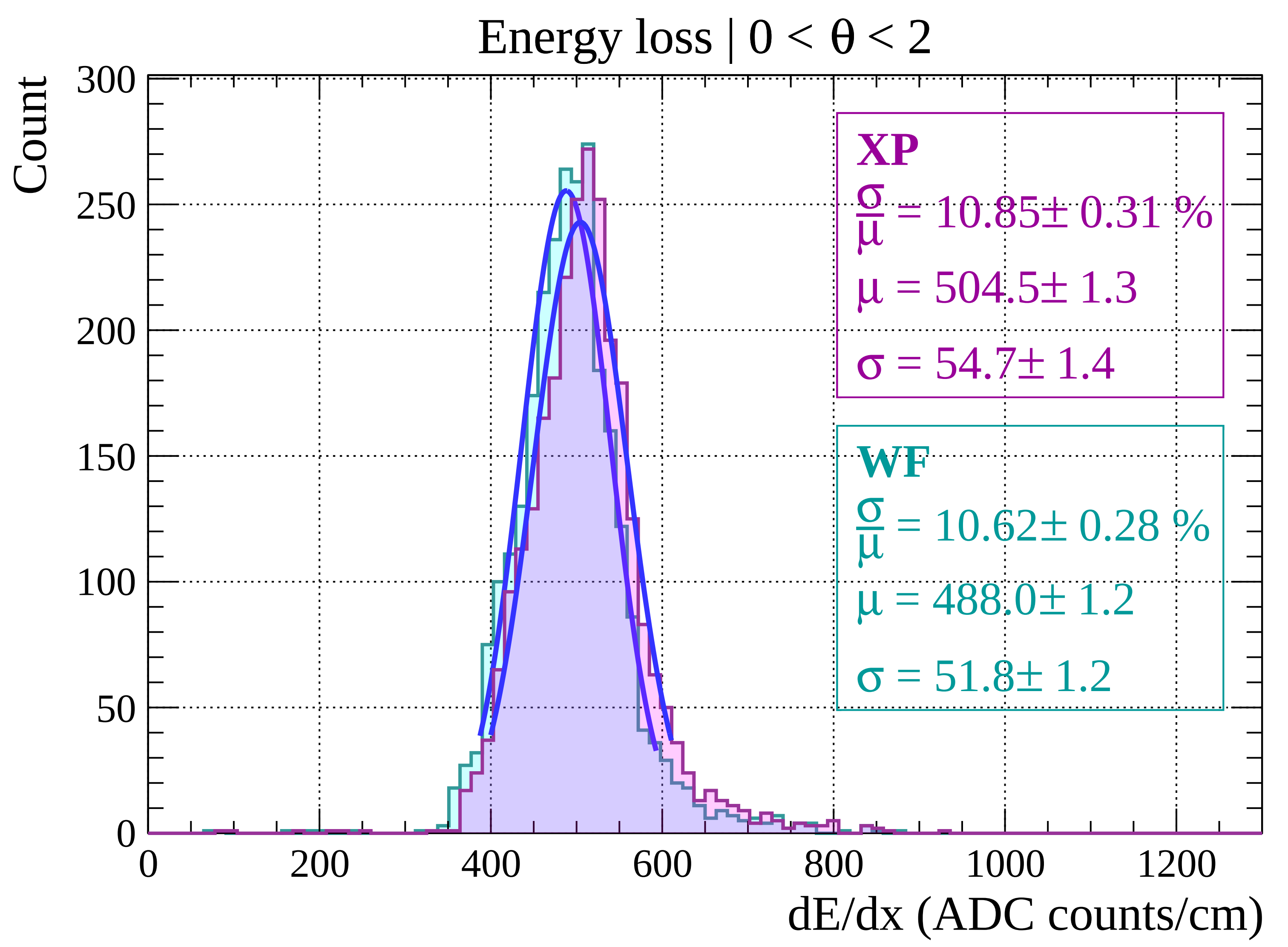


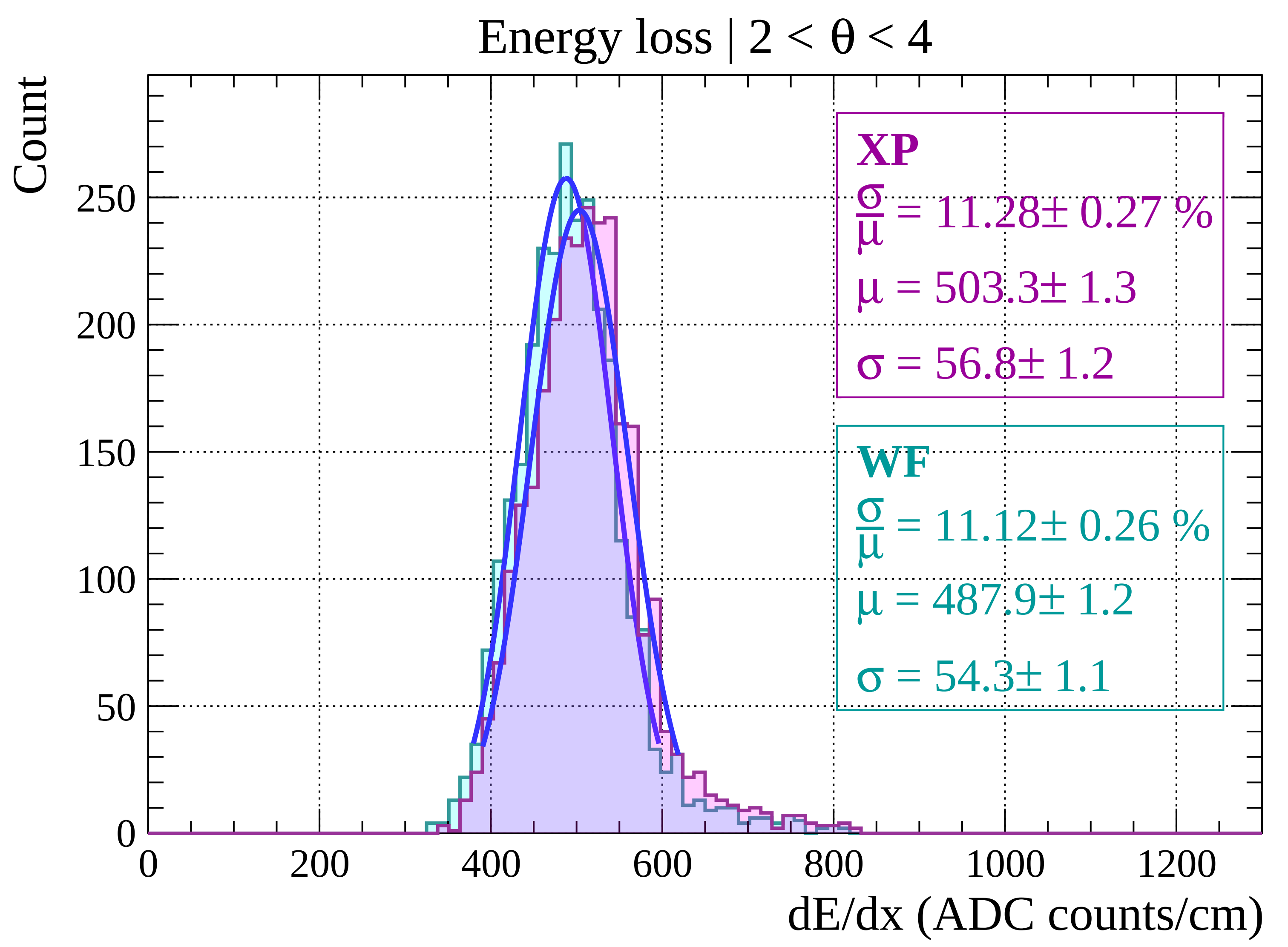
Energy loss | -6 < θ < -4
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 11.49± 0.28 %
μ = 504.5± 1.3
σ = 58.0± 1.3
WF
σ/μ = 11.04± 0.26 %
μ = 488.1± 1.2
σ = 53.9± 1.1



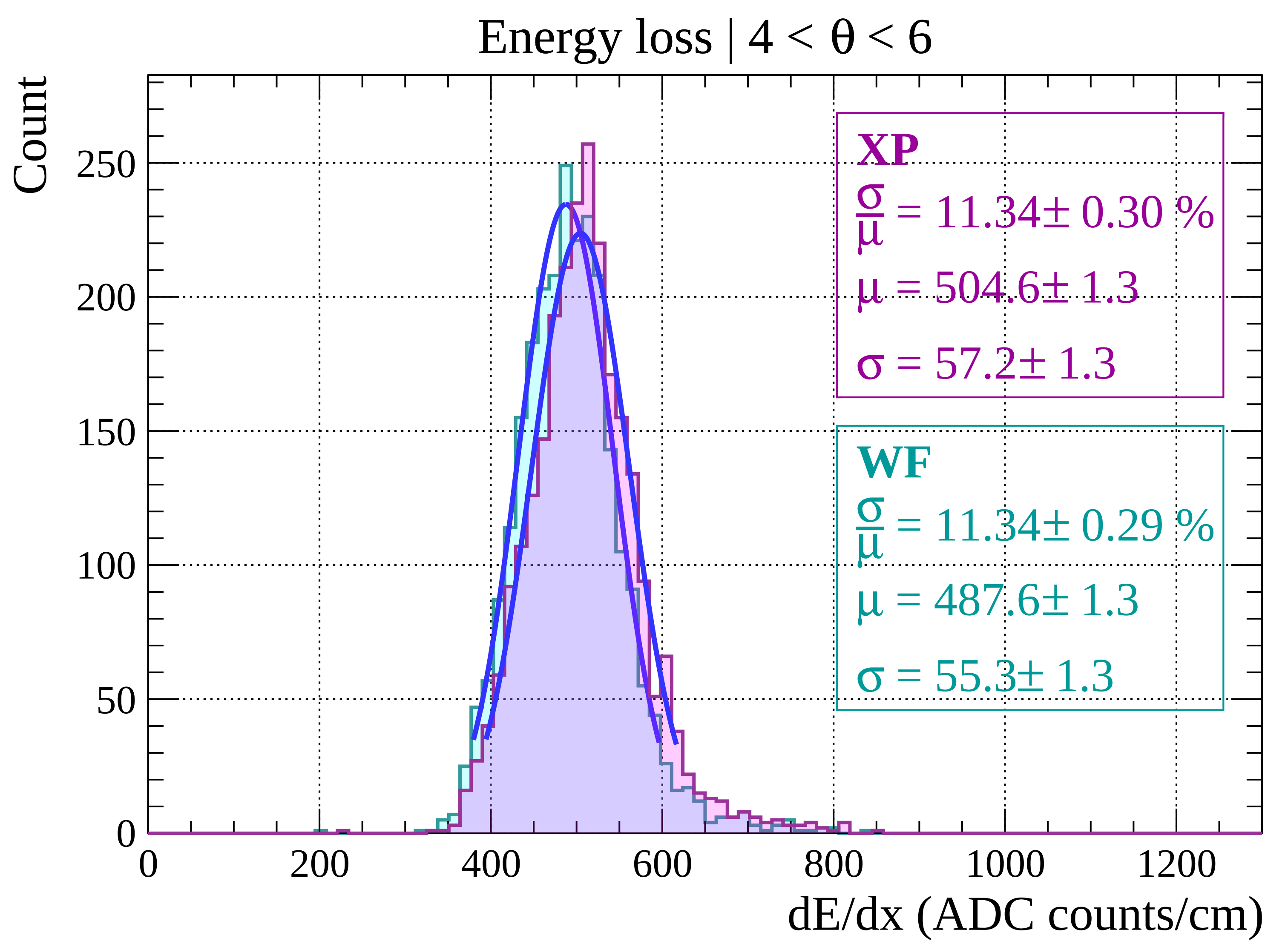
Energy loss | -4 < θ < -2
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 11.11± 0.27 %
μ = 506.3± 1.3
σ = 56.3± 1.2
WF
σ/μ = 11.15± 0.26 %
μ = 489.6± 1.3
σ = 54.6± 1.1

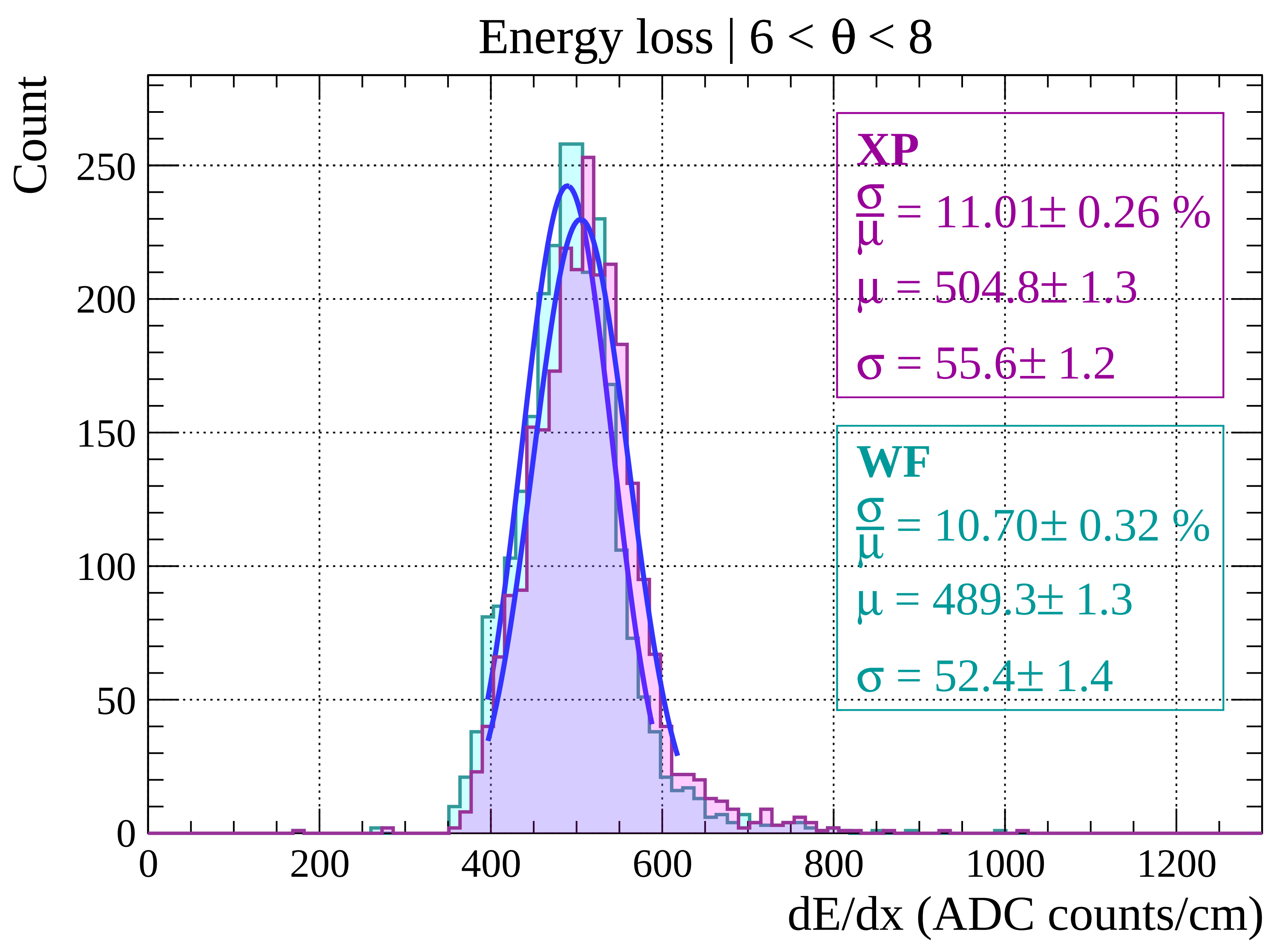

Energy loss | -2 < θ < 0
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.53± 0.27 %
μ = 504.7± 1.2
σ = 53.2± 1.2
WF
$\frac{\sigma}{\mu}$ = 10.60± 0.30 %
μ = 489.1± 1.2
σ = 51.9± 1.3


Energy loss | 0 < θ < 2
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 10.85± 0.31 %
μ = 504.5± 1.3
σ = 54.7± 1.4
WF
σ/μ = 10.62± 0.28 %
μ = 488.0± 1.2
σ = 51.8± 1.2

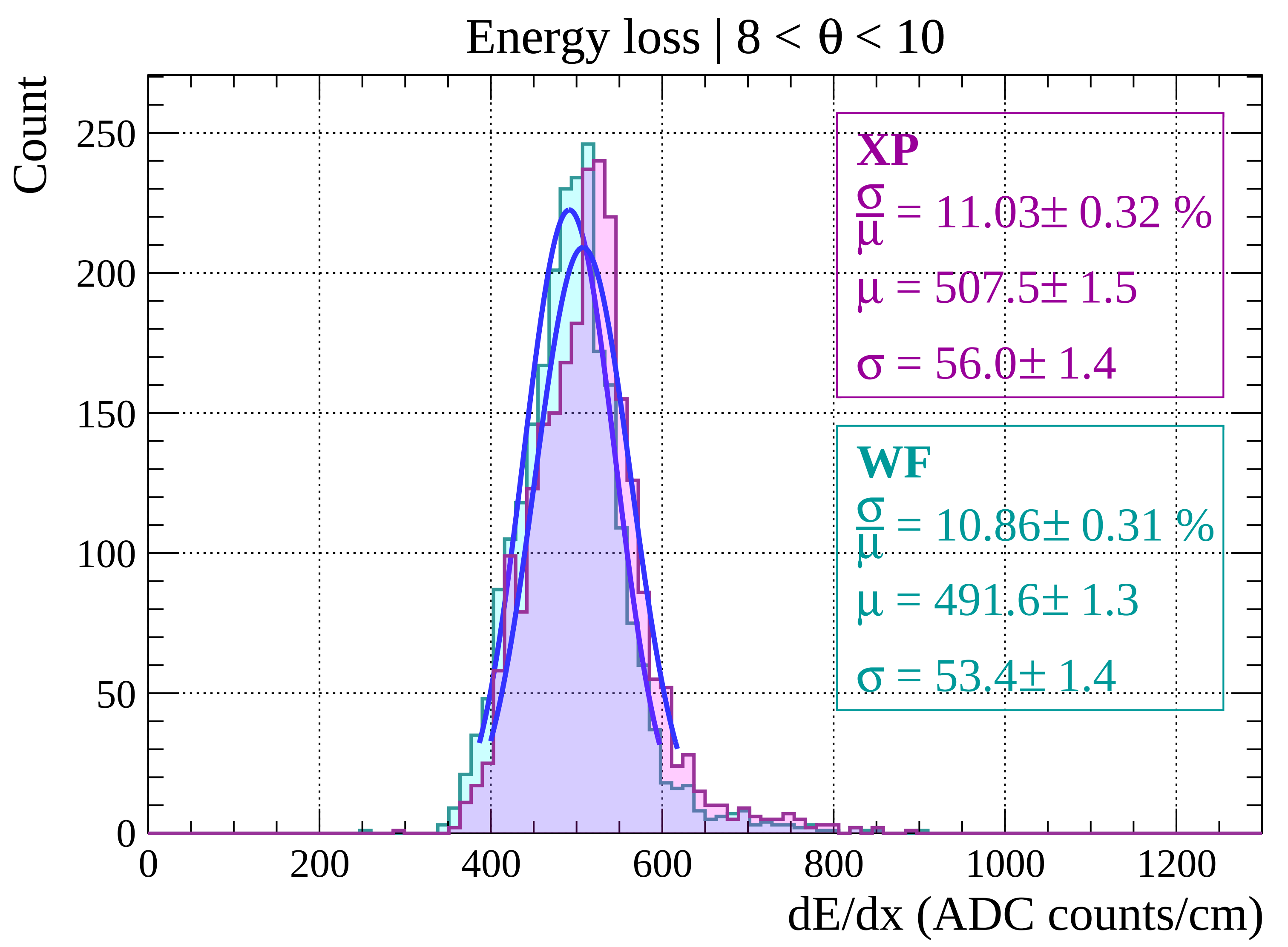


Energy loss | 2 < θ < 4
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 11.28± 0.27 %
μ = 503.3± 1.3
σ = 56.8± 1.2
WF
σ/μ = 11.12± 0.26 %
μ = 487.9± 1.2
σ = 54.3± 1.1

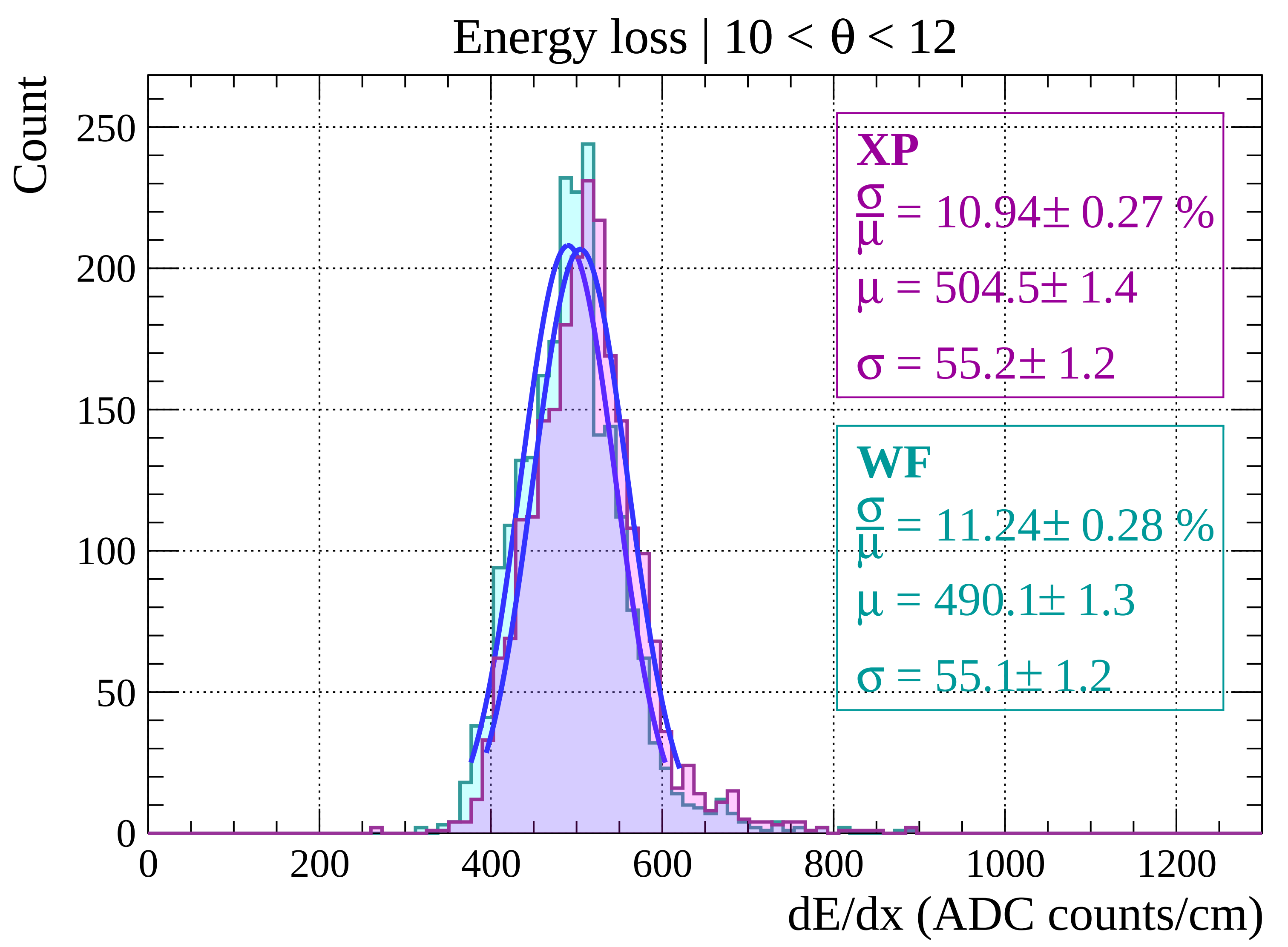

Energy loss | 4 < θ < 6
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.34± 0.30 %
μ = 504.6± 1.3
σ = 57.2± 1.3
WF
σ/μ = 11.34± 0.29 %
μ = 487.6± 1.3
σ = 55.3± 1.3



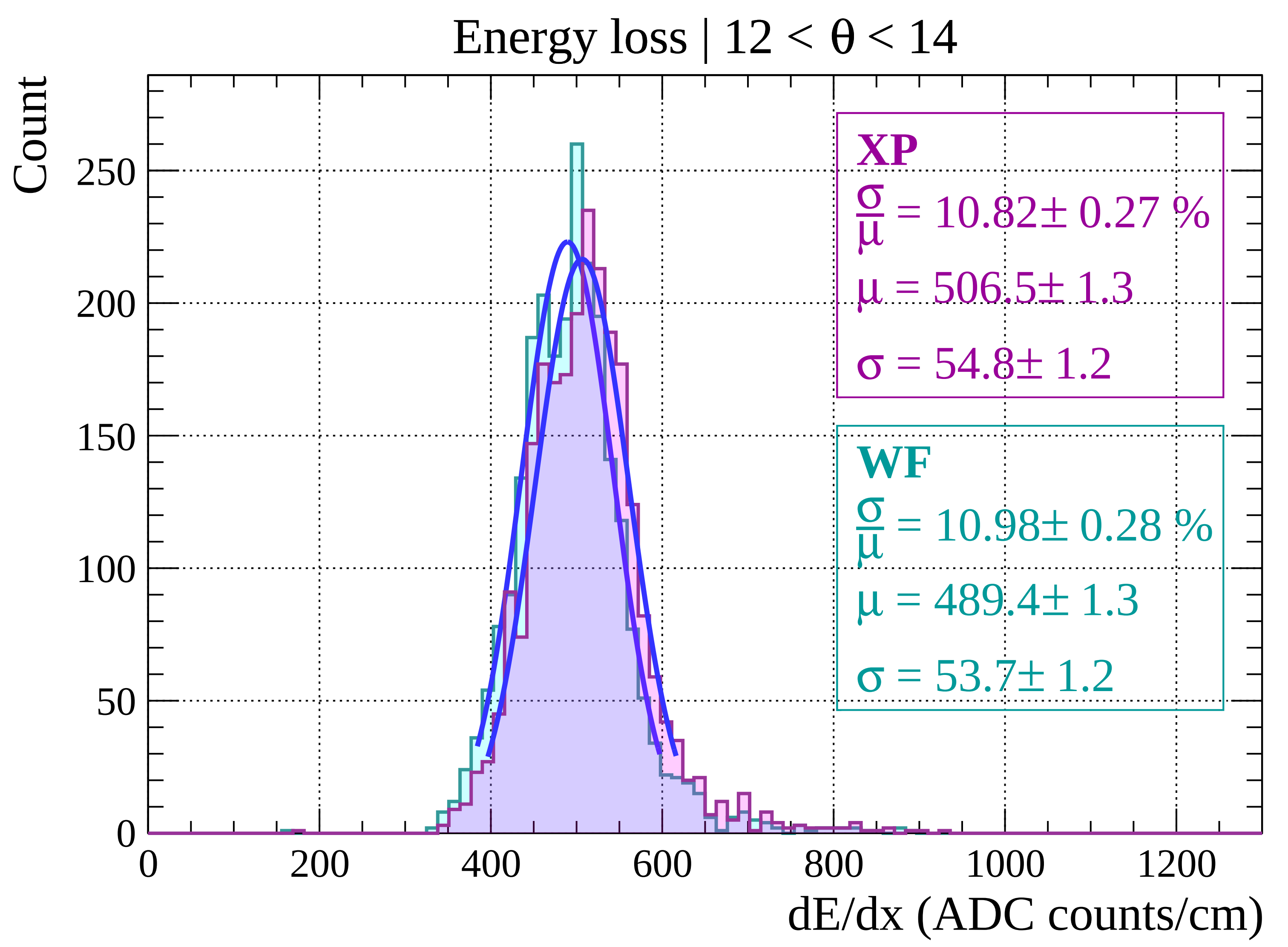
Energy loss | 6 < θ < 8
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.01± 0.26 %
μ = 504.8± 1.3
σ = 55.6± 1.2
WF
σ/μ = 10.70± 0.32 %
μ = 489.3± 1.3
σ = 52.4± 1.4

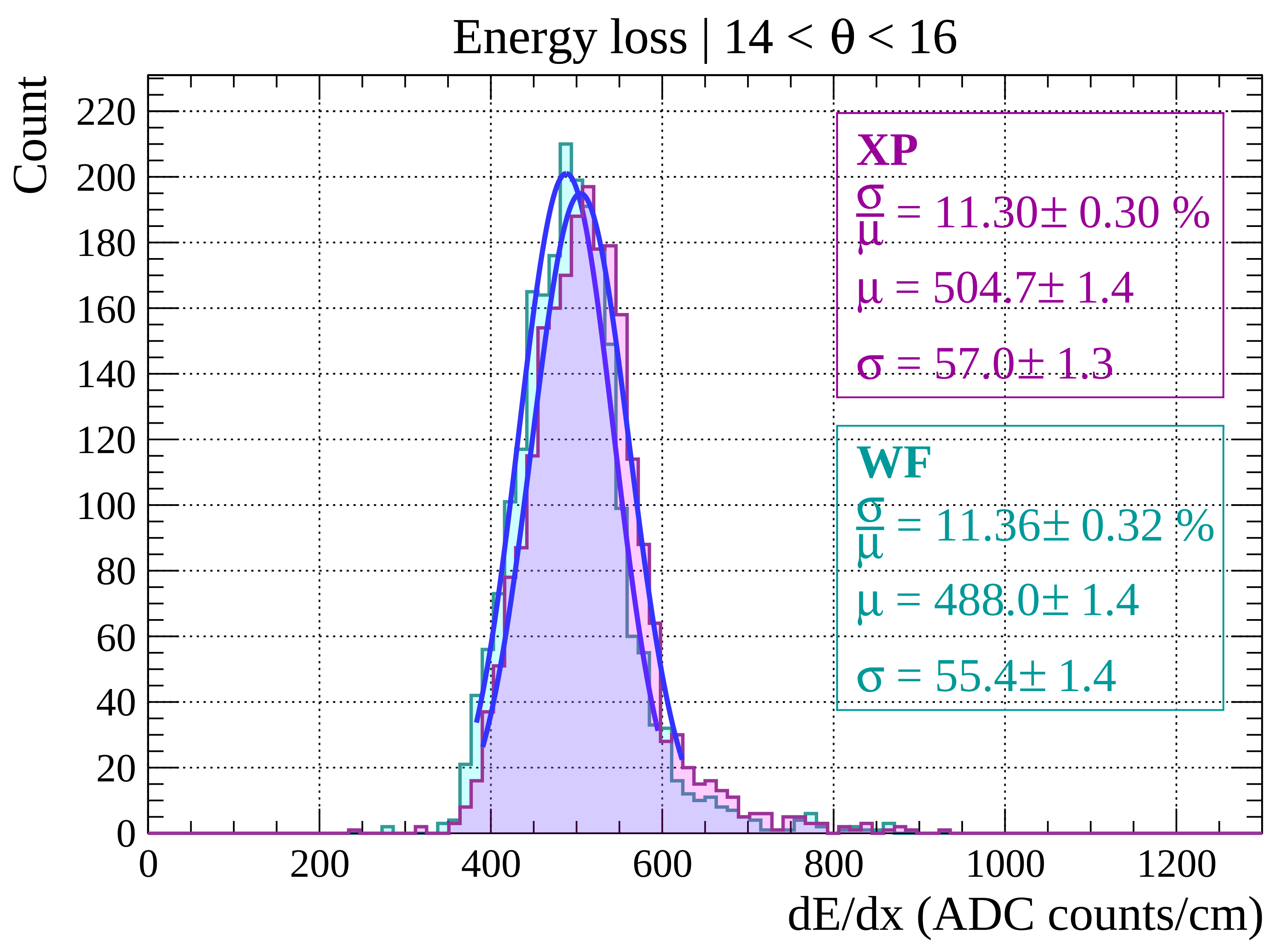


Energy loss | 8 < θ < 10
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.03± 0.32 %
μ = 507.5± 1.5
σ = 56.0± 1.4
WF
σ/μ = 10.86± 0.31 %
μ = 491.6± 1.3
σ = 53.4± 1.4

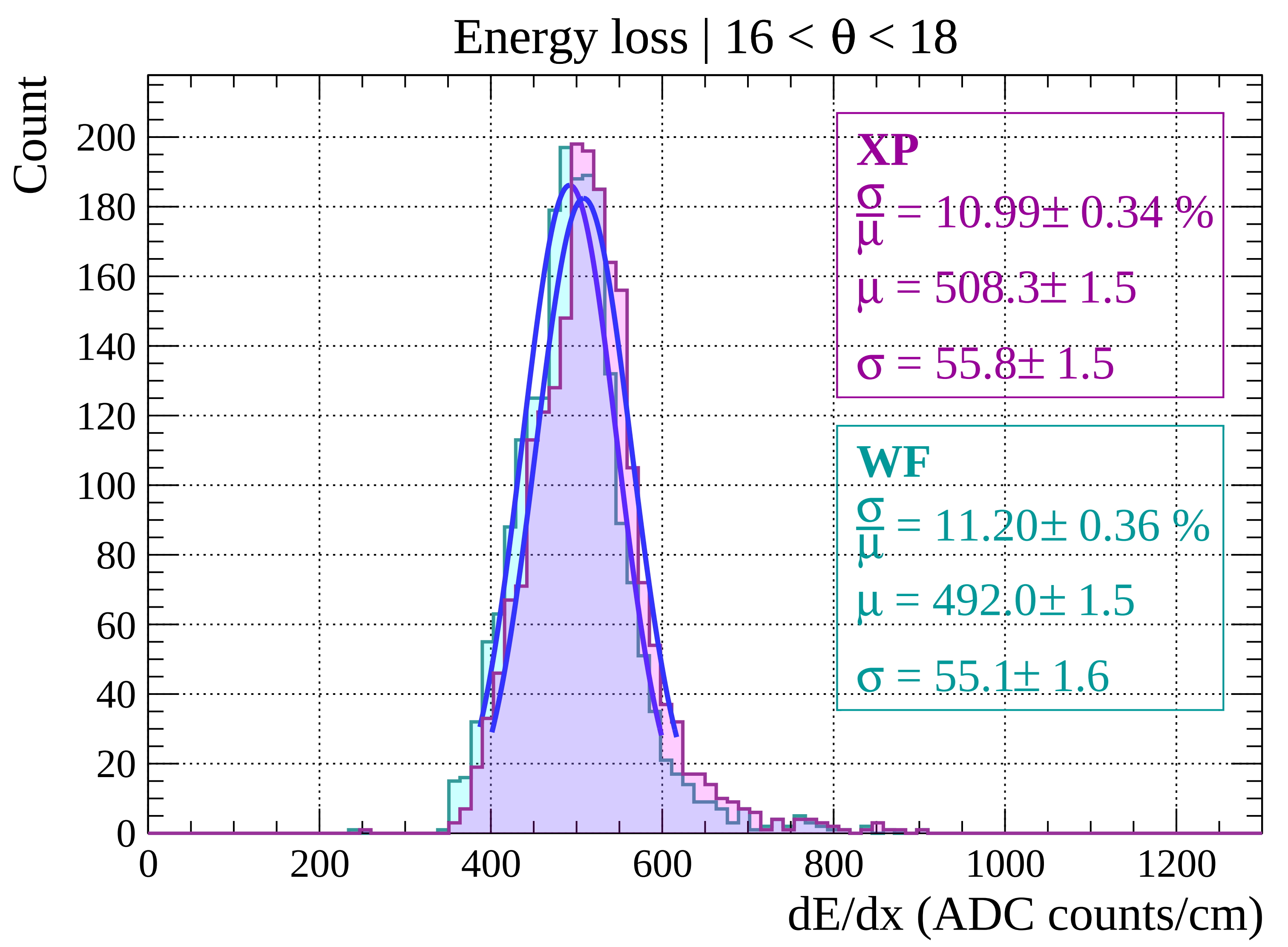


Energy loss | 10 < θ < 12
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP
σ/μ = 10.94± 0.27 %
μ = 504.5± 1.4
σ = 55.2± 1.2
WF
σ/μ = 11.24± 0.28 %
μ = 490.1± 1.3
σ = 55.1± 1.2

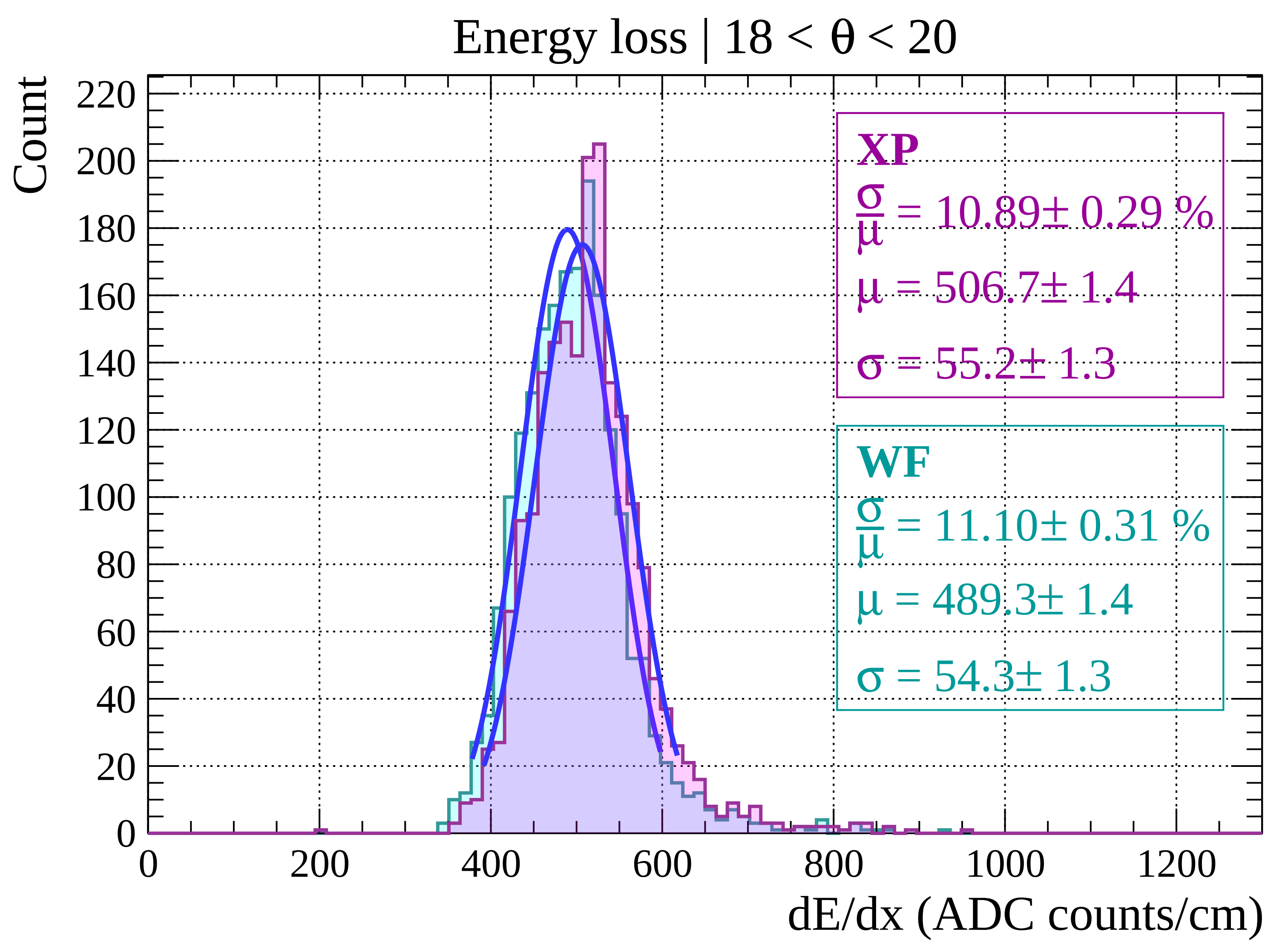

Energy loss | 12 < θ < 14
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 10.82± 0.27 %
μ = 506.5± 1.3
σ = 54.8± 1.2
WF
σ/μ = 10.98± 0.28 %
μ = 489.4± 1.3
σ = 53.7± 1.2



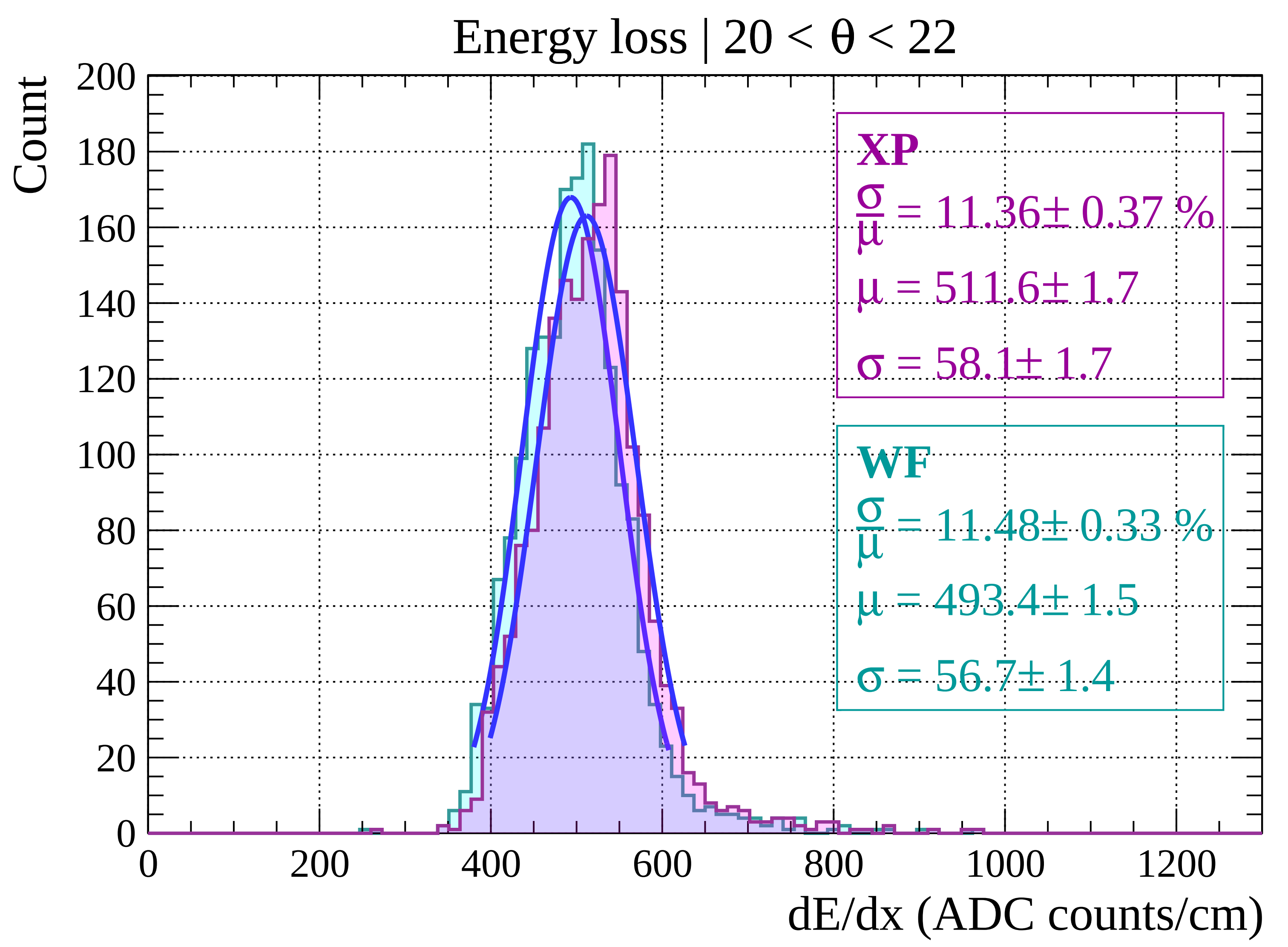
Energy loss | 14 < θ < 16
Count
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.30± 0.30 %
μ = 504.7± 1.4
σ = 57.0± 1.3
WF
$\frac{\sigma}{\mu}$ = 11.36± 0.32 %
μ = 488.0± 1.4
σ = 55.4± 1.4



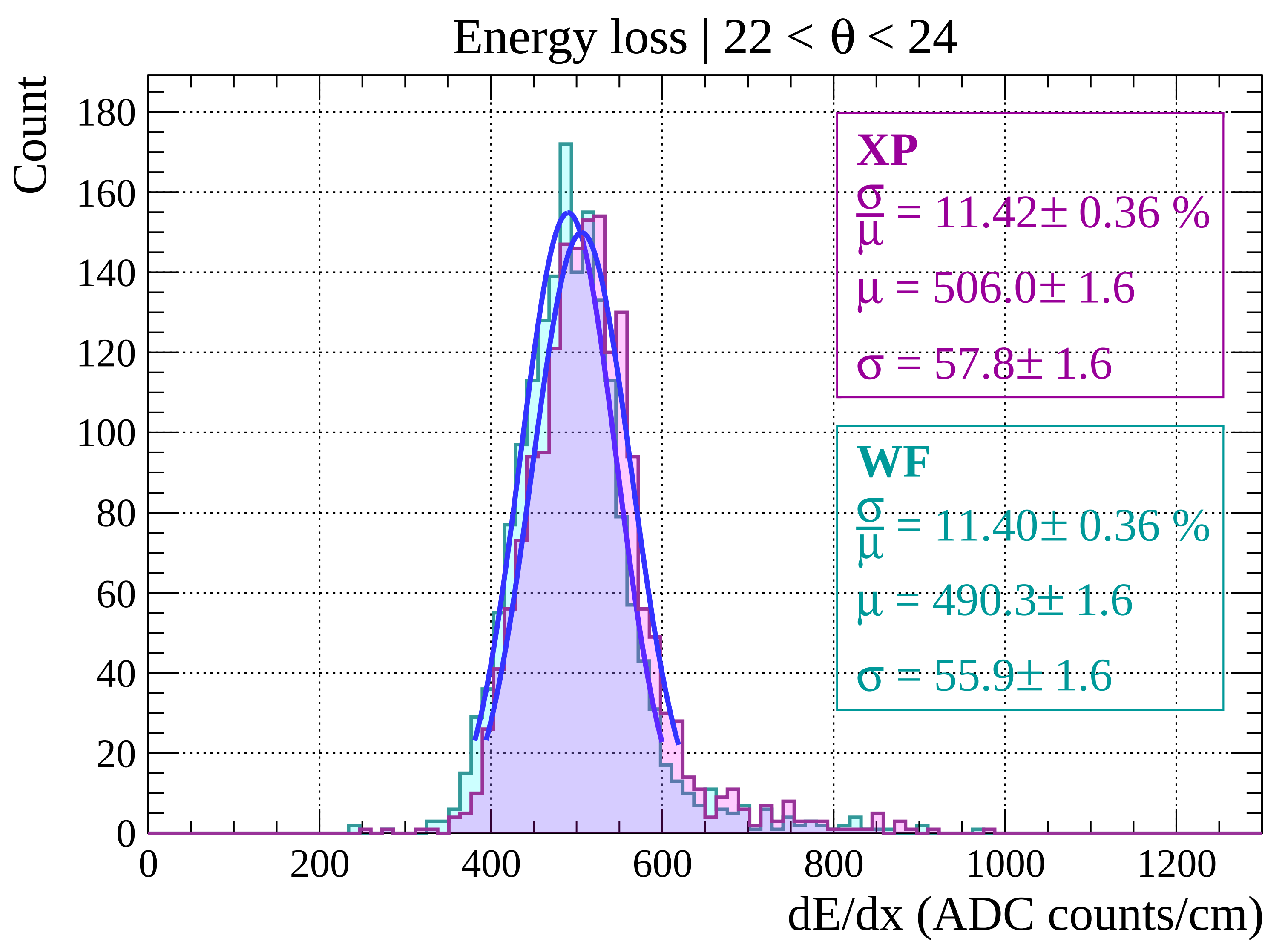
Energy loss | 16 < θ < 18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.99± 0.34 %
μ = 508.3± 1.5
σ = 55.8± 1.5
WF
σ/μ = 11.20± 0.36 %
μ = 492.0± 1.5
σ = 55.1± 1.6
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200

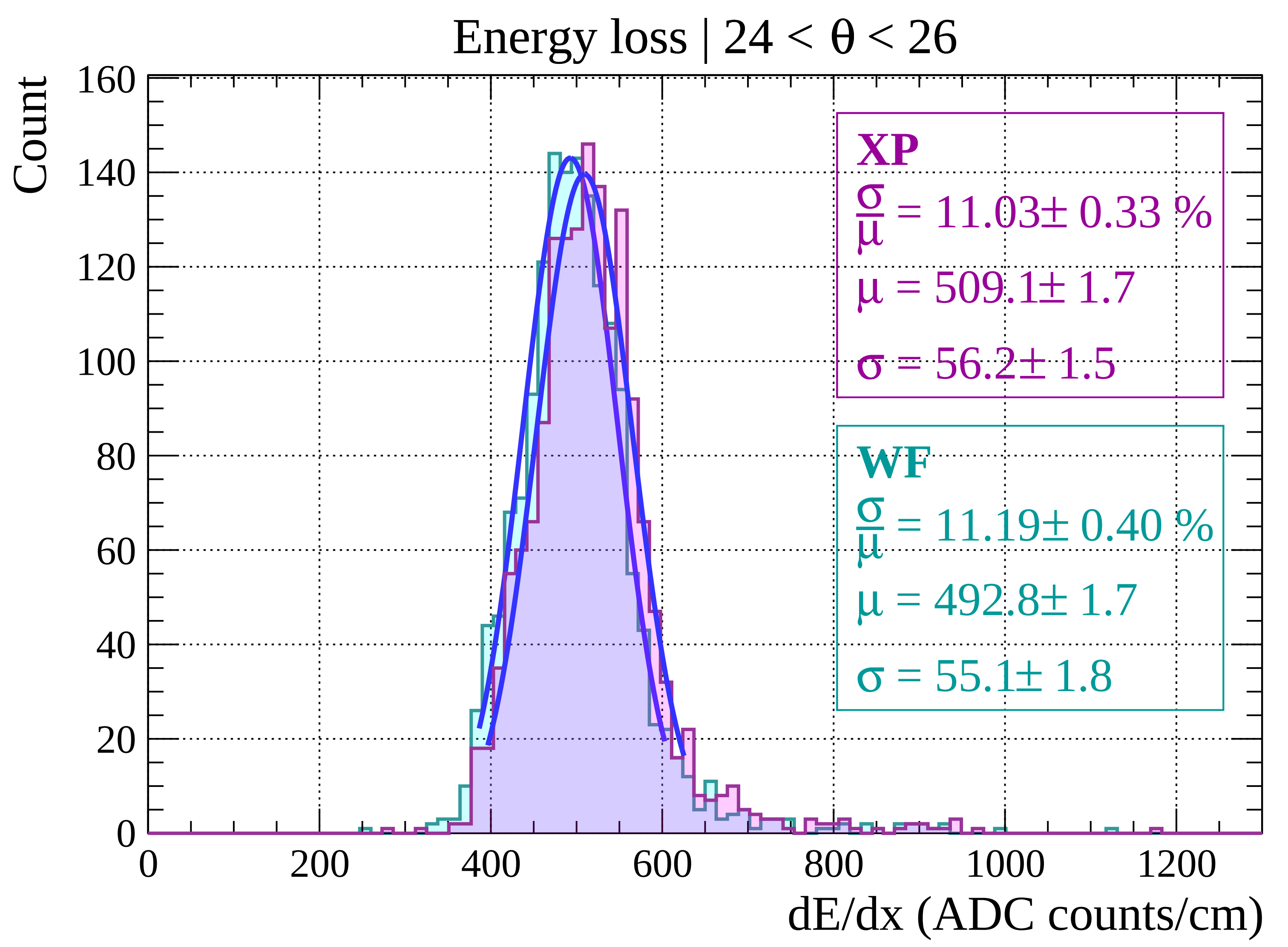


Energy loss | 18 < θ < 20
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
σ/μ = 10.89± 0.29 %
μ = 506.7± 1.4
σ = 55.2± 1.3
WF
σ/μ = 11.10± 0.31 %
μ = 489.3± 1.4
σ = 54.3± 1.3

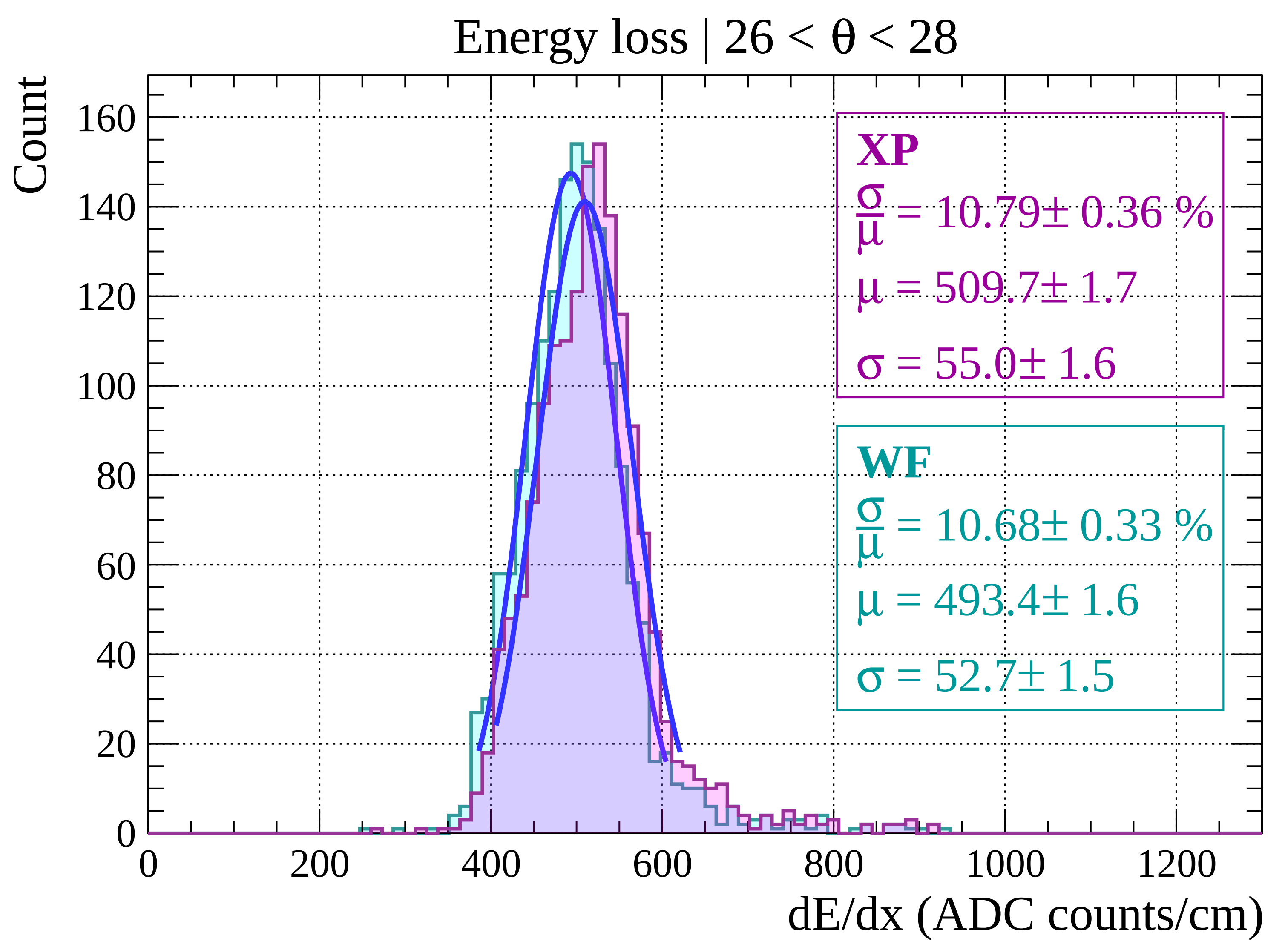


Energy loss | 20 < θ < 22
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
σ/μ = 11.36± 0.37 %
μ = 511.6± 1.7
σ = 58.1± 1.7
WF
σ/μ = 11.48± 0.33 %
μ = 493.4± 1.5
σ = 56.7± 1.4

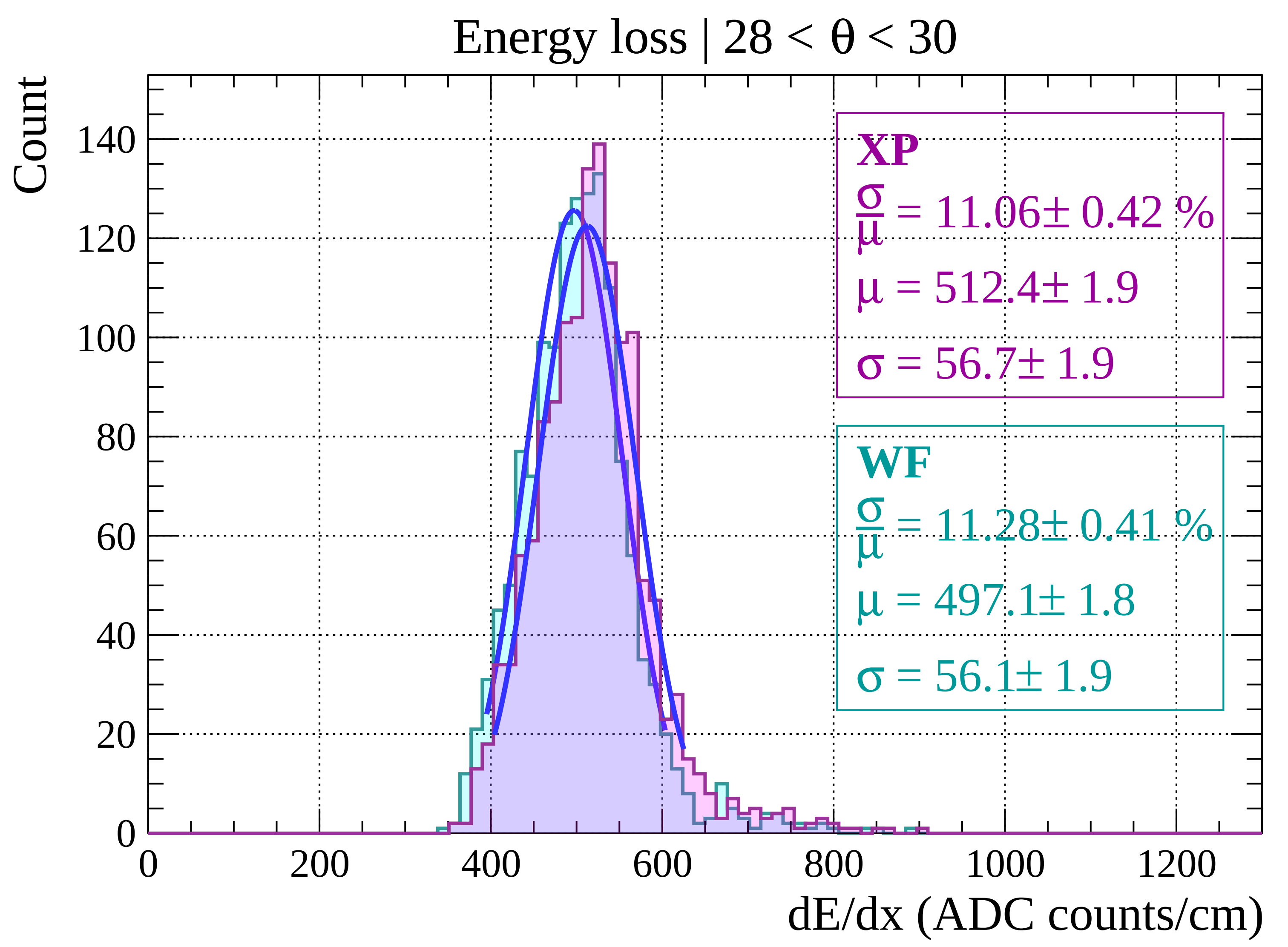

Energy loss | 22 < θ < 24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 11.42± 0.36 %
μ = 506.0± 1.6
σ = 57.8± 1.6
WF
σ/μ = 11.40± 0.36 %
μ = 490.3± 1.6
σ = 55.9± 1.6



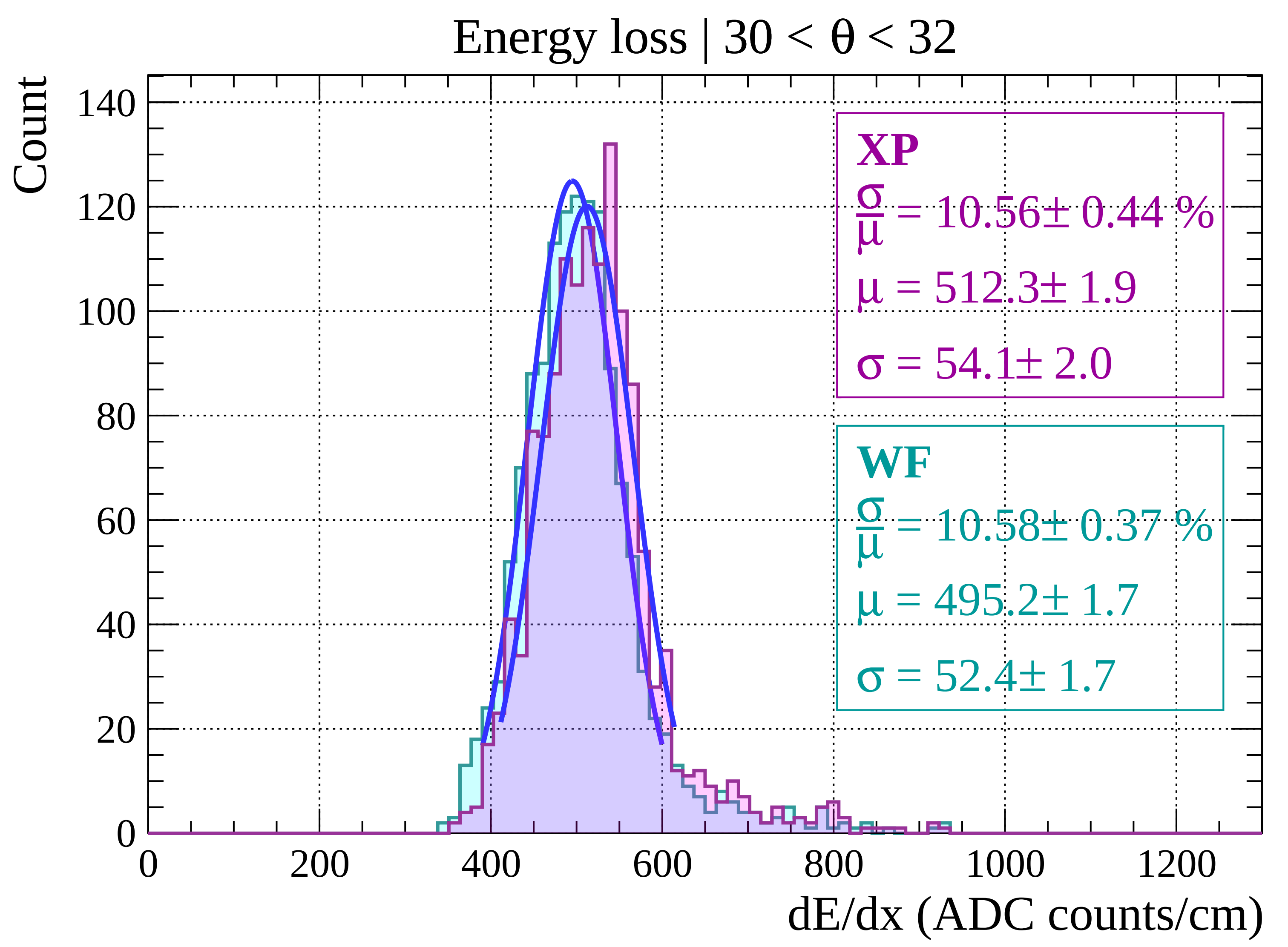
Energy loss | 24 < θ < 26
Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.03± 0.33 %
μ = 509.1± 1.7
σ = 56.2± 1.5
WF
σ/μ = 11.19± 0.40 %
μ = 492.8± 1.7
σ = 55.1± 1.8



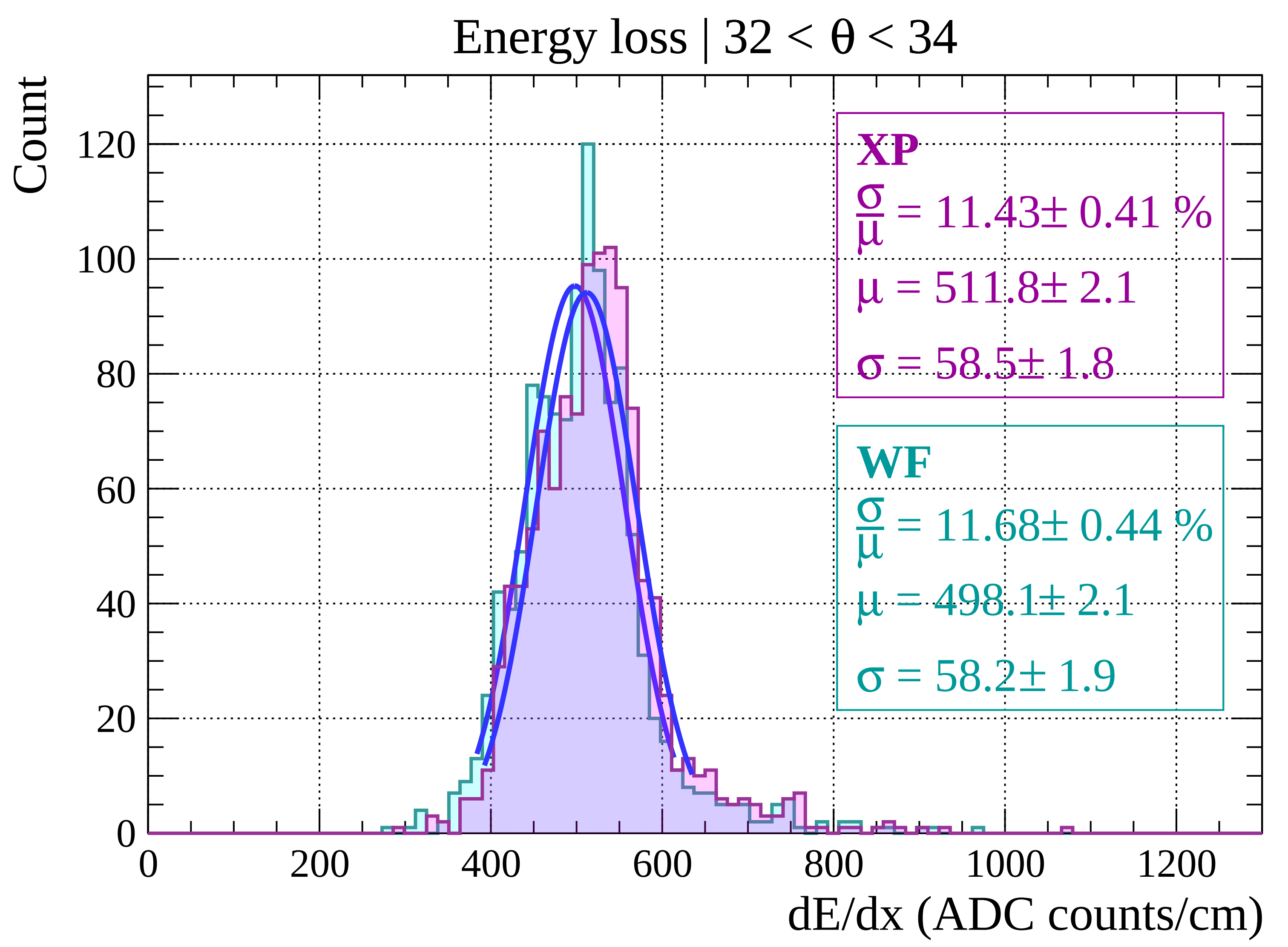
Energy loss | 26 < θ < 28
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.79± 0.36 %
μ = 509.7± 1.7
σ = 55.0± 1.6
WF
σ/μ = 10.68± 0.33 %
μ = 493.4± 1.6
σ = 52.7± 1.5

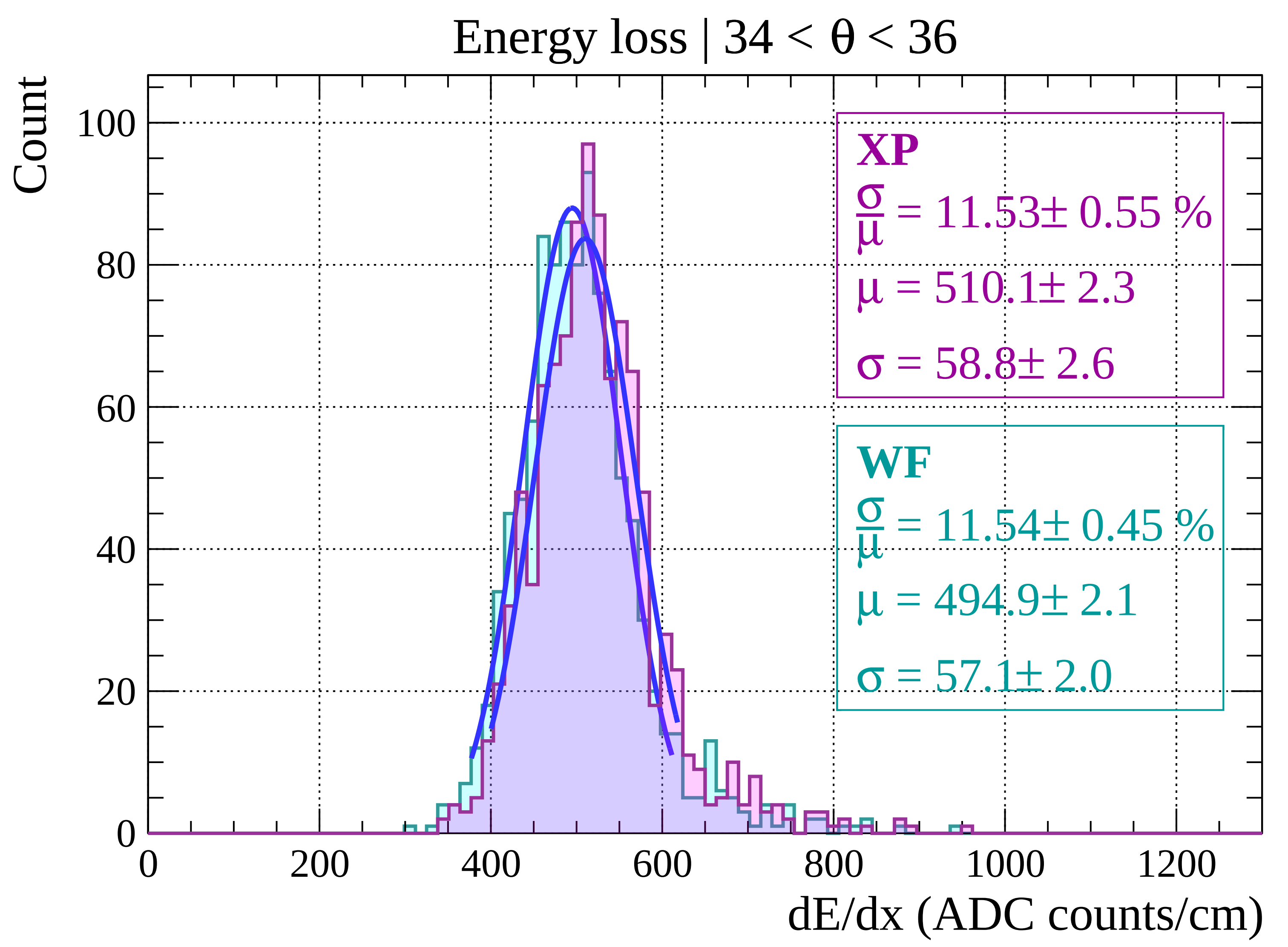


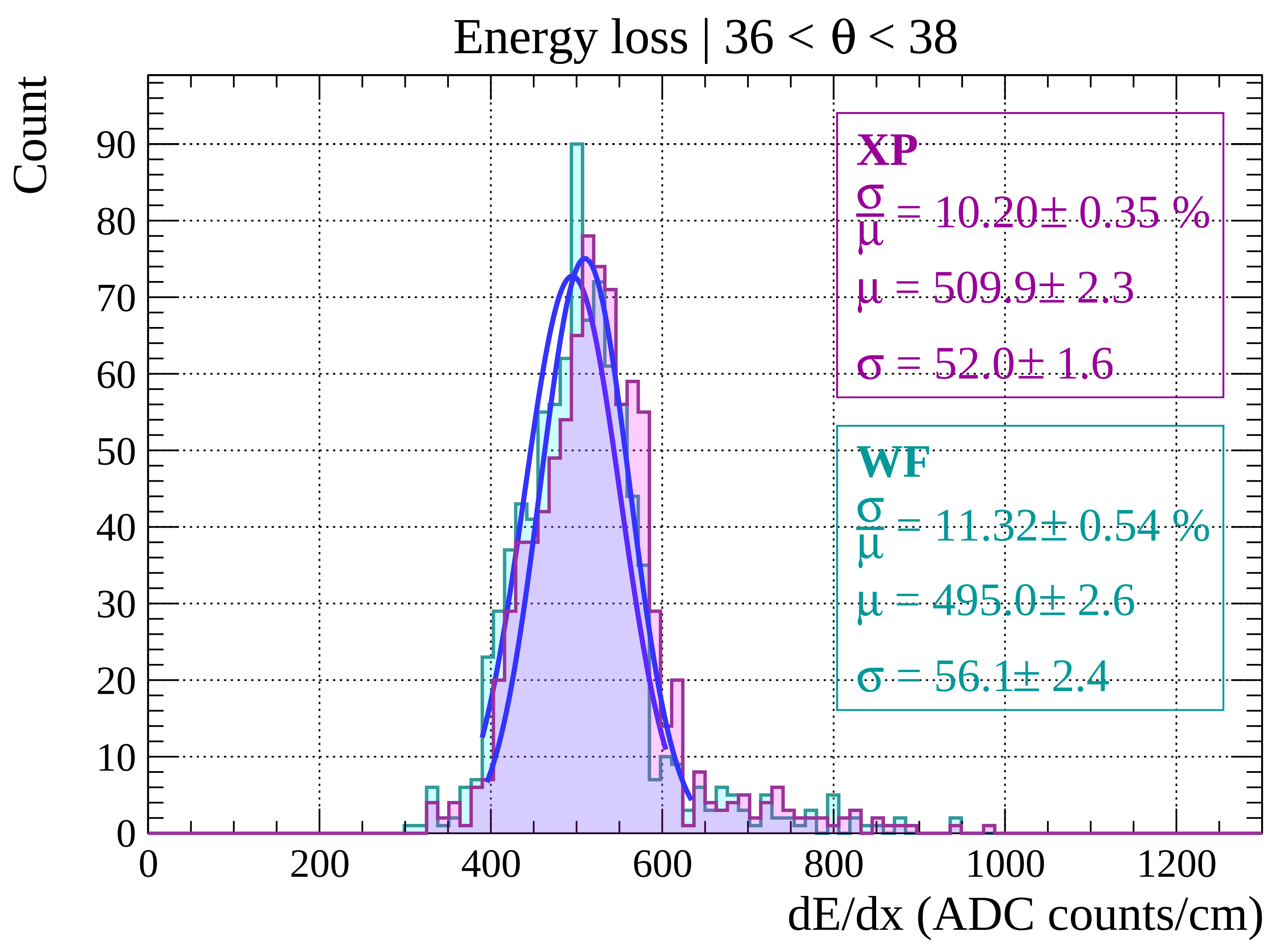
Energy loss | 28 < θ < 30
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
σ/μ = 11.06± 0.42 %
μ = 512.4± 1.9
σ = 56.7± 1.9
WF
σ/μ = 11.28± 0.41 %
μ = 497.1± 1.8
σ = 56.1± 1.9

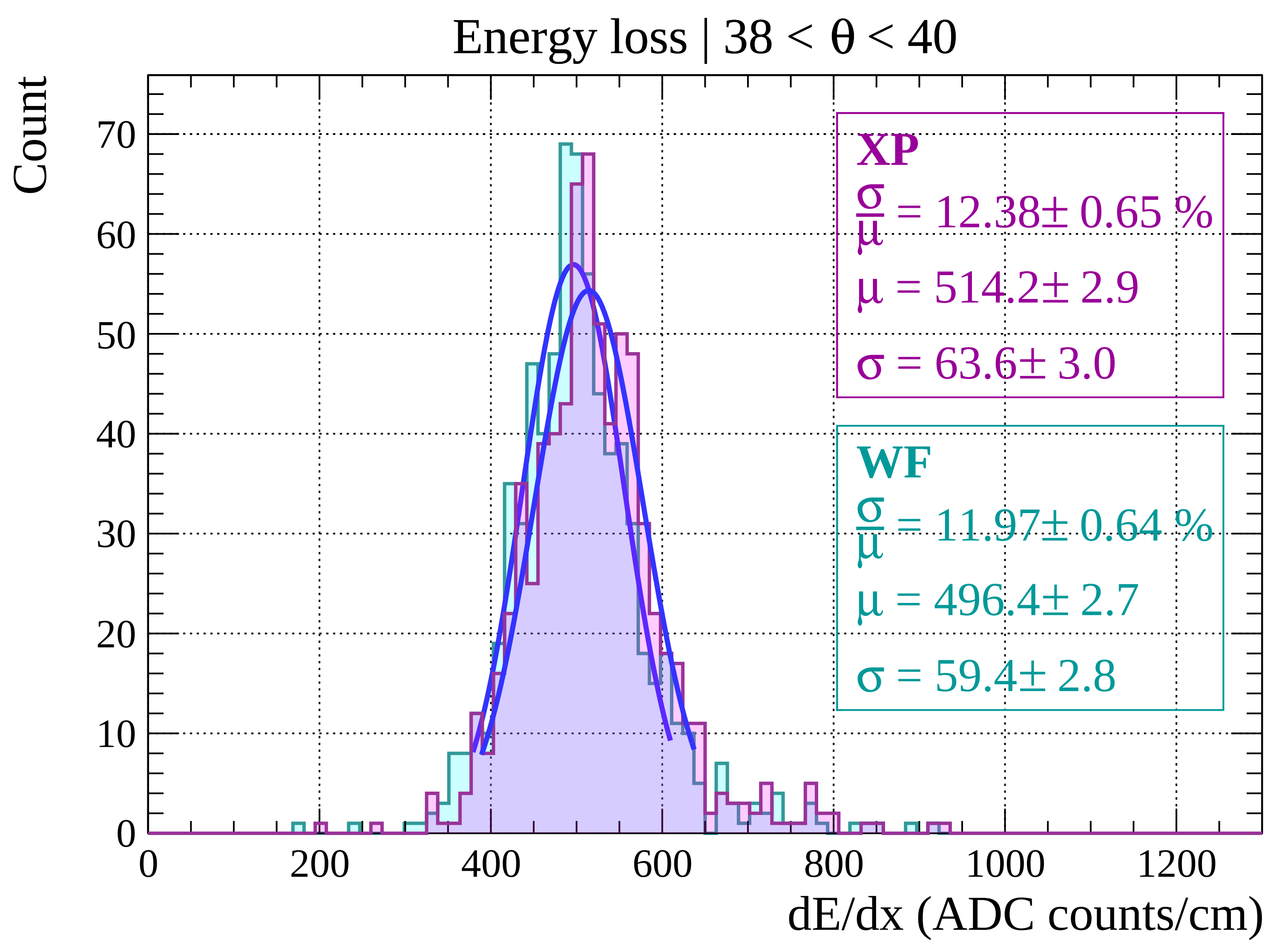


Energy loss | 30 < θ < 32
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.56± 0.44 %
μ = 512.3± 1.9
σ = 54.1± 2.0
WF
$\frac{\sigma}{\mu}$ = 10.58± 0.37 %
μ = 495.2± 1.7
σ = 52.4± 1.7
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200

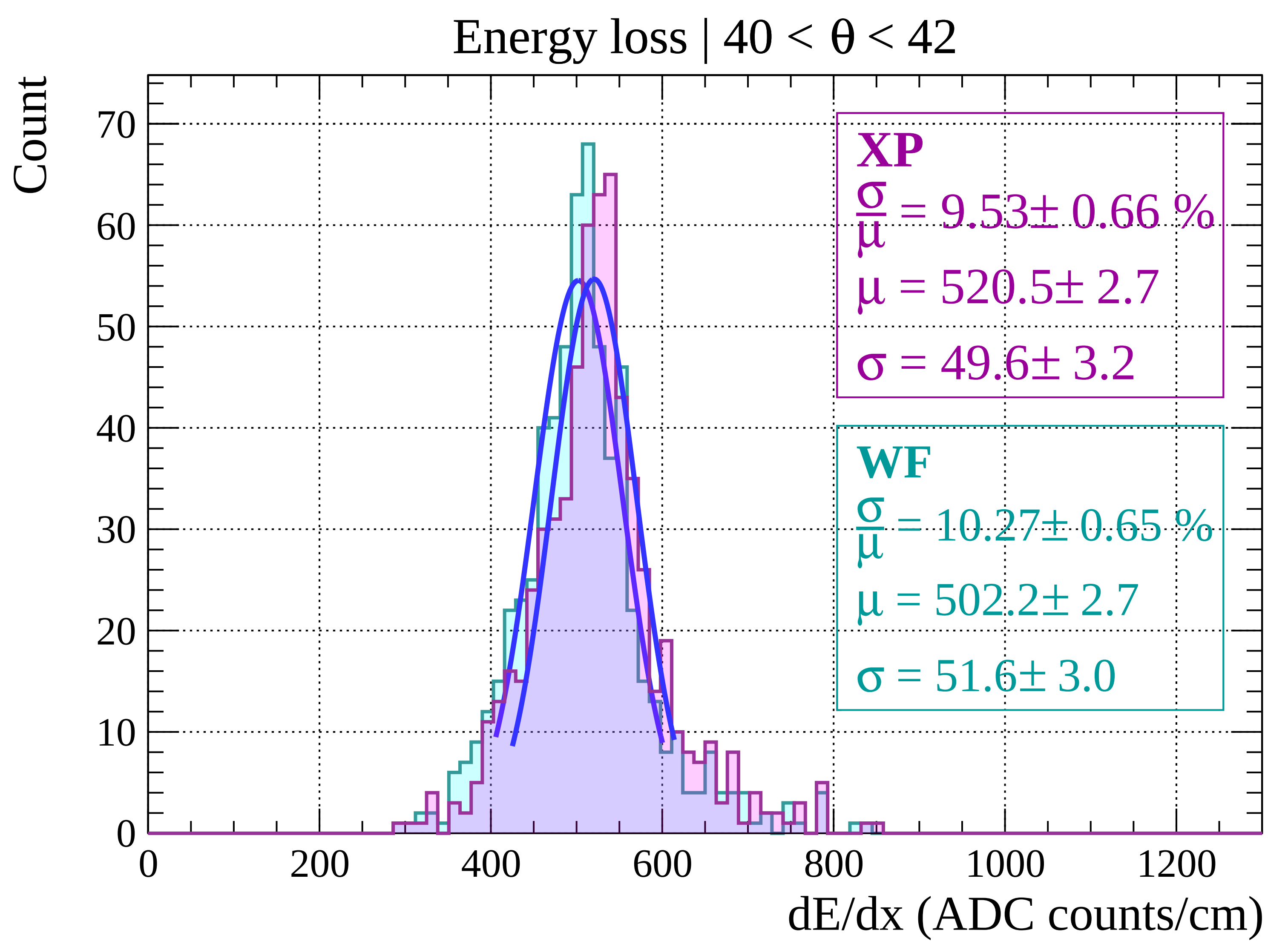

Energy loss | 32 < θ < 34
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.43± 0.41 %
μ = 511.8± 2.1
σ = 58.5± 1.8
WF
σ/μ = 11.68± 0.44 %
μ = 498.1± 2.1
σ = 58.2± 1.9


Energy loss | 34 < θ < 36
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 11.53± 0.55 %
μ = 510.1± 2.3
σ = 58.8± 2.6
WF
σ/μ = 11.54± 0.45 %
μ = 494.9± 2.1
σ = 57.1± 2.0

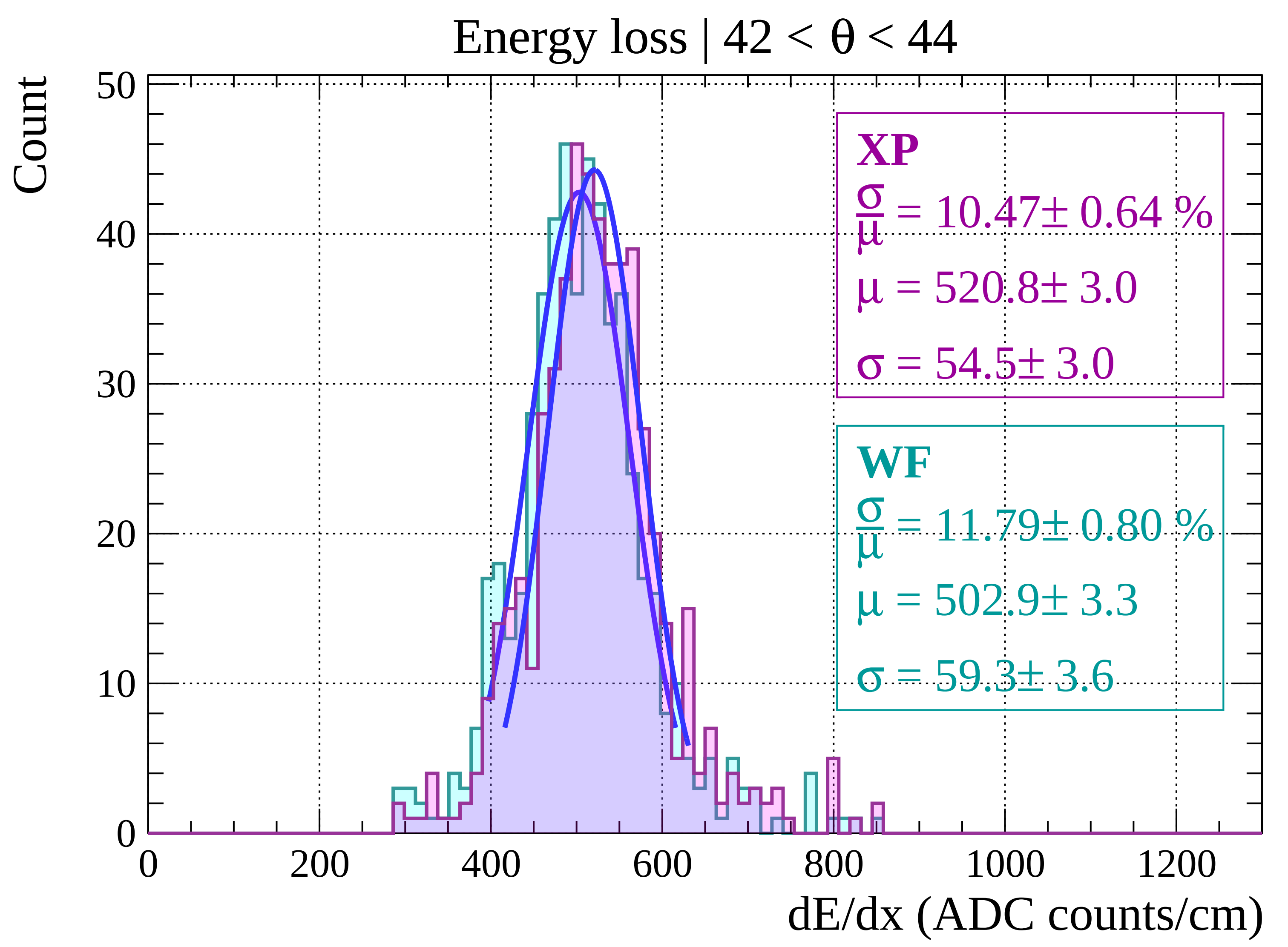

Energy loss | 36 < θ < 38
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP
$\frac{\sigma}{\mu}$ = 10.20± 0.35 %
μ = 509.9± 2.3
σ = 52.0± 1.6
WF
$\frac{\sigma}{\mu}$ = 11.32± 0.54 %
μ = 495.0± 2.6
σ = 56.1± 2.4



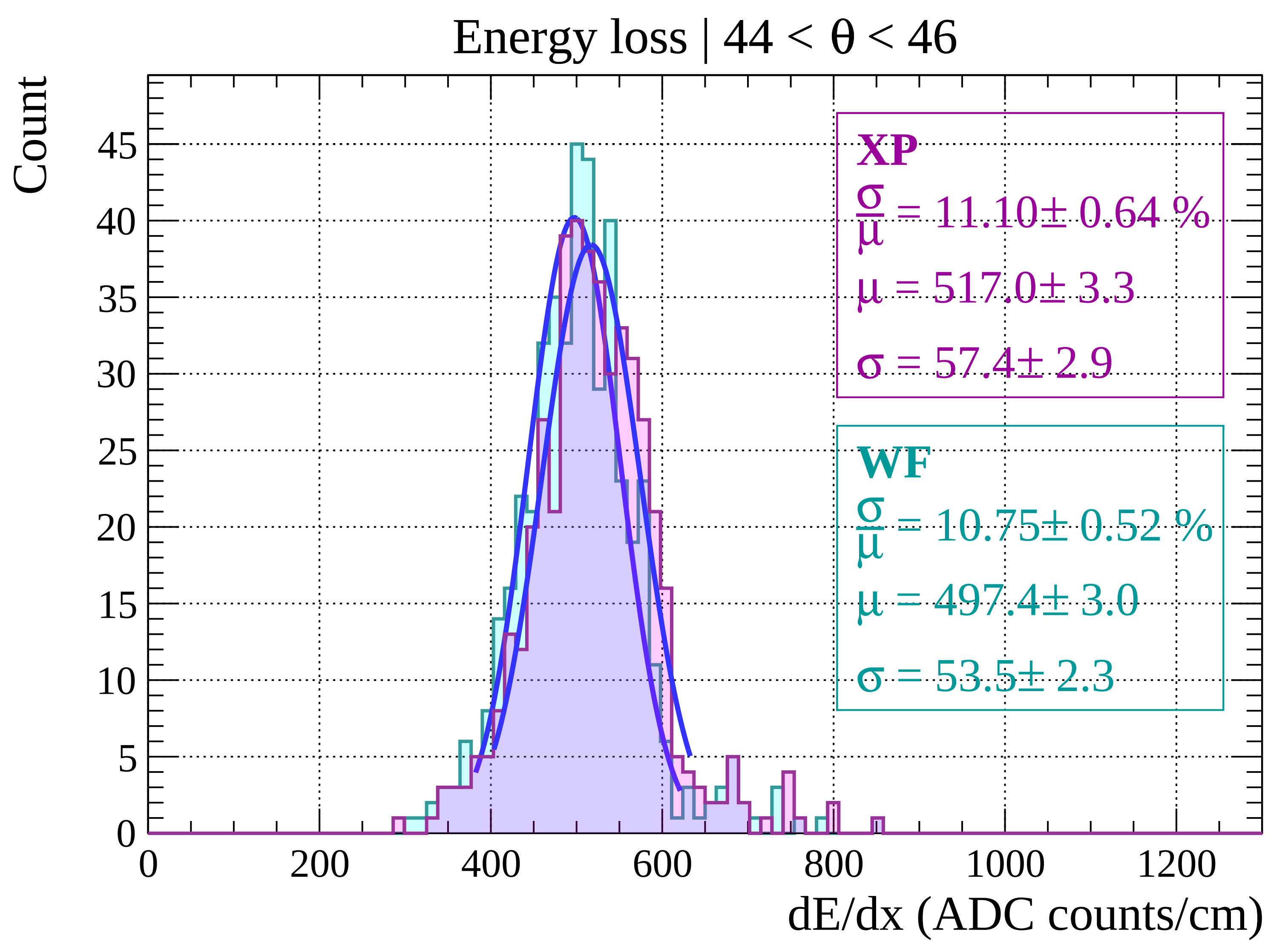
Energy loss | 38 < θ < 40
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP
σ/μ = 12.38± 0.65 %
μ = 514.2± 2.9
σ = 63.6± 3.0
WF
σ/μ = 11.97± 0.64 %
μ = 496.4± 2.7
σ = 59.4± 2.8

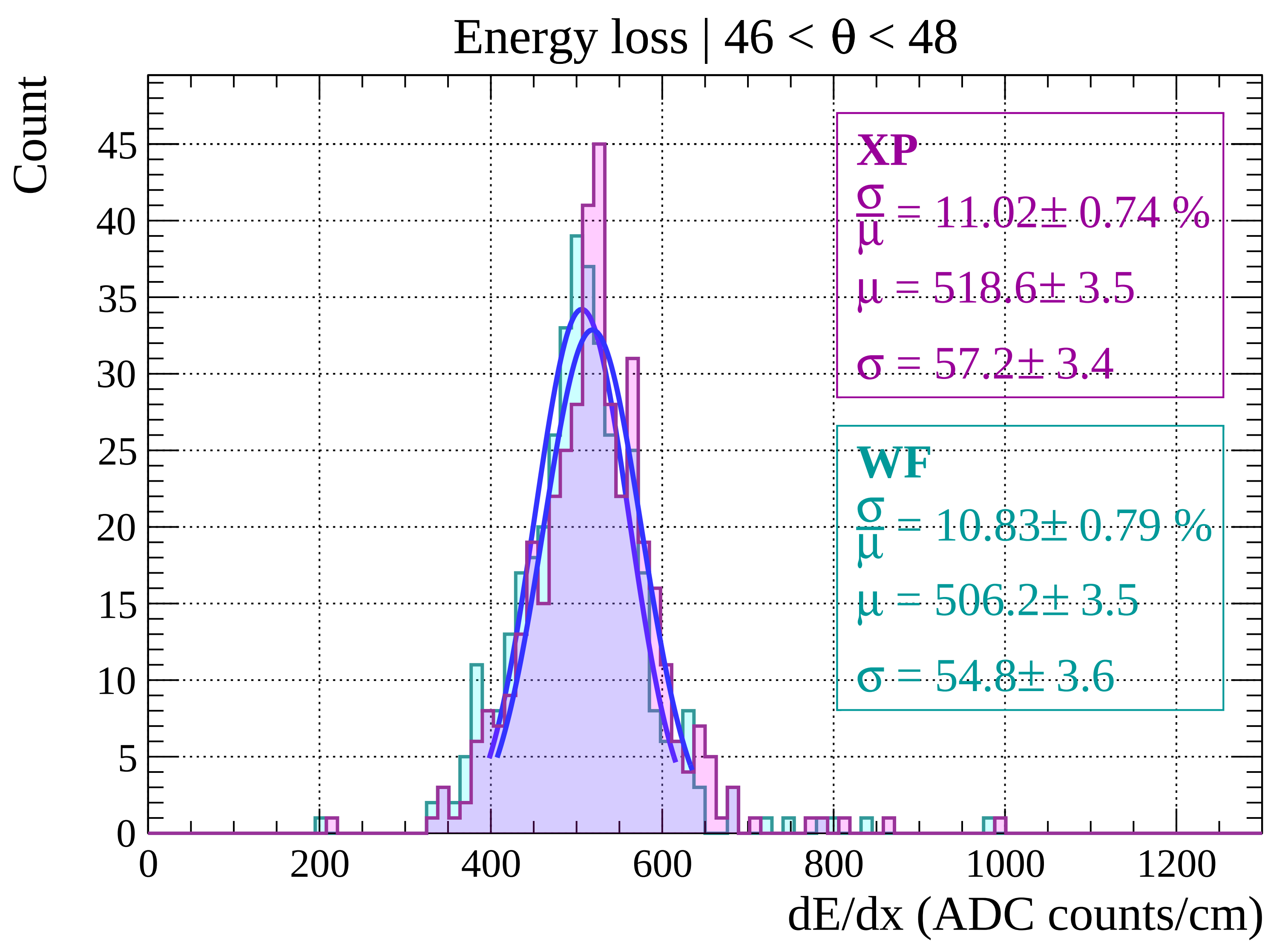


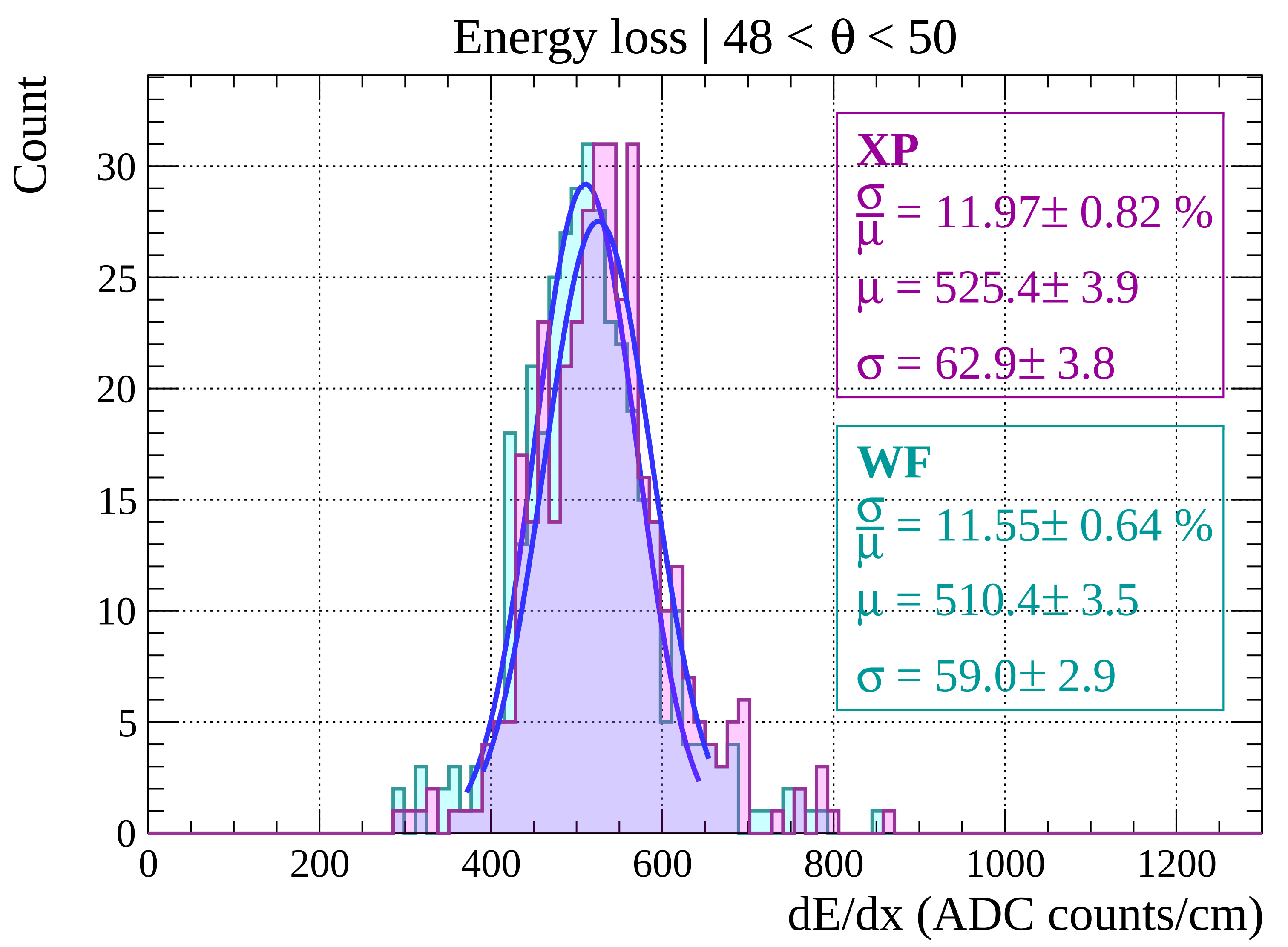
Energy loss | 40 < θ < 42
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP
$\frac{\sigma}{\mu}$ = 9.53± 0.66 %
μ = 520.5± 2.7
σ = 49.6± 3.2
WF
$\frac{\sigma}{\mu}$ = 10.27± 0.65 %
μ = 502.2± 2.7
σ = 51.6± 3.0

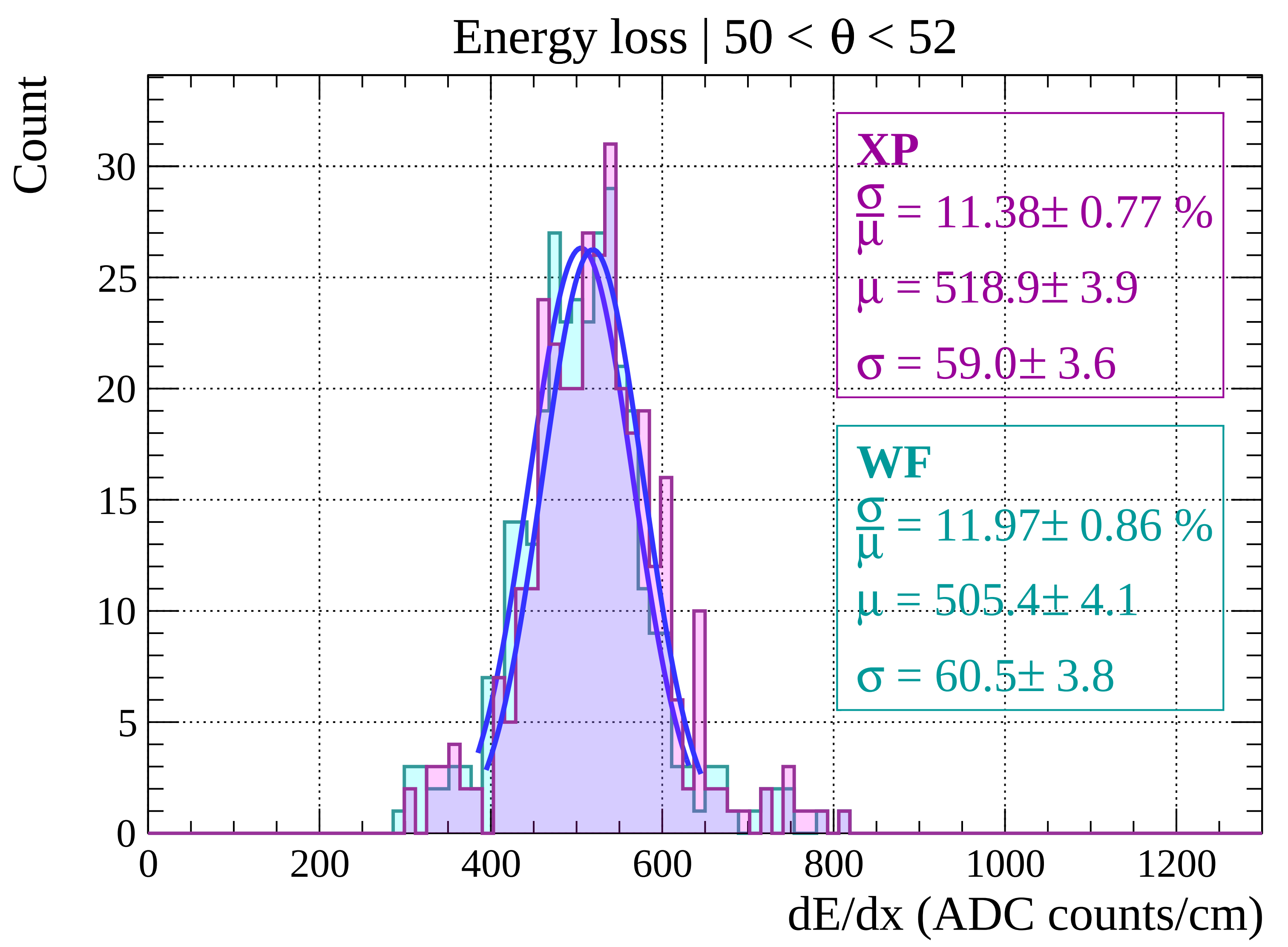


Energy loss | 42 < θ < 44
Count
XP
σ/μ = 10.47± 0.64 %
μ = 520.8± 3.0
σ = 54.5± 3.0
WF
σ/μ = 11.79± 0.80 %
μ = 502.9± 3.3
σ = 59.3± 3.6
0
10
20
30
40
50
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

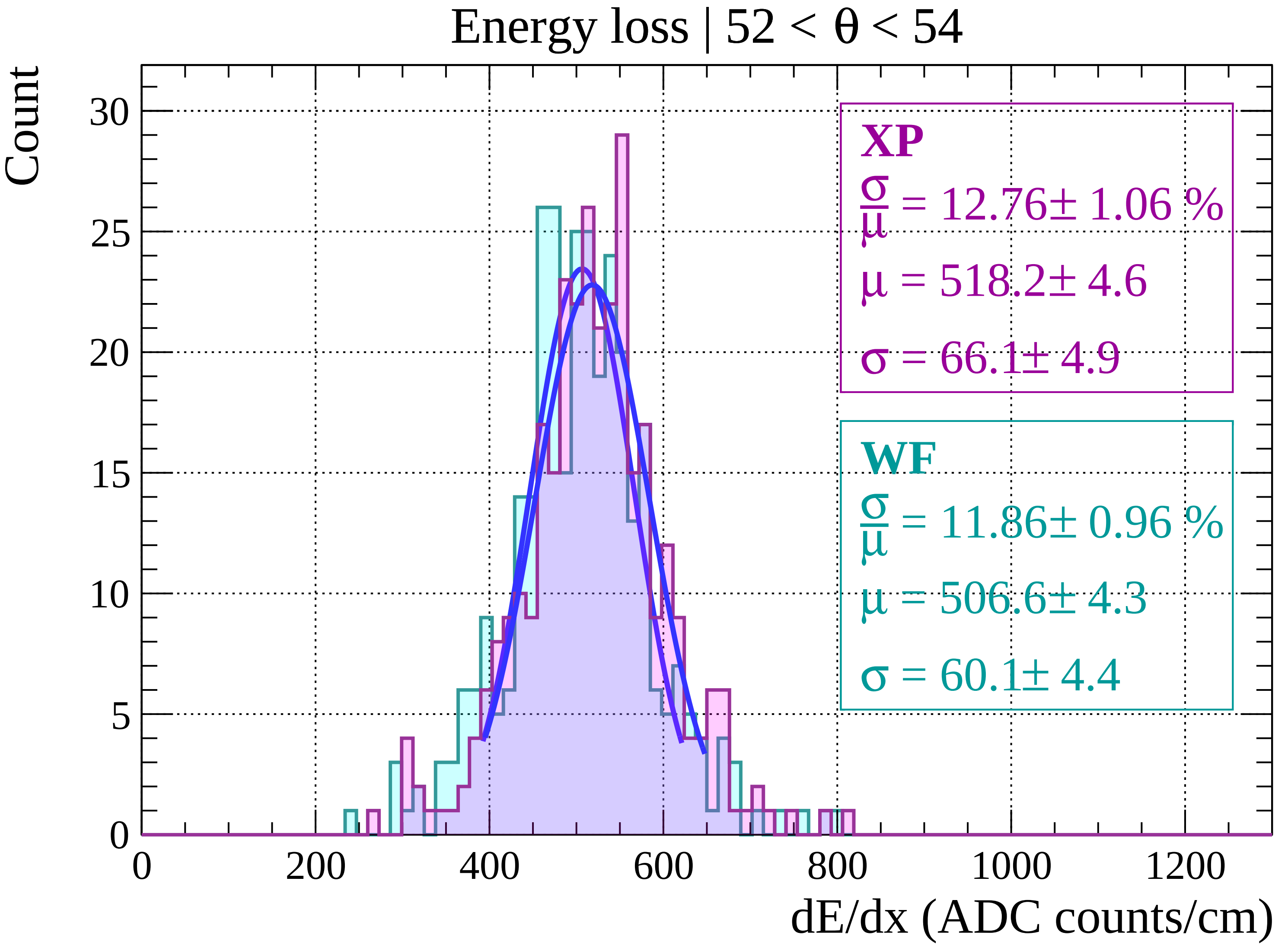

Energy loss | 44 < θ < 46
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.10± 0.64 %
μ = 517.0± 3.3
σ = 57.4± 2.9
WF
σ/μ = 10.75± 0.52 %
μ = 497.4± 3.0
σ = 53.5± 2.3
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200


Energy loss | 46 < θ < 48
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP
σ/μ = 11.02± 0.74 %
μ = 518.6± 3.5
σ = 57.2± 3.4
WF
σ/μ = 10.83± 0.79 %
μ = 506.2± 3.5
σ = 54.8± 3.6

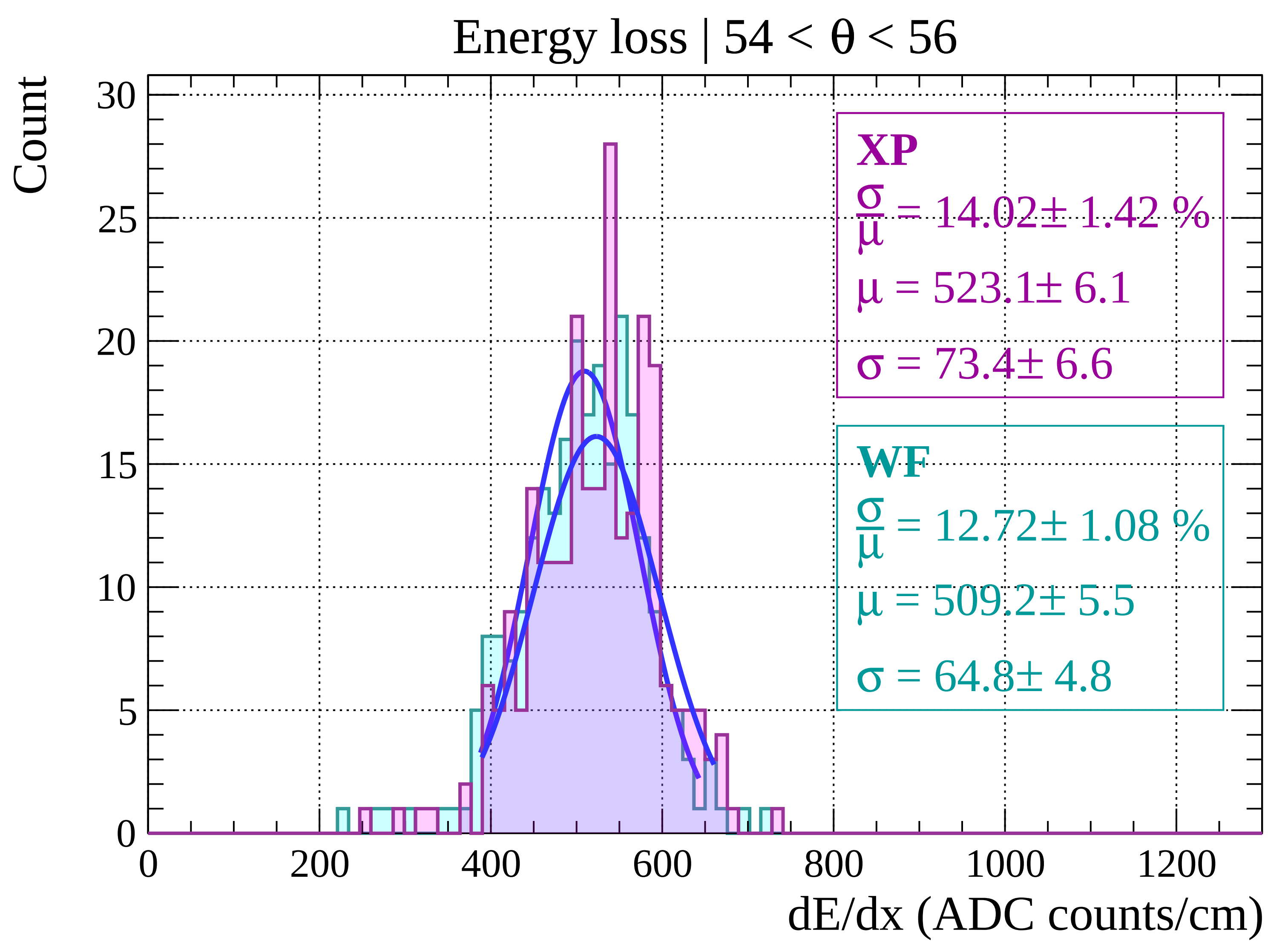

Energy loss | 48 < θ < 50
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.97± 0.82 %
μ = 525.4± 3.9
σ = 62.9± 3.8
WF
σ/μ = 11.55± 0.64 %
μ = 510.4± 3.5
σ = 59.0± 2.9



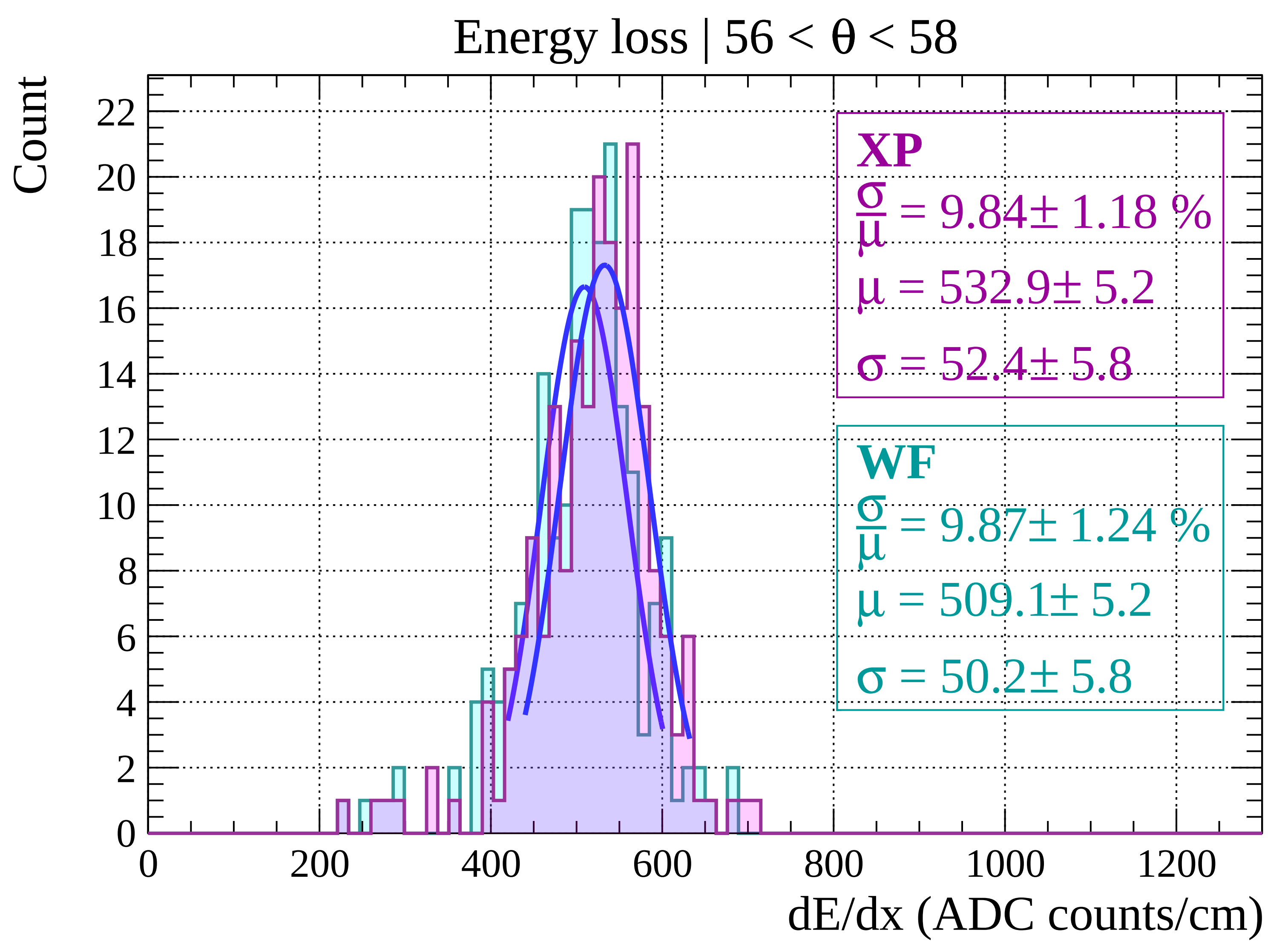
Energy loss | 50 < θ < 52
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP
σ/μ = 11.38± 0.77 %
μ = 518.9± 3.9
σ = 59.0± 3.6
WF
σ/μ = 11.97± 0.86 %
μ = 505.4± 4.1
σ = 60.5± 3.8



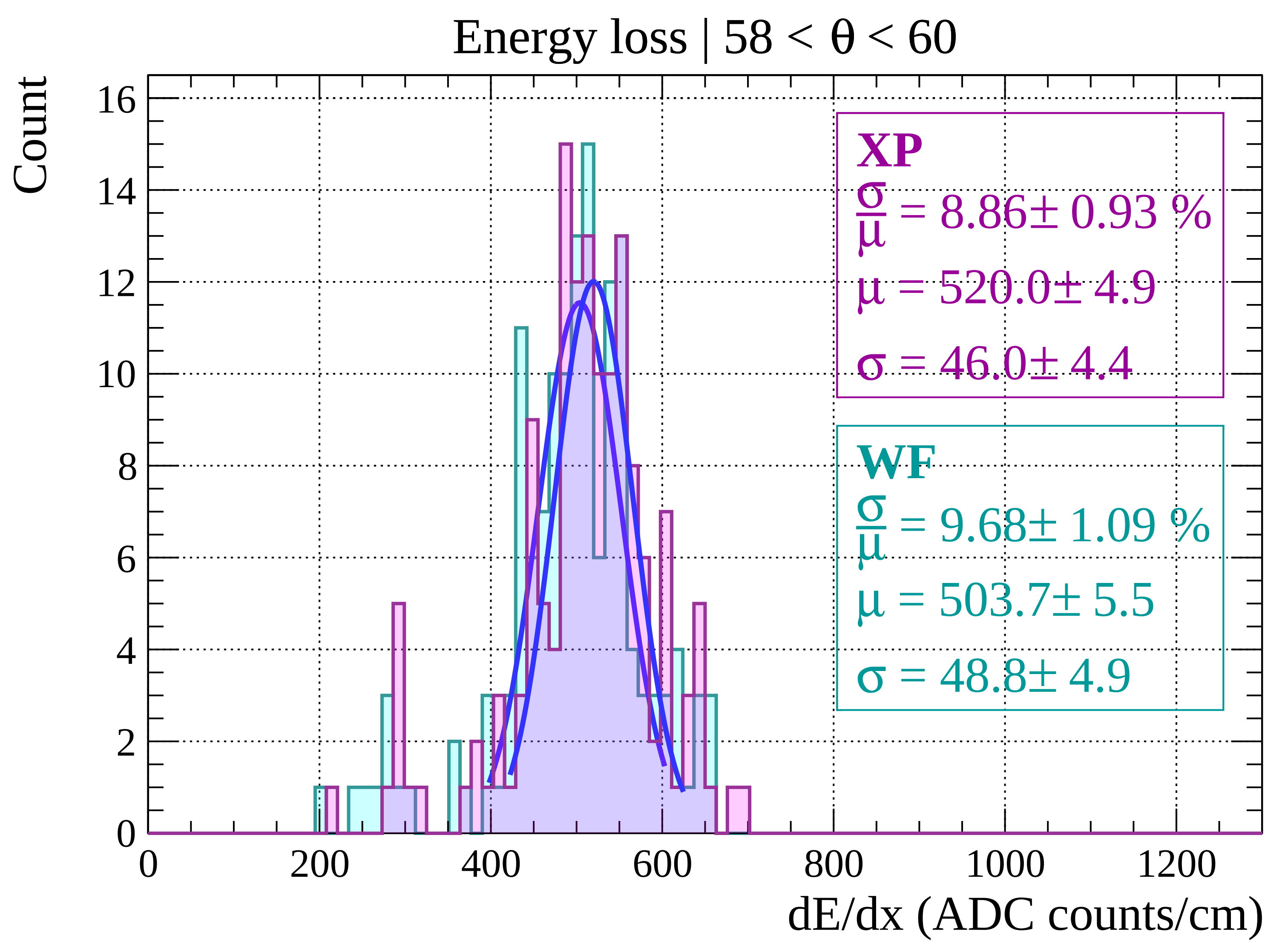
Energy loss | 52 < θ < 54
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP
σ/μ = 12.76 ± 1.06 %
μ = 518.2 ± 4.6
σ = 66.1 ± 4.9
WF
σ/μ = 11.86 ± 0.96 %
μ = 506.6 ± 4.3
σ = 60.1 ± 4.4



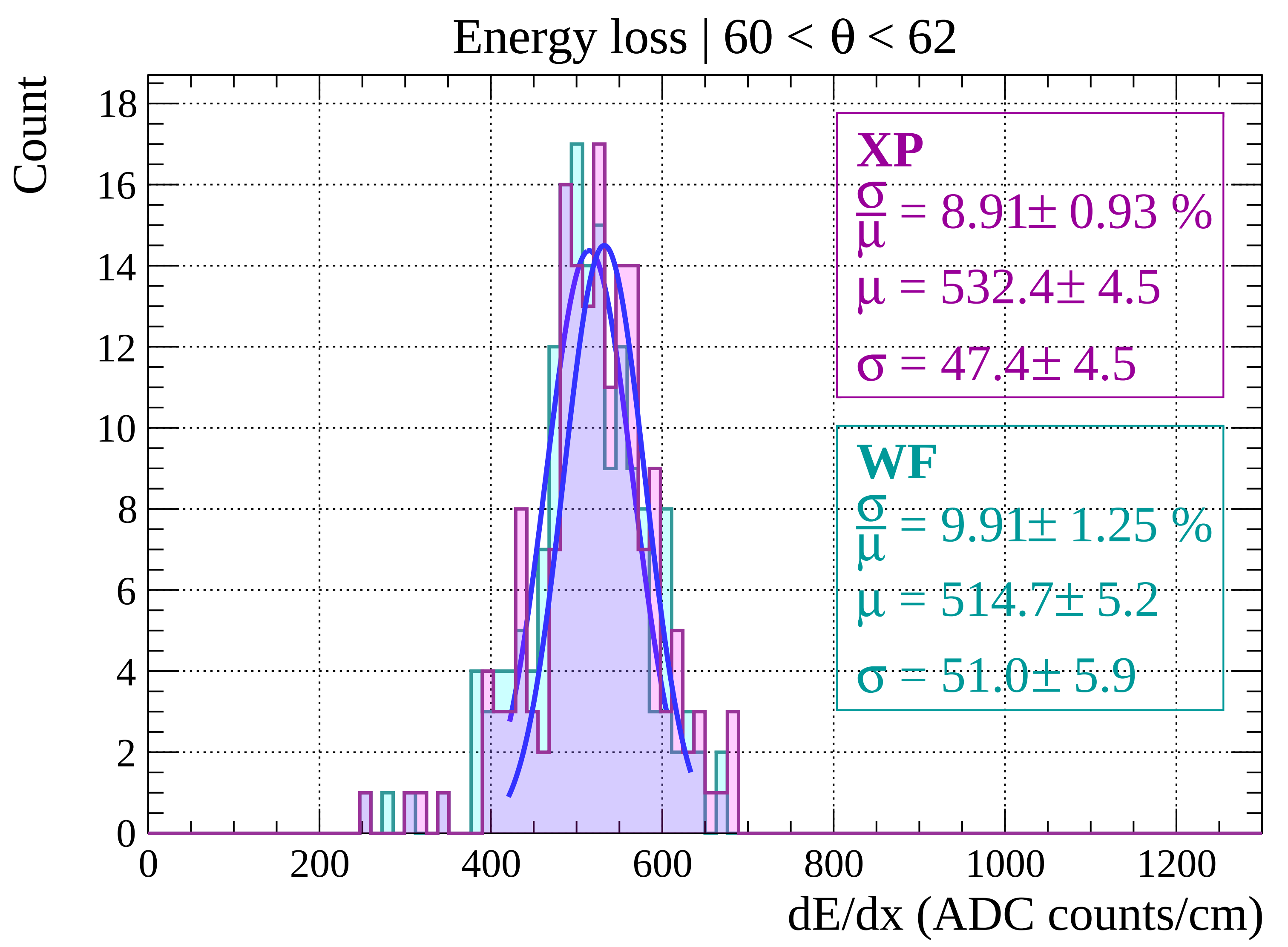
Energy loss | 54 < θ < 56
Count
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.02± 1.42 %
μ = 523.1± 6.1
σ = 73.4± 6.6
WF
σ/μ = 12.72± 1.08 %
μ = 509.2± 5.5
σ = 64.8± 4.8

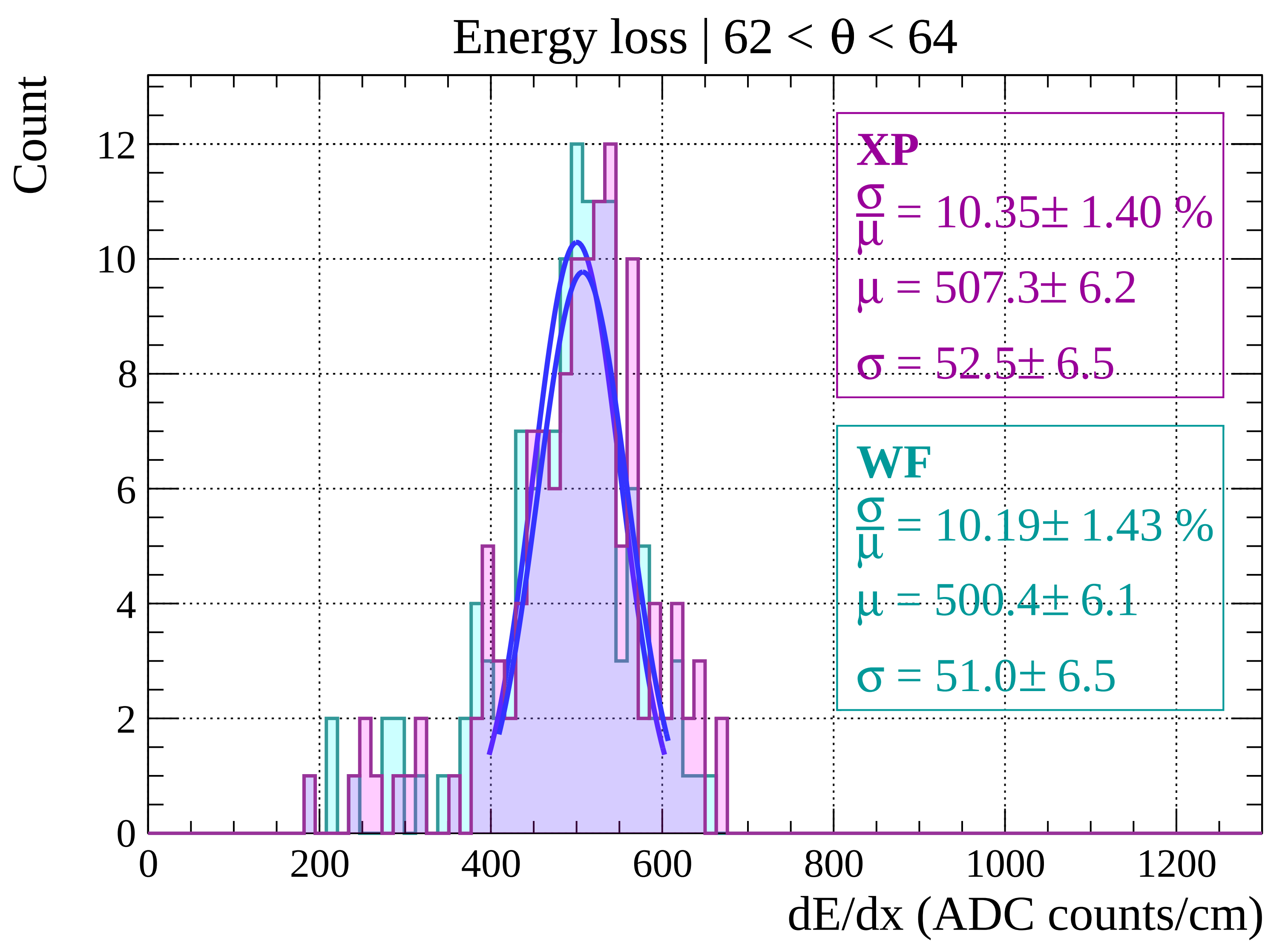


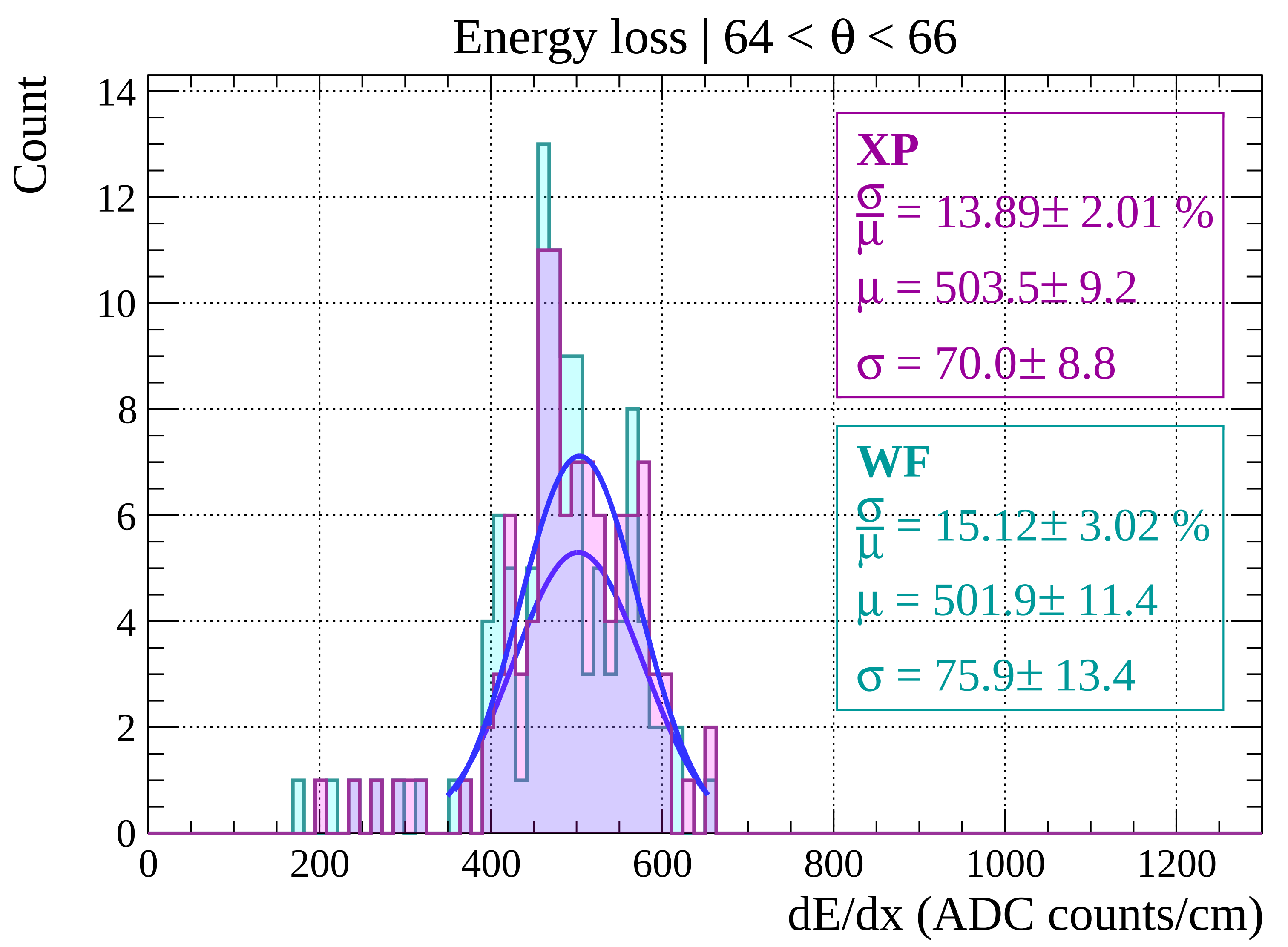
Energy loss | 56 < θ < 58
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.84± 1.18 %
μ = 532.9± 5.2
σ = 52.4± 5.8
WF
σ/μ = 9.87± 1.24 %
μ = 509.1± 5.2
σ = 50.2± 5.8
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
20
22

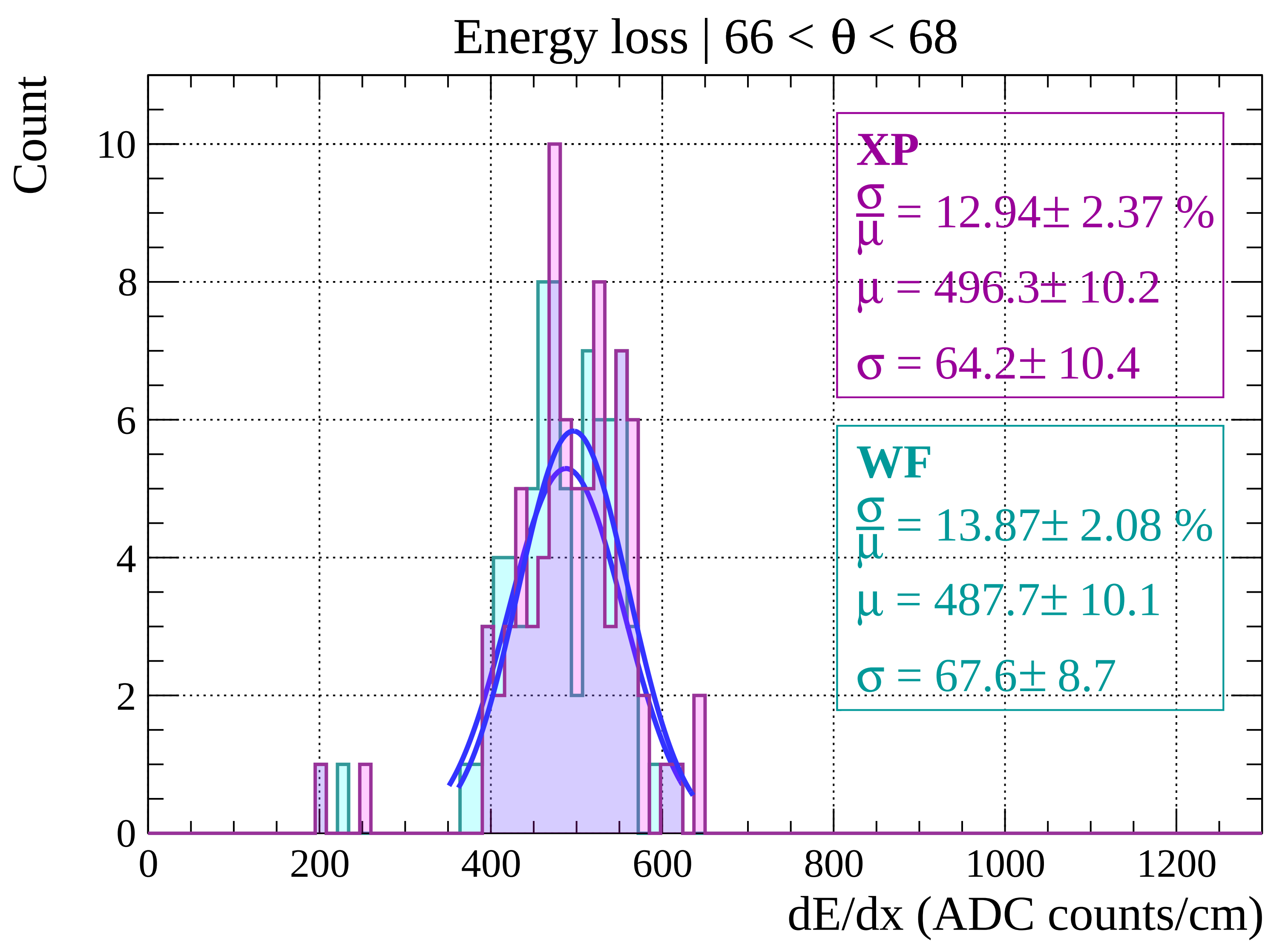


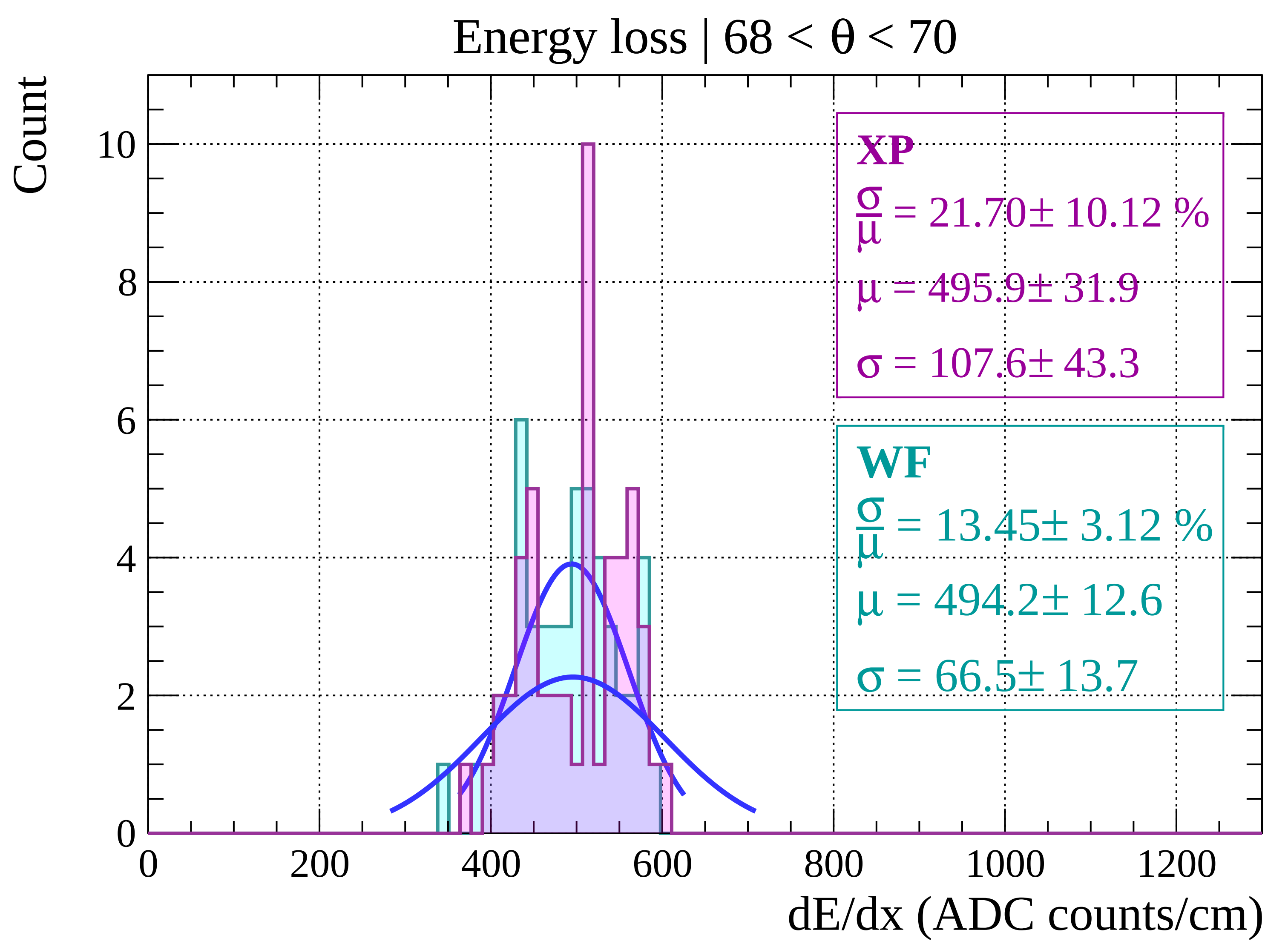
Energy loss | 58 < θ < 60
Count
16
14
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.86± 0.93 %
μ = 520.0± 4.9
σ = 46.0± 4.4
WF
σ/μ = 9.68± 1.09 %
μ = 503.7± 5.5
σ = 48.8± 4.9


Energy loss | 60 < θ < 62
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200
XP
σ/μ = 8.91± 0.93 %
μ = 532.4± 4.5
σ = 47.4± 4.5
WF
σ/μ = 9.91± 1.25 %
μ = 514.7± 5.2
σ = 51.0± 5.9

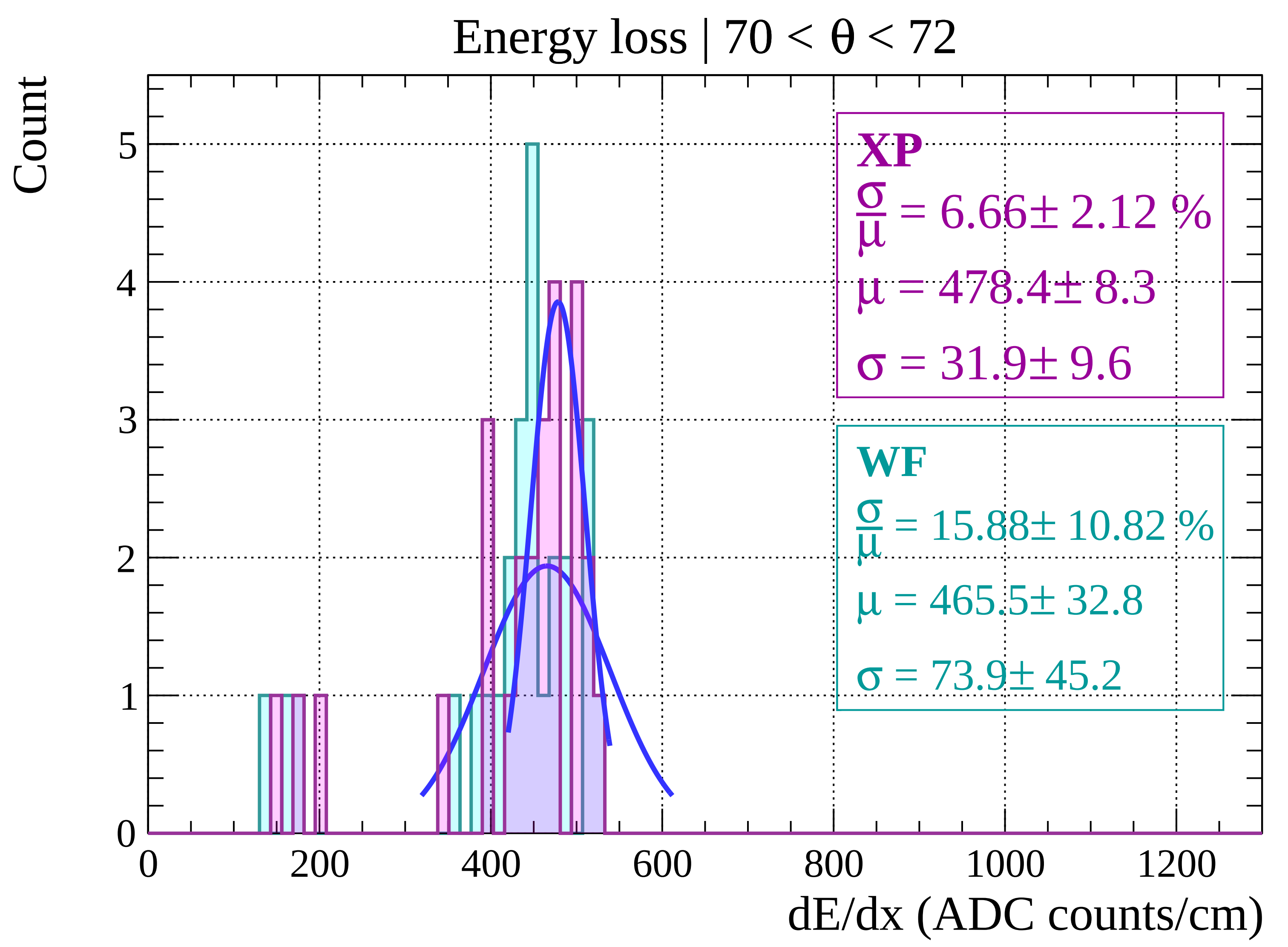

Energy loss | 62 < θ < 64
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 10.35± 1.40 %
μ = 507.3± 6.2
σ = 52.5± 6.5
WF
σ/μ = 10.19± 1.43 %
μ = 500.4± 6.1
σ = 51.0± 6.5


Energy loss | 64 < θ < 66
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 13.89± 2.01 %
μ = 503.5± 9.2
σ = 70.0± 8.8
WF
$\frac{\sigma}{\mu}$ = 15.12± 3.02 %
μ = 501.9± 11.4
σ = 75.9± 13.4

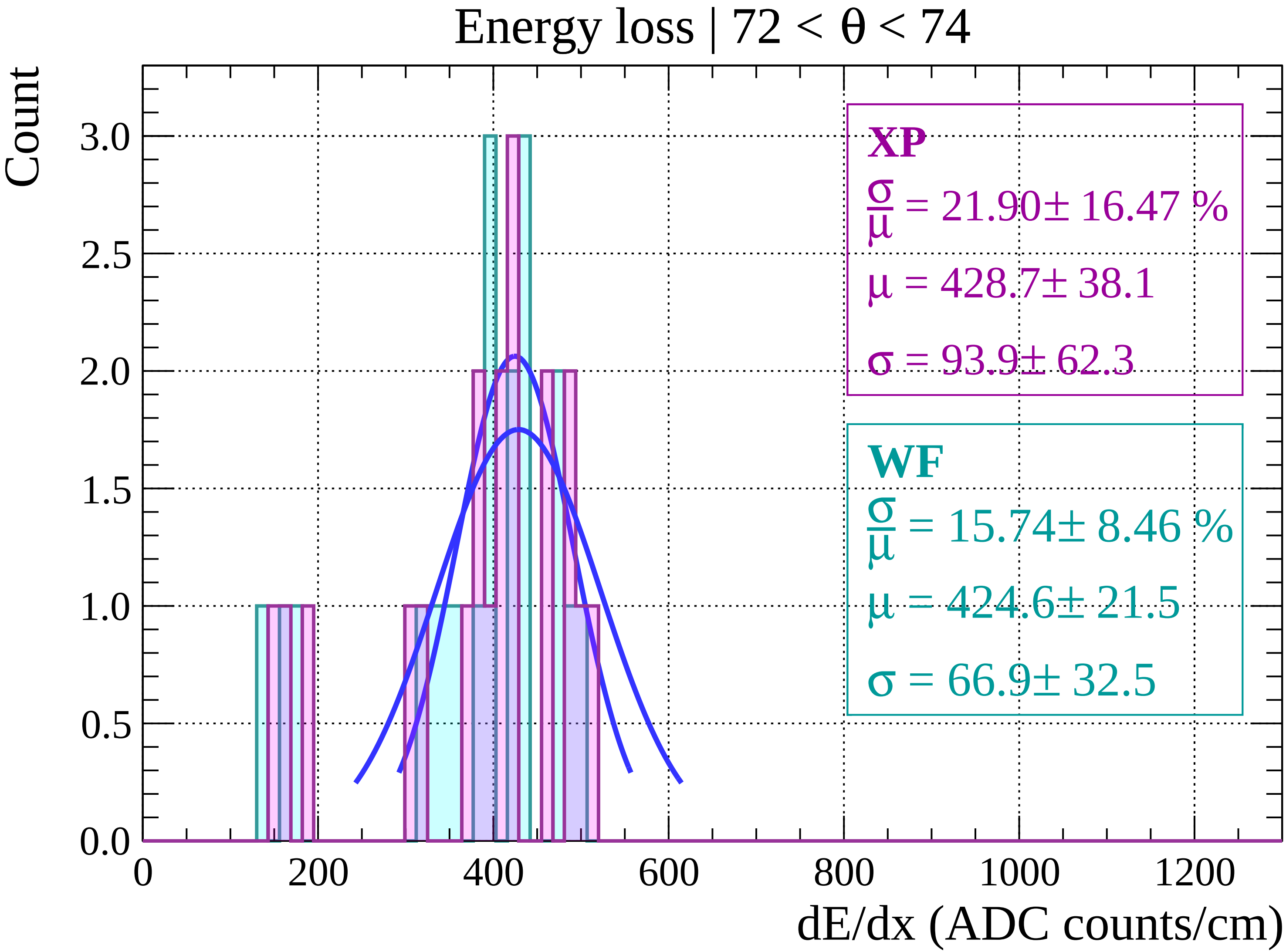

Energy loss | 66 < θ < 68
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
0
200
400
600
800
1000
1200
XP
σ/μ = 12.94± 2.37 %
μ = 496.3± 10.2
σ = 64.2± 10.4
WF
σ/μ = 13.87± 2.08 %
μ = 487.7± 10.1
σ = 67.6± 8.7



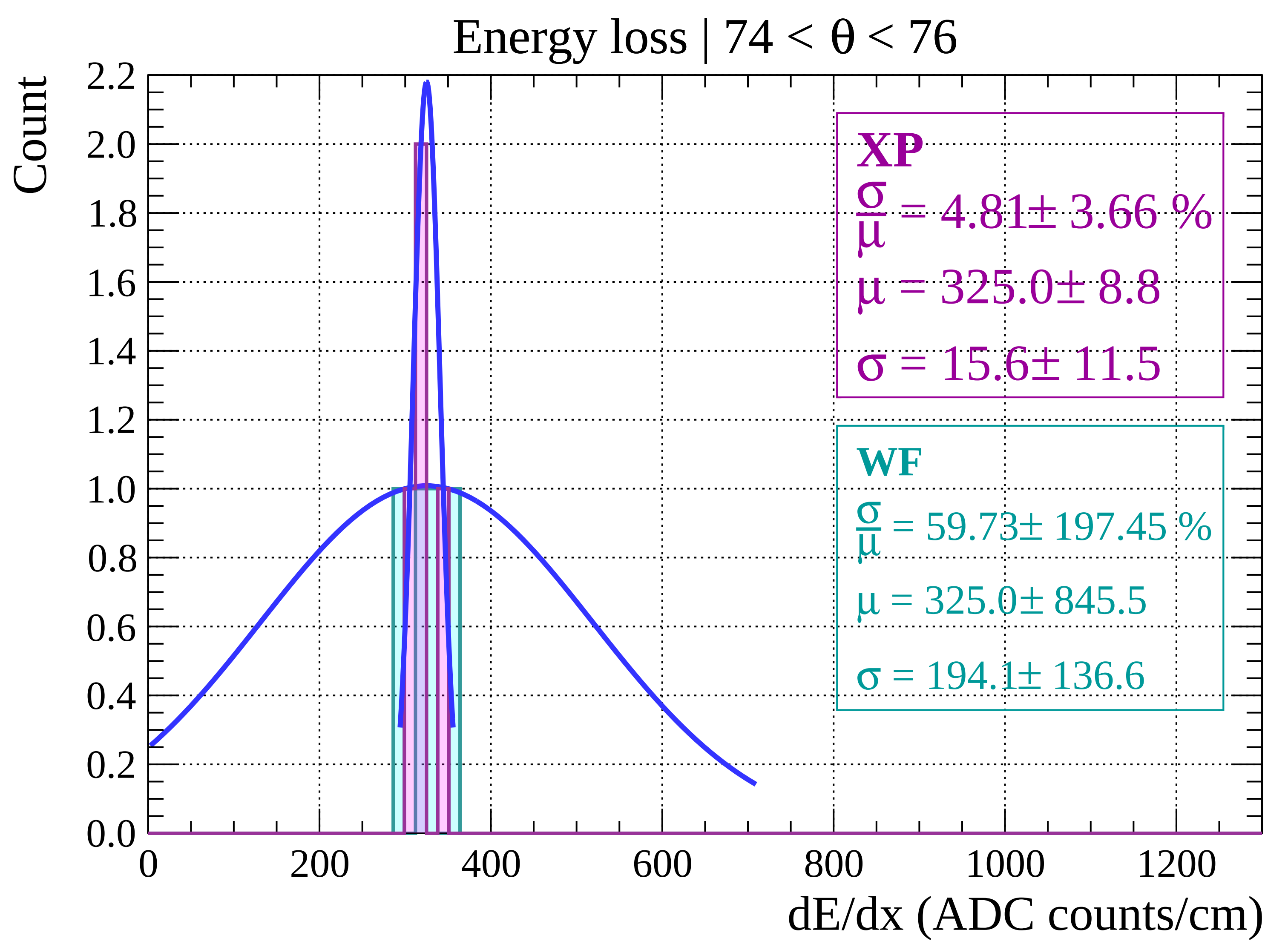
Energy loss | 68 < θ < 70
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
0
200
400
600
800
1000
1200
XP
σ/μ = 21.70± 10.12 %
μ = 495.9± 31.9
σ = 107.6± 43.3
WF
σ/μ = 13.45± 3.12 %
μ = 494.2± 12.6
σ = 66.5± 13.7



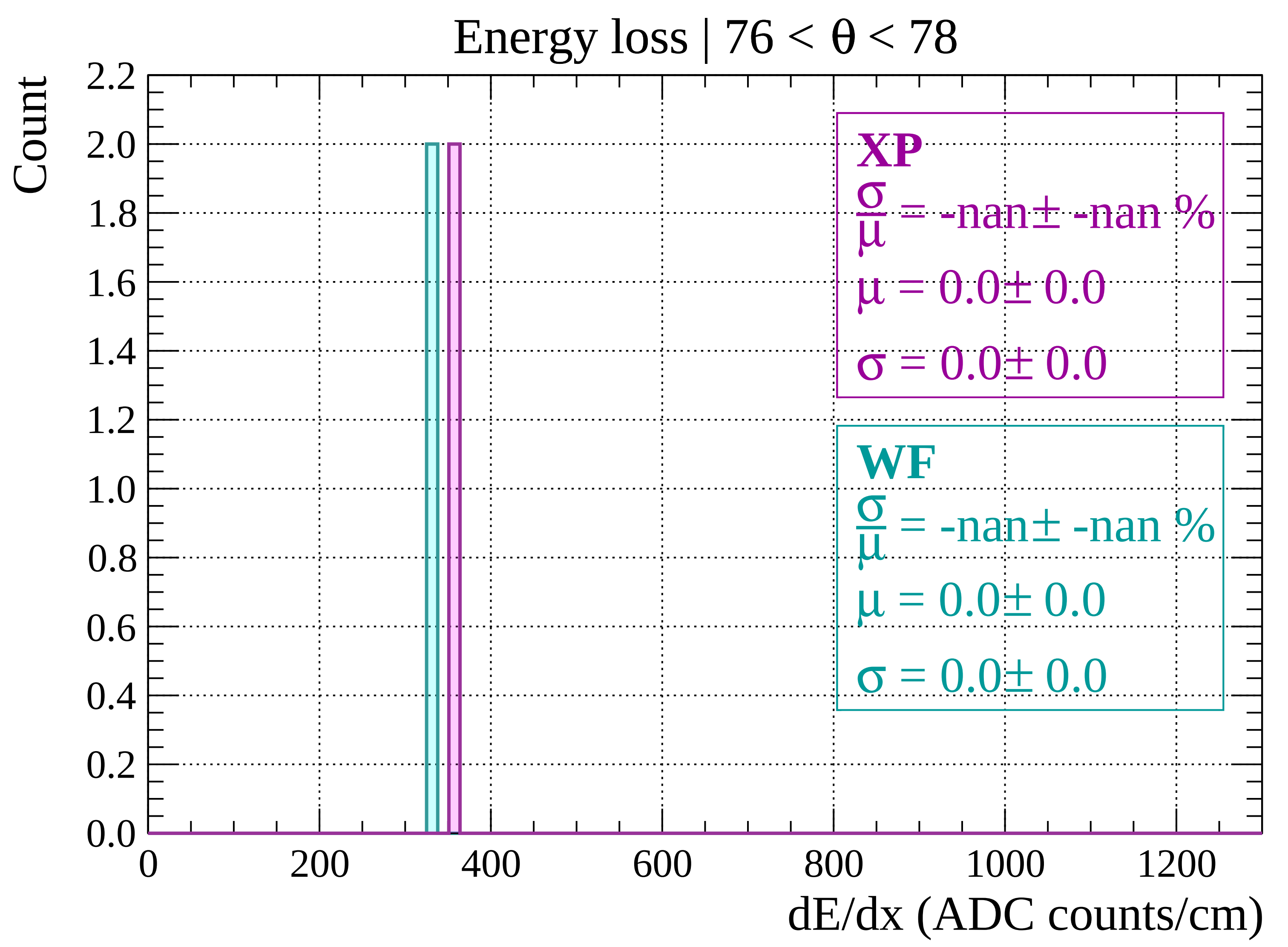
Energy loss | 70 < θ < 72
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
σ/μ = 6.66± 2.12 %
μ = 478.4± 8.3
σ = 31.9± 9.6
WF
σ/μ = 15.88± 10.82 %
μ = 465.5± 32.8
σ = 73.9± 45.2

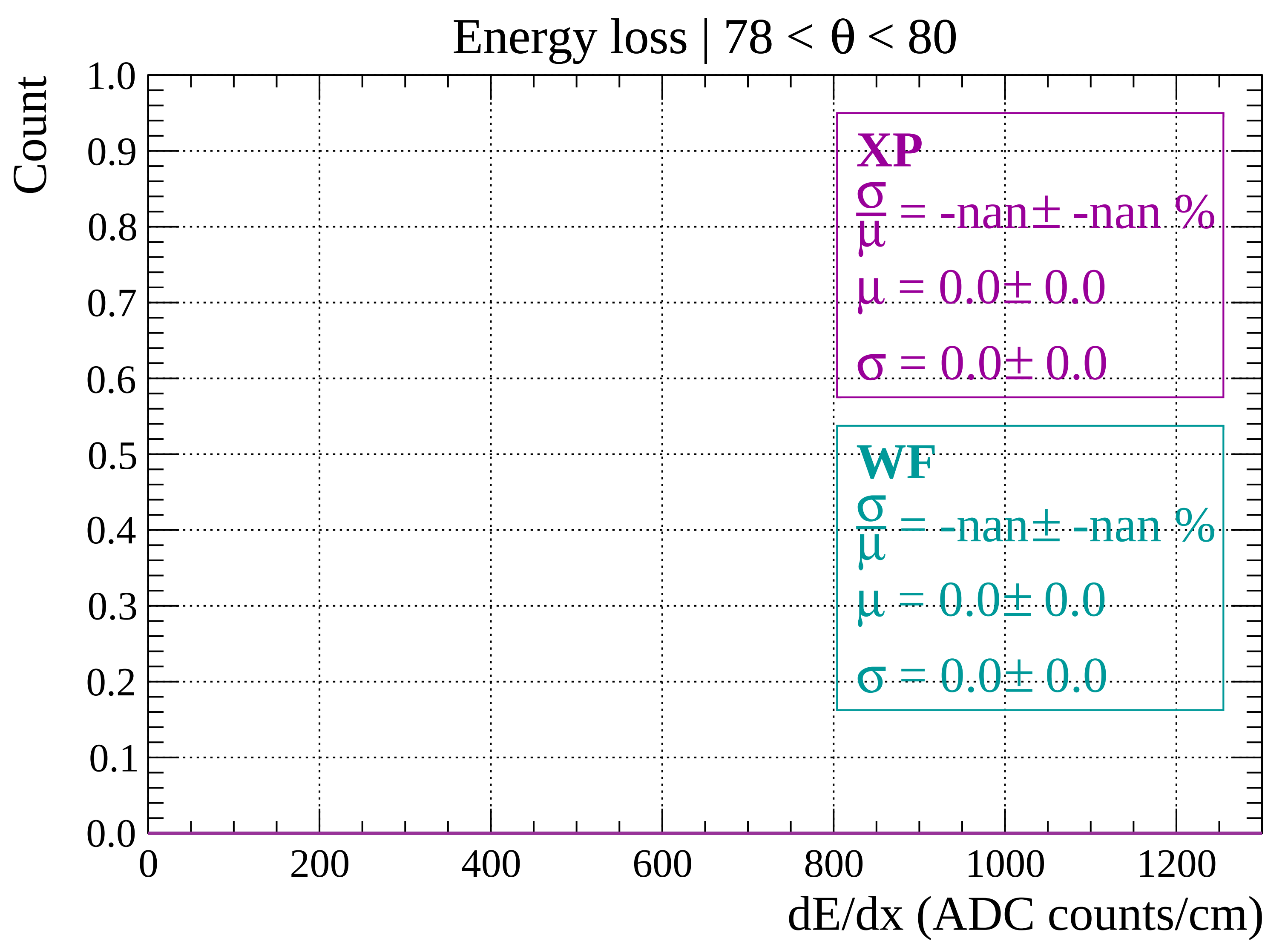


Energy loss | 72 < θ < 74
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 21.90± 16.47 %
μ = 428.7± 38.1
σ = 93.9± 62.3
WF
σ/μ = 15.74± 8.46 %
μ = 424.6± 21.5
σ = 66.9± 32.5

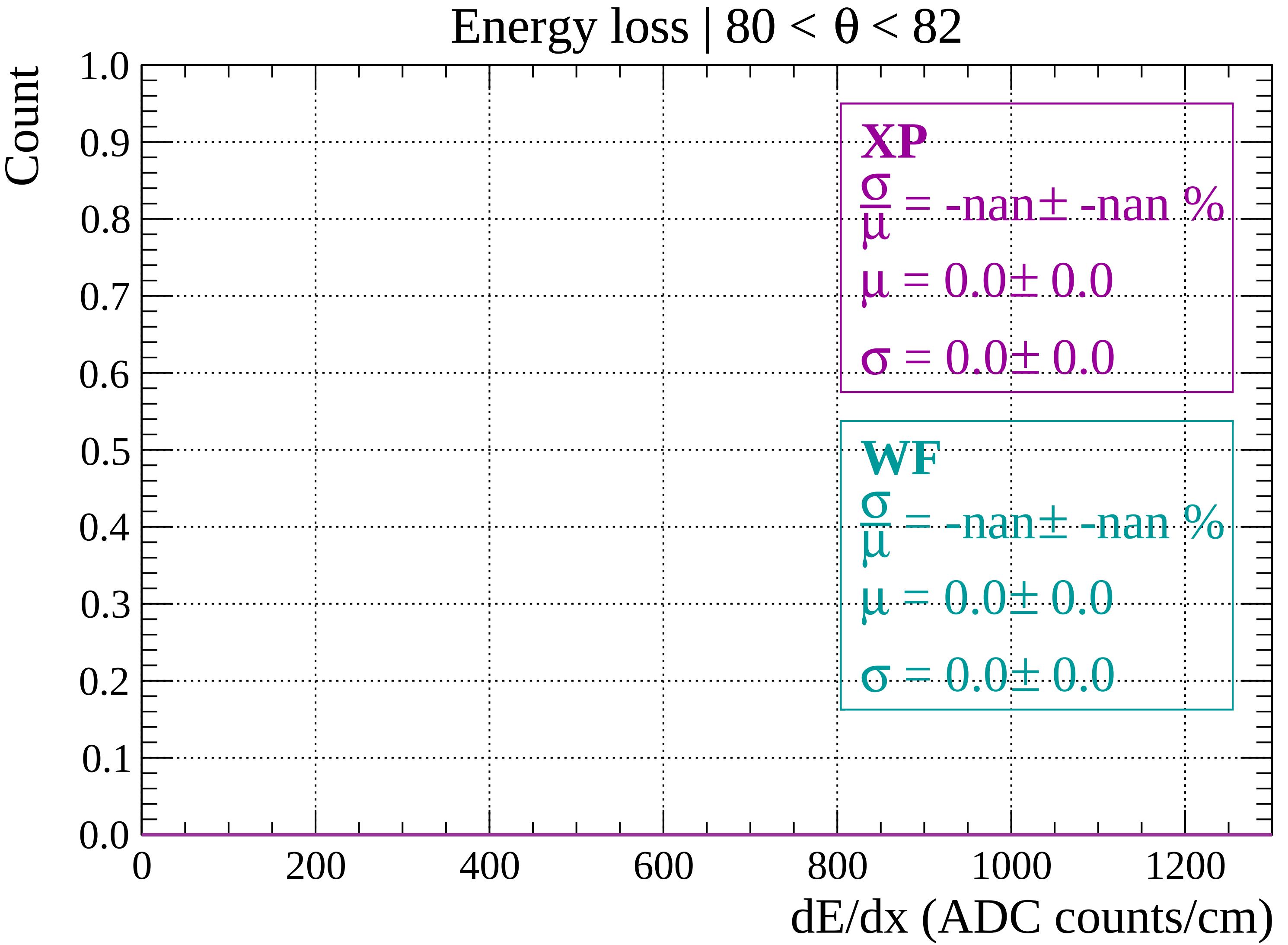


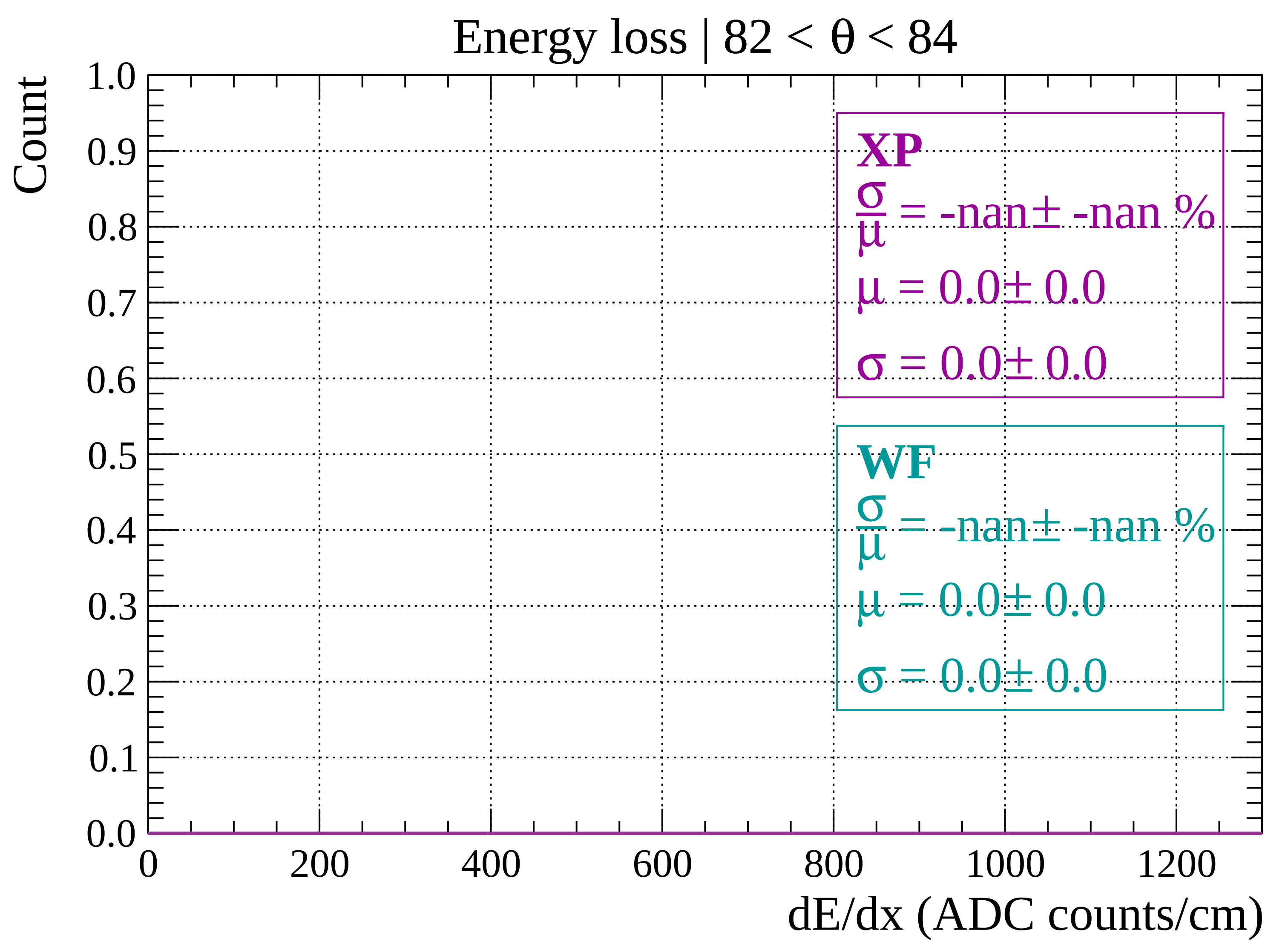
Energy loss | 74 < θ < 76
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 4.81± 3.66 %
μ = 325.0± 8.8
σ = 15.6± 11.5
WF
σ/μ = 59.73± 197.45 %
μ = 325.0± 845.5
σ = 194.1± 136.6

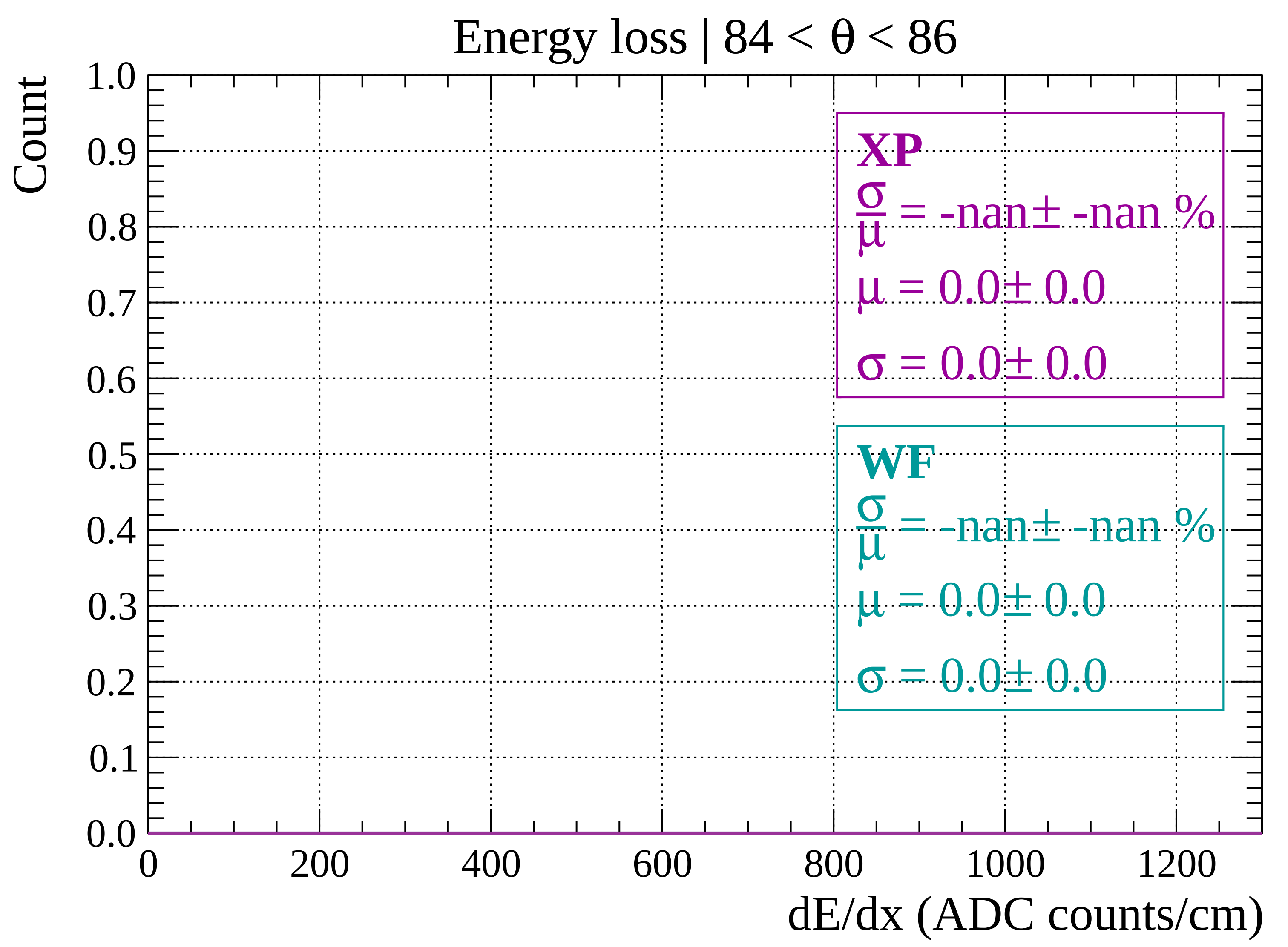

Energy loss | 76 < θ < 78
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



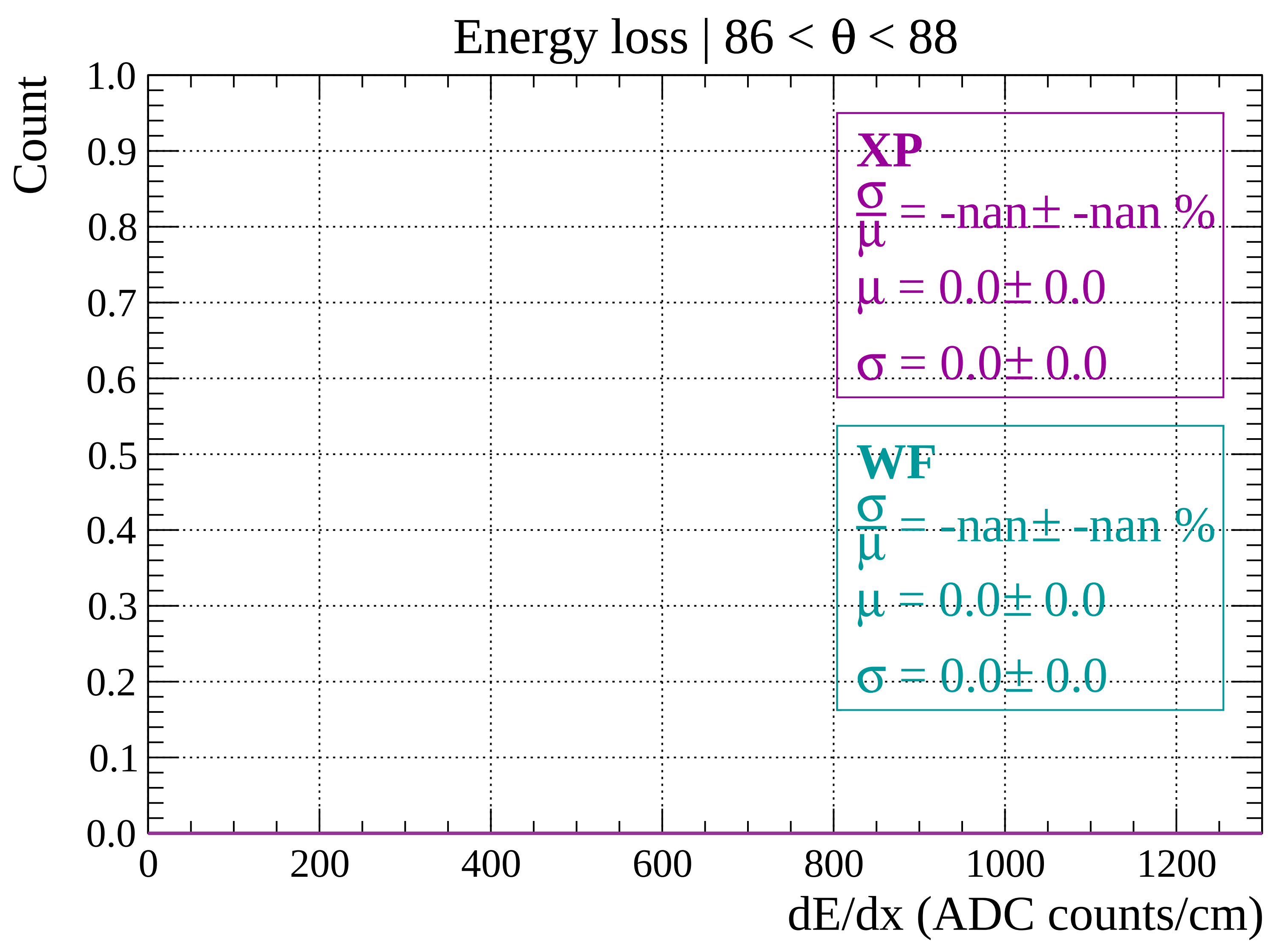
Energy loss | 78 < θ < 80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 80 < θ < 82
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

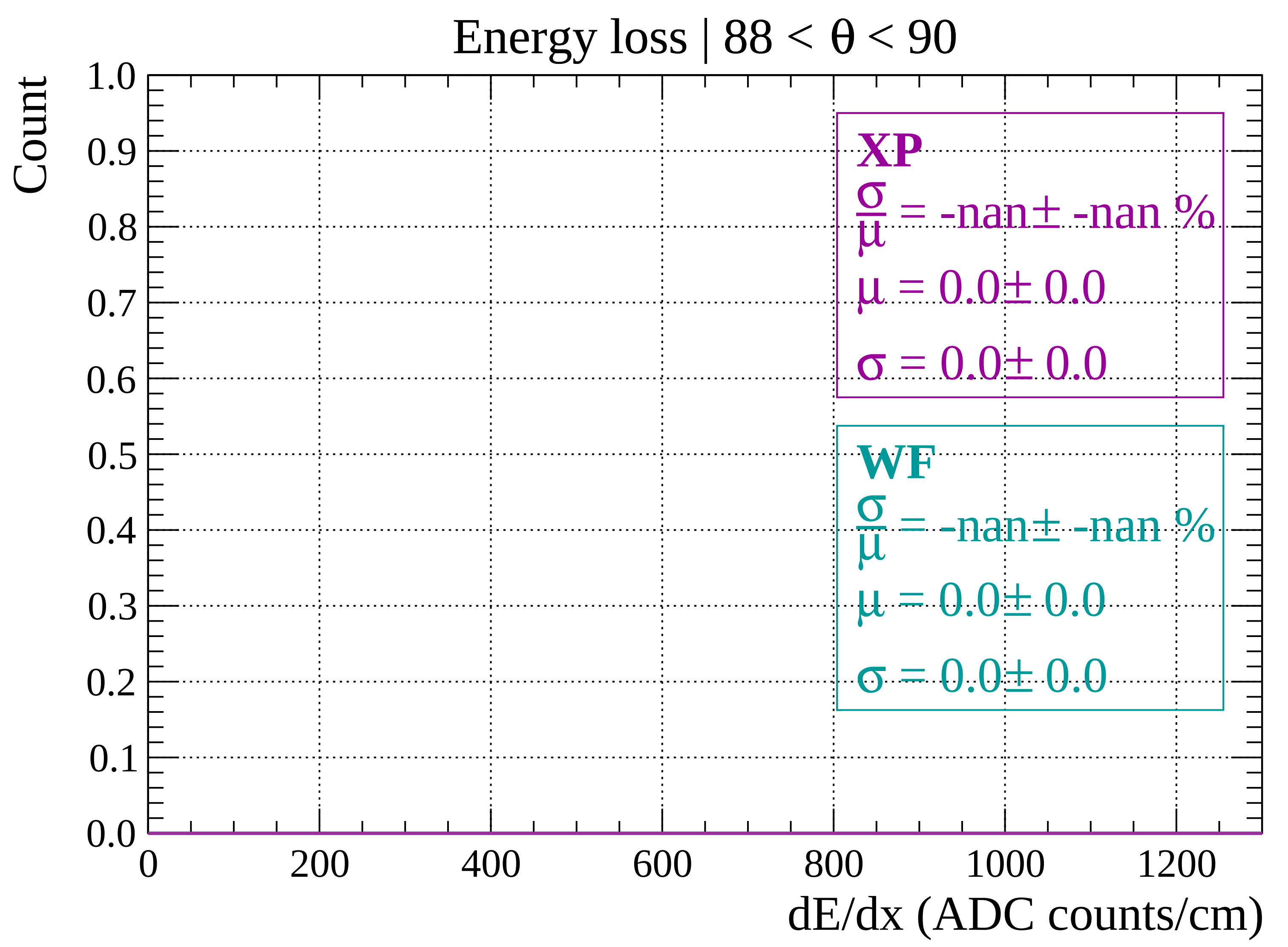

Energy loss | 82 < θ < 84
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200

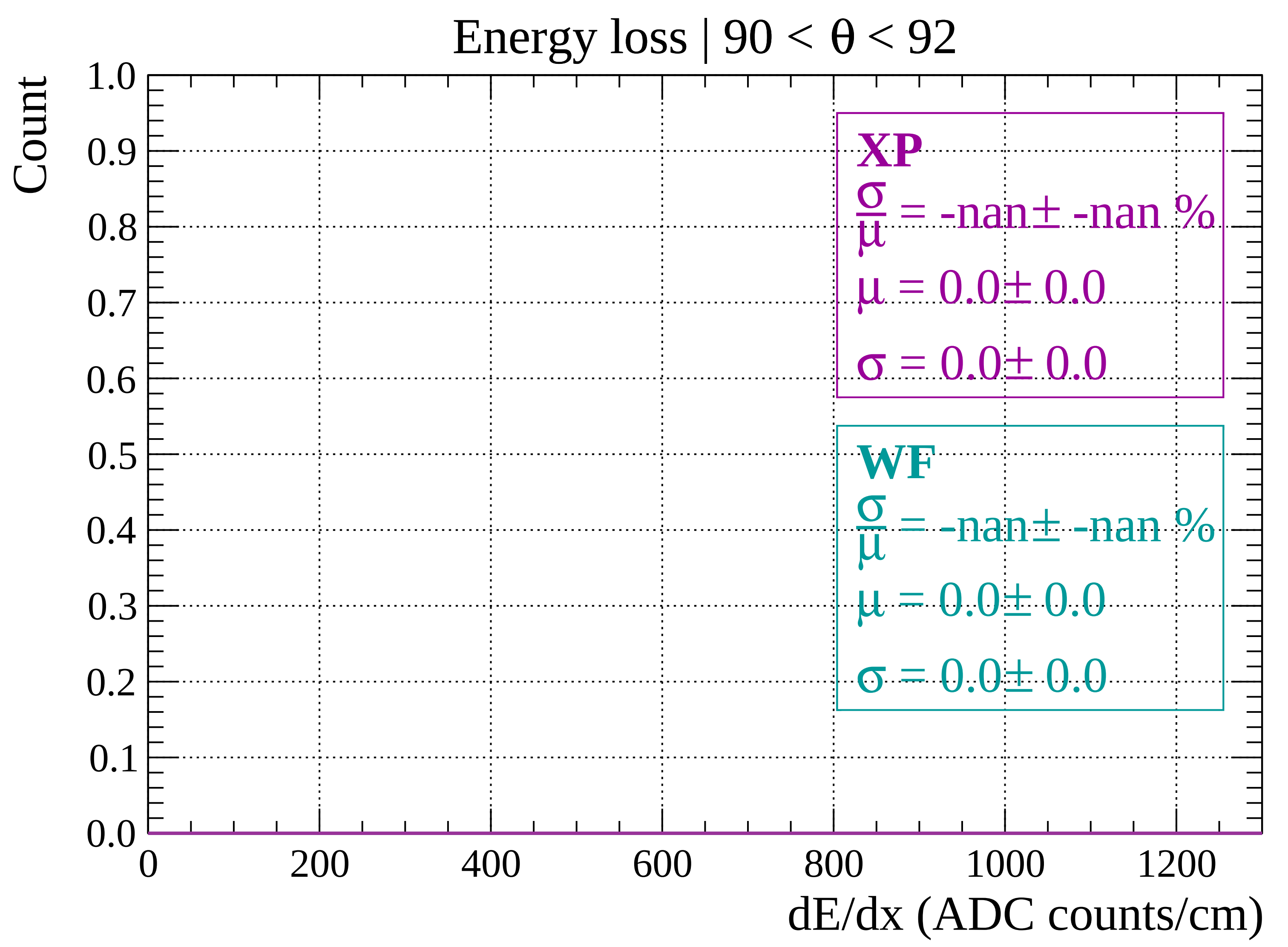


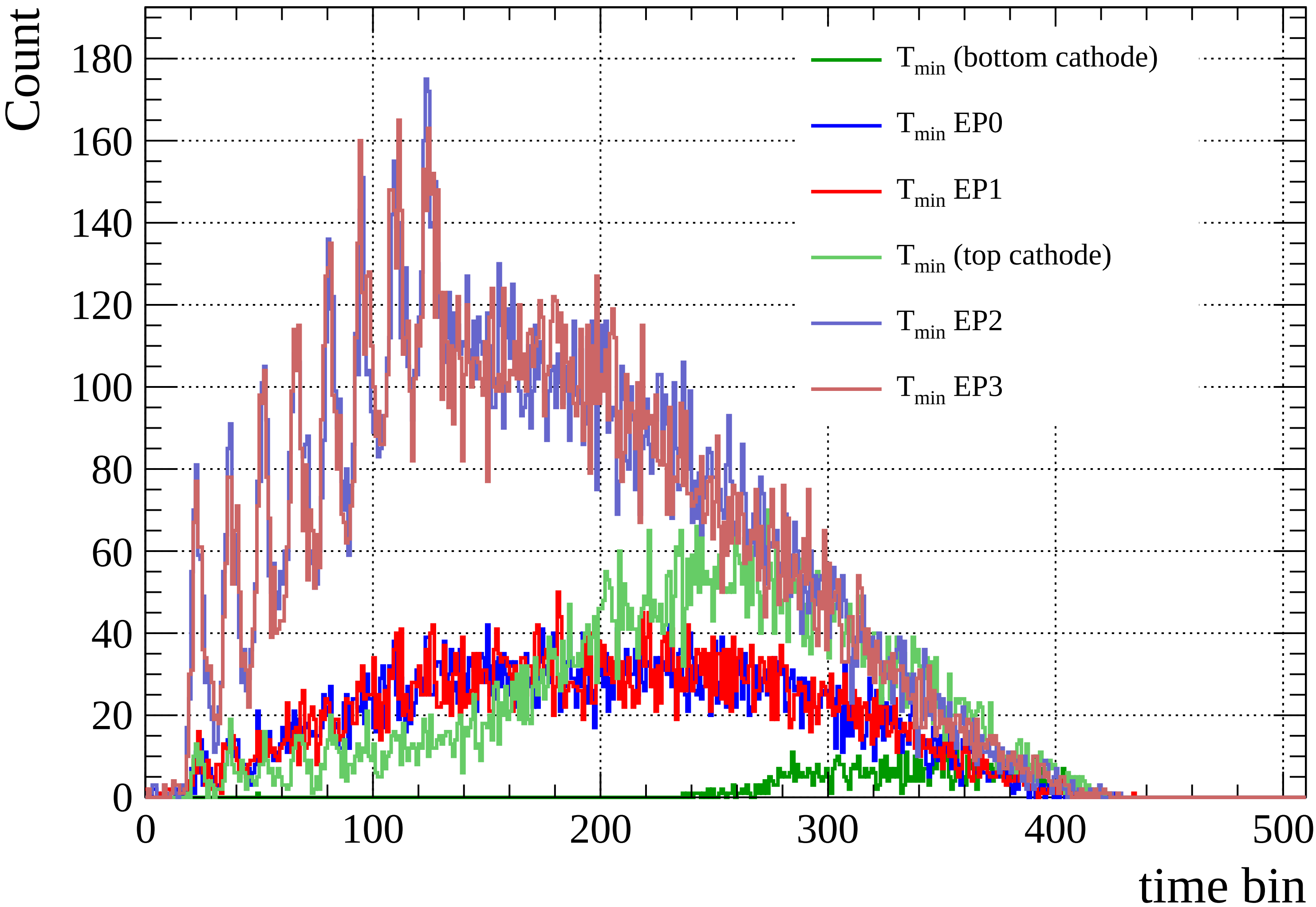
Energy loss | 84 < θ < 86
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



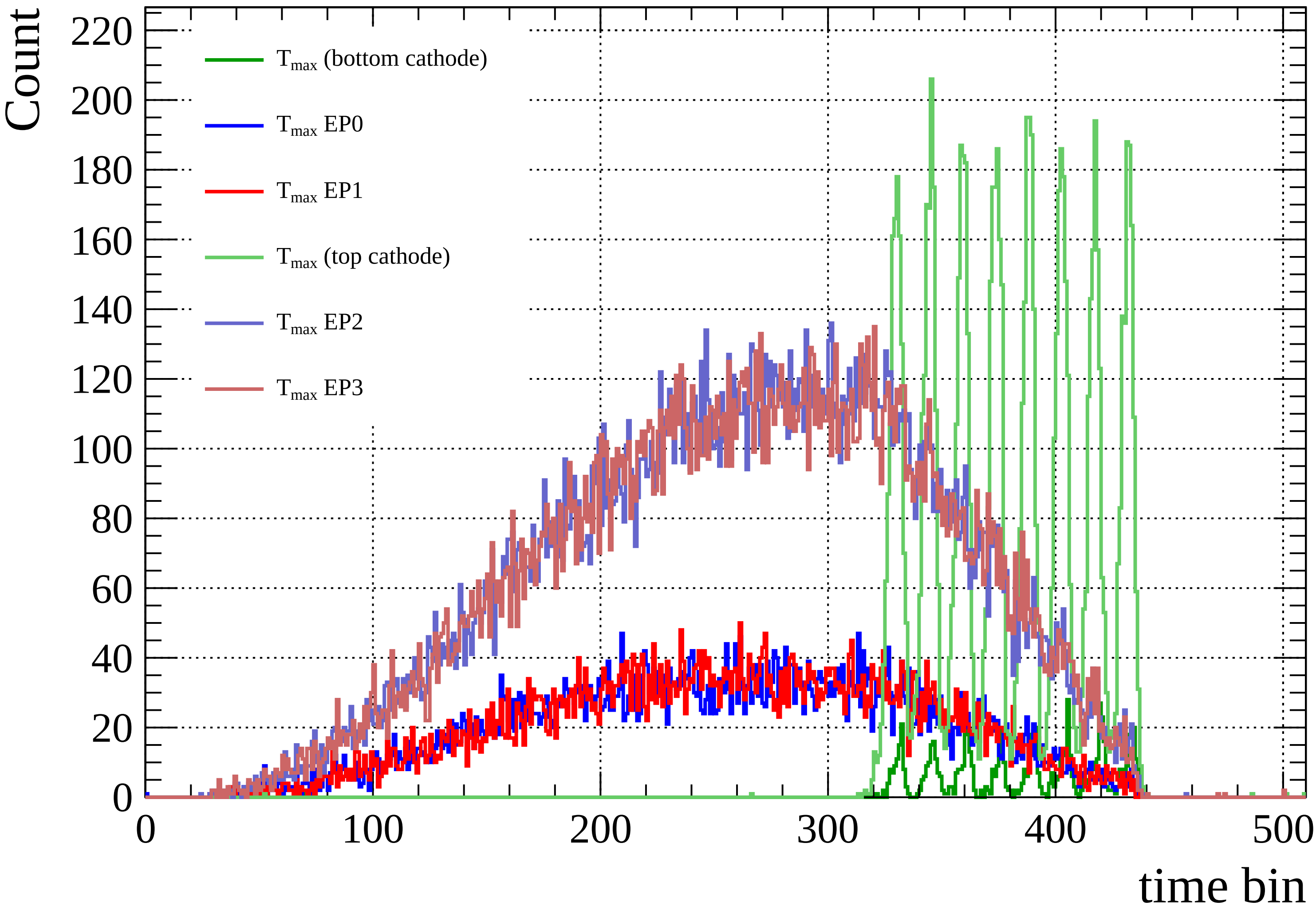
Energy loss | 86 < θ < 88
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 88 < θ < 90
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200


Energy loss | 90 < θ < 92
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
0
20
40
60
80
100
120
140
160
180
0
100
200
300
400
500
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3



Count
time bin
0
20
40
60
80
100
120
140
160
180
200
220
0
100
200
300
400
500
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3